

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, D.C. 20549

FORM 10-K

Annual report pursuant to Section 13 or 15(d) of
the Securities Exchange Act of 1934

For the fiscal year ended
December 31, 2024

Commission file
number 1-5805

JPMorgan Chase & Co.

(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction of
incorporation or organization)

13-2624428
(I.R.S. employer
identification no.)

383 Madison Avenue,
New York, New York
(Address of principal executive offices)

10179
(Zip Code)

Registrant's telephone number, including area code: (212) 270-6000
Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	JPM	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 5.75% Non-Cumulative Preferred Stock, Series DD	JPM PR D	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 6.00% Non-Cumulative Preferred Stock, Series EE	JPM PR C	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 4.75% Non-Cumulative Preferred Stock, Series GG	JPM PR J	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 4.55% Non-Cumulative Preferred Stock, Series JJ	JPM PR K	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 4.625% Non-Cumulative Preferred Stock, Series LL	JPM PR L	The New York Stock Exchange
Depository Shares, each representing a one-four hundredth interest in a share of 4.20% Non-Cumulative Preferred Stock, Series MM	JPM PR M	The New York Stock Exchange
Guarantee of Callable Fixed Rate Notes due June 10, 2032 of JPMorgan Chase Financial Company LLC	JPM/32	The New York Stock Exchange
Guarantee of Alerian MLP Index ETNs due January 28, 2044 of JPMorgan Chase Financial Company LLC	AMJB	NYSE Arca, Inc.

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. ☐ Yes ☒ No

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. ☐ Yes ☒ No

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. ☒ Yes ☐ No

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). ☒ Yes ☐ No

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

☒ Large accelerated filer ☐ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒ Yes ☐ No

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1 (b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). ☐ Yes ☒ No

The aggregate market value of JPMorgan Chase & Co. common stock held by non-affiliates as of June 30, 2024: \$573,443,601,053

Number of shares of common stock outstanding as of January 31, 2025: 2,796,106,099

Documents incorporated by reference: Portions of the registrant's Proxy Statement for the annual meeting of stockholders to be held on May 20, 2025, are incorporated by reference in this Form 10-K in response to Items 10, 11, 12, 13 and 14 of Part III.

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Part I

Item 1. Business.

Overview

JPMorgan Chase & Co. (“JPMorganChase” or the “Firm”, NYSE: JPM), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the United States of America (“U.S.”), with operations worldwide. JPMorganChase had \$4.0 trillion in assets and \$344.8 billion in stockholders’ equity as of December 31, 2024. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Under the J.P. Morgan and Chase brands, the Firm serves millions of customers, predominantly in the U.S., and many of the world’s most prominent corporate, institutional and government clients globally.

JPMorganChase’s principal bank subsidiary is JPMorgan Chase Bank, National Association (“JPMorgan Chase Bank, N.A.”), a national banking association with U.S. branches in 48 states and Washington, D.C. JPMorganChase’s principal non-bank subsidiary is J.P. Morgan Securities LLC (“J.P. Morgan Securities”), a U.S. broker-dealer. The bank and non-bank subsidiaries of JPMorganChase operate nationally as well as through overseas branches and subsidiaries, representative offices and subsidiary foreign banks. The Firm’s principal operating subsidiaries outside the U.S. are J.P. Morgan Securities plc and J.P. Morgan SE (“JPMSE”), which are subsidiaries of JPMorgan Chase Bank, N.A. and are based in the United Kingdom (“U.K.”) and Germany, respectively.

The Firm’s website is www.jpmorganchase.com. JPMorganChase makes available on its website, free of charge, annual reports on Form 10-K, quarterly reports on Form 10-Q and current reports on Form 8-K pursuant to Section 13(a) or Section 15(d) of the Securities Exchange Act of 1934, as soon as reasonably practicable after it electronically files or furnishes such material to the U.S. Securities and Exchange Commission (the “SEC”) at www.sec.gov. JPMorganChase makes new and important information about the Firm available on its website at <https://www.jpmorganchase.com>, including on the Investor Relations section of its website at <https://www.jpmorganchase.com/ir>. Information on the Firm’s website, including documents on the website that are referenced in this Form 10-K, is not incorporated by reference into this Annual Report on Form 10-K for the year ended December 31, 2024 (“2024 Form 10-K” or “Form 10-K”) or the Firm’s other filings with the SEC.

Business segments & Corporate

Effective in the second quarter of 2024, JPMorganChase reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking (“CCB”), Commercial & Investment Bank (“CIB”) and Asset & Wealth Management (“AWM”) – with the remaining activities in Corporate. The Firm’s consumer business segment is CCB, and the Firm’s wholesale business segments are CIB and AWM.

A description of the Firm’s reportable business segments and the products and services that they provide to their respective client bases, as well as a description of Corporate activities, is provided in the Management’s discussion and analysis of financial condition and results of operations section of this Form 10-K (“Management’s discussion and analysis” or “MD&A”) under the heading “Business Segment & Corporate Results,” which begins on page 52, and in Note 32.

Competition

JPMorganChase and its subsidiaries and affiliates operate in highly competitive environments. Competitors include other banks, brokerage firms, investment banking companies, merchant banks, hedge funds, commodity trading companies, private equity firms, insurance companies, mutual fund companies, investment managers, credit card companies, mortgage banking companies, trust companies, securities processing companies, automobile financing companies, leasing companies, e-commerce and other internet-based companies, financial technology companies, and other companies engaged in providing similar and new products and services. The Firm’s businesses generally compete on the basis of the quality and variety of the Firm’s products and services, transaction execution, innovation, reputation and price. Competition also varies based on the types of clients, customers, industries and geographies served. With respect to some of its geographies and products, JPMorganChase competes globally; with respect to others, the Firm competes on a national or regional basis. New competitors in the financial services industry continue to emerge, including firms that offer products and services solely through the internet and non-financial companies that offer products and services that disintermediate traditional banking products and services offered by financial services firms such as JPMorganChase.

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Supervision and regulation

The Firm is subject to extensive and comprehensive regulation under U.S. federal and state laws, as well as the applicable laws of the jurisdictions outside the U.S. in which the Firm does business.

Financial holding company:

Consolidated supervision. JPMorgan Chase & Co. is a bank holding company (“BHC”) and a financial holding company (“FHC”) under U.S. federal law, and is subject to comprehensive consolidated supervision, regulation and examination by the Board of Governors of the Federal Reserve System (the “Federal Reserve”). The Federal Reserve acts as the supervisor of the consolidated operations of BHCs. Certain of JPMorganChase’s subsidiaries are also regulated directly by additional authorities based on the activities or licenses of those subsidiaries.

JPMorganChase’s national bank subsidiary, JPMorgan Chase Bank, N.A., is supervised and regulated by the Office of the Comptroller of the Currency (“OCC”) and, with respect to certain matters, by the Federal Deposit Insurance Corporation (the “FDIC”).

JPMorganChase’s U.S. broker-dealers are supervised and regulated by the Securities and Exchange Commission (“SEC”) and the Financial Industry Regulatory Authority (“FINRA”). Subsidiaries of the Firm that engage in certain futures-related and swaps-related activities are supervised and regulated by the Commodity Futures Trading Commission (“CFTC”). J.P. Morgan Securities plc holds a banking license in the U.K. and is regulated by the U.K. Prudential Regulation Authority (the “PRA”) and the U.K. Financial Conduct Authority (“FCA”).

JPMSE is a Germany-based credit institution jointly regulated by the European Central Bank (“ECB”), the German Financial Supervisory Authority and the German Central Bank, as well as the local regulators in each of the countries in which it operates. The Firm’s other non-U.S. subsidiaries are regulated by the banking, securities, prudential, payments and conduct regulatory authorities, as applicable, in the countries in which they operate.

Permissible business activities. The Bank Holding Company Act restricts BHCs from engaging in business activities other than the business of banking and certain closely-related activities. FHCs are permitted to engage in a broader range of financial activities. The Federal Reserve has the authority to limit an FHC’s ability to conduct otherwise permissible activities if the FHC or any of its depository institution subsidiaries ceases to meet applicable eligibility requirements. The Federal Reserve may also impose corrective capital and/or managerial requirements on the FHC, and if deficiencies are persistent, may require divestiture of the FHC’s depository institutions. If any

depository institution controlled by an FHC fails to maintain a satisfactory rating under the Community Reinvestment Act, the Federal Reserve must prohibit the FHC and its subsidiaries from engaging in any new activities other than those permissible for BHCs, or acquiring a company engaged in such activities.

Capital and liquidity requirements. The Federal Reserve establishes capital, liquidity and leverage requirements for JPMorganChase that are generally consistent with the international Basel III capital and liquidity framework and evaluates the Firm’s compliance with those requirements. The OCC establishes similar requirements for JPMorgan Chase Bank, N.A. Certain of the Firm’s non-U.S. subsidiaries and branches are also subject to local capital and liquidity requirements.

Banking supervisors globally continue to refine and enhance the Basel III capital framework for financial institutions. In July 2023, U.S. banking regulators released a proposal to amend the U.S. risk-based capital framework to incorporate certain elements of the revised international Basel III capital framework. The proposal would significantly revise risk-based capital requirements for banks with assets of \$100 billion or more, including the Firm and other U.S. global systemically important banks (“GSIBs”). Finalization of the proposal, including the required implementation date, is uncertain. The Firm continues to monitor developments and potential impacts.

In the EU and U.K., regulators have finalized the rules implementing their Basel III frameworks. The new rules became effective in the EU beginning January 1, 2025, with market risk aspects delayed until January 1, 2026. In January 2025, the PRA announced that it intends to delay the implementation of the new rules in the U.K. to January 1, 2027. There are certain transitional arrangements applicable in both the EU and U.K. until 2032 and 2030, respectively.

Stress tests. As a large BHC, JPMorganChase is subject to supervisory stress testing administered by the Federal Reserve as part of the Federal Reserve’s annual Comprehensive Capital Analysis and Review (“CCAR”) framework. The Firm must conduct annual company-run stress tests and must also submit an annual capital plan to the Federal Reserve, taking into account the results of separate stress tests designed by each of the Firm and the Federal Reserve. The Federal Reserve uses the results under the severely adverse scenario from its supervisory stress test to determine the Firm’s Stress Capital Buffer (“SCB”) requirement for the coming year, which forms part of the Firm’s applicable capital buffers. The Firm is required to file its annual CCAR submission on April 5, 2025. The Federal Reserve will notify the Firm of its indicative SCB requirement by June 30, 2025 and final SCB requirement by August 31, 2025. The Firm’s final

SCB requirement will become effective on October 1, 2025. The OCC requires JPMorgan Chase Bank, N.A. to perform separate, similar stress tests annually. The Firm publishes each year the results of the annual stress tests for the Firm and JPMorgan Chase Bank, N.A. under the supervisory “severely adverse” scenarios provided by the Federal Reserve and the OCC. In December 2024, the Federal Reserve indicated in a press release that it intends to seek public comment on changes to its stress testing framework. Additionally, there is a pending legal challenge to the manner in which stress testing is administered. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Refer to Capital Risk Management on pages 97–107 and Liquidity Risk Management on pages 108–115 for more information.

Enhanced prudential standards. As part of its mandate to identify and monitor risks to the financial stability of the U.S. posed by large banking organizations, the Financial Stability Oversight Council (“FSOC”) recommends prudential standards and reporting requirements to the Federal Reserve for systemically important financial institutions (“SIFIs”), such as JPMorganChase. The Federal Reserve has adopted several rules to implement those heightened prudential standards, including rules relating to risk management and corporate governance of subject BHCs. JPMorganChase is required under these rules to comply with enhanced liquidity and overall risk management standards, including oversight by the board of directors of risk management activities.

Holding company as a source of strength. JPMorgan Chase & Co. is required to serve as a source of financial strength for its depository institution subsidiaries and to commit resources to support those subsidiaries, including when directed to do so by the Federal Reserve.

Regulation of acquisitions. Acquisitions by BHCs and their banks are subject to requirements, limitations and prohibitions established by law and by the Federal Reserve and the OCC. For example, FHCs and BHCs are required to obtain the approval of the Federal Reserve before they acquire more than 5% of the voting shares of an unaffiliated bank. In addition, acquisitions by financial companies are generally prohibited if, as a result of the acquisition, the total liabilities of the financial company would exceed 10% of the total liabilities of all financial companies, as determined under Federal Reserve regulations. Furthermore, for certain acquisitions, the Firm must provide written notice to the Federal Reserve prior to acquiring direct or indirect ownership or control of any voting shares of any company with over \$10 billion in assets that is engaged in activities that are “financial in nature.” Moreover, while FHCs may engage in a

broader range of activities (including acquisitions) than BHCs, the Federal Reserve has the authority to limit an FHC’s ability to conduct otherwise permissible acquisitions if the FHC or any of its depository institution subsidiaries ceases to meet applicable eligibility requirements.

Ongoing obligations. The Firm is subject to a five-year cooperation obligation under an order issued by the CFTC on September 29, 2020, relating to precious metals and U.S. Treasuries markets investigations. The Firm also remains subject to consent orders entered into in March 2024 with the OCC and the Board of Governors of the Federal Reserve System, and a resolution entered into in May 2024 with the CFTC, which relate to the Firm’s processes to inventory trading venues and confirm the completeness of certain data fed to trade surveillance platforms.

Subsidiary banks:

The activities of JPMorgan Chase Bank, N.A., the Firm’s principal subsidiary bank, are limited to those specifically authorized under the National Bank Act and related interpretations of the OCC. The OCC has authority to bring an enforcement action against JPMorgan Chase Bank, N.A. for unsafe or unsound banking practices, which could include limiting JPMorgan Chase Bank, N.A.’s ability to conduct otherwise permissible activities, or imposing corrective capital or managerial requirements on the bank.

FDIC deposit insurance. The FDIC deposit insurance fund provides insurance coverage for certain deposits and is funded through assessments on banks, including JPMorgan Chase Bank, N.A. The FDIC is required to maintain a minimum reserve ratio, which measures the balance of reserves in the deposit insurance fund against an estimate of FDIC-insured deposits, of 1.35%. The reserve ratio is currently below the statutory minimum and, in October 2022, the FDIC adopted a final rule to raise bank assessments and accelerate the time by which the reserve ratio would meet the statutory minimum. As a result, the FDIC has adopted a restoration plan to bring the reserve ratio up to the required 1.35% by September 30, 2028, with a longer-term target of maintaining a reserve ratio of 2%.

FDIC powers upon a bank insolvency. Upon any insolvency of JPMorgan Chase Bank, N.A., the FDIC could be appointed as conservator or receiver under the Federal Deposit Insurance Act. The FDIC has broad powers to transfer assets and liabilities without the approval of the institution’s creditors.

Prompt corrective action. The Federal Deposit Insurance Corporation Improvement Act of 1991 requires the relevant federal banking regulator to take “prompt corrective action” with respect to a depository institution if that institution does not meet certain

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capital adequacy standards. The Federal Reserve is also authorized to take appropriate action against the parent BHC, such as JPMorgan Chase & Co., based on the undercapitalized status of any bank subsidiary. In certain instances, the BHC would be required to guarantee the performance of the capital restoration plan for its undercapitalized subsidiary.

Heightened Supervisory Standards. In the U.S., the OCC has established guidelines setting forth heightened standards for large banks, including minimum standards for the design and implementation of a risk governance framework for banks. Under these standards, a bank's risk governance framework must ensure that the bank's risk profile is easily distinguished and separate from that of its parent BHC for risk management purposes. The bank's board or risk committee is responsible for approving the bank's risk governance framework, providing active oversight of the bank's risk-taking activities, and holding management accountable for adhering to the risk governance framework.

The Firm's banking entities in the EU and the U.K. are subject to supervisory expectations published by the ECB and the PRA, respectively, addressing bank strategy, governance and risk management in the areas of climate change, operational resilience, reliance on IT systems and third-party services, and resilience from macro-financial and geopolitical shocks.

Restrictions on transactions with affiliates. JPMorgan Chase Bank, N.A. and its subsidiaries are subject to restrictions imposed by federal law on extensions of credit to, investments in stock or securities of, and derivatives, securities lending and certain other transactions with, JPMorgan Chase & Co. and certain other affiliates. These restrictions prevent JPMorgan Chase & Co. and other affiliates from borrowing from JPMorgan Chase Bank, N.A. and its subsidiaries unless the loans are secured in specified amounts and comply with certain other requirements.

Dividend restrictions. Federal law imposes limitations on the payment of dividends by national banks, such as JPMorgan Chase Bank, N.A. Refer to Note 26 for the amount of dividends that JPMorgan Chase Bank, N.A. could pay, at January 1, 2025, to JPMorganChase without the approval of the banking regulators. The OCC and the Federal Reserve also have authority to prohibit or limit the payment of dividends of a bank subsidiary that they supervise if, in the banking regulator's opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the bank.

Depositor preference. Under federal law, the claims of a receiver of an insured depository institution ("IDI") for administrative expense and the claims of holders of

U.S. deposit liabilities (including the FDIC and deposits in non-U.S. branches that are dually payable in the U.S. and in a non-U.S. branch) have priority over the claims of other unsecured creditors of the institution, including depositors in non-U.S. branches and public noteholders.

Consumer supervision and regulation.

JPMorganChase and JPMorgan Chase Bank, N.A. are subject to supervision and regulation in the U.S. by the Consumer Financial Protection Bureau ("CFPB") with respect to federal consumer protection laws, including laws relating to fair lending and the prohibition of unfair, deceptive or abusive acts or practices in connection with the offer, sale or provision of consumer financial products and services. The CFPB also has jurisdiction over small business lending activities with respect to fair lending and the Equal Credit Opportunity Act. As part of its regulatory oversight, the CFPB has authority to take enforcement actions against firms that offer certain products and services to consumers using practices that are deemed to be unfair, deceptive or abusive. In March 2024, the CFPB released a final rule which significantly reduces the late payment fees that large credit card issuers, including the Firm, are permitted to charge to customers (the "CFPB Late Fee Rule"). The final rule is currently stayed pending resolution of an industry-led challenge in federal court. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

In addition, in October 2024, the CFPB issued a final rule that requires data providers, including banks, to make certain consumer data available to consumers and authorized third parties in electronic form beginning in April 2026 (the "CFPB Data Sharing Rule"). Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Separately, in December 2024, the CFPB announced a final rule that would significantly restrict overdraft fees for certain insured depository institutions, including the Firm (the "CFPB Overdraft Rule"). The rule imposes certain requirements on overdraft protections that are similar to those that apply to credit cards, unless the institution prices the overdraft fee at \$5 per transaction or the institution's cost as outlined by the CFPB. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

In October 2023, the Federal Reserve Board proposed to lower the maximum interchange fee that large debit card issuers, including the Firm, would be permitted to receive for a debit card transaction. The proposal would also establish a process for automatically publishing an updated maximum fee amount every other year going forward. The Firm's consumer activities are also subject to regulation under state

statutes which are enforced by the Attorney General or empowered agency of each state.

In the U.K., the Firm operates a retail bank through J.P. Morgan Europe Limited (“JPME”) and provides retail investment management services through Nutmeg Saving and Investment Limited (“Nutmeg”). JPME is regulated by the PRA, and both JPME and Nutmeg are regulated by the FCA with respect to their conduct of financial services in the U.K., including obligations relating to the fair treatment of customers. JPME is also regulated by the U.K. Payment Systems Regulator with respect to its operation and use of payment systems. In addition, the retail businesses of JPME and Nutmeg are subject to U.K. consumer-protection legislation.

Securities and broker-dealer regulation:

The Firm conducts securities underwriting, dealing and brokerage activities in the U.S. through J.P. Morgan Securities LLC and other non-bank broker-dealer subsidiaries, all of which are subject to regulations of the SEC, FINRA and the New York Stock Exchange, among others. The Firm conducts similar securities activities outside the U.S. subject to local regulatory requirements. In the U.K., those activities are primarily conducted by J.P. Morgan Securities plc and in the EU, those activities are primarily conducted by JPMSE. Broker-dealers are subject to laws and regulations covering all aspects of the securities business, including sales and trading practices, securities offerings, publication of research reports, use of customer funds, the financing of client purchases, capital structure, record-keeping and retention, and the conduct of their directors, officers and employees. Refer to Broker-dealer regulatory capital on page 107 for information concerning the capital of J.P. Morgan Securities LLC, J.P. Morgan Securities plc and JPMSE. In addition, the Firm's sales and trading activities, which are conducted through both bank and non-bank subsidiaries, are subject to laws and regulations relating to market conduct, including prohibitions on manipulative or anti-competitive practices.

Investment management regulation:

The Firm's asset and wealth management businesses are subject to significant regulation in jurisdictions around the world relating to, among other things, the safeguarding and management of client assets, offerings of funds and marketing activities. Certain of the Firm's subsidiaries are registered with, and subject to oversight by, the SEC as investment advisers and broker-dealers. The Firm's registered investment advisers in the U.S. are subject to the fiduciary and other obligations imposed under the Investment Advisers Act of 1940 and applicable state and federal law. The Firm's bank fiduciary activities are subject to supervision by the OCC.

The Firm's asset and wealth management businesses continue to be subject to ongoing rule-making and implementation of new regulations and other guidance, including by the SEC and certain U.S. states with respect to enhanced standards of conduct and conflicts of interest. In April 2024, the Department of Labor (“DOL”) finalized a new “fiduciary” rule that would significantly expand the scope for defining who can be deemed investment advice fiduciaries for purposes of retirement plans and individual retirement accounts (“IRAs”) under the Employee Retirement Income Security Act of 1974, as amended (the “Fiduciary Rule”). Among the most significant impacts of the rule and related amendments to prohibited transaction exemptions would be the impact on the fee and compensation practices at financial institutions that offer investment recommendations to retirement clients, including in the context of rollovers from an employer plan to an IRA. The effective date of the Fiduciary Rule has been stayed by two federal courts. Refer to Litigation and regulatory challenges on pages 6-7 for further information.

Derivatives regulation:

The Firm is subject to comprehensive regulation of its derivatives businesses. In the U.S., JPMorgan Chase Bank, N.A., J.P. Morgan Securities LLC and J.P. Morgan Securities plc are registered with the CFTC as “swap dealers”. In addition, JPMorgan Chase Bank, N.A. and J.P. Morgan Securities LLC are registered with the SEC as “security-based swap dealers”. As a result, these entities are subject to a comprehensive regulatory framework applicable to their swap or security-based swap activities, including capital requirements, rules requiring the collateralization of uncleared swaps and security-based swaps, rules regarding segregation of counterparty collateral, business conduct and documentation standards, rules requiring the central clearing of standardized over-the-counter (“OTC”) derivatives, requirements that certain standardized OTC swaps be traded on regulated trading venues, record-keeping and reporting obligations, and anti-fraud and anti-manipulation requirements. Similar requirements have also been established in the European Union (“EU”) under the European Market Infrastructure Regulation (“EMIR”) and the Markets in Financial Instruments Directive (“MiFID II”), as well as in the U.K. and other jurisdictions around the world.

J.P. Morgan Securities LLC is also registered with the CFTC as a futures commission merchant and is a member of the National Futures Association.

Data, privacy, cybersecurity and artificial intelligence regulation:

The Firm and its subsidiaries are subject to laws, rules and regulations globally concerning data, including data protection, consumer protection, privacy, cybersecurity, artificial intelligence and related

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matters. These laws, rules and regulations are constantly evolving, subject to interpretation, remain a focus of regulators globally, may be enforced by private parties or government bodies, and continue to have a significant impact on all of the Firm's businesses and operations.

For example, the Digital Operational Resilience Act (DORA) mandates that the Firm's financial services subsidiaries operating in the EU comply with requirements relating to information and communications technology ("ICT") risk management, reporting, security control testing and ICT third party risks beginning in January 2025. In addition, the EU Artificial Intelligence Act regulates the development and deployment of artificial intelligence systems within the EU, with phased-in requirements that began in February 2025.

The Bank Secrecy Act and Economic Sanctions:

The Bank Secrecy Act ("BSA") requires all financial institutions, including banks and securities broker-dealers, to establish a risk-based system of internal controls reasonably designed to prevent money laundering and the financing of terrorism. The BSA includes a variety of record-keeping and reporting requirements, as well as due diligence/know-your-customer documentation requirements. The Firm is also subject to the regulations and economic sanctions programs administered and enforced by the U.S. Treasury's Office of Foreign Assets Control ("OFAC") and EU and U.K. authorities which target entities or individuals that are, or are located in countries that are, involved in activities including terrorism, hostilities, embezzlement or human rights violations. The Firm is also subject to economic sanctions laws, rules and regulations in other jurisdictions in which it operates, including those that conflict with or prohibit a firm such as JPMorganChase from complying with certain laws, rules and regulations to which it is otherwise subject.

Anti-Corruption:

The Firm is subject to laws and regulations relating to corrupt and illegal payments to government officials and others in the jurisdictions in which it operates, including the U.S. Foreign Corrupt Practices Act and the U.K. Bribery Act.

Compensation practices:

The Firm's compensation practices are subject to oversight by the Federal Reserve, as well as other agencies. The Federal Reserve has jointly issued guidance with the FDIC and the OCC that is designed to ensure that incentive compensation paid by banking organizations does not encourage imprudent risk-taking that threatens the organizations' safety and soundness. The Financial Stability Board ("FSB") has also established standards covering compensation principles for banks. The Firm's compensation

practices are also subject to regulation and oversight by regulators in other jurisdictions, notably the Fifth Capital Requirements Directive ("CRD V"), as implemented in the EU and as largely adopted in the U.K, which includes compensation-related provisions. The European Banking Authority has instituted guidelines on compensation policies including under CRD V which in certain countries (such as Germany) are implemented or supplemented by local regulations or guidelines. The U.K. regulators have also instituted regulations and guidelines on compensation policies, which diverge in certain areas from EU rules. The Firm expects that the implementation of regulatory guidelines regarding compensation in the U.S. and other countries will continue to evolve, and may affect the manner in which the Firm structures its compensation programs and practices.

Sustainability:

Policymakers in the U.K. and the EU have continued to implement and enhance sustainability-related initiatives and disclosure requirements. The Corporate Sustainability Reporting Directive ("CSRD") will replace and significantly expand the scope and content of certain EU ESG reporting requirements, with phased-in requirements expected to start with fiscal year 2024. The implementation of CSRD into local law has been delayed in a number of member states, including in Germany, and the Firm continues to monitor developments and potential impacts. In addition, in July 2024, the EU enacted the Corporate Sustainability Due Diligence Directive ("CSDDD"), which provides for phased-in requirements starting in 2027. The CSDDD sets mandatory due diligence obligations for companies to address actual and potential human rights violations and environmental adverse impacts stemming from their own operations and business relationships, including the activities of certain companies with which they have established business relationships and also requires the adoption of company-specific climate-related transition plans. Both the CSRD and CSDDD will impact certain of the Firm's EU and non-EU entities.

Litigation and regulatory challenges:

Trade organizations representing the financial services industry and others have filed lawsuits challenging various laws, rules and regulations that, if enacted, adopted or implemented, could have significant adverse impacts on the results of operations or compliance costs of financial institutions, including the Firm. These matters include:

- **Stress tests:** The Bank Policy Institute ("BPI"), the U.S. Chamber of Commerce and other trade organizations filed an action against the Federal Reserve in the United States District Court for the Southern District of Ohio in December 2024 challenging the manner in which the annual stress testing process is administered.

- **CFPB Late Fee Rule:** This rule has been stayed pending resolution of an action challenging the rule filed against the CFPB in the United States District Court for the Northern District of Texas in March 2024 by trade organizations including the American Bankers Association and the Consumer Bankers Association.
- **CFPB Data Sharing Rule:** The BPI, the Kentucky Bankers Association and other organizations filed an action against the CFPB in the United States District Court for the Eastern District of Kentucky in October 2024 challenging key aspects of this rule.
- **CFPB Overdraft Rule:** An action filed by trade organizations led by the Mississippi Bankers Association against the CFPB in the United States District Court for the Southern District of Mississippi in December 2024 seeks a preliminary injunction to stay the October 1, 2025 effective date of this rule. The CFPB has consented in part to stay the effective date of the rule by 90 days and to temporarily stay the litigation. The preliminary injunction and the stay of litigation are pending court approval.
- **Fiduciary Rule:** Trade organizations including the Federation of Americans for Consumer Choice and the American Council of Life Insurers filed actions against the DOL seeking to enjoin this rule, and in July 2024, the effective date of the rule was stayed by two United States District Courts.

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Human capital

JPMorganChase believes that its long-term growth and success depend on its ability to attract, develop and retain talented employees and foster an inclusive work environment. The information provided below relates to JPMorganChase's full-time and part-time employees and does not include the Firm's contractors.

Global workforce

As of December 31, 2024, JPMorganChase had 317,233 employees globally, an increase of 7,307 employees from the prior year. The increase was primarily attributable to growth in the number of front office and technology employees. JPMorganChase's employees are located in 66 countries, with 59% of the Firm's employees located in the U.S. The following table presents the distribution of the Firm's global workforce by region and by line of business ("LOB") and Corporate as of December 31, 2024:

Employee Breakdown by Region		Employee Breakdown by LOB and Corporate	
Region	Employees	LOB	Employees
North America	187,179	CCB	144,989
Asia-Pacific	93,941	CIB	93,231
Europe/Middle East/Africa	30,729	AWM	29,403
Latin America/Caribbean	5,384	Corporate	49,610
Total Firm	317,233	Total Firm	317,233

Workforce composition

The following table presents information based on voluntary self-identifications by the Firm's employees, including members of the Firm's Operating Committee and other senior level employees, as well as members of the Board of Directors, as of December 31, 2024. Information on race/ethnicity of employees is categorized based on Equal Employment Opportunity ("EEO") classifications and is presented for U.S. employees who self-identified, and information on gender is presented for global employees who self-identified. Information on race/ethnicity and gender for members of the Operating Committee and the Board of Directors reflects all such members. Information on LGBTQ+ and veteran statuses is based on all U.S. employees, and all members of the Operating Committee and the Board of Directors. Information on disability status is based on all U.S. employees and all members of the Operating Committee.

December 31, 2024	Total employees	Senior level employees ^(e)	Operating Committee	Board of Directors ^(f)
Race/Ethnicity ^(a) :				
White	43 %	74 %	86 %	80 %
Hispanic	21 %	6 %	7 %	—
Asian	20 %	14 %	7 %	—
Black	13 %	5 %	—	20 %
Other ^(b)	3 %	1 %	—	—
Gender ^(c) :				
Men	51 %	71 %	53 %	50 %
Women	49 %	29 %	47 %	50 %
LGBTQ+ ^(d)	4 %	2 %	7 %	—
Military veterans ^(d)	3 %	2 %	—	10 %
People with disabilities ^(d)	5 %	3 %	—	— ^(g)

(a) Based on EEO metrics. Presented as a percentage of the respective populations who self-identified race/ethnicity, which was 97% and 95% of the Firm's total U.S.-based employees and U.S.-based senior level employees, respectively, and all members of the Operating Committee and the Board of Directors. Information for the Operating Committee includes one member who is based outside of the U.S.

(b) Other includes American Indian or Alaskan Native, Native Hawaiian or Other Pacific Islander, and two or more races/ethnicities.

(c) Presented as a percentage of the respective populations who self-identified gender, which was 99% of each of the Firm's total global employees and senior level employees, and all members of the Operating Committee and the Board of Directors.

(d) Presented as a percentage of total U.S.-based employees, total U.S.-based senior level employees, all members of the Operating Committee, and all members of the Board of Directors, respectively.

(e) Senior level employees represents employees with the titles of Managing Director and above.

(f) Excludes Brad D. Smith and Michele G. Buck, who were elected to the Firm's Board of Directors, effective January 21, 2025 and March 17, 2025, respectively.

(g) The Firm has not asked members of the Board of Directors to self-identify disability status.

Attracting and retaining employees

The goal of JPMorganChase's recruitment efforts is to attract and hire highly qualified candidates in all roles and at all career levels. The Firm's hiring practices focus on the skills and qualifications of a candidate relative to the job requirements.

The Firm strives to provide both external candidates and internal employees who are seeking a different role with challenging and stimulating career opportunities. These opportunities range from internship training programs for students to entry-level, management and executive careers. During 2024, approximately 56% of the Firm's employment opportunities were filled by external candidates, with the remainder filled by existing employees.

Developing employees

JPMorganChase supports the professional development and career growth of its employees. The Firm requires that its employees, including new hires, complete a training curriculum which covers, among other topics, compliance with the Firm's Code of Conduct and information concerning Firm policies and standards, including those relating to cybersecurity. In addition, the Firm offers extensive voluntary training programs and educational resources to all employees covering a broad variety of topics such as leadership and management, artificial intelligence, data literacy and operational and professional skills. Leadership Edge, the Firm's global leadership and management development center of excellence, is focused on creating one Firmwide leadership culture.

Compensation and benefits

The Firm provides market-competitive compensation and benefits programs. JPMorganChase's compensation philosophy includes guiding principles that drive compensation-related decisions across the Firm, and includes: pay-for-performance practices designed to attract and retain top talent; responsiveness and alignment with shareholder interests; and reinforcement of the Firm's culture. The Firm follows a disciplined and balanced compensation framework, including the integration of risk, controls and conduct considerations. The Firm's compensation review processes seek to ensure that the Firm's employees are paid fairly and competitively for the work they do.

JPMorganChase offers extensive benefits and wellness packages to support employees and their families, which vary depending on location and include healthcare coverage, retirement benefits, life and disability insurance, access to on-site health and wellness centers, counseling and resources related to mental health, time away policies, child care access and support, tuition assistance, and financial coaching.

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Item 1A. Risk Factors.

The following discussion sets forth the material risk factors that could affect JPMorganChase's financial condition and operations. Readers should not consider any descriptions of these factors to be a complete set of all potential risks that could affect the Firm. Any of the risk factors discussed below could by itself, or combined with other factors, materially and adversely affect JPMorganChase's business, results of operations, financial condition, capital position, liquidity, competitive position or reputation, including by materially increasing expenses or decreasing revenues, which could result in material losses or a decrease in earnings.

Summary

The principal risk factors that could adversely affect JPMorganChase's business, results of operations, financial condition, capital position, liquidity, competitive position or reputation include:

- **Regulatory** risks, including the impact that applicable laws, rules and regulations in the highly-regulated and supervised financial services industry, as well as changes to or in the application, interpretation or enforcement of those laws, rules and regulations, can have on JPMorganChase's business and operations, including JPMorganChase incurring additional costs associated with assessments, levies or other governmental charges; the ways in which differences in financial services regulation and supervision in different jurisdictions or with respect to certain competitors can negatively impact JPMorganChase's business; the ways in which governmental policies that discourage or penalize business relationships with clients in certain industries, or require specific business practices, can negatively affect JPMorganChase's businesses; the penalties and collateral consequences, and higher compliance and operational costs, that JPMorganChase may incur when resolving a regulatory investigation; the ways in which less predictable legal and regulatory frameworks in certain jurisdictions can negatively impact JPMorganChase's operations and financial results; and the losses that security holders and other unsecured creditors will absorb if JPMorganChase were to enter into a resolution.
- **Political** risks, including the potential negative effects on JPMorganChase's businesses due to economic uncertainty or instability caused by political developments.
- **Market** risks, including the effects that economic and market events and conditions, political developments, changes in interest rates and credit spreads, and market fluctuations can have on JPMorganChase's consumer and wholesale businesses and its investment and market-making positions and on JPMorganChase's earnings and its liquidity and capital levels.
- **Credit** risks, including potential negative effects from adverse changes in the financial condition of clients, customers, counterparties, custodians and central counterparties; the potential for losses due to declines in the value of collateral in stressed market conditions; and potential negative impacts from concentrations of credit risk with respect to clients, customers, counterparties and other market participants.
- **Liquidity** risks, including the risk that JPMorganChase's liquidity could be impaired by market-wide illiquidity or disruption, unforeseen liquidity or capital requirements, the inability to sell assets, default by a significant market participant, unanticipated outflows of cash or collateral, or lack of market or customer confidence in JPMorganChase; the dependence of JPMorgan Chase & Co. on the cash flows of its subsidiaries; and the potential adverse effects that any downgrade in any of JPMorganChase's credit ratings may have on its liquidity and cost of funding.
- **Capital** risks, including the risk that any failure by or inability of JPMorganChase to maintain the required level and composition of capital, or unfavorable changes in applicable capital requirements, could limit JPMorganChase's ability to distribute capital to shareholders or to support its business activities.
- **Operational** risks, including risks associated with JPMorganChase's dependence on its operational systems and its employees, as well as the systems and employees of third parties, market participants and service providers; the potential negative effects of failing to identify and address operational risks related to the failure of internal or external operational systems, the introduction of or changes to products, services and delivery platforms or the adoption of new technologies; risks related to safeguarding personal information; the harm that could be caused by a successful cyber attack affecting JPMorganChase or by other extraordinary events; risks associated with JPMorganChase's risk management framework and control environment, its models and estimations and associated judgments used in its stress testing and financial statements, and controls over disclosure and financial reporting; and potential adverse effects of failing to comply with applicable standards for the oversight of vendors and other service providers.
- **Strategic** risks, including the damage to JPMorganChase's competitive standing and results that could occur if management fails to develop and execute effective business strategies; risks associated with the significant and increasing competition that JPMorganChase faces; and the potential adverse impacts of climate change on the

effectiveness of JPMorganChase's existing business strategies with respect to its operations, clients and customers.

- **Conduct** risks, including the negative impact that can result from the actions or misconduct of employees, including any failure of employees to conduct themselves in accordance with JPMorganChase's expectations, policies and practices.
- **Reputation** risks, including the potential adverse effects on JPMorganChase's relationships with its clients, customers, shareholders, regulators and other stakeholders that could arise from employee misconduct, security breaches, inadequate risk management, compliance or operational failures, litigation and regulatory investigations, failure to satisfy expectations concerning environmental, social and governance concerns, failure to effectively manage conflicts of interest or to satisfy fiduciary obligations, or other factors that could damage JPMorganChase's reputation.
- **Country** risks, including potential impacts on JPMorganChase's businesses from an outbreak or escalation of hostilities between countries or within a country or region; and the potential adverse effects of local economic, political, regulatory and social factors on JPMorganChase's business and revenues in certain countries in which it operates.
- **People** risks, including the criticality of attracting and retaining qualified employees; and the potential adverse effects of unfavorable changes in immigration or travel policies on JPMorganChase's workforce.
- **Legal** risks, including those relating to litigation and regulatory and government investigations.

The above summary is subject in its entirety to the discussion of the risk factors set forth below.

Regulatory

JPMorganChase's businesses are highly regulated, and the laws, rules and regulations that apply to JPMorganChase have a significant impact on its business and operations.

JPMorganChase is a financial services firm with operations worldwide. JPMorganChase must comply with the laws, rules and regulations that apply to its operations in all of the jurisdictions around the world in which it does business, and financial services firms such as JPMorganChase are subject to extensive and constantly-evolving regulation and supervision.

The regulation and supervision of JPMorganChase significantly affects the way that it conducts its business and structures its operations, and JPMorganChase could be required to make changes to its business and operations in response to supervisory expectations or decisions or to new or changed laws,

rules and regulations. These types of developments could result in JPMorganChase incurring additional costs or experiencing a reduction in revenues to comply with applicable laws, rules and regulations, which could reduce its profitability. Furthermore, JPMorganChase's entry into or acquisition of a new business or an increase in its principal investments may require JPMorganChase to comply with additional laws, rules, and regulations.

Additionally, JPMorganChase's ability to execute certain business initiatives could become more challenging due to increased regulation in the financial services industry, such as limitations on late payment, overdraft and interchange fees. This could adversely affect JPMorganChase's earnings from its consumer businesses, prompting the reevaluation or adjustment of certain businesses or product offerings, as well as the reallocation of resources and incurrence of restructuring costs, which could impact revenue and profitability in the affected lines of business.

In response to new and existing laws, rules and regulations and expanded supervision, JPMorganChase has in the past been and could in the future be, required to:

- limit the products and services that it offers
- reduce the liquidity that it can provide through its market-making activities
- refrain from engaging in business opportunities that it might otherwise pursue
- pay higher taxes (including as part of any minimum global tax regime), assessments, levies or other governmental charges, including in connection with the resolution of tax examinations
- incur losses, including with respect to fraudulent transactions perpetrated against its customers
- dispose of certain assets, and do so at times or prices that are disadvantageous
- impose restrictions on certain business activities, or
- increase the prices that it charges for products and services, which could reduce the demand for them.

Any failure by JPMorganChase to comply with the laws, rules and regulations to which it is subject could result in:

- increased regulatory and supervisory scrutiny
- regulatory and governmental enforcement actions
- the imposition of fines, penalties or other sanctions
- increased exposure to litigation, or
- harm to its reputation.

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Differences and inconsistencies in financial services regulation and supervision can negatively impact JPMorganChase's businesses, operations and financial results.

The content and application of laws, rules and regulations affecting financial services firms can vary according to factors such as the size of the firm, the jurisdiction in which it is organized or operates, and other criteria. For example:

- larger firms such as JPMorganChase are often subject to more stringent supervision, regulation and regulatory scrutiny
- financial technology companies and other non-traditional competitors may not be subject to banking regulation, or may be supervised by a national or state regulatory agency that does not have the same resources or regulatory priorities as the regulatory agencies that supervise more diversified financial services firms, or
- the financial services regulatory and supervisory framework in a particular jurisdiction may favor financial institutions that are based in that jurisdiction.

These types of differences in the regulatory and supervisory framework can result in JPMorganChase losing market share to competitors that are less regulated or not subject to regulation, especially with respect to unregulated financial products.

There can also be significant differences in the ways that similar regulatory initiatives affecting the financial services industry are implemented in the U.S. and in other countries and regions in which JPMorganChase does business. For example, when adopting rules that are intended to implement a global regulatory or supervisory standard, a national regulator may introduce additional or more restrictive requirements, which can create competitive disadvantages for financial services firms, such as JPMorganChase, that may be subject to those enhanced regulations.

In addition, certain national and multi-national bodies and governmental agencies outside the U.S. have adopted laws, rules or regulations that may conflict with or prohibit JPMorganChase from complying with laws, rules and regulations to which it is otherwise subject, creating conflict of law issues that also increase its risk of non-compliance in those jurisdictions.

Legislative and regulatory initiatives outside the U.S. have required and could in the future require JPMorganChase to make significant modifications to its operations and legal entity structure in the relevant countries or regions in order to comply with those requirements. These include laws, rules and regulations that have been adopted or proposed, as well as regulatory expectations, relating to:

- the establishment of locally-based intermediate holding companies or operating subsidiaries
- requirements to maintain minimum amounts of capital or liquidity in locally-based subsidiaries
- the implementation of processes within locally-based subsidiaries to comply with local regulatory requirements or expectations
- the separation (or “ring fencing”) of core banking products and services from markets activities
- requirements for the orderly resolution of financial institutions
- requirements for executing or settling transactions on exchanges or through central counterparties (“CCPs”), or for depositing funds with other financial institutions or clearing and settlement systems
- position limits and reporting rules for derivatives
- governance and accountability regimes
- conduct of business and control requirements, and
- restrictions on compensation.

These types of differences, inconsistencies and conflicts in financial services regulation have required and could in the future require JPMorganChase to:

- divest assets or restructure its operations
- maintain higher levels of capital and liquidity, or absorb increased capital and liquidity costs
- incur higher operational and compliance costs
- change the prices that it charges for its products and services
- curtail the products and services that it offers to its customers and clients
- curtail other business opportunities, including acquisitions or principal investments, that it otherwise would have pursued
- become subject to regulatory fines, penalties or other sanctions, or
- incur higher costs for complying with different legal and regulatory frameworks.

Any or all of these factors could harm JPMorganChase's ability to compete against other firms that are not subject to the same laws, rules and regulations or supervisory oversight, or harm JPMorganChase's businesses, results of operations and profitability.

Resolving regulatory investigations can subject JPMorganChase to significant penalties and collateral consequences, and could result in higher compliance costs or restrictions on its operations.

JPMorganChase is subject to heightened oversight and scrutiny from regulatory authorities in many jurisdictions. JPMorganChase has paid significant fines, provided other monetary relief, incurred other

penalties and experienced other repercussions in connection with resolving investigations and enforcement actions by governmental agencies. JPMorganChase could become subject to similar regulatory or governmental resolutions or other actions in the future, and addressing the requirements of any such resolutions or actions could result in JPMorganChase incurring higher operational and compliance costs, including devoting substantial resources to the required remediation or needing to comply with other restrictions.

In connection with resolving specific regulatory investigations or enforcement actions, certain regulators have required JPMorganChase and other financial institutions to admit wrongdoing with respect to the activities that gave rise to the resolution. These types of admissions can lead to:

- greater exposure in litigation
- damage to JPMorganChase's reputation
- disqualification from doing business with certain clients or customers, or in specific jurisdictions, or
- other direct and indirect adverse effects.

Furthermore, government officials in the U.S. and other countries have demonstrated a willingness to bring criminal actions against financial institutions and have required that institutions plead guilty to criminal offenses or admit other wrongdoing in connection with resolving regulatory investigations or enforcement actions. Resolutions of this type can have significant collateral consequences for the subject financial institution, including:

- loss of clients, customers and business
- restrictions on offering certain products or services, and
- losing permission to operate certain businesses, either temporarily or permanently.

JPMorganChase expects that:

- it and other financial services firms will continue to be subject to heightened regulatory scrutiny and governmental investigations and enforcement actions
- governmental authorities will continue to require that financial institutions be penalized for actual or deemed violations of law with formal and punitive enforcement actions, including the imposition of significant monetary and other sanctions, rather than resolving these matters through informal supervisory actions, and
- governmental authorities will be more likely to pursue formal enforcement actions and resolutions against JPMorganChase to the extent that it has previously been subject to other governmental investigations or enforcement actions.

If JPMorganChase fails to meet the requirements of any resolution of a governmental investigation or enforcement action, or to maintain risk and control processes that meet the heightened standards and expectations of its regulators, it could be required to, among other things:

- enter into further resolutions of investigations or enforcement actions
- pay additional regulatory penalties or enter into judgments, or
- accept material regulatory restrictions on, or changes in the management of, its businesses.

In these circumstances, JPMorganChase could also become subject to other sanctions, or to prosecution or civil litigation with respect to the conduct that gave rise to an investigation or enforcement action. In addition, JPMorganChase can be subject to higher costs or requests for additional capital in connection with the resolution of governmental investigations and enforcement actions involving newly-acquired businesses, companies in which JPMorganChase has made principal investments, parties to joint ventures with JPMorganChase, and vendors with which JPMorganChase does business.

JPMorganChase's operations and financial results can be negatively impacted in jurisdictions with less predictable legal and regulatory frameworks.

JPMorganChase conducts existing and new business in certain countries, states, municipalities, territories and other jurisdictions in which the application of the rule of law is inconsistent, extralegal or less predictable, including with respect to:

- the absence of a statutory or regulatory basis or guidance for engaging in specific types of business or transactions
- conflicting or ambiguous laws, rules, regulations and judicial orders, or the inconsistent application or interpretation of existing laws, rules, regulations and judicial precedents
- actions by or at the direction of government officials or agencies
- uncertainty concerning the enforceability of intellectual property rights or contractual or other obligations
- difficulty in competing in economies in which the government controls or protects all or a portion of the local economy or specific businesses, or where graft or corruption may be pervasive
- the threat of regulatory investigations, civil litigations or criminal prosecutions that are arbitrary or otherwise contrary to established legal principles in other parts of the world, and

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- the termination of licenses required to operate in the local market or the suspension of business relationships with governmental bodies.

If the application of the laws, rules, regulations and judicial precedents in any jurisdiction is susceptible to producing outcomes that are inconsistent, unexpected or contrary to established legal principles, this can create a more difficult environment in which JPMorganChase conducts its business and could negatively affect JPMorganChase's operations and reduce its earnings with respect to that jurisdiction. For example, JPMorganChase has faced actual and threatened litigation in Russia with respect to payments that JPMorganChase cannot make under, and is contractually excused from paying as a result of, relevant economic sanctions laws. That litigation has also resulted in the seizure of assets. In addition, conducting business in jurisdictions with less predictable legal and regulatory frameworks could require JPMorganChase to devote significant additional resources to understanding local laws, rules and regulations, as well as structuring its operations to comply with local laws, rules and regulations and implementing and administering related internal policies and procedures.

There can be no assurance that JPMorganChase will always be successful in its efforts to fully understand and to conduct its business in compliance with the laws, rules and regulations of all of the jurisdictions in which it operates, and the risk of non-compliance, or of interference with JPMorganChase's businesses, can be greater in jurisdictions that have less predictable legal and regulatory frameworks.

JPMorganChase's businesses may be negatively impacted by governmental policies that either discourage or penalize business with certain industries or require specific business practices.

JPMorganChase's businesses and results of operations may be adversely affected by actions or initiatives by national, state or local governmental authorities that:

- seek to discourage financial institutions from doing business with companies engaged in certain industries, or conversely, to penalize financial institutions that elect not to do business with such companies, or
- mandate specific business practices that companies operating in the relevant jurisdiction must adopt.

Because governmental policies in one jurisdiction may differ or conflict with those in other jurisdictions, JPMorganChase may face negative consequences regardless of the course of action it takes or elects not to take, including:

- restrictions or prohibitions on doing business within a particular jurisdiction, or with governmental entities in a jurisdiction

- the threat of enforcement actions, including under antitrust or other anti-competition laws, rules and regulations, and
- harm to its reputation arising from public criticism, including from politicians, activists and other stakeholders.

JPMorganChase has been prohibited from engaging in certain business activities in specific jurisdictions as a result of these types of governmental actions, and there is no assurance that it will not face similar restrictions on its business and operations in the future.

Requirements for the orderly resolution of JPMorganChase could result in JPMorganChase having to restructure or reorganize its businesses and could increase its funding or operational costs or curtail its businesses.

JPMorganChase is required under Federal Reserve and FDIC rules to prepare and submit periodically to those agencies a detailed plan for rapid and orderly resolution in bankruptcy, without extraordinary government support, in the event of material financial distress or failure. The evaluation of JPMorganChase's resolution plan by these agencies may change, and the requirements for resolution plans may be modified from time to time. Any such determinations or modifications could result in JPMorganChase needing to make changes to its legal entity structure or to certain internal or external activities, which could increase its funding or operational costs, or hamper its ability to serve clients and customers.

If the Federal Reserve and the FDIC were both to determine that a resolution plan submitted by JPMorganChase has deficiencies, they could jointly impose more stringent capital, leverage or liquidity requirements or restrictions on JPMorganChase's growth, activities or operations. The agencies could also require that JPMorganChase restructure, reorganize or divest assets or businesses in ways that could materially and adversely affect JPMorganChase's operations and strategy.

Holders of JPMorgan Chase & Co.'s debt and equity securities will absorb losses if it were to enter into a resolution.

Federal Reserve rules require that JPMorgan Chase & Co. (the "Parent Company") maintain minimum levels of unsecured external long-term debt and other loss-absorbing capacity with specific terms ("eligible LTD") for purposes of recapitalizing JPMorganChase's operating subsidiaries if the Parent Company were to enter into a resolution either:

- in a bankruptcy proceeding under Chapter 11 of the U.S. Bankruptcy Code, or
- in a receivership administered by the FDIC under Title II of the Dodd-Frank Act ("Title II").

If the Parent Company were to enter into a resolution, holders of eligible LTD, other unsecured creditors and holders of equity securities of the Parent Company will absorb the losses of the Parent Company and its subsidiaries.

The preferred “single point of entry” strategy under JPMorganChase’s resolution plan contemplates that the Parent Company would enter bankruptcy proceedings and JPMorganChase’s material subsidiaries would be recapitalized, as needed, so that they could continue normal operations or subsequently be divested or wound down in an orderly manner. As a result, the Parent Company’s losses and any losses incurred by its subsidiaries would be imposed first on holders of the Parent Company’s equity securities and thereafter on its unsecured creditors, including holders of eligible LTD. Claims of the Parent Company’s shareholders and unsecured creditors would have a junior position to the claims of creditors of JPMorganChase’s subsidiaries and to the claims of priority (as determined by statute) and secured creditors of the Parent Company.

Accordingly, in a resolution of the Parent Company in bankruptcy, unsecured creditors of the Parent Company, including holders of eligible LTD of the Parent Company, would realize value only to the extent available to the Parent Company as a shareholder of JPMorgan Chase Bank, N.A. and its other subsidiaries, and only after any claims of priority and secured creditors of the Parent Company have been fully repaid.

The FDIC has similarly indicated that a single point of entry recapitalization model would be its expected strategy to resolve a systemically important financial institution, such as the Parent Company, under Title II. However, the FDIC has not formally adopted or committed to any specific resolution strategy.

If the Parent Company were to approach, or enter into, a resolution, none of the Parent Company, the Federal Reserve or the FDIC is obligated to follow JPMorganChase’s preferred resolution strategy, and losses to unsecured creditors of the Parent Company, including holders of eligible LTD, and to holders of equity securities of the Parent Company, under whatever strategy is ultimately followed, could be greater than they might have been under JPMorganChase’s preferred strategy.

Political

Economic uncertainty or instability caused by political and geopolitical developments can negatively impact JPMorganChase’s businesses.

Political developments in the U.S. and other countries can cause uncertainty in the economic environment and market conditions in which JPMorganChase operates its businesses. Certain governmental policy initiatives, as well as heightened geopolitical tensions,

could significantly affect U.S. and global economic growth and cause higher volatility in the financial markets, including:

- monetary policies and actions taken by the Federal Reserve and other central banks or governmental authorities, including changes in interest rate levels and any sustained large-scale asset purchases or any suspension or reversal of those actions
- fiscal policies, including with respect to taxation and spending
- isolationist foreign policies
- economic or financial sanctions
- the implementation of tariffs and other protectionist trade policies
- changes to immigration policies, or
- actions that the government takes or fails to take in response to the effects of health emergencies, the spread of infectious diseases, epidemics or pandemics.

These types of political developments, and uncertainty about the possible outcomes of these developments, could:

- erode investor or consumer confidence in the U.S. economy and financial markets, which could potentially undermine the status of the U.S. dollar as a safe haven currency
- provoke retaliatory countermeasures by other countries and otherwise heighten tensions in regulatory, enforcement or diplomatic relations
- increase the risk of targeted cyber attacks
- increase concerns about whether the U.S. government will be funded, and its outstanding debt serviced, at any particular time
- result in periodic shutdowns of the U.S. government
- influence investor perceptions concerning government support of certain sectors of the economy or the economy as a whole
- influence monetary policy actions of the Federal Reserve to moderate the economic impact of political developments, including decisions on interest rate levels and asset purchases and sales
- adversely affect the financial condition or credit ratings of clients and counterparties with which JPMorganChase does business, or
- cause JPMorganChase to refrain from engaging in business opportunities that it might otherwise pursue.

These factors could lead to:

- slower growth rates, rising inflation or recession
- disruptions in labor markets
- greater market volatility

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- a contraction of available credit and the widening of credit spreads
- U.S. dollar currency fluctuations
- lower investments in a particular country or sector of the economy
- large-scale sales of government debt and other debt and equity securities
- reduced commercial activity among trading partners or disruptions to supply chains, or
- the possible departure of a country from, or the dissolution or formation of, a political or economic alliance or treaty.

Under certain circumstances, such as geopolitically challenging situations in regions like Russia, the Middle East and China, these various risks could become highly correlated or combine in unprecedented ways.

Any of these potential outcomes could cause JPMorganChase to suffer losses on its market-making positions or in its investment portfolio, reduce its liquidity and capital levels, increase the allowance for credit losses or lead to higher net charge-offs, hamper its ability to deliver products and services to its clients and customers, and weaken its results of operations and financial condition or credit rating.

Market

Economic and market events and conditions can materially affect JPMorganChase's businesses and investment and market-making positions.

JPMorganChase's results of operations can be negatively affected by adverse changes in any of the following:

- investor, consumer and business sentiment
- events that reduce confidence in the financial markets
- inflation, deflation or recession
- high unemployment or, conversely, a tightening labor market
- the availability and cost of capital, liquidity and credit
- levels and volatility of interest rates, credit spreads and market prices for currencies, debt and equity securities and commodities, as well as the duration of any such changes
- the economic effects of an outbreak or escalation of war, hostilities, terrorism or other geopolitical instabilities, cyber attacks, climate change, natural disasters, severe weather conditions, health emergencies, the spread of infectious diseases, epidemics or pandemics or other extraordinary events beyond JPMorganChase's control, and
- the state of the U.S. and global economies.

All of these are affected by global economic, market and political events and conditions, including monetary policies and actions taken by central banks or other governmental authorities, as well as by the regulatory environment.

In addition, JPMorganChase's investment portfolio and market-making businesses can suffer losses due to unanticipated market events, including:

- severe declines in asset values
- unexpected credit events
- unforeseen events or conditions that may cause previously uncorrelated factors to become correlated (and vice versa)
- the inability to effectively hedge risks related to market-making and investment portfolio positions, or
- other market risks that may not have been appropriately taken into account in the development, structuring or pricing of a financial instrument.

If JPMorganChase experiences significant losses in its investment portfolio or from market-making activities, this could reduce JPMorganChase's profitability and its liquidity and capital levels, and thereby constrain the growth of its businesses.

JPMorganChase's consumer businesses can be negatively affected by adverse economic conditions and governmental policies.

JPMorganChase's consumer businesses are particularly affected by U.S. and global economic conditions, including:

- personal and household income distribution
- unemployment or underemployment
- prolonged periods of exceptionally high or low interest rates, or significant changes to interest rates
- changes in the value of collateral such as residential real estate and vehicles
- changes in housing prices
- the level of inflation and its effect on prices for goods and services
- consumer and small business confidence levels, and
- changes in consumer spending or in the level of consumer debt.

Heightened levels of unemployment or underemployment that result in reduced personal and household income could negatively affect consumer credit performance to the extent that consumers are less able to service their debts. In addition, sustained low growth, low or negative interest rates, inflationary pressures or recessionary conditions could diminish customer demand for the products and services offered by JPMorganChase's consumer businesses.

Adverse economic conditions could also lead to an increase in delinquencies, additions to the allowance for credit losses and higher net charge-offs, which can reduce JPMorganChase's earnings. These consequences could be significantly worse in certain geographies, including where declining industrial or manufacturing activity has resulted in or could result in higher levels of unemployment, or where high levels of consumer debt, such as outstanding student loans, could impair the ability of customers to pay their other consumer loan obligations.

JPMorganChase's earnings from its consumer businesses could also be adversely affected by governmental policies and actions that affect consumers, including:

- policies and initiatives relating to medical insurance, education, immigration and housing, or that may impact employment status
- laws, rules and regulations relating specifically to the financial services industry, such as limitations on late payment, overdraft and interchange fees, and
- policies aimed at the economy more broadly, such as higher taxes and increased regulation, which could result in reductions in consumer disposable income.

Unfavorable market and economic conditions can have an adverse effect on JPMorganChase's wholesale businesses.

In JPMorganChase's wholesale businesses, market and economic factors can affect the volume of transactions that JPMorganChase executes for its clients or for which it advises clients, and, therefore, the revenue that JPMorganChase receives from those transactions. These factors can also influence the willingness of other financial institutions and investors to participate in capital markets transactions that JPMorganChase manages, such as loan syndications or securities underwriting. Furthermore, if a significant and sustained deterioration in market conditions were to occur, the profitability of JPMorganChase's businesses engaged in capital markets activities, including loan syndication, securities underwriting and leveraged lending activities, could be reduced to the extent that those businesses:

- earn less fee revenue due to lower transaction volumes, including when clients are unwilling or unable to refinance their outstanding debt obligations in unfavorable market conditions, or
- dispose of portions of credit commitments at a loss, or hold larger residual positions in credit commitments that cannot be sold at favorable prices.

The fees that JPMorganChase earns from managing client assets or holding assets under custody for clients could be diminished by declining asset values

or other adverse macroeconomic conditions. For example, higher interest rates or a downturn in financial markets could affect the valuation of client assets that JPMorganChase manages or holds under custody, which, in turn, could affect JPMorganChase's revenue from fees that are based on the amount of assets under management or custody. Similarly, adverse macroeconomic or market conditions could prompt outflows from JPMorganChase funds or accounts, or cause clients to invest in products that generate lower revenue. Substantial and unexpected withdrawals from a JPMorganChase fund can also hamper the investment performance of the fund, particularly if the outflows create the need for the fund to dispose of fund assets at disadvantageous times or prices, and could lead to further withdrawals based on the weaker investment performance.

An adverse change in market conditions in particular segments of the economy, such as a sudden and severe downturn in oil and gas prices or an increase in commodity prices, severe declines in commercial real estate values, or sustained changes in consumer behavior that affect specific economic sectors, could have a material adverse effect on clients of JPMorganChase whose operations or financial condition are directly or indirectly dependent on the health or stability of those market segments or economic sectors, as well as clients that are engaged in related businesses. JPMorganChase could incur credit losses on its loans and other commitments to clients that operate in, or are dependent on, any sector of the economy that is or comes under stress.

An economic downturn or sustained changes in consumer behavior that results in shifts in consumer and business spending could also have a negative impact on certain of JPMorganChase's wholesale clients, and thereby diminish JPMorganChase's earnings from its wholesale operations. For example, the businesses of certain of JPMorganChase's wholesale clients are dependent on consistent streams of rental income from commercial real estate properties, including offices, which are owned or being built by those clients. Sustained adverse economic conditions or hybrid work models could result in reductions in the rental cash flows that owners or developers receive from their tenants which, in turn, could depress the values of the properties, impair the ability of borrowers to service or refinance their commercial real estate loans and lead to an increase in foreclosures. These consequences could result in JPMorganChase experiencing increases in the allowance for credit losses, higher delinquencies, defaults and charge-offs within its commercial real estate loan portfolio and incurring higher costs for servicing a larger volume of delinquent loans in that portfolio. An increase in foreclosures could result in higher operational risk associated with JPMorganChase owning and managing real property,

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and any inadequacy in governance or control over the foreclosed properties could result in regulatory scrutiny and reputational harm.

Changes in interest rates and credit spreads can adversely affect JPMorganChase's earnings, its liquidity or its capital levels.

When interest rates are high or increasing, JPMorganChase can generally be expected to earn higher net interest income. However, higher interest rates can also lead to:

- fewer originations of commercial and residential real estate loans
- losses on underwriting exposures or incremental client-specific downgrades, or increases in the allowance for credit losses and net charge-offs due to higher financing costs for clients
- the loss of deposits, particularly if customers withdraw deposits because they believe that interest rates offered by JPMorganChase are lower than those of competitors or if JPMorganChase makes incorrect assumptions about depositor behavior
- losses on available-for-sale ("AFS") securities held in the investment securities portfolio
- lower net interest income if central banks introduce interest rate increases more quickly than anticipated and this results in a misalignment in the pricing of short-term and long-term borrowings
- less liquidity in the financial markets, and
- higher funding costs.

All of these outcomes could adversely affect JPMorganChase's earnings or its liquidity and capital levels, and any negative outcomes could be more severe in a prolonged period of high interest rates. Higher interest rates can also negatively affect the payment performance on loans within JPMorganChase's consumer and wholesale loan portfolios that are linked to variable interest rates. If borrowers of variable rate loans are unable to afford higher interest payments, those borrowers may reduce or stop making payments, thereby causing JPMorganChase to incur losses and increased operational costs related to servicing a higher volume of delinquent loans.

On the other hand, a low or negative interest rate environment may cause:

- net interest margins to be compressed, which could reduce the amounts that JPMorganChase earns on its investment securities portfolio to the extent that it is unable to reinvest contemporaneously in higher-yielding instruments
- unanticipated or adverse changes in depositor behavior, which could negatively affect JPMorganChase's broader asset and liability management strategy, and

- a reduction in the value of JPMorganChase's mortgage servicing rights ("MSRs") asset, resulting in decreased revenues.

When credit spreads widen, it becomes more expensive for JPMorganChase to borrow. JPMorganChase's credit spreads may widen or narrow not only in response to events and circumstances that are specific to JPMorganChase but also as a result of general economic and geopolitical events and conditions. Changes in JPMorganChase's credit spreads will affect, positively or negatively, JPMorganChase's earnings on certain liabilities, such as derivatives, that are recorded at fair value.

JPMorganChase's results may be materially affected by market fluctuations and significant changes in the valuation of financial instruments.

The value of securities, derivatives and other financial instruments which JPMorganChase owns or in which it makes markets can be materially affected by market fluctuations. Market volatility, illiquid market conditions and other disruptions in the financial markets may make it extremely difficult to value certain financial instruments. Subsequent valuations of financial instruments in future periods, in light of factors then prevailing, may result in significant changes in the value of these instruments. In addition, at the time of any disposition of these financial instruments, the price that JPMorganChase ultimately realizes will depend on the demand and liquidity in the market at that time and may be materially lower than their current fair value. Any of these factors could cause a decline in the value of financial instruments that JPMorganChase owns or in which it makes markets, which may have an adverse effect on JPMorganChase's results of operations.

JPMorganChase's risk management and monitoring processes, including its stress testing framework, seek to quantify and manage JPMorganChase's exposure to more extreme market moves. However, JPMorganChase's hedging and other risk management strategies may not be effective, and it could incur significant losses, if extreme market events were to occur.

Credit

JPMorganChase can be negatively affected by adverse changes in the financial condition of clients, counterparties, custodians and CCPs.

JPMorganChase routinely executes transactions with clients and counterparties such as corporations, financial institutions, asset managers, hedge funds, securities exchanges and government entities within and outside the U.S. Many of these transactions expose JPMorganChase to the credit risk of its clients and counterparties, and can involve JPMorganChase in disputes and litigation if a client or counterparty defaults. JPMorganChase can also be subject to losses

or liability where a financial institution that it has appointed to provide custodial services for client assets or funds becomes insolvent as a result of fraud or the failure to abide by existing laws and obligations, or where clients are unable to access assets held by JPMorganChase as custodian due to governmental actions or other factors.

A default by, or the financial or operational failure of, a CCP through which JPMorganChase executes contracts would require JPMorganChase to replace those contracts, thereby increasing its operational costs and potentially resulting in losses. In addition, JPMorganChase can be exposed to losses if a member of a CCP in which JPMorganChase is also a member defaults on its obligations to the CCP because of requirements that each member of the CCP absorb a portion of those losses. Furthermore, JPMorganChase can be subject to bearing its share of non-default losses incurred by a CCP, including losses from custodial, settlement or investment activities or due to cyber or other security breaches.

As part of its clearing services activities, JPMorganChase is exposed to the risk of nonperformance by its clients, which it seeks to mitigate by requiring clients to provide adequate collateral. JPMorganChase is also exposed to intra-day credit risk of its clients in connection with providing cash management, clearing, custodial and other transaction services to those clients. If a client for which JPMorganChase provides these services becomes bankrupt or insolvent, JPMorganChase may incur losses, become involved in disputes and litigation with one or more CCPs, the client's bankruptcy estate and other creditors, or be subject to regulatory investigations. All of the foregoing events can increase JPMorganChase's operational and litigation costs, and JPMorganChase may suffer losses to the extent that any collateral that it has received is insufficient to cover those losses.

Transactions with government entities, including national, state, provincial, municipal and local authorities, can expose JPMorganChase to enhanced sovereign, credit, operational and reputation risks. Government entities may, among other things, claim that actions taken by government officials were beyond the legal authority of those officials or repudiate transactions authorized by a previous incumbent government. These types of actions have in the past caused, and could in the future cause, JPMorganChase to suffer losses or hamper its ability to conduct business in the relevant jurisdiction.

In addition, local laws, rules and regulations could limit JPMorganChase's ability to resolve disputes and litigation in the event of a counterparty default or unwillingness to make previously agreed-upon payments, which could subject JPMorganChase to losses.

Disputes may arise with counterparties to derivatives contracts with regard to the terms, the settlement procedures or the value of underlying collateral. The disposition of those disputes could cause JPMorganChase to incur unexpected transaction, operational and legal costs, or result in credit losses. These consequences can also impair JPMorganChase's ability to effectively manage its credit risk exposure from its market activities, or cause harm to JPMorganChase's reputation.

The financial or operational failure of a significant market participant, such as a major financial institution or a CCP, or concerns about the creditworthiness of such a market participant or its ability to fulfill its obligations, can cause substantial and cascading disruption within the financial markets, including in circumstances where coordinated action by multiple other market participants is required to address the failure or disruption. JPMorganChase's businesses could be significantly disrupted by such an event, particularly if it leads to other market participants incurring significant losses, experiencing liquidity issues or defaulting, and JPMorganChase is likely to have significant interrelationships with, and credit exposure to, such a significant market participant.

JPMorganChase may suffer losses if the value of collateral declines in stressed market conditions.

During periods of market stress or illiquidity, JPMorganChase's credit risk may be further increased when:

- JPMorganChase fails to realize the estimated value of the collateral it holds
- collateral is liquidated at prices that are not sufficient to recover the full amount owed to it, or
- counterparties are unable to post collateral, whether for operational or other reasons.

Furthermore, disputes with counterparties concerning the valuation of collateral may increase in times of significant market stress, volatility or illiquidity, and JPMorganChase could suffer losses during these periods if it is unable to realize the fair value of collateral or to manage declines in the value of collateral.

JPMorganChase could incur significant losses arising from concentrations of credit and market risk.

JPMorganChase is exposed to greater credit and market risk to the extent that groupings of its clients or counterparties, or obligors on securities and other financial instruments:

- engage in similar or related businesses, or in businesses in related industries
- do business in the same geographic region, or

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- have business profiles, models or strategies that could cause their ability to meet their obligations to be similarly affected by changes in economic conditions.

For example, a significant deterioration in the credit quality of a counterparty, borrower or other obligor could lead to concerns about the creditworthiness of other counterparties, borrowers or obligors in similar, related or dependent industries. This type of interrelationship could exacerbate JPMorganChase's credit, liquidity and market risk exposure and potentially cause it to incur losses, including fair value losses in its market-making businesses and investment portfolios. In addition, JPMorganChase may be required to increase the allowance for credit losses or establish other reserves with respect to certain clients, industries or country exposures in order to align with directives or expectations of its banking regulators.

Similarly, challenging economic conditions that affect a particular industry or geographic area could lead to concerns about the credit quality of counterparties, borrowers or other obligors not only in that particular industry or geography but in related or dependent industries, wherever located. These conditions could also heighten concerns about the ability of customers of JPMorganChase's consumer businesses who live in those areas or work in those affected industries or related or dependent industries to meet their obligations to JPMorganChase. JPMorganChase regularly monitors various segments of its credit and market risk exposures to assess the potential risks of concentration or contagion, but its ability to diversify or hedge its exposure against those risks may be limited.

JPMorganChase's consumer businesses can also be harmed by an excessive expansion of consumer credit by bank or non-bank competitors. Heightened competition for certain types of consumer loans could prompt industry-wide reactions such as significant reductions in the pricing or margins of those loans or the making of loans to less-creditworthy borrowers. If large numbers of consumers subsequently default on their loans, whether due to weak credit profiles, an economic downturn or other factors, this could impair their ability to repay obligations owed to JPMorganChase and result in higher charge-offs and other credit-related losses. More broadly, widespread defaults on consumer debt could lead to recessionary conditions in the U.S. economy, and JPMorganChase's consumer businesses may earn lower revenues in such an environment.

If JPMorganChase is unable to reduce positions effectively during a market dislocation, this can increase both the market and credit risks associated with those positions and the level of risk-weighted-assets ("RWA") that JPMorganChase holds on its

balance sheet. These factors could adversely affect JPMorganChase's capital position, funding costs and the profitability of its businesses.

Liquidity

JPMorganChase's ability to operate its businesses could be impaired if its liquidity is constrained.

JPMorganChase's liquidity can be impacted at any given time as a result of factors such as:

- market-wide illiquidity or disruption
- changes in liquidity or capital requirements resulting from changes in laws, rules and regulations, including those in response to economic effects of systemic events
- actions taken by the U.S. government or by the Federal Reserve to reduce its balance sheet, which may reduce deposits held by JPMorganChase and other financial institutions
- inability to sell assets, or to sell assets at favorable times or prices
- default by a CCP or other significant market participant
- unanticipated outflows of cash or collateral
- unexpected loss of deposits or higher than anticipated draws on lending-related commitments, and
- lack of market or customer confidence in JPMorganChase or financial institutions in general.

A reduction in JPMorganChase's liquidity may be caused by events over which it has little or no control. For example, periods of market stress, low investor confidence and significant market illiquidity could result in higher funding costs for JPMorganChase and could limit its access to some of its traditional sources of liquidity.

JPMorganChase may need to raise funding from alternative sources if its access to stable and lower-cost sources of funding, such as deposits and borrowings from Federal Home Loan Banks, is reduced. Alternative sources of funding could be more expensive or limited in availability. JPMorganChase's funding costs could also be negatively affected by actions that JPMorganChase may take in order to:

- satisfy applicable liquidity coverage ratio and net stable funding ratio requirements
- address obligations under its resolution plan, or
- satisfy regulatory requirements in jurisdictions outside the U.S. relating to the pre-positioning of liquidity in subsidiaries that are material legal entities.

More generally, if JPMorganChase fails to effectively manage its liquidity, this could constrain its ability to fund or invest in its businesses and subsidiaries, and thereby adversely affect its results of operations.

JPMorgan Chase & Co. is a holding company and depends on the cash flows of its subsidiaries to make payments on its outstanding securities.

JPMorgan Chase & Co. is a holding company that holds the stock of JPMorgan Chase Bank, N.A. and an intermediate holding company, JPMorgan Chase Holdings LLC (the "IHC"). The IHC in turn generally holds the stock of JPMorgan Chase's subsidiaries other than JPMorgan Chase Bank, N.A. and its subsidiaries. The IHC also owns other assets and provides intercompany lending to the Parent Company.

The Parent Company is obligated to contribute to the IHC substantially all the net proceeds received from securities issuances (including issuances of senior and subordinated debt securities and of preferred and common stock).

The ability of JPMorgan Chase Bank, N.A. and the IHC to make payments to the Parent Company is also limited. JPMorgan Chase Bank, N.A. is subject to regulatory restrictions on its dividend distributions, as well as capital adequacy requirements, such as the Supplementary Leverage Ratio ("SLR"), and liquidity requirements and other regulatory restrictions on its ability to make payments to the Parent Company. The IHC is prohibited from paying dividends or extending credit to the Parent Company if certain capital or liquidity thresholds are breached, or if limits are otherwise imposed by the Parent Company's management or Board of Directors.

As a result of these arrangements, the ability of the Parent Company to make various payments is dependent on its receiving dividends from JPMorgan Chase Bank, N.A. and dividends and borrowings from the IHC. These limitations could affect the Parent Company's ability to:

- pay interest on its debt securities
- pay dividends on its equity securities
- redeem or repurchase outstanding securities, and
- fulfill its other payment obligations.

These arrangements could also result in the Parent Company seeking protection under bankruptcy laws or otherwise entering into resolution proceedings at a time earlier than would have been the case absent the existence of the capital and liquidity thresholds to which JPMorgan Chase Bank, N.A. and the IHC are subject.

Reductions in JPMorgan Chase's credit ratings may adversely affect its liquidity and cost of funding.

JPMorgan Chase & Co. and certain of its principal subsidiaries are rated by credit rating agencies. Rating agencies evaluate general, firm-specific and industry-specific factors when determining credit ratings for a particular financial institution, including:

- expected future profitability

- risk management practices
- legal expenses
- ratings differentials between bank holding companies and their bank and non-bank subsidiaries
- regulatory developments
- assumptions about government support, and
- economic and geopolitical developments.

JPMorgan Chase closely monitors and manages, to the extent that it is able, factors that could influence its credit ratings. However, there is no assurance that JPMorgan Chase's credit ratings will not be downgraded in the future. Furthermore, any such downgrade could occur at times of broader market instability when JPMorgan Chase's options for responding to events may be more limited and general investor confidence is low.

A reduction in JPMorgan Chase's credit ratings could curtail JPMorgan Chase's business activities and reduce its profitability in a number of ways, including:

- reducing its access to capital markets
- materially increasing its cost of issuing and servicing securities
- triggering additional collateral or funding requirements, and
- decreasing the number of investors and counterparties that are willing or permitted to do business with or lend to JPMorgan Chase.

Any rating reduction could also increase the credit spreads charged by the market for taking credit risk on JPMorgan Chase & Co. and its subsidiaries. This could, in turn, adversely affect the value of debt and other obligations of JPMorgan Chase & Co. and its subsidiaries.

Capital

Maintaining the required level and composition of capital may impact JPMorgan Chase's ability to support business activities, meet evolving regulatory requirements and distribute capital to shareholders.

JPMorgan Chase is subject to various regulatory capital requirements, including leverage- and risk-based capital requirements. In addition, as a Global Systemically Important Bank ("GSIB"), JPMorgan Chase is required to hold additional capital buffers, including a GSIB surcharge, a Stress Capital Buffer ("SCB"), and a countercyclical buffer, each of which is reassessed at least annually. The amount of capital that JPMorgan Chase is required to hold in order to satisfy these leverage- and risk-based requirements could increase at any given time due to factors such as:

- actions by banking regulators, including changes in laws, rules and regulations

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- changes in the composition of JPMorganChase's balance sheet or developments that could increase RWA, such as increased market risk, customer delinquencies, client credit rating downgrades or other factors, and
- increases in estimated stress losses as determined by the Federal Reserve under the Comprehensive Capital Analysis and Review, which could increase JPMorganChase's SCB.

Any failure by or inability of JPMorganChase to maintain the required level and composition of capital, or unfavorable changes in applicable capital requirements, could have an adverse impact on JPMorganChase's shareholders, such as:

- reducing the amount of common stock that JPMorganChase is permitted to repurchase
- requiring the issuance of, or prohibiting the redemption of, capital instruments in a manner inconsistent with JPMorganChase's capital management strategy
- constraining the amount of dividends that may be paid on common stock, or
- curtailing JPMorganChase's business activities or operations.

In 2023, U.S. banking regulators released a proposal to implement the final Basel III reforms which would have significantly revised the risk-based capital requirements for banks with assets of \$100 billion or more, including JPMorganChase. In addition, in 2023 the Federal Reserve released a proposal to amend the calculation of the GSIB surcharge. Uncertainty remains regarding the content of the final versions of these rule proposals and how they might ultimately apply to JPMorganChase. However, it is possible that the final rules could impact JPMorganChase's decisions concerning the business activities in which it will engage and its levels of capital distributions to its shareholders.

Operational

JPMorganChase's businesses are dependent on the effectiveness of internal and external operational systems.

JPMorganChase's businesses rely on the ability of JPMorganChase's financial, accounting, transaction execution, data processing and other operational systems, including devices supporting those systems, to process, record, monitor and report a large number of transactions on a continuous basis, and to do so accurately, quickly and securely. In addition to proper design, installation, maintenance and training, the effective functioning of JPMorganChase's operational systems depends on:

- the quality of the information contained in those systems, as inaccurate, outdated, incomplete or corrupted data can significantly compromise the

functionality or reliability of a particular system and other systems to which it transmits or from which it receives information, and

- JPMorganChase's ability to continue to maintain and upgrade its systems on a regular and timely basis in line with technological advancements and evolving security requirements, maintain security and operational continuity of its systems, including by carefully managing any changes introduced to its systems, prevent unauthorized access and the misuse of access to its systems, and adhere to all applicable legal and regulatory requirements, particularly in regions where JPMorganChase may face a heightened risk of malicious activity.

JPMorganChase has experienced and expects that it will continue to experience failures and disruptions in the stability of its operational systems, including degraded performance of data processing systems, data quality issues, disruptions of network connectivity and malfunctioning software, as well as disruptions in its ability to access and use the operational systems of third parties and interruptions in service from third-party service providers. These incidents have resulted in various negative effects for customers, including the inability to access account information or transact through ATM, internet or mobile channels, the exfiltration of customer personal data, the recording of duplicative transactions and extended delays for customers requiring services from call centers. There can be no assurance that these and other types of operational failures or disruptions will not occur in the future.

JPMorganChase's ability to effectively manage the stability of its operational systems and infrastructure could be hindered by many factors, any of which could have a negative impact on JPMorganChase and its clients, customers and counterparties, including:

- JPMorganChase's ability to effectively maintain and upgrade systems and infrastructure can become more challenging as the speed, frequency, volume, interconnectivity and complexity of transactions continue to increase
- attempts by third parties to defraud JPMorganChase or its clients and customers continue to increase, evolve and become more complex, and during periods of market disruption or economic uncertainty, these attempts can be expected to further increase in volume
- errors made by JPMorganChase or another market participant, whether inadvertent or malicious, could cause widespread system disruption
- failure to detect weaknesses or shortcomings in operational systems in a timely manner
- isolated or seemingly insignificant errors in operational systems could compound, or migrate to other systems over time, to become larger issues

- disruptions in operational systems or in the ability of systems to communicate with each other could be caused by failures in synchronization or encryption software, or degraded performance of microprocessors, and
- attempts by third parties to block the use of key technology solutions by claiming that the use infringes on their intellectual property rights.

JPMorganChase also depends on its ability to access and use the operational systems of third parties, including its custodians, vendors (such as those that provide data and cloud computing services, and security and technology services) and other market participants (such as clearing and payment systems, CCPs and securities exchanges). These external operational systems with which JPMorgan is connected, whether directly or indirectly, can be sources of operational risk to JPMorganChase. JPMorganChase may be exposed not only to a systems failure or cyber attack that may be experienced by a vendor or market infrastructure with which JPMorganChase is directly connected, but also to a systems breakdown or cyber attack involving another party to which such a vendor or infrastructure is connected. Similarly, retailers, payment systems and processors, data aggregators and other external parties with which JPMorganChase's customers do business can increase JPMorganChase's operational risk. This is particularly the case where activities of customers or other parties are beyond JPMorganChase's security and control systems, including through the use of the internet, cloud computing services, and personal smart phones and other mobile devices or services.

If an external party obtains access to customer account data on JPMorganChase's systems, whether authorized or unauthorized, and that party misappropriates that data, this could result in negative outcomes for JPMorganChase and its clients and customers, including a heightened risk of fraudulent transactions using JPMorganChase's systems, losses from fraudulent transactions and reputational harm arising from the perception that JPMorganChase's systems may not be secure.

As JPMorganChase's interconnectivity with clients, customers and other external parties continues to expand, JPMorganChase increasingly faces the risk of operational failure or cyber attacks with respect to the systems of those parties. Security breaches affecting JPMorganChase's clients or customers, or systems breakdowns or failures, security breaches or human error or misconduct affecting other external parties, may require JPMorganChase to take steps to protect the integrity of its own operational systems or to safeguard confidential information, including restricting the access of customers to their accounts. These actions can increase JPMorganChase's

operational costs and potentially diminish customer satisfaction and confidence in JPMorganChase.

Furthermore, the widespread and expanding interconnectivity among financial institutions, clearing banks, CCPs, payments processors, financial technology companies, securities exchanges, clearing houses and other financial market infrastructures increases the risk that the disruption of an operational system involving one institution or entity, including due to a cyber attack, may cause industry-wide operational disruptions that could materially affect JPMorganChase's ability to conduct business. In addition, the risks associated with the disruption of an operational system of a third party could be exacerbated to the extent that the services provided by that system are used by a significant number or proportion of market participants.

The ineffectiveness, failure or other disruption of operational systems upon which JPMorganChase depends, including due to a systems malfunction, cyber incident or other systems failure, could result in unfavorable ripple effects in the financial markets and for JPMorganChase and its clients and customers, including:

- delays or other disruptions in providing services, including the provision of liquidity or information to clients and customers
- impairment of JPMorganChase's ability to execute transactions, including delays or failures in the confirmation or settlement of transactions or in obtaining access to funds or other assets required for settlement
- the possibility that funds transfers, capital markets trades or other transactions are executed erroneously
- financial losses, including due to loss-sharing requirements of CCPs, payment systems or other market infrastructures, or as possible restitution to clients and customers
- higher operational costs associated with replacing services provided by a system that has experienced a failure or other disruption
- limitations on JPMorganChase's ability to collect data needed for its business and operations
- loss of confidence in the ability of JPMorganChase, or financial institutions generally, to protect against and withstand operational disruptions
- dissatisfaction among JPMorganChase's clients or customers
- significant exposure to litigation and regulatory fines, penalties or other sanctions, and
- harm to JPMorganChase's reputation.

If JPMorganChase's operational systems, or those of acquired businesses or of external parties on which

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JPMorganChase's businesses depend, are unable to meet the requirements of JPMorganChase's businesses and operations or bank regulatory standards, or if they fail or have other significant shortcomings, JPMorganChase could be materially and adversely affected.

A successful cyber attack affecting JPMorganChase could cause significant harm to JPMorganChase and its clients and customers.

JPMorganChase experiences numerous cyber attacks on its computer systems, software, networks and other technology assets on a daily basis from various actors, including groups acting on behalf of hostile countries, cyber-criminals, "hacktivists" (i.e., individuals or groups that use technology to promote a political agenda or social change) and others. These cyber attacks can take many forms, including attempts to introduce computer viruses or malicious code, which are commonly referred to as "malware," into JPMorganChase's systems. These attacks are often designed to:

- obtain unauthorized access to JPMorganChase's systems or to confidential information belonging to JPMorganChase or its clients, customers, counterparties or employees
- manipulate data
- destroy data or systems with the aim of rendering services unavailable
- disrupt, sabotage or degrade service on JPMorganChase's systems
- steal money, or
- extort money through the use of so-called "ransomware."

JPMorganChase also experiences:

- distributed denial-of-service attacks intended to disrupt JPMorganChase's websites, including those that provide online banking and other services,
- a higher volume and complexity of cyber attacks against the backdrop of heightened geopolitical tensions, and
- a high volume of disruptions to internet-based services used by JPMorganChase that are provided by third parties.

JPMorganChase has experienced security breaches due to cyber attacks in the past, and it is inevitable that additional breaches will occur in the future. Any such breach could result in serious and harmful consequences for JPMorganChase or its clients and customers.

A principal reason that JPMorganChase cannot provide absolute security against cyber attacks is that it may not always be possible to anticipate, detect or recognize threats to JPMorganChase's systems, or to

implement effective preventive measures against all breaches due to evolving risks, including:

- the techniques used in cyber attacks evolve frequently and increase in sophistication, and therefore may not be recognized until launched or may go undetected for extended periods
- cyber attacks can originate from a wide variety of sources, including JPMorganChase's own employees, cyber-criminals, hacktivists, groups linked to terrorist organizations or hostile nation-states that can sustain malicious activities for extended periods, or third parties whose objective is to disrupt the operations of financial institutions more generally
- JPMorganChase does not have control over the cybersecurity of the systems of the large number of clients, customers, counterparties and third-party service providers with which it does business, and
- it is possible that a third party, after establishing a foothold on an internal network without being detected, may gain access to other networks and systems.

The risk of a security breach due to a cyber attack could increase in the future due to factors such as:

- JPMorganChase's ongoing expansion of its digital banking and other internet-based product offerings and its internal use of internet-based products and applications, including those that use cloud computing services
- advances in artificial intelligence, such as the use of machine learning, generative artificial intelligence and quantum computing by malicious actors to develop more advanced social engineering attacks, including targeted phishing attacks
- the inability to maintain the security of information transmitted by JPMorganChase due to advances in quantum computing that may counteract or nullify existing information protections, and
- the acquisition and integration of new businesses.

In addition, a third party could misappropriate confidential information obtained by intercepting signals or communications from mobile devices used by JPMorganChase's employees.

The dynamic nature of the cyber threat landscape, including the pace of innovation and increased threat of novel attack methods, necessitates ongoing investment in, as well as enhancement and adaptation of, cybersecurity controls, including the adoption of enhanced security measures in certain jurisdictions. Failure to discover or address emerging threats, known vulnerabilities or shortcomings in cybersecurity controls, or to prioritize or complete enhancements to address them, in each case in a timely manner, may leave JPMorganChase vulnerable to cyber attacks, potentially resulting in data breaches, financial losses,

reputational damage and regulatory penalties, including the failure to prioritize or complete enhancements relating to:

- preventing unauthorized access and protecting against the misuse of access, including the maintenance and enhancement of controls related to secure software development practices and identity and access management, such as those relating to the management of administrative access to systems
- detecting, escalating and addressing effectively and in a timely manner any vulnerabilities that may be present either in internally-developed software or externally-provided software or services, including vulnerabilities that could allow attackers to exploit unknown security flaws in software and hardware (“zero-day vulnerabilities”)
- oversight of third-party vendors and early detection of attacks against those vendors, including ransomware attacks and attacks targeting vulnerabilities in third-party open-source software, in support of the secure development and maintenance of internal systems
- maintaining and enhancing controls related to technology asset management and inventory systems to prevent the risk of undetected vulnerabilities that could undermine JPMorganChase’s ability to operate an effective control process
- upgrading the coverage and capabilities of systems and controls to protect JPMorganChase and its clients and customers from the impact of distributed denial-of-service attacks, or to recover from outages that could be caused by a malware or ransomware attack
- the continuing migration of client-facing services to the cloud, and modernization of those services
- strengthening network security and managing outbound connections to reduce the risk of data loss
- identifying, assessing and mitigating insider threat activities that could lead to the misuse of JPMorganChase’s systems or client and customer information, and
- integrating acquired businesses where system integration may be complex or may require extensive and lengthy remediation or enhancement of controls.

A successful penetration or circumvention of the security of JPMorganChase’s systems or the systems of a vendor, governmental body or another market participant could cause serious negative consequences, including:

- significant disruption of JPMorganChase’s operations and those of its clients, customers and

counterparties, including losing access to operational systems

- misappropriation of confidential information of JPMorganChase or that of its clients, customers, counterparties, employees or regulators
- disruption of or damage to JPMorganChase’s systems and those of its clients, customers and counterparties
- the inability, or extended delays in the ability, to fully recover and restore data that has been stolen, manipulated or destroyed, or the inability to prevent systems from processing fraudulent transactions
- demands that JPMorganChase pay a ransom to a malicious actor that has perpetrated a cybersecurity breach
- unintended violations by JPMorganChase of applicable privacy and other laws
- financial loss to JPMorganChase or to its clients, customers, counterparties or employees
- losses to JPMorganChase in excess of cyber insurance policy coverage
- loss of confidence in JPMorganChase’s cybersecurity and business resiliency measures
- dissatisfaction among JPMorganChase’s clients, customers or counterparties
- significant exposure to litigation and regulatory fines, penalties or other sanctions, and
- harm to JPMorganChase’s reputation.

The extent of a particular cyber attack, the methods and tools used by various actors, and the steps that JPMorganChase may need to take to investigate the attack may not be immediately clear, and it may take a significant amount of time before such an investigation can be completed. While such an investigation is ongoing, JPMorganChase may not necessarily know the full extent of the harm caused by the cyber attack, and that damage may continue to spread. These factors may inhibit JPMorganChase’s ability to provide rapid, full and reliable information about the cyber attack to its clients, customers, counterparties and regulators, as well as the public. Furthermore, it may not be clear how best to contain and remediate the harm caused by the cyber attack, and certain errors or actions could be repeated or compounded before they are discovered and remediated. Any or all of these factors could further increase the costs and consequences of a cyber attack.

JPMorganChase can be negatively affected if it fails to identify and address operational risks associated with the introduction of or changes to products, services and delivery platforms or the adoption of new technologies.

When JPMorganChase launches a new product or service, introduces a new platform for the delivery or

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distribution of products or services (including mobile connectivity, electronic trading and cloud computing), acquires or invests in a business, makes changes to an existing product, service or delivery platform, or adopts a new technology, it may not fully appreciate or identify new operational risks that may arise from those changes, including increased reliance on third party providers, or may fail to implement adequate controls to mitigate the risks associated with those changes. Any significant failure in this regard could diminish JPMorganChase's ability to operate one or more of its businesses or result in:

- potential liability to clients, counterparties and customers
- higher compliance, operational or integration costs
- higher litigation costs, including regulatory fines, penalties and other sanctions
- damage to JPMorganChase's reputation
- impairment of JPMorganChase's liquidity
- regulatory intervention, or
- weaker competitive standing.

Any of the foregoing consequences could materially and adversely affect JPMorganChase's businesses and results of operations.

JPMorganChase's business and operations rely on its ability, and the ability of key external parties, to maintain appropriately-staffed workforces, and on the competence, trustworthiness, health and safety of employees.

JPMorganChase's ability to operate its businesses efficiently and profitably, to offer products and services that meet the expectations of its clients and customers, and to maintain an effective risk management framework is highly dependent on its ability to staff its operations appropriately and on the competence, trustworthiness, health and safety of its employees. JPMorganChase's businesses and operations similarly rely on the workforces of third parties, including employees of vendors, custodians and financial markets infrastructures, and of businesses that it may seek to acquire.

JPMorganChase's businesses could be materially and adversely affected by:

- the ineffective implementation of business decisions
- any failure to institute controls that appropriately address risks associated with business activities, or to appropriately train employees with respect to those risks and controls
- staffing shortages, particularly in tight labor markets
- the possibility that significant portions of JPMorganChase's workforce are unable to work effectively, including because of illness, quarantines, shelter-in-place arrangements, government actions or other restrictions in connection with health

emergencies, the spread of infectious diseases, epidemics or pandemics, or due to extraordinary events beyond JPMorganChase's control such as natural disasters or an outbreak or escalation of hostilities

- a significant operational breakdown or failure, theft, fraud or other unlawful conduct, or
- other negative outcomes caused by human error or misconduct by an employee of JPMorganChase or of another party on which JPMorganChase's businesses or operations rely.

JPMorganChase's operations could also be impaired if the measures taken by it or by governmental authorities to protect the health and safety of its employees are ineffective, or if any external party on which JPMorganChase relies fails to take appropriate and effective actions to protect the health and safety of its employees.

JPMorganChase faces substantial legal and operational risks in the processing and safeguarding of personal information.

JPMorganChase's businesses and operations are subject to complex and evolving laws, rules and regulations, both within and outside the U.S., governing the privacy and protection of personal information of individuals. Governmental authorities around the world have adopted and are considering the adoption of numerous legislative and regulatory initiatives concerning privacy, data protection and security. Litigation or enforcement actions relating to these laws, rules and regulations could result in fines or orders requiring that JPMorganChase change its data-related practices, which could have an adverse effect on JPMorganChase's ability to provide products and otherwise harm its business operations.

Implementing processes relating to JPMorganChase's collection, use, sharing and storage of personal information to comply with all applicable laws, rules and regulations in all relevant jurisdictions, including where the laws of different jurisdictions are in conflict, can:

- increase JPMorganChase's compliance and operating costs
- hinder the development of new products or services, curtail the offering of existing products or services, or affect how products and services are offered to clients and customers
- demand significant oversight by JPMorganChase's management, and
- require JPMorganChase to structure its businesses, operations and systems in less efficient ways.

Not all of JPMorganChase's clients, customers, vendors, counterparties and other external parties may have appropriate controls in place to protect the confidentiality, integrity or availability of the

information exchanged between them and JPMorganChase, particularly where information is transmitted by electronic means. JPMorganChase could be exposed to litigation or regulatory fines, penalties or other sanctions if personal information of clients, customers, employees or others were to be mishandled or misused, such as situations where such information is:

- erroneously provided to parties who are not permitted to have the information, or
- intercepted or otherwise compromised by unauthorized third parties.

The increasing sophistication of artificial intelligence technologies poses a greater risk of identity fraud, as malicious actors may exploit artificial intelligence to create convincing false identities or manipulate verification processes. This challenge necessitates ongoing enhancements to client verification systems and security protocols to prevent unauthorized access and protect sensitive client information. Failure to manage these risks or to implement effective countermeasures could lead to unauthorized transactions, financial losses, reputational damage and increased regulatory scrutiny.

Concerns regarding the effectiveness of JPMorganChase's measures to safeguard personal information, or the perception that those measures are inadequate, could cause JPMorganChase to lose existing or potential clients and customers or employees, and thereby reduce JPMorganChase's revenues. Furthermore, any failure or perceived failure by JPMorganChase to comply with applicable privacy or data protection laws, rules and regulations, or any failure to appropriately calibrate, manage and monitor access by employees or third parties to personal information, could subject JPMorganChase to inquiries, examinations and investigations that could result in requirements to modify or cease certain operations or practices, significant liabilities or regulatory fines, penalties or other sanctions. Any of these could damage JPMorganChase's reputation and otherwise adversely affect its businesses.

In recent years, well-publicized incidents involving the inappropriate collection, use, sharing or storage of personal information have led to expanded governmental scrutiny of practices relating to the processing or safeguarding of personal information by companies in the U.S. and other countries. That scrutiny has in some cases resulted in, and could in the future lead to, the adoption of stricter laws, rules and regulations relating to the collection, use, sharing and storage of personal information. These types of laws, rules and regulations can prohibit or significantly restrict financial services firms such as JPMorganChase from transferring information across national borders or sharing information among affiliates or with third parties such as vendors, thereby

increase compliance costs and operational risk, or restrict JPMorganChase's use of personal information when developing or offering products or services to customers. Some countries are considering or have adopted legislation implementing data protection requirements or requiring local storage and processing of data which could increase the cost and complexity of JPMorganChase's delivery of products and services. These restrictions could also inhibit JPMorganChase's development or marketing of certain products or services, or increase the costs of offering them to customers.

JPMorganChase's operations, results and reputation could be harmed by occurrences of extraordinary events beyond its control.

JPMorganChase's business and operational systems could be seriously disrupted, and its reputation could be harmed, by events or contributing factors that are wholly or partially beyond its control, including material instances of:

- cyber attacks
- security breaches of its physical premises, including threats to health and safety
- power, telecommunications or internet outages, or shutdowns of mass transit
- failure of, or loss of access to, technology or operational systems, including any resulting loss of critical data
- interruption of service from third-party service providers
- damage to or loss of property or assets of JPMorganChase or third parties, and any consequent injuries, including in connection with any construction projects undertaken by JPMorganChase
- effects of climate change
- natural disasters or severe weather conditions
- accidents such as explosions or structural failures
- health emergencies, the spread of infectious diseases, epidemics or pandemics, or
- events arising from local or larger-scale civil or political unrest, any outbreak or escalation of hostilities, or terrorist acts.

JPMorganChase operates a Firmwide resiliency framework that is intended to enable it to prepare for and adapt to changing conditions and withstand and recover from, and address any adverse effects on its operations caused by, disruptions that may impact critical business functions and supporting assets, including its staff, technology, data and facilities and those of third-party service providers.

Although not every form of disruption can be anticipated or defended against, JPMorganChase

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strives for resiliency or recovery in a range of scenarios in the event of a disruption, including due to the occurrence of an extraordinary event beyond its control. There can be no assurance that JPMorganChase's Firmwide resiliency framework will fully mitigate all potential resiliency risks to JPMorganChase, its clients, and customers and third parties with which it does business, or that its resiliency framework will be adequate to address the effects of simultaneous occurrences of multiple or extended disruption events. In addition, JPMorganChase's ability to respond effectively to a disruption event could be hampered to the extent that the members of its workforce, physical assets, systems and other support infrastructure, or those of its third-party service providers, that are needed to address the event are geographically dispersed, or conversely, if such an event were to occur in an area in which they are concentrated. Further, should extraordinary events or the factors that cause or contribute to those events become more chronic, the disruptive effects of those events on JPMorganChase's business and operations, and on its clients, customers, counterparties and employees, could become more significant and long-lasting.

Any significant failure or disruption of JPMorganChase's operations or operational systems, or the occurrence of one or more extraordinary events that are beyond its control, could:

- hinder JPMorganChase's ability to provide services to its clients and customers or to transact with its counterparties
- require it to expend significant resources to correct the failure or disruption or to address the event
- cause it to incur losses or liabilities, including from loss of revenue, damage to or loss of property, or injuries
- disrupt market infrastructure systems on which JPMorganChase's businesses rely
- expose it to litigation or regulatory fines, penalties or other sanctions, and
- harm its reputation.

The occurrence of one or more extraordinary events could also negatively impact the financial condition or creditworthiness of JPMorganChase's clients and customers, and could lead to an increase in delinquencies, additions to the allowance for credit losses and higher net charge-offs, which can reduce JPMorganChase's earnings.

Data quality is essential to JPMorganChase's business and operations, and if JPMorganChase fails to maintain adequate data management processes, this could adversely affect its ability to effectively manage its businesses, comply with applicable laws, rules and regulations, or remain competitive.

JPMorganChase relies on accurate, timely and complete data to effectively operate its systems and processes, including:

- assessing risk exposures and limits
- monitoring and detecting fraudulent transactions and cyber threats
- developing or maintaining models and other analytical and judgment-based estimations, including those that use machine learning or artificial intelligence
- implementing and maintaining compliance programs, and
- preparing financial statements, disclosures and regulatory reports, as well as internal reporting

Any deficiencies in JPMorganChase's data management processes, including with respect to the accuracy or completeness of data, the timeliness of data collection, the analysis or validation of data, or the safeguarding of data could undermine the reliability and effectiveness of its operations, including:

- risk management practices, including inaccurate or untimely risk reporting
- delivery of regulatory reporting or internal or external financial reporting
- compliance practices, such as those relating to transaction monitoring, customer screening, blocking and rejecting transactions, recordkeeping or reporting
- business activities, such as those related to managing JPMorganChase's market-making positions and liquidity and capital levels, including reliance on timely data for informed decision-making
- providing services to clients and customers, including transaction processing, lending services, account management and customer support, or
- fraud detection and prevention processes.

Any or all of these factors could impair the ability of JPMorganChase to make sound business decisions, cause it to incur higher operational and compliance costs, result in operational breakdowns or failure to meet its regulatory requirements, negatively affect clients and customers, or lead to reputational harm.

Enhanced regulatory and other standards for the oversight of vendors and other service providers can result in higher costs and other potential exposures.

JPMorganChase must comply with enhanced regulatory and other standards associated with doing

business with vendors and other service providers, including standards relating to the outsourcing of functions as well as the performance of significant banking and other functions by subsidiaries. JPMorganChase incurs significant costs and expenses in connection with its initiatives to address the risks associated with oversight of its internal and external service providers. JPMorganChase's failure to appropriately assess and manage these relationships, especially those involving significant banking functions, shared services or other critical activities, could materially adversely affect JPMorganChase. Specifically, any such failure could result in:

- potential harm to clients and customers, and any liability associated with that harm
- regulatory fines, penalties or other sanctions
- lower revenues, and the opportunity cost from lost revenues
- increased operational costs, or
- harm to JPMorganChase's reputation.

JPMorganChase's risk management framework and control environment will not be effective in identifying and mitigating every risk to JPMorganChase.

Any inadequacy or lapse in JPMorganChase's risk management framework, governance structure, practices, models or reporting systems, or in its control environment, could expose it to unexpected losses, and its financial condition or results of operations could be materially and adversely affected. Any such inadequacy or lapse could:

- hinder the timely escalation of material risk issues to JPMorganChase's senior management and Board of Directors
- lead to business decisions that have negative outcomes for JPMorganChase
- require significant resources and time to remediate
- lead to non-compliance with laws, rules and regulations
- attract heightened regulatory scrutiny
- expose JPMorganChase to litigation, regulatory investigations or regulatory fines, penalties or other sanctions
- lead to potential harm to customers and clients, and any liability associated with that harm
- harm its reputation, or
- otherwise diminish confidence in JPMorganChase.

Many of JPMorganChase's risk management strategies and techniques consider historical market behavior and to some degree are based on management's subjective judgment or assumptions. For example, many models used by JPMorganChase are based on assumptions regarding historical

correlations among prices of various asset classes or other market indicators. In times of market stress, including difficult or less liquid market environments, or in the event of other unforeseen circumstances, previously uncorrelated indicators may become correlated. Conversely, previously-correlated indicators may become uncorrelated at those times. Sudden market movements and unanticipated market or economic events could, in some circumstances, limit the effectiveness of JPMorganChase's risk management strategies, causing it to incur losses.

JPMorganChase could recognize unexpected losses, its capital levels could be reduced and it could face greater regulatory scrutiny if its models, estimations or judgments, including those used in its financial statements, are inadequate or incorrect.

JPMorganChase has developed and uses a variety of models and other analytical and judgment-based estimations to measure, monitor and implement controls over its market, credit, capital, liquidity, operational and other risks. JPMorganChase also uses internal models and estimations as a basis for its stress testing and in connection with the preparation of its financial statements under U.S. generally accepted accounting principles ("U.S. GAAP").

These models and estimations are based on a variety of assumptions and historical trends, and are periodically reviewed and modified as necessary. The models and estimations that JPMorganChase uses, including those that use machine learning or artificial intelligence, may not be effective in all cases to identify, observe and mitigate risk due to a variety of factors, such as:

- reliance on historical trends that may not persist in the future, including assumptions underlying the models and estimations such as correlations among certain market indicators or asset prices
- inherent limitations associated with forecasting uncertain economic and financial outcomes
- historical trend information may be incomplete, or may not be indicative of severely negative market conditions such as extreme volatility, dislocation or lack of liquidity
- sudden illiquidity in markets or declines in prices of certain loans and securities may make it more difficult to value certain financial instruments
- technology that is introduced to run models or estimations may not perform as expected, or may not be well understood by the personnel using the technology
- models and estimations may contain erroneous data, valuations, formulas or algorithms
- review processes may fail to detect flaws in models and estimations, and

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- models may inadvertently incorporate biases present in data used in the models.

JPMorganChase may experience unexpected losses if models, estimates or judgments used or applied in connection with its risk management activities or the preparation of its financial statements are inadequate or incorrect. For example, where quoted market prices are not available for certain financial instruments that require a determination of their fair value, JPMorganChase may make fair value determinations based on internally developed models or other means which ultimately rely to some degree on management estimates and judgment. In addition, JPMorganChase may experience increased uncertainty in its estimates if assets acquired differ from those used to develop those models, which may lead to unexpected losses.

Similarly, JPMorganChase establishes an allowance for expected credit losses related to its credit exposures which requires significant judgments, including forecasts of how macroeconomic conditions might impair the ability of JPMorganChase's clients and customers to repay their loans or other obligations. These types of estimates and judgments may not prove to be accurate due to a variety of factors, including when the current and forecasted environments are significantly different from the historical environments upon which the models were developed. The increased uncertainty may necessitate a greater degree of judgment and analytics to inform any adjustments that JPMorganChase may make to model outputs than would otherwise be the case.

Some of the models and other analytical and judgment-based estimations used by JPMorganChase in managing risks are subject to review by, and require the approval of, JPMorganChase's regulators. These reviews are required before JPMorganChase may use those models and estimations for calculating market risk RWA, credit risk RWA and operational risk RWA under Basel III. If JPMorganChase's models or estimations are not approved by its regulators, it may be subject to higher capital charges, which could adversely affect its financial results or limit the ability to expand its businesses.

Lapses, weaknesses or deficiencies in controls over disclosure or financial reporting could materially affect JPMorganChase's profitability or reputation.

JPMorganChase's businesses and operations are subject to complex and evolving laws, rules and regulations, both within and outside the U.S., requiring continuous enhancements to various disclosures in its financial statements and regulatory reports.

There can be no assurance that JPMorganChase's disclosure controls and procedures will be effective in every circumstance, or that a material weakness or significant deficiency in internal control over financial reporting will not occur. Any such lapse, weakness or

deficiency could result in inaccurate financial reporting which, in turn, could:

- materially and adversely affect JPMorganChase's business and results of operations or financial condition
- restrict its ability to access the capital markets
- require it to expend significant resources to correct the lapse, weakness or deficiency
- expose it to litigation or regulatory fines, penalties or other sanctions
- harm its reputation, or
- otherwise diminish investor confidence in JPMorganChase.

Strategic

JPMorganChase's results or competitive standing could suffer if its management fails to develop and execute effective business strategies, and to anticipate changes affecting those strategies.

The development and execution of effective business strategies by JPMorganChase's management, along with the ability to anticipate and respond to shifts in the competitive environment, are critical to JPMorganChase's competitive standing and to achieving its strategic objectives. These strategies relate to:

- the products and services that JPMorganChase offers
- the geographies in which it operates
- the types of clients and customers that it serves
- the businesses that it acquires or in which it invests
- the counterparties with which it does business
- the technologies that it adopts or in which it invests, which may include new and currently unproven technologies, and
- the methods, distribution channels and third party service providers by or through which it offers products and services.

If management makes choices about these strategies and goals that prove to be incorrect, are based on incomplete, inaccurate or fraudulent information, do not accurately assess the competitive landscape and industry trends, or fail to address changing regulatory and market environments or the expectations of clients, customers, investors, employees and other stakeholders, then the franchise values and growth prospects of JPMorganChase's businesses may suffer and its earnings could decline.

JPMorganChase's growth prospects also depend on management's ability to develop and execute effective business plans to address these strategic priorities, both in the near term and over longer time horizons. Management's effectiveness in this regard will affect

JPMorganChase's ability to develop and enhance its resources, control expenses and return capital to shareholders. Each of these objectives could be adversely affected by any failure on the part of management to:

- devise effective business plans and strategies
- offer products and services that meet changing expectations of clients and customers
- allocate capital in a manner that promotes long-term stability to enable JPMorganChase to build and invest in market-leading businesses, even in a highly stressed environment
- allocate capital appropriately due to imprecise modeling or subjective judgments made in connection with those allocations
- appropriately assess and monitor principal investments made to enhance or accelerate JPMorganChase's business strategies
- conduct appropriate due diligence on prospective business acquisitions or investments, or effectively integrate newly-acquired businesses
- appropriately address concerns of clients, customers, investors, employees and other stakeholders, including with respect to climate and other ESG matters
- react quickly to changes in market conditions or market structures, or
- develop and enhance the operational, technology, risk, financial and managerial resources and capabilities necessary to grow and manage JPMorganChase's businesses.

Furthermore, JPMorganChase may incur costs in connection with disposing of excess properties, premises and facilities, and those costs could be material to its results of operations.

JPMorganChase faces significant and increasing competition in the rapidly evolving financial services industry.

JPMorganChase operates in a highly competitive environment in which it must evolve and adapt to changes in financial regulation, technological advances, increased public scrutiny and changes in economic conditions. JPMorganChase expects that competition in the U.S. and global financial services industry will continue to be intense. Competitors include:

- other banks and financial institutions
- trading, advisory and investment management firms
- finance companies
- technology companies, and
- other non-bank firms that are engaged in providing similar as well as new products and services.

JPMorganChase cannot provide assurance that the significant competition in the financial services industry will not materially and adversely affect its future results of operations. For example, aggressive or less disciplined lending practices by non-bank competitors could lead to a loss of market share for traditional banks, and in an economic downturn could result in instability in the financial services industry and adversely impact other market participants, including JPMorganChase.

New competitors in the financial services industry continue to emerge. For example, technological advances and the growth of e-commerce have made it possible for non-depository institutions to offer products and services that traditionally were banking products. These advances have also allowed financial institutions and other companies to provide electronic and internet-based financial solutions, including electronic securities and cryptocurrency trading, lending and other extensions of credit to consumers, payments processing and online automated algorithmic-based investment advice. Furthermore, both financial institutions and their non-banking competitors face the risk that payments processing and other products and services, including deposits and other traditional banking products, could be significantly disrupted by the use of new technologies, such as cryptocurrencies and other applications using secure distributed ledgers, that may not require intermediation. New technologies have required and could require JPMorganChase to spend more to modify or adapt its products to attract and retain clients and customers or to match products and services offered by its competitors, including technology companies. In addition, new technologies may be used by customers, or breached or infiltrated by third parties, in unexpected ways, which can increase JPMorganChase's costs for complying with laws, rules and regulations that apply to the offering of products and services through those technologies and reduce the income that JPMorganChase earns from providing products and services through those technologies.

Ongoing or increased competition may put pressure on the pricing for JPMorganChase's products and services or may cause JPMorganChase to lose market share, particularly with respect to traditional banking products. This competition may be based on quality and variety of products and services offered, transaction execution, innovation, reputation and price. The failure of any of JPMorganChase's businesses to meet the expectations of clients and customers, whether due to general market conditions, under-performance, a decision not to offer a particular product or service, changes in client and customer expectations or other factors, could affect JPMorganChase's ability to attract or retain clients and customers. Any such impact could, in turn, reduce

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JPMorganChase's revenues. Increased competition also may require JPMorganChase to make additional capital investments in its businesses, or to extend more of its capital on behalf of its clients to remain competitive.

The effects of climate change could adversely affect JPMorganChase's business and operations, both directly and as a result of impacts on its clients and customers.

JPMorganChase operates in many regions, countries and communities around the world where its business, and the activities of its clients and customers, could be adversely affected by climate change. Climate change could manifest as a financial risk to JPMorganChase either through changes in the physical climate or from the process of transitioning to a lower-carbon economy. Both physical risks and transition risks associated with climate change could have negative impacts on the financial condition or creditworthiness of JPMorganChase's clients and customers, on JPMorganChase's exposure to affected companies and markets, and on the effectiveness of JPMorganChase's existing business strategy with respect to its operations, clients and customers.

Physical risks include the increased frequency or severity of acute weather events, such as floods, wildfires and tropical cyclones, and chronic shifts in the climate, such as rising sea levels, persistent changes in precipitation levels, or increases in average ambient temperatures. Potential adverse impacts of climate-related physical risks to JPMorganChase, its clients or customers include:

- declines in asset values, including due to the destruction or degradation of property
- reduced availability or increased cost of insurance for clients of JPMorganChase
- interruptions to business operations, including supply chain disruption, and
- population migration or unemployment in affected regions.

Transition risks arise from the financial and economic consequences of society's shift towards a lower-carbon economy, such as changes in public policy, adoption of new technologies or changes in consumer preferences towards low-carbon goods and services. These risks could also be influenced by changes in the physical climate. Potential adverse impacts of transition risks to JPMorganChase, its clients or customers include:

- sudden devaluation of assets, including unanticipated write-downs ("stranded assets")
- increased operational and compliance costs driven by changes in climate policy

- increased energy costs driven by governmental actions and initiatives such as emission pricing and accelerated decarbonization policies
- negative consequences to business models, and the need to make changes in response to those consequences, and
- damage to JPMorganChase's reputation, including due to any perception that its business practices are contrary to public policy or the preferences of different stakeholders.

Climate risks can also arise from inconsistencies and conflicts in the manner in which climate policy and financial regulations are implemented in the many regions where JPMorganChase operates, including initiatives to apply and enforce policy and regulation with extraterritorial effect. Additionally, internal models and estimations used in climate risk assessments have an increased level of uncertainty due to limited historical trend information and the absence of standardized, reliable and comprehensive greenhouse gas emissions data, which could lead to inaccurate disclosures or financial reporting.

Conduct

Conduct failure by JPMorganChase employees can harm clients and customers, impact market integrity, damage JPMorganChase's reputation and trigger litigation and regulatory action.

JPMorganChase's employees interact with clients, customers, counterparties and other market and industry participants, and with each other, every day. All employees are expected to demonstrate values and exhibit the behaviors that are an integral part of JPMorganChase's Code of Conduct and Business Principles. JPMorganChase endeavors to embed conduct risk management throughout an employee's life cycle, including recruiting, onboarding, training and development, and performance management. Conduct risk management is also an integral component of JPMorganChase's promotion and compensation processes.

Notwithstanding these expectations, policies and practices, certain employees have engaged in improper or illegal conduct in the past. These instances of misconduct have resulted in litigation, and resolutions of governmental investigations or enforcement actions involving consent orders, deferred prosecution agreements, non-prosecution agreements and other civil or criminal sanctions. There is no assurance that further inappropriate or unlawful actions by employees have not occurred or will not occur, lead to a violation of the terms of these resolutions (and associated consequences), or that any such actions will always be detected, deterred or prevented.

JPMorganChase's reputation could be harmed by, and collateral consequences could result from, a failure by

one or more employees to conduct themselves in accordance with JPMorganChase's expectations, policies and practices, including by acting in ways that harm clients, customers, other market participants, employees or others. Some examples of this include:

- improperly selling and marketing JPMorganChase's products or services
- engaging in insider trading, market manipulation or unauthorized trading
- engaging in improper or fraudulent behavior in connection with government relief programs
- facilitating a transaction where a material objective is to achieve a particular tax, accounting or financial disclosure treatment that may be subject to scrutiny by governmental or regulatory authorities, or where the proposed treatment is unclear or may not reflect the economic substance of the transaction
- failing to fulfill fiduciary obligations or other duties owed to clients or customers
- violating antitrust or anti-competition laws by colluding with other market participants
- using electronic communications channels that have not been approved by JPMorganChase
- engaging in discriminatory behavior or harassment with respect to clients, customers or employees, or acting contrary to JPMorganChase's goal of fostering an inclusive workplace
- managing or reporting risks in ways that subordinate JPMorganChase's risk appetite to business performance goals or employee compensation objectives, and
- misappropriating property, confidential or proprietary information, or technology assets belonging to JPMorganChase, its clients and customers or third parties.

The consequences of any failure by one or more employees to conduct themselves in accordance with JPMorganChase's expectations, policies or practices could include litigation, or regulatory or other governmental investigations or enforcement actions. Any of these proceedings or actions could result in judgments, settlements, fines, penalties or other sanctions, or lead to:

- financial losses
- increased operational and compliance costs
- greater scrutiny by regulators and other parties
- regulatory actions that require JPMorganChase to restructure, curtail or cease certain of its activities
- the need for significant oversight by JPMorganChase's management
- loss of clients or customers, and
- harm to JPMorganChase's reputation.

The foregoing risks could be heightened with respect to newly-acquired businesses if JPMorganChase fails to successfully integrate employees of those businesses or any of those employees do not conduct themselves in accordance with JPMorganChase's expectations, policies and practices.

Reputation

Damage to JPMorganChase's reputation could harm its businesses.

Maintaining trust in JPMorganChase is critical to its ability to attract and retain clients, customers, investors and employees. Damage to JPMorganChase's reputation can therefore cause significant harm to JPMorganChase's business and prospects, and can arise from numerous sources, including:

- employee misconduct, including discriminatory behavior or harassment with respect to clients, customers or employees, or actions that are contrary to JPMorganChase's goal of fostering an inclusive workplace
- security breaches, including as a result of cyber attacks
- failure to safeguard client, customer or employee information
- failure to manage risks associated with its client relationships, or with transactions or business activities in which JPMorganChase or its clients engage, including transactions or activities that may be unpopular among one or more constituencies
- rapid and broad dissemination of misinformation and disinformation across the media landscape, including social networking sites
- incorrect, biased or misleading results or content generated by artificial intelligence, leading to harmful outcomes, including discrimination in lending practices against vulnerable populations, fraud, manipulation of customers, privacy breaches or intellectual property infringement
- deficiencies or perceived failures in managing ESG-related initiatives, including modifying or failing to meet publicly-announced targets
- operational failures
- litigation or regulatory fines, penalties or other sanctions
- actions taken in executing regulatory and governmental requirements during a global or regional health emergency, spread of infectious disease, epidemic or pandemic
- regulatory investigations or enforcement actions, or resolutions of these matters, and
- failure or perceived failure to comply with laws, rules or regulations by JPMorganChase or its clients,

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customers, counterparties or other parties, including newly-acquired businesses, companies in which JPMorganChase has made principal investments, parties to joint ventures with JPMorganChase, and vendors with which JPMorganChase does business.

Social and environmental activists have been targeting JPMorganChase and other financial services firms with public criticism concerning their business practices, including business relationships with clients that are engaged in certain sensitive industries, such as companies:

- whose products are or are perceived to be harmful to human health, or
- whose activities negatively affect or are perceived to negatively affect the environment, workers' rights or communities.

Activists have also taken actions intended to change or influence JPMorganChase's business practices with respect to ESG matters, including public protests at JPMorganChase's headquarters and other properties, and submitting specific ESG-related proposals for a vote by JPMorganChase's shareholders.

In addition, JPMorganChase has been and expects that it will continue to be criticized by activists, politicians and other members of the public concerning business practices or positions taken by JPMorganChase with respect to matters of public policy (such as diversity, equity and inclusion initiatives) or regarding transactions or other business or interactions between JPMorganChase and governmental or regulatory bodies. Furthermore, JPMorganChase's relationships or ability to transact with clients and customers, and with governmental or regulatory bodies in jurisdictions in which JPMorganChase does business, could be adversely affected if its decisions with respect to doing business with companies in certain sensitive industries are perceived to harm those companies or to align with particular political viewpoints. The foregoing types of criticism can be more widespread during election years in various jurisdictions, and could have the effect of focusing attention on a company such as JPMorganChase as part of a wider public debate on public policy matters. Furthermore, JPMorganChase's participation in or association with certain environmental and social industry groups or initiatives could be viewed by activists or governmental authorities as boycotting or other discriminatory business behavior.

These and other types of criticism and actions directed at JPMorganChase could potentially engender dissatisfaction among clients, customers, investors, employees, government officials and other stakeholders. In all of these cases, JPMorganChase's reputation and its business and results of operations could be harmed by:

- greater scrutiny from governmental or regulatory bodies, or further criticism from politicians and other members of the public, including in the form of governmental or regulatory investigations or litigation
- unfavorable coverage or commentary in the media, including through social media campaigns
- certain clients and customers ceasing doing business with JPMorganChase, and encouraging others to do so
- impairment of JPMorganChase's ability to attract new clients and customers, to expand its relationships with existing clients and customers, or to hire or retain employees, or
- certain investors opting to divest from investments in securities of JPMorganChase.

Actions by the financial services industry generally or individuals in the industry can also affect JPMorganChase's reputation. For example, the reputation of the industry as a whole can be damaged by concerns that:

- consumers have been treated unfairly by a financial institution, or
- a financial institution has acted inappropriately with respect to the methods used to offer products to customers.

If JPMorganChase is perceived to have engaged in these types of behaviors, this could weaken its reputation among clients or customers, employees or other stakeholders.

Failure to effectively manage potential conflicts of interest or to satisfy fiduciary obligations can result in litigation and enforcement actions, as well as damage JPMorganChase's reputation.

JPMorganChase's ability to manage potential conflicts of interest is highly complex due to the broad range of its business activities which encompass a variety of transactions, obligations and interests with and among JPMorganChase's clients and customers.

JPMorganChase can become subject to litigation, enforcement actions, and heightened regulatory scrutiny, and its reputation can be damaged, by the failure or perceived failure to:

- adequately address or appropriately disclose conflicts of interest, including potential conflicts of interest that may arise in connection with providing multiple products and services in, or having one or more investments related to, the same transaction
- identify and address any conflict of interest that a third party with which it does business may have with respect to a transaction involving JPMorganChase
- deliver appropriate standards of service and quality

- treat clients and customers fairly and with the appropriate standard of care
- use client and customer data responsibly and in a manner that meets legal requirements and regulatory expectations
- provide fiduciary products or services in accordance with the applicable legal and regulatory standards, or
- handle or use confidential information of customers or clients appropriately and in compliance with applicable data protection and privacy laws, rules and regulations.

A failure or perceived failure to appropriately address conflicts of interest or fiduciary obligations could result in customer dissatisfaction, litigation and regulatory fines, penalties or other sanctions, and heightened regulatory scrutiny and enforcement actions, all of which can lead to lost revenue and higher operating costs and cause serious harm to JPMorganChase's reputation.

Country

An outbreak or escalation of hostilities between countries or within a country or region could have a material adverse effect on the global economy and on JPMorganChase's businesses within the affected region or globally.

Aggressive actions by hostile governments or groups, including armed conflict or intensified cyber attacks, could expand in unpredictable ways by drawing in other countries or escalating into full-scale war with potentially catastrophic consequences, particularly if one or more of the combatants possess nuclear weapons. Depending on the scope of the conflict, the hostilities could result in:

- worldwide economic disruption
- heightened volatility in financial markets
- severe declines in asset values, accompanied by widespread sell-offs of investments
- sudden increases in prices in the energy and commodity markets or for certain safe haven currencies
- substantial depreciation of local currencies, potentially leading to defaults by borrowers and counterparties in the affected region
- disruption of global trade
- diminished consumer, business and investor confidence
- refugee and humanitarian crises, and
- new economic sanctions or other regulatory requirements, including those that introduce exceptional compliance challenges for multinational companies such as JPMorganChase.

Any of the above consequences could have significant negative effects on JPMorganChase's operations and earnings, both in the countries or regions directly affected by the hostilities or globally. Further, if the U.S. were to become directly involved in such a conflict, this could lead to a curtailment of any operations that JPMorganChase may have in the affected countries or region, as well as in any nation that is aligned against the U.S. in the hostilities. JPMorganChase could also experience more numerous and aggressive cyber attacks launched by or under the sponsorship of one or more of the adversaries in such a conflict.

JPMorganChase's business and operations in certain countries can be adversely affected by local economic, political, regulatory and social factors.

Some of the countries in which JPMorganChase conducts business have economies or markets that are less developed and more volatile or may have political, legal and regulatory regimes that are less established or predictable than other countries in which JPMorganChase operates. In addition, in some jurisdictions in which JPMorganChase conducts business, the local economy and business activities are subject to substantial government influence or control. Some of these countries have in the past experienced economic disruptions, including:

- extreme currency fluctuations
- high inflation
- low or negative growth
- defaults or reduced ability to service sovereign debt and
- increased fraud or other misrepresentation of value.

The governments in these countries have sometimes reacted to these developments by imposing restrictive policies that adversely affect the local and regional business environment, such as:

- price, capital or exchange controls, including imposition of punitive transfer and convertibility restrictions or forced currency exchange
- expropriation or nationalization of assets, including client assets, or confiscation of property, including intellectual property, and
- changes in laws, rules and regulations.

The impact of these actions could be accentuated in trading markets that are smaller, less liquid and more volatile than more-developed markets. These types of government actions can negatively affect JPMorganChase's operations in the relevant country, either directly or by suppressing the business activities of local clients or multi-national clients that conduct business in the jurisdiction.

In addition, emerging markets countries, as well as more developed countries, have been susceptible to

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unfavorable social developments arising from poor economic conditions or governmental actions, including:

- widespread demonstrations, civil unrest or general strikes
- crime and corruption
- security and personal safety issues
- an outbreak or escalation of hostilities, or other geopolitical instabilities
- overthrow of incumbent governments
- terrorist attacks, and
- other forms of internal discord.

These economic, political, regulatory and social developments have in the past resulted in, and in the future could lead to, conditions that can adversely affect JPMorganChase's operations in those countries and impair the revenues, growth and profitability of those operations. In addition, any of these events or circumstances in one country can affect JPMorganChase's operations and investments in another country or countries, including in the U.S.

People

JPMorganChase's ability to attract and retain qualified employees is critical to its success.

JPMorganChase's employees are its most important resource, and in many areas of the financial services industry, competition for qualified personnel is intense. JPMorganChase endeavors to attract talented new employees from a variety of backgrounds and retain, develop and motivate its existing employees. JPMorganChase's efforts to hire and retain talented employees could be hindered by factors such as:

- the emerging need for more-skilled workers in an evolving labor and workplace environment, including due to changes in technology
- targeted recruitment of JPMorganChase employees by competitors, and
- modifications to or discontinuation of JPMorganChase's hybrid work models.

JPMorganChase's performance and competitive position could be materially and adversely affected if it is unable to attract or retain qualified employees for its workforce or to devise and execute effective succession planning for key leadership roles, such as the Chief Executive Officer, members of the Operating Committee and other senior leaders.

In addition, advances in technology, such as automation and artificial intelligence, may lead to workforce displacement. This could require JPMorganChase to invest in additional employee training, manage impacts on morale and retention, and compete for employment candidates who possess more advanced technological skills, all of which could

have a negative impact on JPMorganChase's business and operations.

Unfavorable changes in immigration or travel policies could adversely affect JPMorganChase's businesses and operations.

JPMorganChase relies on the skills, knowledge and expertise of employees located throughout the world. Changes in immigration or travel policies in the U.S. and other countries that unduly restrict or otherwise make it more difficult for employees or their family members to work in, or travel to or transfer between, jurisdictions in which JPMorganChase has operations or conducts its business could inhibit JPMorganChase's ability to attract and retain qualified employees, and thereby dilute the quality of its workforce, or could prompt JPMorganChase to make structural changes to its worldwide or regional operating models that cause its operations to be less efficient or more costly.

Legal

JPMorganChase faces significant legal risks from litigation and formal and informal regulatory and government investigations.

JPMorganChase is named as a defendant or is otherwise involved in many legal proceedings, including class actions, derivative actions and other litigation or disputes with third parties, as well as criminal proceedings. Actions currently pending against JPMorganChase may result in judgments, settlements, fines, penalties or other sanctions adverse to JPMorganChase. Any of these matters could materially and adversely affect JPMorganChase's business, financial condition or results of operations, or cause serious reputational harm. As a participant in the financial services industry, it is likely that JPMorganChase will continue to experience a high level of litigation and regulatory and government investigations related to its businesses and operations.

Regulators and other government agencies conduct examinations of JPMorganChase and its subsidiaries both on a routine basis and in targeted exams, and JPMorganChase's businesses and operations are subject to heightened regulatory oversight. This heightened regulatory scrutiny, or the results of such an investigation or examination, may lead to additional regulatory investigations or enforcement actions. There is no assurance that those actions will not result in resolutions or other enforcement actions against JPMorganChase. Furthermore, a single event involving a potential violation of law or regulation may give rise to numerous and overlapping investigations and proceedings, either by multiple federal, state or local agencies and officials in the U.S. or, in some instances, regulators and other governmental officials in non-U.S. jurisdictions.

In addition, if another financial institution violates a law or regulation relating to a particular business activity or practice, this will often give rise to an investigation by regulators and other governmental agencies of the same or similar activity or practice by JPMorganChase.

JPMorganChase could become subject to a significant regulatory investigation and be unable to disclose specific information concerning that investigation to the public if such a disclosure would violate JPMorganChase's obligations under applicable rules and regulations to maintain the confidentiality of confidential supervisory information, even if the resolution of that investigation could have a material adverse effect on JPMorganChase's business, operations, results or financial condition.

Regulatory investigations, examinations or other initiatives by U.S. and non-U.S. governmental authorities may subject JPMorganChase to judgments, settlements, fines, penalties or other sanctions, and may require JPMorganChase to restructure its operations and activities or to cease offering certain products or services. All of these potential outcomes could harm JPMorganChase's reputation or lead to higher operational costs, thereby reducing JPMorganChase's profitability, or result in collateral consequences. In addition, the extent of JPMorganChase's exposure to legal and regulatory matters can be unpredictable and could, in some cases, exceed the amount of reserves that JPMorganChase has established for those matters.

Parts I and II

Item 1B. Unresolved Staff Comments.

None.

Item 1C. Cybersecurity.

Refer to the Operational Risk Management section of Management's discussion and analysis on pages 153–156 for a discussion of cybersecurity risk.

Item 2. Properties.

JPMorganChase's headquarters is located in New York City at 383 Madison Avenue, a 47-story office building that it owns. The demolition of the Firm's former headquarters at 270 Park Avenue in New York City was completed in 2021, and construction of a new headquarters on the same site is nearing completion.

The Firm owned or leased facilities in the following locations at December 31, 2024.

December 31, 2024 (in millions)	Approximate square footage
United States^(a)	
New York City, New York	
383 Madison Avenue, New York, New York	1.1
All other New York City locations	6.0
Total New York City, New York	7.1
Other U.S. locations	
Columbus/Westerville, Ohio	3.5
Chicago, Illinois	2.7
Dallas/Plano/Fort Worth, Texas	2.5
Wilmington/Newark, Delaware	2.1
Houston, Texas	1.6
Jersey City, New Jersey	1.4
Phoenix/Tempe, Arizona	1.3
All other U.S. locations	33.8
Total United States	56.0
Europe, the Middle East and Africa ("EMEA")	
25 Bank Street, London, U.K.	1.4
All other U.K. locations	2.3
All other EMEA locations	1.5
Total EMEA	5.2
Asia-Pacific, Latin America and Canada	
India	6.4
Philippines	1.8
All other locations	2.8
Total Asia-Pacific, Latin America and Canada	11.0
Total	72.2

(a) At December 31, 2024, the Firm owned or leased 4,966 branches in 48 states and Washington D.C.

The premises and facilities occupied by JPMorganChase are collectively used across all of the Firm's business segments and for corporate purposes. JPMorganChase continues to evaluate its current and projected space requirements and may determine from time to time that certain of its properties (including the premises and facilities noted above) are no longer necessary for its operations. There is no assurance that the Firm will be able to dispose of any such excess properties, premises or facilities, or that it will not incur costs in connection with such dispositions. Such disposition costs may be material to the Firm's results of operations in a given period. Refer to the Consolidated Results of Operations on pages 59–62 for information on occupancy expense.

Item 3. Legal Proceedings.

Refer to Note 30 for a description of the Firm's material legal proceedings.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities.

Market for registrant's common equity

JPMorganChase's common stock is listed and traded on the New York Stock Exchange. Refer to "Five-year stock performance," on page 51 for a comparison of the cumulative total return for JPMorganChase common stock with the comparable total return of the S&P 500 Index, the KBW Bank Index and the S&P Financials Index over the five-year period ended December 31, 2024.

Refer to Capital actions in the Capital Risk Management section of Management's discussion and analysis on page 105 for information on the common dividend payout ratio. Refer to Note 21 and Note 26 for discussions of restrictions on dividend payments. On January 31, 2025, there were 196,005 holders of record of JPMorganChase common stock. Refer to Part III, Item 12 on page 43 for information regarding securities authorized for issuance under the Firm's employee share-based incentive plans.

Repurchases under the common share repurchase program

Refer to Capital actions in the Capital Risk Management section of Management's discussion and analysis on page 105 for information regarding repurchases under the Firm's common share repurchase program.

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new \$30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to \$30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

Shares repurchased pursuant to the common share repurchase programs during 2024 were as follows:

Year ended December 31, 2024	Total number of shares of common stock repurchased	Average price paid per share of common stock ^(a)	Aggregate purchase price of common stock repurchases (in millions) ^(a)	Dollar value of remaining authorized repurchase (in millions) ^{(a)(b)}
First quarter	15,869,936	\$ 179.50	\$ 2,849	\$ 16,886
Second quarter	27,019,730	196.83	5,318	11,568 ^(c)
Third quarter	30,343,933	209.61	6,361	23,639
October	6,173,254	218.00	1,345	22,294
November	5,142,243	241.03	1,240	21,054
December	7,170,130	241.10	1,728	19,326
Fourth quarter	18,485,627	233.37	4,313	19,326
Full year	91,719,226	\$ 205.43	\$ 18,841	\$ 19,326

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.

(b) Represents the amount remaining under the \$30 billion repurchase program.

(c) The remaining \$11.6 billion of share repurchase capacity under the prior Board authorization was canceled when the new \$30 billion repurchase program was authorized by the Board of Directors effective July 1, 2024.

Item 6. Reserved

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations.

Management's discussion and analysis of financial condition and results of operations, entitled "Management's discussion and analysis," appears on pages 52–167. Such information should be read in conjunction with the Consolidated Financial Statements and Notes thereto, which appear on pages 172–321.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk.

Refer to the Market Risk Management section of Management's discussion and analysis on pages 141–149 for a discussion of quantitative and qualitative disclosures about market risk.

Parts II and III

Item 8. Financial Statements and Supplementary Data.

The Consolidated Financial Statements, together with the Notes thereto and the report thereon dated February 14, 2025, of PricewaterhouseCoopers LLP, the Firm's independent registered public accounting firm (PCAOB ID 238), appear on pages 169–321.

The "Glossary of Terms and Acronyms" is included on pages 327–333.

Item 9. Changes in and Disagreements With Accountants on Accounting and Financial Disclosure.

None.

Item 9A. Controls and Procedures.

The internal control framework promulgated by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO"), "Internal Control — Integrated Framework" ("COSO 2013"), provides guidance for designing, implementing and conducting internal control and assessing its effectiveness. The Firm used the COSO 2013 framework to assess the effectiveness of the Firm's internal control over financial reporting as of December 31, 2024. Refer to "Management's report on internal control over financial reporting" on page 168.

As of the end of the period covered by this report, an evaluation was carried out under the supervision and with the participation of the Firm's management, including its Chairman and Chief Executive Officer and its Chief Financial Officer, of the effectiveness of its disclosure controls and procedures (as defined in Rule 13a-15(e) under the Securities Exchange Act of 1934). Based on that evaluation, the Chairman and Chief Executive Officer and the Chief Financial Officer concluded that these disclosure controls and procedures were effective. Refer to Exhibits 31.1 and 31.2 for the Certifications furnished by the Chairman and Chief Executive Officer and Chief Financial Officer, respectively.

The Firm is committed to maintaining high standards of internal control over financial reporting. Nevertheless, because of its inherent limitations, internal control over financial reporting may not prevent or detect all misstatements. Deficiencies or lapses in internal controls may occur from time to time, and there can be no assurance that any such deficiencies will not result in significant deficiencies or material weaknesses in internal control in the future and collateral consequences therefrom. Refer to "Management's report on internal control over financial reporting" on page 168 for further information. There was no change in the Firm's internal control over financial reporting (as defined in

Rule 13a-15(f) under the Securities Exchange Act of 1934) that occurred during the three months ended December 31, 2024, that has materially affected, or is reasonably likely to materially affect, the Firm's internal control over financial reporting.

Item 9B. Other Information.

Director and executive officer trading arrangements

The following table provides information concerning Rule 10b5-1 trading arrangements (as defined in Item 408 of Regulation S-K under the Securities Exchange Act of 1934) adopted in the fourth quarter of 2024 by any director or any officer who is subject to the filing requirements of Section 16 of the Securities Exchange Act of 1934 ("Section 16 Director or Officer"). These trading arrangements are intended to satisfy the affirmative defense of Rule 10b5-1(c). Certain of the Firm's Section 16 Directors or Officers may participate in employee stock purchase plans, 401(k) plans or dividend reinvestment plans of the Firm that have been designed to comply with Rule 10b5-1(c). No non-Rule 10b5-1 trading arrangements (as defined in Item 408 of Regulation S-K under the Securities Exchange Act of 1934) were adopted by any Section 16 Director or Officer during the fourth quarter of 2024. Additionally, no Rule 10b5-1 or non-Rule 10b5-1 trading arrangements were terminated by any Section 16 Director or Officer in the fourth quarter of 2024.

Name	Title	Adoption date	Duration ^(c)	Aggregate number of shares to be sold
Lori Beer	Chief Information Officer	November 15, 2024	November 15, 2024 - March 31, 2025	4,105
James Dimon ^(a)	Chairman and CEO	November 7, 2024	November 7, 2024 - August 1, 2025	1,000,000
Robin Leopold	Head of Human Resources	November 4, 2024	November 4, 2024 - December 31, 2025	2,500
Jennifer Piepszak ^(b)	Co-CEO, CIB	October 30, 2024	October 30, 2024 - March 31, 2025	8,545
Troy Rohrbaugh	Co-CEO, CIB	November 15, 2024	November 15, 2024 - June 30, 2025	75,000

(a) Transaction by trusts and an entity of which Mr. Dimon has either a direct or indirect pecuniary interest.

(b) On January 14, 2025, JPMorganChase announced that Ms. Piepszak became a Chief Operating Officer of the Firm, effective January 14, 2025.

(c) Sales under the trading arrangement will not commence until completion of the required cooling off period under Rule 10b5-1. Subject to compliance with Rule 10b5-1, duration could cease earlier than the final date shown above to the extent that the aggregate number of shares to be sold under the trading arrangement have been sold.

Item 9C. Disclosure regarding Foreign Jurisdictions that Prevent Inspections.

Not applicable.

Parts III and IV

Item 10. Directors, Executive Officers and Corporate Governance.

Executive officers of the registrant

Name	Age (at December 31, 2024)	Positions and offices
James Dimon	68	Chairman of the Board since December 2006 and Chief Executive Officer since December 2005.
Ashley Bacon	55	Chief Risk Officer since June 2013.
Jeremy Barnum	52	Chief Financial Officer since May 2021, prior to which he was Head of Global Research for the former Corporate & Investment Bank since February 2021. He previously served as Chief Financial Officer of the former Corporate & Investment Bank from July 2013 until February 2021.
Lori A. Beer	57	Chief Information Officer since September 2017.
Mary Callahan Erdoes	57	Chief Executive Officer of Asset & Wealth Management since September 2009.
Stacey Friedman	56	General Counsel since January 2016.
Marianne Lake	55	Chief Executive Officer of Consumer & Community Banking since January 2024, having previously served as its Co-Chief Executive Officer since May 2021. She was Chief Executive Officer of Consumer Lending from May 2019 until May 2021.
Robin Leopold	60	Head of Human Resources since January 2018.
Jennifer A. Piepszak ^{(a)(b)}	54	Co-Chief Executive Officer of the Commercial & Investment Bank, having previously served as Co-Chief Executive Officer of Consumer & Community Banking since May 2021, prior to which she had been Chief Financial Officer since May 2019.
Daniel E. Pinto ^{(a)(b)}	62	President and Chief Operating Officer since January 2022, Co-President and Co-Chief Operating Officer since January 2018. He also served as Chief Executive Officer of the former Corporate & Investment Bank from March 2014 until January 2024.
Troy Rohrbaugh ^(a)	54	Co-Chief Executive Officer of the Commercial & Investment Bank since January 2024, prior to which he had been the Co-Head of Markets & Securities Services since June 2023. He was Head of Global Markets from January 2019 until June 2023.

Unless otherwise noted, during the five fiscal years ended December 31, 2024, all of JPMorganChase's above-named executive officers have continuously held senior-level positions with JPMorganChase. There are no family relationships among the foregoing executive officers. Information to be provided in Items 10, 11, 12, 13 and 14 of this 2024 Form 10-K and not otherwise included herein is incorporated by reference to the Firm's Definitive Proxy Statement for its 2025 Annual Meeting of Stockholders to be held on May 20, 2025, which will be filed with the SEC within 120 days of the end of the Firm's fiscal year ended December 31, 2024.

- (a) Effective in the second quarter of 2024, JPMorganChase reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. Refer to Business Segment & Corporate Results on page 70 for further information.
- (b) On January 14, 2025, JPMorganChase announced new responsibilities for certain executives: Mr. Pinto will retire at the end of 2026, will relinquish his duties as President and Chief Operating Officer as of June 30, 2025, and will continue to serve as Vice Chairman through the end of 2026; Ms. Piepszak became a Chief Operating Officer of the Firm, effective January 14, 2025; and Doug Petno, Co-Head of Global Banking, succeeded Ms. Piepszak as co-Chief Executive Officer of the Commercial & Investment Bank. Refer to Recent events on page 57 of this 2024 Form 10-K for further information.

Code of Conduct and Code of Ethics

JPMorganChase has adopted, and posted on its website at <https://www.jpmorganchase.com>, a Code of Conduct for all employees of the Firm and a Code of Ethics for its Chairman and Chief Executive Officer, Chief Financial Officer, Principal Accounting Officer and all other professionals of the Firm worldwide serving in a finance, accounting, treasury, tax or investor relations role. The Code of Ethics is also available in print upon request to the Firm's Investor Relations team. Within the time period required by the SEC, JPMorganChase will post on its website any amendment to the Code of Ethics and any waiver applicable to a director or executive officer.

Insider Trading Policy

JPMorganChase has adopted an insider trading policy applicable to its directors, officers and employees, as well as to JPMorganChase itself, governing the purchase, sale and other dispositions of the Firm's securities (the "Insider Trading Policy"). The Firm believes that the Insider Trading Policy is reasonably designed to promote compliance with applicable U.S. federal securities laws and the listing standards of the New York Stock Exchange relating to insider trading. The Insider Trading Policy is filed as Exhibit 19 to this 2024 Form 10-K.

Item 11. Executive Compensation.

Refer to Item 10.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters.

Refer to Item 10 for security ownership of certain beneficial owners and management.

The following table sets forth the total number of shares available for issuance under JPMorganChase's employee share-based incentive plans (including shares available for issuance to non-employee directors). The Firm is not authorized to grant share-based incentive awards to non-employees, other than to non-employee directors.

December 31, 2024	Number of shares to be issued upon exercise of outstanding stock appreciation rights	Weighted-average exercise price of outstanding stock appreciation rights	Number of shares remaining available for future issuance under stock incentive plans
Plan category			
Employee share-based incentive plans approved by shareholders	2,250,000 ^(a)	\$ 152.19	81,151,866 ^(b)
Total	2,250,000	\$ 152.19	81,151,866

(a) Does not include restricted stock units or performance stock units granted under the shareholder-approved Long-Term Incentive Plan ("LTIP"), as amended and restated effective May 21, 2024. Refer to Note 9 for further information.

(b) Represents shares available for future issuance under the shareholder-approved LTIP.

All shares available for future issuance will be issued under the shareholder-approved LTIP. Refer to Note 9 for further discussion.

Item 13. Certain Relationships and Related Transactions, and Director Independence.

Refer to Item 10.

Item 14. Principal Accounting Fees and Services.

Refer to Item 10.

Part IV

Item 15. Exhibits, Financial Statement Schedules.

1	Financial statements	3.7	<u>Certificate of Designations for Fixed-to-Floating Rate Non-Cumulative Preferred Stock, Series II (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed February 24, 2020).</u>
	The Consolidated Financial Statements, the Notes thereto and the report of the Independent Registered Public Accounting Firm thereon listed in Item 8 are set forth commencing on page 169.	3.8	<u>Certificate of Designations for 4.55% Non-Cumulative Preferred Stock, Series JJ (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed March 17, 2021).</u>
2	Financial statement schedules		
3	Exhibits	3.90	<u>Certificate of Designations for 3.65% Fixed-Rate Reset Non-Cumulative Preferred Stock, Series KK (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed May 12, 2021).</u>
3.1	<u>Restated Certificate of Incorporation of JPMorgan Chase & Co., effective April 5, 2006 (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed April 7, 2006).</u>	3.10	<u>Certificate of Designations for 4.625% Non-Cumulative Preferred Stock, Series LL (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed May 20, 2021).</u>
3.2	<u>Amendment to the Restated Certificate of Incorporation of JPMorgan Chase & Co., effective June 7, 2013 (incorporated by reference to Appendix F to the Proxy Statement on Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 10, 2013).</u>	3.11	<u>Certificate of Designations for 4.20% Non-Cumulative Preferred Stock, Series MM (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed July 29, 2021).</u>
3.3	<u>Certificate of Designations for Fixed-to-Floating Rate Non-Cumulative Preferred Stock, Series CC (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed October 20, 2017).</u>	3.12	<u>Certificate of Designations for 6.875% Non-Cumulative Preferred Stock, Series NN (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File 1-5805) filed March 12, 2024).</u>
3.4	<u>Certificate of Designations for 5.75% Non-Cumulative Preferred Stock, Series DD (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed September 21, 2018).</u>	3.13	<u>Certificate of Designations for 6.500% Non-Cumulative Preferred Stock, Series OO (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File 1-5808) filed February 4, 2025).</u>
3.5	<u>Certificate of Designations for 6.00% Non-Cumulative Preferred Stock, Series EE (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 24, 2019).</u>	3.14	<u>By-laws of JPMorgan Chase & Co., as amended, effective September 19, 2023 (incorporated by reference to Exhibit 3.2 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed September 22, 2023).</u>
3.6	<u>Certificate of Designations for 4.75% Non-Cumulative Preferred Stock, Series GG (incorporated by reference to Exhibit 3.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed November 7, 2019).</u>	4.1(a)	<u>Indenture, dated as of October 21, 2010, between JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No.1-5805) filed October 21, 2010).</u>

4.1(b)	<u>First Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee, to the Indenture, dated as of October 21, 2010 (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).</u>	4.6	<u>Description of Securities of JPMorgan Chase & Co. registered pursuant to Section 12 of the Securities Exchange Act of 1934.</u> ^(b) <i>Other instruments defining the rights of holders of long-term debt securities of JPMorgan Chase & Co. and its subsidiaries are omitted pursuant to Section (b)(4)(iii)(A) of Item 601 of Regulation S-K. JPMorgan Chase & Co. agrees to furnish copies of these instruments to the SEC upon request.</i>
4.2(a)	<u>Subordinated Indenture, dated as of March 14, 2014, between JPMorgan Chase & Co. and U.S. Bank Trust National Association, as Trustee (incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No.1-5805) filed March 14, 2014).</u>	10.1	<u>Deferred Compensation Plan for Non-Employee Directors of JPMorgan Chase & Co., as amended and restated July 2001 and as of December 31, 2004 (incorporated by reference to Exhibit 10.1 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2007).</u> ^(a)
4.2(b)	<u>First Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and U.S. Bank Trust National Association, as Trustee, to the Subordinated Indenture, dated as of March 14, 2014 (incorporated by reference to Exhibit 4.2 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).</u>	10.2	<u>2005 Deferred Compensation Plan for Non-Employee Directors of JPMorgan Chase & Co., effective as of January 1, 2005 (incorporated by reference to Exhibit 10.2 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2007).</u> ^(a)
4.3(a)	<u>Indenture, dated as of May 25, 2001, between JPMorgan Chase & Co. and Bankers Trust Company (succeeded by Deutsche Bank Trust Company Americas), as Trustee (incorporated by reference to Exhibit 4(a)(1) to the Registration Statement on Form S-3 of JPMorgan Chase & Co. (File No. 333-52826) filed June 13, 2001).</u>	10.3	<u>2005 Deferred Compensation Program of JPMorgan Chase & Co., restated effective as of December 31, 2008 (incorporated by reference to Exhibit 10.4 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).</u> ^(a)
4.3(b)	<u>Sixth Supplemental Indenture, dated as of January 13, 2017, between JPMorgan Chase & Co. and Bankers Trust Company (succeeded by Deutsche Bank Trust Company Americas), as Trustee, to the Indenture, dated as of May 25, 2001 (incorporated by reference to Exhibit 4.3 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed January 13, 2017).</u>	10.4	<u>JPMorgan Chase & Co. Amended and Restated Long-Term Incentive Plan, effective May 21, 2024 (incorporated by reference to the Appendix of the Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 8, 2024).</u> ^(a)
4.4	<u>Indenture, dated as of February 19, 2016, among JPMorgan Chase Financial Company LLC, JPMorgan Chase & Co. and Deutsche Bank Trust Company Americas, as Trustee (incorporated by reference to Exhibit 4(a)(7) to the Registration Statement on Form S-3 of JPMorgan Chase & Co. and JPMorgan Chase Financial Company LLC (File No. 333-209682) filed February 24, 2016).</u>	10.5	<u>Key Executive Performance Plan of JPMorgan Chase & Co., as amended and restated effective January 1, 2014 (incorporated by reference to Appendix G of the Schedule 14A of JPMorgan Chase & Co. (File No. 1-5805) filed April 10, 2013).</u> ^(a)
4.5	<u>Form of Deposit Agreement (incorporated by reference to Exhibit 4.3 to the Registration Statement on Form S-3 of JPMorgan Chase & Co. (File No. 333-191692) filed October 11, 2013).</u>	10.6	<u>Excess Retirement Plan of JPMorgan Chase & Co., restated and amended as of December 31, 2008, as amended (incorporated by reference to Exhibit 10.7 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2009).</u> ^(a)
		10.7	<u>Executive Retirement Plan of JPMorgan Chase & Co., as amended and restated December 31, 2008 (incorporated by reference to Exhibit 10.9 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).</u> ^(a)

Part IV

10.8	<u>Bank One Corporation Supplemental Savings and Investment Plan, as amended and restated effective December 31, 2008 (incorporated by reference to Exhibit 10.13 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2008).</u>^(a)	10.14	<u>Form of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for stock appreciation rights for Chairman/Chief Executive Officer, dated July 20, 2021 (incorporated by reference to Exhibit 99 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed July 20, 2021).</u>^(a)
10.9	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for performance share units and restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 17, 2017 (incorporated by reference to Exhibit 10.23 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2016).</u>^(a)	10.15	<u>Form of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for stock appreciation rights for President and Chief Operating Officer, dated December 14, 2021 (incorporated by reference to Exhibit 99 to the Current Report on Form 8-K of JPMorgan Chase & Co. (File No. 1-5805) filed December 15, 2021).</u>^(a)
10.10	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for performance share units and restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 16, 2018 (incorporated by reference to Exhibit 10.19 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2017).</u>^(a)	10.16	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 18, 2022 (incorporated by reference to Exhibit 10.20 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2021).</u>^(a)
10.11	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units for Operating Committee members (U.S. and U.K.), dated as of January 15, 2019 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2018).</u>^(a)	10.17	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 17, 2023 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2022).</u>^(a)
10.12	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 21, 2020 (incorporated by reference to Exhibit 10.18 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2019).</u>^(a)	10.18	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 16, 2024 (incorporated by reference to Exhibit 10.19 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2023).</u>^(a)
10.13	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 19, 2021 (incorporated by reference to Exhibit 10.17 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5805) for the year ended December 31, 2020).</u>^(a)	10.19	<u>Forms of JPMorgan Chase & Co. Long-Term Incentive Plan Terms and Conditions for restricted stock units and performance share unit awards for Operating Committee members (U.S. and U.K.), dated as of January 21, 2025.</u>^{(a)(b)}

10.20	<u>Employee Stock Purchase Plan of JPMorgan Chase & Co., as amended and restated effective as of January 1, 2019 (incorporated by reference to Exhibit 10.1 to the Quarterly Report on Form 10-Q of JPMorgan Chase & Co. (File No. 1-5805) for the quarter ended September 30, 2019).</u>
10.21	<u>Form of JPMorgan Chase & Co. Performance-Based Incentive Compensation Plan, effective as of January 1, 2021, as amended (incorporated by reference to Exhibit 10.23 to the Annual Report on Form 10-K of JPMorgan Chase & Co. (File No. 1-5808) for the year ended December 31, 2021).</u> ^(a)
19	<u>Insider Trading Policy - Firmwide</u> ^(b)
21	<u>List of subsidiaries of JPMorgan Chase & Co.</u> ^(b)
22.1	Annual Report on Form 11-K of The JPMorgan Chase 401(k) Savings Plan for the year ended December 31, 2024 (to be filed pursuant to Rule 15d-21 under the Securities Exchange Act of 1934).
22.2	<u>Subsidiary Guarantors and Issuers of Guaranteed Securities.</u> ^(b)
23	<u>Consent of independent registered public accounting firm.</u> ^(b)
31.1	<u>Certification.</u> ^(b)
31.2	<u>Certification.</u> ^(b)
32	<u>Certification pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.</u> ^(c)
97	<u>Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation.</u> ^(b)
101.INS	The instance document does not appear in the interactive data file because its XBRL tags are embedded within the Inline XBRL document. ^(d)
101.SCH	XBRL Taxonomy Extension Schema Document. ^(b)
101.CAL	XBRL Taxonomy Extension Calculation Linkbase Document. ^(b)
101.DEF	XBRL Taxonomy Extension Definition Linkbase Document. ^(b)
101.LAB	XBRL Taxonomy Extension Label Linkbase Document. ^(b)
101.PRE	XBRL Taxonomy Extension Presentation Linkbase Document. ^(b)

104 Cover Page Interactive Data File (embedded within the Inline XBRL document and included in Exhibit 101).

-
- (a) This exhibit is a management contract or compensatory plan or arrangement.
- (b) Filed herewith.
- (c) Furnished herewith. This exhibit shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, or otherwise subject to the liability of that Section. Such exhibit shall not be deemed incorporated into any filing under the Securities Act of 1933 or the Securities Exchange Act of 1934.
- (d) Pursuant to Rule 405 of Regulation S-T, includes the following financial information included in the Firm's Form 10-K for the year ended December 31, 2024, formatted in XBRL (eXtensible Business Reporting Language) interactive data files: (i) the Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, (ii) the Consolidated statements of comprehensive income for the years ended December 31, 2024, 2023 and 2022, (iii) the Consolidated balance sheets as of December 31, 2024 and 2023, (iv) the Consolidated statements of changes in stockholders' equity for the years ended December 31, 2024, 2023 and 2022, (v) the Consolidated statements of cash flows for the years ended December 31, 2024, 2023 and 2022, and (vi) the Notes to Consolidated Financial Statements.

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THREE-YEAR SUMMARY OF CONSOLIDATED FINANCIAL HIGHLIGHTS (unaudited)

As of or for the year ended December 31, (in millions, except per share, ratio, employee data and where otherwise noted)		2024	2023	2022
Selected income statement data				
Total net revenue	\$	177,556 ^(e)	\$ 158,104	\$ 128,695
Total noninterest expense		91,797 ^(e)	87,172	76,140
Pre-provision profit^(a)		85,759	70,932	52,555
Provision for credit losses		10,678	9,320	6,389
Income before income tax expense		75,081	61,612	46,166
Income tax expense		16,610	12,060	8,490
Net income	\$	58,471	\$ 49,552	\$ 37,676
Earnings per share data				
Net income:				
Basic	\$	19.79	\$ 16.25	\$ 12.10
Diluted		19.75	16.23	12.09
Average shares:				
Basic		2,873.9	2,938.6	2,965.8
Diluted		2,879.0	2,943.1	2,970.0
Market and per common share data				
Market capitalization		670,618	489,320	393,484
Common shares at period-end		2,797.6	2,876.6	2,934.2
Book value per share		116.07	104.45	90.29
Tangible book value per share ("TBVPS") ^(a)		97.30	86.08	73.12
Cash dividends declared per share		4.80	4.10	4.00
Selected ratios and metrics				
Return on common equity ("ROE")		18 %	17 %	14 %
Return on tangible common equity ("ROTCE") ^(a)		22	21	18
Return on assets ("ROA")		1.43	1.30	0.98
Overhead ratio		52	55	59
Loans-to-deposits ratio		56	55	49
Firm Liquidity coverage ratio ("LCR") (average) ^(b)		113	113	112
JPMorgan Chase Bank, N.A. LCR (average) ^(b)		124	129	151
Common equity Tier 1 ("CET1") capital ratio ^{(c)(d)}		15.7	15.0	13.2
Tier 1 capital ratio ^{(c)(d)}		16.8	16.6	14.9
Total capital ratio ^{(c)(d)}		18.5	18.5	16.8
Tier 1 leverage ratio ^{(b)(c)}		7.2	7.2	6.6
Supplementary leverage ratio ("SLR") ^{(b)(c)}		6.1	6.1	5.6
Selected balance sheet data (period-end)				
Trading assets	\$	637,784	\$ 540,607	\$ 453,799
Investment securities, net of allowance for credit losses		681,320	571,552	631,162
Loans		1,347,988	1,323,706	1,135,647
Total assets		4,002,814	3,875,393	3,665,743
Deposits		2,406,032	2,400,688	2,340,179
Long-term debt		401,418	391,825	295,865
Common stockholders' equity		324,708	300,474	264,928
Total stockholders' equity		344,758	327,878	292,332
Employees		317,233	309,926	293,723
Credit quality metrics				
Allowances for credit losses	\$	26,866	\$ 24,765	\$ 22,204
Allowance for loan losses to total retained loans		1.87 %	1.75 %	1.81 %
Nonperforming assets	\$	9,300	\$ 7,597	\$ 7,247
Net charge-offs		8,638	6,209	2,853
Net charge-off rate		0.68 %	0.52 %	0.27 %

- (a) Pre-provision profit, TBVPS and ROTCE are each non-GAAP financial measures. Tangible common equity ("TCE") is also a non-GAAP financial measure. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 67–69 for a discussion of these measures.
- (b) For the years ended December 31, 2024, 2023 and 2022, the percentage represents average ratios for the three months ended December 31, 2024, 2023 and 2022.
- (c) The ratios reflect the Current Expected Credit Losses ("CECL") capital transition provisions. Refer to Note 27 for additional information.
- (d) Reflects the Firm's ratios under the Basel III Standardized approach. Refer to Capital Risk Management on pages 97–107 for additional information.
- (e) Total net revenue included a \$7.9 billion net gain related to Visa shares, and total noninterest expense included a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation, both recorded in the second quarter of 2024. Refer to Executive Overview on pages 54–58, and Notes 2 and 6 for additional information on the exchange offer for Visa Class B-1 common stock.

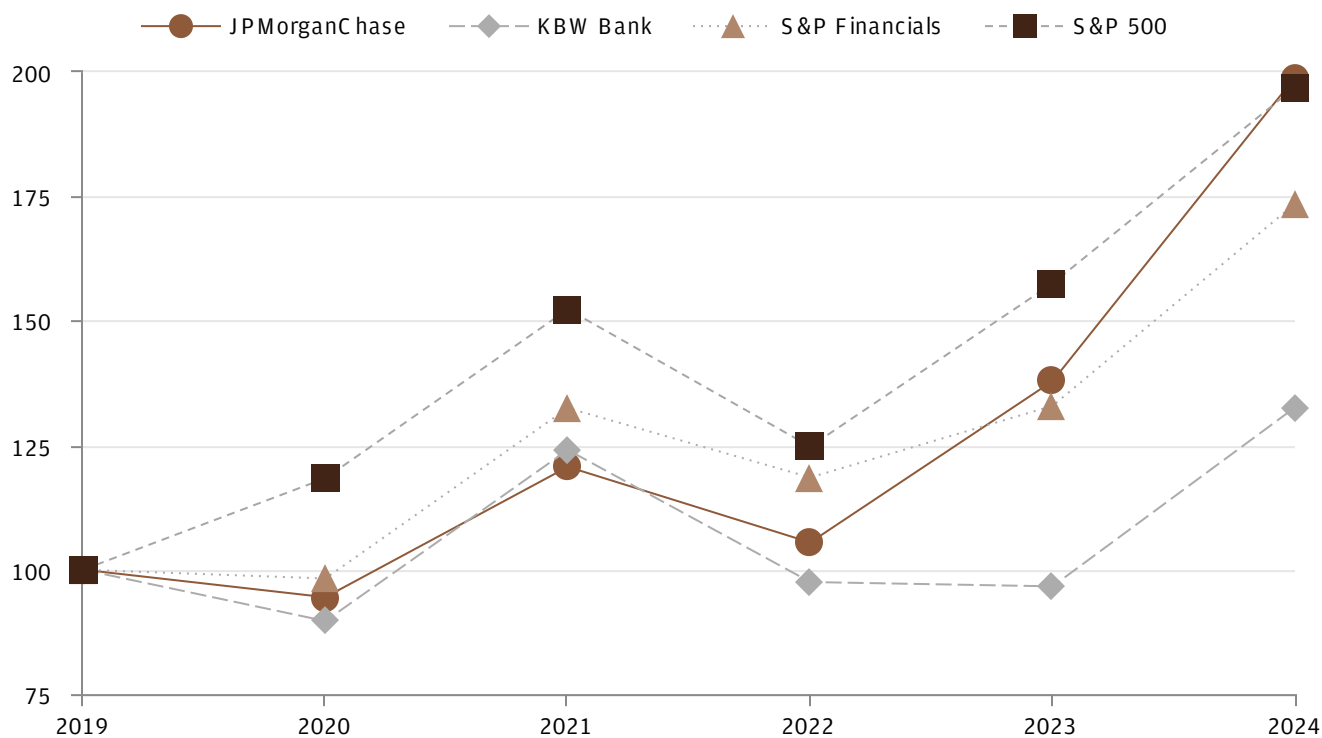
FIVE-YEAR STOCK PERFORMANCE

The following table and graph compare the five-year cumulative total return for JPMorgan Chase & Co. ("JPMorganChase" or the "Firm") common stock with the cumulative return of the S&P 500 Index, the KBW Bank Index and the S&P Financials Index. The S&P 500 Index is a commonly referenced equity benchmark in the United States of America ("U.S."), consisting of leading companies from different economic sectors. The KBW Bank Index seeks to reflect the performance of banks and thrifts that are publicly traded in the U.S. and is composed of leading national money center and regional banks and thrifts. The S&P Financials Index is an index of financial companies, all of which are components of the S&P 500. The Firm is a component of all three industry indices.

The following table and graph assume simultaneous investments of \$100 on December 31, 2019, in JPMorganChase common stock and in each of the above indices. The comparison assumes that all dividends were reinvested.

December 31, (in dollars)	2019	2020	2021	2022	2023	2024
JPMorganChase	\$ 100.00	\$ 94.48	\$ 120.68	\$ 105.48	\$ 137.91	\$ 198.96
KBW Bank Index	100.00	89.69	124.08	97.53	96.66	132.62
S&P Financials Index	100.00	98.24	132.50	118.54	132.94	173.57
S&P 500 Index	100.00	118.39	152.34	124.75	157.54	196.96

December 31,
(in dollars)



Management's discussion and analysis

The following is Management's discussion and analysis of the financial condition and results of operations ("MD&A") of JPMorganChase for the year ended December 31, 2024. The MD&A is included in both JPMorganChase's Annual Report for the year ended December 31, 2024 ("Annual Report") and its Annual Report on Form 10-K for the year ended December 31, 2024 ("2024 Form 10-K" or "Form 10-K") filed with the Securities and Exchange Commission ("SEC"). Refer to the Glossary of terms and acronyms on pages 327–333 for definitions of terms and acronyms used throughout the Annual Report and the 2024 Form 10-K.

This Form 10-K contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. These forward-looking statements are based on the current beliefs and expectations of JPMorganChase's management, speak only as of the date of this Form 10-K and are subject to significant risks and uncertainties. Refer to Forward-looking Statements on page 167 and Part 1, Item 1A: Risk Factors in this Form 10-K on pages 10-37 for a discussion of certain of those risks and uncertainties and the factors that could cause JPMorganChase's actual results to differ materially because of those risks and uncertainties. There is no assurance that actual results will be in line with any outlook information set forth herein, and the Firm does not undertake to update any forward-looking statements.

INTRODUCTION

JPMorgan Chase & Co. (NYSE: JPM), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the United States of America ("U.S."), with operations worldwide. JPMorganChase had \$4.0 trillion in assets and \$344.8 billion in stockholders' equity as of December 31, 2024. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Under the J.P. Morgan and Chase brands, the Firm serves millions of customers, predominantly in the U.S., and many of the world's most prominent corporate, institutional and government clients globally.

JPMorganChase's principal bank subsidiary is JPMorgan Chase Bank, National Association ("JPMorgan Chase Bank, N.A."), a national banking association with U.S. branches in 48 states and Washington, D.C. JPMorganChase's principal non-bank subsidiary is J.P. Morgan Securities LLC ("J.P. Morgan Securities"), a U.S. broker-dealer. The bank and non-bank subsidiaries of JPMorganChase operate nationally as well as through overseas branches and subsidiaries, representative offices and subsidiary foreign banks. The Firm's principal operating subsidiaries outside the U.S. are J.P. Morgan Securities plc and J.P. Morgan SE ("JPMSE"), which are subsidiaries of JPMorgan Chase Bank, N.A. and are based in the United Kingdom ("U.K.") and Germany, respectively.

Business segments & Corporate: Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking ("CCB"), Commercial & Investment Bank ("CIB") and Asset & Wealth Management ("AWM") – with the remaining activities in Corporate. The Firm's consumer business segment is CCB, and the Firm's wholesale business segments are CIB and AWM.

A description of each of the Firm's reportable business segments, and the products and services that they provide to their respective client bases, as well as a description of Corporate activities, is provided in the Management's discussion and analysis of financial condition and results of operations section of this Form 10-K ("Management's discussion and analysis" or "MD&A") under the heading "Business Segment & Corporate Results," which begins on page 70, and in Note 32.

First Republic: On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation ("FDIC"). References in this Form 10-K to "associated with First Republic," "impact of First Republic" or similar expressions refer to the relevant effects of the First Republic acquisition, as well as subsequent related business and activities, as applicable. Refer to Note 34 for additional information.

The Firm's website is www.jpmorganchase.com. JPMorganChase makes available on its website, free of charge, annual reports on Form 10-K, quarterly reports on Form 10-Q and current reports on Form 8-K pursuant to Section 13(a) or Section 15(d) of the Securities Exchange Act of 1934, as soon as reasonably practicable after it electronically files or furnishes such material to the U.S. Securities and Exchange Commission (the "SEC") at www.sec.gov. JPMorganChase makes new and important information about the Firm available on its website at <https://www.jpmorganchase.com>, including on the Investor Relations section of its website at <https://www.jpmorganchase.com/ir>. Information on the Firm's website, including documents on the website that are referenced in this Form 10-K, is not incorporated by reference into this 2024 Form 10-K or the Firm's other filings with the SEC.

EXECUTIVE OVERVIEW

This executive overview of the MD&A highlights selected information and does not contain all of the information that is important to readers of the Firm's 2024 Form 10-K. For a complete description of the trends and uncertainties, as well as the risks and critical accounting estimates affecting the Firm, the 2024 Form 10-K should be read in its entirety.

Financial performance of JPMorganChase

Year ended December 31, (in millions, except per share data and ratios)			
	2024	2023	Change
Selected income statement data			
Noninterest revenue	\$ 84,973	\$ 68,837	23%
Net interest income	92,583	89,267	4
Total net revenue	177,556	158,104	12
Total noninterest expense	91,797	87,172	5
Pre-provision profit	85,759	70,932	21
Provision for credit losses	10,678	9,320	15
Net income	58,471	49,552	18
Diluted earnings per share	19.75	16.23	22
Selected ratios and metrics			
Return on common equity	18 %	17 %	
Return on tangible common equity	22	21	
Book value per share	\$ 116.07	\$ 104.45	11
Tangible book value per share	97.3	86.08	13
Capital ratios^{(a)(b)}			
CET1 capital	15.7 %	15.0 %	
Tier 1 capital	16.8	16.6	
Total capital	18.5	18.5	
Memo:			
NII excluding Markets ^(c)	\$ 92,419	\$ 90,041	3
NIR excluding Markets ^(c)	58,167	44,361	31
Markets ^(c)	30,007	27,964	7
Total net revenue - managed basis	\$ 180,593	\$ 162,366	11

(a) The ratios reflect the CECL capital transition provisions. Refer to Note 27 for additional information.

(b) Reflects the Firm's ratios under the Basel III Standardized approach. Refer to Capital Risk Management on pages 97–107 for additional information.

(c) NII and NIR refer to net interest income and noninterest revenue, respectively. Markets consists of CIB's Fixed Income Markets and Equity Markets businesses.

Visa shares: On April 8, 2024, Visa Inc. commenced an initial exchange offer for its Class B-1 common shares. On May 6, 2024, the Firm announced that Visa had accepted the Firm's tender of its 37.2 million Visa Class B-1 common shares in exchange for a combination of Visa Class B-2 common shares and Visa Class C common shares ("Visa C shares"), resulting in a \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. As of September 30, 2024, the Firm had disposed of all of its Visa C shares through sales and through a contribution to the Firm's Foundation. Refer to Market Risk Management on pages 141–149, and Notes 2 and 6 for additional information.

First Republic: JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank from the FDIC on May 1, 2023. As a result, the year-to-date results include the twelve-month impact of First Republic compared with eight months in the prior-year period. Where meaningful to the results, this is referred to in this Form 10-K as the "timing impact" of First Republic. Refer to Notes 6 and 34 for additional information.

Comparisons noted in the sections below are for the full year of 2024 versus the full year of 2023, unless otherwise specified.

Firmwide overview

JPMorganChase reported net income of \$58.5 billion for 2024, up 18%, earnings per share of \$19.75, ROE of 18% and ROTCE of 22%.

- **Total net revenue** was \$177.6 billion, up 12%, reflecting:
 - **Net interest income** ("NII") of \$92.6 billion, up 4%, driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, higher revolving balances in Card Services, the timing impact of First Republic, higher wholesale deposit balances, and higher Markets net interest income, largely offset by deposit margin compression across the lines of business and lower average deposit balances in CCB. NII excluding Markets was \$92.4 billion, up 3%.
 - **Noninterest revenue** ("NIR") was \$85.0 billion, up 23%, predominantly driven by a \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024, higher asset management fees in AWM and CCB, higher investment banking fees, and lower net investment securities losses in Treasury and CIO.

The prior year included the estimated bargain purchase gain of \$2.8 billion associated with First Republic.

- **Noninterest expense** was \$91.8 billion, up 5%, driven by higher compensation expense, including higher revenue-related compensation and growth in the number of employees, as well as a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, partially offset by lower FDIC-related expense, reflecting a \$2.9 billion special assessment recognized in the fourth quarter of 2023, compared with a \$725 million increase to the FDIC special assessment recognized in the first quarter of 2024.
- The **provision for credit losses** was \$10.7 billion, reflecting \$8.6 billion of net charge-offs and a net addition to the allowance for credit losses of \$2.0 billion. Net charge-offs increased by \$2.4 billion, driven by Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization, and balance growth. The net addition to the allowance for credit losses included a net addition of \$2.1 billion in **consumer**, driven by Card Services, and a net reduction of \$19 million in **wholesale**.
The provision in the prior year was \$9.3 billion, reflecting \$6.2 billion of net charge-offs and a \$3.1 billion net addition to the allowance for credit losses.
- The total **allowance for credit losses** was \$26.9 billion at December 31, 2024. The Firm had an allowance for loan losses to retained loans coverage ratio of 1.87%, compared with 1.75% in the prior year.
- The Firm's **nonperforming assets** totaled \$9.3 billion at December 31, 2024, up 22%, driven by higher wholesale nonaccrual loans, which reflected downgrades in Real Estate, concentrated in Office, partially offset by lower consumer nonaccrual loans, which included loan sales. Refer to Wholesale Credit Portfolio and Consumer Credit Portfolio on pages 126–136 and pages 120–125, respectively, for additional information.
- Firmwide **average loans** of \$1.3 trillion were up 6%, driven by higher loans across the lines of business.
- Firmwide **average deposits** of \$2.4 trillion were up 1%, reflecting:
 - net inflows in Payments and net issuances of structured notes in Markets,
 - the timing impact of First Republic, and
 - growth in balances in new and existing client accounts in AWM,
 predominantly offset by
 - a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending.
 Refer to Liquidity Risk Management on pages 108–115 for additional information.

Selected capital and other metrics

- **CET1 capital** was \$275.5 billion, and the Standardized and Advanced CET1 ratios were 15.7% and 15.8%, respectively.
- **SLR** was 6.1%.
- **TBVPS** grew 13.0%, ending 2024 at \$97.30.
- As of December 31, 2024, the Firm had eligible end-of-period **High Quality Liquid Assets** (“HQLA”) of approximately \$834 billion and **unencumbered marketable securities** with a fair value of approximately \$594 billion, resulting in approximately \$1.4 trillion of liquidity sources. Refer to Liquidity Risk Management on pages 108–115 for additional information.

Refer to Consolidated Results of Operations and Consolidated Balance Sheets Analysis on pages 59–62 and pages 63–65, respectively, for a further discussion of the Firm's results, including the provision for credit losses, and Note 34 for additional information on the First Republic acquisition.

Pre-provision profit, ROTCE, TCE, TBVPS, NII and NIR excluding Markets, and total net revenue on a managed basis are non-GAAP financial measures. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 67–69 for a further discussion of each of these measures.

Business segment highlights

Selected business metrics for each of the Firm's lines of business ("LOB") are presented below for the full year of 2024.

CCB ROE 32%	- Average deposits down 6%; client investment assets up 14% - Average loans up 9%; Card Services net charge-off rate of 3.34% - Debit and credit card sales volume^(a) up 8% - Active mobile customers^(b) up 7%
CIB ^(c) ROE 18%	- Investment Banking fees up 37%; #1 ranking for Global Investment Banking fees with 9.3% wallet share for the year - Markets revenue up 7%, with Fixed Income Markets up 5% and Equity Markets up 13% - Average Banking & Payments loans up 2%; average client deposits^(d) up 5%
AWM ROE 34%	- Assets under management ("AUM") of \$4.0 trillion, up 18% - Average loans up 3%; average deposits up 9% including the transfer of First Republic deposits to AWM in 4Q23^(e)

(a) Excludes Commercial Card.

(b) Users of all mobile platforms who have logged in within the past 90 days.

(c) Reflects the reorganization of the Firm's business segments. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.

(d) Represents client deposits and other third-party liabilities pertaining to the Payments and Securities Services businesses.

(e) In the fourth quarter of 2023, CCB transferred certain deposits associated with First Republic to AWM and CIB.

Refer to the Business Segment & Corporate Results on pages 70–90 for a detailed discussion of results by business segment.

Credit provided and capital raised

JPMorganChase continues to support consumers, businesses and communities around the globe. The Firm provided new and renewed credit and raised capital for wholesale and consumer clients during 2024, consisting of approximately:

\$2.8 trillion	Total credit provided and capital raised (including loans and commitments)
\$250 billion	Credit for consumers
\$40 billion	Credit for U.S. small businesses
\$2.4 trillion	Credit and capital for corporations and non-U.S. government entities ^(a)
\$65 billion	Credit and capital for nonprofit and U.S. government entities ^(b)

(a) Includes Individuals and Individual Entities primarily consisting of Global Private Bank clients within AWM.

(b) Includes states, municipalities, hospitals and universities.

Recent events

- On January 14, 2025, JPMorganChase announced new responsibilities for several of its senior executives:
 - Daniel Pinto, President and Chief Operating Officer (“COO”), will retire at the end of 2026. Mr. Pinto will relinquish his responsibilities as President and COO as of June 30, 2025. He will continue to serve the Firm as Vice Chairman through the end of 2026.
 - Jennifer Piepszak, Co-Chief Executive Officer of the Commercial & Investment Bank (“CIB”), was named a COO of the Firm, effective January 14, 2025.
 - Doug Petno, Co-head of Global Banking, succeeded Ms. Piepszak as Co-Chief Executive Officer of CIB.
- On December 12, 2024, the Firm announced that Michele G. Buck, 63, had been elected as a director of the Firm, effective March 17, 2025. Ms. Buck is Chairman of the Board, President and CEO of The Hershey Company.

Outlook

These current expectations are forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Such forward-looking statements are based on the current beliefs and expectations of JPMorganChase’s management, speak only as of the date of this Form 10-K, and are subject to significant risks and uncertainties. Refer to Forward-Looking Statements on page 167 and Part I, Item 1A: Risk Factors on pages 10-37 of this Form 10-K for a further discussion of certain of those risks and uncertainties and the other factors that could cause JPMorganChase’s actual results to differ materially because of those risks and uncertainties. There is no assurance that actual results in 2025 will be in line with the outlook information set forth below, and the Firm does not undertake to update any forward-looking statements.

JPMorganChase’s current outlook for full-year 2025 should be viewed against the backdrop of the global and U.S. economies, financial markets activity, the geopolitical environment, the competitive environment, client and customer activity levels, and regulatory and legislative developments in the U.S. and other countries where the Firm does business. Each of these factors will affect the performance of the Firm. The Firm will continue to make appropriate adjustments to its businesses and operations in response to ongoing developments in the business, economic, regulatory and legal environments in which it operates.

Full-year 2025

- Management expects net interest income to be approximately \$94.0 billion and net interest income excluding Markets to be approximately \$90.0 billion, market dependent.
- Management expects adjusted expense to be approximately \$95.0 billion, market dependent.
- Management expects the net charge-off rate in Card Services to be approximately 3.60%.

Net interest income excluding Markets and adjusted expense are non-GAAP financial measures. Refer to Explanation and Reconciliation of the Firm’s Use of Non-GAAP Financial Measures on pages 67–69.

Business Developments

First Republic acquisition

On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation ("FDIC"), as receiver.

As of December 31, 2024, the Firm had substantially completed the conversion of operations, and the integration of clients, products and services, associated with the First Republic acquisition to align with the Firm's businesses and operations.

Refer to Note 34 for additional information on First Republic.

CONSOLIDATED RESULTS OF OPERATIONS

This section provides a comparative discussion of JPMorganChase's Consolidated Results of Operations on a reported basis for the two-year period ended December 31, 2024, unless otherwise specified. Refer to Consolidated Results of Operations on pages 54-57 of the Firm's Annual Report on Form 10-K for the year ended December 31, 2023 (the "2023 Form 10-K") for a discussion of the 2023 versus 2022 results. Factors that relate primarily to a single business segment or Corporate are discussed in more detail in the results of that segment or Corporate. Refer to pages 161-164 for a discussion of the Critical Accounting Estimates Used by the Firm that affect the Consolidated Results of Operations.

Revenue

Year ended December 31, (in millions)	2024	2023	2022
Investment banking fees	\$ 8,910	\$ 6,519	\$ 6,686
Principal transactions	24,787	24,460	19,912
Lending- and deposit-related fees	7,606	7,413	7,098
Asset management fees	17,801	15,220	14,096
Commissions and other fees	7,530	6,836	6,581
Investment securities losses	(1,021)	(3,180)	(2,380)
Mortgage fees and related income	1,401	1,176	1,250
Card income	5,497	4,784	4,420
Other income ^{(a)(b)}	12,462 ^(c)	5,609 ^(d)	4,322
Noninterest revenue	84,973	68,837	61,985
Net interest income	92,583	89,267	66,710
Total net revenue	\$177,556	\$ 158,104	\$ 128,695

- (a) Included operating lease income of \$2.8 billion, \$2.8 billion and \$3.7 billion for the years ended December 31, 2024, 2023 and 2022, respectively. Refer to Note 6 for additional information.
- (b) Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments that was previously recognized in other income is now recognized in income tax expense. Refer to Notes 1, 6, 14 and 25 for additional information.
- (c) Included the net gain related to Visa shares of \$7.9 billion recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
- (d) Included the estimated bargain purchase gain of \$2.8 billion for the year ended December 31, 2023 associated with the First Republic acquisition. Refer to Notes 6 and 34 for additional information.

2024 compared with 2023

Investment banking fees increased, reflecting in CIB the benefit of favorable market conditions, which resulted in:

- higher debt underwriting fees predominantly driven by higher industry-wide issuances in leveraged loans, and in high-grade and high-yield bonds,
- higher equity underwriting fees driven by higher industry-wide fees and wallet share gains in IPOs, and in follow-on and convertible securities offerings, and
- higher advisory fees driven by higher industry-wide mergers and acquisitions ("M&A") activity and wallet share gains.

Refer to CIB segment results on pages 77-83 and Note 6 for additional information.

Principal transactions revenue increased driven by CIB, reflecting:

- higher Equity Markets revenue in Prime Finance and Equity Derivatives, predominantly offset by
- lower Fixed Income Markets revenue, reflecting the net impact of declines in revenue across macro businesses and higher revenue in Securitized Products.

Principal transactions revenue in CIB generally has offsets across other revenue lines, including net interest income. The Firm assesses the performance of its Markets business on a total net revenue basis.

The increase in CIB was partially offset by a net decrease in Corporate, reflecting lower revenue in Treasury and CIO, and gains compared with a net loss on certain legacy private equity investments in the prior year.

Refer to CIB segment and Corporate results on pages 77-83 and pages 88-90, respectively, and Note 6 for additional information.

Lending- and deposit-related fees increased, reflecting, in CIB, higher deposit-related fees, including cash management fees in Payments, on higher volume; and higher lending-related fees, including loan commitment fees. These factors were largely offset by a decline in the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic, primarily in AWM, as certain of the commitments have expired.

Refer to CIB and AWM segment results on pages 77-83 and pages 84-87, respectively, and Note 6 for additional information.

Asset management fees increased, reflecting, in AWM and CCB, higher average market levels and net inflows, as well as higher performance fees in AWM; and in CCB, the timing impact of First Republic. Refer to CCB and AWM segment results on pages 73-76 and pages 84-87, respectively, and Note 6 for additional information.

Commissions and other fees increased, predominantly due to higher brokerage commissions and fees on higher volume, and higher custody fees, in both CIB and AWM, as well as higher annuity sales commissions in CCB. Refer to CCB, CIB and AWM segment results on pages 73–76, pages 77–83 and pages 84–87, respectively, and Note 6 for additional information.

Investment securities losses decreased, reflecting lower losses on sales of securities, primarily U.S. Treasuries and U.S. GSE and government agency MBS, associated with repositioning the investment securities portfolio in Treasury and CIO. Refer to Corporate results on pages 88–90 and Note 10 for additional information.

Mortgage fees and related income increased in Home Lending, reflecting higher production revenue, which included the timing impact of First Republic. Refer to CCB segment results on pages 73–76, and Note 6 and 15 for additional information.

Card income increased, reflecting higher net interchange on increased debit and credit card sales volume, as well as higher annual fees in CCB, partially offset by an increase in amortization related to new account origination costs. Refer to CCB segment results on pages 73–76 and Note 6 for additional information.

Other income increased, reflecting:

- in Corporate:
 - the \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024, partially offset by
 - the absence of the prior-year \$2.8 billion estimated bargain purchase gain associated with the First Republic acquisition, and
- in CIB:
 - the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024, resulting in the amortization of certain of the Firm's alternative energy tax-oriented investments previously recognized in other income which is now recognized in income tax expense.

Both periods included impairment losses related to certain equity investments.

The prior year included a gain of \$339 million on the original minority interest in China International Fund Management ("CIFM"), partially offset by net investment valuation losses, both in AWM.

Refer to AWM segment results on pages 84–87 for additional information on CIFM; Notes 1, 6, 14 and 25 for additional information on the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance; Notes 2 and 6 for additional information on Visa shares; and Notes 6 and 34 for additional information on the First Republic acquisition.

Net interest income increased driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, higher revolving balances in Card Services, the timing impact of First Republic, higher wholesale deposit balances and higher Markets net interest income. These factors were largely offset by deposit margin compression across the lines of business and lower average deposit balances in CCB.

The Firm's average interest-earning assets were \$3.5 trillion, up \$212 billion, and the yield was 5.50%, up 36 bps. The net yield on these assets, on an FTE basis, was 2.63%, a decrease of 7 bps. The net yield excluding Markets was 3.84%, relatively flat when compared to the prior year.

Refer to the Consolidated average balance sheets, interest and rates schedule on pages 322–326 for additional information. Net yield excluding Markets is a non-GAAP financial measure. Refer to Explanation and Reconciliation of the Firm's Use of Non-GAAP Financial Measures on pages 67–69 for an additional discussion of net yield excluding Markets.

Provision for credit losses

Year ended December 31, (in millions)	2024	2023	2022
Consumer, excluding credit card	\$ 631	\$ 935	\$ 506
Credit card	9,292	6,048	3,353
Total consumer	9,923	6,983	3,859
Wholesale	731	2,299	2,476
Investment securities	24	38	54
Total provision for credit losses	\$10,678	\$ 9,320	\$ 6,389

2024 compared with 2023

The **provision for credit losses** was \$10.7 billion, reflecting \$8.6 billion of net charge-offs and a \$2.0 billion net addition to the allowance for credit losses.

Net charge-offs included \$7.8 billion in **consumer**, predominantly driven by Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth, and \$0.8 billion in **wholesale**, primarily in Real Estate, largely concentrated in Office.

The net addition to the allowance for credit losses consisted of:

- \$2.1 billion in **consumer**, reflecting:
 - a \$2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years, partially offset by
 - a \$125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of \$19 million in **wholesale**, reflecting:
 - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans, predominantly offset by
 - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

The provision in the prior year was \$9.3 billion, reflecting net charge-offs of \$6.2 billion and a \$3.1 billion net addition to the allowance for credit losses, which included \$1.2 billion to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Refer to CCB, CIB and AWM segment and Corporate results on pages 73–76, pages 77–83, pages 84–87, and pages 88–90, respectively; Allowance for Credit Losses on pages 137–139; Critical Accounting Estimates Used by the Firm on pages 161–164; and Notes 12 and 13 for additional information on the credit portfolio and the allowance for credit losses.

Noninterest expense

Year ended December 31, (in millions)	2024	2023	2022
Compensation expense	\$ 51,357	\$ 46,465	\$ 41,636
Noncompensation expense:			
Occupancy	5,026	4,590	4,696
Technology, communications and equipment ^(a)	9,831	9,246	9,358
Professional and outside services	11,057	10,235	10,174
Marketing	4,974	4,591	3,911
Other expense	9,552 ^(c)	12,045	6,365
Total noncompensation expense	40,440	40,707	34,504
Total noninterest expense	\$ 91,797	\$ 87,172	\$ 76,140
Certain components of other expense^(b)			
Legal expense	\$ 740	\$ 1,436	\$ 266
FDIC-related expense	1,893	4,203	860
Operating losses	1,417	1,228	1,101

(a) Includes depreciation expense associated with auto operating lease assets. Refer to Note 18 for additional information.

(b) Refer to Note 6 for additional information.

(c) Included a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

2024 compared with 2023

Compensation expense increased driven by:

- higher revenue-related compensation across the LOBs,
- growth in the number of employees across the LOBs and Corporate, primarily in front office and technology, and
- the impact of First Republic, predominantly in CCB, reflecting timing and the classification of the prior-year expense, which was recognized in other expense in Corporate in the second quarter of 2023 as the individuals associated with First Republic were not employees of the Firm until July 2023.

Noncompensation expense decreased as a result of:

- lower FDIC-related expense recognized in Corporate, which included the impact of a \$2.9 billion special assessment recognized in the fourth quarter of 2023, compared with a \$725 million increase to the FDIC special assessment recognized in the first quarter of 2024, and
- lower legal expense, reflecting the net impact of declines in CCB, CIB and Corporate, and an increase in AWM,

predominantly offset by

- a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024 in Corporate,

- higher investments in technology in the businesses, as well as marketing, predominantly in CCB,
- higher occupancy expense, which included the impact of net additions to the Firm's properties,
- higher distribution fees in AWM and brokerage expense in CIB, and
- the timing impact associated with First Republic, offset by the alignment of expense to compensation expense, as noted above.

Refer to Notes 2 and 6 for additional information on Visa shares; Note 6 for additional information on other expense; and Note 34 for additional information on the First Republic acquisition.

Income tax expense

Year ended December 31, (in millions, except rate)	2024	2023	2022
Income before income tax expense	\$ 75,081	\$ 61,612	\$ 46,166
Income tax expense	16,610 ^(a)	12,060	8,490
Effective tax rate	22.1 %	19.6 %	18.4 %

(a) Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments is now recognized in income tax expense. Refer to Notes 1, 6, 14 and 25 for additional information.

2024 compared with 2023

The **effective tax rate** increased predominantly driven by:

- the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance on January 1, 2024, and
- a higher level of pretax income and changes in the mix of income and expenses subject to U.S. federal, state and local taxes, including the impact of the net gain on Visa shares and the contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024.

The prior year included the impact to income tax expense associated with the First Republic acquisition that was reflected in the estimated bargain purchase gain, and an income tax benefit related to the finalization of certain income tax regulations, both of which resulted in a reduction in the Firm's effective tax rate.

Refer to Note 25 for additional information.

CONSOLIDATED BALANCE SHEETS AND CASH FLOWS ANALYSIS

Consolidated balance sheets analysis

The following is a discussion of the significant changes between December 31, 2024 and 2023. Refer to pages 161–164 for a discussion of the Critical Accounting Estimates Used by the Firm that affect the Consolidated Balance Sheets.

Selected Consolidated balance sheets data

December 31, (in millions)	2024	2023	Change
Assets			
Cash and due from banks	\$ 23,372	\$ 29,066	(20)%
Deposits with banks	445,945	595,085	(25)
Federal funds sold and securities purchased under resale agreements	295,001	276,152	7
Securities borrowed	219,546	200,436	10
Trading assets	637,784	540,607	18
Available-for-sale securities	406,852	201,704	102
Held-to-maturity securities	274,468	369,848	(26)
Investment securities, net of allowance for credit losses	681,320	571,552	19
Loans	1,347,988	1,323,706	2
Allowance for loan losses	(24,345)	(22,420)	9
Loans, net of allowance for loan losses	1,323,643	1,301,286	2
Accrued interest and accounts receivable	101,223	107,363	(6)
Premises and equipment	32,223	30,157	7
Goodwill, MSRs and other intangible assets	64,560	64,381	—
Other assets	178,197	159,308	12
Total assets	\$ 4,002,814	\$ 3,875,393	3 %

Cash and due from banks and deposits with banks decreased driven by higher investment securities in Treasury and CIO, and Markets activities in CIB.

Federal funds sold and securities purchased under resale agreements increased driven by Markets, reflecting higher client-driven market-making activities.

Securities borrowed increased driven by Markets, reflecting a higher demand for securities to cover short positions.

Refer to Note 11 for additional information on securities purchased under resale agreements and securities borrowed.

Trading assets increased predominantly due to higher levels of debt and equity instruments in Markets related to client-driven market-making activities. Refer to Notes 2 and 5 for additional information.

Investment securities increased due to:

- higher available-for-sale ("AFS") securities, reflecting net purchases, primarily U.S. Treasuries and non-U.S. government debt securities, partially offset by maturities and paydowns, and
- lower held-to-maturity ("HTM") securities driven by maturities and paydowns.

Refer to Corporate results on pages 88–90,

Investment Portfolio Risk Management on page 140, and Notes 2 and 10 for additional information.

Loans increased, reflecting:

- higher outstanding balances in Card Services driven by growth in new accounts and normalization of revolving balances,
- higher wholesale loans in CIB, and
- higher securities-based lending in AWM due to higher client demand,

partially offset by

- a decline in Home Lending as paydowns and loan sales outpaced originations.

The **allowance for loan losses** increased, reflecting a net addition to the allowance for loan losses of \$1.9 billion, consisting of:

- \$2.1 billion net addition in **consumer**, primarily in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years, partially offset by a net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of \$176 million in **wholesale**, reflecting:
 - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in

Markets, and a reduction due to charge-offs largely from collateral-dependent loans, predominantly offset by

- net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

There was also a \$128 million net addition to the allowance for lending-related commitments recognized in other liabilities on the Consolidated balance sheets.

Refer to Consolidated Results of Operations and Credit and Investment Risk Management on pages 59–62 and pages 117–140, respectively, Critical Accounting Estimates Used by the Firm on pages 161–164, and Notes 2, 3, 12 and 13 for additional information on loans and the total allowance for credit losses.

Accrued interest and accounts receivable decreased primarily driven by lower receivables in Payments related to the timing of processing payment activities, due to December 31, 2023 falling on a weekend, as well as lower client receivables related to client-driven activities in Markets.

Premises and equipment increased primarily as a result of the construction-in-process associated with the Firm's headquarters, and purchases of properties. Refer to Notes 16 and 18 for additional information.

Goodwill, MSRs and other intangibles: Refer to Note 15 for additional information.

Other assets increased primarily due to higher cash collateral placed with central counterparties ("CCP") in Markets, the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024, and higher auto operating lease assets in CCB. Refer to Notes 1, 6, 14 and 25 for additional information on updates to the accounting guidance.

Selected Consolidated balance sheets data (continued)

December 31, (in millions)	2024	2023	Change
Liabilities			
Deposits	\$ 2,406,032	\$ 2,400,688	—
Federal funds purchased and securities loaned or sold under repurchase agreements	296,835	216,535	37
Short-term borrowings	52,893	44,712	18
Trading liabilities	192,883	180,428	7
Accounts payable and other liabilities	280,672	290,307	(3)
Beneficial interests issued by consolidated variable interest entities ("VIEs")	27,323	23,020	19
Long-term debt	401,418	391,825	2
Total liabilities	3,658,056	3,547,515	3
Stockholders' equity	344,758	327,878	5
Total liabilities and stockholders' equity	\$ 4,002,814	\$ 3,875,393	3 %

Deposits increased, reflecting:

- an increase in CIB due to net inflows predominantly in Payments, largely offset by net maturities of structured notes in Markets,
- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending and migration into higher-yielding investments, predominantly offset by new accounts.

Federal funds purchased and securities loaned or sold under repurchase agreements increased driven by Markets, reflecting higher client-driven market-making activities and higher secured financing of trading assets.

Short-term borrowings increased driven by Markets, reflecting net issuance of structured notes due to client demand, and higher financing requirements. Refer to Liquidity Risk Management on pages 108–115 for additional information on deposits, federal funds purchased and securities loaned or sold under repurchase agreements, and short-term borrowings; Notes 2 and 17 for deposits; and Note 11 for federal funds purchased and securities loaned or sold under repurchase agreements.

Trading liabilities increased due to client-driven market-making activities primarily in Fixed Income Markets, which resulted in higher levels of short positions in debt instruments. Refer to Notes 2 and 5 for additional information.

Accounts payable and other liabilities decreased primarily driven by lower payables in Payments related to the timing of processing payment activities, due to December 31, 2023 falling on a weekend, as well as lower client payables related to client-driven activities in Markets, partially offset by the impact of the adoption of updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024. Refer to Note 19 for additional information on accounts payable; and Notes 1, 6, 14 and 25 for additional information on updates to the accounting guidance.

Beneficial interests issued by consolidated VIEs increased driven by the issuance of credit card securitizations in Treasury and CIO, and activity in municipal bond vehicles in CIB.

Refer to Liquidity Risk Management on pages 108–115; and Notes 14 and 28 for additional information on Firm-sponsored VIEs and loan securitization trusts.

Long-term debt increased, primarily driven by:

- net issuances of structured notes in Markets due to client demand,

partially offset by

- a decline in Treasury and CIO, reflecting the net impact of lower FHLB advances and higher long-term debt from net issuances.

Refer to Liquidity Risk Management on pages 108–115 and Note 34 for additional information on the First Republic acquisition.

Stockholders' equity increased, reflecting:

- net income,

largely offset by

- the impact of capital actions, including repurchases of common shares, the declaration of common and preferred stock dividends, and net redemption of preferred stock, and
- net unrealized losses in AOCI, including the impact of higher interest rates on cash flow hedges in Treasury and CIO.

Refer to Consolidated Statements of changes in stockholders' equity on page 175, Capital Actions on page 105, and Note 24 for additional information.

Consolidated cash flows analysis

The following is a discussion of cash flow activities during the years ended December 31, 2024 and 2023. Refer to Consolidated cash flows analysis on page 61 of the Firm's 2023 Form 10-K for a discussion of the 2022 activities.

(in millions)	Year ended December 31,		
	2024	2023	2022
Net cash provided by/ (used in)			
Operating activities	\$ (42,012)	\$ 12,974	\$ 107,119
Investing activities	(163,403)	67,643	(137,819)
Financing activities	63,447	(25,571)	(126,257)
Effect of exchange rate changes on cash	(12,866)	1,871	(16,643)
Net increase/(decrease) in cash and due from banks and deposits with banks	\$(154,834)	\$ 56,917	\$(173,600)

Operating activities

JPMorganChase's operating assets and liabilities primarily support the Firm's lending and capital markets activities. These assets and liabilities can vary significantly in the normal course of business due to the amount and timing of cash flows, which are affected by client-driven and risk management activities and market conditions. The Firm believes that cash flows from operations, available cash and other liquidity sources, and its capacity to generate cash through secured and unsecured sources, are sufficient to meet its operating liquidity needs.

- In 2024, cash used resulted from higher trading assets and higher securities borrowed, largely offset by net income.
- In 2023, cash provided primarily reflected net income, lower other assets, and accrued interest and accounts receivable, predominantly offset by higher trading assets, lower accounts payable and other liabilities, and higher securities borrowed.

Investing activities

The Firm's investing activities predominantly include originating held-for-investment loans, investing in the investment securities portfolio and other short-term instruments.

- In 2024, cash used resulted from net purchases of investment securities, net loan originations and higher securities purchased under resale agreements, partially offset by proceeds from sales and securitizations of loans held-for-investment.
- In 2023, cash provided resulted from net proceeds from investment securities, proceeds from sales and securitizations of loans held-for-investment, and lower securities purchased under resale agreements, largely offset by net originations of loans and net cash used in the First Republic Bank acquisition.

Financing activities

The Firm's financing activities include acquiring customer deposits and issuing long-term debt and preferred stock.

- In 2024, cash provided primarily reflected higher securities loaned or sold under repurchase agreements and net proceeds from long- and short-term borrowings, partially offset by net redemption of preferred stock.
- In 2023, cash used reflected lower deposits, which included the impact of the repayment of the deposits provided to First Republic Bank by the consortium of large U.S. banks that the Firm assumed as part of the First Republic acquisition, partially offset by higher securities loaned under repurchase agreements and net proceeds from long- and short-term borrowings.
- For both periods, cash was used for repurchases of common stock and cash dividends on common and preferred stock.

* * *

Refer to Consolidated Balance Sheets Analysis on pages 63–65, Capital Risk Management on pages 97–107, and Liquidity Risk Management on pages 108–115, and the Consolidated Statements of Cash Flows on page 176 for a further discussion of the activities affecting the Firm's cash flows.

EXPLANATION AND RECONCILIATION OF THE FIRM'S USE OF NON-GAAP FINANCIAL MEASURES

Non-GAAP financial measures

The Firm prepares its Consolidated Financial Statements in accordance with U.S. GAAP; these financial statements appear on pages 172–176. That presentation, which is referred to as “reported” basis, provides the reader with an understanding of the Firm’s results that can be tracked consistently from year-to-year and enables a comparison of the Firm’s performance with the U.S. GAAP financial statements of other companies.

In addition to analyzing the Firm’s results on a reported basis, management reviews Firmwide results, including the overhead ratio, on a “managed” basis; these Firmwide managed basis results are non-GAAP financial measures. The Firm also reviews the results of the lines of business on a managed basis. The Firm’s definition of managed basis starts, in each case, with the reported U.S. GAAP results and includes certain reclassifications to present total net revenue for the Firm as a whole, and for each of the reportable business segments and Corporate, on an FTE basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. These financial measures

allow management to assess the comparability of revenue from year-to-year arising from both taxable and tax-exempt sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense. These adjustments have no impact on net income as reported by the Firm as a whole or by each of the lines of business and Corporate.

Management also uses certain non-GAAP financial measures at the Firm and business-segment levels because these other non-GAAP financial measures provide information to investors about the underlying operational performance and trends of the Firm or of the particular business segment, as the case may be, and therefore facilitate a comparison of the Firm or the business segment with the performance of its relevant competitors. Refer to Business Segment & Corporate Results on pages 70–90 for additional information on these non-GAAP measures. Non-GAAP financial measures used by the Firm may not be comparable to similarly named non-GAAP financial measures used by other companies.

The following summary table provides a reconciliation from the Firm’s reported U.S. GAAP results to managed basis.

Year ended December 31, (in millions, except ratios)	2024			2023			2022		
	Reported	Fully taxable- equivalent adjustments ^(b)	Managed basis	Reported	Fully taxable- equivalent adjustments ^(b)	Managed basis	Reported	Fully taxable- equivalent adjustments ^(b)	Managed basis
Other income	\$12,462 ^(a)	\$ 2,560 ^(a)	\$15,022	\$ 5,609	\$ 3,782	\$ 9,391	\$ 4,322	\$ 3,148	\$ 7,470
Total noninterest revenue	84,973	2,560	87,533	68,837	3,782	72,619	61,985	3,148	65,133
Net interest income	92,583	477	93,060	89,267	480	89,747	66,710	434	67,144
Total net revenue	177,556	3,037	180,593	158,104	4,262	162,366	128,695	3,582	132,277
Total noninterest expense	91,797	NA	91,797	87,172	NA	87,172	76,140	NA	76,140
Pre-provision profit	85,759	3,037	88,796	70,932	4,262	75,194	52,555	3,582	56,137
Provision for credit losses	10,678	NA	10,678	9,320	NA	9,320	6,389	NA	6,389
Income before income tax expense	75,081	3,037	78,118	61,612	4,262	65,874	46,166	3,582	49,748
Income tax expense	16,610 ^(a)	3,037 ^(a)	19,647	12,060	4,262	16,322	8,490	3,582	12,072
Net income	\$58,471	NA	\$58,471	\$49,552	NA	\$49,552	\$37,676	NA	\$37,676
Overhead ratio	52 %	NM	51 %	55 %	NM	54 %	59 %	NM	58 %

(a) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.

(b) Predominantly recognized in CIB and Corporate.

Net interest income, net yield, and noninterest revenue excluding Markets

In addition to reviewing net interest income, net yield, and noninterest revenue on a managed basis, management also reviews these metrics excluding Markets, as shown below. Markets consists of CIB's Fixed Income Markets and Equity Markets. These metrics, which exclude Markets, are non-GAAP financial measures. Management reviews these metrics to assess the performance of the Firm's lending, investing (including asset-liability management) and deposit-raising activities, apart from any volatility associated with Markets activities. In addition, management also assesses Markets business performance on a total revenue basis as offsets may occur across revenue lines. Management believes that these measures provide investors and analysts with alternative measures to analyze the revenue trends of the Firm.

Year ended December 31, (in millions, except rates)	2024	2023	2022
Net interest income – reported^(a)	\$ 92,583	\$ 89,267	\$ 66,710
Fully taxable-equivalent adjustments	477	480	434
Net interest income – managed basis	\$ 93,060	\$ 89,747	\$ 67,144
Less: Markets net interest income ^(b)	641	(294)	4,789
Net interest income excluding Markets	\$ 92,419	\$ 90,041	\$ 62,355
Average interest-earning assets^(a)	\$3,537,567	\$3,325,708	\$3,349,079
Less: Average Markets interest-earning assets ^(b)	1,128,153	985,777	953,195
Average interest-earning assets excluding Markets	\$2,409,414	\$2,339,931	\$2,395,884
Net yield on average interest-earning assets – managed basis	2.63 %	2.70 %	2.00 %
Net yield on average Markets interest-earning assets ^(b)	0.06	(0.03)	0.50
Net yield on average interest-earning assets excluding Markets	3.84 %	3.85 %	2.60 %
Noninterest revenue – reported^(c)	\$ 84,973	\$ 68,837	\$ 61,985
Fully taxable-equivalent adjustments ^(c)	2,560	3,782	3,148
Noninterest revenue – managed basis	\$ 87,533	\$ 72,619	\$ 65,133
Less: Markets noninterest revenue ^{(b)(d)}	29,366	28,258	24,373
Noninterest revenue excluding Markets	\$ 58,167	\$ 44,361	\$ 40,760
Memo: Total Markets net revenue^(b)	\$ 30,007	\$ 27,964	\$ 29,162

- (a) Includes the effect of derivatives that qualify for hedge accounting. Taxable-equivalent amounts are used, also where applicable. Refer to Note 5 for additional information on hedge accounting.
- (b) Refer to pages 81–82 for further information on Markets.
- (c) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investment in Tax Credit Stricture guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.
- (d) Includes the market-related revenues of the former Commercial Banking business segment. Prior-period amounts have been revised to conform with the current presentation.

Calculation of certain U.S. GAAP and non-GAAP financial measures

Certain U.S. GAAP and non-GAAP financial measures are calculated as follows:

Book value per share (“BVPS”)

Common stockholders' equity at period-end /
Common shares at period-end

Overhead ratio

Total noninterest expense / Total net revenue

ROA

Reported net income / Total average assets

ROE

Net income* / Average common stockholders' equity

ROTCE

Net income* / Average tangible common equity

TBVPS

Tangible common equity at period-end / Common shares at period-end

* Represents net income applicable to common equity

In addition, the Firm reviews other non-GAAP measures such as:

- Adjusted expense, which represents noninterest expense excluding Firmwide legal expense, and
- Pre-provision profit, which represents total net revenue less total noninterest expense.

Management believes that these measures help investors to understand the effect of these items on reported results and provide an alternative presentation of the Firm's performance.

TCE, ROTCE and TBVPS

TCE, ROTCE and TBVPS are each non-GAAP financial measures. TCE represents the Firm's common stockholders' equity (i.e., total stockholders' equity less preferred stock) less goodwill and identifiable intangible assets (other than MSRs), net of related deferred tax liabilities. ROTCE measures the Firm's net income applicable to common equity as a percentage of average TCE. TBVPS represents the Firm's TCE at period-end divided by common shares at period-end. TCE, ROTCE and TBVPS are utilized by the Firm, as well as investors and analysts, in assessing the Firm's use of equity.

The following summary table provides a reconciliation from the Firm's common stockholders' equity to TCE.

(in millions, except per share and ratio data)	Period-end		Average		
	Dec 31, 2024	Dec 31, 2023	Year ended December 31,		
			2024	2023	2022
Common stockholders' equity	\$ 324,708	\$ 300,474	\$ 312,370	\$ 282,056	\$ 253,068
Less: Goodwill	52,565	52,634	52,627	52,258	50,952
Less: Other intangible assets	2,874	3,225	3,042	2,572	1,112
Add: Certain deferred tax liabilities ^(a)	2,943	2,996	2,970	2,883	2,505
Tangible common equity	\$ 272,212	\$ 247,611	\$ 259,671	\$ 230,109	\$ 203,509
Return on tangible common equity	NA	NA	22 %	21 %	18 %
Tangible book value per share	\$ 97.30	\$ 86.08	NA	NA	NA

(a) Represents deferred tax liabilities related to tax-deductible goodwill and to identifiable intangibles created in nontaxable transactions, which are netted against goodwill and other intangibles when calculating TCE.

BUSINESS SEGMENT & CORPORATE RESULTS

The Firm is managed on an LOB basis. Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank (“CIB”). As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking, Commercial & Investment Bank, and Asset & Wealth Management – with the remaining activities in Corporate.

The business segments are determined based on the products and services provided, or the type of customer served, and they reflect the manner in which financial information is evaluated by the Firm’s Operating Committee, whose members act collectively as the Firm’s chief operating decision maker. Segment results are presented on a managed basis. Refer to Explanation and Reconciliation of the Firm’s use of Non-GAAP Financial Measures, on pages 67–69 for a definition of managed basis.

The following table depicts the Firm’s reportable business segments.

JPMorganChase					
Consumer Business Segment			Wholesale Business Segments		
Consumer & Community Banking			Commercial & Investment Bank		Asset & Wealth Management
Banking & Wealth Management	Home Lending	Card Services & Auto	Banking & Payments	Markets & Securities Services	
• Consumer Banking • Business Banking • J.P. Morgan Wealth Management	• Home Lending Production • Home Lending Servicing • Real Estate Portfolios	• Card Services • Auto	• Investment Banking • Payments • Lending • Other	• Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	• Asset Management • Global Private Bank

Description of business segment reporting methodology

Results of the reportable business segments are intended to present each segment as if it were a stand-alone business. The management reporting process that derives business segment results includes the allocation of certain income and expense items. The Firm periodically assesses the assumptions, methodologies and reporting classifications used for segment reporting, and therefore further refinements may be implemented in future periods. The Firm also assesses the level of capital required for each LOB on at least an annual basis. The Firm’s LOBs also provide various business metrics which are utilized by the Firm and its investors and analysts in assessing performance.

Revenue sharing

When business segments or businesses within each segment join efforts to sell products and services to the Firm’s clients and customers, the participating businesses may agree to share revenue from those transactions. Revenue is generally recognized in the segment responsible for the related product or service, with allocations to the other segments or businesses involved in the transaction. The segment and business results reflect these revenue-sharing agreements.

Expense allocation

Where business segments use services provided by Corporate support units, or another business segment, the costs of those services are allocated to the respective business segments. The expense is generally allocated based on the actual cost and use of services provided. In contrast, certain costs and investments related to Corporate that are not currently utilized by any LOB are not allocated to the business segments and are retained in Corporate. Expense retained in Corporate generally includes costs that would not be incurred if the segments were stand-alone businesses, and other items not solely aligned with a particular reportable business segment.

Funds transfer pricing

Funds transfer pricing (“FTP”) is the process by which the Firm allocates interest income and expense to the LOBs and Other Corporate and transfers the primary interest rate risk and liquidity risk to Treasury and CIO.

The funds transfer pricing process considers the interest rate and liquidity risk characteristics of assets and liabilities and off-balance sheet products. Periodically, the methodology and assumptions utilized in the FTP process are adjusted to reflect economic conditions and other factors, which may impact the allocation of net interest income to the segments. Effective in the fourth quarter of 2024, the Firm updated its FTP with respect to consumer deposits, which resulted in an increase in the funding benefit reflected within CCB’s net interest income that is fully offset in Corporate, with no effect on the Firm’s net interest income.

As a result of higher average interest rates, the cost of funds for assets and the FTP benefit earned for liabilities have generally increased in the current year, impacting the net interest income of the business segments. During the period ended December 31, 2024, this has resulted in a higher cost of funds for loans and Markets activities. In addition, rates paid to deposit holders increased more than the FTP benefit increase during the year, resulting in deposit margin compression.

Foreign exchange risk

Foreign exchange risk is transferred from the LOBs and Other Corporate to Treasury and CIO for certain revenues and expenses. Treasury and CIO manages these risks centrally and reports the impact of foreign exchange rate movements related to the transferred risk in its results. Refer to Market Risk Management on page 149 for additional information.

Debt expense and preferred stock dividend allocation

As part of the FTP process, almost all of the cost of the credit spread component of outstanding unsecured long-term debt and preferred stock dividends is allocated to the reportable business segments, while the balance of the cost is retained in Corporate. The methodology to allocate the cost of unsecured long-term debt and preferred stock dividends to the business segments is aligned with the relevant regulatory capital requirements and funding needs of the LOBs, as applicable. The allocated cost of unsecured long-term debt is included in a business segment’s net interest income, and net income is reduced by preferred stock dividends, to arrive at a business segment’s net income applicable to common equity. Refer to Capital Risk Management on pages 97–107 for additional information.

Capital allocation

The amount of capital assigned to each LOB and Corporate is referred to as equity. The Firm’s current equity allocation methodology incorporates Basel III Standardized risk-weighted assets (“RWA”) and the global systemically important banks (“GSIB”) surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change. Refer to Line of business and Corporate equity on page 104 for additional information on capital allocation.

Segment & Corporate Results – Managed Basis

The following tables summarize the Firm's results by business segments and Corporate for the periods indicated.

Year ended December 31, (in millions, except ratios)	Consumer & Community Banking			Commercial & Investment Bank			Asset & Wealth Management		
	2024	2023	2022	2024	2023	2022	2024	2023	2022
Total net revenue	\$ 71,507	\$ 70,148	\$ 54,814 ^(a)	\$ 70,114	\$ 64,353	\$ 59,635 ^(a)	\$ 21,578	\$ 19,827	\$ 17,748
Total noninterest expense	38,036	34,819	31,208 ^(a)	35,353	33,972	32,069 ^(a)	14,414	12,780	11,829
Pre-provision profit/(loss)	33,471	35,329	23,606	34,761	30,381	27,566	7,164	7,047	5,919
Provision for credit losses	9,974	6,899	3,813	762	2,091	2,426	(68)	159	128
Net income/(loss)	17,603	21,232	14,916 ^(a)	24,846	20,272	19,138 ^(a)	5,421	5,227	4,365
Return on equity ("ROE")	32 %	38 %	29 %	18 %	14 %	14 %	34 %	31 %	25 %

Year ended December 31, (in millions, except ratios)	Corporate			Total		
	2024	2023	2022	2024	2023	2022
Total net revenue	\$ 17,394 ^(b)	\$ 8,038	\$ 80	\$180,593 ^(b)	\$162,366	\$ 132,277
Total noninterest expense	3,994 ^(c)	5,601	1,034	91,797 ^(c)	87,172	76,140
Pre-provision profit/(loss)	13,400	2,437	(954)	88,796	75,194	56,137
Provision for credit losses	10	171	22	10,678	9,320	6,389
Net income/(loss)	10,601	2,821	(743)	58,471	49,552	37,676
Return on equity ("ROE")	NM	NM	NM	18 %	17 %	14 %

(a) In the first quarter of 2023, the allocations of revenue and expense to CCB associated with a Merchant Services revenue sharing agreement were discontinued and are now retained in Payments in CIB. Prior-period amounts have been revised to conform with the current presentation.

(b) Included a \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

(c) Included a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

Refer to Note 32 for further details on total net revenue and total noninterest expense.

The following sections provide a comparative discussion of the Firm's results by business segments and Corporate as of or for the years ended December 31, 2024 and 2023, unless otherwise specified.

CONSUMER & COMMUNITY BANKING

Consumer & Community Banking offers products and services to consumers and small businesses through bank branches, ATMs, digital (including mobile and online) and telephone banking. CCB is organized into Banking & Wealth Management (including Consumer Banking, Business Banking and J.P. Morgan Wealth Management), Home Lending (including Home Lending Production, Home Lending Servicing and Real Estate Portfolios) and Card Services & Auto. Banking & Wealth Management offers deposit, investment and lending products, cash management, payments and services. Home Lending includes mortgage origination and servicing activities, as well as portfolios consisting of residential mortgages and home equity loans. Card Services issues credit cards and offers travel services. Auto originates and services auto loans and leases.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2024	2023	2022
Revenue			
Lending- and deposit-related fees	\$ 3,387	\$ 3,356	\$ 3,316
Asset management fees	4,014	3,282	2,734
Mortgage fees and related income	1,378	1,175	1,236
Card income	3,139	2,532	2,469
All other income ^(a)	4,731	4,773	5,131
Noninterest revenue	16,649	15,118	14,886
Net interest income	54,858	55,030	39,928
Total net revenue	71,507	70,148	54,814
Provision for credit losses	9,974	6,899	3,813
Noninterest expense			
Compensation expense	17,045	15,171	13,092
Noncompensation expense ^(b)	20,991	19,648	18,116
Total noninterest expense	38,036	34,819	31,208
Income before income tax expense	23,497	28,430	19,793
Income tax expense	5,894	7,198	4,877
Net income	\$17,603	\$ 21,232	\$ 14,916
Revenue by business			
Banking & Wealth Management	\$40,943	\$ 43,199	\$ 30,059
Home Lending	5,097	4,140	3,674
Card Services & Auto	25,467	22,809	21,081
Mortgage fees and related income details:			
Production revenue	627	421	497
Net mortgage servicing revenue ^(c)	751	754	739
Mortgage fees and related income	\$ 1,378	\$ 1,175	\$ 1,236
Financial ratios			
Return on equity	32 %	38 %	29 %
Overhead ratio	53	50	57

(a) Primarily includes operating lease income and commissions and other fees. Operating lease income was \$2.8 billion, \$2.8 billion and \$3.6 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

(b) Included depreciation expense on leased assets of \$1.7 billion, \$1.7 billion and \$2.4 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

(c) Included MSR risk management results of \$159 million, \$131 million and \$93 million for the years ended December 31, 2024, 2023 and 2022, respectively.

(d) In the second quarter of 2023, substantially all of the expense associated with First Republic was reported in Corporate. Commencing in the third quarter of 2023, the expense was aligned to the appropriate LOB.

2024 compared with 2023

Net income was \$17.6 billion, down 17%.

Net revenue was \$71.5 billion, up 2%.

Net interest income was \$54.9 billion, flat when compared with the prior year, reflecting:

- lower NII in Banking & Wealth Management ("BWM"), predominantly driven by deposit margin compression and lower average deposits,

offset by

- higher Card Services NII, predominantly driven by higher revolving balances, and
- the timing impact of First Republic in Home Lending.

Noninterest revenue was \$16.6 billion, up 10%, predominantly driven by:

- higher asset management fees reflecting higher average market levels, including the timing impact of First Republic and, to a lesser extent, net inflows, as well as higher commissions from annuity sales in BWM,
- higher card income, driven by higher net interchange reflecting increased debit and credit card sales volume, and higher annual fees, partially offset by an increase in amortization related to new account origination costs, as well as
- higher production revenue in Home Lending, including the timing impact of First Republic.

Refer to Note 6 for additional information on card income, asset management fees, and commissions and other fees; and Critical Accounting Estimates on pages 161–164 for additional information on the credit card rewards liability.

Refer to Executive Overview on page 54 and Note 34 for additional information on First Republic.

Noninterest expense was \$38.0 billion, up 9%, reflecting First Republic-related expense that was aligned to CCB from Corporate starting in the third quarter of 2023, impacting both compensation and noncompensation expense.

The increase in expense also reflected:

- higher compensation expense, largely driven by higher revenue-related compensation predominantly for advisors and bankers, and an increase in the number of employees, including in technology, and
- higher noncompensation expense, largely driven by continued investments in technology and marketing, as well as higher operating losses, partially offset by lower legal expense.

The provision for credit losses was \$10.0 billion, reflecting:

- net charge-offs of \$7.9 billion, up \$2.6 billion, including \$2.4 billion in Card Services, reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth, and
- a \$2.0 billion net addition to the allowance for credit losses, consisting of:

- \$2.2 billion in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years,

partially offset by

- a \$125 million net reduction in Home Lending, primarily due to improvements in the outlook for home prices in the first quarter of 2024.

The provision in the prior year was \$6.9 billion, reflecting net charge-offs of \$5.3 billion, a \$1.2 billion net addition to the allowance for credit losses, predominantly driven by Card Services, and a \$408 million net addition to the allowance for credit losses to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Refer to Credit and Investment Risk Management on pages 117–140 and Allowance for Credit Losses on pages 137–139 for a further discussion of the credit portfolios and the allowance for credit losses.

Selected metrics

As of or for the year ended
December 31,
(in millions, except
employees)

	2024	2023	2022
Selected balance sheet data (period-end)			
Total assets	\$650,268	\$ 642,951	\$ 514,085
Loans:			
Banking & Wealth Management	33,221	31,142	29,008
Home Lending ^(a)	246,498	259,181	172,554
Card Services	233,016	211,175	185,175
Auto	73,619	77,705	68,191
Total loans	586,354	579,203	454,928
Deposits ^(b)	1,056,652	1,094,738	1,131,611
Equity	54,500	55,500	50,000
Selected balance sheet data (average)			
Total assets	\$ 631,648	\$ 584,367	\$ 497,263
Loans:			
Banking & Wealth Management	31,544	30,142	31,545
Home Lending ^(c)	252,542	232,115	176,285
Card Services	214,139	191,424	163,335
Auto	75,009	72,674	68,098
Total loans	573,234	526,355	439,263
Deposits ^(b)	1,064,215	1,126,552	1,162,680
Equity	54,500	54,349	50,000
Employees	144,989	141,640	135,347

- (a) At December 31, 2024, 2023 and 2022, Home Lending loans held-for-sale and loans at fair value were \$8.1 billion, \$3.4 billion and \$3.0 billion, respectively.
- (b) In the fourth quarter of 2023, CCB transferred approximately \$18.8 billion of deposits associated with First Republic to AWM and CIB.
- (c) Average Home Lending loans held-for-sale and loans at fair value were \$7.1 billion, \$4.8 billion and \$7.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

Selected metrics

As of or for the year ended
December 31,
(in millions, except ratio
data)

Credit data and quality statistics			
Nonaccrual loans ^(a)	\$ 3,366	\$3,740	\$ 3,899
Net charge-offs/(recoveries)			
Banking & Wealth Management	442	340	370
Home Lending	(106)	(56)	(229)
Card Services	7,148	4,699	2,403
Auto	444	357	144
Total net charge-offs/(recoveries)	\$ 7,928	\$5,340	\$ 2,688
Net charge-off/(recovery) rate			
Banking & Wealth Management	1.40 %	1.13 %	1.17 %
Home Lending	(0.04)	(0.02)	(0.14)
Card Services	3.34	2.45	1.47
Auto	0.59	0.49	0.21
Total net charge-off/(recovery) rate	1.40 %	1.02 %	0.62 %
30+ day delinquency rate			
Home Lending ^(b)	0.78 %	0.66 %	0.83 %
Card Services	2.17	2.14	1.45
Auto	1.43	1.19	1.01
90+ day delinquency rate - Card Services	1.14 %	1.05 %	0.68 %
Allowance for loan losses			
Banking & Wealth Management	\$ 764	\$ 685	\$ 722
Home Lending	447	578	867
Card Services	14,608	12,453	11,200
Auto	692	742	715
Total allowance for loan losses	\$ 16,511	\$14,458	\$13,504

- (a) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024, 2023 and 2022, mortgage loans 90 or more days past due and insured by U.S. government agencies were \$84 million, \$123 million and \$187 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance.
- (b) At December 31, 2024, 2023 and 2022, excluded mortgage loans insured by U.S. government agencies of \$122 million, \$176 million and \$258 million, respectively, that are 30 or more days past due. These amounts have been excluded based upon the government guarantee.

Selected metrics

As of or for the year ended December 31, (in billions, except ratios and where otherwise noted)	2024	2023	2022
Business Metrics			
CCB Consumer customers (in millions)	84.4	82.1	79.2
CCB Small business customers (in millions)	7.0	6.4	5.7
Number of branches	4,966	4,897	4,787
Active digital customers (in thousands) ^(a)	70,813	66,983	63,136
Active mobile customers (in thousands) ^(b)	57,821	53,828	49,710
Debit and credit card sales volume	\$1,805.4	\$1,678.6	\$1,555.4
Total payments transaction volume (in trillions) ^(c)	6.4	5.9	5.6
Banking & Wealth Management			
Average deposits	\$1,049.3	\$ 1,111.7	\$ 1,145.7
Deposit margin	2.66 %	2.84 %	1.71 %
Business Banking average loans	\$ 19.5	\$ 19.6	\$ 22.3
Business Banking origination volume	4.5	4.8	4.3
Client investment assets ^(d)	1,087.6	951.1	647.1
Number of client advisors	5,755	5,456	5,029
Home Lending			
Mortgage origination volume by channel			
Retail	\$ 25.5	\$ 22.4	\$ 38.5
Correspondent	15.3	12.7	26.9
Total mortgage origination volume^(e)	\$ 40.8	\$ 35.1	\$ 65.4
Third-party mortgage loans serviced (period-end)	\$ 648.0	\$ 631.2	\$ 584.3
MSR carrying value (period-end)	9.1	8.5	8.0
Card Services			
Sales volume, excluding commercial card	\$1,259.3	\$ 1,163.6	\$1,064.7
Net revenue rate	10.03 %	9.72 %	9.87 %
Net yield on average loans	9.73	9.61	9.77
New credit card accounts opened (in millions)	10.0	10.0	9.6
Auto			
Loan and lease origination volume	\$ 40.3	\$ 41.3	\$ 30.4
Average auto operating lease assets	11.1	10.9	14.3

- (a) Users of all web and/or mobile platforms who have logged in within the past 90 days.
- (b) Users of all mobile platforms who have logged in within the past 90 days.
- (c) Total payments transaction volume includes debit and credit card sales volume and gross outflows of ACH, ATM, teller, wires, BillPay, PayChase, Zelle, person-to-person and checks.
- (d) Includes assets invested in managed accounts and J.P. Morgan mutual funds where AWM is the investment manager. Refer to AWM segment results on pages 84–87 for additional information.
- (e) Firmwide mortgage origination volume was \$47.4 billion, \$41.4 billion and \$81.8 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

COMMERCIAL & INVESTMENT BANK^(a)

The Commercial & Investment Bank is comprised of the Banking & Payments and Markets & Securities Services businesses. These businesses offer investment banking, lending, payments, market-making, financing, custody and securities products and services to a global base of corporate and institutional clients. Banking & Payments offers products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, and loan origination and syndication. Banking & Payments also provides services that enable clients to manage payments globally across liquidity and account solutions, commerce solutions, clearing, trade, and working capital. Markets & Securities Services includes Markets, which is a global market-maker across products, including cash and derivative instruments, and also offers sophisticated risk management solutions, lending, prime brokerage, clearing and research. Markets & Securities Services also includes Securities Services, a leading global custodian that provides custody, fund services, liquidity and trading services, and data solutions products.

(a) Reflects the reorganization of the Firm's business segments in the second quarter of 2024. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.

Selected income statement data

Year ended December 31, (in millions)	2024	2023	2022
Revenue			
Investment banking fees	\$ 9,116	\$ 6,631	\$ 6,977
Principal transactions	24,382	23,794	19,792
Lending- and deposit-related fees	3,914	3,423	3,662
Commissions and other fees	5,278	4,879	5,113
Card income	2,310	2,213	1,934
All other income	3,253	2,869	2,060
Noninterest revenue	48,253	43,809	39,538
Net interest income	21,861	20,544	20,097
Total net revenue^(a)	70,114	64,353	59,635
Provision for credit losses	762	2,091	2,426
Noninterest expense			
Compensation expense	18,191	17,105	16,214
Noncompensation expense	17,162	16,867	15,855
Total noninterest expense	35,353	33,972	32,069
Income before income tax expense	33,999	28,290	25,140
Income tax expense	9,153	8,018	6,002
Net income	\$24,846	\$20,272	\$ 19,138

(a) Included tax equivalent adjustments primarily from income tax credits from investments in alternative energy, affordable housing and new markets, income from tax-exempt securities and loans, and the related amortization and other tax benefits of the investments in alternative energy and affordable housing of \$2.8 billion, \$4.0 billion and \$3.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively. Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2024	2023	2022
Financial ratios			
Return on equity	18 %	14 %	14 %
Overhead ratio	50	53	54
Compensation expense as percentage of total net revenue	26	27	27
Revenue by business			
Investment Banking	\$ 9,636	\$ 7,076	\$ 7,205
Payments	18,085	17,818	13,490
Lending	7,470	6,896	5,882
Other	76	107	244
Total Banking & Payments	35,267	31,897	26,821
Fixed Income Markets ^(a)	20,066	19,180	19,074
Equity Markets ^(a)	9,941	8,784	10,088
Securities Services	5,084	4,772	4,488
Credit Adjustments & Other ^(b)	(244)	(280)	(836)
Total Markets & Securities Services	34,847	32,456	32,814
Total net revenue	\$70,114	\$64,353	\$59,635

(a) In the fourth quarter of 2024, certain net funding costs that were previously allocated to Fixed Income Markets were reclassified to Equity Markets. Prior-period amounts have been revised to conform with the current presentation.

(b) Consists primarily of centrally-managed credit valuation adjustments ("CVA"), funding valuation adjustments ("FVA") on derivatives, other valuation adjustments, and certain components of fair value option elected liabilities, which are primarily reported in principal transactions revenue. Results are presented net of associated hedging activities and net of CVA and FVA amounts allocated to Fixed Income Markets and Equity Markets. Refer to Notes 2, 3 and 24 for additional information.

Banking & Payments Revenue by Client Coverage Segment: ^(a)

Global Corporate Banking & Global Investment

Banking provides banking products and services generally to large corporations, financial institutions and merchants.

Commercial Banking provides banking products and services generally to middle market clients, including start-ups, small and mid-sized companies, local governments, municipalities, and nonprofits, as well as to commercial real estate clients.

Other includes amounts related to credit protection purchased against certain retained loans and lending-related commitments in Lending, the impact of equity investments in Payments and revenues not aligned with a primary client coverage segment.

(a) Global Banking is a client coverage view within the Banking & Payments business and is comprised of the Global Corporate Banking, Global Investment Banking and Commercial Banking client coverage segments.

Selected income statement data

Year ended December 31, (in millions)	2024		2023		2022
Banking & Payments revenue by client coverage segment					
Global Corporate Banking & Global Investment Banking	\$	24,549	\$	21,700	\$ 19,325
Commercial Banking		11,487		11,050	7,906
Middle Market Banking		7,759		7,740	5,443
Commercial Real Estate Banking		3,728		3,310	2,463
Other		(769)		(853)	(410)
Total Banking & Payments revenue	\$	35,267	\$	31,897	\$ 26,821

2024 compared with 2023

Net income was \$24.8 billion, up 23%.

Net revenue was \$70.1 billion, up 9%.

Banking & Payments revenue was \$35.3 billion, up 11%.

- Investment Banking revenue was \$9.6 billion, up 36%. Investment Banking fees were up 37%, driven by higher fees across products. The Firm ranked #1 for Global Investment Banking fees, according to Dealogic.
 - Debt underwriting fees were \$4.1 billion, up 55%, predominantly driven by higher industry-wide issuances in leveraged loans, and in high-grade and high-yield bonds.
 - Equity underwriting fees were \$1.7 billion, up 47%, driven by increased industry-wide fees and wallet share gains in IPOs, and in follow-on and convertible securities offerings.
 - Advisory fees were \$3.3 billion, up 17%, driven by increased industry-wide M&A activity and wallet share gains.
- Payments revenue was \$18.1 billion, up 1%, driven by fee growth on higher volumes as well as higher average deposits, predominantly offset by deposit margin compression, reflecting higher rates paid, and higher deposit-related client credits.
- Lending revenue was \$7.5 billion, up 8%, predominantly driven by the impacts of higher rates and the First Republic acquisition.

Markets & Securities Services revenue was \$34.8 billion, up 7%. Markets revenue was \$30.0 billion, up 7%.

- Equity Markets revenue was \$9.9 billion, up 13%, driven by higher revenue in Equity Derivatives and Prime Finance.
- Fixed Income Markets revenue was \$20.1 billion, up 5%, driven by higher revenue in the Securitized Products Group, Currencies & Emerging Markets, and Credit, largely offset by lower revenue in Rates and Commodities.
- Securities Services revenue was \$5.1 billion, up 7%, predominantly driven by fee growth on higher client activity and market levels.
- Credit Adjustments & Other was a loss of \$244 million, compared with a loss of \$280 million in the prior year.

Noninterest expense was \$35.4 billion, up 4%, driven by higher compensation expense, including revenue-related compensation and an increase in the number of employees, as well as higher technology and brokerage expense partially offset by lower legal expense.

The provision for credit losses was \$762 million, reflecting:

- net charge-offs of \$617 million, primarily in Real Estate, largely concentrated in Office, and
- a \$145 million net addition to the allowance for credit losses, driven by
 - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024, predominantly offset by
 - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs predominantly from collateral-dependent loans.

The provision in the prior year was \$2.1 billion, reflecting a \$1.5 billion net addition to the allowance for credit losses, which included \$608 million to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023, and net charge-offs of \$588 million.

Selected metrics

As of or for the year ended December 31, (in millions, except employees)	2024	2023	2022
Selected balance sheet data (period-end)			
Total assets	\$1,773,194	\$1,638,493	\$1,591,402
Loans:			
Loans retained	483,043	475,186	421,521
Loans held-for-sale and loans at fair value ^(a)	40,324	39,464	43,011
Total loans	523,367	514,650	464,532
Equity	132,000	138,000	128,000
Banking & Payments loans by client coverage segment (period-end)^(b)			
Global Corporate Banking & Global Investment Banking	\$ 125,083	\$ 128,097	\$ 128,165
Commercial Banking	217,674	221,550	180,624
Middle Market Banking	72,814	78,043	72,625
Commercial Real Estate Banking	144,860	143,507	107,999
Other	187	526	122
Total Banking & Payments loans	342,944	350,173	308,911
Selected balance sheet data (average)			
Total assets	\$1,912,466	\$ 1,716,755	\$1,649,358
Trading assets-debt and equity instruments	624,032	508,792	405,948
Trading assets-derivative receivables	57,028	63,862	77,822
Loans:			
Loans retained	\$ 475,426	\$ 457,886	\$ 395,015
Loans held-for-sale and loans at fair value ^(a)	43,621	40,891	48,196
Total loans	\$ 519,047	\$ 498,777	\$ 443,211
Deposits ^(c)	1,061,488	996,295	1,033,880
Equity	132,000	137,507	128,000
Banking & Payments loans by client coverage segment (average)^(b)			
Global Corporate Banking & Global Investment Banking	\$ 128,142	\$ 131,230	\$ 122,174
Commercial Banking	220,285	209,244	173,289
Middle Market Banking	75,605	77,130	67,830
Commercial Real Estate Banking	144,680	132,114	105,459
Other	354	331	168
Total Banking & Payments loans	\$ 348,781	\$ 340,805	\$ 295,631
Employees	93,231	92,271	88,139

- (a) Loans held-for-sale and loans at fair value primarily reflect lending-related positions originated and purchased in Markets, including loans held for securitization.
- (b) Refer to page 78 for a description of each of the client coverage segments.
- (c) In the fourth quarter of 2023, certain deposits associated with First Republic were transferred to CIB from CCB.

Selected metrics

As of or for the year ended December 31, (in millions, except ratios)	2024	2023	2022
Credit data and quality statistics			
Net charge-offs/(recoveries)	\$ 689	^(d) \$ 588	\$ 166
Nonperforming assets:			
Nonaccrual loans:			
Nonaccrual loans retained ^(a)	\$3,258	\$ 1,675	\$ 1,484
Nonaccrual loans held- for-sale and loans at fair value ^(a)	1,502	828	848
Total nonaccrual loans	4,760	2,503	2,332
Derivative receivables	145	364	296
Assets acquired in loan satisfactions	213	169	87
Total nonperforming assets	\$ 5,118	\$ 3,036	\$ 2,715
Allowance for credit losses:			
Allowance for loan losses	\$7,294	\$ 7,326	\$ 5,616
Allowance for lending- related commitments	1,976	1,849	2,278
Total allowance for credit losses	\$9,270	\$ 9,175	\$ 7,894
Net charge-off/(recovery) rate ^(c)	0.14 %	0.13 %	0.04 %
Allowance for loan losses to period-end loans retained	1.51	1.54	1.33
Allowance for loan losses to nonaccrual loans retained ^(a)	224	437	378
Nonaccrual loans to total period-end loans	0.91	0.49	0.50

- (a) Allowance for loan losses of \$435 million, \$251 million and \$257 million were held against these nonaccrual loans at December 31, 2024, 2023 and 2022, respectively.
- (b) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024, 2023 and 2022, mortgage loans 90 or more days past due and insured by U.S. government agencies were \$37 million, \$59 million and \$115 million, respectively.
- (c) Loans held-for-sale and loans at fair value were excluded when calculating the net charge-off/(recovery) rate.
- (d) Includes \$72 million related to a purchased credit deteriorated ("PCD") loan that was charged off in the fourth quarter of 2024.

Investment banking fees

Year ended December 31, (in millions)	2024		2023		2022
Advisory	\$	3,290	\$	2,814	\$ 3,051
Equity underwriting		1,692		1,151	1,034
Debt underwriting ^(a)		4,134		2,666	2,892
Total investment banking fees	\$	9,116	\$	6,631	\$ 6,977

(a) Represents long-term debt and loan syndications.

League table results – wallet share

Year ended December 31,	2024		2023		2022	
	Rank	Share	Rank	Share	Rank	Share
Based on fees ^(a)						
M&A^(b)						
Global	# 1	9.6 %	# 2	9.0 %	# 2	7.9 %
U.S.	1	11.4	2	10.9	2	8.9
Equity and equity-related^(c)						
Global	1	11.0	1	7.7	2	5.7
U.S.	1	14.7	1	14.4	1	14.0
Long-term debt^(d)						
Global	1	7.6	1	7.0	1	6.9
U.S.	1	11.4	1	10.9	1	12.1
Loan syndications						
Global	1	10.2	1	11.9	1	11.0
U.S.	1	11.8	1	15.1	1	12.9
Global investment banking fees^(e)	# 1	9.3 %	# 1	8.6 %	# 1	7.8 %

(a) Source: Dealogic as of January 2, 2025. Reflects the ranking of revenue wallet and market share.

(b) Global M&A excludes any withdrawn transactions. U.S. M&A revenue wallet represents wallet from client parents based in the U.S.

(c) Global equity and equity-related ranking includes rights offerings and Chinese A-Shares.

(d) Long-term debt rankings include investment-grade, high-yield, supranationals, sovereigns, agencies, covered bonds, asset-backed securities ("ABS") and mortgage-backed securities ("MBS"); and exclude money market, short-term debt and U.S. municipal securities.

(e) Global investment banking fees exclude money market, short-term debt and shelf securities.

Markets revenue

The following table summarizes selected income statement data for the Markets businesses. Markets includes both Fixed Income Markets and Equity Markets. Markets revenue consists of principal transactions, fees, commissions and other income, as well as net interest income. The Firm assesses its Markets business performance on a total revenue basis, as offsets generally occur across revenue line items. For example, securities that generate net interest income may be risk-managed by derivatives that are reflected at fair value in principal transactions revenue. Refer to Notes 6 and 7 for a description of the composition of these income statement line items.

Principal transactions reflects revenue on financial instruments and commodities transactions that arise from client-driven market-making activity. Principal transactions revenue includes amounts recognized upon executing new transactions with market participants, as well as "inventory-related revenue," which is revenue recognized from gains and losses on derivatives and other instruments that the Firm has been holding in anticipation of, or in response to, client demand, and changes in the fair value of instruments

used by the Firm to actively manage the risk exposure arising from such inventory. Principal transactions revenue recognized upon executing new transactions with market participants is affected by many factors including the level of client activity, the bid-offer spread (which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa), market liquidity and volatility. These factors are interrelated and sensitive to the same factors that drive inventory-related revenue, which include general market conditions, such as interest rates, foreign exchange rates, credit spreads, and equity and commodity prices, as well as other macroeconomic conditions.

For the periods presented below, the primary source of principal transactions revenue was the amount recognized upon executing new transactions.

Year ended December 31, (in millions, except where otherwise noted)	2024			2023			2022		
	Fixed Income Markets	Equity Markets	Total Markets	Fixed Income Markets ^(c)	Equity Markets ^(c)	Total Markets	Fixed Income Markets ^(c)	Equity Markets ^(c)	Total Markets
Principal transactions	\$ 10,603	\$ 13,526	\$ 24,129	\$ 13,198	\$ 10,380	\$ 23,578	\$ 12,244	\$ 8,284	\$ 20,528
Lending- and deposit-related fees	391	100	491	307	40	347	303	22	325
Commissions and other fees	605	2,086	2,691	596	1,908	2,504	550	1,975	2,525
All other income	2,120	(65)	2,055	1,908	(79)	1,829	1,083	(88)	995
Noninterest revenue	13,719	15,647	29,366	16,009	12,249	28,258	14,180	10,193	24,373
Net interest income ^(a)	6,347	(5,706)	641	3,171	(3,465)	(294)	4,894	(105)	4,789
Total net revenue	\$ 20,066	\$ 9,941	\$ 30,007	\$ 19,180	\$ 8,784	\$ 27,964	\$ 19,074	\$ 10,088	\$ 29,162
Loss days^(b)	1			2			7		

(a) The decline in Equity Markets net interest income was driven by higher funding costs.

(b) Markets consists of Fixed Income Markets and Equity Markets. Loss days represent the number of days for which Markets recorded losses in total net revenue, which includes revenue related to both trading and non-trading positions. The loss days determined under this measure differ from the measure used to determine backtesting gains and losses. Daily backtesting gains and losses include positions in the Firm's Risk Management value-at-risk ("VaR") measure and exclude certain components of total net revenue, which may more than offset backtesting gains or losses on a particular day. For more information on daily backtesting gains and losses, refer to the VaR discussion on pages 143–145.

(c) In the fourth quarter of 2024, certain net funding costs that were previously allocated to Fixed Income Markets were reclassified to Equity Markets. Prior-period amounts have been revised to conform with the current presentation.

Selected metrics

As of or for the year ended December 31, (in millions, except where otherwise noted)	2024	2023	2022
Assets under custody ("AUC") by asset class (period-end) (in billions):			
Fixed Income	\$ 16,409	\$ 15,543	\$ 14,361
Equity	14,848	12,927	10,748
Other ^(a)	4,023	3,922	3,526
Total AUC	\$ 35,280	\$ 32,392	\$ 28,635
Client deposits and other third-party liabilities (average) ^(b)	\$ 961,646	\$ 912,859	\$ 981,653

(a) Consists of mutual funds, unit investment trusts, currencies, annuities, insurance contracts, options and other contracts.

(b) Client deposits and other third-party liabilities pertain to the Payments and Securities Services businesses.

International metrics

As of or for the year ended December 31, (in millions, except where otherwise noted)	2024	2023	2022
Total net revenue^(a)			
Europe/Middle East/Africa	\$ 15,191	\$ 14,418	\$ 15,716
Asia-Pacific	8,867	7,891	8,043
Latin America/Caribbean	2,427	2,161	2,288
Total international net revenue	26,485	24,470	26,047
North America	43,629	39,883	33,588
Total net revenue	\$ 70,114	\$ 64,353	\$ 59,635
Loans retained (period-end)^(a)			
Europe/Middle East/Africa	\$ 44,374	\$ 44,793	\$ 40,715
Asia-Pacific	16,107	15,506	16,764
Latin America/Caribbean	10,331	8,610	8,866
Total international loans	70,812	68,909	66,345
North America	412,231	406,277	355,176
Total loans retained	\$ 483,043	\$ 475,186	\$ 421,521
Client deposits and other third-party liabilities (average)^(b)			
Europe/Middle East/Africa	\$ 264,227	\$ 247,804	\$ 265,061
Asia-Pacific	141,042	135,388	136,539
Latin America/Caribbean	42,716	39,861	40,531
Total international	\$ 447,985	\$ 423,053	\$ 442,131
North America	513,661	489,806	539,522
Total client deposits and other third-party liabilities	\$ 961,646	\$ 912,859	\$ 981,653
AUC (period-end)^(b) (in billions)			
North America	\$ 23,845	\$ 21,792	\$ 19,219
All other regions	11,435	10,600	9,416
Total AUC	\$ 35,280	\$ 32,392	\$ 28,635

(a) Total net revenue and loans retained (excluding loans held-for-sale and loans at fair value) are based on the location of the trading desk, booking location, or domicile of the client, as applicable.

(b) Client deposits and other third-party liabilities pertaining to the Payments and Securities Services businesses, and AUC, are based on the domicile of the client or booking location, as applicable.

ASSET & WEALTH MANAGEMENT

Asset & Wealth Management, with client assets of \$5.9 trillion, is a global leader in investment and wealth management.

Asset Management

Offers multi-asset investment management solutions across equities, fixed income, alternatives and money market funds to institutional and retail investors providing for a broad range of clients' investment needs.

Global Private Bank

Provides retirement products and services, brokerage, custody, estate planning, lending, deposits and investment management to high net worth clients.

The majority of AWM's client assets are in actively managed portfolios.

Selected income statement data

Year ended December 31, (in millions, except ratios)	2024	2023	2022
Revenue			
Asset management fees	\$13,693	\$11,826	\$11,510
Commissions and other fees	874	697	\$ 662
All other income	456 ^(a)	1,037 ^{(a)(b)}	335
Noninterest revenue	15,023	13,560	12,507
Net interest income	6,555	6,267	5,241
Total net revenue	21,578	19,827	17,748
Provision for credit losses	(68)	159	128
Noninterest expense			
Compensation expense	7,984	7,115	6,336
Noncompensation expense	6,430	5,665	5,493
Total noninterest expense	14,414	12,780	11,829
Income before income tax expense	7,232	6,888	5,791
Income tax expense	1,811	1,661	1,426
Net income	\$ 5,421	\$ 5,227	\$4,365
Revenue by line of business			
Asset Management	\$10,175	\$ 9,129	\$ 8,818
Global Private Bank	11,403	10,698	8,930
Total net revenue	\$21,578	\$19,827	\$17,748
Financial ratios			
Return on equity	34 %	31 %	25 %
Overhead ratio	67	64	67
Pre-tax margin ratio:			
Asset Management	31	31	30
Global Private Bank	35	38	35
Asset & Wealth Management	34	35	33

(a) Includes the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic. The discount is deferred in other liabilities and recognized on a straight-line basis over the commitment period and was largely recognized in 2023 as the commitments were generally short term. Refer to Note 34 for additional information.

(b) Includes the gain on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% interest in the entity.

2024 compared with 2023

Net income was \$5.4 billion, up 4%.

Net revenue was \$21.6 billion, up 9%. Net interest income was \$6.6 billion, up 5%. Noninterest revenue was \$15.0 billion, up 11%.

Revenue from Asset Management was \$10.2 billion, up 11%, driven by:

- higher asset management fees, reflecting higher average market levels and strong net inflows, as well as
- higher performance fees.

The prior year included a gain of \$339 million on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% interest in the entity.

Revenue from Global Private Bank was \$11.4 billion, up 7%, driven by:

- higher noninterest revenue, reflecting:
 - higher management fees on strong net inflows and higher average market levels, as well as higher brokerage fees, partially offset by
 - a decline in the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic that have expired, and
- higher net interest income driven by:
 - higher average deposits associated with First Republic, which were transferred to AWM from CCB in the fourth quarter of 2023, as well as wider spreads on loans and higher average loans, largely offset by
 - deposit margin compression reflecting higher rates paid.

The prior year included net investment valuation losses.

Noninterest expense was \$14.4 billion, up 13%, predominantly driven by:

- higher compensation, including revenue-related compensation, and continued growth in private banking advisor teams, and
- higher distribution fees and legal expense,

The provision for credit losses was a net benefit of \$68 million.

The provision in the prior year was \$159 million, reflecting a \$146 million addition to the allowance for credit losses to establish the allowance for the First Republic loans and lending-related commitments in the second quarter of 2023.

Asset Management has two high-level measures of its overall fund performance.

• **Percentage of active mutual fund and active ETF assets under management in funds rated 4- or 5-star:** Mutual fund rating services rank funds based on their risk adjusted performance over various periods. A 5-star rating is the best rating and represents the top 10% of industry-wide ranked funds. A 4-star rating represents the next 22.5% of industry-wide ranked funds. A 3-star rating represents the next 35% of industry-wide ranked funds. A 2-star rating represents the next 22.5% of industry-wide ranked funds. A 1-star rating is the worst rating and represents the bottom 10% of industry-wide ranked funds. An overall Morningstar rating is derived from a weighted average of the performance associated with a fund's three-, five and ten-year (if applicable) Morningstar Rating metrics. For U.S.-domiciled funds, separate star ratings are provided at the individual share class level. The Nomura "star rating" is based on three-year risk-adjusted performance only. Funds with fewer than three years of history are not rated and hence excluded from these rankings. All ratings, the assigned peer categories and the asset values used to derive these rankings are sourced from the applicable fund rating provider. Where applicable, the fund rating providers redenominate asset values into U.S. dollars. The percentage of AUM is based on star ratings at the share class level for U.S.-domiciled funds, and at a "primary share class" level to represent the star rating of all other funds, except for Japan, for which Nomura provides ratings at the fund level. The performance data may have been different if all share classes had been included. Past performance is not indicative of future results.

• **Percentage of active mutual fund and active ETF assets under management in funds ranked in the 1st or 2nd quartile (one, three and five years):** All quartile rankings, the assigned peer categories and the asset values used to derive these rankings are sourced from the fund rating providers. Quartile rankings are based on the net-of-fee absolute return of each fund. Where applicable, the fund rating providers redenominate asset values into U.S. dollars. The percentage of AUM is based on fund performance and associated peer rankings at the share class level for U.S.-domiciled funds, at a "primary share class" level to represent the quartile ranking for U.K., Luxembourg and Hong Kong SAR funds and at the fund level for all other funds. The performance data may have been different if all share classes had been included. Past performance is not indicative of future results.

"Primary share class" means the C share class for European funds and Acc share class for Hong Kong SAR and Taiwan funds. If these share classes are not available, the oldest share class is used as the primary share class.

Selected metrics

As of or for the year ended December 31, (in millions, except ranking data, ratios and employees)	2024	2023	2022
% of JPM mutual fund assets and ETFs rated as 4- or 5-star ^(a)	69 %	69 %	73 %
% of JPM mutual fund assets and ETFs ranked in 1 st or 2 nd quartile: ^(b)			
1 year	73	40	68
3 years	75	67	76
5 years	77	71	81
Selected balance sheet data (period-end)^(c)			
Total assets	\$255,385	\$245,512	\$232,037
Loans	236,303	227,929	214,006
Deposits	248,287	233,232 ^(d)	233,130
Equity	15,500	17,000	17,000
Selected balance sheet data (average)^(c)			
Total assets	\$246,254	\$240,222	\$232,438
Loans	227,676	220,487	215,582
Deposits	235,146	216,178 ^(d)	261,489
Equity	15,500	16,671	17,000
Employees	29,403	28,485	26,041
Number of Global Private Bank client advisors	3,775	3,515	3,137
Credit data and quality statistics^(c)			
Net charge-offs/ (recoveries)	\$ 21	\$ 13	\$ (7)
Nonaccrual loans	700	650	459
Allowance for credit losses:			
Allowance for loan losses	\$ 539	\$ 633	\$ 494
Allowance for lending-related commitments	35	28	20
Total allowance for credit losses	\$ 574	\$ 661	\$ 514
Net charge-off/(recovery) rate	0.01 %	0.01 %	— %
Allowance for loan losses to period-end loans	0.23	0.28	0.23
Allowance for loan losses to nonaccrual loans	77	97	108
Nonaccrual loans to period-end loans	0.30	0.29	0.21

(a) Represents the Morningstar Rating for all domiciled funds except for Japan domiciled funds which use Nomura. Includes only Asset Management retail active open-ended mutual funds and active ETFs that have a rating. Excludes money market funds, Undiscovered Managers Fund, and Brazil domiciled funds.

- (b) Quartile ranking sourced from Morningstar, Lipper and Nomura based on country of domicile. Includes only Asset Management retail active open-ended mutual funds and active ETFs that are ranked by the aforementioned sources. Excludes money market funds, Undiscovered Managers Fund, and Brazil domiciled funds.
- (c) Loans, deposits and related credit data and quality statistics relate to the Global Private Bank business.
- (d) In the fourth quarter of 2023, certain deposits associated with First Republic were transferred to AWM from CCB.

Client assets

2024 compared with 2023

Assets under management were \$4.0 trillion and client assets were \$5.9 trillion, each up 18%, driven by continued net inflows and higher market levels.

Client assets

December 31, (in billions)	2024	2023	2022
Assets by asset class			
Liquidity	\$ 1,083	\$ 926	\$ 654
Fixed income	851	751	638
Equity	1,128	868	670
Multi-asset	764	680	603
Alternatives	219	197	201
Total assets under management	4,045	3,422	2,766
Custody/brokerage/ administration/deposits	1,887	1,590	1,282
Total client assets^(a)	\$ 5,932	\$ 5,012	\$ 4,048

Assets by client segment

Private Banking	\$ 1,234	\$ 974	\$ 751
Global Institutional	1,692	1,488	1,252
Global Funds	1,119	960	763
Total assets under management	\$ 4,045	\$ 3,422	\$ 2,766
Private Banking	\$ 2,974	\$ 2,452	\$ 1,964
Global Institutional	1,820	1,594	1,314
Global Funds	1,138	966	770
Total client assets^(a)	\$ 5,932	\$ 5,012	\$ 4,048

- (a) Includes CCB client investment assets invested in managed accounts and J.P. Morgan mutual funds where AWM is the investment manager.

Client assets (continued)

Year ended December 31, (in billions)	2024	2023	2022
Assets under management rollforward			
Beginning balance	\$ 3,422	\$ 2,766	\$ 3,113
Net asset flows:			
Liquidity	140	242	(55)
Fixed income	91	70	13
Equity	114	70	35
Multi-asset	19	1	(9)
Alternatives	10	(1)	8
Market/performance/other impacts	249	274	(339)
Ending balance, December 31	\$ 4,045	\$ 3,422	\$ 2,766

Client assets rollforward

Beginning balance	\$ 5,012	\$ 4,048	\$ 4,295
Net asset flows	486	490	49
Market/performance/other impacts	434	474	(296)
Ending balance, December 31	\$ 5,932	\$ 5,012	\$ 4,048

Selected Metrics

	As of December 31,		
	2024	2023	Change
Firmwide Wealth Management			
Client assets (in billions) ^(a)	\$ 3,756	\$ 3,177	18 %
Number of client advisors	9,530	8,971	6
Stock Plan Administration^(b)			
Number of stock plan participants (in thousands)	1,327	974	36
Client assets (in billions)	\$ 270	\$ 230	17 %

- (a) Consists of Global Private Bank in AWM and client investment assets in J.P. Morgan Wealth Management in CCB.
- (b) Relates to an equity plan administration business which was acquired in 2022 with the Firm's purchase of Global Shares. The increase in 2024 includes the impact of onboarding participants in the Firm's employee stock plans during the fourth quarter of 2024.

International metrics

Year ended December 31, (in billions, except where otherwise noted)	2024	2023	2022
Total net revenue (in millions)^(a)			
Europe/Middle East/Africa	\$ 3,563	\$ 3,377	\$ 3,240
Asia-Pacific	2,023	1,876	1,836
Latin America/Caribbean	1,065	985	967
Total international net revenue	6,651	6,238	6,043
North America	14,927	13,589	11,705
Total net revenue	\$ 21,578	\$ 19,827	\$ 17,748
Assets under management			
Europe/Middle East/Africa	\$ 604	\$ 539	\$ 487
Asia-Pacific	302	263	218
Latin America/Caribbean	106	86	69
Total international assets under management	1,012	888	774
North America	3,033	2,534	1,992
Total assets under management	\$ 4,045	\$ 3,422	\$ 2,766
Client assets			
Europe/Middle East/Africa	\$ 841	\$ 740	\$ 610
Asia-Pacific	482	406	331
Latin America/Caribbean	254	232	189
Total international client assets	1,577	1,378	1,130
North America	4,355	3,634	2,918
Total client assets	\$ 5,932	\$ 5,012	\$ 4,048

(a) Regional revenue is based on the domicile of the client.

CORPORATE

Corporate consists of Treasury and Chief Investment Office (“CIO”) and Other Corporate. Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm’s liquidity, funding, capital, structural interest rate and foreign exchange risks.

Other Corporate includes staff functions and expense that is centrally managed as well as certain Firm initiatives and activities not solely aligned to a specific LOB. The major Other Corporate functions include Real Estate, Technology, Legal, Corporate Finance, Human Resources, Internal Audit, Risk Management, Compliance, Control Management, Corporate Responsibility and various Other Corporate groups.

Selected income statement and balance sheet data

Year ended December 31, (in millions, except employees)	2024	2023	2022
Revenue			
Principal transactions	\$ 152	\$ 302	\$ (227)
Investment securities losses	(1,020)	(3,180)	(2,380)
All other income	8,476 ^(c)	3,010 ^(f)	809
Noninterest revenue	7,608	132	(1,798)
Net interest income	9,786	7,906	1,878
Total net revenue ^(a)	17,394	8,038	80
Provision for credit losses	10	171	22
Noninterest expense	3,994 ^{(d)(e)}	5,601 ^{(e)(g)}	1,034
Income/(loss) before income tax expense/(benefit)	13,390	2,266	(976)
Income tax expense/(benefit)	2,789	(555) ^(h)	(233)
Net income/(loss)	\$ 10,601	\$ 2,821	\$ (743)
Total net revenue			
Treasury and CIO	9,638	6,072	(439)
Other Corporate	7,756	1,966	519
Total net revenue	\$ 17,394	\$ 8,038	\$ 80
Net income/(loss)			
Treasury and CIO	7,013	4,206	(197)
Other Corporate	3,588 ^(e)	(1,385) ^(e)	(546)
Total net income/(loss)	\$ 10,601	\$ 2,821	\$ (743)
Total assets (period-end)	\$1,323,967	\$1,348,437	\$1,328,219
Loans (period-end)	1,964	1,924	2,181
Deposits ^(b)	27,581	21,826	14,203
Employees	49,610	47,530	44,196

(a) Included tax-equivalent adjustments, predominantly driven by tax-exempt income from municipal bonds, of \$182 million, \$211

million and \$235 million for the years ended December 31, 2024, 2023 and 2022, respectively.

- (b) Predominantly relates to the Firm's international consumer initiatives.
- (c) Included the net gain related to Visa shares of \$7.9 billion recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
- (d) Included a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
- (e) The first quarter of 2024 included an increase of \$725 million to the FDIC special assessment reflecting the FDIC's revised estimate of Deposit Insurance Fund losses. The fourth quarter of 2023 included the \$2.9 billion FDIC special assessment.
- (f) Included the estimated bargain purchase gain of \$2.8 billion for the year ended December 31, 2023 associated with the First Republic acquisition. Refer to Notes 6 and 34 for additional information.
- (g) In the second quarter of 2023, substantially all of the expense associated with First Republic was reported in Corporate. Commencing in the third quarter of 2023, the expense was aligned to the appropriate LOBs.
- (h) Income taxes associated with the First Republic acquisition were reflected in the estimated bargain purchase gain.

2024 compared with 2023

Net income was \$10.6 billion, compared with \$2.8 billion in the prior year.

Net revenue was \$17.4 billion, compared with \$8.0 billion in the prior year.

Net interest income was \$9.8 billion, up 24%, driven by the impact of balance sheet actions, primarily reinvestments in the investment securities portfolio, partially offset by the net impact of rates.

Noninterest revenue was \$7.6 billion, compared with \$132 million in the prior year. Excluding the \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024 and the prior-year \$2.8 billion estimated bargain purchase gain associated with the First Republic acquisition, revenue was up \$2.4 billion, predominantly driven by lower investment securities losses, primarily on sales of U.S. Treasuries and U.S. GSE and government agency MBS, associated with repositioning the investment securities portfolio in Treasury and CIO.

Noninterest expense was \$4.0 billion, down 29%, driven by:

- a lower FDIC special assessment,
 - lower expense associated with the First Republic acquisition as the prior year expense in Corporate included individuals associated with First Republic who were not employees of the Firm until July 2023, and this expense was subsequently aligned to the appropriate LOBs starting in the third quarter of 2023, and
 - lower legal expense,
- partially offset by
- a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, and
 - higher costs associated with the Firm's international consumer initiatives.

The provision for credit losses was \$10 million.

The provision in the prior year was \$171 million, reflecting a net addition to the allowance for credit losses related to a single name exposure, which was subsequently charged off upon the restructuring of a loan.

Refer to Note 10 and Note 13 for additional information on the investment securities portfolio and the allowance for credit losses.

The current period income tax expense was driven by changes in the level and mix of income and expenses subject to U.S. federal and state and local taxes, including the impact of the net gain on Visa shares and the contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024, partially offset by benefits related to tax audit settlements.

Other Corporate includes the Firm's international consumer initiatives, which primarily consists of Chase U.K., Nutmeg, and an ownership stake in C6 Bank.

The deposits within Corporate relate to the Firm's international consumer initiatives and have increased as a result of growth in client accounts, reflecting the impact of additional product offerings.

Treasury and CIO overview

Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks. The risks managed by Treasury and CIO arise from the activities undertaken by the Firm's three reportable business segments to serve their respective client bases, which generate both on- and off-balance sheet assets and liabilities.

Treasury and CIO seeks to achieve the Firm's asset-liability management objectives generally by investing in high quality securities that are managed for the longer-term as part of the Firm's investment securities portfolio. Treasury and CIO also uses derivatives to meet the Firm's asset-liability management objectives. Refer to Note 5 for further information on derivatives. In addition, Treasury and CIO manages the Firm's cash position primarily through deposits at central banks and investments in short-term instruments. Refer to Liquidity Risk Management on pages 108–115 for further information on liquidity and funding risk. Refer to Market Risk Management on pages 141–149 for information on interest rate and foreign exchange risks.

The investment securities portfolio predominantly consists of U.S. and non-U.S. government securities, U.S. GSE and government agency and nonagency mortgage-backed securities, collateralized loan obligations, obligations of U.S. states and municipalities and other ABS. At December 31, 2024, the Treasury and CIO investment securities portfolio, net of the allowance for credit losses, was \$678.3 billion, and the average credit rating of the securities comprising the portfolio was AA+ (based upon external ratings where available and, where not available, based primarily upon internal risk ratings). Refer to Note 10 for further information on the Firm's investment securities portfolio and internal risk ratings.

Selected income statement and balance sheet data

As of or for the year ended December 31, (in millions)	2024	2023	2022
Investment securities losses	\$ (1,020)	\$ (3,180)	\$ (2,380)
Available-for-sale securities (average)	\$287,260	\$200,708	\$239,924
Held-to-maturity securities (average) ^(a)	321,384	402,010	412,180
Investment securities portfolio (average)	\$608,644	\$602,718	\$652,104
Available-for-sale securities (period-end)	\$403,796	\$199,354	\$203,981
Held-to-maturity securities (period-end) ^(a)	274,468	369,848	425,305
Investment securities portfolio, net of allowance for credit losses (period- end) ^(b)	\$678,264	\$569,202	\$629,286

- (a) Effective January 1, 2023, the Firm adopted the portfolio layer method hedge accounting guidance. As permitted by the guidance, the Firm elected to transfer \$7.1 billion of investment securities from HTM to AFS. During 2022, the Firm transferred \$78.3 billion of investment securities from AFS to HTM for capital management purposes. Refer to Note 1 and Note 10 for additional information on the portfolio layer method hedge accounting guidance.
- (b) As of December 31, 2024, 2023 and 2022, the allowance for credit losses on investment securities was \$105 million, \$94 million and \$67 million, respectively.

FIRMWIDE RISK MANAGEMENT

Risk is an inherent part of JPMorganChase's business activities. When the Firm extends a consumer or wholesale loan, advises customers and clients on their investment decisions, makes markets in securities, or offers other products or services, the Firm takes on some degree of risk. The Firm's overall objective is to manage its business, and the associated risks, in a manner that balances serving the interests of its clients, customers and investors, and protecting the safety and soundness of the Firm.

The Firm believes that effective risk management requires, among other things:

- Acceptance of responsibility, including identification and escalation of risks by all individuals within the Firm;
- Ownership of risk identification, assessment, data and management within each of the LOBs and Corporate; and
- A Firmwide risk governance and oversight structure.

The Firm follows a disciplined and balanced compensation framework with strong internal governance and independent oversight by the Board of Directors (the "Board"). The impact of risk and control issues is carefully considered in the Firm's performance evaluation and incentive compensation processes.

Risk governance framework

The Firm's risk governance framework involves understanding drivers of risks, types of risks and impacts of risks.



Drivers of risks are factors that cause a risk to exist. Drivers of risks include the economic environment, regulatory or government policy, competitor or market evolution, business decisions, process or judgment error, deliberate wrongdoing, dysfunctional markets and natural disasters.

Types of risks are categories by which risks manifest themselves. The Firm's risks are generally categorized in the following four risk types:

- Strategic risk is the risk to earnings, capital, liquidity or reputation associated with poorly-designed or failed business plans or an inadequate response to changes in the operating environment.

- Credit and investment risk is the risk associated with the default or change in credit profile of a client, counterparty or customer; or loss of principal or a reduction in expected returns on investments, including consumer credit risk, wholesale credit risk and investment portfolio risk.
- Market risk is the risk associated with the effect of changes in market factors, such as interest and foreign exchange rates, equity and commodity prices, credit spreads or implied volatilities, on the value of assets and liabilities held for both the short and long term.
- Operational risk is the risk of an adverse outcome resulting from inadequate or failed internal processes or systems; human factors; or external events impacting the Firm's processes or systems. Operational risk includes cybersecurity, compliance, conduct, legal, and estimations and model risk.

Impacts of risks are consequences of risks, both quantitative and qualitative. There may be many consequences when risks manifest themselves, including quantitative impacts such as a reduction in earnings and capital, liquidity outflows, and fines or penalties, or qualitative impacts such as damage to the Firm's reputation, loss of clients and customers, and regulatory and enforcement actions.

The Firm's risk governance framework is managed on a Firmwide basis. The Firm has an Independent Risk Management ("IRM") function, which is comprised of Risk Management and Compliance. The Firm's Chief Executive Officer ("CEO") appoints, subject to approval by the Risk Committee of the Board of Directors (the "Board Risk Committee"), the Firm's Chief Risk Officer ("CRO") to lead the IRM function and maintain the risk governance framework of the Firm. The framework is subject to approval by the Board Risk Committee through its review and approval of the Risk Governance and Oversight Policy.

The Firm's CRO oversees and delegates authority to the Firmwide Risk Executives ("FREs"), the Chief Risk Officers of the LOBs and Corporate ("LOB CROs"), and the Firm's Chief Compliance Officer ("CCO"), who, in turn, establish Risk Management and Compliance organizations, develop the Firm's risk governance policies and standards, and define and oversee the implementation of the Firm's risk governance framework. The LOB CROs oversee risks that arise in their LOBs and Corporate, while FREs oversee risks that span across the LOBs and Corporate, as well as functions and regions. Each area of the Firm that gives rise to risk is expected to operate within the parameters identified by the IRM function, and within the risk and control standards established by its own management.

Management's discussion and analysis

Three lines of defense

The Firm's "three lines of defense" are as follows:

The first line of defense consists of each LOB, Treasury and CIO, and certain Other Corporate initiatives, including their aligned Operations, Technology and Control Management. The first line of defense owns the risks, and identification of risks, associated with their respective activities and the design and execution of controls to manage those risks.

Responsibilities also include adherence to applicable laws, rules and regulations and implementation of the risk governance framework established by IRM, which may include policies, standards, limits, thresholds and controls.

The second line of defense is the IRM function, which is separate from the first line of defense and is responsible for independently measuring risk, as well as assessing and challenging the risk management activities of the first line of defense. IRM is also responsible for the identification of risks within its organization, its own adherence to applicable laws, rules and regulations and for the development and implementation of policies and standards with respect to its own processes.

The third line of defense is Internal Audit, an independent function that provides objective assessment of the adequacy and effectiveness of Firmwide processes, controls, governance and risk management. The Internal Audit function is led by the General Auditor, who reports to the Audit Committee and administratively to the CEO.

In addition, there are other functions that contribute to the Firmwide control environment but are not considered part of a particular line of defense, including Corporate Finance, Human Resources and Legal. These other functions are responsible for the identification of risks within their respective organizations, adherence to applicable laws, rules and regulations and implementation of the risk governance framework established by IRM.

Risk identification and ownership

The LOBs and Corporate are responsible for the identification of risks within their respective organizations, as well as the design and execution of controls, including IRM-specified controls, to manage those risks. To support this activity, the Firm has a risk identification framework designed to facilitate the responsibility of each LOB and Corporate to identify material risks inherent to the Firm's businesses and operational activities, catalog them in a central repository and review material risks on a regular basis. The IRM function reviews and challenges the risks identified by each LOB and Corporate, maintains the central repository and provides the consolidated Firmwide results to the Firmwide Risk Committee ("FRC") and the Board Risk Committee.

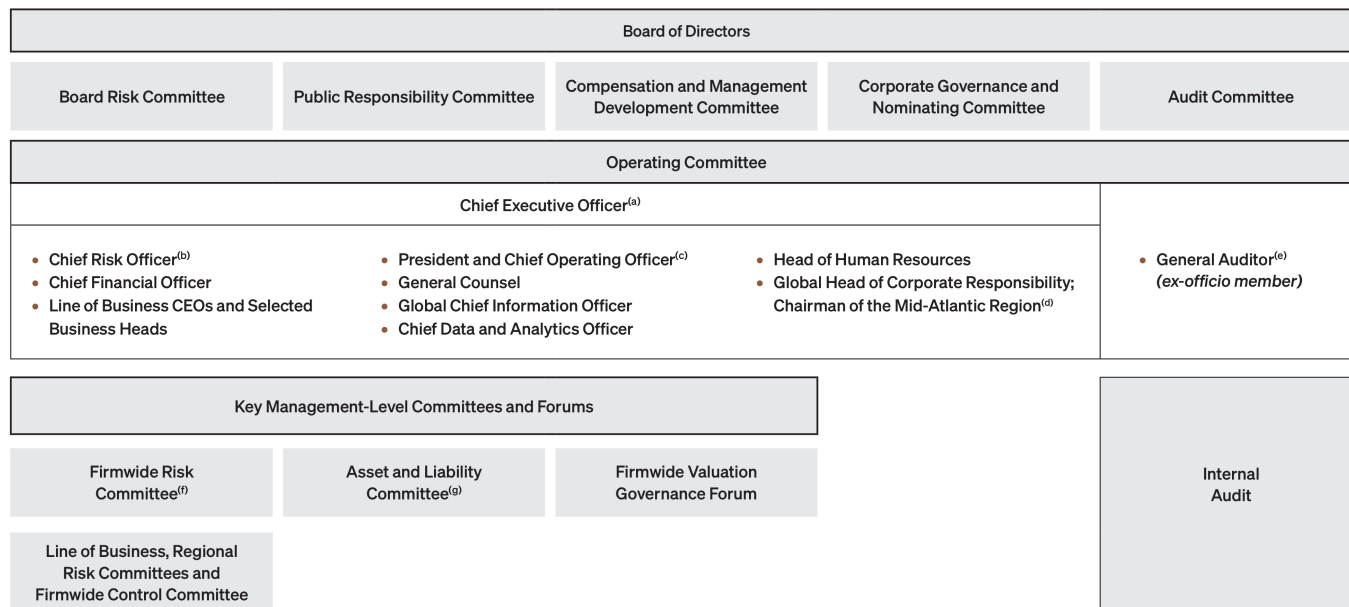
Risk appetite

The Firm's overall appetite for risk is governed by Risk Appetite frameworks for quantitative and qualitative risks. The Firm's risk appetite is periodically set and approved by senior management (including the CEO and CRO) and approved by the Board Risk Committee. Quantitative and qualitative risks are assessed to monitor and measure the Firm's capacity to take risk consistent with its stated risk appetite. Risk appetite results are reported to the Board Risk Committee.

Risk governance and oversight structure

The independent status of the IRM function is supported by a risk governance and oversight structure that provides channels for the escalation of risks and issues to senior management, the FRC and the Board of Directors, as appropriate.

The chart below illustrates the principal standing committees of the Board of Directors and key senior management-level committees in the Firm's risk governance and oversight structure. In addition, there are other committees, forums and channels of escalation that support the oversight of risk that are not shown in the chart below or described in this Form 10-K.



(a) The Firm's CEO is also the Chairman of the Board of Directors.

(b) The Firm's CRO reports to the Firm's CEO and the Board Risk Committee. The Firm's CRO may escalate directly to the Board of Directors (including its committees), as appropriate.

(c) Effective January 14, 2025, the co-CEO of the Commercial & Investment Bank became a Chief Operating Officer of the Firm. Refer to Recent events on page 57 for further information.

(d) Effective February 12, 2024, the Global Head of Corporate Responsibility and Chairman of the Mid-Atlantic Region became a member of the Operating Committee.

(e) The Firm's General Auditor reports to the Audit Committee and administratively to the Firm's CEO.

(f) The Firmwide Risk Committee escalates to the Board Risk Committee, as appropriate.

(g) The Asset and Liability Committee escalates to the Firm's CEO or the Board of Directors (including its committees), as appropriate.

The Firm's Operating Committee, which consists of the Firm's CEO, CRO, Chief Financial Officer ("CFO"), General Counsel, CEOs of the LOBs and other senior executives, is accountable to and may refer matters to the Firm's Board of Directors. The Operating Committee and certain other members of senior management are responsible for escalating to the Board the information necessary to facilitate the Board's exercise of its duties.

Board oversight

The Firm's Board of Directors actively oversees the business and affairs of the Firm. This includes monitoring the Firm's financial performance and condition and reviewing the strategic objectives and plans of the Firm. The Board carries out a significant portion of its oversight responsibilities through its principal standing committees, each of which consists solely of independent members of the Board. The Board Risk Committee is the principal committee that oversees risk matters. The Audit Committee oversees the control environment, and the Compensation & Management Development Committee oversees compensation and other management-related

matters. Each committee of the Board oversees reputation risks, conduct risks, and environmental, social and governance ("ESG") matters within its scope of responsibility.

The JPMorgan Chase Bank, N.A. Board of Directors is responsible for the oversight of management of the bank, which it discharges both acting directly and through the principal standing committees of the Firm's Board of Directors. Risk and control oversight on behalf of JPMorgan Chase Bank N.A. is primarily the responsibility of the Board Risk Committee and the Audit Committee, respectively, and, with respect to compensation and other management-related matters, the Compensation & Management Development Committee.

The Board Risk Committee assists the Board in its oversight of management's responsibility to implement a global risk management framework reasonably designed to identify, assess and manage the Firm's risks. The Board Risk Committee's responsibilities include approval of applicable primary risk policies and review of certain associated frameworks, analysis and reporting established by

Management's discussion and analysis

management. Breaches in risk appetite and parameters, issues that may have a material adverse impact on the Firm, including capital and liquidity issues, and other significant risk-related matters are escalated to the Board Risk Committee, as appropriate.

The Audit Committee assists the Board in its oversight of management's responsibilities to ensure that there is an effective system of controls reasonably designed to safeguard the Firm's assets and income, ensure the integrity of the Firm's financial statements, and maintain compliance with the Firm's ethical standards, policies, plans and procedures, and with laws and regulations. It also assists the Board in its oversight of the qualifications, independence and performance of the Firm's independent registered public accounting firm, and of the performance of the Firm's Internal Audit function.

The Compensation & Management Development Committee ("CMDC") assists the Board in its oversight of the Firm's compensation principles and practices. The CMDC reviews and approves the Firm's compensation and qualified benefits programs. The Committee reviews the performance of Operating Committee members against their goals, and approves their compensation awards. In addition, the CEO's compensation award is subject to ratification by the independent directors of the Board. The CMDC also reviews the development of and succession for key executives. As part of the Board's role of reinforcing, demonstrating and communicating the "tone at the top," the CMDC oversees the Firm's culture, including reviewing updates from management regarding significant conduct issues and any related actions with respect to employees, including compensation actions.

The Public Responsibility Committee oversees and reviews the Firm's positions and practices on public responsibility matters such as community investment, fair lending, sustainability, consumer practices and other public policy issues that reflect the Firm's values and character and could impact the Firm's reputation among its stakeholders. The Committee also provides guidance on these matters to management and the Board, as appropriate.

The Corporate Governance & Nominating Committee exercises general oversight with respect to the governance of the Board of Directors. It reviews the qualifications of and recommends to the Board proposed nominees for election to the Board. The Committee evaluates and recommends to the Board corporate governance practices applicable to the Firm. It also reviews the framework for assessing the Board's performance and self-evaluation.

Management oversight

The Firm's senior management-level committees that are primarily responsible for key risk-related functions include:

The Firmwide Risk Committee ("FRC") is the Firm's highest management-level risk committee. It oversees the risks inherent in the Firm's business and provides a forum for discussion of risk-related and other topics and issues that are raised or escalated by its members and other committees.

The Firmwide Control Committee ("FCC") is an escalation committee for senior management to review and discuss the Firmwide compliance and operational risk environment, including identified issues, compliance and operational risk metrics and significant events that have been escalated.

Line of Business and Regional Risk Committees are responsible for overseeing the governance, limits and controls that have been established within the scope of their respective activities. These committees review the ways in which the particular LOB or the businesses operating in a particular region could be exposed to adverse outcomes, with a focus on identifying, accepting, escalating and/or requiring remediation of matters brought to these committees.

Line of Business and Corporate Function Control

Committees oversee the risk and control environment of their respective business or function, inclusive of Operational Risk, Compliance and Conduct Risks. As part of that mandate, they are responsible for reviewing indicators of elevated or emerging risks and other data that may impact the level of compliance and operational risk in a business or function, addressing key compliance and operational risk issues, with an emphasis on processes with control concerns, and overseeing control remediation.

The Asset and Liability Committee (“ALCO”) is responsible for overseeing the Firm’s asset and liability management (“ALM”), including the activities and frameworks supporting management of the balance sheet, liquidity risk, interest rate risk and capital risk.

The Firmwide Valuation Governance Forum (“VGF”) is composed of senior finance and risk executives and is responsible for overseeing the management of risks arising from valuation activities conducted across the Firm.

Risk governance and oversight functions

The Firm monitors and measures its risk through risk governance and oversight functions. The scope of a particular function or business activity may include one or more drivers, types and/or impacts of risk. For example, Country Risk Management oversees country risk which may be a driver of risk or an aggregation of exposures that could give rise to multiple risk types such as credit or market risk.

The following sections discuss the risk governance and oversight functions that have been established to oversee the risks inherent in the Firm’s business activities.

Risk governance and oversight functions	Page
Strategic Risk	96
Capital Risk	97-107
Liquidity Risk	108-115
Reputation Risk	116
Consumer Credit Risk	120-125
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STRATEGIC RISK MANAGEMENT

Strategic risk is the risk to earnings, capital, liquidity or reputation associated with poorly-designed or failed business plans or an inadequate response to changes in the operating environment.

Management and oversight

The Operating Committee, together with the senior leadership of each LOB and Corporate, are responsible for managing the Firm's most significant strategic risks. IRM engages regularly in strategic business discussions and decision-making, including participation in relevant business reviews and senior management meetings, risk and control committees and other relevant governance forums, and review of acquisitions and new business initiatives. The Board of Directors oversees management's strategic decisions, and the Board Risk Committee oversees IRM and the Firm's risk governance framework.

In the process of developing business plans and strategic initiatives, LOB and Corporate senior management identify the associated risks that are incorporated into the Firmwide Risk Identification framework and their impact on risk appetite.

In addition, IRM conducts a qualitative assessment of the LOB and Corporate strategic initiatives to assess their impact on the risk profile of the Firm.

The Firm's strategic planning process, which includes the development of the Firm's strategic plan and other strategic initiatives, is one component of managing the Firm's strategic risk. The strategic plan outlines the Firm's strategic framework and initiatives, and includes components such as budget, risk appetite, capital, earnings and asset-liability management objectives. Guided by the Firm's Business Principles, the Operating Committee and senior management teams in each LOB and Corporate review and update the strategic plan periodically, including evaluating the strategic framework and performance of strategic initiatives, assessing the operating environment, refining existing strategies and developing new strategies.

The Firm's strategic plan, together with IRM's assessment, are provided to the Board as part of its review and approval of the Firm's strategic plan, and the plan is also reflected in the Firm's budget.

The Firm's balance sheet strategy, which focuses on risk-adjusted returns, strong capital and robust liquidity, is also a component in the management of strategic risk. Refer to Capital Risk Management on pages 97–107 for further information on capital risk. Refer to Liquidity Risk Management on pages 108–115 for further information on liquidity risk. Refer to Reputation Risk Management on page 116 for further information on reputation risk.

CAPITAL RISK MANAGEMENT

Capital risk is the risk that the Firm has an insufficient level or composition of capital to support the Firm's business activities and associated risks during normal economic environments and under stressed conditions.

A strong capital position is essential to the Firm's business strategy and competitive position. Maintaining a strong balance sheet to manage through economic volatility is a strategic imperative of the Firm's Board of Directors, CEO and Operating Committee. The Firm's "fortress balance sheet" philosophy focuses on risk-adjusted returns, strong capital and robust liquidity. The Firm's capital risk management strategy focuses on maintaining long-term stability to enable the Firm to build and invest in market-leading businesses, including in highly stressed environments. Senior management considers the implications on the Firm's capital prior to making significant decisions that could impact future business activities. In addition to considering the Firm's earnings outlook, senior management evaluates all sources and uses of capital with a view to ensuring the Firm's capital strength.

Capital risk management

The Firm has a Capital Risk Management function whose primary objective is to provide independent oversight of capital risk across the Firm.

Capital Risk Management's responsibilities include:

- Defining, monitoring and reporting capital risk metrics;
- Establishing, calibrating and monitoring capital risk limits and indicators, including capital risk appetite;
- Developing processes to classify, monitor and report capital limit breaches;
- Performing assessments of the Firm's capital management activities, including changes made to the Contingency Capital Plan described below; and
- Conducting assessments of the Firm's regulatory capital framework intended to ensure compliance with applicable regulatory capital rules.

Capital management

Treasury and CIO is responsible for capital management.

The primary objectives of the Firm's capital management are to:

- Maintain sufficient capital in order to continue to build and invest in the Firm's businesses through normal economic cycles and in stressed environments;
- Retain flexibility to take advantage of future investment opportunities;

- Promote the Parent Company's ability to serve as a source of strength to its subsidiaries;
- Ensure the Firm operates above the minimum regulatory capital ratios as well as maintain "well-capitalized" status for the Firm and its principal insured depository institution ("IDI") subsidiary, JPMorgan Chase Bank, N.A., at all times under applicable regulatory capital requirements;
- Meet capital distribution objectives; and
- Maintain sufficient capital resources to operate throughout a resolution period in accordance with the Firm's preferred resolution strategy.

The Firm addresses these objectives through:

- Establishing internal minimum capital requirements and maintaining a strong capital governance framework. The internal minimum capital levels consider the Firm's regulatory capital requirements as well as an internal assessment of capital adequacy, in normal economic cycles and in stress events;
- Retaining flexibility in order to react to a range of potential events; and
- Regularly monitoring the Firm's capital position and following prescribed escalation protocols, both at the Firm and material legal entity levels.

Governance

Committees responsible for overseeing the Firm's capital management include the Capital Governance Committee, the Firmwide ALCO as well as regional ALCOs, and the CIO, Treasury and Corporate ("CTC") Risk Committee. In addition, the Board Risk Committee periodically reviews the Firm's capital risk tolerance. Refer to Firmwide Risk Management on pages 91–95 for additional discussion of the Firmwide ALCO and other risk-related committees.

Capital planning and stress testing

Comprehensive Capital Analysis and Review

The Federal Reserve requires the Firm, as a large Bank Holding Company ("BHC"), to submit at least annually a capital plan that has been reviewed and approved by the Board of Directors. The Federal Reserve uses Comprehensive Capital Analysis and Review ("CCAR") and other stress testing processes to assess whether large BHCs, such as the Firm, have sufficient capital during periods of economic and financial stress, and have robust, forward-looking capital assessment and planning processes in place that address each BHC's unique risks to enable it to absorb losses under certain stress scenarios. Through CCAR, the Federal Reserve evaluates each BHC's capital adequacy and internal capital adequacy assessment processes ("ICAAP"), as well as its plans to make capital distributions, such as dividend payments or stock repurchases. The Federal

Management's discussion and analysis

Reserve uses results under the severely adverse scenario from its supervisory stress test to determine each firm's Stress Capital Buffer ("SCB") requirement for the coming year.

The Firm's current SCB requirement is 3.3% and will remain in effect until September 30, 2025. The Firm's Standardized CET1 capital ratio requirement, including regulatory buffers, was 12.3% as of December 31, 2024.

Refer to Capital actions on page 105 for information on actions taken by the Firm's Board of Directors.

Internal Capital Adequacy Assessment Process

Annually, the Firm prepares the ICAAP, which informs the Board of Directors of the ongoing assessment of the Firm's processes for managing the sources and uses of capital as well as compliance with supervisory expectations for capital planning and capital adequacy. The Firm's ICAAP integrates stress testing protocols with capital planning. The Firm's Audit Committee is responsible for reviewing and approving the capital planning framework.

Stress testing assesses the potential impact of alternative economic and business scenarios on the Firm's earnings and capital. Economic scenarios, and the parameters underlying those scenarios, are defined centrally and applied uniformly across the businesses. These scenarios are articulated in terms of macroeconomic factors, which are key drivers of business results; global market shocks, which generate short-term but severe trading losses; and idiosyncratic operational risk events. The scenarios are intended to capture and stress key vulnerabilities and idiosyncratic risks facing the Firm. In addition to CCAR and other periodic stress testing, management also considers tailored stress scenarios and sensitivity analyses, as necessary.

Contingency Capital Plan

The Firm's Contingency Capital Plan establishes the capital management framework for the Firm and specifies the principles underlying the Firm's approach towards capital management in normal economic conditions and in stressed environments. The Contingency Capital Plan defines how the Firm calibrates its targeted capital levels and meets minimum capital requirements, monitors the ongoing appropriateness of planned capital distributions, and sets out the capital contingency actions that are expected to be taken or considered at various levels of capital depletion during a period of stress.

Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards, for the Firm as a consolidated financial holding company. The Office of the Comptroller of the Currency ("OCC") establishes similar minimum capital requirements and standards for the Firm's principal IDI subsidiary, JPMorgan Chase Bank, N.A. The U.S. capital requirements generally follow the Capital Accord of the Basel Committee, as amended from time to time.

Basel III Overview

The capital rules under Basel III establish minimum capital ratios and overall capital adequacy standards for large and internationally active U.S. BHCs and banks, including the Firm and JPMorgan Chase Bank, N.A. The minimum amount of regulatory capital that must be held by BHCs and banks is determined by calculating RWA, which are on-balance sheet assets and off-balance sheet exposures, weighted according to risk. Under the rules currently in effect, two comprehensive approaches are prescribed for calculating RWA: a standardized approach ("Basel III Standardized"), and an advanced approach ("Basel III Advanced").

For each of these risk-based capital ratios, the capital adequacy of the Firm is evaluated against the lower of the Standardized or Advanced approaches compared to their respective regulatory capital ratio requirements.

The current Basel III rules establish capital requirements for calculating credit risk RWA and market risk RWA, and in the case of Basel III Advanced, operational risk RWA. Key differences in the calculation of credit risk RWA between the Standardized and Advanced approaches are that for Basel III Advanced, credit risk RWA is based on risk-sensitive approaches which largely rely on the use of internal credit models and parameters, whereas for Basel III Standardized, credit risk RWA is generally based on supervisory risk-weightings which vary primarily by counterparty type and asset class. Market risk RWA is generally calculated consistently between Basel III Standardized and Basel III Advanced. In addition to the RWA calculated under these approaches, the Firm may supplement such amounts to incorporate management judgment and feedback from its regulators.

As of December 31, 2024, the Firm's Basel III Standardized ratios risk-based ratios were more binding than the Basel III Advanced risk-based ratios.

Additionally, Basel III requires that Advanced Approaches banking organizations, including the Firm, calculate their SLRs.

Refer to page 104 for additional information on SLR.

Key Regulatory Developments

U.S. Basel III Finalization

In July 2023, the Federal Reserve, the OCC and the FDIC released a proposal to amend the risk-based capital framework, entitled "Regulatory capital rule: Amendments applicable to large banking organizations and to banking organizations with significant trading activity", which is referred to in this Form 10-K as the "U.S. Basel III proposal". Under this proposal, changes to the framework would include replacement of the Advanced approach with an expanded risk-based approach for the calculation of RWA. In addition, the stress capital buffer requirement would be applicable to both the expanded risk-based approach and the Standardized approach.

GSIB Surcharge and TLAC and Eligible LTD Requirements

In July 2023, the Federal Reserve released a proposal to amend the calculation of the GSIB surcharge. Under the proposal, the annual GSIB surcharge would be based on an average of the quarterly surcharge calculations throughout the calendar year, with daily averaging required for certain measures. The proposal would also reduce surcharge increments from 50 bps to 10 bps and includes other technical amendments to the "Method 2" calculation. The proposed changes would revise risk-based capital requirements for the Firm and other U.S. GSIBs. Refer to Risk-based Capital Regulatory Requirements on page 100 for further information on the GSIB surcharge.

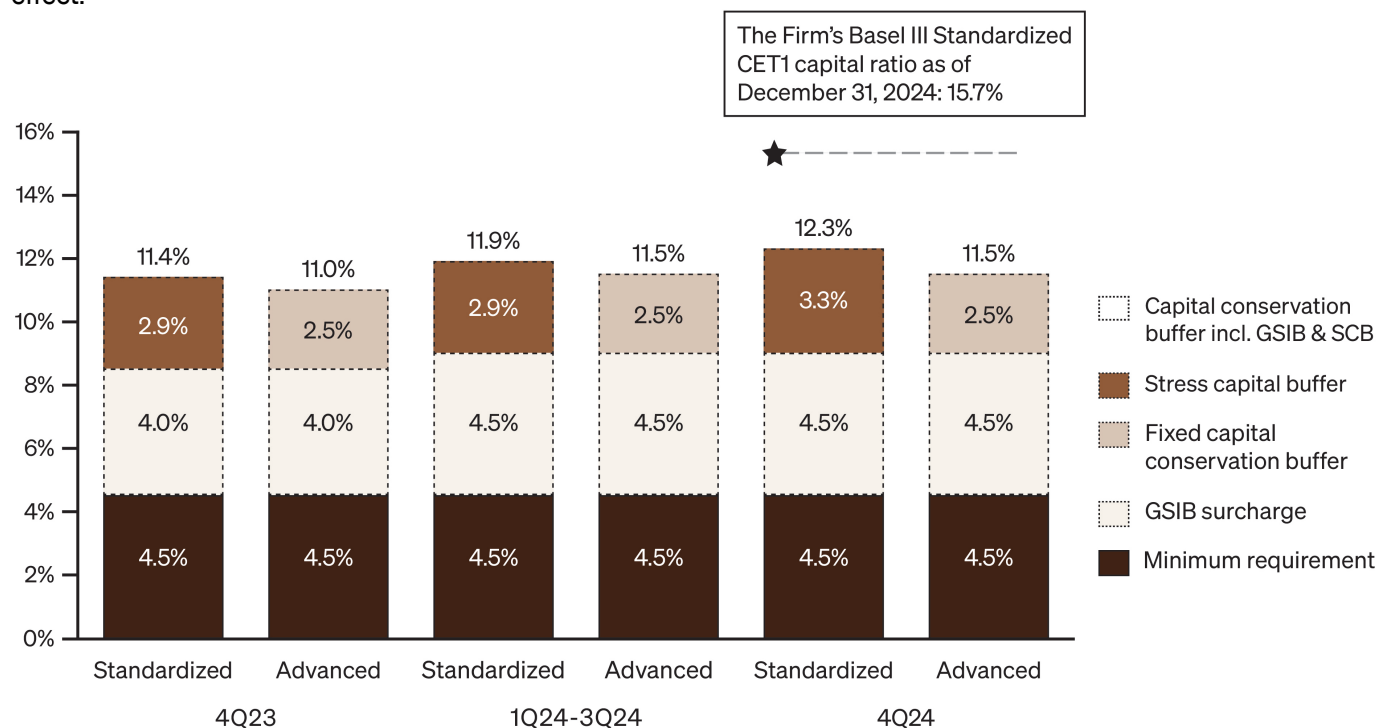
Additionally, in August 2023, the Federal Reserve, the FDIC and the OCC released a proposal to expand the eligible long-term debt ("eligible LTD") and clean holding company requirements under the existing total loss-absorbing capacity ("TLAC") rule to apply to non-GSIB banks with \$100 billion or more in total consolidated assets. The proposal would also reduce the amount of LTD with remaining maturities of less than two years that count towards a U.S. GSIB's TLAC requirement and expand the existing capital deduction framework for LTD issued by GSIBs to include LTD issued by non-GSIB banks subject to the LTD requirements.

Finalization of the above proposals, including the required implementation dates, is uncertain. The Firm continues to monitor developments and potential impacts.

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Risk-based Capital Regulatory Requirements

The following chart presents the Firm's Basel III CET1 capital ratio requirements under the Basel III rules currently in effect.



All banking institutions are currently required to have a minimum CET1 capital ratio of 4.5% of risk-weighted assets.

Certain banking organizations, including the Firm, are required to hold additional levels of capital to serve as a “capital conservation buffer”. The capital conservation buffer incorporates a GSIB surcharge, a discretionary countercyclical capital buffer and a fixed capital conservation buffer of 2.5% for Advanced regulatory capital requirements, as well as a variable SCB requirement, floored at 2.5%, for Standardized regulatory capital requirements.

Under the Federal Reserve's GSIB rule, the Firm is required to assess its GSIB surcharge on an annual basis under two separately prescribed methods based on data for the previous fiscal year-end, and is subject to the higher of the two. “Method 1” reflects the GSIB surcharge as prescribed by the Basel Committee's assessment methodology, and is calculated across five criteria: size, cross-jurisdictional activity, interconnectedness, complexity and substitutability. “Method 2” modifies the Method 1 requirements to include a measure of short-term wholesale funding in place of substitutability, and introduces a GSIB score “multiplication factor”.

The following table presents the Firm's effective GSIB surcharge for the years ended December 31, 2024 and 2023. For 2025, the Firm's effective regulatory minimum GSIB surcharge calculated under both

Method 1 and Method 2 remains unchanged at 2.5% and 4.5%, respectively.

	2024	2023
Method 1	2.5 %	2.5 %
Method 2	4.5 %	4.0 %

The U.S. federal regulatory capital standards include a framework for setting a discretionary countercyclical capital buffer taking into account the macro financial environment in which large, internationally active banks function. As of December 31, 2024, the U.S. countercyclical capital buffer remained at 0%. The Federal Reserve will continue to review the buffer at least annually. The buffer can be increased if the Federal Reserve, the FDIC and the OCC determine that systemic risks are meaningfully above normal and can be calibrated up to an additional 2.5% of RWA subject to a 12-month implementation period.

Failure to maintain regulatory capital equal to or in excess of the risk-based regulatory capital minimum plus the capital conservation buffer (inclusive of the GSIB surcharge) and any countercyclical buffer will result in limitations to the amount of capital that the Firm may distribute, such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

Total Loss-Absorbing Capacity

The Federal Reserve's TLAC rule requires the U.S. GSIB top-tier holding companies, including the Firm, to maintain minimum levels of external TLAC and eligible LTD. Refer to TLAC on page 106 for additional information.

Leverage-based Capital Regulatory Requirements Supplementary leverage ratio

Banking organizations subject to the Basel III Advanced approach are currently required to have a minimum SLR of 3.0%. Certain banking organizations, including the Firm, are also required to hold an additional 2.0% leverage buffer. The SLR is defined as Tier 1 capital under Basel III divided by the Firm's total leverage exposure. Total leverage exposure is calculated by taking the Firm's total average on-balance sheet assets, less amounts permitted to be deducted for Tier 1 capital, and adding certain off-balance sheet exposures, as defined in regulatory capital rules. Refer to SLR on page 104 for additional information.

Failure to maintain an SLR equal to or greater than the regulatory requirement will result in limitations on the amount of capital that the Firm may distribute such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

Other regulatory capital

In addition to meeting the capital ratio requirements of Basel III, the Firm and its principal IDI subsidiary, JPMorgan Chase Bank, N.A., must also maintain minimum capital and leverage ratios in order to be "well-capitalized" under the regulations issued by the Federal Reserve and the Prompt Corrective Action requirements of the FDIC Improvement Act, respectively. Refer to Note 27 for additional information.

Additional information regarding the Firm's capital ratios, as well as the U.S. federal regulatory capital standards to which the Firm is subject, is presented in Note 27. Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm's website, for further information on the Firm's current capital measures.

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Selected capital and RWA data

The following tables present the Firm's risk-based capital metrics under both the Basel III Standardized and Advanced approaches and leverage-based capital metrics. Refer to Note 27 for JPMorgan Chase Bank, N.A.'s risk-based and leverage-based capital metrics. First Republic Bank was not subject to Advanced approach regulatory capital requirements. As a result, for certain exposures associated with the First Republic acquisition, Advanced RWA and any impact on Advanced Total capital is calculated under the Standardized approach as permitted by the transition provisions in the U.S. capital rules. Refer to Note 34 for additional information on the First Republic acquisition.

(in millions, except ratios)	Standardized			Advanced		
	December 31, 2024	December 31, 2023	Capital ratio requirements ^(b)	December 31, 2024	December 31, 2023	Capital ratio requirements ^(b)
Risk-based capital metrics:^(a)						
CET1 capital	\$ 275,513	\$ 250,585		\$ 275,513	\$ 250,585	
Tier 1 capital	294,881	277,306		294,881	277,306	
Total capital	325,589	308,497		311,898 ^(c)	295,417 ^(c)	
Risk-weighted assets	1,757,460	1,671,995		1,740,429 ^(c)	1,669,156 ^(c)	
CET1 capital ratio	15.7 %	15.0 %	12.3 %	15.8 %	15.0 %	11.5 %
Tier 1 capital ratio	16.8	16.6	13.8	16.9	16.6	13.0
Total capital ratio	18.5	18.5	15.8	17.9	17.7	15.0

- (a) The capital metrics reflect the CECL capital transition provisions. As of December 31, 2024, CET1 capital reflected the remaining \$720 million CECL benefit and were fully phased in as of January 1, 2025; as of December 31, 2023, CET1 capital reflected a \$1.4 billion benefit. Refer to Note 27 for additional information.
- (b) Represents minimum requirements and regulatory buffers applicable to the Firm for the period ended December 31, 2024. For the period ended December 31, 2023, the Basel III Standardized CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.4%, 12.9%, and 14.9%, respectively; the Basel III Advanced CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.0%, 12.5%, and 14.5%, respectively. Refer to Note 27 for additional information.
- (c) Includes the impacts of certain assets associated with First Republic to which the Standardized approach has been applied as permitted by the transition provisions in the U.S. capital rules.

Three months ended (in millions, except ratios)	December 31, 2024		December 31, 2023	Capital ratio requirements ^(c)
Leverage-based capital metrics: ^(a)				
Adjusted average assets ^(b)	\$	4,070,499	\$	3,831,200
Tier 1 leverage ratio		7.2 %		7.2 %
Total leverage exposure	\$	4,837,568	\$	4,540,465
SLR		6.1 %		6.1 %
				4.0 %

- (a) The capital metrics reflect the CECL capital transition provisions. Refer to Note 27 for additional information.
- (b) Adjusted average assets, for purposes of calculating the leverage ratios, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, and other intangible assets.
- (c) Represents minimum requirements and regulatory buffers applicable to the Firm. Refer to Note 27 for additional information.

Capital components

The following table presents reconciliations of total stockholders' equity to Basel III CET1 capital, Tier 1 capital and Total capital as of December 31, 2024 and 2023.

(in millions)	December 31, 2024	December 31, 2023
Total stockholders' equity	\$ 344,758	\$ 327,878
Less: Preferred stock	20,050	27,404
Common stockholders' equity	324,708	300,474
Add:		
Certain deferred tax liabilities ^(a)	2,943	2,996
Other CET1 capital adjustments ^(b)	4,499	4,717
Less:		
Goodwill ^(c)	53,763	54,377
Other intangible assets	2,874	3,225
Standardized/Advanced CET1 capital	275,513	250,585
Add: Preferred stock	20,050	27,404
Less: Other Tier 1 adjustments	682	683
Standardized/Advanced Tier 1 capital	\$ 294,881	\$ 277,306
Long-term debt and other instruments qualifying as Tier 2 capital	\$ 10,312	\$ 11,779
Qualifying allowance for credit losses ^(d)	20,992	20,102
Other	(596)	(690)
Standardized Tier 2 capital	\$ 30,708	\$ 31,191
Standardized Total capital	\$ 325,589	\$ 308,497
Adjustment in qualifying allowance for credit losses for Advanced Tier 2 capital ^{(e)(f)}	(13,691)	(13,080)
Advanced Tier 2 capital	\$ 17,017	\$ 18,111
Advanced Total capital	\$ 311,898	\$ 295,417

- (a) Represents deferred tax liabilities related to tax-deductible goodwill and to identifiable intangibles created in nontaxable transactions, which are netted against goodwill and other intangibles when calculating CET1 capital.
- (b) As of December 31, 2024 and 2023, included a net benefit associated with cash flow hedges and debit valuation adjustments ("DVA") related to structured notes recorded in AOCI of \$5.2 billion and \$4.3 billion and the benefit from the CECL capital transition provisions of \$720 million and \$1.4 billion, respectively.
- (c) Goodwill deducted from capital includes goodwill associated with equity method investments in nonconsolidated financial institutions based on regulatory requirements. Refer to page 140 for additional information on principal investment risk.
- (d) Represents the allowance for credit losses eligible for inclusion in Tier 2 capital up to 1.25% of credit risk RWA, including the impact of the CECL capital transition provision with any excess deducted from RWA. Refer to Note 27 for additional information on the CECL capital transition.
- (e) Represents an adjustment to qualifying allowance for credit losses for the excess of eligible credit reserves over expected credit losses up to 0.6% of credit risk RWA, including the impact of the CECL capital transition provision with any excess deducted from RWA.
- (f) As of December 31, 2024 and 2023, included an incremental \$541 million and \$655 million allowance for credit losses, respectively, on certain assets associated with First Republic to which the Standardized approach has been applied, as permitted by the transition provisions in the U.S. capital rules.

Capital rollforward

The following table presents the changes in Basel III CET1 capital, Tier 1 capital and Tier 2 capital for the year ended December 31, 2024.

Year ended December 31, (in millions)	2024
Standardized/Advanced CET1 capital at December 31, 2023	\$ 250,585
Net income applicable to common equity	57,212
Dividends declared on common stock	(13,786)
Net purchase of treasury stock	(17,801)
Changes in additional paid-in capital	783
Changes related to AOCI applicable to capital:	
Unrealized gains/(losses) on investment securities	(87)
Translation adjustments, net of hedges ^(a)	(858)
Fair value hedges	(87)
Defined benefit pension and other postretirement employee benefit ("OPEB") plans	(63)
Changes related to other CET1 capital adjustments ^(b)	(385)
Change in Standardized/Advanced CET1 capital	24,928
Standardized/Advanced CET1 capital at December 31, 2024	\$ 275,513
Standardized/Advanced Tier 1 capital at December 31, 2023	\$ 277,306
Change in CET1 capital ^(b)	24,928
Net redemptions of noncumulative perpetual preferred stock	(7,354)
Other	1
Change in Standardized/Advanced Tier 1 capital	17,575
Standardized/Advanced Tier 1 capital at December 31, 2024	\$ 294,881
Standardized Tier 2 capital at December 31, 2023	\$ 31,191
Change in long-term debt and other instruments qualifying as Tier 2	(1,467)
Change in qualifying allowance for credit losses ^(b)	890
Other	94
Change in Standardized Tier 2 capital	(483)
Standardized Tier 2 capital at December 31, 2024	\$ 30,708
Standardized Total capital at December 31, 2024	\$ 325,589
Advanced Tier 2 capital at December 31, 2023	\$ 18,111
Change in long-term debt and other instruments qualifying as Tier 2	(1,467)
Change in qualifying allowance for credit losses ^{(b)(c)}	279
Other	94
Change in Advanced Tier 2 capital	(1,094)
Advanced Tier 2 capital at December 31, 2024	\$ 17,017
Advanced Total capital at December 31, 2024	\$ 311,898

- (a) Includes foreign currency translation adjustments and the impact of related derivatives.
- (b) Includes the impact of the CECL capital transition provisions and the cumulative effect of changes in accounting principles. Refer to Note 1 for additional information on changes in accounting principles and Note 27 for additional information on the CECL capital transition provisions.
- (c) As of December 31, 2024 and 2023, included an incremental \$541 million and \$655 million allowance for credit losses, respectively, on certain assets associated with First Republic to which the Standardized approach has been applied, as permitted by the transition provisions in the U.S. capital rules.

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RWA rollforward

The following table presents changes in the components of RWA under Basel III Standardized and Advanced approaches for the year ended December 31, 2024. The amounts in the rollforward categories are estimates, based on the predominant driver of the change.

Year ended December 31, 2024 (in millions)	Standardized			Advanced			
	Credit risk RWA ^(c)	Market risk RWA	Total RWA	Credit risk RWA ^{(c)(d)}	Market risk RWA	Operational risk RWA	Total RWA
December 31, 2023	\$ 1,603,851	\$ 68,144	\$ 1,671,995	\$ 1,155,261	\$ 68,603	\$ 445,292	\$ 1,669,156
Model & data changes ^(a)	4,743	(366)	4,377	4,811	(366)	—	4,445
Movement in portfolio levels ^(b)	64,169	16,919	81,088	57,933	16,895	(8,000)	66,828
Changes in RWA	68,912	16,553	85,465	62,744	16,529	(8,000)	71,273
December 31, 2024	\$ 1,672,763	\$ 84,697	\$ 1,757,460	\$ 1,218,005	\$ 85,132	\$ 437,292	\$ 1,740,429

- (a) Model & data changes refer to material movements in levels of RWA as a result of revised methodologies and/or treatment per regulatory guidance (exclusive of rule changes).
- (b) Movement in portfolio levels (inclusive of rule changes) refers to: for Credit risk RWA, changes in book size, changes in composition and credit quality, market movements, and deductions for excess eligible allowances for credit losses not eligible for inclusion in Tier 2 capital; for Market risk RWA, changes in position and market movements; and for Operational risk RWA, updates to cumulative losses and macroeconomic model inputs.
- (c) As of December 31, 2024 and 2023, the Basel III Standardized Credit risk RWA included wholesale and retail off balance-sheet RWA of \$208.0 billion and \$208.5 billion, respectively; and the Basel III Advanced Credit risk RWA included wholesale and retail off balance-sheet RWA of \$192.1 billion and \$188.5 billion, respectively.
- (d) As of December 31, 2024 and 2023, Credit risk RWA reflected approximately \$43.3 billion and \$52.4 billion, respectively, of RWA calculated under the Standardized approach for certain assets associated with First Republic as permitted by the transition provisions in the U.S. capital rules.

Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm's website, for further information on Credit risk RWA, Market risk RWA and Operational risk RWA.

Supplementary leverage ratio

The following table presents the components of the Firm's SLR.

Three months ended (in millions, except ratio)	December 31, 2024	December 31, 2023
Tier 1 capital	\$ 294,881	\$ 277,306
Total average assets	4,125,167	3,885,632
Less: Regulatory capital adjustments ^(a)	54,668	54,432
Total adjusted average assets ^(b)	4,070,499	3,831,200
Add: Off-balance sheet exposures ^(c)	767,069	709,265
Total leverage exposure	\$ 4,837,568	\$ 4,540,465
SLR	6.1 %	6.1 %

- (a) For purposes of calculating the SLR, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, other intangible assets and adjustments for the CECL capital transition provisions. Refer to Note 27 for additional information on the CECL capital transition.
- (b) Adjusted average assets used for the calculation of Tier 1 leverage ratio.
- (c) Off-balance sheet exposures are calculated as the average of the three month-end spot balances on applicable regulatory exposures during the reporting quarter. Refer to the Firm's Pillar 3 Regulatory Capital Disclosures reports for additional information.

Line of business and Corporate equity

Each LOB and Corporate is allocated capital by taking into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. ROE is measured and internal targets for expected returns are established as key measures of an LOB's performance.

The Firm's current equity allocation methodology incorporates Basel III Standardized RWA and the GSIB surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change. As of January 1, 2025, changes to the Firm's capital allocations are primarily a result of updates to the Firm's current capital requirements and changes in RWA for each LOB under rules currently in effect. Any capital that the Firm has accumulated in excess of these current requirements, including the capital required to meet the potential increased requirements of the U.S. Basel III proposal, has been retained in Corporate in addition to its allocated balance.

The following table presents the capital allocated to each LOB and Corporate.

(in billions)	January 1, 2025	December 31,	
		2024	2023
Consumer & Community Banking	\$ 56.0	\$ 54.5	\$ 55.5
Commercial & Investment Bank	149.5	132.0	138.0
Asset & Wealth Management	16.0	15.5	17.0
Corporate	103.2	122.7	90.0
Total common stockholders' equity	\$ 324.7	\$ 324.7	\$ 300.5

Capital actions

Common stock dividends

The Firm's common stock dividends are planned as part of the Capital Management governance framework in line with the Firm's capital management objectives.

On December 9, 2024, the Firm announced that its Board of Directors had declared a quarterly common stock dividend of \$1.25 per share, payable on January 31, 2025. The Firm's dividends are subject to approval by the Board of Directors on a quarterly basis.

Refer to Note 21 and Note 26 for information regarding dividend restrictions.

The following table shows the common dividend payout ratio based on net income applicable to common equity.

Year ended December 31,	2024	2023	2022
Common dividend payout ratio	24 %	25 %	33 %

Common stock

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new \$30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to \$30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

The following table sets forth the Firm's repurchases of common stock for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024	2023	2022 ^(b)
Total number of shares of common stock repurchased	91.7	69.5	23.1
Aggregate purchase price of common stock repurchases ^(a)	\$ 18,841	\$ 9,898	\$ 3,122

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.

(b) In the second half of 2022, the Firm temporarily suspended share repurchases, which it resumed in the first quarter of 2023 under its common share repurchase program.

The Board of Directors' authorization to repurchase common shares is utilized at management's discretion. The \$30 billion common share repurchase program approved by the Board of Directors does not establish specific price targets or timetables.

Management determines the amount and timing of common share repurchases based on various factors, including market conditions; legal and regulatory considerations affecting the amount and timing of repurchase activity; the Firm's capital position (taking into account goodwill and intangibles); organic capital generation; current and proposed future capital requirements; and other investment opportunities. The amount of common shares that the Firm repurchases in any period may be substantially more or less than the amounts estimated or actually repurchased in prior periods, reflecting the dynamic nature of the decision-making process. The Firm's common share repurchases may be suspended by management at any time; and may be executed through open market purchases or privately negotiated transactions, or utilizing Rule 10b5-1 plans, which are written trading plans that the Firm may enter into from time to time under Rule 10b5-1 of the Securities Exchange Act of 1934 and which allow the Firm to repurchase its common shares during periods when it may otherwise not be repurchasing common shares — for example, during internal trading blackout periods.

Refer to capital planning and stress testing on pages 97–98 for additional information.

Refer to Part II, Item 5: Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities on page 39 of this 2024 Form 10-K for additional information regarding repurchases of the Firm's equity securities.

Preferred stock

Preferred stock dividends were \$1.3 billion, \$1.5 billion, and \$1.6 billion for the years ended December 31, 2024, 2023, and 2022, respectively.

During the year ended and subsequent to December 31, 2024, the Firm issued and redeemed certain series of non-cumulative preferred stock. Refer to Note 21 for additional information on the Firm's preferred stock, including the issuance and redemption of preferred stock.

Subordinated Debt

Refer to Long-term funding on page 114 and Note 20 for additional information on the Firm's subordinated debt.

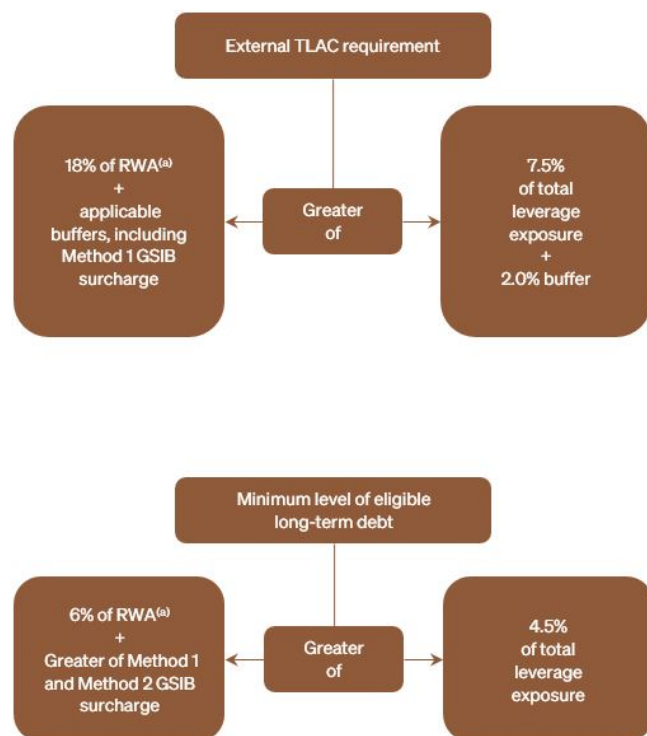
Management's discussion and analysis

Other capital requirements

Total Loss-Absorbing Capacity

The Federal Reserve's TLAC rule requires the U.S. GSIB top-tier holding companies, including the Firm, to maintain minimum levels of external TLAC and eligible long-term debt.

The external TLAC requirements and the minimum level of eligible long-term debt requirements are shown below:



(a) RWA is the greater of Standardized and Advanced compared to their respective regulatory capital ratio requirements.

Failure to maintain TLAC equal to or in excess of the regulatory minimum plus applicable buffers will result in limitations on the amount of capital that the Firm may distribute, such as through dividends and common share repurchases, as well as on discretionary bonus payments for certain executive officers.

The following table presents the eligible external TLAC and eligible LTD amounts, as well as a representation of these amounts as a percentage of the Firm's total RWA and total leverage exposure applying the impact of the CECL capital transition provisions as of December 31, 2024 and 2023.

	December 31, 2024		December 31, 2023	
(in billions, except ratio)	External TLAC	LTD	External TLAC	LTD
Total eligible amount	\$ 546.6	\$ 236.8	\$ 513.8	\$ 222.6
% of RWA	31.1 %	13.5 %	30.7 %	13.3 %
Regulatory requirements	23.0	10.5	23.0	10.0
Surplus/(shortfall)	\$ 142.3	\$ 52.3	\$ 129.2	\$ 55.4
% of total leverage exposure	11.3 %	4.9 %	11.3 %	4.9 %
Regulatory requirements	9.5	4.5	9.5	4.5
Surplus/(shortfall)	\$ 87.0	\$ 19.2	\$ 82.5	\$ 18.3

Effective January 1, 2024, the Firm's regulatory requirement for its eligible LTD to RWA ratio increased by 50 bps to 10.5%, due to the increase in the Firm's GSIB Method 2 requirements. The Firm's regulatory requirement for its TLAC to RWA ratio remained at 23.0%. Refer to Risk-based Capital Regulatory Requirements on pages 100–101 for further information on the GSIB surcharge.

Refer to Liquidity Risk Management on pages 108–115 for further information on long-term debt issued by the Parent Company.

Refer to Part I, Item 1A: Risk Factors on pages 10–37 of this 2024 Form 10-K for information on the financial consequences to holders of the Firm's debt and equity securities in a resolution scenario.

U.S. broker-dealer regulatory capital

J.P. Morgan Securities

JPMorganChase's principal U.S. broker-dealer subsidiary is J.P. Morgan Securities. J.P. Morgan Securities is subject to the regulatory capital requirements of Rule 15c3-1 under the Securities Exchange Act of 1934 (the "Net Capital Rule"). J.P. Morgan Securities is also registered as a futures commission merchant and is subject to regulatory capital requirements, including those imposed by the SEC, the Commodity Futures Trading Commission ("CFTC"), the Financial Industry Regulatory Authority ("FINRA") and the National Futures Association ("NFA").

J.P. Morgan Securities has elected to compute its minimum net capital requirements in accordance with the "Alternative Net Capital Requirements" of the Net Capital Rule.

The following table presents J.P. Morgan Securities' net capital.

December 31, 2024		
(in millions)	Actual	Minimum
Net Capital	\$ 24,980	\$ 5,999

J.P. Morgan Securities is registered with the SEC as a security-based swap dealer and with the CFTC as a swap dealer. As a result of additional SEC and CFTC capital and financial reporting requirements for security-based swap dealers and swap dealers, J.P. Morgan Securities is subject to alternative minimum net capital requirements and required to hold "tentative net capital" in excess of \$5.0 billion. J.P. Morgan Securities is also required to notify the SEC and CFTC in the event that its tentative net capital is less than \$6.0 billion. Tentative net capital is net capital before deducting market and credit risk charges as defined by the Net Capital Rule. As of December 31, 2024, J.P. Morgan Securities maintained tentative net capital in excess of the minimum and notification requirements.

Non-U.S. subsidiary regulatory capital

J.P. Morgan Securities plc

J.P. Morgan Securities plc is a wholly-owned subsidiary of JPMorgan Chase Bank, N.A. and has authority to engage in banking, investment banking and broker-dealer activities. J.P. Morgan Securities plc is jointly regulated in the U.K. by the Prudential Regulation Authority ("PRA") and the Financial Conduct Authority ("FCA"). J.P. Morgan Securities plc is subject to the Capital Requirements Regulation ("CRR"), as adopted and amended in the U.K., and the capital rules in the PRA Rulebook. These requirements collectively represent the U.K.'s implementation of the Basel III standards. The PRA announced that it intends to delay the U.K.'s implementation of the final Basel III

standards until January 1, 2027, with a three-year transitional period for certain aspects.

The Bank of England requires that U.K. banks, including U.K. regulated subsidiaries of overseas groups, maintain minimum requirements for own funds and eligible liabilities ("MREL"). As of December 31, 2024, J.P. Morgan Securities plc was compliant with its MREL requirements.

The following table presents J.P. Morgan Securities plc's risk-based and leverage-based capital metrics.

December 31, 2024		
(in millions, except ratios)	Actual	Regulatory Minimum ratios ^(a)
Total capital	\$ 53,120	
CET1 capital ratio	17.0 %	4.5 %
Tier 1 capital ratio	22.1	6.0
Total capital ratio	27.1	8.0
Tier 1 leverage ratio	7.1	3.3 ^(b)

(a) Represents minimum Pillar 1 requirements specified by the PRA. J.P. Morgan Securities plc's capital ratios as of December 31, 2024 exceeded the minimum requirements, including the additional capital requirements specified by the PRA.

(b) At least 75% of the Tier 1 leverage ratio minimum must be met with CET1 capital.

J.P. Morgan SE

JPMSE is a wholly-owned subsidiary of JPMorgan Chase Bank, N.A. and has authority to engage in banking, investment banking and markets activities. JPMSE is regulated by the European Central Bank ("ECB"), the German Financial Supervisory Authority and the German Central Bank, as well as the local regulators in each of the countries in which it operates, and it is subject to EU capital requirements under Basel III. JPMSE is subject to the EU implementation of the final Basel III standards. Those standards became effective beginning on January 1, 2025, with the exception of market risk aspects for which the effective date is January 1, 2026.

JPMSE is required by the EU Single Resolution Board to maintain MREL. As of December 31, 2024, JPMSE was compliant with its MREL requirements.

The following table presents JPMSE's risk-based and leverage-based capital metrics.

December 31, 2024		
(in millions, except ratios)	Actual	Regulatory Minimum ratios ^(a)
Total capital	\$ 43,298	
CET1 capital ratio	20.0 %	4.5 %
Tier 1 capital ratio	20.0	6.0
Total capital ratio	34.8	8.0
Tier 1 leverage ratio	6.1	3.0

(a) Represents minimum Pillar 1 requirements specified by the EU CRR. J.P. Morgan SE's capital and leverage ratios as of December 31, 2024 exceeded the minimum requirements, including the additional capital requirements specified by EU regulators.

LIQUIDITY RISK MANAGEMENT

Liquidity risk is the risk that the Firm will be unable to meet its cash and collateral needs as they arise or that it does not have the appropriate amount, composition and tenor of funding and liquidity to support its assets and liabilities.

Liquidity risk management

The Firm has a Liquidity Risk Management ("LRM") function whose primary objective is to provide independent oversight of liquidity risk across the Firm. Liquidity Risk Management's responsibilities include:

- Defining, monitoring and reporting liquidity risk metrics;
- Independently establishing and monitoring limits and indicators, including liquidity risk appetite;
- Developing a process to classify, monitor and report limit breaches;
- Performing an independent review of liquidity risk management processes to evaluate their adequacy and effectiveness;
- Monitoring and reporting internal Firmwide and legal entity liquidity stress tests, regulatory defined metrics, as well as liquidity positions, balance sheet variances and funding activities; and
- Approving or escalating for review new or updated liquidity stress assumptions.

Liquidity management

Treasury and CIO is responsible for liquidity management.

The primary objectives of the Firm's liquidity management are to:

- Ensure that the Firm's core businesses and material legal entities are able to operate in support of client needs and meet contractual and contingent financial obligations through normal economic cycles as well as during stress events, and
- Manage an optimal funding mix and availability of liquidity sources.

The Firm addresses these objectives through:

- Analyzing and understanding the liquidity characteristics of the assets and liabilities of the Firm, LOBs, legal entities, as well as currencies, taking into account legal, regulatory, and operational restrictions;
- Developing and maintaining internal liquidity stress testing assumptions;
- Defining and monitoring Firmwide and legal entity-specific liquidity strategies, policies, reporting and contingency funding plans;
- Managing liquidity within the Firm's approved limits and indicators, including liquidity risk appetite tolerances;

- Managing compliance with regulatory requirements related to funding and liquidity risk; and
- Setting FTP in accordance with underlying liquidity characteristics of balance sheet assets and liabilities as well as certain off-balance sheet items.

As part of the Firm's overall liquidity management strategy, the Firm manages liquidity and funding using a centralized, global approach designed to:

- Optimize liquidity sources and uses;
- Monitor exposures;
- Identify constraints on the transfer of liquidity between the Firm's legal entities; and
- Maintain the appropriate amount of surplus liquidity at a Firmwide and legal entity level, where relevant.

Governance

Committees responsible for liquidity governance include the Firmwide ALCO, as well as regional ALCOs, the Treasurer Committee, and the CTC Risk Committee. In addition, the Board Risk Committee reviews and recommends to the Board of Directors, for approval, the Firm's liquidity risk tolerances, liquidity strategy, and liquidity policy. Refer to Firmwide Risk Management on pages 91–95 for further discussion of ALCO and other risk-related committees.

Internal stress testing

The Firm conducts internal liquidity stress testing to monitor liquidity positions at the Firm and its material legal entities under a variety of adverse scenarios, including scenarios analyzed as part of the Firm's resolution and recovery planning. Internal stress tests are produced on a daily basis, and other stress tests are performed in response to specific market events or concerns. Liquidity stress tests assume all of the Firm's contractual financial obligations are met and take into consideration:

- Varying levels of access to unsecured and secured funding markets;
- Estimated non-contractual and contingent cash outflows;
- Credit rating downgrades;
- Collateral haircuts; and
- Potential impediments to the availability and transferability of liquidity between jurisdictions and material legal entities such as regulatory, legal or other restrictions.

Liquidity outflows are modeled across a range of time horizons and currency dimensions and contemplate both market and idiosyncratic stresses.

Results of stress tests are considered in the formulation of the Firm's funding plan and assessment of its liquidity position. The Parent Company acts as a source of funding for the Firm through equity and

long-term debt issuances, and its intermediate holding company, JPMorgan Chase Holdings LLC (the “IHC”), provides funding to support the ongoing operations of the Parent Company and its subsidiaries. The Firm manages liquidity at the Parent Company, the IHC, and operating subsidiaries at levels sufficient to comply with liquidity risk tolerances and minimum liquidity requirements, and to manage through periods of stress when access to normal funding sources may be disrupted.

Contingency funding plan

The Firm’s Contingency Funding Plan (“CFP”) sets out the strategies for addressing and managing liquidity resource needs during a liquidity stress event and incorporates liquidity risk limits, indicators and risk appetite tolerances. The CFP also identifies the alternative contingent funding and liquidity resources available to the Firm and its legal entities in a period of stress.

LCR and HQLA

The LCR rule requires that the Firm and JPMorgan Chase Bank, N.A. maintain an amount of eligible HQLA that is sufficient to meet their respective estimated total net cash outflows over a prospective 30 calendar-day period of significant stress. Eligible HQLA, for purposes of calculating the LCR, is the amount of unencumbered HQLA that satisfy certain operational considerations as defined in the LCR rule. HQLA primarily consist of cash and certain high-quality liquid securities as defined in the LCR rule.

Under the LCR rule, the amount of eligible HQLA held by JPMorgan Chase Bank, N.A. that is in excess of its stand-alone 100% minimum LCR requirement, and that is not transferable to non-bank affiliates, must be excluded from the Firm’s reported eligible HQLA.

Estimated net cash outflows are based on standardized stress outflow and inflow rates prescribed in the LCR rule, which are applied to the balances of the Firm’s assets, sources of funds, and obligations. The LCR for both the Firm and JPMorgan Chase Bank, N.A. is required to be a minimum of 100%.

The following table summarizes the Firm and JPMorgan Chase Bank, N.A.’s average LCR for the three months ended December 31, 2024, September 30, 2024 and December 31, 2023 based on the Firm’s interpretation of the LCR framework.

Average amount (in millions)	Three months ended		
	December 31, 2024	September 30, 2024	December 31, 2023
JPMorgan Chase & Co.:			
HQLA			
Eligible cash ^(a)	\$ 396,123	\$ 412,389	\$ 485,263
Eligible securities ^{(b)(c)}	464,877	453,899	313,365
Total HQLA^(d)	\$ 861,000	\$ 866,288	\$ 798,628
Net cash outflows	\$ 763,648	\$ 762,072	\$ 704,857
LCR	113 %	114 %	113 %
Net excess eligible HQLA^(d)	\$ 97,352	\$ 104,216	\$ 93,771
JPMorgan Chase Bank, N.A.:			
LCR	124 %	121 %	129 %
Net excess eligible HQLA	\$ 193,682	\$ 168,137	\$ 215,190

- (a) Represents cash on deposit at central banks, primarily the Federal Reserve Banks.
- (b) Eligible HQLA securities may be reported in securities borrowed or purchased under resale agreements, trading assets, or investment securities on the Firm’s Consolidated balance sheets. For purposes of calculating the LCR, HQLA securities are included at fair value, which may differ from the accounting treatment under U.S. GAAP.
- (c) Predominantly U.S. Treasuries, U.S. GSE and government agency MBS, and sovereign bonds net of regulatory haircuts under the LCR rule.
- (d) Excludes average excess eligible HQLA at JPMorgan Chase Bank, N.A. that are not transferable to non-bank affiliates.

JPMorgan Chase Bank, N.A.’s average LCR increased during the three months ended December 31, 2024, compared with the three months ended September 30, 2024, driven by activities in CIB Markets, partially offset by lower market values of HQLA-eligible investment securities and funding maturities.

JPMorgan Chase Bank, N.A.’s average LCR for the three months ended December 31, 2024 decreased compared with the three months ended December 31, 2023, driven by dividend payments to the Parent Company and lending activity, largely offset by higher market values of HQLA-eligible investment securities, a reduction in unencumbered non-HQLA AFS securities, activities in CIB Markets, and long-term debt issuances.

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Each of the Firm and JPMorgan Chase Bank, N.A.'s average LCR may fluctuate from period to period due to changes in their respective eligible HQLA and estimated net cash outflows as a result of ongoing business activity and from the impacts of Federal Reserve actions as well as other factors. Refer to the Firm's U.S. LCR Disclosure reports, which are available on the Firm's website, for a further discussion of the Firm's LCR.

Liquidity sources

In addition to the assets reported in the Firm's eligible HQLA discussed above, the Firm had unencumbered marketable securities, such as equity and debt securities, that the Firm believes would be available to raise liquidity. This includes excess eligible HQLA securities at JPMorgan Chase Bank, N.A. that are not transferable to non-bank affiliates. The fair value of these securities was approximately \$594 billion and \$649 billion as of December 31, 2024 and 2023, respectively, although the amount of liquidity that could be raised at any particular time would be dependent on prevailing market conditions. The decrease compared to December 31, 2023, was driven by reductions in unencumbered AFS securities in Treasury and CIO, excess eligible HQLA securities at JPMorgan Chase Bank, N.A., and unencumbered CIB trading assets.

The Firm had approximately \$1.4 trillion of available cash and securities as of both December 31, 2024 and 2023. For each respective period, the amount was comprised of eligible end-of-period HQLA, excluding the impact of regulatory haircuts, of approximately \$834 billion and \$798 billion, and unencumbered marketable securities with a fair value of approximately \$594 billion and \$649 billion.

The Firm also had available borrowing capacity at the Federal Home Loan Banks ("FHLBs") and the discount window at the Federal Reserve Banks as a result of collateral pledged by the Firm to such banks of approximately \$413 billion and \$340 billion as of December 31, 2024 and 2023, respectively. This borrowing capacity excludes the benefit of cash and securities reported in the Firm's eligible HQLA or other unencumbered securities that are currently pledged at the Federal Reserve Banks discount window and other central banks. Available borrowing capacity increased from December 31, 2023 primarily due to a higher amount of commercial loans and credit card receivables pledged at the Federal Reserve Banks. Although available, the Firm does not view this borrowing capacity at the Federal Reserve Banks discount window and the other central banks as a primary source of liquidity.

NSFR

The net stable funding ratio ("NSFR") is a liquidity requirement for large banking organizations that is intended to measure the adequacy of "available" stable funding that is sufficient to meet their "required" amounts of stable funding over a one-year horizon.

For the three months ended December 31, 2024, both the Firm and JPMorgan Chase Bank, N.A. were compliant with the 100% minimum NSFR requirement, based on the Firm's interpretation of the final NSFR rule. Refer to the Firm's U.S. NSFR Disclosure report on the Firm's website for additional information.

Funding

Sources of funds

Management believes that the Firm's unsecured and secured funding capacity is sufficient to meet its on- and off-balance sheet obligations, which includes both short- and long-term cash requirements.

The Firm funds its global balance sheet through diverse sources of funding including deposits, secured and unsecured funding in the capital markets and stockholders' equity. Deposits are the primary funding source for JPMorgan Chase Bank, N.A. Additionally, JPMorgan Chase Bank, N.A. may access funding through short- or long-term secured borrowings, the issuance of unsecured long-term debt, or from

borrowings from the IHC. The Firm's non-bank subsidiaries are primarily funded from long-term unsecured borrowings and short-term secured borrowings which are primarily securities loaned or sold under repurchase agreements. Excess funding is invested by Treasury and CIO in the Firm's investment securities portfolio or deployed in cash or other short-term liquid investments based on their interest rate and liquidity risk characteristics.

Refer to Note 28 for additional information on off-balance sheet obligations.

Deposits

The table below summarizes, by LOB and Corporate, the period-end and average deposit balances as of and for the years ended December 31, 2024 and 2023.

As of or for the year ended December 31, (in millions)			Average	
	2024	2023	2024	2023
Consumer & Community Banking ^(a)	\$ 1,056,652	\$ 1,094,738	\$ 1,064,215	\$ 1,126,552
Commercial & Investment Bank ^(a)	1,073,512	1,050,892	1,061,488	996,295
Asset & Wealth Management ^(a)	248,287	233,232	235,146	216,178
Corporate	27,581	21,826	25,793	20,042
Total Firm	\$ 2,406,032	\$ 2,400,688	\$ 2,386,642	\$ 2,359,067

(a) In the fourth quarter of 2023, CCB transferred certain deposits associated with First Republic to AWM and CIB.

The Firm believes that deposits provide a stable source of funding and reduce the Firm's reliance on the wholesale funding markets. A significant portion of the Firm's deposits are consumer deposits and wholesale operating deposits, which are both considered to be stable sources of liquidity. Wholesale operating deposits are generally considered to be stable sources of liquidity because they are generated from customers that maintain operating service relationships with the Firm.

The Firm believes that average deposit balances are generally more representative of deposit trends than period-end deposit balances. However, during periods of market disruption, average deposit trends may be impacted.

The following discussion excludes the impact of the transfer of certain First Republic deposits in the fourth quarter of 2023 from CCB to the other LOBs as the transfers had no net impact on Firmwide deposits.

Average deposits increased for the year ended December 31, 2024 compared to the year ended December 31, 2023, reflecting:

- an increase in CIB due to net inflows predominantly in Payments and net issuances of structured notes as a result of client demand in Markets, partially offset by deposit attrition, which included actions taken to reduce certain deposits,
- the timing impact of First Republic,

- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending, largely offset by new accounts.

Period-end deposits increased from December 31, 2023, reflecting:

- an increase in CIB due to net inflows predominantly in Payments, largely offset by net maturities of structured notes in Markets,
- an increase in AWM as a result of growth in balances in new and existing client accounts, reflecting the impact of higher-yielding product offerings, largely offset by continued migration into other investments, and
- a decline in CCB primarily driven by a decrease in balances in existing accounts due to increased customer spending and migration into higher-yielding investments, predominantly offset by new accounts.

Refer to the Firm's Consolidated Balance Sheets Analysis and the Business Segment & Corporate Results on pages 63–65 and pages 70–90, respectively, for further information on deposit and liability balance trends, as well as Executive Overview

Management's discussion and analysis

on pages 54–58 and Note 34 for additional information on the First Republic acquisition. Refer to Note 3 for further information on structured notes.

Certain deposits are covered by insurance protection that provides additional funding stability and results in a benefit to the LCR. Deposit insurance protection may be available to depositors in the countries in which the deposits are placed. For example, the FDIC provides deposit insurance protection for deposits placed in a U.S. depository institution. At December 31, 2024 and 2023^(a), the Firmwide estimated uninsured deposits were \$1,414.0 billion and \$1,347.8 billion, respectively, primarily reflecting wholesale operating deposits.

Total uninsured deposits include time deposits. The table below presents an estimate of uninsured U.S. and non-U.S. time deposits, and their remaining maturities. The Firm's estimates of its uninsured U.S. time deposits are based on data that the Firm calculates periodically under applicable FDIC regulations. For purposes of this presentation, all non-U.S. time deposits are deemed to be uninsured.

(in millions)	December 31, 2024		December 31, 2023	
	U.S.	Non-U.S.	U.S.	Non-U.S.
Three months or less	\$ 119,333	\$ 77,253	\$ 98,606 ^(a)	\$ 77,466
Over three months but within 6 months	11,040	12,229	17,736	5,358
Over six months but within 12 months	7,056	1,542	10,294	4,820
Over 12 months	823	1,924	710	2,543
Total	\$138,252	\$ 92,948	\$127,346 ^(a)	\$ 90,187

(a) Prior-period amounts have been revised to include cash collateral for certain derivatives to align with a change in the methodology for calculating uninsured U.S. time deposits.

The table below shows the loan and deposit balances, the loans-to-deposits ratios, and deposits as a percentage of total liabilities, as of December 31, 2024 and 2023.

As of December 31, (in billions except ratios)	2024	2023
Deposits	\$ 2,406.0	\$ 2,400.7
Deposits as a % of total liabilities	66 %	68 %
Loans	\$ 1,348.0	\$ 1,323.7
Loans-to-deposits ratio	56 %	55 %

The following table provides a summary of the average balances and average interest rates of JPMorganChase's deposits for the years ended December 31, 2024, 2023, and 2022.

Year ended December 31, (in millions, except interest rates)	Average balances			Average interest rates		
	2024	2023	2022	2024	2023	2022
U.S. offices						
Noninterest-bearing	\$ 611,734	\$ 635,791	\$ 691,206	NA	NA	NA
Interest-bearing						
Demand ^(a)	282,533	279,725	324,512	3.90 %	3.50 %	0.92 %
Savings ^(b)	800,964	864,558	971,788	1.39	1.10	0.28
Time	223,503	145,827	62,022	4.93	4.74	2.07
Total interest-bearing deposits	1,307,000	1,290,110	1,358,322	2.54	2.03	0.52
Total deposits in U.S. offices	1,918,734	1,925,901	2,049,528	1.73	1.36	0.34
Non-U.S. offices						
Noninterest-bearing	26,858	24,747	28,043	NA	NA	NA
Interest-bearing						
Demand	346,179	321,976	324,740	3.13	2.71	0.57
Time	94,871	86,443	65,604	5.86	5.82	1.85
Total interest-bearing deposits	441,050	408,419	390,344	3.72	3.37	0.78
Total deposits in non-U.S. offices	467,908	433,166	418,387	3.50	3.18	0.73
Total deposits	\$ 2,386,642	\$ 2,359,067	\$ 2,467,915	2.08 %	1.70 %	0.41 %

(a) Includes Negotiable Order of Withdrawal accounts, and certain trust accounts.

(b) Includes Money Market Deposit Accounts.

Refer to Note 17 for additional information on deposits.

The following table summarizes short-term and long-term funding, excluding deposits, as of December 31, 2024 and 2023, and average balances for the years ended December 31, 2024 and 2023. Refer to the Consolidated Balance Sheets Analysis on pages 63–65 and Note 11 for additional information.

Sources of funds (excluding deposits)

As of or for the year ended December 31, (in millions)			Average	
	2024	2023	2024	2023
Commercial paper	\$ 14,932	\$ 14,737	\$ 11,398	\$ 12,675
Other borrowed funds	13,018	8,200	12,040	9,712
Federal funds purchased	567	787	1,547	1,754
Total short-term unsecured funding	\$ 28,517	\$ 23,724	\$ 24,985	\$ 24,141
Securities sold under agreements to repurchase ^(a)	\$ 291,500	\$ 212,804	\$ 357,144	\$ 249,661
Securities loaned ^(a)	4,768	2,944	5,129	4,671
Other borrowed funds	24,943	21,775	25,504	22,010
Obligations of Firm-administered multi-seller conduits ^(b)	18,228	17,781	18,620	14,918
Total short-term secured funding	\$ 339,439	\$ 255,304	\$ 406,397	\$ 291,260
Senior notes	\$ 203,639	\$ 191,202	\$ 199,908	\$ 181,803
Subordinated debt	16,060	19,708	18,614	20,374
Structured notes ^(c)	98,792	86,056	93,483	76,574
Total long-term unsecured funding	\$ 318,491	\$ 296,966	\$ 312,005	\$ 278,751
Credit card securitization ^(b)	\$ 5,312	\$ 2,998	\$ 5,138	\$ 1,634
FHLB advances	29,257	41,246	35,040 ^(g)	28,865
Purchase Money Note ^(d)	49,207	\$ 48,989	\$ 49,090	\$ 32,829
Other long-term secured funding ^(e)	4,463	4,624	4,676	4,513
Total long-term secured funding	\$ 88,239	\$ 97,857	\$ 93,944	\$ 67,841
Preferred stock^(f)	\$ 20,050	\$ 27,404	\$ 24,054	\$ 27,404
Common stockholders' equity^(f)	\$ 324,708	\$ 300,474	\$ 312,370	\$ 282,056

(a) Primarily consists of short-term securities loaned or sold under agreements to repurchase.

(b) Included in beneficial interests issued by consolidated variable interest entities on the Firm's Consolidated balance sheets.

(c) Includes certain TLAC-eligible long-term unsecured debt issued by the Parent Company.

(d) Reflects the Purchase Money Note associated with the First Republic acquisition on May 1, 2023. Refer to Note 34 for additional information.

(e) Includes long-term structured notes that are secured.

(f) Refer to Capital Risk Management on pages 97–107, Consolidated statements of changes in stockholders' equity on page 175, Note 21 and Note 22 for additional information on preferred stock and common stockholders' equity.

(g) Includes the timing impact of First Republic. Refer to the Executive Overview on pages 54–58 and Note 34 for additional information.

Short-term funding

The Firm's primary source of short-term secured funding is securities sold under agreements to repurchase. These instruments are secured predominantly by high-quality securities collateral, including government-issued debt and U.S. GSE and government agency MBS. Securities sold under agreements to repurchase increased at December 31, 2024, compared with December 31, 2023, driven by Markets, reflecting higher client-driven market-making activities and higher secured financing of trading assets.

The increase in secured other borrowed funds at December 31, 2024 from December 31, 2023, as well as the increase for the average year ended December 31, 2024, compared to the prior year period, were both due to higher financing requirements in Markets, partially offset by FHLB maturities in Treasury and CIO.

The balances associated with securities loaned or sold under agreements to repurchase fluctuate over time due to investment and financing activities of clients,

the Firm's demand for financing, the ongoing management of the mix of the Firm's liabilities, including its secured and unsecured financing (for both the investment securities and market-making portfolios), and other market and portfolio factors.

The Firm's primary sources of short-term unsecured funding consist of issuances of wholesale commercial paper and other borrowed funds.

The decrease in average commercial paper for the year ended December 31, 2024 compared to the prior year period was due to lower issuances primarily as a result of short-term liquidity management.

The increase in unsecured other borrowed funds at December 31, 2024 from December 31, 2023, was predominantly driven by net issuances of structured notes in Markets.

Management's discussion and analysis

Long-term funding

Long-term funding provides an additional source of stable funding and liquidity for the Firm. The Firm's long-term funding plan is driven primarily by expected client activity, liquidity considerations and regulatory requirements, including TLAC. Long-term funding objectives include maintaining diversification, maximizing market access and optimizing funding costs. The Firm evaluates various funding markets, tenors and currencies in creating its optimal long-term funding plan.

Unsecured funding and issuance

The significant majority of the Firm's total outstanding long-term debt has been issued by the Parent Company to provide flexibility in support of the funding needs of both bank and non-bank subsidiaries. The Parent Company advances substantially all net funding proceeds to its subsidiary, the IHC. The IHC does not issue debt to external counterparties. The increase in structured notes at December 31, 2024 from December 31, 2023, and for the average year ended December 31, 2024, compared to the prior year period, was primarily driven by net issuances of structured notes in Markets due to client demand.

The following table summarizes long-term unsecured issuance and maturities or redemptions for the years ended December 31, 2024 and 2023. Refer to Note 20 for additional information on the IHC and long-term debt.

Long-term unsecured funding

Year ended December 31, (Notional in millions)	2024		2023	
	Parent Company		Subsidiaries	
Issuance				
Senior notes issued in the U.S. market	\$ 37,000	\$ 14,256	\$ —	\$ 3,750
Senior notes issued in non-U.S. markets	4,079	2,141	—	—
Total senior notes	41,079	16,397	—	3,750
Structured notes ^(a)	3,944	3,013	54,993	35,281
Total long-term unsecured funding – issuance	\$ 45,023	\$ 19,410	\$ 54,993	\$ 39,031
Maturities/redemptions				
Senior notes	\$ 25,765	\$ 21,483	\$ 65	\$ 67
Subordinated debt	3,097	2,090	250	—
Structured notes	892	1,532	47,425	28,777
Total long-term unsecured funding – maturities/redemptions	\$ 29,754	\$ 25,105	\$ 47,740	\$ 28,844

(a) Includes certain TLAC-eligible long-term unsecured debt issued by the Parent Company.

Secured funding and issuance

The Firm can also raise secured long-term funding through securitization of consumer credit card loans and FHLB advances. The following table summarizes the securitization issuance, the FHLB advances and their respective maturities or redemptions, as applicable for the years ended December 31, 2024 and 2023.

Long-term secured funding

Year ended December 31, (in millions)	Issuance		Maturities/Redemptions	
	2024	2023	2024	2023
Credit card securitization	\$ 2,348	\$ 1,998	\$ —	\$ 1,000
FHLB advances	6,000	39,775 ^(c)	18,050	9,485
Purchase Money Note ^(a)	—	50,000	—	—
Other long-term secured funding ^(b)	1,578	991	1,049	432
Total long-term secured funding	\$ 9,926	\$ 92,764	\$ 19,099	\$ 10,917

(a) Reflects the Purchase Money Note associated with the First Republic acquisition. Refer to Note 34 for additional information.

(b) Includes long-term structured notes that are secured.

(c) Includes FHLB advances associated with the First Republic acquisition on May 1, 2023. Refer to Note 34 for additional information.

The Firm's wholesale businesses also securitize loans for client-driven transactions; those client-driven loan securitizations are not considered to be a source of funding for the Firm and are not included in the table above. Refer to Note 14 for a further description of client-driven loan securitizations.

Credit ratings

The cost and availability of financing are influenced by credit ratings. Reductions in these ratings could have an adverse effect on the Firm's access to liquidity sources, increase the cost of funds, trigger additional collateral or funding requirements and decrease the number of investors and counterparties willing to lend to the Firm. The nature and magnitude of the impact of ratings downgrades depends on numerous contractual and behavioral factors, which the Firm

believes are incorporated in its liquidity risk and stress testing metrics. The Firm believes that it maintains sufficient liquidity to withstand a potential decrease in funding capacity due to ratings downgrades.

Additionally, the Firm's funding requirements for VIEs and other third-party commitments may be adversely affected by a decline in credit ratings. Refer to Notes 5 and 14 for additional information.

The credit ratings of the Parent Company and certain of its principal subsidiaries as of December 31, 2024 were as follows:

December 31, 2024	JPMorgan Chase & Co.			JPMorgan Chase Bank, N.A.			J.P. Morgan Securities LLC J.P. Morgan Securities plc J.P. Morgan SE ^(a)		
	Long-term issuer	Short-term issuer	Outlook	Long-term issuer	Short-term issuer	Outlook	Long-term issuer	Short-term issuer	Outlook
Moody's Investors Service ^(a)	A1	P-1	Positive	Aa2	P-1	Developing	Aa3	P-1	Positive
Standard & Poor's ^(b)	A	A-1	Stable	AA-	A-1+	Stable	AA-	A-1+	Stable
Fitch Ratings	AA-	F1+	Stable	AA	F1+	Stable	AA	F1+	Stable

(a) On November 11, 2024, Moody's (i) affirmed the credit ratings of the Parent Company, JPMorgan Chase Bank, N.A. and the other subsidiaries listed above; (ii) revised its outlook for the Parent Company, J.P. Morgan Securities LLC and J.P. Morgan Securities plc from stable to positive; (iii) revised its outlook for JPMorgan Chase Bank, N.A. from negative to developing, reflecting its view with respect to possible support from the U.S. government; and (iv) assessed its outlook for J.P. Morgan SE as negative with an "(m)" modifier, reflecting a negative outlook for long-term bank deposits and a positive outlook for the long-term issuer rating.

(b) The credit ratings of the Parent Company, JPMorgan Chase Bank, N.A. and the other subsidiaries presented in the table reflect ratings upgrades by Standard & Poor's on November 15, 2024. Standard & Poor's also revised its outlook for the Parent Company and such subsidiaries from positive to stable.

JPMorganChase's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings, or stock price.

Critical factors in maintaining high credit ratings include a stable and diverse earnings stream, strong capital and liquidity ratios, strong credit quality and risk management controls, and diverse funding sources. Rating agencies continue to evaluate economic and geopolitical trends, regulatory developments, future profitability, risk management practices, and litigation matters, as well as their broader ratings methodologies. Changes in any of these factors could lead to changes in the Firm's credit ratings.

REPUTATION RISK MANAGEMENT

Reputation risk is the risk that an action or inaction may negatively impact perception of the Firm's integrity and reduce confidence in the Firm's competence by various stakeholders, including clients, counterparties, customers, communities, investors, regulators, or employees.

The types of events that may result in reputation risk are wide-ranging and can be introduced by the Firm's employees, business strategies and activities, clients, customers and counterparties with which the Firm does business. These events could contribute to financial losses, litigation, regulatory enforcement actions, fines, penalties or other sanctions, as well as other harm to the Firm.

Organization and management

Reputation Risk Management is an independent risk management function that establishes the governance framework for managing reputation risk across the Firm's LOBs and Corporate. Reputation risk is inherently challenging to identify, manage, and quantify.

The Firm's reputation risk management function includes the following activities:

- Maintaining a Firmwide Reputation Risk Governance policy and a standard consistent with the reputation risk framework
- Providing oversight of the governance framework through processes and infrastructure to support consistent identification, escalation and monitoring of reputation risk issues Firmwide

Governance and oversight

The Reputation Risk Governance policy establishes the principles for managing reputation risk for the Firm. It is the responsibility of each LOB and Corporate, and the Firm's employees, to consider the reputation of the Firm when deciding whether to offer a new product, engage in a transaction or client relationship, enter a new jurisdiction, initiate a business process or consider any other activity. Environmental impacts and social concerns are important considerations in assessing the Firm's reputation risk, and are a component of the Firm's reputation risk governance.

CREDIT AND INVESTMENT RISK MANAGEMENT

Credit and investment risk is the risk associated with the default or change in credit profile of a client, counterparty or customer; or loss of principal or a reduction in expected returns on investments, including consumer credit risk, wholesale credit risk, and investment portfolio risk.

Credit risk management

Credit risk is the risk associated with the default or change in credit profile of a client, counterparty or customer. The Firm provides credit to a variety of clients and customers, ranging from large corporate and institutional clients to individual consumers and small businesses. In its consumer businesses, the Firm is exposed to credit risk primarily through its home lending, credit card, auto, and business banking businesses. In its wholesale businesses, the Firm is exposed to credit risk through its underwriting, lending, market-making, and hedging activities with and for clients and counterparties, as well as through its operating services activities (such as cash management and clearing activities), and securities financing activities. The Firm is also exposed to credit risk through its investment securities portfolio and cash placed with banks.

Credit Risk Management monitors and measures credit risk throughout the Firm, and defines credit risk policies, procedures and limits. The Firm's credit risk management governance includes the following activities:

- Maintaining a credit risk policy framework
- Monitoring and measuring credit risk across all portfolio segments, including transaction and exposure approval
- Setting industry and geographic concentration limits, as appropriate, and setting guidelines for credit review and analysis
- Assigning and maintaining credit approval authorities in connection with the approval of credit exposure
- Monitoring and independent assessment of criticized exposures and delinquent loans, and
- Estimating credit losses and supporting appropriate credit risk-based capital management

Risk identification and measurement

To measure credit risk, the Firm employs several methodologies for estimating the likelihood of obligor or counterparty default. Methodologies for measuring credit risk vary depending on several factors, including type of asset (e.g., consumer versus wholesale), risk measurement parameters (e.g., delinquency status and borrower's credit score versus wholesale risk-rating) and risk management and collection processes (e.g., retail collection center versus centrally managed workout groups). Credit risk measurement is based on the probability of default of an obligor or counterparty, the loss severity given a default event and the exposure at default.

Based on these factors and the methodology and estimates described in Note 13 and Note 10, the Firm estimates credit losses for its exposures. The allowance for loan losses reflects estimated credit losses related to the consumer and wholesale held-for-investment loan portfolios, the allowance for lending-related commitments reflects estimated credit losses related to the Firm's lending-related commitments and the allowance for investment securities reflects estimated credit losses related to the investment securities portfolio. Refer to Note 13, Note 10 and Critical Accounting Estimates used by the Firm on pages 161–164 for further information.

In addition, potential and unexpected credit losses are reflected in the allocation of credit risk capital and represent the potential volatility of actual losses relative to the established allowances for loan losses and lending-related commitments. The analyses for these losses include stress testing that considers alternative economic scenarios as described below.

Stress testing

Stress testing is important in measuring and managing credit risk in the Firm's credit portfolio. The stress testing process assesses the potential impact of alternative economic and business scenarios on estimated credit losses for the Firm. Economic scenarios and the underlying parameters are defined centrally, articulated in terms of macroeconomic factors and applied across the businesses. The stress test results may indicate credit migration, changes in delinquency trends and potential losses in the credit portfolio. In addition to the periodic stress testing processes, management also considers additional stresses outside these scenarios, including industry and country-specific stress scenarios, as appropriate. The Firm uses stress testing to inform decisions on setting risk appetite both at a Firm and LOB level, as well as to assess the impact of stress on individual counterparties.

Management's discussion and analysis

Risk monitoring and management

The Firm has developed policies and practices that are designed to preserve the independence and integrity of the approval and decision-making process for extending credit so that credit risks are assessed accurately, approved properly, monitored regularly and managed actively at both the transaction and portfolio levels. The policy framework establishes credit approval authorities, concentration limits, risk-rating methodologies, portfolio review parameters and guidelines for management of distressed exposures. In addition, certain models, assumptions and inputs used in evaluating and monitoring credit risk are independently validated by groups that are separate from the LOBs.

Consumer credit risk is monitored for delinquency and other trends, including any concentrations at the portfolio level, as certain of these trends can be addressed through changes in underwriting policies and portfolio guidelines. Consumer Risk Management evaluates delinquency and other trends against business expectations, current and forecasted economic conditions, and industry benchmarks. Historical and forecasted economic performance and trends are incorporated into the modeling of estimated consumer credit losses and are part of the monitoring of the credit risk profile of the portfolio.

Wholesale credit risk is monitored regularly at an aggregate portfolio, industry, and individual client and counterparty level with established concentration limits that are reviewed and revised periodically as deemed appropriate by management. Industry and counterparty limits, as measured in terms of exposure and economic risk appetite, are subject to stress-based loss constraints.

Management of the Firm's wholesale credit risk exposure is accomplished through a number of means, including:

- Loan underwriting and credit approval processes
- Loan syndications and participations
- Loan sales and securitizations
- Credit derivatives
- Master netting agreements, and
- Collateral and other risk-reduction techniques

In addition to Credit Risk Management, an independent Credit Review function is responsible for:

- Independently assessing risk grades assigned to exposures in the Firm's wholesale credit portfolio and the timeliness of risk grade changes initiated by responsible business units; and
- Evaluating the effectiveness of the credit management processes of the LOBs and Corporate, including the adequacy of credit analyses and risk grading/loss given default ("LGD") rationales, proper monitoring and management of credit exposures, and compliance with applicable grading policies and underwriting guidelines.

Refer to Note 12 for further discussion of consumer and wholesale loans.

Risk reporting

To enable monitoring of credit risk and effective decision-making, aggregate credit exposure, credit quality forecasts, concentration levels and risk profile changes are reported regularly to senior members of Credit Risk Management. Detailed portfolio reporting of industry, clients, counterparties and customers, product and geography are prepared, and the appropriateness of the allowance for credit losses is reviewed by senior management at least on a quarterly basis. Through the risk reporting and governance structure, credit risk trends and limit exceptions are provided regularly to, and discussed with, risk committees, senior management and the Board of Directors.

CREDIT PORTFOLIO

Credit risk is the risk associated with the default or change in credit profile of a client, counterparty or customer.

In the following tables, total loans include loans retained (i.e., held-for-investment); loans held-for-sale; and certain loans accounted for at fair value. The following tables do not include loans which the Firm accounts for at fair value and classifies as trading assets; refer to Notes 2 and 3 for further information regarding these loans. Refer to Notes 12, 28, and 5 for additional information on the Firm's loans, lending-related commitments and derivative receivables, including the Firm's related accounting policies.

Refer to Note 10 for information regarding the credit risk inherent in the Firm's investment securities portfolio; and refer to Note 11 for information regarding credit risk inherent in the securities financing portfolio. Refer to Consumer Credit Portfolio on pages 120–125 and Note 12 for further discussions of the consumer credit environment, consumer loans and nonperforming exposure. Refer to Wholesale Credit Portfolio on pages 126–136 and Note 12 for further discussions of the wholesale credit environment and wholesale loans.

Total credit portfolio

December 31, (in millions)	Credit exposure		Nonperforming ^(c)	
	2024	2023	2024	2023
Loans retained	\$1,299,590	\$1,280,870	\$ 7,175	\$ 5,989
Loans held-for-sale	7,048	3,985	160	184
Loans at fair value	41,350	38,851	1,502	744
Total loans	1,347,988	1,323,706	8,837	6,917
Derivative receivables	60,967	54,864	145	364
Receivables from customers ^(a)	51,929	47,625	—	—
Total credit-related assets	1,460,884	1,426,195	8,982	7,281
Assets acquired in loan satisfactions				
Real estate owned	NA	NA	284	274
Other	NA	NA	34	42
Total assets acquired in loan satisfactions	NA	NA	318	316
Lending-related commitments	1,577,622	1,497,847	737	464
Total credit portfolio	\$3,038,506	\$2,924,042	\$10,037	\$ 8,061
Credit derivatives and credit-related notes used in credit portfolio management activities ^(b)	\$ (41,367)	\$ (37,779)	\$ —	\$ —
Liquid securities and other cash collateral held against derivatives	(28,160)	(22,461)	NA	NA

- (a) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM; these are reported within accrued interest and accounts receivable on the Consolidated balance sheets.
- (b) Represents the net notional amount of protection purchased and sold through credit derivatives and credit-related notes used to manage credit exposures.
- (c) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were \$121 million and \$182 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance.

The following table provides information on Firmwide nonaccrual loans to total loans.

December 31, (in millions, except ratios)	2024	2023
Total nonaccrual loans	\$ 8,837	\$ 6,917
Total loans	1,347,988	1,323,706
Firmwide nonaccrual loans to total loans outstanding	0.66 %	0.52 %

The following table provides information about the Firm's net charge-offs and recoveries.

December 31, (in millions, except ratios)	2024	2023
Net charge-offs	\$ 8,638	\$ 6,209
Average retained loans	1,271,344	1,202,348
Net charge-off rates	0.68 %	0.52 %

CONSUMER CREDIT PORTFOLIO

The Firm's retained consumer portfolio consists primarily of loans and lending-related commitments for residential real estate, credit card, scored auto and business banking. The consumer credit portfolio also includes loans at fair value, predominantly in residential real estate. The Firm's focus is on serving primarily the prime segment of the consumer credit market.

Originated mortgage loans are retained in the residential real estate portfolio, securitized or sold to U.S. government agencies and U.S. government-sponsored enterprises; other types of consumer loans are typically retained on the balance sheet. Refer to Note 12 for further information on the consumer loan portfolio. Refer to Note 28 for further information on lending-related commitments.

The following tables present consumer credit-related information with respect to the scored credit portfolio held in CCB, AWM, CIB and Corporate.

Consumer credit portfolio

December 31, (in millions)	Credit exposure		Nonaccrual loans ⁽ⁱ⁾	
	2024	2023	2024	2023
Consumer, excluding credit card				
Residential real estate ^(a)	\$ 309,513	\$ 326,409	\$ 2,984	\$ 3,466
Auto and other ^{(b)(c)}	66,821	70,866	249	177
Total loans - retained	376,334	397,275	3,233	3,643
Loans held-for-sale	945	487	155	95
Loans at fair value ^(d)	15,531	12,331	538	465
Total consumer, excluding credit card loans	392,810	410,093	3,926	4,203
Lending-related commitments ^(e)	44,844	45,403		
Total consumer exposure, excluding credit card	437,654	455,496		
Credit card				
Loans retained ^(f)	232,860	211,123	NA	NA
Total credit card loans	232,860	211,123	NA	NA
Lending-related commitments ^{(e)(g)}	1,001,311	915,658		
Total credit card exposure	1,234,171	1,126,781		
Total consumer credit portfolio	\$ 1,671,825	\$ 1,582,277	\$ 3,926	\$ 4,203
Credit-related notes used in credit portfolio management activities ^(h)	\$ (479)	\$ (790)		

(in millions, except ratios)	Year ended December 31,					
	Net charge-offs/ (recoveries)		Average loans - retained		Net charge-off/(recovery) rate ⁽ⁱ⁾	
	2024	2023	2024	2023	2024	2023
Consumer, excluding credit card						
Residential real estate	\$ (101)	\$ (52)	\$ 316,042	\$ 296,515	(0.03)%	(0.02)%
Auto and other	775	684	67,959	67,546	1.14	1.01
Total consumer, excluding credit card - retained	674	632	384,001	364,061	0.18	0.17
Credit card - retained	7,142	4,698	214,033	191,412	3.34	2.45
Total consumer - retained	\$ 7,816	\$ 5,330	\$ 598,034	\$ 555,473	1.31 %	0.96 %

(a) Includes scored mortgage and home equity loans held in CCB and AWM.

(b) At December 31, 2024 and 2023, excluded operating lease assets of \$12.8 billion and \$10.4 billion, respectively. These operating lease assets are included in other assets on the Firm's Consolidated balance sheets. Refer to Note 18 for further information.

(c) Includes scored auto and business banking loans, and overdrafts.

(d) Includes scored mortgage loans held in CCB and CIB, and other consumer unsecured loans in CIB.

(e) Credit card, home equity and certain business banking lending-related commitments represent the total available lines of credit for these products. The Firm has not experienced, and does not anticipate, that all available lines of credit would be used at the same time. Refer to Note 28 for further information.

(f) Includes billed interest and fees.

(g) Also includes commercial card lending-related commitments primarily in CIB.

(h) Represents the notional amount of protection obtained through the issuance of credit-related notes that reference certain pools of residential real estate and auto loans in the retained consumer portfolio.

(i) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were \$121 million and \$182 million, respectively. In addition, the Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status, as permitted by regulatory guidance.

(j) Average consumer loans held-for-sale and loans at fair value were \$17.2 billion and \$12.9 billion for the years ended December 31, 2024 and 2023, respectively. These amounts were excluded when calculating net charge-off/(recovery) rates.

Management's discussion and analysis

Maturities and sensitivity to changes in interest rates

The table below sets forth loan maturities by scheduled repayments, by class of loan and the distribution between fixed and floating interest rates based on the stated terms of the loan agreements. The Firm estimated the principal repayment amounts for both the residential real estate and auto and other loan classes by calculating the weighted-average loan balance and interest rates for loan pools based on remaining loan term. Refer to Note 12 for further information on loan classes.

December 31, 2024 (in millions)	Within 1 year ^(a)	1-5 years	5-15 years	After 15 years	Total
Consumer, excluding credit card					
Residential real estate	\$ 21,442	\$ 26,712	\$109,608	\$ 166,715	\$324,477
Auto and other	19,404 ^(b)	43,701	5,224	4	68,333
Total consumer, excluding credit card loans	\$ 40,846	\$ 70,413	\$ 114,832	\$ 166,719	\$392,810
Total credit card loans	\$ 231,799	\$ 1,048	\$ 13	\$ —	\$232,860
Total consumer loans	\$272,645	\$ 71,461	\$ 114,845	\$ 166,719	\$625,670
Loans due after one year at fixed interest rates					
Residential real estate		\$ 19,639	\$ 57,351	\$ 77,865	
Auto and other		43,565	2,957	4	
Credit card		1,048	13	—	
Loans due after one year at variable interest rates					
Residential real estate		\$ 7,073	\$ 52,257	\$ 88,850	
Auto and other		136	2,267	—	
Total consumer loans		\$ 71,461	\$ 114,845	\$ 166,719	

(a) Includes loans held-for-sale and loans at fair value.

(b) Includes overdrafts.

Consumer, excluding credit card

Portfolio analysis

Loans decreased from December 31, 2023 driven by residential real estate loans and scored auto loans.

The following discussions provide information concerning individual loan products. Refer to Note 12 for further information about this portfolio, including information about delinquencies, loan modifications and other credit quality indicators.

Residential real estate: The residential real estate portfolio, including loans held-for-sale and loans at fair value, predominantly consists of prime mortgage loans and home equity lines of credit.

Retained loans decreased compared to December 31, 2023, predominantly driven by paydowns and loan sales, net of originations. Retained nonaccrual loans decreased compared to December 31, 2023, predominantly driven by loan sales. Net recoveries were higher for the year ended December 31, 2024 compared to the prior year, driven by loan sales.

Loans held-for-sale and nonaccrual loans held-for-sale increased from December 31, 2023, predominantly driven by transfers of certain retained loans in anticipation of securitization and loan sales, respectively.

Loans at fair value increased from December 31, 2023, predominantly driven by higher Home Lending loans, as originations outpaced warehouse loan sales. Nonaccrual loans at fair value increased compared to December 31, 2023, driven by CIB.

At December 31, 2024 and 2023, the carrying values of retained interest-only residential mortgage loans were \$88.9 billion and \$90.6 billion, respectively. These loans have an interest-only payment period generally followed by an adjustable-rate or fixed-rate fully amortizing payment period to maturity and are typically originated as higher-balance loans to higher-income borrowers. The credit performance of this portfolio is comparable to the performance of the broader prime mortgage portfolio.

The carrying value of retained home equity lines of credit outstanding was \$14.5 billion at December 31, 2024, including \$3.8 billion of HELOCs that have recast from interest-only to fully amortizing payments or have been modified, and \$3.6 billion of interest-only balloon HELOCs, which primarily mature after 2030. The Firm manages the risk of HELOCs during their revolving period by reducing or canceling the undrawn line in accordance with the contract or to the extent otherwise permitted by law, including when there has been a demonstrable decline in the creditworthiness of the borrower or significant decrease in the value of the underlying property.

The following table provides a summary of the Firm's residential mortgage portfolio insured and/or guaranteed by U.S. government agencies, predominantly loans held-for-sale and loans at fair value. The Firm monitors its exposure to certain potential unrecoverable claim payments related to government-insured loans and considers this exposure in estimating the allowance for loan losses.

(in millions)	December 31, 2024	December 31, 2023
Current	\$ 462	\$ 446
30-89 days past due	72	102
90 or more days past due	121	182
Total government guaranteed loans	\$ 655	\$ 730

Geographic composition and current estimated loan-to-value ratio of residential real estate loans

At December 31, 2024, \$217.7 billion, or 70% of the total retained residential real estate loan portfolio, was concentrated in California, New York, Florida, Texas and Massachusetts, compared to \$228.4 billion, or 70% at December 31, 2023.

Average current estimated loan-to-value ("LTV") ratios have improved, reflecting an increase in home prices.

Refer to Note 12 for information on the geographic composition and current estimated LTVs of the Firm's residential real estate loans.

Management's discussion and analysis

Auto and other: The auto and other loan portfolio, including loans at fair value, generally consists of prime-quality scored auto and business banking loans, other consumer unsecured loans, and overdrafts. The portfolio decreased when compared to December 31, 2023, predominantly due to loan securitizations. Net charge-offs increased compared to the prior year, predominantly due to net charge-offs of scored auto loans of \$445 million compared to \$357 million for the year ended December 31, 2023, reflecting a decline in used vehicle valuations. Refer to Note 14 for further information on securitization activity.

Nonperforming assets

The following table presents information as of December 31, 2024 and 2023, about consumer, excluding credit card, nonperforming assets.

Nonperforming assets^(a)

December 31, (in millions)	2024	2023
Nonaccrual loans		
Residential real estate	\$ 3,665	\$ 4,015
Auto and other	261	188
Total nonaccrual loans	3,926	4,203
Assets acquired in loan satisfactions		
Real estate owned	78	120
Other	34	42
Total assets acquired in loan satisfactions	112	162
Total nonperforming assets	\$ 4,038	\$ 4,365

(a) Excludes mortgage loans past due and insured by U.S. government agencies, which are primarily 90 or more days past due. These loans have been excluded based upon the government guarantee. At December 31, 2024 and 2023, mortgage loans 90 or more days past due and insured by U.S. government agencies were \$121 million and \$182 million, respectively.

Nonaccrual loans

The following table presents changes in consumer, excluding credit card, nonaccrual loans for the years ended December 31, 2024 and 2023.

Nonaccrual loan activity

Year ended December 31, (in millions)	2024	2023
Beginning balance	\$ 4,203	\$ 4,325
Additions:	3,225	2,894
Reductions:		
Principal payments and other	894	1,030
Sales	803	276
Charge-offs	665	472
Returned to performing status	963	1,052
Foreclosures and other liquidations	177	186
Total reductions	3,502	3,016
Net changes	(277)	(122)
Ending balance	\$ 3,926	\$ 4,203

Refer to Note 12 for further information about the consumer credit portfolio, including information about delinquencies, other credit quality indicators and loans that were in the process of active or suspended foreclosure.

Credit card

Total credit card loans increased from December 31, 2023 reflecting growth from new accounts and revolving balances. The December 31, 2024 30+ and 90+ day delinquency rates of 2.17% and 1.14%, respectively, increased compared to the December 31, 2023 30+ and 90+ day delinquency rates of 2.14% and 1.05%, respectively, in line with the Firm's expectations. Net charge-offs increased for the year ended December 31, 2024 compared to the prior year reflecting the seasoning of vintages originated in recent years, credit normalization and balance growth.

Consistent with the Firm's policy, all credit card loans typically remain on accrual status until charged off. However, the Firm's allowance for loan losses includes the estimated uncollectible portion of accrued and billed interest and fee income.

Geographic and FICO composition of credit card loans

At December 31, 2024, \$109.0 billion, or 47% of the total retained credit card loan portfolio, was concentrated in California, Texas, New York, Florida and Illinois, compared to \$98.1 billion, or 46%, at December 31, 2023.

Refer to Note 12 for further information about this portfolio, including information about delinquencies, geographic and FICO composition.

WHOLESALE CREDIT PORTFOLIO

In its wholesale businesses, the Firm is exposed to credit risk primarily through its underwriting, lending, market-making, and hedging activities with and for clients and counterparties, as well as through various operating services (such as cash management and clearing activities), securities financing activities and cash placed with banks. A portion of the loans originated or acquired by the Firm's wholesale businesses is generally retained on the balance sheet. The Firm distributes a significant percentage of the loans that it originates into the market as part of its syndicated loan business and to manage portfolio concentrations and credit risk. The wholesale portfolio is actively managed, in part by conducting ongoing, in-depth reviews of client credit quality and transaction structure, inclusive of collateral where applicable, and of industry, product and client concentrations. Refer to the industry discussion on pages 128–131 for further information.

The Firm's wholesale credit portfolio includes exposure held in CIB, AWM and Corporate, and risk-rated exposure held in CCB, for which the wholesale methodology is applied when determining the allowance for loan losses.

As of December 31, 2024, loans increased \$19.8 billion, driven by higher loans in CIB and higher securities-based lending in AWM. Lending-related commitments decreased \$5.3 billion, with decreases in AWM and CCB, largely offset by higher commitments in CIB.

As of December 31, 2024, nonperforming exposure increased by \$2.3 billion, predominantly driven by Real Estate, concentrated in Office, Healthcare and Consumer & Retail, in each case resulting from downgrades.

For the year ended December 31, 2024, wholesale net charge-offs were \$822 million, largely driven by Real Estate, concentrated in Office, and client-specific charge-offs across multiple industries including Consumer & Retail and Individuals.

Wholesale credit portfolio

December 31, (in millions)	Credit exposure		Nonperforming	
	2024	2023	2024	2023
Loans retained	\$ 690,396	\$ 672,472	\$ 3,942	\$ 2,346
Loans held-for-sale	6,103	3,498	5	89
Loans at fair value	25,819	26,520	964	279
Loans	722,318	702,490	4,911	2,714
Derivative receivables	60,967	54,864	145	364
Receivables from customers ^(a)	51,929	47,625	—	—
Total wholesale credit-related assets	835,214	804,979	5,056	3,078
Assets acquired in loan satisfactions				
Real estate owned	NA	NA	206	154
Other	NA	NA	—	—
Total assets acquired in loan satisfactions	NA	NA	206	154
Lending-related commitments	531,467	536,786	737	464
Total wholesale credit portfolio	\$1,366,681	\$1,341,765	\$ 5,999	\$ 3,696
Credit derivatives and credit-related notes used in credit portfolio management activities ^(b)	\$ (40,888)	\$ (36,989)	\$ —	\$ —
Liquid securities and other cash collateral held against derivatives	(28,160)	(22,461)	NA	NA

(a) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM; these are reported within accrued interest and accounts receivable on the Consolidated balance sheets.

(b) Represents the net notional amount of protection purchased and sold through credit derivatives and credit-related notes used to manage both performing and nonperforming wholesale credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP. Refer to Credit derivatives on page 136 and Note 5 for additional information.

Wholesale credit exposure – maturity and ratings profile

The following tables present the maturity and internal risk ratings profiles of the wholesale credit portfolio as of December 31, 2024 and 2023. The Firm generally considers internal ratings with qualitative characteristics equivalent to BBB-/Baa3 or higher as investment grade, and takes into consideration collateral and structural support when determining the internal risk rating for each credit facility. Refer to Note 12 for further information on internal risk ratings.

December 31, 2024 (in millions, except ratios)	Maturity profile ^(d)				Ratings profile			
	1 year or less	After 1 year through 5 years	After 5 years	Total	Investment-grade	Noninvestment-grade	Total	Total % of IG
Loans retained	\$ 225,982	\$ 289,199	\$ 175,215	\$ 690,396	\$ 471,670	\$ 218,726	\$ 690,396	68 %
Derivative receivables				60,967			60,967	
Less: Liquid securities and other cash collateral held against derivatives				(28,160)			(28,160)	
Total derivative receivables, net of collateral	11,515	7,418	13,874	32,807	24,707	8,100	32,807	75
Lending-related commitments	121,283	384,529	25,655	531,467	352,082	179,385	531,467	66
Subtotal	358,780	681,146	214,744	1,254,670	848,459	406,211	1,254,670	68
Loans held-for-sale and loans at fair value ^(a)				31,922			31,922	
Receivables from customers				51,929			51,929	
Total exposure – net of liquid securities and other cash collateral held against derivatives				\$1,338,521			\$ 1,338,521	
Credit derivatives and credit-related notes used in credit portfolio management activities ^{(b)(c)}	\$ (5,442)	\$ (33,751)	\$ (1,695)	\$ (40,888)	\$ (31,691)	\$ (9,197)	\$ (40,888)	78 %

December 31, 2023 (in millions, except ratios)	Maturity profile ^(d)				Ratings profile			
	1 year or less	After 1 year through 5 years	After 5 years	Total	Investment-grade	Noninvestment-grade	Total	Total % of IG
Loans retained	\$ 211,104	\$ 280,821	\$ 180,547	\$ 672,472	\$ 458,838	\$ 213,634	\$ 672,472	68 %
Derivative receivables				54,864			54,864	
Less: Liquid securities and other cash collateral held against derivatives				(22,461)			(22,461)	
Total derivative receivables, net of collateral	8,007	8,970	15,426	32,403	24,919	7,484	32,403	77
Lending-related commitments	143,337	368,646	24,803	536,786	341,611	195,175	536,786	64
Subtotal	362,448	658,437	220,776	1,241,661	825,368	416,293	1,241,661	66
Loans held-for-sale and loans at fair value ^(a)				30,018			30,018	
Receivables from customers				47,625			47,625	
Total exposure – net of liquid securities and other cash collateral held against derivatives				\$1,319,304			\$ 1,319,304	
Credit derivatives and credit-related notes used in credit portfolio management activities ^{(b)(c)}	\$ (3,311)	\$ (28,353)	\$ (5,325)	\$ (36,989)	\$ (28,869)	\$ (8,120)	\$ (36,989)	78 %

(a) Loans held-for-sale are primarily related to syndicated loans and loans transferred from the retained portfolio.

(b) These derivatives do not qualify for hedge accounting under U.S. GAAP.

(c) The notional amounts are presented on a net basis by underlying reference entity and the ratings profile shown is based on the ratings of the reference entity on which protection has been purchased. Predominantly all of the credit derivatives entered into by the Firm where it has purchased protection used in credit portfolio management activities are executed with investment-grade counterparties. In addition, the Firm obtains credit protection against certain loans in the retained loan portfolio through the issuance of credit-related notes.

(d) The maturity profile of retained loans, lending-related commitments and derivative receivables is generally based on remaining contractual maturity. Derivative contracts that are in a receivable position at December 31, 2024, may become payable prior to maturity based on their cash flow profile or changes in market conditions.

Management's discussion and analysis

Wholesale credit exposure – industry exposures

The Firm focuses on the management and diversification of its industry exposures, and pays particular attention to industries with actual or potential credit concerns.

Exposures that are deemed to be criticized align with the U.S. banking regulators' definition of criticized exposures, which consist of the special mention, substandard and doubtful categories. Total criticized exposure, excluding loans held-for-sale and loans at fair value, was \$44.7 billion and \$41.4 billion at December 31, 2024 and 2023, representing approximately 3.5% and 3.3% of total wholesale credit exposure, respectively; of the \$44.7 billion, \$39.9 billion was performing. The increase in criticized exposure was driven by Real Estate resulting from downgrades, primarily in Multifamily and Office, and new commitments in Technology and Media, partially offset by Consumer & Retail resulting from net portfolio activity and upgrades.

The table below summarizes by industry the Firm's exposures as of December 31, 2024 and 2023. The industry of risk category is generally based on the client or counterparty's primary business activity. Refer to Note 4 for additional information on industry concentrations.

Wholesale credit exposure – industries^(a)

As of or for the year ended December 31, 2024 (in millions)	Credit exposure ^{(f)(g)}	Investment- grade	Noninvestment-grade			Selected metrics			
			Noncriticized	Criticized performing	Criticized nonperforming	30 days or more past due and accruing loans	Net charge-offs/ (recoveries)	Credit derivative and credit- related notes ^(h)	Liquid securities and other cash collateral held against derivative receivables
Real Estate	\$ 207,050	\$ 143,803	\$ 50,865	\$ 10,858	\$ 1,524	\$ 913	\$ 345	\$ (584)	\$ —
Individuals and Individual Entities ^(b)	144,145	118,650	24,831	217	447	831	122	—	—
Asset Managers	135,541	101,150	34,148	206	37	375	2	—	(9,194)
Consumer & Retail	129,815	62,800	60,141	6,055	819	252	123	(4,320)	—
Technology, Media & Telecommunications	84,716	45,021	28,629	10,592	474	79	94	(4,800)	—
Industrials	72,530	37,572	30,912	3,807	239	185	91	(2,312)	—
Healthcare	64,224	44,135	17,062	2,219	808	245	56	(3,286)	(34)
Banks & Finance Companies	61,287	36,884	24,119	257	27	36	—	(702)	(729)
Utilities	35,871	24,205	10,256	1,273	137	1	—	(2,700)	—
State & Municipal Govt ^(c)	35,039	33,303	1,711	9	16	90	—	(2)	(1)
Automotive	34,336	22,015	11,353	931	37	121	1	(997)	—
Oil & Gas	31,724	19,053	12,479	188	4	9	(3)	(1,711)	(2)
Insurance	24,267	17,847	6,198	222	—	2	—	(1,077)	(9,184)
Chemicals & Plastics	20,782	11,013	8,152	1,521	96	31	14	(1,164)	—
Transportation	17,019	9,462	7,135	391	31	17	(20)	(658)	—
Metals & Mining	15,860	7,373	7,860	590	37	9	—	(246)	(2)
Central Govt	13,862	13,580	157	125	—	4	—	(1,490)	(2,051)
Securities Firms	9,443	5,424	4,014	5	—	—	—	(13)	(2,635)
Financial Markets Infrastructure	4,446	4,201	245	—	—	—	—	(1)	—
All other ^(d)	140,873	117,986	22,398	398	91	10	(3)	(14,825)	(4,328)
Subtotal	\$ 1,282,830	\$ 875,477	\$ 362,665	\$ 39,864	\$ 4,824	\$ 3,210	\$ 822	\$ (40,888)	\$ (28,160)
Loans held-for-sale and loans at fair value	31,922								
Receivables from customers	51,929								
Total^(e)	\$ 1,366,681								

As of or for the year ended December 31, 2023 (in millions)							Selected metrics			
	Credit exposure ^{(f)(g)}	Investment- grade	Noninvestment-grade			30 days or more past due and accruing loans	Net charge-offs/ (recoveries)	Credit derivative and credit- related notes ^(h)	Liquid securities and other cash collateral held against derivative receivables	
			Noncriticized	Criticized performing	Criticized nonperforming					
Real Estate	\$ 208,261	\$ 148,866	\$ 50,190	\$ 8,558	\$ 647	\$ 717	\$ 275	\$ (574)	\$ —	
Individuals and Individual Entities ^(b)	145,849	110,673	34,261	334	581	861	10	—	—	
Asset Managers	129,574	83,857	45,623	90	4	201	1	—	(7,209)	
Consumer & Retail	127,086	60,168	58,606	7,863	449	318	161	(4,204)	—	
Technology, Media & Telecommunications	77,296	40,468	27,094	9,388	346	36	81	(4,287)	—	
Industrials	75,092	40,951	30,586	3,419	136	213	31	(2,949)	—	
Healthcare	65,025	43,163	18,396	3,005	461	130	17	(3,070)	—	
Banks & Finance Companies	57,177	33,881	22,744	545	7	9	277	(511)	(412)	
Utilities	36,061	25,242	9,929	765	125	1	(3)	(2,373)	—	
State & Municipal Govt ^(c)	35,986	33,561	2,390	27	8	31	—	(4)	—	
Automotive	33,977	23,152	10,060	640	125	59	—	(653)	—	
Oil & Gas	34,475	18,276	16,076	111	12	45	11	(1,927)	(5)	
Insurance	20,501	14,503	5,700	298	—	2	—	(961)	(6,898)	
Chemicals & Plastics	20,773	11,353	8,352	916	152	106	2	(1,045)	—	
Transportation	16,060	8,865	5,943	1,196	56	23	(26)	(574)	—	
Metals & Mining	15,508	8,403	6,514	536	55	12	44	(229)	—	
Central Govt	17,704	17,264	312	127	1	—	—	(3,490)	(2,085)	
Securities Firms	8,689	4,570	4,118	1	—	—	—	(14)	(2,765)	
Financial Markets Infrastructure	4,251	4,052	199	—	—	—	—	—	—	
All other ^(d)	134,777	115,711	18,618	439	9	21	(2)	(10,124)	(3,087)	
Subtotal	\$ 1,264,122	\$ 846,979	\$ 375,711	\$ 38,258	\$ 3,174	\$ 2,785	\$ 879	\$ (36,989)	\$ (22,461)	
Loans held-for-sale and loans at fair value	30,018									
Receivables from customers	47,625									
Total^(e)	\$ 1,341,765									

- (a) The industry rankings presented in the table as of December 31, 2023, are based on the industry rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.
- (b) Individuals and Individual Entities predominantly consists of Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB, and includes exposure to personal investment companies and personal and testamentary trusts.
- (c) In addition to the credit risk exposure to states and municipal governments (both U.S. and non-U.S.) at December 31, 2024 and 2023, noted above, the Firm held: \$6.1 billion and \$5.9 billion, respectively, of trading assets; \$17.9 billion and \$21.4 billion, respectively, of AFS securities; and \$9.3 billion and \$9.9 billion, respectively, of HTM securities, issued by U.S. state and municipal governments. Refer to Note 2 and Note 10 for further information.
- (d) All other includes: SPEs and Private education and civic organizations, representing approximately 94% and 6%, respectively, at both December 31, 2024 and 2023.
- (e) Excludes cash placed with banks of \$459.2 billion and \$614.1 billion, at December 31, 2024 and 2023, respectively, which is predominantly placed with various central banks, primarily Federal Reserve Banks.
- (f) Credit exposure is net of risk participations and excludes the benefit of credit derivatives and credit-related notes used in credit portfolio management activities held against derivative receivables or loans and liquid securities and other cash collateral held against derivative receivables.
- (g) Credit exposure includes held-for-sale and fair value option elected lending-related commitments.
- (h) Represents the net notional amounts of protection purchased and sold through credit derivatives and credit-related notes used to manage the credit exposures; these derivatives do not qualify for hedge accounting under U.S. GAAP. The All other category includes purchased credit protection on certain credit indices.

Management's discussion and analysis

Presented below is additional detail on certain of the Firm's industry exposures.

Real Estate

Real Estate exposure was \$207.1 billion as of December 31, 2024. Criticized exposure increased by \$3.2 billion from \$9.2 billion at December 31, 2023 to \$12.4 billion at December 31, 2024, predominantly driven by downgrades, primarily in Multifamily and Office.

(in millions, except ratios)	December 31, 2024				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(d)
Multifamily ^(a)	\$ 124,074	\$ 7	\$ 124,081	77 %	92 %
Industrial	19,092	17	19,109	65	72
Other Income Producing Properties ^(b)	16,411	158	16,569	50	63
Office	16,331	29	16,360	47	81
Services and Non Income Producing	14,047	57	14,104	62	46
Retail	12,230	23	12,253	77	75
Lodging	4,555	19	4,574	31	53
Total Real Estate Exposure^(c)	\$ 206,740	\$ 310	\$ 207,050	69 %	82 %

(in millions, except ratios)	December 31, 2023				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(d)
Multifamily ^(a)	\$ 121,946	\$ 21	\$ 121,967	79 %	90 %
Industrial	20,254	18	20,272	70	72
Other Income Producing Properties ^(b)	15,542	208	15,750	55	63
Office	16,462	32	16,494	51	81
Services and Non Income Producing	16,145	74	16,219	62	46
Retail	12,763	48	12,811	75	73
Lodging	4,729	19	4,748	30	48
Total Real Estate Exposure	\$ 207,841	\$ 420	\$ 208,261	71 %	80 %

(a) Total Multifamily exposure is approximately 99% performing. Multifamily exposure is largely in California.

(b) Other Income Producing Properties consists of clients with diversified property types or other property types outside of categories listed in the table above.

(c) Real Estate exposure is approximately 84% secured; unsecured exposure is largely investment-grade primarily to Real Estate Investment Trusts ("REITs") and Real Estate Operating Companies ("REOCs") whose underlying assets are generally diversified.

(d) Represents drawn exposure as a percentage of credit exposure.

Consumer & Retail

Consumer & Retail exposure was \$129.8 billion as of December 31, 2024. Criticized exposure decreased by \$1.4 billion from \$8.3 billion at December 31, 2023 to \$6.9 billion at December 31, 2024, driven by net portfolio activity and upgrades, largely offset by downgrades.

(in millions, except ratios)	December 31, 2024				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(d)
Food and Beverage	\$ 34,774	\$ 683	\$ 35,457	61 %	34 %
Retail	34,917	261	35,178	51	31
Business and Consumer Services ^(a)	34,534	412	34,946	42	41
Consumer Hard Goods	13,796	208	14,004	43	35
Leisure ^(b)	10,186	44	10,230	26	43
Total Consumer & Retail^(c)	\$ 128,207	\$ 1,608	\$ 129,815	48 %	36 %

(in millions, except ratios)	December 31, 2023				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(d)
Food and Beverage	\$ 32,256	\$ 930	\$ 33,186	57 %	36 %
Retail	36,042	334	36,376	51	30
Business and Consumer Services ^(a)	34,822	392	35,214	42	42
Consumer Hard Goods	13,169	197	13,366	43	33
Leisure ^(b)	8,784	160	8,944	25	47
Total Consumer & Retail	\$ 125,073	\$ 2,013	\$ 127,086	47 %	36 %

(a) Retail consists of Home Improvement & Specialty Retailers, Restaurants, Discount & Drug Stores, Specialty Apparel, Department Stores and Supermarkets.

(b) Leisure consists of Arts & Culture, Travel Services, Gaming and Sports & Recreation. As of December 31, 2024, approximately 90% of the noninvestment-grade Leisure portfolio is secured.

(c) Consumer & Retail exposure is approximately 57% secured; unsecured exposure is approximately 80% investment-grade.

(d) Represents drawn exposure as a percent of credit exposure.

Oil & Gas

Oil & Gas exposure was \$31.7 billion as of December 31, 2024. Criticized exposure was \$192 million and \$123 million at December 31, 2024 and 2023, respectively.

(in millions, except ratios)	December 31, 2024				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(c)
Exploration & Production ("E&P") and Oil field Services	\$ 14,265	\$ 848	\$ 15,113	55 %	27 %
Other Oil & Gas ^(a)	16,306	305	16,611	65	19
Total Oil & Gas^(b)	\$ 30,571	\$ 1,153	\$ 31,724	60 %	23 %

(in millions, except ratios)	December 31, 2023				
	Loans and Lending-related Commitments	Derivative Receivables	Credit exposure	% Investment-grade	% Drawn ^(c)
Exploration & Production ("E&P") and Oil field Services	\$ 18,121	\$ 536	\$ 18,657	51 %	26 %
Other Oil & Gas ^(a)	15,649	169	15,818	55	22
Total Oil & Gas	\$ 33,770	\$ 705	\$ 34,475	53 %	25 %

(a) Other Oil & Gas includes Integrated Oil & Gas companies, Midstream/Oil Pipeline companies and refineries.

(b) Oil & Gas exposure is approximately 33% secured, and includes reserve-based lending to the Exploration & Production sub-sector; unsecured exposure is approximately 69% investment-grade.

(c) Represents drawn exposure as a percent of credit exposure.

Management's discussion and analysis

Loans

In its wholesale businesses, the Firm provides loans to a variety of clients, ranging from large corporate and institutional clients to high-net-worth individuals. Refer to Note 12 for a further discussion on loans, including information about delinquencies, loan modifications and other credit quality indicators.

The following table presents the change in the nonaccrual loan portfolio for the years ended December 31, 2024 and 2023. Since December 31, 2023, nonaccrual loan exposure increased by \$2.2 billion, predominantly driven by Real Estate, concentrated in Office, Healthcare and Consumer & Retail, in each case resulting from downgrades.

Wholesale nonaccrual loan activity

Year ended December 31, (in millions)	2024	2023
Beginning balance	\$ 2,714	\$ 2,395
Additions	5,841	3,543
Reductions:		
Paydowns and other	2,387	1,336
Gross charge-offs	780	965
Returned to performing status	392	616
Sales	85	307
Total reductions	3,644	3,224
Net changes	2,197	319
Ending balance	\$ 4,911	\$ 2,714

The following table presents net charge-offs/ recoveries, which are defined as gross charge-offs less recoveries, for the years ended December 31, 2024 and 2023. The amounts in the table below do not include gains or losses from sales of nonaccrual loans recognized in noninterest revenue.

Wholesale net charge-offs/(recoveries)

Year ended December 31, (in millions, except ratios)	2024	2023
Loans		
Average loans retained	\$ 673,310	\$ 646,875
Gross charge-offs	1,022	1,011
Gross recoveries collected	(200)	(132)
Net charge-offs/(recoveries)	822	879
Net charge-off/(recovery) rate	0.12 %	0.14 %

Maturities and sensitivity to changes in interest rates

The table below sets forth wholesale loan maturities and the distribution between fixed and floating interest rates based on the stated terms of the loan agreements by loan class. Refer to Note 12 for further information on loan classes.

December 31, 2024 (in millions, except ratios)	1 year or less ^(b)	After 1 year through 5 years	After 5 years through 15 years	After 15 years	Total
Wholesale loans:					
Secured by real estate	\$ 12,474	\$ 57,125	\$ 57,967	\$ 42,597	\$ 170,163
Commercial and industrial	55,731	109,839	8,587	94	174,251
Other	182,722	150,346	36,281	8,555	377,904
Total wholesale loans	\$250,927	\$ 317,310	\$ 102,835	\$ 51,246	\$ 722,318
Loans due after one year at fixed interest rates					
Secured by real estate		\$ 13,119	\$ 17,943	\$ 935	
Commercial and industrial		3,964	1,231	7	
Other		26,929	15,542	5,824	
Loans due after one year at variable interest rates^(a)					
Secured by real estate		\$ 44,006	\$ 40,024	\$ 41,662	
Commercial and industrial		105,875	7,356	87	
Other		123,417	20,739	2,731	
Total wholesale loans		\$ 317,310	\$ 102,835	\$ 51,246	

(a) Includes loans that have an initial fixed interest rate that resets to a variable rate as the variable rate will be the prevailing rate over the life of the loan.

(b) Includes loans held-for-sale, demand loans and overdrafts.

The following table presents net charge-offs/recoveries, average retained loans and net charge-off/recovery rate by loan class for the years ended December 31, 2024 and 2023.

(in millions, except ratios)	Year ended December 31,							
	Secured by real estate		Commercial and industrial		Other		Total	
	2024	2023	2024	2023	2024	2023	2024	2023
Net charge-offs/(recoveries)	\$ 313	\$ 178	\$ 381	\$ 370	\$ 128	\$ 331	\$ 822	\$ 879
Average retained loans	162,653	151,214	169,363	170,503	341,294	325,158	673,310	646,875
Net charge-off/(recovery) rate	0.19 %	0.12 %	0.22 %	0.22 %	0.04 %	0.10 %	0.12 %	0.14 %

Management's discussion and analysis

Lending-related commitments

The Firm uses lending-related financial instruments, such as commitments (including revolving credit facilities) and guarantees, to address the financing needs of its clients. The contractual amounts of these financial instruments represent the maximum possible credit risk should the clients draw down on these commitments or when the Firm fulfills its obligations under these guarantees, and the clients subsequently fail to perform according to the terms of these contracts. Most of these commitments and guarantees have historically been refinanced, extended, cancelled, or expired without being drawn upon or a default occurring. As a result, the Firm does not believe that the total contractual amount of these wholesale lending-related commitments is representative of the Firm's expected future credit exposure or funding requirements. Refer to Note 28 for further information on wholesale lending-related commitments.

Receivables from customers

Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM that are collateralized by assets maintained in the clients' brokerage accounts (including cash on deposit, and primarily liquid and readily marketable debt or equity securities). To manage its credit risk, the Firm establishes margin requirements and monitors the required margin levels on an ongoing basis, and requires clients to deposit additional cash or other collateral, or to reduce positions, when appropriate. Credit risk arising from lending activities subject to collateral maintenance requirements is generally mitigated by factors such as the short-term nature of the activity, the fair value of collateral held and the Firm's right to call for, and the borrower's obligation to provide, additional margin when the fair value of the collateral declines. Because of these mitigating factors, these receivables generally do not require an allowance for credit losses. However, if in management's judgment, an allowance for credit losses is required, the Firm estimates expected credit losses based on the value of the collateral and probability of borrower default. These receivables are reported within accrued interest and accounts receivable on the Firm's Consolidated balance sheets.

Refer to Note 13 for further information on the Firm's accounting policies for the allowance for credit losses.

Derivative contracts

Derivatives enable clients and counterparties to manage risk, including credit risk and risks arising from fluctuations in interest rates, foreign exchange and equities and commodities prices. The Firm makes markets in derivatives in order to meet these needs and uses derivatives to manage certain risks associated with net open risk positions from its market-making activities, including the counterparty credit risk arising from derivative receivables. The Firm also uses derivative instruments to manage its own credit risk and other market risk exposure. The nature of the counterparty and the settlement mechanism of the

derivative affect the credit risk to which the Firm is exposed. For over-the-counter ("OTC") derivatives, the Firm is exposed to the credit risk of the derivative counterparty. For exchange-traded derivatives ("ETD"), such as futures and options, and cleared over-the-counter ("OTC-cleared") derivatives, the Firm can also be exposed to the credit risk of the relevant CCP. Where possible, the Firm seeks to mitigate its credit risk exposures arising from derivative contracts through the use of legally enforceable master netting arrangements and collateral agreements. The percentage of the Firm's OTC derivative transactions subject to collateral agreements — excluding foreign exchange spot trades, which are not typically covered by collateral agreements due to their short maturity and centrally cleared trades that are settled daily — was approximately 86% and 87% at December 31, 2024 and 2023, respectively. Refer to Note 5 for additional information on the Firm's use of collateral agreements and for a further discussion of derivative contracts, counterparties and settlement types.

The fair value of derivative receivables reported on the Consolidated balance sheets was \$61.0 billion and \$54.9 billion at December 31, 2024 and 2023, respectively. The increase was primarily as a result of market movements. Derivative receivables represent the fair value of the derivative contracts after giving effect to legally enforceable master netting agreements and the related cash collateral held by the Firm.

In addition, the Firm holds liquid securities and other cash collateral that may be used as security when the fair value of the client's exposure is in the Firm's favor. For these purposes, the definition of liquid securities is consistent with the definition of high quality liquid assets as defined in the LCR rule.

In management's view, the appropriate measure of current credit risk should also take into consideration other collateral, which generally represents securities that do not qualify as high quality liquid assets under the LCR rule. The benefits of these additional collateral amounts for each counterparty are subject to a legally enforceable master netting agreement and limited to the net amount of the derivative receivables for each counterparty.

The Firm also holds additional collateral (primarily cash, G7 government securities, other liquid government agency and guaranteed securities, and corporate debt and equity securities) delivered by clients at the initiation of transactions, as well as collateral related to contracts that have a non-daily call frequency and collateral that the Firm has agreed to return but has not yet settled as of the reporting date. Although this collateral does not reduce the receivables balances and is not included in the tables below, it is available as security against potential exposure that could arise should the fair value of the client's derivative contracts move in the Firm's favor. Refer to Note 5 for additional information on the Firm's use of collateral agreements for derivative transactions.

The following tables summarize the net derivative receivables and the internal ratings profile for the periods presented.

Derivative receivables

December 31, (in millions)	2024		2023	
Total, net of cash collateral	\$	60,967	\$	54,864
Liquid securities and other cash collateral held against derivative receivables		(28,160)		(22,461)
Total, net of liquid securities and other cash collateral	\$	32,807	\$	32,403
Other collateral held against derivative receivables		(1,021)		(993)
Total, net of collateral	\$	31,786	\$	31,410

Ratings profile of derivative receivables

December 31, (in millions, except ratios)	2024		2023	
	Exposure net of collateral	% of exposure net of collateral	Exposure net of collateral	% of exposure net of collateral
Investment-grade	\$ 23,783	75 %	\$ 24,004	76 %
Noninvestment-grade	8,003	25	7,406	24
Total	\$ 31,786	100 %	\$ 31,410	100 %

While useful as a current view of credit exposure, the net fair value of the derivative receivables does not capture the potential future variability of that credit exposure. To capture this variability, the Firm calculates, on a client-by-client basis, three measures of potential derivatives-related credit loss: Peak, Derivative Risk Equivalent (“DRE”), and Average exposure (“AVG”). These measures all incorporate netting and collateral benefits, where applicable.

Peak represents a conservative measure of potential derivative exposure, including the benefit of collateral, to a counterparty calculated in a manner that is broadly equivalent to a 97.5% confidence level over the life of the transaction. Peak is the primary measure used by the Firm for setting credit limits for derivative contracts, senior management reporting and derivatives exposure management.

DRE exposure is a measure that expresses the risk of derivative exposure, including the benefit of collateral, on a basis intended to be equivalent to the risk of loan exposures. DRE is a less extreme measure of potential credit loss than Peak.

Finally, AVG is a measure of the expected fair value of the Firm’s derivative exposures, including the benefit of collateral, at future time periods. AVG over the total life of the derivative contract is used as the primary metric for pricing purposes and is used to calculate credit risk capital and CVA, as further described below.

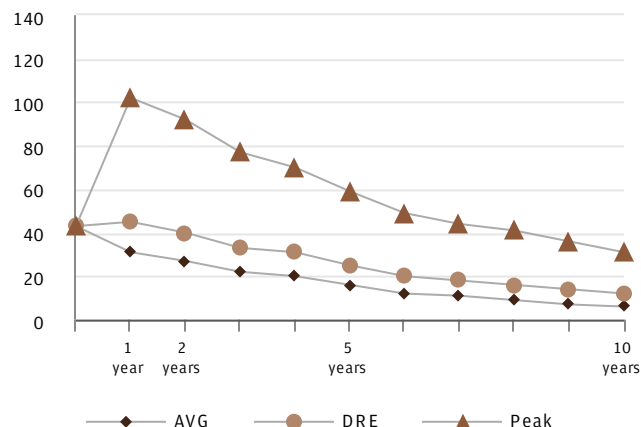
The fair value of the Firm’s derivative receivables incorporates CVA to reflect the credit quality of counterparties. CVA is based on the Firm’s AVG to a counterparty and the counterparty’s credit spread in the credit derivatives market. The Firm believes that active risk management is essential to controlling the dynamic credit risk in the derivatives portfolio. In addition, the Firm’s risk management process for derivatives exposures takes into consideration the potential impact of wrong-way risk, which is broadly

defined as the risk that exposure to a counterparty is positively correlated with the impact of a default by the same counterparty, which could cause exposure to increase at the same time as the counterparty’s capacity to meet its obligations is decreasing. Many factors may influence the nature and magnitude of these correlations over time. To the extent that these correlations are identified, the Firm may adjust the CVA associated with a particular counterparty’s AVG. The Firm risk manages exposure to changes in CVA by entering into credit derivative contracts, as well as interest rate, foreign exchange, equity and commodity derivative contracts.

The below graph shows exposure profiles to the Firm’s current derivatives portfolio over the next 10 years as calculated by the Peak, DRE and AVG metrics. The three measures generally show that exposure will decline after the first year, if no new trades are added to the portfolio.

Exposure profile of derivatives measures

December 31, 2024
(in billions)



Management's discussion and analysis

Credit derivatives

The Firm uses credit derivatives for two primary purposes: first, in its capacity as a market-maker, and second, as an end-user to manage the Firm's own credit risk associated with various exposures.

Credit portfolio management activities

Included in the Firm's end-user activities are credit derivatives used to mitigate the credit risk associated with traditional lending activities (loans and lending-related commitments) and derivatives counterparty exposure in the Firm's wholesale businesses (collectively, "credit portfolio management activities"). Information on credit portfolio management activities is provided in the table below.

The Firm also uses credit derivatives as an end-user to manage other exposures, including credit risk arising from certain securities held in the Firm's market-making businesses. These credit derivatives are not included in credit portfolio management activities.

Credit derivatives and credit-related notes used in credit portfolio management activities

December 31, (in millions)	Notional amount of protection purchased and sold ^(a)	
	2024	2023
Credit derivatives and credit-related notes used to manage:		
Loans and lending-related commitments	\$ 25,216	\$ 24,157
Derivative receivables	15,672	12,832
Credit derivatives and credit-related notes used in credit portfolio management activities	\$ 40,888	\$ 36,989

(a) Amounts are presented net, considering the Firm's net protection purchased or sold with respect to each underlying reference entity or index.

The credit derivatives used in credit portfolio management activities do not qualify for hedge accounting under U.S. GAAP; these derivatives are reported at fair value, with gains and losses recognized in principal transactions revenue. In contrast, the loans and lending-related commitments being risk-managed are accounted for on an accrual basis. This asymmetry in accounting treatment, between loans and lending-related commitments and the credit derivatives used in credit portfolio management activities, causes earnings volatility that is not representative, in the Firm's view, of the true changes in value of the Firm's overall credit exposure.

The effectiveness of credit default swaps ("CDS") as a hedge against the Firm's exposures may vary depending on a number of factors, including the named reference entity (i.e., the Firm may experience losses on specific exposures that are different than the named reference entities in the purchased CDS); the contractual terms of the CDS (which may have a defined credit event that does not align with an actual loss realized by the Firm); and the maturity of the Firm's CDS protection (which in some cases may be shorter than the Firm's exposures). However, the Firm generally seeks to purchase credit protection with a maturity date that is the same or similar to the maturity date of the exposure for which the protection was purchased, and remaining differences in maturity are actively monitored and managed by the Firm. Refer to Credit derivatives in Note 5 for further information on credit derivatives and derivatives used in credit portfolio management activities.

ALLOWANCE FOR CREDIT LOSSES

The Firm's allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The Firm's allowance for credit losses generally consists of:

- the allowance for loan losses, which covers the Firm's retained loan portfolios (scored and risk-rated) and is presented separately on the Consolidated balance sheets,
- the allowance for lending-related commitments, which is reflected in accounts payable and other liabilities on the Consolidated balance sheets, and
- the allowance for credit losses on investment securities, which is reflected in investment securities on the Consolidated balance sheets.

Discussion of changes in the allowance

The allowance for credit losses as of December 31, 2024 was \$26.9 billion, reflecting a net addition of \$2.1 billion from December 31, 2023.

The net addition to the allowance for credit losses included:

- \$2.1 billion in **consumer**, reflecting:
 - a \$2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years, partially offset by
 - a \$125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of \$30 million in **wholesale**, reflecting:
 - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans, predominantly offset by
 - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

The Firm's qualitative adjustments continued to include additional weight placed on the adverse scenarios to reflect ongoing uncertainties and downside risks related to the geopolitical and macroeconomic environment.

The Firm's allowance for credit losses is estimated using a weighted average of five internally developed macroeconomic scenarios. The adverse scenarios incorporate more punitive macroeconomic factors than the central case assumptions provided in the table below, resulting in:

- a weighted average U.S. unemployment rate peaking at 5.5% in the fourth quarter of 2025, and
- a weighted average U.S. real GDP level that is 1.9% lower than the central case at the end of the second quarter of 2026.

The following table presents the Firm's central case assumptions for the periods presented:

	Central case assumptions at December 31, 2024		
	2Q25	4Q25	2Q26
U.S. unemployment rate ^(a)	4.5 %	4.3 %	4.3 %
YoY growth in U.S. real GDP ^(b)	2.0 %	1.9 %	1.8 %
	Central case assumptions at December 31, 2023		
	2Q24	4Q24	2Q25
U.S. unemployment rate ^(a)	4.1 %	4.4 %	4.1 %
YoY growth in U.S. real GDP ^(b)	1.8 %	0.7 %	1.0 %

(a) Reflects quarterly average of forecasted U.S. unemployment rate.

(b) The year over year growth in U.S. real GDP in the forecast horizon of the central scenario is calculated as the percentage change in U.S. real GDP levels from the prior year.

Subsequent changes to this forecast and related estimates will be reflected in the provision for credit losses in future periods.

Refer to Consumer Credit Portfolio on pages 120–125, Wholesale Credit Portfolio on pages 126–136 and Note 12 for additional information on the consumer and wholesale credit portfolios.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 for further information on the allowance for credit losses and related management judgments.

Management's discussion and analysis

Allowance for credit losses and related information

Year ended December 31, (in millions, except ratios)	2024				2023			
	Consumer, excluding credit card	Credit card	Wholesale	Total	Consumer, excluding credit card	Credit card	Wholesale	Total
Allowance for loan losses								
Beginning balance at January 1,	\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420	\$ 2,040	\$ 11,200	\$ 6,486	\$ 19,726
Cumulative effect of a change in accounting principle ^(a)	NA	NA	NA	NA	(489)	(100)	2	(587)
Gross charge-offs	1,299	8,198	1,022	10,519	1,151	5,491	1,011	7,653
Gross recoveries collected	(625)	(1,056)	(200)	(1,881)	(519)	(793)	(132)	(1,444)
Net charge-offs	674	7,142	822	8,638	632	4,698	879	6,209
Provision for loan losses	624	9,292	578	10,494	936	6,048	2,484	9,468
Other	1	—	68	69	1	—	21	22
Ending balance at December 31,	\$ 1,807	\$ 14,600	\$ 7,938	\$ 24,345	\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420
Allowance for lending-related commitments								
Beginning balance at January 1,	\$ 75	\$ —	\$ 1,899	\$ 1,974	\$ 76	\$ —	\$ 2,306	\$ 2,382
Provision for lending-related commitments	7	—	121	128	(1)	—	(407)	(408)
Other	—	—	(1)	(1)	—	—	—	—
Ending balance at December 31,	\$ 82	\$ —	\$ 2,019	\$ 2,101	\$ 75	\$ —	\$ 1,899	\$ 1,974
Impairment methodology								
Asset-specific ^(b)	\$ (728)	\$ —	\$ 526	\$ (202)	\$ (876)	\$ —	\$ 392	\$ (484)
Portfolio-based	2,535	14,600	7,412	24,547	2,732	12,450	7,722	22,904
Total allowance for loan losses	\$ 1,807	\$ 14,600	\$ 7,938	\$ 24,345	\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420
Impairment methodology								
Asset-specific	\$ —	\$ —	\$ 109	\$ 109	\$ —	\$ —	\$ 89	\$ 89
Portfolio-based	82	—	1,910	1,992	75	—	1,810	1,885
Total allowance for lending-related commitments	\$ 82	\$ —	\$ 2,019	\$ 2,101	\$ 75	\$ —	\$ 1,899	\$ 1,974
Total allowance for investment securities	NA	NA	NA	\$ 152	NA	NA	NA	\$ 128
Total allowance for credit losses^(c)	\$ 1,889	\$ 14,600	\$ 9,957	\$ 26,598	\$ 1,931	\$ 12,450	\$ 10,013	\$ 24,522
Memo:								
Retained loans, end of period	\$376,334	\$232,860	\$690,396	\$1,299,590	\$397,275	\$ 211,123	\$ 672,472	\$1,280,870
Retained loans, average	384,001	214,033	673,310	1,271,344	364,061	191,412	646,875	1,202,348
Credit ratios								
Allowance for loan losses to retained loans	0.48 %	6.27 %	1.15 %	1.87 %	0.47 %	5.90 %	1.21 %	1.75 %
Allowance for loan losses to retained nonaccrual loans ^(d)	56	NA	201	339	51	NA	346	374
Allowance for loan losses to retained nonaccrual loans excluding credit card	56	NA	201	136	51	NA	346	166
Net charge-off rates	0.18	3.34	0.12	0.68	0.17	2.45	0.14	0.52

(a) Represents the impact to the allowance for loan losses upon the adoption of the Financial Instruments - Credit Losses: Troubled Debt Restructurings accounting guidance. Refer to Note 1 for further information.

(b) Includes collateral-dependent loans, including those for which foreclosure is deemed probable, and nonaccrual risk-rated loans.

(c) At December 31, 2024 and 2023, in addition to the allowance for credit losses in the table above, the Firm also had an allowance for credit losses of \$268 million and \$243 million, respectively, associated with certain accounts receivable in CIB.

(d) The Firm's policy is generally to exempt credit card loans from being placed on nonaccrual status as permitted by regulatory guidance.

Allocation of allowance for loan losses

The table below presents a breakdown of the allowance for loan losses by loan class. Refer to Note 12 for further information on loan classes.

December 31, (in millions, except ratios)	2024		2023	
	Allowance for loan losses	Percent of retained loans to total retained loans	Allowance for loan losses	Percent of retained loans to total retained loans
Residential real estate	\$ 666	24 %	\$ 817	25 %
Auto and other	1,141	5	1,039	6
Consumer, excluding credit card	1,807	29	1,856	31
Credit card	14,600	18	12,450	16
Total consumer	16,407	47	14,306	47
Secured by real estate	2,978	12	2,997	13
Commercial and industrial	3,350	13	3,519	13
Other	1,610	28	1,598	27
Total wholesale	7,938	53	8,114	53
Total	\$ 24,345	100 %	\$ 22,420	100 %

INVESTMENT PORTFOLIO RISK MANAGEMENT

Investment portfolio risk is the risk associated with the loss of principal or a reduction in expected returns on investments arising from the investment securities portfolio or from principal investments. The investment securities portfolio is predominantly held by Treasury and CIO in connection with the Firm's balance sheet and asset-liability management objectives. Principal investments are predominantly privately-held financial instruments and are managed in the LOBs and Corporate. Investments are typically intended to be held over extended periods and, accordingly, the Firm has no expectation for short-term realized gains with respect to these investments.

Investment securities risk

Investment securities risk includes the exposure associated with a default in the payment of principal and interest. This risk is mitigated given that the investment securities portfolio held by Treasury and CIO predominantly consists of high-quality securities. At December 31, 2024, the Treasury and CIO investment securities portfolio, net of the allowance for credit losses, was \$678.3 billion, and the average credit rating of the securities comprising the portfolio was AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings). Refer to Corporate results on pages 88–90 and Note 10 for further information on the investment securities portfolio and internal risk ratings. Refer to Liquidity Risk Management on pages 108–115 for further information on related liquidity risk. Refer to Market Risk Management on pages 141–149 for further information on the market risk inherent in the portfolio.

Governance and oversight

Investment securities risks are governed by the Firm's Risk Appetite framework, and reviewed at the CTC Risk Committee with regular updates provided to the Board Risk Committee.

The Firm's independent control functions are responsible for reviewing the appropriateness of the carrying value of investment securities in accordance with relevant policies. Approved levels for investment securities are established for each risk category, including capital and credit risks.

Principal investment risk

Principal investments are typically privately-held financial instruments representing ownership interests or other forms of junior capital. In general, principal investments include tax-oriented investments and investments made to enhance or accelerate the Firm's business strategies and exclude those that are consolidated on the Firm's balance sheets. These investments are made by dedicated investing businesses or as part of a broader business strategy. The Firm's principal investments are managed by the LOBs and Corporate and are reflected within their respective financial results. The Firm's investments will continue to evolve based on market circumstances and in line with its strategic initiatives.

The table below presents the aggregate carrying values of the principal investment portfolios as of December 31, 2024 and 2023.

(in billions)	December 31, 2024	December 31, 2023
Tax-oriented investments, primarily in alternative energy and affordable housing ^(a)	\$ 33.3	\$ 28.8
Private equity, various debt and equity instruments, and real assets	9.1	10.5
Total carrying value	\$ 42.4	\$ 39.3

(a) Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance. Refer to Notes 1, 6, 14 and 25 for additional information.

Governance and oversight

The Firm's approach to managing principal investment risk is consistent with the Firm's risk governance structure. The Firm has established a Firmwide risk policy framework for all principal investing activities that includes approval by executives who are independent from the investing businesses, as appropriate.

The Firm's independent control functions are responsible for reviewing the appropriateness of the carrying value of investments in accordance with relevant policies. As part of the risk governance structure, approved levels for investments are established and monitored for each relevant business or segment in order to manage the overall size of the portfolios. The Firm also conducts stress testing on these portfolios using specific scenarios that estimate losses based on significant market moves and/or other risk events.

MARKET RISK MANAGEMENT

Market risk is the risk associated with the effect of changes in market factors such as interest and foreign exchange rates, equity and commodity prices, credit spreads or implied volatilities, on the value of assets and liabilities held for both the short and long term.

Market Risk Management

Market Risk Management monitors market risks throughout the Firm and defines market risk policies and procedures.

Market Risk Management seeks to facilitate efficient risk/return decisions, reduce volatility in operating performance and provide transparency into the Firm's market risk profile for senior management, the Board of Directors and regulators. Market Risk Management is responsible for the following functions:

- Maintaining a market risk policy framework
- Independently measuring and monitoring LOB, Corporate, and Firmwide market risk
- Defining, approving and monitoring limits
- Performing stress testing and qualitative risk assessments

Risk measurement

Measures used to capture market risk

There is no single measure to capture market risk and therefore Market Risk Management uses various metrics, both statistical and nonstatistical, to assess risk including:

- Value-at-risk
- Stress testing
- Profit and loss drawdowns
- Earnings-at-risk
- Economic Value Sensitivity
- Other sensitivity-based measures

Risk monitoring and control

Market risk exposure is managed primarily through a series of limits set in the context of the market environment and business strategy. In setting limits, Market Risk Management takes into consideration factors such as market volatility, product liquidity, accommodation of client business, and management judgment. Market Risk Management maintains different levels of limits. Firm level limits include VaR and stress limits. Similarly, LOB and Corporate limits include VaR and stress limits and may be supplemented by certain nonstatistical risk measures such as profit and loss drawdowns. Limits may also be set within the LOBs and Corporate, as well as at the legal entity level.

Market Risk Management sets limits and regularly reviews and updates them as appropriate. Senior management is responsible for reviewing and approving certain of these risk limits on an ongoing basis. Limits that have not been reviewed within specified time periods by Market Risk Management are reported to senior management. The LOBs and Corporate are responsible for adhering to established limits against which exposures are monitored and reported.

Limit breaches are required to be reported in a timely manner to limit approvers, which include Market Risk Management and senior management. In the event of a breach, Market Risk Management consults with senior members of appropriate groups within the Firm to determine the suitable course of action required to return the applicable positions to compliance, which may include a reduction in risk in order to remedy the breach or granting a temporary increase in limits to accommodate an expected increase in client activity and/or market volatility. Firm, Corporate or LOB-level limit breaches are escalated as appropriate.

Models used to measure market risk are inherently imprecise and are limited in their ability to measure certain risks or to predict losses. This imprecision may be heightened when sudden or severe shifts in market conditions occur. For additional discussion on model uncertainty refer to Estimations and Model Risk Management on page 160.

Market Risk Management periodically reviews the Firm's existing market risk measures to identify opportunities for enhancement, and to the extent appropriate, will calibrate those measures accordingly over time.

Management's discussion and analysis

The following table summarizes the predominant business activities and related market risks, as well as positions which give rise to market risk and certain measures used to capture those risks, for each LOB and Corporate.

In addition to the predominant business activities, each LOB and Corporate may engage in principal investing activities. To the extent principal investments are deemed market risk sensitive, they are reflected in relevant risk measures and captured in the table below. Refer to Investment Portfolio Risk Management on page 140 for additional discussion on principal investments.

LOBs and Corporate	Predominant business activities	Related market risks	Positions included in Risk Management VaR	Positions included in earnings-at-risk	Positions included in other sensitivity-based measures
CCB	- Originates and services mortgage loans - Originates loans and takes deposits	- Risk from changes in the probability of newly originated mortgage commitments closing - Interest rate risk and prepayment risk	- Mortgage commitments, classified as derivatives - Warehouse loans that are fair value option elected, classified as loans – debt instruments - MSRs - Hedges of mortgage commitments, warehouse loans and MSRs, classified as derivatives - Interest-only and mortgage-backed securities, classified as trading assets debt instruments, and related hedges, classified as derivatives - Fair value option elected liabilities^(b)	- Retained and held-for-sale loan portfolios - Deposits	Fair value option elected liabilities DVA^(b)
CIB^(a)	- Makes markets and services clients across fixed income, foreign exchange, equities and commodities - Originates loans and takes deposits	- Risk of loss from adverse movements in market prices and implied volatilities across interest rate, foreign exchange, credit, commodity and equity instruments - Basis and correlation risk from changes in the way asset values move relative to one another - Interest rate risk and prepayment risk	- Trading assets/liabilities – debt and marketable equity instruments, and derivatives, including hedges of the retained loan portfolio - Certain securities purchased, loaned or sold under resale agreements and securities borrowed - Fair value option elected liabilities^(b) - Certain fair value option elected loans - Derivative CVA and associated hedges - Marketable equity investments	- Retained and held-for-sale loan portfolios - Deposits	- Privately held equity and other investments measured at fair value; and certain real estate-related fair value option elected loans - Derivatives FVA and fair value option elected liabilities DVA^(b) - Credit risk component of CVA and associated hedges for counterparties with credit spreads that have widened to elevated levels
AWM	- Provides initial capital investments in products such as mutual funds and capital invested alongside third-party investors - Originates loans and takes deposits	- Risk from adverse movements in market factors (e.g., market prices, rates and credit spreads) - Interest rate risk and prepayment risk	- Debt securities held in advance of distribution to clients, classified as trading assets - debt instruments - Trading assets/liabilities - derivatives that hedge the retained loan portfolio	- Retained and held-for-sale loan portfolios - Deposits	- Initial seed capital investments and related hedges, classified as derivatives - Certain deferred compensation and related hedges, classified as derivatives - Capital invested alongside third-party investors, typically in privately distributed collective vehicles managed by AWM (i.e., co-investments), as well as in third-party funds
Corporate	Manages the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks	- Structural interest rate risk from the Firm's traditional banking activities - Structural non-USD foreign exchange risks	- Derivative positions measured through noninterest revenue in earnings - Marketable equity investments	- Deposits with banks and financing activities - Investment securities portfolio and related interest rate hedges - Cash flow hedges on retained loan portfolios in the LOBs - Long-term and short-term funding and related interest rate hedges - Deposits - Foreign exchange hedges of non-U.S. dollar capital investments	- Privately held equity and other investments measured at fair value - Foreign exchange exposure related to Firm-issued non-USD long-term debt ("LTD") and related hedges

(a) Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank ("CIB"). Refer to Business Segment & Corporate Results on pages 70–90 for additional information.

(b) Reflects structured notes in Risk Management VaR and the DVA on structured notes in other sensitivity-based measures.

Value-at-risk

JPMorganChase utilizes value-at-risk (“VaR”), a statistical risk measure, to estimate the potential loss from adverse market moves in the current market environment. The Firm has a single VaR framework used as a basis for calculating Risk Management VaR and Regulatory VaR.

The framework is employed across the Firm using historical simulation based on data for the previous 12 months. The framework’s approach assumes that historical changes in market values are representative of the distribution of potential outcomes in the immediate future. The Firm believes the use of Risk Management VaR provides a daily measure of risk that is closely aligned to risk management decisions made by the LOBs and Corporate and, along with other market risk measures, provides the appropriate information needed to respond to risk events.

The Firm’s Risk Management VaR is calculated assuming a one-day holding period and an expected tail-loss methodology which approximates a 95% confidence level. Risk Management VaR provides a consistent framework to measure risk profiles and levels of diversification across product types and is used for aggregating risks and monitoring limits across businesses. VaR results are reported as appropriate to various groups including senior management, the Board Risk Committee and regulators.

Underlying the overall VaR model framework are individual VaR models that simulate historical market returns for individual risk factors and/or product types. To capture material market risks as part of the Firm’s risk management framework, comprehensive VaR model calculations are performed daily for businesses whose activities give rise to market risk. These VaR models are granular and incorporate numerous risk factors and inputs to simulate daily changes in market values over the historical period; inputs are selected based on the risk profile of each portfolio, as sensitivities and historical time series used to generate daily market values may be different across product types or risk management systems. The VaR model results across all portfolios are aggregated at the Firm level.

As VaR is based on historical data, it is an imperfect measure of market risk exposure and potential future losses. In addition, based on their reliance on available historical data, limited time horizons, and other factors, VaR measures are inherently limited in their ability to measure certain risks and to predict losses, particularly those associated with market illiquidity and sudden or severe shifts in market conditions.

For certain products, specific risk parameters are not captured in VaR due to the lack of liquidity and availability of appropriate historical data. The Firm uses proxies to estimate the VaR for these and other products when daily time series are not available. It is likely that using an actual price-based time series for these products, if available, would affect the VaR results presented. The Firm therefore considers other nonstatistical measures such as stress testing, in addition to VaR, to capture and manage its market risk positions.

As VaR model calculations require daily data and a consistent source for valuation, the daily market data used may be different than the independent third-party data collected for VCG price testing in its monthly valuation process. For example, in cases where market prices are not observable, or where proxies are used in VaR historical time series, the data sources may differ. Refer to Valuation process in Note 2 for further information on the Firm’s valuation process.

The Firm’s VaR model calculations are periodically evaluated and enhanced in response to changes in the composition of the Firm’s portfolios, changes in market conditions, improvements in the Firm’s modeling techniques and measurements, and other factors. Such changes may affect historical comparisons of VaR results. Refer to Estimations and Model Risk Management on page 160 for information regarding model reviews and approvals.

The Firm calculates separately a daily aggregated VaR in accordance with regulatory rules (“Regulatory VaR”), which is used to derive the Firm’s regulatory VaR-based capital requirements under Basel III capital rules. This Regulatory VaR model framework currently assumes a ten business-day holding period and an expected tail-loss methodology which approximates a 99% confidence level. Regulatory VaR is applied to “covered” positions as defined by Basel III capital rules, which may be different than the positions included in the Firm’s Risk Management VaR. For example, credit derivative hedges of accrual loans are included in the Firm’s Risk Management VaR, while Regulatory VaR excludes these credit derivative hedges. In addition, in contrast to the Firm’s Risk Management VaR, Regulatory VaR currently excludes the diversification benefit for certain VaR models.

Refer to JPMorganChase’s Basel III Pillar 3 Regulatory Capital Disclosures reports, which are available on the Firm’s website, for additional information on Regulatory VaR and the other components of market risk regulatory capital for the Firm (e.g., VaR-based measure, stressed VaR-based measure and the respective backtesting).

Management's discussion and analysis

The table below shows the results of the Firm's Risk Management VaR measure using a 95% confidence level. VaR can vary significantly as positions change, market volatility fluctuates, and diversification benefits change.

Total VaR

As of or for the year ended December 31, (in millions)	2024			2023		
	Avg.	Min	Max	Avg.	Min	Max
CIB trading VaR by risk type^(a)						
Fixed income	\$ 34	\$ 26	\$ 53	\$ 49	\$ 31	\$ 71
Foreign exchange	15	7	23	12	6	26
Equities	8	4	15	7	3	11
Commodities and other	8	6	13	11	6	19
Diversification benefit to CIB trading VaR ^(b)	(32)	NM	NM	(42)	NM	NM
CIB trading VaR	33	27	42	37	24	55
Credit Portfolio VaR ^(c)	22	18	28	14	8	26
Diversification benefit to CIB VaR ^(b)	(16)	NM	NM	(11)	NM	NM
CIB VaR	39	27	52	40	23	58
CCB VaR	3	1	6	7	1	15
AWM VaR ^(d)	9	5	10	1	—	10
Corporate VaR ^{(d)(e)}	23	7	102	12	9	17
Diversification benefit to other VaR ^(b)	(10)	NM	NM	(6)	NM	NM
Other VaR	25	10	101	14	9	22
Diversification benefit to CIB and other VaR ^(b)	(17)	NM	NM	(11)	NM	NM
Total VaR	\$ 47	\$ 30	\$ 91	\$ 43	\$ 26	\$ 57

- (a) The impact of the business segment reorganization in the second quarter of 2024 was not material to Total CIB VaR. Prior periods have not been revised. Refer to Business Segment & Corporate Results on pages 70–90 for additional information.
- (b) Diversification benefit represents the difference between the portfolio VaR and the sum of its individual components. This reflects the non-additive nature of VaR due to imperfect correlation across LOBs, Corporate, and risk types. For maximum and minimum VaR, diversification benefit is not meaningful as the maximum and minimum VaR for each portfolio may have occurred on different trading days than the components.
- (c) Includes the derivative CVA, hedges of the CVA and credit protection purchased against certain retained loans and lending-related commitments, which are reported in principal transactions revenue. This VaR does not include the retained loan portfolio, which is not reported at fair value. In line with the Firm's internal model governance, the credit risk component of CVA related to certain counterparties was removed from Credit Portfolio VaR due to the widening of the credit spreads for those counterparties to elevated levels. The related hedges were also removed to maintain consistency. This exposure is now reflected in other sensitivity-based measures.
- (d) In the second quarter of 2024, the presentation of Corporate and other LOB VaR was updated to disaggregate AWM VaR due to the increase associated with credit protection purchased against certain retained loans and lending-related commitments. The VaR does not include the retained loan portfolio, which is not reported at fair value.
- (e) Includes a legacy private equity position which is publicly traded, as well as Visa C shares which the Firm disposed of in the second and third quarters of 2024. The impact of Visa C shares resulted in elevated average and maximum Corporate VaR, Other VaR and Total VaR. Refer to Executive Overview on pages 54–58 for additional information.

2024 compared with 2023

Average Total VaR increased by \$4 million for the year ended December 31, 2024 when compared with the prior year. The increase was predominantly driven by the impact of the Firm's receipt of Visa C shares on Corporate VaR and increases associated with credit

protection purchased against certain retained loans and lending-related commitments within Credit Portfolio VaR and AWM VaR, largely offset by market volatility rolling out of the one-year historical look-back period impacting the Fixed income risk type.

The following graph presents daily Risk Management VaR for the four trailing quarters. The increase in VaR and subsequent decline observed in the second quarter of 2024 was primarily driven by changes in Visa C share exposure in the Firm's Corporate VaR.



VaR backtesting

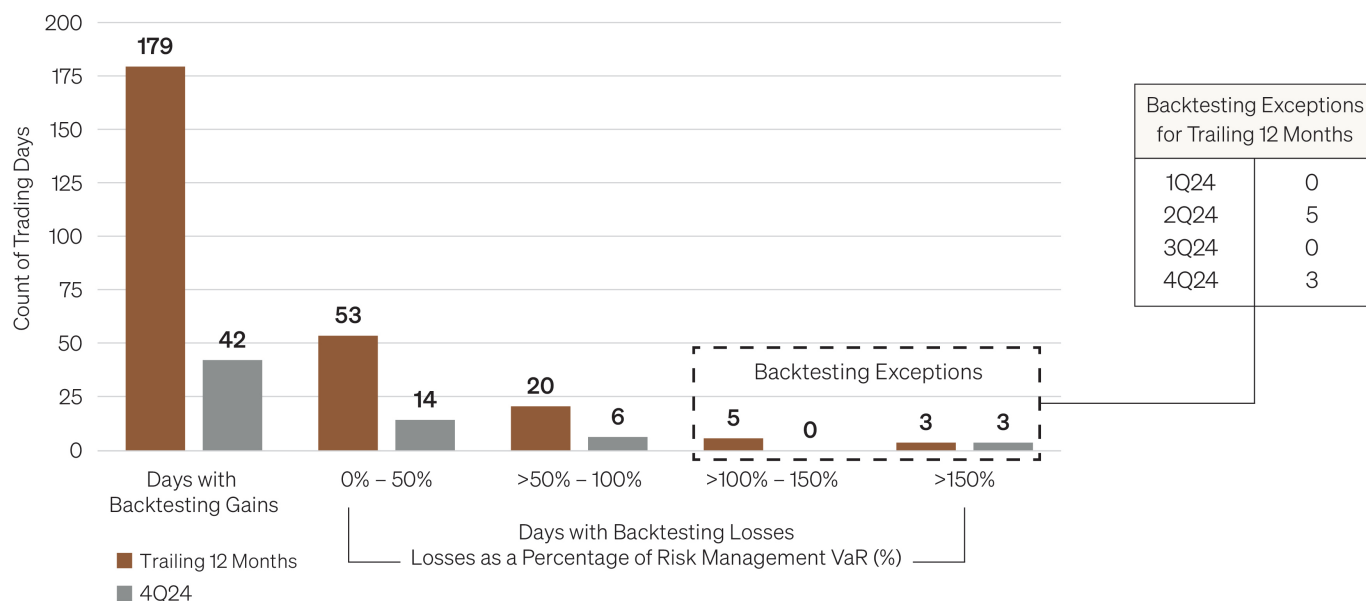
The Firm performs daily VaR model backtesting, which compares the daily Risk Management VaR results with the daily gains and losses that are utilized for VaR backtesting purposes. The gains and losses depicted in the chart below do not reflect the Firm's reported revenue as they exclude certain components of total net revenue, such as those associated with the execution of new transactions (i.e., intraday client-driven trading and intraday risk management activities), fees, commissions, other valuation adjustments and net interest income. These excluded components of total net revenue may more than offset the backtesting gain or loss on a particular day. The definition of backtesting gains and losses above is consistent with the requirements for backtesting under Basel III capital rules.

A backtesting exception occurs when the daily backtesting loss exceeds the daily Risk Management VaR for the prior day. Under the Firm's Risk Management VaR methodology, assuming current changes in market values are consistent with the historical changes used in the simulation, the Firm would expect to incur VaR backtesting exceptions five times every 100 trading days on average. The number of VaR backtesting exceptions observed can differ from the statistically expected number of backtesting exceptions if the current level of market volatility is materially different from the level of market volatility during the 12 months of historical data used in the VaR calculation.

For the 12 months ended December 31, 2024, the Firm posted backtesting gains on 179 of the 260 days, and observed eight VaR backtesting exceptions, of which three were in the three months ended December 31, 2024. Firmwide backtesting loss days can differ from the loss days for which Fixed Income Markets and Equity Markets posted losses, as disclosed in CIB Markets revenue, as the population of positions which comprise each metric are different and due to the exclusion of certain components of total net revenue in backtesting gains and losses as described above.

The following chart presents the distribution of Firmwide daily backtesting gains and losses for the trailing 12 months and three months ended December 31, 2024. The daily backtesting losses are displayed as a percentage of the corresponding daily Risk Management VaR. The count of days with backtesting losses are shown in aggregate, in fifty percentage point intervals. Backtesting exceptions are displayed within the intervals that are greater than one hundred percent. The results in the chart below differ from the results of backtesting disclosed in the Market Risk section of the Firm's Basel III Pillar 3 Regulatory Capital Disclosures reports, which are based on Regulatory VaR applied to the Firm's covered positions.

Distribution of Daily Backtesting Gains and Losses



Other risk measures

Stress testing

Along with VaR, stress testing is an important tool used to assess risk. While VaR reflects the risk of loss due to adverse changes in markets using recent historical market behavior, stress testing reflects the risk of loss from hypothetical changes in the value of market risk sensitive positions applied simultaneously. Stress testing measures the Firm's vulnerability to losses under a range of stressed but possible economic and market scenarios. The results are used to understand the exposures responsible for those potential losses and are measured against limits.

The Firm's stress framework covers market risk sensitive positions in the LOBs and Corporate. The framework is used to calculate multiple magnitudes of potential stress for both market rallies and market sell-offs, assuming significant changes in market factors such as credit spreads, equity prices, interest rates, currency rates and commodity prices, and combines them in multiple ways to capture an array of hypothetical economic and market scenarios.

The Firm generates a number of scenarios that focus on tail events in specific asset classes and geographies, including how the event may impact multiple market factors simultaneously. Scenarios also incorporate specific idiosyncratic risks and stress basis risk between different products. The flexibility in the stress framework allows the Firm to construct new scenarios that can test the outcomes against possible future stress events. Stress testing results are reported periodically to senior management of the Firm, as appropriate.

Stress methodologies are governed by the overall stress framework, under the oversight of Market Risk Management. The Firmwide Market Risk Stress Methodology Committee reviews and approves changes to stress testing methodology and scenarios across the Firm. Significant changes to the framework are escalated to senior management, as appropriate. In addition, stress methodology and the models to calculate the stress results are subject to the Firm's Estimations and Model Risk Management Policy.

The Firm's stress testing framework is utilized in calculating the Firm's CCAR and other stress test results, which are reported periodically to the Board of Directors. In addition, stress testing results are incorporated into the Firm's Risk Appetite framework, and are reported periodically to the Board Risk Committee.

Profit and loss drawdowns

Profit and loss drawdowns are used to highlight trading losses above certain levels of risk tolerance. A profit and loss drawdown is a decline in revenue from its year-to-date peak level.

Structural interest rate risk management

The effect of interest rate exposure on the Firm's reported net income is important as interest rate risk represents one of the Firm's significant market risks. Interest rate risk arises not only from trading activities which are included in VaR, but also from the Firm's traditional banking activities, which include extension of loans and credit facilities, taking deposits, issuing debt, as well as the investment securities portfolio, and associated derivative instruments. Refer to the table on page 142 for a summary by LOB and Corporate identifying positions included in earnings-at-risk.

Governance

The CTC Risk Committee establishes the Firm's interest rate risk management policy and related limits, which are subject to approval by the Board Risk Committee. Treasury and CIO, working in partnership with the LOBs, calculates the Firm's structural interest rate risk profile and reviews it with senior management, including the CTC Risk Committee. In addition, oversight of structural interest rate risk is managed through a dedicated risk function reporting to the CTC CRO. This risk function is responsible for providing independent oversight and governance around assumptions and establishing and monitoring limits for structural interest rate risk, including limits related to Earnings-at-Risk and Economic Value Sensitivity. The Firm manages structural interest rate risk generally through its investment securities portfolio and interest rate derivatives.

Key risk drivers and risk management process

Structural interest rate risk can arise due to a variety of factors, including:

- Differences in timing among the maturity or repricing of assets, liabilities and off-balance sheet instruments
- Differences in the amounts of assets, liabilities and off-balance sheet instruments that are maturing or repricing at the same time
- Differences in the amounts by which short-term and long-term market interest rates change (for example, changes in the slope of the yield curve)
- The impact of changes in the maturity of various assets, liabilities or off-balance sheet instruments as interest rates change

The Firm manages interest rate exposure related to its assets and liabilities on a consolidated, Firmwide basis. Business units transfer their interest rate risk to Treasury and CIO through funds transfer pricing, which takes into account the elements of interest rate exposure that can be risk-managed in financial markets. These elements include asset and liability balances and contractual rates of interest, contractual principal payment schedules, expected prepayment

experience, interest rate reset dates and maturities, rate indices used for repricing, and any interest rate ceilings or floors for adjustable rate products.

Earnings-at-Risk

One way that the Firm evaluates its structural interest rate risk is through earnings-at-risk. Earnings-at-risk estimates the Firm's interest rate exposure for a given interest rate scenario. It is presented as a sensitivity to a baseline, which includes net interest income and certain interest rate sensitive fees. The baseline uses market interest rates and, in the case of deposits, pricing assumptions. The Firm conducts simulations of changes to this baseline for interest rate-sensitive assets and liabilities denominated in U.S. dollars and other currencies ("non-U.S. dollar" currencies). These simulations primarily include retained and held-for-sale loans, deposits, deposits with banks and financing activities, investment securities, long-term debt, related interest rate hedges, and funds transfer pricing of other positions in risk management VaR and other sensitivity-based measures as described on page 142. Beginning in the fourth quarter of 2024, these simulations also include hedges of non-U.S. dollar foreign exchange exposures arising from capital investments. Refer to non-U.S. dollar foreign exchange risk on page 149 for more information.

Earnings-at-risk scenarios estimate the potential change to a baseline over the following 12 months utilizing multiple assumptions. These scenarios include a parallel shift involving changes to both short-term and long-term rates by an equal amount; a steeper yield curve involving holding short-term rates constant and increasing long-term rates; and a flatter yield curve involving increasing short-term rates and holding long-term rates constant or holding short-term rates constant and decreasing long-term rates. These scenarios consider many different factors, including:

- The impact on exposures as a result of instantaneous changes in interest rates from baseline rates.
- Forecasted balance sheet, as well as modeled prepayment and reinvestment behavior, but excluding assumptions about actions that could be taken by the Firm or its clients and customers in response to instantaneous rate changes. Mortgage prepayment assumptions are based on the interest rates used in the scenarios compared with underlying contractual rates, the time since origination, and other factors which are updated periodically based on historical experience. Deposit forecasts are a key assumption in the Firm's earnings-at-risk. The baseline reflects certain assumptions relating to the Federal Reserve's balance sheet policy (e.g., quantitative tightening and usage at the Reverse Repurchase Facility) that

require management judgment. The amount of deposits that the Firm holds at any given time may be influenced by Federal Reserve actions, as well as broader monetary conditions and competition for deposits.

- The pricing sensitivity of deposits, known as deposit betas, represent the amount by which deposit rates paid could change upon a given change in market interest rates. Actual deposit rates paid may differ from the modeled assumptions, primarily due to customer behavior and competition for deposits.

The Firm performs sensitivity analyses of the assumptions used in earnings-at-risk scenarios, including with respect to deposit betas and forecasts of deposit balances, both of which are especially significant in the case of consumer deposits. The results of these sensitivity analyses are reported to the CTC Risk Committee and the Board Risk Committee.

The Firm's earnings-at-risk scenarios are periodically evaluated and enhanced in response to changes in the composition of the Firm's balance sheet, changes in market conditions, improvements in the Firm's simulation and other factors. In the second quarter of 2024, the Firm updated certain deposit rates paid assumptions which take into account observed pricing and client and customer behavior during the most recent economic cycle. These updated deposit rates paid assumptions impacted the U.S. dollar scenarios, resulting in an increase in positive sensitivity in higher interest rate scenarios, and an increase in negative sensitivity in lower interest rate scenarios.

The Firm's earnings-at-risk sensitivities are measures of the Firm's interest rate exposure. The Firm's actual net interest income for the rate changes presented may differ as the earnings-at-risk scenarios are modelled as instantaneous changes and exclude any actions that could be taken by the Firm or its clients or customers in response to rate changes. Other significant assumptions in the earnings-at-risk scenarios, including mortgage prepayments and deposit rates paid, may also differ from actual results. The Firm's forecast for net interest income is included in the Firm's outlook on page 57.

Management's discussion and analysis

The Firm's sensitivities are presented in the table below.

December 31, (in billions)	2024 ^(a)	2023 ^(b)
Parallel shift:		
+100 bps shift in rates	\$ 2.3	\$ 3.1
-100 bps shift in rates	(2.5)	(2.8)
+200 bps shift in rates	4.6	6.2
-200 bps shift in rates	(4.9)	(6.1)
Steeper yield curve:		
+100 bps shift in long-term rates	1.0	0.6
-100 bps shift in short-term rates	(1.4)	(2.2)
Flatter yield curve:		
+100 bps shift in short-term rates	1.2	2.5
-100 bps shift in long-term rates	(1.1)	(0.6)

(a) Reflects the simultaneous shift of U.S. dollar and non-U.S. dollar rates, and the inclusion of the hedges of non-U.S. dollar capital investments. This inclusion had no impact on total sensitivities but increased U.S. dollar and decreased non-U.S. dollar sensitivities. Subsequent to this change, non-U.S. dollar sensitivities were insignificant.

(b) At December 31, 2023, represents the total of the Firm's U.S. dollar and non-U.S. dollar sensitivities as previously reported.

The change in the Firm's sensitivities as of December 31, 2024, compared to December 31, 2023, were primarily driven by Treasury and CIO balance sheet actions where the Firm added duration through investment securities activity, cash flow hedges of retained loans and fair value hedges of Firm debt. The impact on the sensitivities of the Treasury and CIO balance sheet actions were largely offset by the impact of deposits, primarily from the second quarter of 2024 update of the deposit rates paid assumptions for certain consumer and wholesale deposit products. Additionally, the results as of December 31, 2024 reflected the update to include hedges of the Firm's non-U.S. dollar capital investments. Although total results were not impacted, these hedges increase U.S. dollar sensitivities and decrease non-U.S. dollar sensitivities. In the absence of these updates the Firm's sensitivities as of December 31, 2024, would have been different by the amounts reported in the following table:

December 31, 2024 (in billions)	Amounts by which reported sensitivities would have been different	
	Impact from update in the second quarter of 2024	Impact from update in the fourth quarter of 2024
U.S. dollar:		
Parallel shift:		
+100 bps shift in rates	\$ (1.0)	\$ (0.6)
-100 bps shift in rates	0.9	0.6
+200 bps shift in rates	(1.9)	(1.3)
-200 bps shift in rates	1.5	1.3
Steeper yield curve:		
+100 bps shift in long-term rates	—	—
-100 bps shift in short-term rates	0.9	0.6
Flatter yield curve:		
+100 bps shift in short-term rates	(1.0)	(0.6)
-100 bps shift in long-term rates	—	—
Non-U.S. dollar:		
Parallel shift:		
+100 bps shift in rates	—	0.6
-100 bps shift in rates	—	(0.6)

Economic Value Sensitivity

In addition to earnings-at-risk, which is measured as a sensitivity to a baseline of earnings over the next 12 months, the Firm also measures Economic Value Sensitivity ("EVS"). EVS stress tests the longer-term economic value of equity by measuring the sensitivity of the Firm's current balance sheet, primarily retained loans, deposits, debt and investment securities as well as related hedges, under various interest rate scenarios. The Firm's pricing and cash flow assumptions associated with deposits, as well as prepayment assumptions for loans and securities, are significant factors in the EVS measure. In accordance with the CTC interest rate risk management policy, the Firm has established limits on EVS as a percentage of TCE.

Certain assumptions used in the EVS measure may differ from those required in the fair value measurement note to the Consolidated Financial Statements. For example, certain assets and liabilities with no stated maturity, such as credit card receivables and deposits, have longer assumed durations in the EVS measure. Additional information on long-term debt and held to maturity investment securities is disclosed on page 201 in Note 2.

Non-U.S. dollar foreign exchange risk

Non-U.S. dollar FX risk is the risk that changes in foreign exchange rates affect the value of the Firm's assets or liabilities or future results. The Firm has structural non-U.S. dollar FX exposures arising from capital investments, forecasted expense and revenue, the investment securities portfolio and non-U.S. dollar-denominated debt issuance. Treasury and CIO, working in partnership with the LOBs, primarily manage these risks on behalf of the Firm. Treasury and CIO may hedge certain of these risks using derivatives. Refer to Business Segment & Corporate Results on page 71 for additional information.

Other sensitivity-based measures

The Firm quantifies the market risk of certain debt and equity and credit and funding-related exposures by assessing the potential impact on net revenue, other comprehensive income ("OCI") and noninterest expense due to changes in relevant market variables. Refer to the predominant business activities that give rise to market risk on page 142 for additional information on the positions captured in other sensitivity-based measures.

The table below represents the potential impact to net revenue, OCI or noninterest expense for market risk sensitive instruments that are not included in VaR or earnings-at-risk. Where appropriate, instruments used for hedging purposes are reported net of the positions being hedged. The sensitivities disclosed in the table below may not be representative of the actual gain or loss that would have been realized at December 31, 2024 and 2023, as the movement in market parameters across maturities may vary and are not intended to imply management's expectation of future changes in these sensitivities.

Gain/(loss) (in millions)				
Activity	Description	Sensitivity measure	December 31, 2024	December 31, 2023
Debt and equity^(a)				
Asset Management activities	Consists of seed capital and related hedges; fund co-investments ^(c) ; and certain deferred compensation and related hedges ^(d)	10% decline in market value	\$ (53)	\$ (61)
Other debt and equity	Consists of certain real estate-related fair value option elected loans, privately held equity and other investments held at fair value ^(e)	10% decline in market value	(1,030)	(1,044)
Credit- and funding-related exposures				
Non-USD LTD cross-currency basis	Represents the basis risk on derivatives used to hedge the foreign exchange risk on the non-USD LTD ^(e)	1 basis point parallel tightening of cross currency basis	(10)	(12)
Non-USD LTD hedges foreign currency ("FX") exposure	Primarily represents the foreign exchange revaluation on the fair value of the derivative hedges ^(e)	10% depreciation of currency	28	16
Derivatives – funding spread risk	Impact of changes in the spread related to derivatives FVA ^(e)	1 basis point parallel increase in spread	(2)	(3)
CVA - counterparty credit risk ^(b)	Credit risk component of CVA and associated hedges	10% credit spread widening	—	—
Fair value option elected liabilities - funding spread risk	Impact of changes in the spread related to fair value option elected liabilities DVA ^(e)	1 basis point parallel increase in spread	47	46

(a) Excludes equity securities without readily determinable fair values that are measured under the measurement alternative. Refer to Note 2 for additional information.

(b) In line with the Firm's internal model governance, the credit risk component of CVA related to certain counterparties was removed from Credit Portfolio VaR due to the widening of the credit spreads for those counterparties to elevated levels. The related hedges were also removed to maintain consistency. This exposure is now reflected in the table above.

(c) Impact recognized through net revenue.

(d) Impact recognized through noninterest expense.

(e) Impact recognized through OCI.

COUNTRY RISK MANAGEMENT

The Firm, through its LOBs and Corporate, may be exposed to country risk resulting from financial, economic, political or other significant developments which adversely affect the value of the Firm's exposures related to a particular country or set of countries. The Country Risk Management group actively monitors the various portfolios which may be impacted by these developments and measures the extent to which the Firm's exposures are diversified given the Firm's strategy and risk tolerance relative to a country.

Organization and management

Country Risk Management is an independent risk management function that assesses and monitors exposure to country risk across the Firm.

The Firm's country risk management function includes the following activities:

- Maintaining policies, procedures and standards consistent with a comprehensive country risk framework
- Assigning sovereign ratings, assessing country risks and establishing risk tolerance relative to a country
- Measuring and monitoring country risk exposure and stress across the Firm
- Managing and approving country limits and reporting trends and limit breaches to senior management
- Developing surveillance tools, such as signaling models and ratings indicators, for early identification of potential country risk concerns
- Providing country risk scenario analysis

Sources and measurement

The Firm is exposed to country risk through its lending and deposits, investing, and market-making activities, whether cross-border or locally funded. Country exposure includes activity with both government and private-sector entities in a country.

Under the Firm's internal country risk management approach, attribution of exposure to an individual country is based on the country where the largest proportion of the assets of the counterparty, issuer, obligor or guarantor are located or where the largest proportion of its revenue is derived, which may be different than the domicile (i.e. legal residence) or country of incorporation.

Individual country exposures reflect an aggregation of the Firm's risk to an immediate default, with zero recovery, of the counterparties, issuers, obligors or guarantors attributed to that country. Activities which result in contingent or indirect exposure to a country are not included in the country exposure measure (for example, providing clearing services or secondary exposure to collateral on securities financing receivables).

Assumptions are sometimes required in determining the measurement and allocation of country exposure, particularly in the case of certain non-linear or index products, or where the nature of the counterparty, issuer, obligor or guarantor is not suitable for attribution to an individual country. The use of different measurement approaches or assumptions could affect the amount of reported country exposure.

Under the Firm's internal country risk measurement framework:

- Deposits with banks are measured as the cash balances placed with central banks, commercial banks, and other financial institutions
- Lending exposures are measured at the total committed amount (funded and unfunded), net of the allowance for credit losses and eligible cash and marketable securities collateral received
- Securities financing exposures are measured at their receivable balance, net of eligible collateral received
- Debt and equity securities are measured at the fair value of all positions, including both long and short positions
- Counterparty exposure on derivative receivables is measured at the derivative's fair value, net of the fair value of the eligible collateral received
- Credit derivatives exposure is measured at the net notional amount of protection purchased or sold for the same underlying reference entity, inclusive of the fair value of the derivative receivable or payable, reflecting the manner in which the Firm manages these exposures

The Firm's internal country risk reporting differs from the reporting provided under the FFIEC bank regulatory requirements.

Stress testing

Stress testing is an important component of the Firm's country risk management framework, which aims to estimate and limit losses arising from a country crisis by measuring the impact of adverse asset price movements to a country based on market shocks combined with counterparty specific assumptions. Country Risk Management periodically designs and runs tailored stress scenarios to test vulnerabilities to individual countries or sets of countries in response to specific or potential market events, sector performance concerns, sovereign actions and geopolitical risks. These tailored stress results are used to inform potential risk reduction across the Firm, as necessary.

Risk reporting

Country exposure and stress are measured and reported regularly, and used by Country Risk Management to identify trends and monitor high usages and breaches against limits.

For country risk management purposes, the Firm may report exposure to jurisdictions that are not fully autonomous, including dependent territories and Special Administrative Regions ("SAR") such as Hong Kong SAR, separately from the independent sovereign states with which they are associated.

The following table presents the Firm's top 20 exposures by country (excluding the U.S.) as of December 31, 2024, and their comparative exposures as of December 31, 2023. The top 20 country exposures represent the Firm's largest total exposures by individual country. Country exposures may fluctuate from period to period due to a variety of factors, including client activity, market flows and liquidity management activities undertaken by the Firm.

The increase in exposure to Germany when compared to December 31, 2023, was driven by an increase in cash placed with the central bank of Germany, predominantly due to client-driven market-making activities and higher client deposits.

The increase in exposure to Japan when compared to December 31, 2023, was driven by an increase in cash placed with the central bank of Japan as a result of client-driven market-making activities.

The Firm continues to monitor its exposure to Russia, which corresponds to cash placed with the central bank, but which excludes deposits placed on behalf of clients at the Deposit Insurance Agency of Russia. The Firm currently believes that its remaining exposure to Russia is not material. Refer to Note 30 on pages 310–311 for information concerning Russian litigation.

Top 20 country exposures (excluding the U.S.)^(a)

December 31, (in billions)	2024					2023 ^(f)
	Deposits with banks ^(b)	Lending ^(c)	Trading and investing ^(d)	Other ^(e)	Total exposure	Total exposure
Germany	\$ 89.7	\$ 12.6	\$ 0.9	\$ 0.7	\$ 103.9	\$ 84.8
United Kingdom	24.6	22.4	27.7	1.4	76.1	77.1
Japan	55.1	3.1	4.5	0.4	63.1	36.0
France	0.6	12.3	4.2	0.9	18.0	10.1
Canada	1.6	10.6	2.7	0.2	15.1	16.0
Brazil	3.5	4.2	7.0	—	14.7	16.7
Australia	5.0	7.4	1.9	—	14.3	18.3
Switzerland	4.7	4.2	1.4	3.3	13.6	10.9
Mainland China	3.1	6.2	4.1	—	13.4	14.0
India	1.1	5.2	4.1	0.9	11.3	9.7
Italy	0.1	8.2	1.8	0.3	10.4	6.0
South Korea	0.6	2.9	6.3	0.5	10.3	7.8
Saudi Arabia	0.8	5.7	2.9	—	9.4	7.7
Singapore	1.5	2.0	3.5	0.4	7.4	9.8
Mexico	1.3	4.4	1.5	—	7.2	8.2
Spain	0.2	4.6	1.2	0.1	6.1	6.3
Netherlands	—	6.6	(0.9)	0.2	5.9	5.6
Belgium	4.0	1.3	0.1	—	5.4	8.0
Malaysia	2.1	0.2	1.0	0.3	3.6	4.2
Luxembourg	0.9	1.7	1.0	—	3.6	4.0

- (a) Country exposures presented in the table reflect 89% and 88% of total Firmwide non-U.S. exposure, where exposure is attributed to an individual country based on the Firm's internal country risk management approach, at December 31, 2024 and 2023, respectively.
- (b) Predominantly represents cash placed with central banks.
- (c) Includes loans and accrued interest receivable, lending-related commitments (net of eligible collateral and the allowance for credit losses). Excludes intra-day and operating exposures, such as those from settlement and clearing activities.
- (d) Includes market-making positions and hedging, investment securities, and counterparty exposure on derivative and securities financings net of eligible collateral. Market-making positions and hedging includes exposure from single reference entity ("single-name"), index and other multiple reference entity transactions for which one or more of the underlying reference entities is in a country listed in the above table.
- (e) Includes physical commodities inventory and clearing house guarantee funds.
- (f) The country rankings presented in the table as of December 31, 2023, are based on the country rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.

CLIMATE RISK MANAGEMENT

Climate risk refers to the potential threats posed by climate change to the Firm and its clients, customers, operations and business strategy. Climate change is viewed as a driver of risk that may impact existing types of risks managed by the Firm. Climate risk is categorized into physical risk and transition risk.

Physical risk involves economic costs and financial losses due to a changing climate. Acute physical risk drivers include the increased frequency or severity of climate and weather events, such as floods, wildfires and tropical cyclones. Chronic physical risk drivers include more gradual shifts in the climate, such as sea level rise, persistent changes in precipitation levels and increases in average ambient temperatures.

Transition risk involves the financial and economic consequences of society's shift toward a lower-carbon economy. Transition risk drivers include possible changes in public policy, adoption of new technologies and shifts in consumer preferences. Transition risks may also be influenced by changes in the physical climate.

Organization and management

The Firm has a Climate Risk Management function that is responsible for establishing and maintaining the Firmwide framework and strategy for managing climate risks that may impact the Firm.

Other responsibilities of Climate Risk Management include:

- Setting policies, standards, procedures and processes to support identification, escalation, monitoring and management of climate risk across the Firm
- Developing metrics, scenarios and stress testing mechanisms designed to assess the range of potential climate-related financial and economic impacts to the Firm
- Establishing a Firmwide climate risk data strategy and the supporting climate risk technology infrastructure

The LOBs and Corporate are responsible for the identification, assessment and management of climate risks present in their business activities and for the adherence to applicable climate-related laws, rules and regulations.

Governance and oversight

The Firm's framework and strategy for managing climate risk is integrated into the Firm's risk governance structure. This framework allows for the escalation of significant climate risk-related issues to LOB Risk Committees. The Board Risk Committee also receives information on significant climate risks and climate-related initiatives, as appropriate.

OPERATIONAL RISK MANAGEMENT

Operational risk is the risk of an adverse outcome resulting from inadequate or failed internal processes or systems; human factors; or external events impacting the Firm's processes or systems.

Operational Risk includes compliance, conduct, legal, and estimations and model risk. Operational risk is inherent in the Firm's activities and can manifest itself in various ways, including fraudulent acts, business disruptions (including those caused by extraordinary events beyond the Firm's control), cyber attacks, inappropriate employee behavior, failure to comply with applicable laws, rules and regulations or failure of vendors or other third party providers to perform in accordance with their agreements. Operational Risk Management attempts to manage operational risk at appropriate levels in light of the Firm's financial position, the characteristics of its businesses, and the markets and regulatory environments in which it operates.

Operational Risk Management Framework

The Firm's Compliance, Conduct, and Operational Risk ("CCOR") Management Framework is designed to enable the Firm to govern, identify, measure, monitor and test, manage and report on the Firm's operational risk.

Operational Risk Governance

The LOBs and Corporate are responsible for the management of operational risk. The Control Management Organization, which consists of control managers within each LOB and Corporate, is responsible for the day-to-day execution of the CCOR Framework.

The Firm's Global Chief Compliance Officer ("CCO") and FRE for Operational Risk and Qualitative Risk Appetite is responsible for defining the CCOR Management Framework and establishing the minimum standards for its execution. The LOB and Corporate aligned CCOR Lead Officers report to the Global CCO and FRE for Operational Risk and Qualitative Risk Appetite and are independent of the respective businesses or functions they oversee. The CCOR Management Framework is included in the Risk Governance and Oversight Policy that is reviewed and approved by the Board Risk Committee periodically.

Operational Risk Identification

The Firm utilizes a structured risk and control self-assessment process that is executed by the LOBs and Corporate. As part of this process, the LOBs and Corporate evaluate the effectiveness of their respective control environment to assess circumstances in which controls have failed, and to determine where remediation efforts may be required. The Firm's Operational Risk and Compliance organization ("Operational Risk and Compliance")

provides oversight of and challenge to these evaluations and may also perform independent assessments of significant operational risk events and areas of concentrated or emerging risk.

Operational Risk Measurement

Operational Risk and Compliance performs an independent assessment of the operational risks inherent within the LOBs and Corporate, which includes evaluating the effectiveness of the control environments and reporting the results to senior management.

In addition, Operational Risk and Compliance assesses operational risks through quantitative means, including operational risk-based capital and estimation of operational risk losses under both baseline and stressed conditions.

The primary component of the operational risk-based capital estimate is the Loss Distribution Approach ("LDA") statistical model, which simulates the projected frequency and severity of operational risk losses based on historical data. The LDA model is used to estimate an aggregate operational risk loss over a one-year time horizon, at a 99.9% confidence level. The LDA model incorporates actual internal operational risk losses in the quarter following the period in which those losses were realized, and the calculation generally continues to reflect such losses even after the issues or business activities giving rise to the losses have been remediated or reduced.

As required under the Basel III capital framework, the Firm's operational risk capital methodology, which uses the Advanced Measurement Approach ("AMA"), incorporates internal and external losses as well as management's view of tail risk captured through operational risk scenario analysis, and evaluation of key business environment and internal control metrics. The Firm does not reflect the impact of insurance in its AMA estimate of operational risk capital.

The Firm considers the impact of stressed economic conditions on operational risk losses and develops a forward looking view of material operational risk events that may occur in a stressed environment. The Firm's operational risk stress testing framework is utilized in calculating results for the Firm's CCAR and other stress testing processes.

Refer to Capital Risk Management on pages 97–107 for information related to operational risk RWA, and CCAR.

Operational Risk Monitoring and Testing

The results of risk assessments performed by Operational Risk and Compliance are used in connection with their independent monitoring and testing compliance of the LOBs and Corporate with

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laws, rules and regulations. Through monitoring and testing, Operational Risk and Compliance independently identify areas of heightened operational risk and tests the effectiveness of controls within the LOBs and Corporate.

Management of Operational Risk

The operational risk areas or issues identified through monitoring and testing are escalated to the LOBs and Corporate to be remediated through action plans, as needed, to mitigate operational risk. Operational Risk and Compliance may advise the LOBs and Corporate in the development and implementation of action plans.

Operational Risk Reporting

All employees of the Firm are expected to escalate risks appropriately. Risks identified by Operational Risk and Compliance are escalated to the appropriate LOB and Corporate Control Committees, as needed. Operational Risk and Compliance has established standards designed to ensure that consistent operational risk reporting and operational risk reports are produced on a Firmwide basis as well as by the LOBs and Corporate. Reporting includes the evaluation of key risk and performance indicators against established thresholds as well as the assessment of different types of operational risk against stated risk appetite. The standards establish escalation protocols to senior management and to the Board of Directors.

Insurance

One of the ways in which operational risk may be mitigated is through insurance maintained by the Firm. The Firm purchases insurance from commercial insurers and maintains a wholly-owned captive insurer, Park Assurance Company. Insurance may also be required by third parties with whom the Firm does business.

Subcategories and examples of operational risks

Operational risk can manifest itself in various ways. Operational risk subcategories include Compliance risk, Conduct risk, Legal risk, and Estimations and Model risk. Refer to pages 157, 158, 159 and 160, respectively for more information on Compliance, Conduct, Legal, and Estimations and Model risk. Details on other select examples of operational risks such as business and technology resiliency, payment fraud and third-party outsourcing, as well as cybersecurity, are provided below.

Firmwide resiliency risk

Disruptions of the Firm's business and operations can occur due to forces beyond the Firm's control such as the spread of infectious diseases or pandemics, severe weather, natural disasters, the effects of climate change, power or telecommunications loss, failure of a third party to provide expected services, cyberattacks, civil or political unrest or terrorism. The Firm's resiliency framework is intended to enable the Firm to prepare for and adapt to changing conditions and withstand and recover from, and address adverse effects on its operations caused by, disruptions that may impact critical business functions and supporting assets, including its staff, technology, data and facilities, as well as those of third-party service providers. The framework includes governance, awareness training, planning and testing of recovery strategies, as well as strategic and tactical initiatives to identify, assess, and manage resiliency risks. The framework operates in accordance with the Firm's overall approach to Operational Risk Management, including alignment with technology, cybersecurity, data, physical security, crisis management, real estate and outsourcing programs.

Payment fraud risk

Payment fraud risk is the risk of external and internal parties unlawfully obtaining personal monetary benefit through misdirected or otherwise improper payment. The Firm employs various controls for managing payment fraud risk as well as providing employee and client education and awareness trainings.

Third-party outsourcing risk

The Firm's Third-Party Oversight ("TPO") and Inter-affiliates Oversight ("IAO") frameworks assist the LOBs and Corporate in selecting, documenting, onboarding, monitoring and managing their supplier relationships including services provided by affiliates. The objectives of the TPO framework are to hold suppliers and other third parties to an appropriate standard of operational performance and to mitigate key risks, including data loss and business disruptions. The Corporate Third-Party Oversight group is responsible for Firmwide training, monitoring, reporting and standards with respect to third-party outsourcing risks.

Cybersecurity risk

Cybersecurity risk is the risk of harm or loss resulting from misuse or abuse of technology or the unauthorized disclosure of data.

Overview

Cybersecurity risk is an important and continuously evolving focus for the Firm. Significant resources are devoted to protecting and enhancing the security of computer systems, software, networks, storage devices, and other technology. The Firm's security efforts are designed to protect against, among other things, cybersecurity attacks that can result in unauthorized access to confidential information, the destruction of data, disruptions to or degradations of service, the sabotaging of systems or other damage.

The Firm has experienced, and expects that it will continue to experience, a higher volume and complexity of cyber attacks against the backdrop of heightened geopolitical tensions. The Firm has implemented measures and controls reasonably designed to address this evolving environment, including enhanced threat monitoring. In addition, the Firm continues to review and enhance its capabilities to address associated risks, such as those relating to the management of administrative access to systems.

Third parties with which the Firm does business, that facilitate the Firm's business activities (e.g., vendors, supply chain, exchanges, clearing houses, central depositories, and financial intermediaries) or that the Firm has acquired are also sources of cybersecurity risk to the Firm. Third party incidents such as system breakdowns or failures, misconduct by the employees of such parties, or cyber attacks, including ransomware and supply-chain compromises, could have a material adverse effect on the Firm, including in circumstances in which an affected third party is unable to deliver a product or service to the Firm or where the incident delivers compromised software to the Firm or results in lost or compromised information of the Firm or its clients or customers.

Clients and customers are also sources of cybersecurity risk to the Firm and its information assets, particularly when their activities and systems are beyond the Firm's own security and control systems. The Firm engages in periodic discussions with its clients, customers and other external parties concerning cybersecurity risks including opportunities to improve cybersecurity.

Risks from cybersecurity threats, including any previous cybersecurity events, have not materially affected the Firm or its business strategy, results of operations or financial condition. Notwithstanding the comprehensive approach that the Firm takes to address cybersecurity risk, the Firm may not be successful in preventing or mitigating a future

cybersecurity incident that could have a material adverse effect on the Firm or its business strategy, results of operations or financial condition.

Organization and management

The Global Chief Information Security Officer ("CISO") reports to the Global Chief Information Officer, and is a member of key cybersecurity governance forums. The CISO leads the Global Cybersecurity and Technology Controls organization, which is responsible for identifying technology and cybersecurity risks and for implementing and maintaining controls to manage cybersecurity threats. The CISO and the members of senior management within Global Technology and the Cybersecurity and Technology Controls organizations all have relevant expertise and experience in cybersecurity and information technology risk management, including relevant experience at the Firm, at other financial services companies or in other highly-regulated industries.

The CISO is responsible for the Firm's Information Security Program, which is designed to prevent, detect and respond to cyber attacks in order to help safeguard the confidentiality, integrity and availability of the Firm's infrastructure, resources and information. The program includes managing the Firm's global cybersecurity operations centers, providing training, conducting cybersecurity event simulation exercises, implementing the Firm's policies and standards relating to technology risk and cybersecurity management, and enhancing, as needed, the Firm's cybersecurity capabilities.

The Firm's Information Security Program includes the following functions:

Cyber Operations, which is responsible for implementing and maintaining controls designed to detect and defend the Firm against cyber attacks, and includes a dedicated function for incident response and ongoing monitoring for cybersecurity threats and vulnerabilities, including those among the Firm's third-party suppliers.

Technology Governance, Risk & Controls, which is responsible for operationalizing technology risk and control frameworks, analyzing regulatory developments that may impact the Firm, and developing control catalogs and assessments of controls, as well as overseeing governance and reporting of technology and cybersecurity risk.

Security Awareness, which provides awareness and training that reinforces information risk and security management practices and compliance with the Firm's policies, standards and practices. The training is mandatory for all employees globally on a periodic basis, and it is supplemented by Firmwide testing initiatives, including periodic phishing tests. The Firm also provides specialized security training to

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employees in specific roles, such as application developers. The Firm's Global Privacy Program requires all employees to take periodic training on data privacy that focuses on confidentiality and security, as well as responding to unauthorized access to or use of information.

Technology Resiliency, which establishes control requirements for planning and testing the prioritized recovery of technology services in the event of degradation or outage, including incident response planning, data backup and retention, and recovery readiness in support of the Firmwide Business Resiliency Program and operational risk management practices.

The Firm has a cybersecurity incident response plan designed to enable the Firm to respond to attempted cybersecurity incidents, coordinate as appropriate with law enforcement and other government agencies, notify clients and customers, as applicable, and recover from such incidents. In addition, the Firm actively partners with appropriate government and law enforcement agencies and peer industry forums, participating in discussions and simulations to assist in understanding the full spectrum of cybersecurity risks and in enhancing defenses and improving resiliency in the Firm's operating environment.

Governance and oversight

The governance structure for the Global Cybersecurity and Technology Controls organization is designed to appropriately identify, escalate and mitigate cybersecurity risks. Cybersecurity risk management and its governance and oversight are integrated into the Firm's operational risk management framework, including through the escalation of key risk and control issues to management and the development of risk mitigation plans for heightened risk and control issues. IRM independently assesses and challenges the activities and risk management practices of the Global Cybersecurity and Technology Controls organization related to the identification, assessment, measurement and mitigation of cybersecurity risk. As needed, the Firm engages third-party assessors or auditing firms with industry-recognized expertise on cybersecurity matters to review specific aspects of the Firm's cybersecurity risk management framework, processes and controls.

The governance and oversight for cybersecurity risk management includes governance forums that inform management of key areas of concern regarding the prevention, detection, mitigation and remediation of cybersecurity risks.

The Cybersecurity and Technology Controls Operating Committee ("CTOC") is the principal management committee that oversees the Firm's assessment and management of cybersecurity risk, including oversight of the implementation and maintenance of appropriate controls in support of the Firm's Information Security Program. The membership of the CTOC includes senior representatives from the Global Cybersecurity and Technology Controls organization and relevant corporate functions, including IRM and Internal Audit.

The CTOC escalates key operational risk and control issues, as appropriate, to the Global Technology Operating Committee ("GTOC") or its business control committee or to the appropriate LOB and Corporate Control Committees. The GTOC is responsible for the governance of the Firmwide Global Technology organization, including oversight of Firmwide technology strategies, the delivery of technology and technology operations, the effective use of information technology resources, and monitoring and resolving key operational risk and control matters arising in the Global Technology organization.

As part of its oversight of management's implementation and maintenance of the Firm's risk management framework, the Firm's Board of Directors receives periodic updates from the CIO, the CISO and senior members of the CTOC concerning cybersecurity matters. These updates generally include information regarding cybersecurity and technology developments, the Firm's Information Security Program and recommended changes to that program, cybersecurity policies and practices, and ongoing initiatives to improve information security, as well as any significant cybersecurity incidents and the Firm's efforts to address those incidents. The Audit Committee and the Risk Committee assist the Board in this oversight.

COMPLIANCE RISK MANAGEMENT

Compliance risk, a subcategory of operational risk, is the risk of failing to comply with laws, rules, regulations or codes of conduct and standards of self-regulatory organizations.

Overview

Each of the LOBs and Corporate hold primary ownership of and accountability for managing their compliance risk. The Firm's Operational Risk and Compliance Organization ("Operational Risk and Compliance"), which is independent of the LOBs and Corporate, provides independent review, monitoring and oversight of business operations with a focus on compliance with the laws, rules, and regulations applicable to the delivery of the Firm's products and services to clients and customers.

These compliance risks relate to a wide variety of laws, rules and regulations across the LOBs and Corporate, and jurisdictions, and include risks related to financial products and services, relationships and interactions with clients and customers, and employee activities. For example, compliance risks include those associated with anti-money laundering compliance, trading activities, market conduct, and complying with the laws, rules, and regulations related to the offering of products and services across jurisdictional borders. Compliance risk is also inherent in the Firm's fiduciary activities, including the failure to exercise the applicable standard of care to act in the best interest of fiduciary clients and customers or to treat fiduciary clients and customers fairly.

Other functions provide oversight of significant regulatory obligations that are specific to their respective areas of responsibility.

Operational Risk and Compliance implements policies and standards designed to govern, identify, measure, monitor and test, manage, and report on compliance risk.

Governance and oversight

Operational Risk and Compliance is led by the Firm's Global CCO and FRE for Operational Risk and Qualitative Risk Appetite.

The Firm maintains oversight and coordination of its compliance risk through the CCOR Management Framework. The Firm's Global CCO and FRE for Operational Risk and Qualitative Risk Appetite also provides regular updates to the Board Risk Committee and the Audit Committee on significant compliance risk issues, as appropriate.

Code of Conduct

The Firm has a Code of Conduct (the "Code") that sets forth the Firm's expectation that employees will conduct themselves with integrity, at all times. The Code provides the principles that help govern employee conduct with clients, customers, suppliers, vendors, shareholders, regulators, other employees, as well as with the markets and communities in which the Firm operates. The Code requires employees to promptly report any potential or actual violation of the Code, Firm policies, or laws, rules or regulations applicable to the Firm's business. It also requires employees to report any illegal or unethical conduct, or conduct that violates the underlying principles of the Code, by any of the Firm's employees, consultants, clients, customers, suppliers, contract or temporary workers, or business partners or agents. Training is assigned to newly hired employees after joining the Firm, and to current employees periodically thereafter. Employees are required to affirm their compliance with the Code annually.

Employees can report any potential or actual violations of the Code through the Firm's Conduct Hotline (the "Hotline") by phone, mobile device or the internet. The Hotline is anonymous, where permitted by law, is available at all times globally, has translation services, and is administered by an outside service provider. The Code prohibits retaliation against anyone who raises an issue or concern in good faith or assists with an inquiry or investigation. Periodically, the Audit Committee receives reports on the Code of Conduct program.

CONDUCT RISK MANAGEMENT

Conduct risk, a subcategory of operational risk, is the risk that any action or misconduct by an employee could lead to unfair client or customer outcomes, impact the integrity of the markets in which the Firm operates, harm employees or the Firm, or compromise the Firm's reputation.

Overview

Each LOB and Corporate is accountable for identifying and managing its conduct risk to provide appropriate engagement, ownership and sustainability of a culture consistent with the Firm's Business Principles. The Business Principles serve as a guide for how employees are expected to conduct themselves. With the Business Principles serving as a guide, the Firm's Code sets out the Firm's expectations for each employee and provides information and resources to help employees conduct business ethically and in compliance with applicable laws, rules and regulations everywhere the Firm operates. Refer to Compliance Risk Management on page 157 for further discussion of the Code.

Governance and oversight

The Firm's oversight and coordination of conduct risk is managed in the same manner as Compliance risk. Refer to Compliance Risk Management on page 157 for further information.

Conduct risk management encompasses various aspects of people management practices throughout the employee life cycle, including recruiting, onboarding, training and development, performance management, promotion and compensation processes. Each LOB, Treasury and CIO, and each designated corporate function completes an assessment of conduct risk periodically, reviews metrics and issues which may involve conduct risk, and provides conduct education as appropriate.

LEGAL RISK MANAGEMENT

Legal risk, a subcategory of operational risk, is the risk of loss primarily caused by the actual or alleged failure to meet legal obligations that arise from the rule of law in jurisdictions in which the Firm operates, agreements with clients and customers, and products and services offered by the Firm.

Overview

The global Legal function (“Legal”) provides legal services and advice to the Firm. Legal is responsible for managing the Firm’s exposure to legal risk by:

- managing actual and potential litigation and enforcement matters, including internal reviews and investigations related to such matters
- advising on products and services, including contract negotiation and documentation
- advising on offering and marketing documents and new business initiatives
- managing dispute resolution
- interpreting existing laws, rules and regulations, and advising on changes to them
- advising on advocacy in connection with contemplated and proposed laws, rules and regulations, and
- providing legal advice to the LOBs, Corporate and the Board.

Legal selects, engages and manages outside counsel for the Firm on all matters in which outside counsel is engaged. In addition, Legal advises the Firm’s Conflicts Office which reviews the Firm’s wholesale transactions that may have the potential to create conflicts of interest for the Firm.

Governance and oversight

The Firm’s General Counsel reports to the CEO and is a member of the Operating Committee, the Firmwide Risk Committee and the Firmwide Control Committee. The Firm’s General Counsel and other members of Legal report on significant legal matters to the Firm’s Board of Directors and to the Audit Committee.

Legal serves on and advises various committees and advises the Firm’s LOBs and Corporate on potential reputation risk issues.

ESTIMATIONS AND MODEL RISK MANAGEMENT

Estimations and Model risk, a subcategory of operational risk, is the potential for adverse consequences from decisions based on incorrect or misused estimation outputs.

The Firm uses models and other analytical and judgment-based estimations, including those based upon machine learning or artificial intelligence techniques, across various businesses and functions. The estimation methods are of varying levels of sophistication and are used for many purposes, such as the valuation of positions and measurement of risk, assessing regulatory capital requirements, conducting stress testing, evaluating the allowance for credit losses and making business decisions. A dedicated independent function, Model Risk Governance and Review ("MRGR"), defines and governs the Firm's policies relating to the management of model risk and risks associated with certain analytical and judgment-based estimations, such as those used in risk management, budget forecasting and capital planning and analysis.

Model risks are owned by the users of the models within the LOBs and Corporate based on the specific purposes of such models. Users and developers of models are responsible for developing, implementing and testing their models, as well as referring models to MRGR for review and approval. Once models have been approved, model users and developers are responsible for maintaining a robust operating environment, and must monitor and evaluate the performance of the models on an ongoing basis. Model users and developers may seek to enhance models in response to changes in the relevant portfolios and in product and market developments, as well as to capture improvements in available modeling techniques and systems capabilities.

Models are tiered based on an internal standard according to their complexity, the exposure associated with the model and the Firm's reliance on the model. This tiering is subject to the approval of MRGR. In its review of a model, MRGR considers whether the model is suitable for the specific purposes for which it will be used. When reviewing a model, MRGR analyzes and challenges the model methodology and the reasonableness of model assumptions, and may perform or require additional testing, including back-testing of model outcomes. Model reviews are approved by the appropriate level of management within MRGR based on the relevant model tier.

Under the Firm's Estimations and Model Risk Management Policy, MRGR reviews and approves new models, as well as material changes to existing models, prior to their use. In certain circumstances, exceptions may be granted to the Firm's policy to allow a model to be used prior to review or approval. MRGR may also require the user to take appropriate actions to mitigate the model risk if it is to be used in the interim. These actions will depend on the model and may include, for example, limitation of trading activity.

While models are inherently imprecise, the degree of imprecision or uncertainty can be heightened by the market or economic environment. This is particularly true when the current and forecasted environments are significantly different from the historical environments upon which the models were developed. This increased uncertainty may necessitate a greater degree of judgment and analytics to inform any adjustments that the Firm may make to model outputs than would otherwise be the case. In addition, the Firm may experience increased uncertainty in its estimates if assets acquired differ from those used to develop the models.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 and Note 2 for a summary of model-based valuations and other valuation techniques.

CRITICAL ACCOUNTING ESTIMATES USED BY THE FIRM

JPMorganChase's accounting policies and use of estimates are integral to understanding its reported results. The Firm's most complex accounting estimates require management's judgment to ascertain the appropriate carrying value of assets and liabilities. The Firm has established policies and control procedures intended to ensure that estimation methods, including any judgments made as part of such methods, are well-controlled, independently reviewed and applied consistently from period to period. The methods used and judgments made reflect, among other factors, the nature of the assets or liabilities and the related business and risk management strategies, which may vary across the Firm's businesses and portfolios. In addition, the policies and procedures are intended to ensure that the process for changing methodologies occurs in an appropriate manner. The Firm believes its estimates for determining the carrying value of its assets and liabilities are appropriate. The following is a brief description of the Firm's critical accounting estimates involving significant judgments.

Allowance for credit losses

The Firm's allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The allowance for credit losses generally comprises:

- The allowance for loan losses, which covers the Firm's retained loan portfolios (scored and risk-rated),
- The allowance for lending-related commitments, and
- The allowance for credit losses on investment securities.

The allowance for credit losses involves significant judgment on a number of matters including development and weighting of macroeconomic forecasts, incorporation of historical loss experience, assessment of risk characteristics, assignment of risk ratings, valuation of collateral, and the determination of remaining expected life. Refer to Note 10 and Note 13 for further information on these judgments as well as the Firm's policies and methodologies used to determine the Firm's allowance for credit losses.

One of the most significant judgments involved in estimating the Firm's allowance for credit losses relates to the macroeconomic forecasts used to estimate credit losses over the eight-quarter forecast period within the Firm's methodology. The eight-quarter forecast incorporates hundreds of macroeconomic variables ("MEVs") that are relevant for exposures across the Firm, with modeled credit

losses being driven primarily by a subset of less than twenty variables. The specific variables that have the greatest effect on the modeled losses vary by portfolio and geography.

- Key MEVs for the consumer portfolio include regional U.S. unemployment rates and U.S. HPI.
- Key MEVs for the wholesale portfolio include U.S. unemployment, U.S. real GDP, U.S. equity prices, U.S. interest rates, U.S. corporate credit spreads, oil prices, U.S. commercial real estate prices and U.S. HPI.

Changes in the Firm's assumptions and forecasts of economic conditions could significantly affect its estimate of expected credit losses in the portfolio at the balance sheet date or lead to significant changes in the estimate from one reporting period to the next.

As a result of the First Republic acquisition, the Firm recorded an allowance for credit losses for the loans acquired and lending-related commitments assumed as of May 1, 2023. Due to differences in risk rating methodologies for the First Republic portfolio and the ongoing integration of products and systems, the allowance for credit losses for the acquired wholesale portfolio was initially measured based on similar risk characteristics from other facilities underwritten by the Firm. Starting in the second quarter of 2024, the acquired portfolio was incorporated into the Firm's modeled credit loss estimates and is now reflected in the wholesale sensitivity analysis below. Refer to Note 34 for additional information on the First Republic acquisition.

It is difficult to estimate how potential changes in any one factor or input might affect the overall allowance for credit losses because management considers a wide variety of factors and inputs in estimating the allowance for credit losses. Changes in the factors and inputs considered may not occur at the same rate and may not be consistent across all geographies or product types, and changes in factors and inputs may be directionally inconsistent, such that improvement in one factor or input may offset deterioration in others.

To consider the impact of a hypothetical alternate macroeconomic forecast, the Firm compared the modeled credit losses determined using its central and relative adverse macroeconomic scenarios, which are two of the five scenarios considered in estimating the allowances for loan losses and lending-related commitments. The central and relative adverse scenarios each included a full suite of MEVs, but differed in the levels, paths and peaks/troughs of those variables over the eight-quarter forecast period.

For example, compared to the Firm's central scenario shown on page 137 and in Note 13, the Firm's relative adverse scenario assumes an elevated U.S.

Management's discussion and analysis

unemployment rate, averaging approximately 2.1% higher over the eight-quarter forecast, with a peak difference of approximately 3.0% in the fourth quarter of 2025.

This analysis is not intended to estimate expected future changes in the allowance for credit losses, for a number of reasons, including:

- The allowance as of December 31, 2024, reflects credit losses beyond those estimated under the central scenario due to the weight placed on the adverse scenarios.
- The impacts of changes in many MEVs are both interrelated and nonlinear, so the results of this analysis cannot be simply extrapolated for more severe changes in macroeconomic variables.
- Expectations of future changes in portfolio composition and borrower behavior can significantly affect the allowance for credit losses.

To demonstrate the sensitivity of credit loss estimates to macroeconomic forecasts as of December 31, 2024, the Firm compared the modeled estimates under its relative adverse scenario to its central scenario. Without considering offsetting or correlated effects in other qualitative components of the Firm's allowance for credit losses, the comparison between these two scenarios for the exposures below reflect the following differences:

- An increase of approximately \$850 million for residential real estate loans and lending-related commitments
- An increase of approximately \$3.7 billion for credit card loans
- An increase of approximately \$4.1 billion for wholesale loans and lending-related commitments

This analysis relates only to the modeled credit loss estimates and is not intended to estimate changes in the overall allowance for credit losses as it does not reflect any potential changes in other adjustments to the quantitative calculation, which would also be influenced by the judgment management applies to the modeled lifetime loss estimates to reflect the uncertainty and imprecision of these modeled lifetime loss estimates based on then-current circumstances and conditions.

Recognizing that forecasts of macroeconomic conditions are inherently uncertain, the Firm believes that its process to consider the available information and associated risks and uncertainties is appropriately governed and that its estimates of expected credit losses were reasonable and appropriate for the period ended December 31, 2024.

Fair value

JPMorganChase carries a portion of its assets and liabilities at fair value. The majority of such assets and liabilities are measured at fair value on a recurring basis, including derivatives, structured note products and certain securities financing agreements. Certain assets and liabilities are measured at fair value on a nonrecurring basis, including certain mortgage, home equity and other loans, where the carrying value is based on the fair value of the underlying collateral.

Assets measured at fair value

The following table includes the Firm's assets measured at fair value and the portion of such assets that are classified within level 3 of the fair value hierarchy. Refer to Note 2 for further information.

December 31, 2024 (in millions, except ratios)	Total assets at fair value	Total level 3 assets
Federal funds sold and securities purchased under resale agreements	\$ 286,771	\$ —
Securities borrowed	83,962	—
Trading assets:		
Trading-debt and equity instruments	576,817	2,442
Derivative receivables ^(a)	60,967	8,452
Total trading assets	637,784	10,894
AFS securities	406,852	8
Loans	41,350	2,416
MSRs	9,121	9,121
Other	14,073	1,344
Total assets measured at fair value on a recurring basis	1,479,913	23,783
Total assets measured at fair value on a nonrecurring basis	2,489	1,742
Total assets measured at fair value	\$ 1,482,402	\$ 25,525
Total Firm assets	\$ 4,002,814	
Level 3 assets at fair value as a percentage of total Firm assets ^(a)		1%
Level 3 assets at fair value as a percentage of total Firm assets at fair value ^(a)		2%

(a) For purposes of the table above, the derivative receivables total reflects the impact of netting adjustments; however, the \$8.5 billion of derivative receivables classified as level 3 does not reflect the netting adjustment as such netting is not relevant to a presentation based on the transparency of inputs to the valuation of an asset. The level 3 balances would be reduced if netting were applied, including the netting benefit associated with cash collateral.

Valuation

Details of the Firm's processes for determining fair value are set out in Note 2. Estimating fair value requires the application of judgment. The type and level of judgment required is largely dependent on the amount of observable market information available to the Firm. For instruments valued using internally developed valuation models and other valuation techniques that use significant unobservable inputs and are therefore classified within level 3 of the fair value hierarchy, judgments used to estimate fair value are more significant than those required when estimating the fair value of instruments classified within levels 1 and 2.

In arriving at an estimate of fair value for an instrument within level 3, management must first determine the appropriate valuation model or other valuation technique to use. Second, the lack of observability of certain significant inputs requires management to assess relevant empirical data in deriving valuation inputs including, for example, transaction details, yield curves, interest rates, prepayment speeds, default rates, volatilities, correlations, prices (such as commodity, equity or debt prices), valuations of comparable instruments, foreign exchange rates and credit curves. Refer to Note 2 for a further discussion of the valuation of level 3 instruments, including unobservable inputs used.

For instruments classified in levels 2 and 3, management judgment must be applied to assess the appropriate level of valuation adjustments to reflect counterparty credit quality, the Firm's creditworthiness, market funding rates, liquidity considerations, unobservable parameters, and for portfolios that meet specified criteria, the size of the net open risk position. The judgments made are typically affected by the type of product and its specific contractual terms, and the level of liquidity for the product or within the market as a whole. In periods of heightened market volatility and uncertainty judgments are further affected by the wider variation of reasonable valuation estimates, particularly for positions that are less liquid. Refer to Note 2 for a further discussion of valuation adjustments applied by the Firm.

Imprecision in estimating unobservable market inputs or other factors can affect the amount of gain or loss recorded for a particular position. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with those of other market participants, the methods and assumptions used reflect management judgment and may vary across the Firm's businesses and portfolios.

The Firm uses various methodologies and assumptions in the determination of fair value. The use of methodologies or assumptions different than those used by the Firm could result in a different estimate of fair value at the reporting date. Refer to Note 2 for a detailed discussion of the Firm's valuation process and

hierarchy, and its determination of fair value for individual financial instruments.

Goodwill impairment

Under U.S. GAAP, goodwill must be allocated to reporting units and tested for impairment at least annually. The Firm's process and methodology used to conduct goodwill impairment testing is described in Note 15.

Management applies significant judgment when testing goodwill for impairment. The goodwill associated with each business combination is allocated to the related reporting units for goodwill impairment testing.

For the year ended December 31, 2024, the Firm reviewed current economic conditions, estimated market cost of equity, as well as actual business results and projections of business performance. Based on such reviews, the Firm has concluded that goodwill was not impaired as of December 31, 2024. For each of the reporting units, fair value exceeded carrying value by at least 10% and there was no indication of a significant risk of goodwill impairment based on current projections and valuations.

The projections for the Firm's reporting units are consistent with management's current business outlook assumptions in the short term, and the Firm's best estimates of long-term growth and return on equity in the longer term. Where possible, the Firm uses third-party and peer data to benchmark its assumptions and estimates.

Refer to Note 15 for additional information on goodwill, including the goodwill impairment assessment as of December 31, 2024.

Credit card rewards liability

JPMorganChase offers credit cards with various rewards programs which allow cardholders to earn rewards points based on their account activity and the terms and conditions of the rewards program. Generally, there are no limits on the points that an eligible cardholder can earn, nor do the points expire, and the points can be redeemed for a variety of rewards, including cash (predominantly in the form of account credits), gift cards and travel. The Firm maintains a rewards liability which represents the estimated cost of rewards points earned and expected to be redeemed by cardholders. The liability is accrued as the cardholder earns the benefit and is reduced when the cardholder redeems points. This liability was \$14.4 billion and \$13.2 billion at December 31, 2024 and 2023, respectively, and is recorded in accounts payable and other liabilities on the Consolidated balance sheets. The increase in the liability was driven by continued growth in rewards points earned on higher spend and promotional offers outpacing redemptions throughout 2024.

The rewards liability is sensitive to redemption rate ("RR") and cost per point ("CPP") assumptions. The RR

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assumption is used to estimate the number of points earned by customers that will be redeemed over the life of the account. The CPP assumption is used to estimate the cost of future point redemptions. These assumptions are evaluated periodically considering historical actuals, cardholder redemption behavior and management judgment. Updates to these assumptions will impact the rewards liability. As of December 31, 2024, a combined increase of 25 basis points in RR and 1 basis point in CPP would increase the rewards liability by approximately \$442 million.

Income taxes

JPMorganChase is subject to the income tax laws of the various jurisdictions in which it operates, including U.S. federal, state and local, and non-U.S. jurisdictions. These laws are often complex and may be subject to different interpretations. To determine the financial statement impact of accounting for income taxes, including the provision for income tax expense and unrecognized tax benefits, JPMorganChase must make assumptions and judgments about how to interpret and apply these complex tax laws to numerous transactions and business events, as well as make judgments regarding the timing of when certain items may affect taxable income in the U.S. and non-U.S. tax jurisdictions.

JPMorganChase's interpretations of tax laws around the world are subject to review and examination by the various taxing authorities in the jurisdictions where the Firm operates, and disputes may occur regarding its view on a tax position. These disputes over interpretations with the various taxing authorities may be settled by audit, administrative appeals or adjudication in the court systems of the tax jurisdictions in which the Firm operates. JPMorganChase regularly reviews whether it may be assessed additional income taxes as a result of the resolution of these matters, and the Firm records additional unrecognized tax benefits, as appropriate. In addition, the Firm may revise its estimate of income taxes due to changes in income tax laws, legal interpretations, and business strategies. It is possible that revisions in the Firm's estimate of income taxes may materially affect the Firm's results of operations in any reporting period.

Deferred taxes arise from differences between assets and liabilities measured for financial reporting versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely than not. Deferred taxes are measured using enacted tax rates expected to apply to taxable income in the years in which those temporary differences are expected to be recovered or settled. The effect on deferred taxes of a change in tax rates is recognized within the provision for income taxes in the period enacted.

The Firm has also recognized deferred tax assets in connection with certain tax attributes, including net operating loss ("NOL") carryforwards and foreign tax

credit ("FTC") carryforwards. The Firm performs regular reviews to ascertain whether its deferred tax assets are realizable. These reviews include management's estimates and assumptions regarding future taxable income, including foreign source income, and may incorporate various tax planning strategies, including strategies that may be available to utilize NOLs and FTCs before they expire. In connection with these reviews, if it is determined that a deferred tax asset is not realizable, a valuation allowance is established. The valuation allowance may be reversed in a subsequent reporting period if the Firm determines that, based on revised estimates of future taxable income or changes in tax planning strategies, it is more likely than not that all or part of the deferred tax asset will become realizable. As of December 31, 2024, management has determined it is more likely than not that the Firm will realize its deferred tax assets, net of the existing valuation allowance.

The Firm adjusts its unrecognized tax benefits as necessary when new information becomes available, including changes in tax law and regulations, and interactions with taxing authorities. Uncertain tax positions that meet the more-likely-than-not recognition threshold are measured to determine the amount of benefit to recognize. An uncertain tax position is measured at the largest amount of benefit that management believes is more likely than not to be realized upon settlement. It is possible that the reassessment of JPMorganChase's unrecognized tax benefits may have a material impact on its effective income tax rate in the period in which the reassessment occurs. Although the Firm believes that its estimates are reasonable, the final tax amount could be different from the amounts reflected in the Firm's income tax provisions and accruals. To the extent that the final outcome of these amounts is different than the amounts recorded, such differences will generally impact the Firm's provision for income taxes in the period in which such a determination is made.

The Firm's provision for income taxes is composed of current and deferred taxes. The current and deferred tax provisions are calculated based on estimates and assumptions that could differ from the actual results reflected in income tax returns filed during the subsequent year. Adjustments based on filed returns are generally recorded in the period when the tax returns are filed and the global tax implications are known, which could impact the Firm's effective tax rate.

Refer to Note 25 for additional information on income taxes.

Litigation reserves

Refer to Note 30 for a description of the significant estimates and judgments associated with establishing litigation reserves.

ACCOUNTING AND REPORTING DEVELOPMENTS

Financial Accounting Standards Board (“FASB”) Standards Adopted since January 1, 2024

Standard	Summary of guidance	Effects on financial statements
Fair Value Measurement: Fair Value Measurement of Equity Securities Subject to Contractual Sale Restrictions <i>Issued June 2022</i>	- Clarifies that a contractual sale restriction is not considered part of the unit of account of the equity security and, therefore, is not considered in measuring fair value. - Requires disclosure for investments in equity securities subject to contractual sale restrictions, including: 1) fair value of these investments, 2) nature and remaining duration of the restriction(s) and 3) circumstances that could cause a lapse in the restriction(s).	Adopted prospectively on January 1, 2024, with no impact to the Firm's Consolidated Financial Statements.
Investments - Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method <i>Issued March 2023</i>	- Expands the ability to elect proportional amortization on a program-by-program basis, for additional types of tax-oriented investments (beyond affordable housing tax credit investments). - May be adopted using a full retrospective method, or a modified retrospective method wherein the effect of adoption is reflected as an adjustment to retained earnings at the effective date.	- Adopted under the modified retrospective method on January 1, 2024. - Refer to Note 1 for further information.
Segment Reporting: Improvements to Reportable Segment Disclosures <i>Issued November 2023</i>	- Requires disclosure of significant segment expenses that are readily provided to the chief operating decision maker (“CODM”) and included in segment profit or loss. - Requires disclosure of the composition and aggregate amount of other segment items, which represent the difference between profit or loss and segment revenues less significant segment expenses. - Requires disclosure of the title and position of the CODM and an explanation of how the CODM uses the reported segment measures in assessing segment performance and deciding how to allocate resources.	- Adopted retrospectively for the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024.^(a) - The adoption of this guidance resulted in additional reportable segment disclosures, primarily relating to significant segment expenses and the CODM. Refer to Note 32 for further information.

(a) The accounting standards update applies to the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024, and interim financial statements thereafter.

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FASB Standards Issued but not yet Adopted as of December 31, 2024

Standard	Summary of guidance	Effects on financial statements
Income Taxes: Improvements to Income Tax Disclosures <i>Issued December 2023</i>	- Requires disclosure of income taxes paid disaggregated by 1) federal, state, and foreign taxes and 2) individual jurisdiction on the basis of a quantitative threshold of equal to or greater than 5 percent of total income taxes paid (net of refunds received). - Requires disclosure of the effective tax rate reconciliation by specific categories, at a minimum, with accompanying qualitative disclosures, and separate disclosure of reconciling items based on quantitative thresholds. - Requires categories within the effective tax rate reconciliation to be further disaggregated if quantitative thresholds are met.	- Required effective date: Annual financial statements for the year ending December 31, 2025. - The guidance can be applied on a prospective basis with the option to apply the standard retrospectively. - The Firm is evaluating the potential impact on the Consolidated Financial Statements disclosures.
Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures: Disaggregation of Income Statement Expenses <i>Issued November 2024</i>	Requires additional annual and interim disclosures about specific types of expenses presented in the Consolidated statements of income.	- Required effective date: Annual financial statements for the year ending December 31, 2027, and interim financial statements for the year ending December 31, 2028.^(a) - The guidance can be applied on a prospective basis with the option to apply the standard retrospectively. - The Firm is evaluating the potential impact on the Consolidated Financial Statements disclosures, as well as the Firm's planned date of adoption.

(a) Early adoption is permitted.

FORWARD-LOOKING STATEMENTS

From time to time, the Firm has made and will make forward-looking statements. These statements can be identified by the fact that they do not relate strictly to historical or current facts. Forward-looking statements often use words such as “anticipate,” “target,” “expect,” “estimate,” “intend,” “plan,” “goal,” “believe,” or other words of similar meaning. Forward-looking statements provide JPMorganChase’s current expectations or forecasts of future events, circumstances, results or aspirations. JPMorganChase’s disclosures in this 2024 Form 10-K contain forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. The Firm also may make forward-looking statements in its other documents filed or furnished with the SEC. In addition, the Firm’s senior management may make forward-looking statements orally to investors, analysts, representatives of the media and others.

All forward-looking statements are, by their nature, subject to risks and uncertainties, many of which are beyond the Firm’s control. JPMorganChase’s actual future results may differ materially from those set forth in its forward-looking statements. While there is no assurance that any list of risks and uncertainties or risk factors is complete, below are certain factors which could cause actual results to differ from those in the forward-looking statements:

- Local, regional and global business, economic and political conditions and geopolitical events, including geopolitical tensions and hostilities;
- Changes in laws, rules and regulatory requirements, including capital and liquidity requirements affecting the Firm’s businesses, and the ability of the Firm to address those requirements;
- Heightened regulatory and governmental oversight and scrutiny of JPMorganChase’s business practices, including dealings with retail customers;
- Changes in trade, monetary and fiscal policies and laws;
- Changes in the level of inflation;
- Changes in income tax laws, rules, and regulations;
- Changes in FDIC assessments;
- Securities and capital markets behavior, including changes in market liquidity and volatility;
- Changes in investor sentiment or consumer spending or savings behavior;
- Ability of the Firm to manage effectively its capital and liquidity;
- Changes in credit ratings assigned to the Firm or its subsidiaries;
- Damage to the Firm’s reputation;
- Ability of the Firm to appropriately address social, environmental and sustainability concerns that may arise, including from its business activities;
- Ability of the Firm to deal effectively with an economic slowdown or other economic or market disruption, including, but not limited to, in the interest rate environment;
- Technology changes instituted by the Firm, its counterparties or competitors;
- The effectiveness of the Firm’s control agenda;
- Ability of the Firm to develop or discontinue products and services, and the extent to which products or services previously sold by the Firm require the Firm to incur liabilities or absorb losses not contemplated at their initiation or origination;
- Acceptance of the Firm’s new and existing products and services by the marketplace and the ability of the Firm to innovate and to increase market share;
- Ability of the Firm to attract and retain qualified employees;
- Ability of the Firm to control expenses;
- Competitive pressures;
- Changes in the credit quality of the Firm’s clients, customers and counterparties;
- Adequacy of the Firm’s risk management framework, disclosure controls and procedures and internal control over financial reporting;
- Adverse judicial or regulatory proceedings;
- Ability of the Firm to determine accurate values of certain assets and liabilities;
- Occurrence of natural or man-made disasters or calamities, including health emergencies, the spread of infectious diseases, epidemics or pandemics, an outbreak or escalation of hostilities or other geopolitical instabilities, the effects of climate change or extraordinary events beyond the Firm’s control, and the Firm’s ability to deal effectively with disruptions caused by the foregoing;
- Ability of the Firm to maintain the security of its financial, accounting, technology, data processing and other operational systems and facilities;
- Ability of the Firm to withstand disruptions that may be caused by any failure of its operational systems or those of third parties;
- Ability of the Firm to effectively defend itself against cyber attacks and other attempts by unauthorized parties to access information of the Firm or its customers or to disrupt the Firm’s systems; and
- The other risks and uncertainties detailed in Part I, Item 1A: Risk Factors in JPMorganChase’s 2024 Form 10-K.

Any forward-looking statements made by or on behalf of the Firm speak only as of the date they are made, and JPMorganChase does not undertake to update any forward-looking statements. The reader should, however, consult any further disclosures of a forward-looking nature the Firm may make in any subsequent Annual Reports on Form 10-Ks, Quarterly Reports on Form 10-Qs, or Current Reports on Form 8-K.

Management's report on internal control over financial reporting

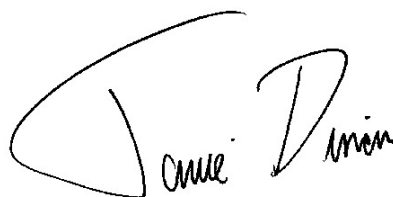
Management of JPMorgan Chase & Co. ("JPMorganChase" or the "Firm") is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, the Firm's principal executive and principal financial officers, or persons performing similar functions, and effected by JPMorganChase's Board of Directors, management and other personnel, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with accounting principles generally accepted in the United States of America ("U.S. GAAP").

JPMorganChase's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records, that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the Firm's assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with U.S. GAAP, and that receipts and expenditures of the Firm are being made only in accordance with authorizations of JPMorganChase's management and directors; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the Firm's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate. Management has completed an assessment of the effectiveness of the Firm's internal control over financial reporting as of December 31, 2024. In making the assessment, management used the "Internal Control — Integrated Framework" ("COSO 2013") promulgated by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO").

Based upon the assessment performed, management concluded that as of December 31, 2024, JPMorganChase's internal control over financial reporting was effective based upon the COSO 2013 framework. Additionally, based upon management's assessment, the Firm determined that there were no material weaknesses in its internal control over financial reporting as of December 31, 2024.

The effectiveness of the Firm's internal control over financial reporting as of December 31, 2024, has been audited by PricewaterhouseCoopers LLP, an independent registered public accounting firm, as stated in their report which appears herein.



James Dimon
Chairman and Chief Executive Officer



Jeremy Barnum
Executive Vice President and Chief Financial Officer

February 14, 2025



To the Board of Directors and Shareholders of JPMorgan Chase & Co.:

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of JPMorgan Chase & Co. and its subsidiaries (the “Firm”) as of December 31, 2024 and 2023, and the related consolidated statements of income, comprehensive income, changes in stockholders’ equity and cash flows for each of the three years in the period ended December 31, 2024, including the related notes (collectively referred to as the “consolidated financial statements”). We also have audited the Firm’s internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Firm as of December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Firm maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on criteria established in *Internal Control - Integrated Framework* (2013) issued by the COSO.

Basis for Opinions

The Firm’s management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management’s report on internal control over financial reporting. Our responsibility is to express opinions on the Firm’s consolidated financial statements and on the Firm’s internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Firm in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial

statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company’s internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company’s internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company’s assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee and that (i) relate to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Allowance for Loan Losses – Portfolio-based component of Wholesale Loan and Credit Card Loan Portfolios

As described in Note 13 to the consolidated financial statements, the allowance for loan losses for the portfolio-based component of the wholesale and credit card loan portfolios was \$22.0 billion on total portfolio-based retained loans of \$919.3 billion at December 31, 2024. The Firm's allowance for loan losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's loan portfolios and considers expected future changes in macroeconomic conditions. The portfolio-based component of the Firm's allowance for loan losses for the wholesale and credit card retained loan portfolios begins with a quantitative calculation of expected credit losses over the expected life of the loan by applying credit loss factors to the estimated exposure at default. The credit loss factors applied are determined based on the weighted average of five internally developed macroeconomic scenarios that take into consideration the Firm's economic outlook as derived through forecast macroeconomic variables, the most significant of which are U.S. unemployment and U.S. real gross domestic product. This quantitative calculation is further adjusted to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate.

The principal considerations for our determination that performing procedures relating to the allowance for loan losses for the portfolio-based component of the wholesale and credit card loan portfolios is a critical audit matter are (i) the significant judgment and estimation by management in the forecast of macroeconomic variables, specifically U.S. unemployment and U.S. real gross domestic product, as the Firm's forecasts of economic conditions significantly affect its estimate of expected credit losses at the balance sheet date, (ii) the significant

judgment and estimation by management in determining the quantitative calculation utilized in their credit loss estimates and the adjustments to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate, which in turn led to a high degree of auditor judgment, subjectivity, and effort in performing procedures and in evaluating audit evidence obtained relating to the credit loss estimates and the appropriateness of the adjustments to the credit loss estimates, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the Firm's allowance for loan losses, including controls over model validation and generation of macroeconomic scenarios. These procedures also included, among others, testing management's process for estimating the allowance for loan losses, which involved (i) evaluating the appropriateness of the models and methodologies used in quantitative calculations; (ii) evaluating the reasonableness of forecasts of U.S. unemployment and U.S. real gross domestic product; (iii) testing the completeness and accuracy of data used in the estimate; and (iv) evaluating the reasonableness of management's adjustments to the quantitative output for the impacts of model imprecision, emerging risk assessments, trends and other subjective factors that are not yet otherwise reflected in the credit loss estimate. These procedures also included the use of professionals with specialized skill and knowledge to assist in evaluating the appropriateness of certain models, methodologies and macroeconomic variables.

Fair Value of Certain Level 3 Financial Instruments

As described in Notes 2 and 3 to the consolidated financial statements, the Firm carries \$1.5 trillion of its assets and \$586.2 billion of its liabilities at fair value on a recurring basis. Included in these balances are \$10.9 billion of trading assets and \$51.8 billion of liabilities measured at fair value on a recurring basis, collectively financial instruments, which are classified as level 3 as they contain one or more inputs to valuation which are unobservable and significant to their fair value measurement. The Firm utilized internally developed valuation models and unobservable inputs to estimate fair value of the level 3 financial instruments. The unobservable inputs used by management to estimate the fair value of certain of these financial instruments include interest rate volatility, equity volatility, Bermudan switch value and correlation relating to interest rates, interest rate-to-foreign exchange,

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equity prices, equity-to-foreign exchange and equity-to-interest rate.

The principal considerations for our determination that performing procedures relating to the fair value of certain level 3 financial instruments is a critical audit matter are (i) the significant judgment and estimation by management in determining the inputs to estimate fair value, which in turn led to a high degree of auditor judgment, subjectivity, and effort in performing procedures and in evaluating audit evidence obtained related to the fair value of these financial instruments, and (ii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the Firm's determination of the fair value, including controls over models, inputs, and data. These procedures also included, among others, the involvement of professionals with specialized skill and knowledge to assist in developing an independent estimate of fair value for a sample of these financial instruments and comparing management's estimate to the independently developed estimate of fair value. Developing the independent estimate involved testing the completeness and accuracy of data provided by management, developing independent inputs and, as appropriate, evaluating and utilizing management's aforementioned unobservable inputs.

A handwritten signature in black ink that reads "PricewaterhouseCoopers LLP". The signature is written in a cursive, flowing style, with the "P" being particularly large and stylized.

New York, New York
February 14, 2025

We have served as the Firm's auditor since 1965.

JPMorgan Chase & Co.
Consolidated statements of income

Year ended December 31, (in millions, except per share data)	2024	2023	2022
Revenue			
Investment banking fees	\$ 8,910	\$ 6,519	\$ 6,686
Principal transactions	24,787	24,460	19,912
Lending- and deposit-related fees	7,606	7,413	7,098
Asset management fees	17,801	15,220	14,096
Commissions and other fees	7,530	6,836	6,581
Investment securities losses	(1,021)	(3,180)	(2,380)
Mortgage fees and related income	1,401	1,176	1,250
Card income	5,497	4,784	4,420
Other income	12,462	5,609	4,322
Noninterest revenue	84,973	68,837	61,985
Interest income	193,933	170,588	92,807
Interest expense	101,350	81,321	26,097
Net interest income	92,583	89,267	66,710
Total net revenue	177,556	158,104	128,695
Provision for credit losses	10,678	9,320	6,389
Noninterest expense			
Compensation expense	51,357	46,465	41,636
Occupancy expense	5,026	4,590	4,696
Technology, communications and equipment expense	9,831	9,246	9,358
Professional and outside services	11,057	10,235	10,174
Marketing	4,974	4,591	3,911
Other expense	9,552	12,045	6,365
Total noninterest expense	91,797	87,172	76,140
Income before income tax expense	75,081	61,612	46,166
Income tax expense	16,610	12,060	8,490
Net income	\$ 58,471	\$ 49,552	\$ 37,676
Net income applicable to common stockholders	\$ 56,868	\$ 47,760	\$ 35,892
Net income per common share data			
Basic earnings per share	\$ 19.79	\$ 16.25	\$ 12.10
Diluted earnings per share	19.75	16.23	12.09
Weighted-average basic shares	2,873.9	2,938.6	2,965.8
Weighted-average diluted shares	2,879.0	2,943.1	2,970.0

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated statements of comprehensive income

Year ended December 31, (in millions)	2024		2023		2022
Net income	\$	58,471	\$	49,552	\$ 37,676
Other comprehensive income/(loss), after-tax					
Unrealized gains/(losses) on investment securities		(87)		5,381	(11,764)
Translation adjustments, net of hedges		(858)		329	(611)
Fair value hedges		(87)		(101)	98
Cash flow hedges		(882)		1,724	(5,360)
Defined benefit pension and OPEB plans		(63)		373	(1,241)
DVA on fair value option elected liabilities		(36)		(808)	1,621
Total other comprehensive income/(loss), after-tax		(2,013)		6,898	(17,257)
Comprehensive income	\$	56,458	\$	56,450	\$ 20,419

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.

Consolidated balance sheets

December 31, (in millions, except share data)	2024	2023
Assets		
Cash and due from banks	\$ 23,372	\$ 29,066
Deposits with banks	445,945	595,085
Federal funds sold and securities purchased under resale agreements (included \$286,771 and \$259,813 at fair value)	295,001	276,152
Securities borrowed (included \$83,962 and \$70,086 at fair value)	219,546	200,436
Trading assets (included assets pledged of \$136,070 and \$128,994)	637,784	540,607
Available-for-sale securities (amortized cost of \$411,045 and \$205,456; included assets pledged of \$10,162 and \$9,219)	406,852	201,704
Held-to-maturity securities	274,468	369,848
Investment securities, net of allowance for credit losses	681,320	571,552
Loans (included \$41,350 and \$38,851 at fair value)	1,347,988	1,323,706
Allowance for loan losses	(24,345)	(22,420)
Loans, net of allowance for loan losses	1,323,643	1,301,286
Accrued interest and accounts receivable	101,223	107,363
Premises and equipment	32,223	30,157
Goodwill, MSRs and other intangible assets	64,560	64,381
Other assets (included \$15,122 and \$12,306 at fair value and assets pledged of \$6,288 and \$6,764)	178,197	159,308
Total assets^(a)	\$ 4,002,814	\$ 3,875,393
Liabilities		
Deposits (included \$33,768 and \$78,384 at fair value)	\$ 2,406,032	\$ 2,400,688
Federal funds purchased and securities loaned or sold under repurchase agreements (included \$226,329 and \$169,003 at fair value)	296,835	216,535
Short-term borrowings (included \$26,521 and \$20,042 at fair value)	52,893	44,712
Trading liabilities	192,883	180,428
Accounts payable and other liabilities (included \$5,893 and \$5,637 at fair value)	280,672	290,307
Beneficial interests issued by consolidated VIEs (included \$1 and \$1 at fair value)	27,323	23,020
Long-term debt (included \$100,780 and \$87,924 at fair value)	401,418	391,825
Total liabilities^(a)	3,658,056	3,547,515
Commitments and contingencies (refer to Notes 28, 29 and 30)		
Stockholders' equity		
Preferred stock (\$1 par value; authorized 200,000,000 shares; issued 2,005,375 and 2,740,375 shares)	20,050	27,404
Common stock (\$1 par value; authorized 9,000,000,000 shares; issued 4,104,933,895 shares)	4,105	4,105
Additional paid-in capital	90,911	90,128
Retained earnings	376,166	332,901
Accumulated other comprehensive losses	(12,456)	(10,443)
Treasury stock, at cost (1,307,313,494 and 1,228,275,301 shares)	(134,018)	(116,217)
Total stockholders' equity	344,758	327,878
Total liabilities and stockholders' equity	\$ 4,002,814	\$ 3,875,393

(a) The following table presents information on assets and liabilities related to VIEs that are consolidated by the Firm at December 31, 2024 and 2023. The assets of the consolidated VIEs are used to settle the liabilities of those entities. The holders of the beneficial interests do not have recourse to the general credit of JPMorganChase. The assets and liabilities in the table below include third-party assets and liabilities of consolidated VIEs and exclude intercompany balances that eliminate in consolidation. Refer to Note 14 for a further discussion.

December 31, (in millions)	2024	2023
Assets		
Trading assets	\$ 3,885	\$ 2,170
Loans	36,510	37,611
All other assets	681	591
Total assets	\$ 41,076	\$ 40,372
Liabilities		
Beneficial interests issued by consolidated VIEs	\$ 27,323	\$ 23,020
All other liabilities	454	263
Total liabilities	\$ 27,777	\$ 23,283

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.
Consolidated statements of changes in stockholders' equity

Year ended December 31, (in millions, except per share data)	2024	2023	2022
Preferred stock			
Balance at January 1	\$ 27,404	\$ 27,404	\$ 34,838
Issuance	2,496	—	—
Redemption	(9,850)	—	(7,434)
Balance at December 31	20,050	27,404	27,404
Common stock			
Balance at January 1 and December 31	4,105	4,105	4,105
Additional paid-in capital			
Balance at January 1	90,128	89,044	88,415
Shares issued and commitments to issue common stock for employee share-based compensation awards, and related tax effects	768	1,084	629
Other	15	—	—
Balance at December 31	90,911	90,128	89,044
Retained earnings			
Balance at January 1	332,901	296,456	272,268
Cumulative effect of change in accounting principles	(161)	449	—
Net income	58,471	49,552	37,676
Preferred stock dividends	(1,259)	(1,501)	(1,595)
Common stock dividends (\$4.80, \$4.10 and \$4.00 per share for 2024, 2023 and 2022, respectively)	(13,786)	(12,055)	(11,893)
Balance at December 31	376,166	332,901	296,456
Accumulated other comprehensive income/(loss)			
Balance at January 1	(10,443)	(17,341)	(84)
Other comprehensive income/(loss), after-tax	(2,013)	6,898	(17,257)
Balance at December 31	(12,456)	(10,443)	(17,341)
Treasury stock, at cost			
Balance at January 1	(116,217)	(107,336)	(105,415)
Repurchase	(19,007)	(9,980)	(3,122)
Reissuance	1,206	1,099	1,201
Balance at December 31	(134,018)	(116,217)	(107,336)
Total stockholders' equity	\$ 344,758	\$ 327,878	\$ 292,332

Effective January 1, 2024, the Firm adopted the Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method accounting guidance. Effective January 1, 2023, the Firm adopted the Financial Instruments – Credit Losses: Troubled Debt Restructurings, and Derivatives and Hedging: Fair Value Hedging – Portfolio Layer Method accounting guidance. Refer to Note 1 for further information.

The Notes to Consolidated Financial Statements are an integral part of these statements.

JPMorgan Chase & Co.

Consolidated statements of cash flows

Year ended December 31, (in millions)	2024	2023	2022
Operating activities			
Net income	\$ 58,471	\$ 49,552	\$ 37,676
Adjustments to reconcile net income to net cash provided by operating activities:			
Provision for credit losses	10,678	9,320	6,389
Depreciation and amortization	7,938	7,512	7,051
Deferred tax (benefit)/expense	2,004	(4,534)	(2,738)
Estimated bargain purchase gain associated with the First Republic acquisition	(103)	(2,775)	—
Initial gain on the Visa share exchange	(7,990)	—	—
Other	1,985	4,301	5,174
Originations and purchases of loans held-for-sale	(212,238)	(115,245)	(149,167)
Proceeds from sales, securitizations and paydowns of loans held-for-sale	205,303	116,430	167,709
Net change in:			
Trading assets	(95,729)	(74,091)	(31,449)
Securities borrowed	(18,762)	(14,902)	20,203
Accrued interest and accounts receivable	5,735	19,928	(22,970)
Other assets	(7,650)	32,970	(2,882)
Trading liabilities	2,276	5,315	11,170
Accounts payable and other liabilities	(90)	(25,388)	58,614
Other operating adjustments	6,160	4,581	2,339
Net cash (used in)/provided by operating activities	(42,012)	12,974	107,119
Investing activities			
Net change in:			
Federal funds sold and securities purchased under resale agreements	(18,706)	39,740	(54,278)
Held-to-maturity securities:			
Proceeds from paydowns and maturities	99,363	53,056	48,626
Purchases	(4,709)	(4,141)	(33,676)
Available-for-sale securities:			
Proceeds from paydowns and maturities	38,499	53,744	39,159
Proceeds from sales	104,625	108,434	84,616
Purchases	(352,712)	(115,499)	(126,258)
Proceeds from sales and securitizations of loans held-for-investment	57,921	47,312	44,892
Other changes in loans, net	(83,176)	(88,343)	(128,968)
Net cash used in First Republic Acquisition	(2,362)	(9,920)	—
All other investing activities, net	(2,146)	(16,740)	(11,932)
Net cash (used in)/provided by investing activities	(163,403)	67,643	(137,819)
Financing activities			
Net change in:			
Deposits	3,299	(32,196)	(136,895)
Federal funds purchased and securities loaned or sold under repurchase agreements	80,288	13,801	8,455
Short-term borrowings	7,439	(1,934)	(8,984)
Beneficial interests issued by consolidated VIEs	1,543	9,029	2,205
Proceeds from long-term borrowings	109,915	75,417	78,442
Payments of long-term borrowings	(96,605)	(64,880)	(45,556)
Proceeds from issuance of preferred stock	2,500	—	—
Redemption of preferred stock	(9,850)	—	(7,434)
Treasury stock repurchased	(18,830)	(9,824)	(3,162)
Dividends paid	(14,783)	(13,463)	(13,562)
All other financing activities, net	(1,469)	(1,521)	234
Net cash provided by/(used in) financing activities	63,447	(25,571)	(126,257)
Effect of exchange rate changes on cash and due from banks and deposits with banks	(12,866)	1,871	(16,643)
Net increase/(decrease) in cash and due from banks and deposits with banks	(154,834)	56,917	(173,600)
Cash and due from banks and deposits with banks at the beginning of the period	624,151	567,234	740,834
Cash and due from banks and deposits with banks at the end of the period	\$ 469,317	\$ 624,151	\$ 567,234
Cash interest paid	\$ 99,642	\$ 77,114	\$ 23,143
Cash income taxes paid, net	11,715	9,908	4,355

The Notes to Consolidated Financial Statements are an integral part of these statements.

Note 1 – Basis of presentation

JPMorgan Chase & Co. (“JPMorganChase” or the “Firm”), a financial holding company incorporated under Delaware law in 1968, is a leading financial services firm based in the U.S., with operations worldwide. The Firm is a leader in investment banking, financial services for consumers and small businesses, commercial banking, financial transaction processing and asset management. Refer to Note 32 for further discussion of the Firm's reportable business segments.

The accounting and financial reporting policies of JPMorganChase and its subsidiaries conform to U.S. GAAP. Additionally, where applicable, the policies conform to the accounting and reporting guidelines prescribed by regulatory authorities.

Consolidation

The Consolidated Financial Statements include the accounts of JPMorganChase and other entities in which the Firm has a controlling financial interest. All material intercompany balances and transactions have been eliminated.

Assets held for clients in an agency or fiduciary capacity by the Firm are not assets of JPMorganChase and are not included on the Consolidated balance sheets.

The Firm determines whether it has a controlling financial interest in an entity by first evaluating whether the entity is a voting interest entity or a variable interest entity.

Voting interest entities

Voting interest entities are entities that have sufficient equity and provide the equity investors voting rights that enable them to make significant decisions relating to the entity's operations. For these types of entities, the Firm's determination of whether it has a controlling interest is primarily based on the amount of voting equity interests held. Entities in which the Firm has a controlling financial interest, through ownership of the majority of the entities' voting equity interests, or through other contractual rights that give the Firm control, are consolidated by the Firm.

Investments in companies in which the Firm has significant influence over operating and financing decisions (but does not own a majority of the voting equity interests) are accounted for (i) in accordance with the equity method of accounting, or (ii) at fair value if the fair value option was elected. These investments are generally included in other assets, with income or loss included in noninterest revenue.

Certain Firm-sponsored asset management funds are structured as limited partnerships or limited liability companies. For many of these entities, the Firm is the

general partner or managing member, but the non-affiliated partners or members have the ability to remove the Firm as the general partner or managing member without cause (i.e., kick-out rights), based on a simple majority vote, or the non-affiliated partners or members have rights to participate in important decisions. Accordingly, the Firm does not consolidate these voting interest entities. However, in the limited cases where the non-managing partners or members do not have substantive kick-out or participating rights, the Firm evaluates the funds as VIEs and consolidates the funds if the Firm is the general partner or managing member and has both power and a potentially significant interest.

The Firm's investment companies and asset management funds have investments in both publicly-held and privately-held entities, including investments in buyouts, growth equity and venture opportunities. These investments are accounted for under investment company guidelines and, accordingly, irrespective of the percentage of equity ownership interests held, are carried on the Consolidated balance sheets at fair value, and are recorded in other assets, with income or loss included in noninterest revenue. If consolidated, the Firm retains the accounting under such specialized investment company guidelines.

Variable interest entities

VIEs are entities that, by design, either (1) lack sufficient equity to permit the entity to finance its activities without additional subordinated financial support from other parties, or (2) have equity investors that do not have the ability to make significant decisions relating to the entity's operations through voting rights, or do not have the obligation to absorb the expected losses, or do not have the right to receive the residual returns of the entity.

The most common type of VIE is an SPE. SPEs are commonly used in securitization transactions in order to isolate certain assets and distribute the cash flows from those assets to investors. The basic SPE structure involves a company selling assets to the SPE; the SPE funds the purchase of those assets by issuing securities to investors. The legal documents that govern the transaction specify how the cash earned on the assets must be allocated to the SPE's investors and other parties that have rights to those cash flows. SPEs are generally structured to insulate investors from claims on the SPE's assets by creditors of other entities, including the creditors of the seller of the assets.

The primary beneficiary of a VIE (i.e., the party that has a controlling financial interest) is required to consolidate the assets and liabilities of the VIE. The

Notes to consolidated financial statements

primary beneficiary is the party that has both (1) the power to direct the activities of the VIE that most significantly impact the VIE's economic performance; and (2) through its interests in the VIE, the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE.

To assess whether the Firm has the power to direct the activities of a VIE that most significantly impact the VIE's economic performance, the Firm considers all the facts and circumstances, including its role in establishing the VIE and its ongoing rights and responsibilities. This assessment includes, first, identifying the activities that most significantly impact the VIE's economic performance; and second, identifying which party, if any, has power over those activities. In general, the parties that make the most significant decisions affecting the VIE (such as asset managers, collateral managers, servicers, or owners of call options or liquidation rights over the VIE's assets) or have the right to unilaterally remove those decision-makers are deemed to have the power to direct the activities of a VIE.

To assess whether the Firm has the obligation to absorb losses of the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE, the Firm considers all of its economic interests, including debt and equity investments, servicing fees, and derivatives or other arrangements deemed to be variable interests in the VIE. This assessment requires that the Firm apply judgment in determining whether these interests, in the aggregate, are considered potentially significant to the VIE. Factors considered in assessing significance include: the design of the VIE, including its capitalization structure; subordination of interests; payment priority; relative share of interests held across various classes within the VIE's capital structure; and the reasons why the interests are held by the Firm.

The Firm performs on-going reassessments of: (1) whether entities previously evaluated under the majority voting-interest framework have become VIEs, based on certain events, and are therefore subject to the VIE consolidation framework; and (2) whether changes in the facts and circumstances regarding the Firm's involvement with a VIE cause the Firm's consolidation conclusion to change.

Refer to Note 14 for further discussion of Firm-sponsored VIEs.

Revenue recognition

Interest income

The Firm recognizes interest income on loans, debt securities, and other debt instruments, generally on a level-yield basis, based on the underlying contractual rate. Refer to Note 7 for further information.

Revenue from contracts with customers

JPMorganChase recognizes noninterest revenue from certain contracts with customers, in investment banking fees, deposit-related fees, asset management fees, commissions and other fees, and components of card income, when the Firm's related performance obligations are satisfied. Refer to Note 6 for further discussion of the Firm's revenue from contracts with customers.

Principal transactions revenue

JPMorganChase carries a portion of its assets and liabilities at fair value. Changes in fair value are reported primarily in principal transactions revenue. Refer to Notes 2 and 3 for further discussion of fair value measurement. Refer to Note 6 for further discussion of principal transactions revenue.

Use of estimates in the preparation of consolidated financial statements

The preparation of the Consolidated Financial Statements requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities, revenue and expense, and disclosures of contingent assets and liabilities. Actual results could be different from these estimates.

Foreign currency translation

JPMorganChase revalues assets, liabilities, revenue and expense denominated in non-U.S. currencies into U.S. dollars using applicable exchange rates.

Gains and losses relating to translating functional currency financial statements for U.S. reporting are included in the Consolidated statements of comprehensive income. Gains and losses relating to nonfunctional currency transactions, including non-U.S. operations where the functional currency is the U.S. dollar, are reported in the Consolidated statements of income.

Offsetting assets and liabilities

U.S. GAAP permits entities to present derivative receivables and derivative payables with the same counterparty and the related cash collateral receivables and payables on a net basis on the Consolidated balance sheets when a legally enforceable master netting agreement exists. U.S. GAAP also permits securities sold and purchased under repurchase agreements and securities borrowed or loaned under securities loan agreements to be presented net when specified conditions are met, including the existence of a legally enforceable master netting agreement. The Firm has elected to net such balances where it has determined that the specified conditions are met.

The Firm uses master netting agreements to mitigate counterparty credit risk in certain transactions, including derivative contracts, resale, repurchase, securities borrowed and securities loaned

agreements. A master netting agreement is a single agreement with a counterparty that permits multiple transactions governed by that agreement to be terminated or accelerated and settled through a single payment in a single currency in the event of a default (e.g., bankruptcy, failure to make a required payment or securities transfer or deliver collateral or margin when due). Upon the exercise of derivatives termination rights by the non-defaulting party (i) all transactions are terminated, (ii) all transactions are valued and the positive values of “in the money” transactions are netted against the negative values of “out of the money” transactions and (iii) the only remaining payment obligation is of one of the parties to pay the netted termination amount. Upon exercise of default rights under repurchase agreements and securities loan agreements in general (i) all transactions are terminated and accelerated, (ii) all values of securities or cash held or to be delivered are calculated, and all such sums are netted against each other and (iii) the only remaining payment obligation is of one of the parties to pay the netted termination amount.

Typical master netting agreements for these types of transactions also often contain a collateral/margin agreement that provides for a security interest in, or title transfer of, securities or cash collateral/margin to the party that has the right to demand margin (the “demanding party”). The collateral/margin agreement typically requires a party to transfer collateral/margin to the demanding party with a value equal to the amount of the margin deficit on a net basis across all transactions governed by the master netting agreement, less any threshold. The collateral/margin agreement grants to the demanding party, upon default by the counterparty, the right to set-off any amounts payable by the counterparty against any posted collateral or the cash equivalent of any posted collateral/margin. It also grants to the demanding party the right to liquidate collateral/margin and to apply the proceeds to an amount payable by the counterparty.

Refer to Note 5 for further discussion of the Firm’s derivative instruments. Refer to Note 11 for further discussion of the Firm’s securities financing agreements.

Statements of cash flows

For JPMorganChase’s Consolidated statements of cash flows, cash is defined as those amounts included in cash and due from banks and deposits with banks on the Consolidated balance sheets.

Accounting standard adopted January 1, 2024 **Equity Method and Joint Ventures: Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method**

The guidance expanded the types of tax-oriented investments, beyond affordable housing tax credit investments, that the Firm can elect on a program by program basis, to be accounted for using the proportional amortization method. This method requires the cost of eligible investments, within an elected program, to be amortized in proportion to the tax benefits received with the resulting amortization reported directly in income tax expense, which aligns with the associated tax credits and other tax benefits. Eligible investments must meet certain criteria, including that substantially all of the return is from income tax credits and other income tax benefits.

This guidance was adopted on January 1, 2024 under the modified retrospective method. The adoption of this guidance resulted in a change to the classification and timing of the amortization associated with certain of the Firm’s alternative energy tax-oriented investments. As a result of the adoption, the amortization of these investments that was previously recognized in other income is now recognized in income tax expense. The change in accounting resulted in a decrease to retained earnings of \$161 million and increased the Firm’s income tax expense and the effective tax rate by approximately \$450 million and two percentage points, respectively, in the first quarter of 2024, with no material impact to net income.

The guidance requires additional disclosure for all investments that generate income tax credits and other income tax benefits from a tax-oriented investment program for which the Firm has elected to apply the proportional amortization method. The guidance also requires a reevaluation of eligible investments when significant modifications or events occur that result in a change in the nature of the investment or a change in the Firm’s relationship with the underlying project.

Refer to Notes 6, 14 and 25 for additional information.

Accounting standards adopted January 1, 2023 **Derivatives and Hedging: Fair Value Hedging – Portfolio Layer Method**

The adoption of this guidance expanded the ability to hedge a portfolio of fixed-rate assets in a qualifying hedge accounting relationship. As permitted by the guidance, the Firm elected to transfer HTM securities to AFS and designated those securities in a portfolio layer method hedge upon adoption. The adoption impact of the transfer on retained earnings was not material.

Notes to consolidated financial statements

Financial Instruments – Credit Losses: Troubled Debt Restructurings (“TDRs”)

The adoption of this guidance eliminated the requirement to measure the allowance for TDRs using a discounted cash flow (“DCF”) methodology and allowed the option of a non-DCF portfolio-based approach for modified loans to troubled borrowers. The Firm elected this option for all portfolios of modified loans to troubled borrowers except collateral-dependent loans and nonaccrual risk-rated loans, for which the Firm elected to continue applying a DCF methodology. The adoption of this guidance under the modified retrospective method on January 1, 2023, resulted in a \$446 million increase to retained earnings.

Significant accounting policies

The following table identifies JPMorganChase’s other significant accounting policies and the Note and page where a detailed description of each policy can be found.

Fair value measurement	Note 2	page 181
Fair value option	Note 3	page 203
Derivative instruments	Note 5	page 209
Noninterest revenue and noninterest expense	Note 6	page 225
Interest income and Interest expense	Note 7	page 229
Pension and other postretirement employee benefit plans	Note 8	page 230
Employee share-based incentives	Note 9	page 233
Investment securities	Note 10	page 235
Securities financing activities	Note 11	page 240
Loans	Note 12	page 243
Allowance for credit losses	Note 13	page 266
Variable interest entities	Note 14	page 271
Goodwill, mortgage servicing rights, and other intangible assets	Note 15	page 280
Premises and equipment	Note 16	page 285
Leases	Note 18	page 286
Accounts payable & other liabilities	Note 19	page 288
Long-term debt	Note 20	page 289
Earnings per share	Note 23	page 294
Income taxes	Note 25	page 296
Off-balance sheet lending-related financial instruments, guarantees and other commitments	Note 28	page 302
Litigation	Note 30	page 309

Note 2 – Fair value measurement

JPMorganChase carries a portion of its assets and liabilities at fair value. These assets and liabilities are predominantly carried at fair value on a recurring basis (i.e., assets and liabilities that are measured and reported at fair value on the Firm's Consolidated balance sheets). Certain assets, liabilities and unfunded lending-related commitments are measured at fair value on a nonrecurring basis; that is, they are not measured at fair value on an ongoing basis but are subject to fair value adjustments only in certain circumstances (for example, when there is evidence of impairment).

Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. Fair value is based on quoted market prices or inputs, where available. If prices or quotes are not available, fair value is based on valuation models and other valuation techniques that consider relevant transaction characteristics (such as maturity) and use, as inputs, observable or unobservable market parameters, including yield curves, interest rates, volatilities, prices (such as commodity, equity or debt prices), correlations, foreign exchange rates and credit curves. Fair value may also incorporate valuation adjustments.

The level of precision in estimating unobservable market inputs or other factors can affect the amount of gain or loss recorded for a particular position. Furthermore, while the Firm believes its valuation methods are appropriate and consistent with those of other market participants, the methods and assumptions used reflect management judgment and may vary across the Firm's businesses and portfolios.

The Firm uses various methodologies and assumptions in the determination of fair value. The use of different methodologies or assumptions by other market participants compared with those used by the Firm could result in the Firm deriving a different estimate of fair value at the reporting date.

Valuation process

Risk-taking functions are responsible for providing fair value estimates for assets and liabilities carried on the Consolidated balance sheets at fair value. The Firm's Valuation Control Group ("VCG"), which is part of the Firm's Finance function and independent of the risk-taking functions, is responsible for verifying these estimates and determining any fair value adjustments that may be required to ensure that the Firm's positions are recorded at fair value. In addition, the Firm's Valuation Governance Forum ("VGF"), which is composed of senior finance and risk executives, is responsible for overseeing the management of risks arising from valuation activities conducted across the Firm. The Firmwide VGF is chaired by the Firmwide

head of the VCG (under the direction of the Firm's Controller), and includes sub-forums covering the CIB, CCB, AWM and certain corporate functions including Treasury and CIO.

Price verification process

The VCG verifies fair value estimates provided by the risk-taking functions by leveraging independently derived prices, valuation inputs and other market data, where available. Where independent prices or inputs are not available, the VCG performs additional review to ensure the reasonableness of the estimates. The additional review may include evaluating the limited market activity including client unwinds, benchmarking valuation inputs to those used for similar instruments, decomposing the valuation of structured instruments into individual components, comparing expected to actual cash flows, reviewing profit and loss trends, and reviewing trends in collateral valuation. There are also additional levels of management review for more significant or complex positions.

The VCG determines any valuation adjustments that may be required to the estimates provided by the risk-taking functions. No adjustments to quoted prices are applied for instruments classified within level 1 of the fair value hierarchy (refer to the discussion of the fair value hierarchy on page 182 for further information). For other positions, judgment is required to assess the need for valuation adjustments to appropriately reflect liquidity considerations, unobservable parameters, and, for certain portfolios that meet specified criteria, the size of the net open risk position. The determination of such adjustments follows a consistent framework across the Firm:

- Liquidity valuation adjustments are considered where an observable external price or valuation parameter exists but is of lower reliability, potentially due to lower market activity. Liquidity valuation adjustments are made based on current market conditions. Factors that may be considered in determining the liquidity adjustment include analysis of: (1) the estimated bid-offer spread for the instrument being traded; (2) alternative pricing points for similar instruments in active markets; and (3) the range of reasonable values that the price or parameter could take.
- The Firm manages certain portfolios of financial instruments on the basis of net open risk exposure and, as permitted by U.S. GAAP, has elected to estimate the fair value of such portfolios on the basis of a transfer of the entire net open risk position in an orderly transaction. Where this is the case, valuation adjustments may be necessary to reflect the cost of exiting a larger-than-normal market-size net open risk position. Where applied, such adjustments are

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based on factors that a relevant market participant would consider in the transfer of the net open risk position, including the size of the adverse market move that is likely to occur during the period required to sufficiently reduce the net open risk position.

- Uncertainty adjustments related to unobservable parameters may be made when positions are valued using prices or input parameters to valuation models that are unobservable due to a lack of market activity or because they cannot be implied from observable market data. Such prices or parameters must be estimated and are, therefore, subject to management judgment. Adjustments are made to reflect the uncertainty inherent in the resulting valuation estimate.
- Where appropriate, the Firm also applies adjustments to its estimates of fair value in order to appropriately reflect counterparty credit quality (CVA), the Firm's own creditworthiness (DVA) and the impact of funding (FVA), using a consistent framework across the Firm. Refer to Credit and funding adjustments on page 198 of this Note for more information on such adjustments.

Valuation model review and approval

If prices or quotes are not available for an instrument or a similar instrument, fair value is generally determined using valuation models that consider relevant transaction terms such as maturity and use as inputs market-based or independently sourced parameters. Where this is the case the price verification process described above is applied to the inputs in those models.

Under the Firm's Estimations and Model Risk Management Policy, MRGR reviews and approves new models, as well as material changes to existing models, prior to implementation in the operating environment. In certain circumstances exceptions may be granted to the Firm's policy to allow a model to be used prior to review or approval. MRGR may also require the user to take appropriate actions to mitigate the model risk if it is to be used in the interim. These actions will depend on the model and may include, for example, limitation of trading activity.

Fair value hierarchy

A three-level fair value hierarchy has been established under U.S. GAAP for disclosure of fair value measurements. The fair value hierarchy is based on the observability of inputs to the valuation of an asset or liability as of the measurement date. The three levels are defined as follows.

- Level 1 – inputs to the valuation methodology are quoted prices (unadjusted) for identical assets or liabilities in active markets.
- Level 2 – inputs to the valuation methodology include quoted prices for similar assets and liabilities in active markets, and inputs that are observable for the asset or liability, either directly or indirectly, for substantially the full term of the financial instrument.
- Level 3 – one or more inputs to the valuation methodology are unobservable and significant to the fair value measurement.

A financial instrument's categorization within the fair value hierarchy is based on the lowest level of input that is significant to the fair value measurement.

The following table describes the valuation methodologies generally used by the Firm to measure its significant products/instruments at fair value, including the general classification of such instruments pursuant to the fair value hierarchy.

Product/instrument	Valuation methodology	Classifications in the fair value hierarchy
Securities financing agreements	Valuations are based on discounted cash flows, which consider: • Derivative features: refer to the discussion of derivatives below for further information • Market rates for the respective maturity • Collateral characteristics	Predominantly level 2
Loans and lending-related commitments — wholesale Loans carried at fair value (trading loans and non-trading loans) and associated lending-related commitments	Where observable market data is available, valuations are based on: • Observed market prices (circumstances are infrequent) • Relevant broker quotes • Observed market prices for similar instruments Where observable market data is unavailable or limited, valuations are based on discounted cash flows, which consider the following: • Credit spreads derived from the cost of CDS; or benchmark credit curves developed by the Firm, by industry and credit rating • Prepayment speed • Collateral characteristics	Level 2 or 3
Loans — consumer Loans carried at fair value — residential mortgage loans expected to be sold	Fair value is based on observable market prices for mortgage-backed securities with similar collateral and incorporates adjustments to these prices to account for differences between the securities and the value of the underlying loans, which include credit characteristics, portfolio composition, and liquidity.	Predominantly level 2
Investment and trading securities	Quoted market prices In the absence of quoted market prices, securities are valued based on: • Observable market prices for similar securities • Relevant broker quotes • Discounted cash flows In addition, the following inputs to discounted cash flows are used for the following products: Mortgage- and asset-backed securities specific inputs: • Collateral characteristics • Deal-specific payment and loss allocations • Current market assumptions related to yield, prepayment speed, conditional default rates and loss severity Collateralized loan obligations (“CLOs”) specific inputs: • Collateral characteristics • Deal-specific payment and loss allocations • Expected prepayment speed, conditional default rates, loss severity • Credit spreads • Credit rating data	Level 1 Level 2 or 3
Physical commodities	Valued using observable market prices or data.	Predominantly Level 1 or 2

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Product/instrument	Valuation methodology	Classifications in the fair value hierarchy
Derivatives	Actively traded derivatives, e.g., exchange-traded derivatives, that are valued using quoted prices.	Level 1
	Derivatives that are valued using models such as the Black-Scholes option pricing model, simulation models, or a combination of models that may use observable or unobservable valuation inputs as well as considering the contractual terms. The key valuation inputs used will depend on the type of derivative and the nature of the underlying instruments and may include equity prices, commodity prices, foreign exchange rates, volatilities, correlations, CDS spreads, recovery rates and prepayment speed. In addition, specific inputs used for derivatives that are valued based on models with significant unobservable inputs are as follows: Interest rate and FX exotic derivatives specific inputs include: • Interest rate curve • Interest rate volatility • Interest rate spread volatility • Bermudan switch value • Interest rate correlation • Interest rate-FX correlation • Foreign exchange correlation Credit derivatives specific inputs include: • Credit correlation between the underlying debt instruments Equity derivatives specific inputs include: • Forward equity price • Equity volatility • Equity correlation • Equity-FX correlation • Equity-IR correlation Commodity derivatives specific inputs include: • Forward commodity price • Commodity volatility • Commodity correlation Additionally, adjustments are made to reflect counterparty credit quality (CVA) and the impact of funding (FVA). Refer to page 198 of this Note.	Level 2 or 3
Mortgage servicing rights	Refer to Mortgage servicing rights in Note 15.	Level 3
Private equity direct investments	Fair value is estimated using all available information; the range of potential inputs include: • Transaction prices • Trading multiples of comparable public companies • Operating performance of the underlying portfolio company • Adjustments as required, since comparable public companies are not identical to the company being valued, and for company-specific issues including lack of liquidity • Additional available inputs relevant to the investment	Level 2 or 3

Product/instrument	Valuation methodology	Classification in the fair value hierarchy
Fund investments (e.g., mutual/collective investment funds, private equity funds, hedge funds, and real estate funds)	Net asset value - NAV is supported by the ability to redeem and purchase at the NAV level - Adjustments to the NAV as required, for restrictions on redemption (e.g., lock-up periods or withdrawal limitations) or where observable activity is limited	Level 1 Level 2 or 3^(a)
Beneficial interests issued by consolidated VIEs	Valued using observable market information, where available. In the absence of observable market information, valuations are based on the fair value of the underlying assets held by the VIE.	Level 2 or 3
Structured notes (included in deposits, short-term borrowings and long-term debt)	Valuations are based on discounted cash flow analyses that consider the embedded derivative and the terms and payment structure of the note. The embedded derivative features are considered using models such as the Black-Scholes option pricing model, simulation models, or a combination of models that may use observable or unobservable valuation inputs, depending on the embedded derivative. The specific inputs used vary according to the nature of the embedded derivative features, as described in the discussion above regarding derivatives valuation. Adjustments are then made to this base valuation to reflect the Firm's own credit risk (DVA). Refer to page 198 of this Note.	Level 2 or 3

(a) Excludes certain investments that are measured at fair value using the net asset value per share (or its equivalent) as a practical expedient.

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The following table presents the assets and liabilities reported at fair value as of December 31, 2024 and 2023, by major product category and fair value hierarchy.

Assets and liabilities measured at fair value on a recurring basis

	Fair value hierarchy			Derivative netting adjustments ^(a)	Total fair value
	Level 1	Level 2	Level 3		
December 31, 2024 (in millions)					
Federal funds sold and securities purchased under resale agreements	\$ —	\$ 286,771	\$ —	\$ —	\$ 286,771
Securities borrowed	—	83,962	—	—	83,962
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. GSEs and government agencies ^(a)	—	104,312	488	—	104,800
Residential – nonagency	—	2,282	5	—	2,287
Commercial – nonagency	—	1,283	10	—	1,293
Total mortgage-backed securities	—	107,877	503	—	108,380
U.S. Treasury, GSEs and government agencies ^(a)	150,580	11,702	—	—	162,282
Obligations of U.S. states and municipalities	—	6,100	1	—	6,101
Certificates of deposit, bankers' acceptances and commercial paper	—	3,950	—	—	3,950
Non-U.S. government debt securities	34,108	54,335	152	—	88,595
Corporate debt securities	—	33,591	390	—	33,981
Loans	—	10,228	1,088	—	11,316
Asset-backed securities	—	2,813	10	—	2,823
Total debt instruments	184,688	230,596	2,144	—	417,428
Equity securities	130,307	1,359	62	—	131,728
Physical commodities ^(b)	5,957	1,533	26	—	7,516
Other	—	19,935	210	—	20,145
Total debt and equity instruments^(c)	320,952	253,423	2,442	—	576,817
Derivative receivables:					
Interest rate	4,934	282,019	3,781	(265,789)	24,945
Credit	—	10,379	708	(10,273)	814
Foreign exchange	196	261,520	1,204	(237,608)	25,312
Equity	—	82,855	2,365	(79,935)	5,285
Commodity	—	15,232	394	(11,015)	4,611
Total derivative receivables	5,130	652,005	8,452	(604,620)	60,967
Total trading assets^(d)	326,082	905,428	10,894	(604,620)	637,784
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. GSEs and government agencies ^(a)	—	91,893	—	—	91,893
Residential – nonagency	—	4,811	—	—	4,811
Commercial – nonagency	—	4,057	8	—	4,065
Total mortgage-backed securities	—	100,761	8	—	100,769
U.S. Treasury and government agencies	234,491	288	—	—	234,779
Obligations of U.S. states and municipalities	—	17,913	—	—	17,913
Non-U.S. government debt securities	23,973	12,272	—	—	36,245
Corporate debt securities	—	70	—	—	70
Asset-backed securities:					
Collateralized loan obligations	—	14,943	—	—	14,943
Other ^(a)	—	2,133	—	—	2,133
Total available-for-sale securities	258,464	148,380	8	—	406,852
Loans ^(e)	—	38,934	2,416	—	41,350
Mortgage servicing rights	—	—	9,121	—	9,121
Other assets ^(d)	5,732	6,997	1,344	—	14,073
Total assets measured at fair value on a recurring basis	\$ 590,278	\$ 1,470,472	\$ 23,783	\$ (604,620)	\$ 1,479,913
Deposits	\$ —	\$ 31,583	\$ 2,185	\$ —	\$ 33,768
Federal funds purchased and securities loaned or sold under repurchase agreements	—	226,329	—	—	226,329
Short-term borrowings	—	23,045	3,476	—	26,521
Trading liabilities:					
Debt and equity instruments ^(c)	120,719	32,457	46	—	153,222
Derivative payables:					
Interest rate	3,981	266,767	3,480	(264,989)	9,239
Credit	—	12,725	1,071	(11,898)	1,898
Foreign exchange	187	253,196	1,184	(238,970)	15,597
Equity	—	90,908	5,231	(87,491)	8,648
Commodity	—	14,021	467	(10,209)	4,279
Total derivative payables	4,168	637,617	11,433	(613,557)	39,661
Total trading liabilities	124,887	670,074	11,479	(613,557)	192,883
Accounts payable and other liabilities	3,100	2,717	76	—	5,893
Beneficial interests issued by consolidated VIEs	—	1	—	—	1
Long-term debt	—	66,216	34,564	—	100,780
Total liabilities measured at fair value on a recurring basis	\$ 127,987	\$ 1,019,965	\$ 51,780	\$ (613,557)	\$ 586,175

December 31, 2023 (in millions)	Fair value hierarchy			Derivative netting adjustments ^(e)	Total fair value
	Level 1	Level 2	Level 3		
Federal funds sold and securities purchased under resale agreements	\$ —	\$ 259,813	\$ —	\$ —	\$ 259,813
Securities borrowed	—	70,086	—	—	70,086
Trading assets:					
Debt instruments:					
Mortgage-backed securities:					
U.S. GSEs and government agencies ^(a)	—	73,840	758	—	74,598
Residential – nonagency	—	1,921	5	—	1,926
Commercial – nonagency	—	1,362	12	—	1,374
Total mortgage-backed securities	—	77,123	775	—	77,898
U.S. Treasury, GSEs and government agencies ^(a)	133,997	9,998	—	—	143,995
Obligations of U.S. states and municipalities	—	5,858	10	—	5,868
Certificates of deposit, bankers' acceptances and commercial paper	—	756	—	—	756
Non-U.S. government debt securities	24,846	55,557	179	—	80,582
Corporate debt securities	—	32,854	484	—	33,338
Loans	—	7,872	684	—	8,556
Asset-backed securities	—	2,199	6	—	2,205
Total debt instruments	158,843	192,217	2,138	—	353,198
Equity securities	107,926	679	127	—	108,732
Physical commodities ^(b)	2,479	3,305	7	—	5,791
Other	—	17,879	101	—	17,980
Total debt and equity instruments^(c)	269,248	214,080	2,373	—	485,701
Derivative receivables:					
Interest rate	2,815	243,578	4,298	(224,367)	26,324
Credit	—	8,644	1,010	(9,103)	551
Foreign exchange	149	204,737	889	(187,756)	18,019
Equity	—	55,167	2,522	(52,761)	4,928
Commodity	—	15,234	205	(10,397)	5,042
Total derivative receivables	2,964	527,360	8,924	(484,384)	54,864
Total trading assets^(d)	272,212	741,440	11,297	(484,384)	540,565
Available-for-sale securities:					
Mortgage-backed securities:					
U.S. GSEs and government agencies ^(a)	—	85,170	—	—	85,170
Residential – nonagency	—	3,639	—	—	3,639
Commercial – nonagency	—	2,803	—	—	2,803
Total mortgage-backed securities	—	91,612	—	—	91,612
U.S. Treasury and government agencies	57,683	122	—	—	57,805
Obligations of U.S. states and municipalities	—	21,367	—	—	21,367
Non-U.S. government debt securities	13,095	8,187	—	—	21,282
Corporate debt securities	—	100	—	—	100
Asset-backed securities:					
Collateralized loan obligations	—	6,752	—	—	6,752
Other ^(a)	—	2,786	—	—	2,786
Total available-for-sale securities	70,778	130,926	—	—	201,704
Loans ^(e)	—	35,772	3,079	—	38,851
Mortgage servicing rights	—	—	8,522	—	8,522
Other assets ^(d)	6,635	3,929	758	—	11,322
Total assets measured at fair value on a recurring basis	\$ 349,625	\$ 1,241,966	\$ 23,656	\$ (484,384)	\$ 1,130,863
Deposits	\$ —	\$ 76,551	\$ 1,833	\$ —	\$ 78,384
Federal funds purchased and securities loaned or sold under repurchase agreements	—	169,003	—	—	169,003
Short-term borrowings	—	18,284	1,758	—	20,042
Trading liabilities:					
Debt and equity instruments ^(c)	107,292	32,252	37	—	139,581
Derivative payables:					
Interest rate	4,409	232,277	3,796	(228,586)	11,896
Credit	—	11,293	745	(10,949)	1,089
Foreign exchange	147	211,289	827	(199,643)	12,620
Equity	—	60,887	4,924	(56,443)	9,368
Commodity	—	15,894	484	(10,504)	5,874
Total derivative payables	4,556	531,640	10,776	(506,125)	40,847
Total trading liabilities	111,848	563,892	10,813	(506,125)	180,428
Accounts payable and other liabilities	3,968	1,617	52	—	5,637
Beneficial interests issued by consolidated VIEs	—	1	—	—	1
Long-term debt	—	60,198	27,726	—	87,924
Total liabilities measured at fair value on a recurring basis	\$ 115,816	\$ 889,546	\$ 42,182	\$ (506,125)	\$ 541,419

- (a) At December 31, 2024 and 2023, included total U.S. GSE obligations of \$120.1 billion and \$78.5 billion, respectively, which were mortgage-related.
- (b) Physical commodities inventories are generally accounted for at the lower of cost or net realizable value. "Net realizable value" is a term defined in U.S. GAAP as not exceeding fair value less costs to sell ("transaction costs"). Transaction costs for the Firm's physical commodities inventories are either not applicable or immaterial to the value of the inventory. Therefore, net realizable value approximates fair value for the Firm's physical commodities inventories. When fair value hedging has been applied (or when net realizable value is below cost), the carrying value of physical commodities approximates fair value, because under fair value hedge accounting, the cost basis is adjusted for changes in fair value. Refer to Note 5 for a further discussion of the Firm's hedge accounting relationships. To provide consistent fair value disclosure information, all physical commodities inventories have been included in each period presented.

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- (c) Balances reflect the reduction of securities owned (long positions) by the amount of identical securities sold but not yet purchased (short positions).
- (d) Certain investments that are measured at fair value using the net asset value per share (or its equivalent) as a practical expedient are not required to be classified in the fair value hierarchy. At both December 31, 2024 and 2023, the fair values of these investments, which include certain hedge funds, private equity funds, real estate and other funds, were \$1.0 billion, primarily reported in other assets.
- (e) As permitted under U.S. GAAP, the Firm has elected to net derivative receivables and derivative payables and the related cash collateral received and paid when a legally enforceable master netting agreement exists. The level 3 balances would be reduced if netting were applied, including the netting benefit associated with cash collateral.

Level 3 valuations

The Firm has established well-structured processes for determining fair value, including for instruments where fair value is estimated using significant unobservable inputs (level 3). Refer to pages 181–185 of this Note for further information on the Firm’s valuation process and a detailed discussion of the determination of fair value for individual financial instruments.

Estimating fair value requires the application of judgment. The type and level of judgment required is largely dependent on the amount of observable market information available to the Firm. For instruments valued using internally developed valuation models and other valuation techniques that use significant unobservable inputs and are therefore classified within level 3 of the fair value hierarchy, judgments used to estimate fair value are more significant than those required when estimating the fair value of instruments classified within levels 1 and 2.

In arriving at an estimate of fair value for an instrument within level 3, management must first determine the appropriate valuation model or other valuation technique to use. Second, due to the lack of observability of significant inputs, management must assess relevant empirical data in deriving valuation inputs including transaction details, yield curves, interest rates, prepayment speed, default rates, volatilities, correlations, prices (such as commodity, equity or debt prices), valuations of comparable instruments, foreign exchange rates and credit curves.

The following table presents the Firm’s primary level 3 financial instruments, the valuation techniques used to measure the fair value of those financial instruments, the significant unobservable inputs, the range of values for those inputs and the weighted or arithmetic averages of such inputs. While the determination to classify an instrument within level 3 is based on the significance of the unobservable inputs to the overall fair value measurement, level 3 financial instruments typically include observable components (that is, components that are actively quoted and can be validated to external sources) in addition to the unobservable components. The level 1 and/or level 2 inputs are not included in the table. In addition, the Firm manages the risk of the observable components of level 3 financial instruments using securities and derivative positions that are classified within levels 1 or 2 of the fair value hierarchy.

The range of values presented in the table is representative of the highest and lowest level input used to value the significant groups of instruments within a product/instrument classification. Where provided, the weighted averages of the input values presented in the table are calculated based on the fair value of the instruments that the input is being used to value.

In the Firm’s view, the input range, weighted and arithmetic average values do not reflect the degree of input uncertainty or an assessment of the reasonableness of the Firm’s estimates and assumptions. Rather, they reflect the characteristics of the various instruments held by the Firm and the relative distribution of instruments within the range of characteristics. For example, two option contracts may have similar levels of market risk exposure and valuation uncertainty, but may have significantly different implied volatility levels because the option contracts have different underlyings, tenors, or strike prices. The input range and weighted and arithmetic average values will therefore vary from period-to-period and parameter-to-parameter based on the characteristics of the instruments held by the Firm at each balance sheet date.

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Level 3 inputs^(a)

December 31, 2024

Product/Instrument	Fair value (in millions)	Principal valuation technique	Unobservable inputs ^(g)	Range of input values		Average ⁽ⁱ⁾
Residential mortgage-backed securities and loans ^(b)	\$ 861	Discounted cash flows	Yield	0%	103%	8%
			Prepayment speed	3%	13%	8%
			Conditional default rate	0%	7%	0%
			Loss severity	0%	110%	5%
Commercial mortgage-backed securities and loans ^(c)	1,424	Market comparables	Price	\$0	\$90	\$81
Corporate debt securities	390	Market comparables	Price	\$0	\$148	\$95
Loans ^(d)	1,730	Market comparables	Price	\$0	\$107	\$79
Non-U.S. government debt securities	152	Market comparables	Price	\$0	\$103	\$95
Net interest rate derivatives	293	Option pricing	Interest rate volatility	9bps	1,097bps	115bps
			Interest rate spread volatility	37bps	77bps	64bps
			Bermudan switch value	0%	45%	17%
			Interest rate correlation	(82)%	97%	64%
			IR-FX correlation	(35)%	60%	8%
	8	Discounted cash flows	Prepayment speed	0%	21%	7%
Net credit derivatives	(393)	Discounted cash flows	Credit correlation	31%	79%	47%
			Credit spread	0bps	2,717bps	331bps
			Recovery rate	10%	90%	61%
	30	Market comparables	Price	\$0	\$115	\$74
Net foreign exchange derivatives	62	Option pricing	IR-FX correlation	(40)%	60%	22%
	(42)	Discounted cash flows	Prepayment speed		11%	11%
			Interest rate curve	1%	27%	7%
Net equity derivatives	(2,866)	Option pricing	Forward equity price ^(h)	76%	153%	100%
			Equity volatility	5%	135%	32%
			Equity correlation	17%	100%	56%
			Equity-FX correlation	(80)%	65%	(32)%
			Equity-IR correlation	5%	25%	14%
Net commodity derivatives	(73)	Option pricing	Oil commodity forward	\$87 / BBL	\$291 / BBL	\$160 / BBL
			Natural gas commodity forward	\$2 / MMBTU	\$7 / MMBTU	\$4 / MMBTU
			Commodity volatility	2%	43%	5%
			Commodity correlation	(35)%	98%	(9)%
MSRs	9,121	Discounted cash flows	Refer to Note 15			
Long-term debt, short-term borrowings, and deposits ^(e)	38,901	Option pricing	Interest rate volatility	9bps	1,097bps	115bps
			Bermudan switch value	0%	45%	17%
			Interest rate correlation	(82)%	97%	64%
			IR-FX correlation	(35)%	60%	8%
			Equity volatility	2%	137%	28%
			Equity correlation	17%	100%	56%
			Equity-FX correlation	(80)%	65%	(32)%
			Equity-IR correlation	5%	25%	14%
	1,324	Discounted cash flows	Credit correlation	30%	57%	46%
			Credit spread	2bps	270bps	83bps
			Recovery rate	20%	40%	35%
			Yield	5%	20%	10%
			Loss severity	0%	100%	50%
Other level 3 assets and liabilities, net ^(f)	1,531					

(a) The categories presented in the table have been aggregated based upon the product type, which may differ from their classification on the Consolidated balance sheets. Furthermore, the inputs presented for each valuation technique in the table are, in some cases, not applicable to every instrument valued using the technique as the characteristics of the instruments can differ.

(b) Comprises U.S. GSE and government agency securities of \$488 million, nonagency securities of \$5 million and non-trading loans of \$368 million.

(c) Comprises nonagency securities of \$18 million, trading loans of \$65 million and non-trading loans of \$1.3 billion.

(d) Comprises trading loans of \$1.0 billion and non-trading loans of \$707 million.

(e) Long-term debt, short-term borrowings and deposits include structured notes issued by the Firm that are financial instruments that typically contain embedded derivatives. The estimation of the fair value of structured notes includes the derivative features embedded within the instrument. The significant unobservable inputs are broadly consistent with those presented for derivative receivables.

(f) Includes equity securities of \$734 million including \$672 million in Other assets, for which quoted prices are not readily available and the fair value is generally based on internal valuation techniques such as EBITDA multiples and comparable analysis. All other level 3 assets and liabilities are insignificant both individually and in aggregate.

(g) Price is a significant unobservable input for certain instruments. When quoted market prices are not readily available, reliance is generally placed on price-based internal valuation techniques. The price input is expressed assuming a par value of \$100.

(h) Forward equity price is expressed as a percentage of the current equity price.

(i) Amounts represent weighted averages except for derivative related inputs where arithmetic averages are used.

Changes in and ranges of unobservable inputs

The following discussion provides a description of the impact on a fair value measurement of a change in each unobservable input in isolation, and the interrelationship between unobservable inputs, where relevant and significant. The impact of changes in inputs may not be independent, as a change in one unobservable input may give rise to a change in another unobservable input. Where relationships do exist between two unobservable inputs, those relationships are discussed below. Relationships may also exist between observable and unobservable inputs (for example, as observable interest rates rise, unobservable prepayment rates decline); such relationships have not been included in the discussion below. In addition, for each of the individual relationships described below, the inverse relationship would also generally apply.

The following discussion also provides a description of attributes of the underlying instruments and external market factors that affect the range of inputs used in the valuation of the Firm's positions.

Yield – The yield of an asset is the interest rate used to discount future cash flows in a discounted cash flow calculation. An increase in the yield, in isolation, would result in a decrease in a fair value measurement.

Credit spread – The credit spread is the amount of additional annualized return over the market interest rate that a market participant would demand for taking exposure to the credit risk of an instrument. The credit spread for an instrument forms part of the discount rate used in a discounted cash flow calculation. Generally, an increase in the credit spread would result in a decrease in a fair value measurement.

The yield and the credit spread of a particular mortgage-backed security primarily reflect the risk inherent in the instrument. The yield is also impacted by the absolute level of the coupon paid by the instrument (which may not correspond directly to the level of inherent risk). Therefore, the range of yield and credit spreads reflects the range of risk inherent in various instruments owned by the Firm. The risk inherent in mortgage-backed securities is driven by the subordination of the security being valued and the characteristics of the underlying mortgages within the collateralized pool, including borrower FICO scores, LTV ratios for residential mortgages and the nature of the property and/or any tenants for commercial mortgages. For corporate debt securities, obligations of U.S. states and municipalities and other similar instruments, credit spreads reflect the credit quality of the obligor and the tenor of the obligation.

Prepayment speed – The prepayment speed is a measure of the voluntary unscheduled principal repayments of a prepayable obligation in a collateralized pool. Prepayment speeds generally decline as borrower delinquencies rise. An increase in prepayment speeds, in isolation, would result in a decrease in a fair value measurement of assets valued at a premium to par and an increase in a fair value measurement of assets valued at a discount to par.

Prepayment speeds may vary from collateral pool to collateral pool, and are driven by the type and location of the underlying borrower, and the remaining tenor of the obligation as well as the level and type (e.g., fixed or floating) of interest rate being paid by the borrower. Typically collateral pools with higher borrower credit quality have a higher prepayment rate than those with lower borrower credit quality, all other factors being equal.

Conditional default rate – The conditional default rate is a measure of the reduction in the outstanding collateral balance underlying a collateralized obligation as a result of defaults. While there is typically no direct relationship between conditional default rates and prepayment speeds, collateralized obligations for which the underlying collateral has high prepayment speeds will tend to have lower conditional default rates. An increase in conditional default rates would generally be accompanied by an increase in loss severity and an increase in credit spreads. An increase in the conditional default rate, in isolation, would result in a decrease in a fair value measurement. Conditional default rates reflect the quality of the collateral underlying a securitization and the structure of the securitization itself. Based on the types of securities owned in the Firm's market-making portfolios, conditional default rates are most typically at the lower end of the range presented.

Loss severity – The loss severity (the inverse concept is the recovery rate) is the expected amount of future realized losses resulting from the ultimate liquidation of a particular loan, expressed as the net amount of loss relative to the outstanding loan balance. An increase in loss severity is generally accompanied by an increase in conditional default rates. An increase in the loss severity, in isolation, would result in a decrease in a fair value measurement.

The loss severity applied in valuing a mortgage-backed security depends on factors relating to the underlying mortgages, including the LTV ratio, the nature of the lender's lien on the property and other instrument-specific factors.

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Correlation – Correlation is a measure of the relationship between the movements of two variables. Correlation is a pricing input for a derivative product where the payoff is driven by one or more underlying risks. Correlation inputs are related to the type of derivative (e.g., interest rate, credit, equity, foreign exchange and commodity) due to the nature of the underlying risks. When parameters are positively correlated, an increase in one parameter will result in an increase in the other parameter. When parameters are negatively correlated, an increase in one parameter will result in a decrease in the other parameter. An increase in correlation can result in an increase or a decrease in a fair value measurement. Given a short correlation position, an increase in correlation, in isolation, would generally result in a decrease in a fair value measurement.

The level of correlation used in the valuation of derivatives with multiple underlying risks depends on a number of factors including the nature of those risks. For example, the correlation between two credit risk exposures would be different than that between two interest rate risk exposures. Similarly, the tenor of the transaction may also impact the correlation input, as the relationship between the underlying risks may be different over different time periods. Furthermore, correlation levels are dependent on market conditions and could have a relatively wide range of levels within or across asset classes over time, particularly in volatile market conditions.

Volatility – Volatility is a measure of the variability in possible returns for an instrument, parameter or market index given how much the particular instrument, parameter or index changes in value over time. Volatility is a pricing input for options, including equity options, commodity options, and interest rate options. Given a long position in an option, an increase in volatility, in isolation, would generally result in an increase in a fair value measurement.

The level of volatility used in the valuation of a particular option-based derivative depends on a number of factors, including the nature of the risk underlying the option (e.g., the volatility of a particular equity security may be significantly different from that of a particular commodity index), the tenor of the derivative as well as the strike price of the option.

Bermudan switch value – The switch value is the difference between the overall value of a Bermudan swaption, which can be exercised at multiple points in time, and its most expensive European swaption and reflects the additional value that the multiple exercise dates provide the holder. Switch values are dependent on market conditions and can vary greatly depending on a number of factors, such as the tenor of the underlying swap as well as the strike price of the option. An increase in switch value, in isolation, would generally result in an increase in a fair value measurement.

Interest rate curve – The interest rate curve represents the relationship of interest rates over differing tenors. The interest rate curve is used to set interest rate and foreign exchange derivative cash flows and is also a pricing input used in the discounting of any derivative cash flow.

Forward price – The forward price is the price at which the buyer agrees to purchase the asset underlying a forward contract on the predetermined future delivery date, and is such that the value of the contract is zero at inception.

The forward price is used as an input in the valuation of certain derivatives and depends on a number of factors including interest rates, the current price of the underlying asset, and the expected income to be received and costs to be incurred by the seller as a result of holding that asset until the delivery date. An increase in the forward can result in an increase or a decrease in a fair value measurement.

Changes in level 3 recurring fair value measurements

The following tables include a rollforward of the Consolidated balance sheets amounts (including changes in fair value) for financial instruments classified by the Firm within level 3 of the fair value hierarchy for the years ended December 31, 2024, 2023 and 2022. When a determination is made to classify a financial instrument within level 3, the determination is based on the significance of the unobservable inputs to the overall fair value measurement. However, level 3 financial instruments typically include, in addition to the unobservable or level 3 components, observable components (that is, components that are actively quoted and can be validated to external sources); accordingly, the gains and losses in the table below include changes in fair value due in part to observable factors that are part of the valuation methodology. The Firm risk-manages the observable components of level 3 financial instruments using securities and derivative positions that are classified within level 1 or 2 of the fair value hierarchy; as these level 1 and level 2 risk management instruments are not included below, the gains or losses in the following tables do not reflect the effect of the Firm's risk management activities related to such level 3 instruments.

Year ended December 31, 2024 (in millions)	Fair value measurements using significant unobservable inputs								Fair value at Dec. 31, 2024	Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2024
	Fair value at Jan. 1, 2024	Total realized/ unrealized gains/ (losses)	Purchases ^(g)	Sales	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3			
Assets:^(a)										
Federal funds sold and securities purchased under resale agreements	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
Trading assets:										
Debt instruments:										
Mortgage-backed securities:										
U.S. GSEs and government agencies	758	18	46	(260)	(81)	7	—	488	(3)	
Residential – nonagency	5	7	—	(5)	(2)	4	(4)	5	—	
Commercial – nonagency	12	(2)	—	—	—	—	—	10	(1)	
Total mortgage-backed securities	775	23	46	(265)	(83)	11	(4)	503	(4)	
Obligations of U.S. states and municipalities	10	—	—	—	(3)	—	(6)	1	—	
Non-U.S. government debt securities	179	(6)	175	(183)	—	17	(30)	152	(10)	
Corporate debt securities	484	36	459	(354)	(181)	13	(67)	390	45	
Loans	684	63	800	(642)	(74)	839	(582)	1,088	29	
Asset-backed securities	6	—	9	(5)	(8)	8	—	10	—	
Total debt instruments	2,138	116	1,489	(1,449)	(349)	888	(689)	2,144	60	
Equity securities	127	(21)	138	(123)	(1)	85	(143)	62	(308)	
Physical commodities	7	17	3	—	(1)	—	—	26	16	
Other	101	144	53	—	(68)	28	(48)	210	108	
Total trading assets – debt and equity instruments	2,373	256^(c)	1,683	(1,572)	(419)	1,001	(880)	2,442	(124)^(c)	
Net derivative receivables: ^(b)										
Interest rate	502	745	387	(197)	(608)	(172)	(356)	301	(362)	
Credit	265	(208)	(2)	(17)	(333)	(61)	(7)	(363)	(265)	
Foreign exchange	62	248	178	(538)	(30)	128	(28)	20	353	
Equity	(2,402)	(321)	904	(2,488)	953	(91)	579	(2,866)	783	
Commodity	(279)	64	32	(215)	310	15	—	(73)	102	
Total net derivative receivables	(1,852)	528^(c)	1,499	(3,455)	292	(181)	188	(2,981)	611^(c)	
Available-for-sale securities:										
Mortgage-backed securities:										
Commercial – nonagency	—	—	—	—	—	8	—	8	—	
Corporate debt securities	—	—	—	—	—	—	—	—	—	
Total available-for-sale securities	—	—^(d)	—	—	—	8	—	8	—^(d)	
Loans	3,079	266 ^(c)	431	(756)	(993)	816	(427)	2,416	251 ^(c)	
Mortgage servicing rights	8,522	762 ^(e)	926	(21)	(1,068)	—	—	9,121	762 ^(e)	
Other assets	758	105 ^(c)	623	(62)	(58)	5	(27)	1,344	88 ^(c)	
Fair value measurements using significant unobservable inputs										
Year ended December 31, 2024 (in millions)	Fair value at Jan. 1, 2024	Total realized/ unrealized (gains)/ losses	Purchases	Sales	Issuances	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2024	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2024
Liabilities:^(a)										
Deposits	\$ 1,833	\$ (14) ^{(c)(f)}	\$ —	\$ —	\$ 2,006	\$ (1,522)	\$ 34	\$ (152)	\$ 2,185	\$ (44) ^{(c)(f)}
Short-term borrowings	1,758	180 ^{(c)(f)}	—	—	7,752	(6,230)	23	(7)	3,476	58 ^{(c)(f)}
Trading liabilities – debt and equity instruments	37	(47) ^(c)	(45)	70	—	—	48	(17)	46	18 ^(c)
Accounts payable and other liabilities	52	(6) ^(c)	(35)	63	—	—	5	(3)	76	(6) ^(c)
Long-term debt	27,726	1,475 ^{(c)(f)}	—	—	23,920	(18,432)	738	(863)	34,564	1,212 ^{(c)(f)}

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Year ended December 31, 2023 (in millions)	Fair value measurements using significant unobservable inputs								Fair value at Dec. 31, 2023	Change in unrealized gains/(losses) related to financial instruments held at Dec. 31, 2023
	Fair value at Jan. 1, 2023	Total realized/ unrealized gains/ (losses)	Purchases ^(a)	Sales	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3			
Assets:^(a)										
Federal funds sold and securities purchased under resale agreements	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —	\$ —
Trading assets:										
Debt instruments:										
Mortgage-backed securities:										
U.S. GSEs and government agencies	759	4	249	(133)	(107)	—	(14)	758	1	
Residential – nonagency	5	6	—	(6)	(1)	1	—	5	1	
Commercial – nonagency	7	6	—	—	(1)	8	(8)	12	7	
Total mortgage-backed securities	771	16	249	(139)	(109)	9	(22)	775	9	
Obligations of U.S. states and municipalities	7	—	1	—	(1)	3	—	10	—	
Non-U.S. government debt securities	155	74	217	(254)	—	22	(35)	179	74	
Corporate debt securities	463	36	322	(172)	(41)	114	(238)	484	35	
Loans	759	(15)	1,027	(499)	(441)	382	(529)	684	30	
Asset-backed securities	23	—	7	(12)	(1)	5	(16)	6	—	
Total debt instruments	2,178	111	1,823	(1,076)	(593)	535	(840)	2,138	148	
Equity securities	665	(53)	164	(239)	(384)	192	(218)	127	(422)	
Physical commodities	2	—	7	—	(2)	—	—	7	—	
Other	64	(58)	141	—	(5)	1	(42)	101	(28)	
Total trading assets – debt and equity instruments	2,909	—	2,135	(1,315)	(984)	728	(1,100)	2,373	(302)^(c)	
Net derivative receivables: ^(b)										
Interest rate	701	556	251	(255)	654	(1,117)	(288)	502	419	
Credit	13	304	(60)	(25)	47	15	(29)	265	230	
Foreign exchange	489	31	151	(144)	(187)	144	(422)	62	(80)	
Equity	(384)	191	928	(1,931)	(1,306)	700	(600)	(2,402)	(646)	
Commodity	(146)	(59)	59	(290)	(51)	(11)	219	(279)	(144)	
Total net derivative receivables	673	1,023^(c)	1,329	(2,645)	(843)	(269)	(1,120)	(1,852)	(221)^(c)	
Available-for-sale securities:										
Mortgage-backed securities:										
Commercial – nonagency	—	—	—	—	—	—	—	—	—	
Corporate debt securities	239	24	—	(225)	—	—	(38)	—	—	
Total available-for-sale securities	239	24^(d)	—	(225)	—	—	(38)	—	—^(d)	
Loans	1,418	289 ^(c)	2,398	(120)	(1,147)	1,306	(1,065)	3,079	293 ^(c)	
Mortgage servicing rights	7,973	467 ^(e)	1,281	(188)	(1,011)	—	—	8,522	467 ^(e)	
Other assets	405	(36) ^(c)	525	(20)	(147)	45	(14)	758	(82) ^(c)	
Fair value measurements using significant unobservable inputs										
Year ended December 31, 2023 (in millions)	Fair value at Jan. 1, 2023	Total realized/ unrealized (gains)/ losses	Purchases	Sales	Issuances	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2023	Change in unrealized (gains)/losses related to financial instruments held at Dec. 31, 2023
Liabilities:^(a)										
Deposits	\$ 2,162	\$ 95 ^{(c)(f)}	\$ —	\$ —	\$ 940	\$ (1,043)	\$ —	\$ (321)	\$ 1,833	\$ 73 ^{(c)(f)}
Short-term borrowings	1,401	201 ^{(c)(f)}	—	—	4,522	(4,345)	3	(24)	1,758	14 ^{(c)(f)}
Trading liabilities – debt and equity instruments	84	(21) ^(c)	(32)	9	—	(2)	19	(20)	37	—
Accounts payable and other liabilities	53	(4) ^(c)	(16)	24	—	—	8	(13)	52	(4) ^(c)
Long-term debt	24,092	3,010 ^{(c)(f)}	—	—	12,679	(11,555)	229	(729)	27,726	2,870 ^{(c)(f)}

	Fair value measurements using significant unobservable inputs								Change in unrealized gains/ (losses) related to financial instruments held at Dec. 31, 2022	
Year ended December 31, 2022 (in millions)	Fair value at Jan. 1, 2022	Total realized/ unrealized gains/ (losses)	Purchases ^(a)	Sales	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3	Fair value at Dec. 31, 2022		
Assets:^(a)										
Federal funds sold and securities purchased under resale agreements	\$ —	\$ —	\$ 1	\$ (1)	\$ (1)	\$ 1	\$ —	\$ —	\$ —	
Trading assets:										
Debt instruments:										
Mortgage-backed securities:										
U.S. GSEs and government agencies	265	31	673	(125)	(84)	4	(5)	759	29	
Residential – nonagency	28	(1)	7	(5)	(12)	—	(12)	5	—	
Commercial – nonagency	10	—	—	(1)	—	3	(5)	7	—	
Total mortgage-backed securities	303	30	680	(131)	(96)	7	(22)	771	29	
Obligations of U.S. states and municipalities	7	—	—	—	—	—	—	7	—	
Non-U.S. government debt securities	81	(92)	494	(338)	(4)	84	(70)	155	(153)	
Corporate debt securities	332	(30)	404	(178)	(100)	357	(322)	463	(48)	
Loans	708	(51)	652	(605)	(230)	925	(640)	759	(26)	
Asset-backed securities	26	5	19	(24)	(1)	5	(7)	23	1	
Total debt instruments	1,457	(138)	2,249	(1,276)	(431)	1,378	(1,061)	2,178	(197)	
Equity securities	662	(1,036)	473	(377)	(2)	1,066	(121)	665	(840)	
Physical commodities	—	(1)	3	—	—	—	—	2	(1)	
Other	160	93	37	—	(221)	1	(6)	64	46	
Total trading assets – debt and equity instruments	2,279	(1,082) ^(c)	2,762	(1,653)	(654)	2,445	(1,188)	2,909	(992) ^(c)	
Net derivative receivables: ^(b)										
Interest rate	(16)	187	325	(483)	329	732	(373)	701	332	
Credit	74	226	17	(9)	(271)	5	(29)	13	170	
Foreign exchange	(419)	726	215	(114)	83	3	(5)	489	459	
Equity	(3,626)	5,016	1,226	(2,530)	96	(656)	90	(384)	3,435	
Commodity	(907)	571	110	(331)	350	5	56	(146)	369	
Total net derivative receivables	(4,894)	6,726 ^(c)	1,893	(3,467)	587	89	(261)	673	4,765 ^(c)	
Available-for-sale securities:										
Mortgage-backed securities:										
Commercial – nonagency	—	—	—	—	—	—	—	—	—	
Corporate debt securities	161	5	88	—	(15)	—	—	239	5	
Total available-for-sale securities	161	5 ^(d)	88	—	(15)	—	—	239	5 ^(d)	
Loans	1,933	(158) ^(c)	568	(261)	(886)	1,053	(831)	1,418	(76) ^(c)	
Mortgage servicing rights	5,494	2,039 ^(e)	2,198	(822)	(936)	—	—	7,973	2,039 ^(e)	
Other assets	306	194 ^(c)	50	(38)	(103)	2	(6)	405	191 ^(c)	

Year ended December 31, 2022 (in millions)	Fair value measurements using significant unobservable inputs								Fair value at Dec. 31, 2022	Change in unrealized (gains)/ losses related to financial instruments held at Dec. 31, 2022
	Fair value at Jan. 1, 2022	Total realized/ unrealized (gains)/ losses	Purchases	Sales	Issuances	Settlements ^(h)	Transfers into level 3	Transfers (out of) level 3		
Liabilities: ^(a)										
Deposits	\$ 2,317	\$ (292) ^{(c)(f)}	\$ —	\$ —	\$ 531	\$ (114)	\$ —	\$ (280)	\$ 2,162	\$ (76) ^{(c)(f)}
Short-term borrowings	2,481	(358) ^{(c)(f)}	—	—	3,963	(4,685)	15	(15)	1,401	90 ^{(c)(f)}
Trading liabilities – debt and equity instruments	30	(31) ^(c)	(41)	77	—	—	57	(8)	84	101 ^(c)
Accounts payable and other liabilities	69	(16) ^(c)	(37)	42	—	—	1	(6)	53	(16) ^(c)
Long-term debt	24,374	(3,869) ^{(c)(f)}	—	—	12,714	(8,876)	793	(1,044)	24,092	(3,447) ^{(c)(f)}

Notes to consolidated financial statements

- (a) Level 3 assets at fair value as a percentage of total Firm assets at fair value (including assets measured at fair value on a nonrecurring basis) were 2% at December 31, 2024, 2023 and 2022. Level 3 liabilities at fair value as a percentage of total Firm liabilities at fair value (including liabilities measured at fair value on a nonrecurring basis) were 9% at December 31, 2024, and 8% at both December 31, 2023 and 2022.
- (b) All level 3 derivatives are presented on a net basis, irrespective of the underlying counterparty.
- (c) Primarily reported in principal transactions revenue, except for changes in fair value for CCB mortgage loans and lending-related commitments originated with the intent to sell, and mortgage loan purchase commitments, which are reported in mortgage fees and related income.
- (d) Realized gains/(losses) on AFS securities are reported in investment securities gains/(losses). Unrealized gains/(losses) are reported in OCI. Realized and unrealized gains/(losses) recorded on level 3 AFS securities were not material for the years ended December 31, 2024, 2023 and 2022.
- (e) Changes in fair value for MSRs are reported in mortgage fees and related income.
- (f) Realized (gains)/losses due to DVA for fair value option elected liabilities are reported in principal transactions revenue, and were not material for the years ended December 31, 2024, 2023 and 2022. Unrealized (gains)/losses are reported in OCI, and were \$(50) million, \$(158) million and \$(529) million for the years ended December 31, 2024, 2023 and 2022, respectively.
- (g) Loan originations are included in purchases.
- (h) Includes financial assets and liabilities that have matured, been partially or fully repaid, impacts of modifications, deconsolidations associated with beneficial interests in VIEs and other items.

Level 3 analysis

Consolidated balance sheets changes

The following describes significant changes to level 3 assets since December 31, 2023, for those items measured at fair value on a recurring basis. Refer to Assets and liabilities measured at fair value on a nonrecurring basis on page 199 for further information on changes impacting items measured at fair value on a nonrecurring basis.

For the year ended December 31, 2024

Level 3 assets were \$23.8 billion at December 31, 2024, reflecting an increase of \$127 million from December 31, 2023.

The increase for the year ended December 31, 2024 was driven by:

- \$599 million increase in MSRs.
- \$586 million increase in other assets primarily due to purchases,

offset by:

- \$472 million decrease in gross derivative receivables due to sales and settlements predominantly offset by gains, purchases and net transfers.
- \$663 million decrease in non-trading loans due to sales and settlements largely offset by gains, purchases and net transfers.

Refer to Note 15 for information on MSRs.

Refer to the sections below for additional information.

Transfers between levels for instruments carried at fair value on a recurring basis

During the year ended December 31, 2024, significant transfers from level 2 into level 3 included the following:

- \$1.0 billion of total debt and equity instruments, predominantly trading loans, driven by a decrease in observability.
- \$959 million of gross interest rate derivative receivables and \$1.1 billion of gross interest rate derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- \$1.6 billion of gross equity derivative receivables and \$1.7 billion of gross equity derivative payables as a

result of a decrease in observability and an increase in the significance of unobservable inputs.

- \$816 million of non-trading loans driven by a decrease in observability.

During the year ended December 31, 2024, significant transfers from level 3 into level 2 included the following:

- \$880 million of total debt and equity instruments, predominantly trading loans and equity securities, driven by an increase in observability.
- \$1.4 billion of gross equity derivative receivables and \$2.0 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- \$863 million of long-term debt as a result of an increase in observability and a decrease in the significance of unobservable inputs.

During the year ended December 31, 2023, significant transfers from level 2 into level 3 included the following:

- \$951 million of gross interest rate derivative receivables as a result of a decrease in observability and an increase in the significance of unobservable inputs and \$2.1 billion of gross interest rate derivative payables as a result of transition to term SOFR for certain interest rate options.
- \$1.5 billion of gross equity derivative receivables and \$829 million of gross equity derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- \$1.3 billion of non-trading loans driven by a decrease in observability.

During the year ended December 31, 2023, significant transfers from level 3 into level 2 included the following:

- \$1.1 billion of total debt and equity instruments, partially due to trading loans, driven by an increase in observability.
- \$921 million of gross interest rate derivative receivables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- \$2.3 billion of gross equity derivative receivables and \$1.7 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- \$1.1 billion of non-trading loans as a result of an increase in observability and a decrease in the significance of unobservable inputs.

During the year ended December 31, 2022, significant transfers from level 2 into level 3 included the following:

- \$2.4 billion of total debt and equity instruments, predominantly due to equity securities of \$1.1 billion driven by a decrease in observability predominantly as a result of restricted access to certain markets and trading loans of \$925 million driven by a decrease in observability.
- \$1.6 billion of gross interest rate derivative receivables and \$878 million of gross interest rate derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- \$1.6 billion of gross equity derivative receivables and \$2.3 billion of gross equity derivative payables as a result of a decrease in observability and an increase in the significance of unobservable inputs.
- \$1.1 billion of non-trading loans driven by a decrease in observability.
- \$793 million of long-term debt driven by a decrease in observability and an increase in the significance of unobservable inputs for structured notes.

During the year ended December 31, 2022, significant transfers from level 3 into level 2 included the following:

- \$1.2 billion of total debt and equity instruments, largely due to trading loans, driven by an increase in observability.
- \$1.2 billion of gross interest rate derivative receivables and \$807 million of gross interest rate derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.
- \$2.2 billion of gross equity derivative receivables and \$2.3 billion of gross equity derivative payables as a result of an increase in observability and a decrease in the significance of unobservable inputs.

- \$831 million of non-trading loans driven by an increase in observability.
- \$1.0 billion of long-term debt driven by an increase in observability and a decrease in the significance of unobservable inputs for structured notes.

All transfers are based on changes in the observability and/or significance of the valuation inputs and are assumed to occur at the beginning of the quarterly reporting period in which they occur.

Gains and losses

The following describes significant components of total realized/unrealized gains/(losses) for instruments measured at fair value on a recurring basis for the years ended December 31, 2024, 2023 and 2022. These amounts exclude any effects of the Firm's risk management activities where the financial instruments are classified as level 1 and 2 of the fair value hierarchy. Refer to Changes in level 3 recurring fair value measurements rollforward tables on pages 192–196 for further information on these instruments.

2024

- \$1.9 billion of net gains on assets, predominantly driven by gains in net interest rate derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- \$1.6 billion of net losses on liabilities, predominantly driven by losses in long-term debt due to market movements.

2023

- \$1.8 billion of net gains on assets, largely driven by gains in net interest rate derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- \$3.3 billion of net losses on liabilities, predominantly driven by losses in long-term debt due to market movements.

2022

- \$7.7 billion of net gains on assets, predominantly driven by gains in net equity derivative receivables due to market movements and gains in MSRs reflecting lower prepayment speeds on higher rates.
- \$4.6 billion of net gains on liabilities, predominantly driven by a decline in the fair value of long-term debt due to market movements.

Refer to Note 15 for information on MSRs.

Credit and funding adjustments – derivatives

Derivatives are generally valued using models that use as their basis observable market parameters. These market parameters generally do not consider factors such as counterparty nonperformance risk, the Firm's own credit quality, and funding costs. Therefore, it is generally necessary to make adjustments to the base estimate of fair value to reflect these factors.

CVA represents the adjustment, relative to the relevant benchmark interest rate, necessary to reflect counterparty nonperformance risk. The Firm estimates CVA using a scenario analysis to estimate the expected positive credit exposure across all of the Firm's existing positions with each counterparty, and then estimates losses based on the probability of default and estimated recovery rate as a result of a counterparty credit event considering contractual factors designed to mitigate the Firm's credit exposure, such as collateral and legal rights of offset. The key inputs to this methodology are (i) the probability of a default event occurring for each counterparty, as derived from observed or estimated CDS spreads; and (ii) estimated recovery rates implied by CDS spreads, adjusted to consider the differences in recovery rates as a derivative creditor relative to those reflected in CDS spreads, which generally reflect senior unsecured creditor risk.

FVA represents the adjustment to reflect the impact of funding and is recognized where there is evidence that a market participant in the principal market would incorporate it in a transfer of the instrument. The Firm's FVA framework, applied to uncollateralized (including partially collateralized) over-the-counter ("OTC") derivatives incorporates key inputs such as: (i) the expected funding requirements arising from the Firm's positions with each counterparty and collateral arrangements; and (ii) the estimated market funding cost in the principal market which, for derivative liabilities, considers the Firm's credit risk (DVA). For collateralized derivatives, the fair value is estimated by discounting expected future cash flows at the relevant overnight indexed swap rate given the underlying collateral agreement with the counterparty, and therefore a separate FVA is not necessary.

The following table provides the impact of credit and funding adjustments on principal transactions revenue in the respective periods, excluding the effect of any associated hedging activities. The FVA presented below includes the impact of the Firm's own credit quality on the inception value of liabilities as well as the impact of changes in the Firm's own credit quality over time.

Year ended December 31, (in millions)	2024	2023	2022
Credit and funding adjustments:			
Derivatives CVA	\$ 29	\$ 221	\$ 22
Derivatives FVA	99	114	42

Valuation adjustments on fair value option elected liabilities

The valuation of the Firm's liabilities for which the fair value option has been elected requires consideration of the Firm's own credit risk. DVA on fair value option elected liabilities reflects changes (subsequent to the issuance of the liability) in the Firm's probability of default and LGD, which are estimated based on changes in the Firm's credit spread observed in the bond market. Realized (gains)/losses due to DVA for fair value option elected liabilities are reported in principal transactions revenue. Unrealized (gains)/losses are reported in OCI. Refer to page 196 in this Note and Note 24 for further information.

Assets and liabilities measured at fair value on a nonrecurring basis

The following tables present the assets held as of December 31, 2024 and 2023, for which nonrecurring fair value adjustments were recorded during the years ended December 31, 2024 and 2023, by major product category and fair value hierarchy. There were no liabilities measured at fair value on a nonrecurring basis at both December 31, 2024 and 2023.

December 31, 2024 (in millions)	Fair value hierarchy			Total fair value
	Level 1	Level 2	Level 3	
Loans	\$ —	\$ 738	\$ 694	\$ 1,432
Other assets ^(a)	—	9	1,048	1,057
Total assets measured at fair value on a nonrecurring basis	\$ —	\$ 747	\$ 1,742	\$ 2,489

December 31, 2023 (in millions)	Fair value hierarchy			Total fair value
	Level 1	Level 2	Level 3	
Loans	\$ —	\$ 599	\$ 1,156	\$ 1,755
Other assets	—	52	1,334	1,386
Total assets measured at fair value on a nonrecurring basis	\$ —	\$ 651	\$ 2,490	\$ 3,141

(a) Included equity securities without readily determinable fair values that were adjusted based on observable price changes in orderly transactions from an identical or similar investment of the same issuer (measurement alternative). Of the \$1.0 billion in level 3 assets measured at fair value on a nonrecurring basis as of December 31, 2024, \$668 million related to equity securities adjusted based on the measurement alternative. These equity securities are classified as level 3 due to the infrequency of the observable prices and/or the restrictions on the shares. Also, included impairments on certain equity method investments.

Nonrecurring fair value changes

The following table presents the total change in value of assets and liabilities for which fair value adjustments have been recognized for the years ended December 31, 2024, 2023 and 2022, related to assets and liabilities held at those dates.

December 31, (in millions)	2024	2023	2022
Loans	\$ (302)	\$ (276)	\$ (55)
Other assets ^(a)	(610)	(789)	(409)
Accounts payable and other liabilities	—	—	(83)
Total nonrecurring fair value gains/(losses)	\$ (912)	\$ (1,065)	\$ (547)

(a) Included \$(197) million, \$(232) million and \$(338) million for the years ended December 31, 2024, 2023 and 2022, respectively, of net gains/(losses) as a result of the measurement alternative. The current period also included impairments on certain equity method investments.

Notes to consolidated financial statements

Equity securities without readily determinable fair values

The Firm measures certain equity securities without readily determinable fair values at cost less impairment (if any), plus or minus observable price changes from an identical or similar investment of the same issuer (i.e., measurement alternative), with such changes recognized in other income.

In its determination of the new carrying values upon observable price changes, the Firm may adjust the prices if deemed necessary to arrive at the Firm's estimated fair values. Such adjustments may include adjustments to reflect the different rights and obligations of similar securities, and other adjustments that are consistent with the Firm's valuation techniques for private equity direct investments.

The following table presents the carrying value of equity securities without readily determinable fair values held as of December 31, 2024 and 2023, that are measured under the measurement alternative and the related adjustments recorded during the periods presented for those securities with observable price changes. These securities are included in the nonrecurring fair value tables when applicable price changes are observable.

As of or for the year ended December 31, (in millions)	2024	2023
Other assets		
Carrying value ^(a)	\$ 3,737	\$ 4,457
Upward carrying value changes ^(b)	89	93
Downward carrying value changes/impairment ^(c)	(286)	(325)

(a) The period-end carrying values reflect cumulative purchases and sales in addition to upward and downward carrying value changes.

(b) The cumulative upward carrying value changes between January 1, 2018 and December 31, 2024 were \$1.1 billion.

(c) The cumulative downward carrying value changes/impairment between January 1, 2018 and December 31, 2024 were \$(1.5) billion.

Included in other assets above is the Firm's interest in approximately 18.6 million Visa Class B-2 common shares ("Visa B-2 shares") and 37.2 million Visa Class B common shares ("Visa B shares") reflected in the Firm's principal investment portfolio as of December 31, 2024 and 2023, respectively.

The Visa B shares were redenominated to Visa Class B-1 common shares ("Visa B-1 shares") on January 24, 2024. On April 8, 2024, Visa commenced an initial exchange offer for any and all outstanding Visa B-1 shares. On May 6, 2024, the Firm announced that Visa had accepted the Firm's tender of its 37.2 million Visa B-1 shares in exchange for a combination of Visa B-2 shares and Visa Class C common shares ("Visa C shares"), resulting in an initial gain of \$8.0 billion based on the fair value of the Visa C shares. In addition, the second quarter of 2024 also reflected other Visa-related activity, including the fair value changes of the Visa C shares and derivative instruments, as well as dividends, resulting in the \$7.9 billion net gain on Visa shares in the quarter. As of September 30, 2024, the Firm had disposed of all of its Visa C shares through sales in the second and third quarters of 2024 and through a \$1.0 billion contribution to the Firm's Foundation in the second quarter of 2024.

The Visa B-2 shares are subject to certain transfer restrictions and are convertible into Visa Class A common shares ("Visa A shares") at a specified conversion rate upon final resolution of certain litigation matters involving Visa. The conversion rate of Visa B-2 shares to Visa A shares was 1.5430 at December 31, 2024 and may be adjusted by Visa depending on developments related to the litigation matters. The outcome of those litigation matters, and the effect that the resolution of those matters may have on the conversion rate, is unknown. Accordingly, as of December 31, 2024, there is significant uncertainty regarding when the transfer restrictions on Visa B-2 shares may be terminated and what the final conversion rate for the Visa B-2 shares will be. As a result of these considerations, as well as differences in voting rights, Visa B-2 shares are not considered to be similar to Visa A shares, and are held at their nominal carryover basis.

Separately, in connection with sales of Visa B shares prior to 2024, the Firm has entered into derivative instruments with the purchasers of the shares under which the Firm retains the risk associated with changes in the conversion rate. Under the terms of the derivative instruments, the Firm will (a) make or receive payments based on subsequent changes in the conversion rate and (b) make periodic interest payments to the purchasers of the Visa B shares. The payments under the derivative instruments will continue as long as the Visa B-2 shares associated with the previously sold Visa B shares remain subject to transfer restrictions. The derivative instruments are accounted for at fair value using a discounted cash flow methodology based upon the Firm's estimate of the timing and magnitude of final resolution of the litigation matters. The derivative instruments are recorded in trading liabilities, and changes in fair value are recognized in other income. The notional amount of shares associated with those derivative instruments has been adjusted as a result of the Visa exchange offer. As of December 31, 2024, the Firm held derivative instruments associated with 11.6 million Visa B-2 shares related to Visa B share sales prior to 2024, which are all subject to similar terms and conditions.

Additional disclosures about the fair value of financial instruments that are not carried on the Consolidated balance sheets at fair value

U.S. GAAP requires disclosure of the estimated fair value of certain financial instruments, which are included in the following table. However, this table does not include other items, such as nonfinancial assets, intangible assets, certain financial instruments, and customer relationships. In the opinion of management, these items, in the aggregate, add significant value to JPMorganChase.

Financial instruments for which carrying value approximates fair value

Certain financial instruments that are not carried at fair value on the Consolidated balance sheets are carried

at amounts that approximate fair value, due to their short-term nature and generally negligible credit risk. These instruments include cash and due from banks, deposits with banks, federal funds sold, securities purchased under resale agreements and securities borrowed, short-term receivables and accrued interest receivable, short-term borrowings, federal funds purchased, securities loaned and sold under repurchase agreements, accounts payable, and accrued liabilities. In addition, U.S. GAAP requires that the fair value of deposit liabilities with no stated maturity (i.e., demand, savings and certain money market deposits) be equal to their carrying value; recognition of the inherent funding value of these instruments is not permitted.

The following table presents, by fair value hierarchy classification, the carrying values and estimated fair values at December 31, 2024 and 2023, of financial assets and liabilities, excluding financial instruments that are carried at fair value on a recurring basis, and their classification within the fair value hierarchy.

(in billions)	December 31, 2024					December 31, 2023				
	Estimated fair value hierarchy					Estimated fair value hierarchy				
	Carrying value	Level 1	Level 2	Level 3	Total estimated fair value	Carrying value	Level 1	Level 2	Level 3	Total estimated fair value
Financial assets										
Cash and due from banks	\$ 23.4	\$ 23.4	\$ —	\$ —	\$ 23.4	\$ 29.1	\$ 29.1	\$ —	\$ —	\$ 29.1
Deposits with banks	445.9	445.8	0.1	—	445.9	595.1	594.6	0.5	—	595.1
Accrued interest and accounts receivable	101.1	—	101.0	0.1	101.1	107.1	—	107.0	0.1	107.1
Federal funds sold and securities purchased under resale agreements	8.2	—	8.2	—	8.2	16.3	—	16.3	—	16.3
Securities borrowed	135.6	—	135.6	—	135.6	130.3	—	130.3	—	130.3
Investment securities, held-to-maturity	274.5	97.4	150.5	—	247.9	369.8	160.6	182.2	—	342.8
Loans, net of allowance for loan losses ^(a)	1,282.3	—	268.7	1,007.8	1,276.5	1,262.5	—	285.6	964.6	1,250.2
Other	82.7	—	81.3	1.6	82.9	76.1	—	74.9	1.4	76.3
Financial liabilities										
Deposits	\$2,372.3	\$ —	\$2,372.5	\$ —	\$ 2,372.5	\$ 2,322.3	\$ —	\$ 2,322.6	\$ —	\$ 2,322.6
Federal funds purchased and securities loaned or sold under repurchase agreements	70.5	—	70.5	—	70.5	47.5	—	47.5	—	47.5
Short-term borrowings	26.4	—	26.3	—	26.3	24.7	—	24.7	—	24.7
Accounts payable and other liabilities ^(b)	232.8	—	219.6	12.6	232.2	241.8	—	233.3	8.1	241.4
Beneficial interests issued by consolidated VIEs	27.3	—	27.4	—	27.4	23.0	—	23.0	—	23.0
Long-term debt	300.6	—	251.2	50.7	301.9	303.9	—	252.2	51.3	303.5

(a) Fair value is typically estimated using a discounted cash flow model that incorporates the characteristics of the underlying loans (including principal, contractual interest rate and contractual fees) and other key inputs, including expected lifetime credit losses, interest rates, prepayment rates, and primary origination or secondary market spreads. For certain loans, the fair value is measured based on the value of the underlying collateral. Carrying value of the loan takes into account the loan's allowance for loan losses, which represents the loan's expected credit losses over its remaining expected life. The difference between the estimated fair value and carrying value of a loan is generally attributable to changes in market interest rates, including credit spreads, market liquidity premiums and other factors that affect the fair value of a loan but do not affect its carrying value.

(b) Excludes lending-related commitments disclosed in the table below.

Notes to consolidated financial statements

The majority of the Firm's lending-related commitments are not carried at fair value on a recurring basis on the Consolidated balance sheets. The carrying value and the estimated fair value of these wholesale lending-related commitments were as follows for the periods indicated.

(in billions)	December 31, 2024					December 31, 2023				
	Estimated fair value hierarchy					Estimated fair value hierarchy				
	Carrying value ^{(a)(b)(c)}	Level 1	Level 2	Level 3	Total estimated fair value	Carrying value ^{(a)(b)(c)}	Level 1	Level 2	Level 3	Total estimated fair value
Wholesale lending-related commitments	\$ 2.7	\$ —	\$ —	\$ 4.4	\$ 4.4	\$ 3.0	\$ —	\$ —	\$ 4.8	\$ 4.8

(a) Excludes the current carrying values of the guarantee liability and the offsetting asset, each of which is recognized at fair value at the inception of the guarantees.

(b) Includes the wholesale allowance for lending-related commitments.

(c) As of December 31, 2024 and 2023, included fair value adjustments associated with First Republic for other unfunded commitments to extend credit totaling \$699 million and \$1.1 billion, respectively, recorded in accounts payable and other liabilities on the Consolidated balance sheets. Refer to Notes 28 and 34 for additional information.

The Firm does not estimate the fair value of consumer off-balance sheet lending-related commitments. In many cases, the Firm can reduce or cancel these commitments with or without notice to the borrower, as permitted by law, or in accordance with the contract. Refer to page 183 of this Note for a further discussion of the valuation of lending-related commitments.

Note 3 – Fair value option

The fair value option provides an option to elect fair value for selected financial assets, financial liabilities, unrecognized firm commitments, and written loan commitments.

The Firm has elected to measure certain instruments at fair value for several reasons including to mitigate income statement volatility caused by the differences between the measurement basis of elected instruments (e.g., certain instruments that otherwise would be accounted for on an accrual basis) and the associated risk management arrangements that are accounted for on a fair value basis, as well as to better reflect those instruments that are managed on a fair value basis.

The Firm's election of fair value includes the following instruments:

- Loans purchased or originated as part of securitization warehousing activity, subject to bifurcation accounting, or managed on a fair value basis, including lending-related commitments
- Certain securities financing agreements
- Owned beneficial interests in securitized financial assets that contain embedded credit derivatives, which would otherwise be required to be separately accounted for as a derivative instrument
- Structured notes and other hybrid instruments, which are predominantly financial instruments that contain embedded derivatives, that are issued or transacted as part of client-driven activities
- Certain long-term beneficial interests issued by CIB's consolidated securitization trusts where the underlying assets are carried at fair value

Notes to consolidated financial statements

Changes in fair value under the fair value option election

The following table presents the changes in fair value included in the Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, for items for which the fair value option was elected. The profit and loss information presented below only includes the financial instruments that were elected to be measured at fair value; related risk management instruments, which are required to be measured at fair value, are not included in the table.

December 31, (in millions)	2024			2023			2022		
	Principal transactions	All other income	Total changes in fair value recorded ^(e)	Principal transactions	All other income	Total changes in fair value recorded ^(e)	Principal transactions	All other income	Total changes in fair value recorded ^(e)
Federal funds sold and securities purchased under resale agreements	\$ 144	\$ —	\$ 144	\$ 300	\$ —	\$ 300	\$ (384)	\$ —	\$ (384)
Securities borrowed	347	—	347	164	—	164	(499)	—	(499)
Trading assets:									
Debt and equity instruments, excluding loans	7,205	—	7,205	3,656	—	3,656	(1,703)	—	(1,703)
Loans reported as trading assets:									
Changes in instrument-specific credit risk	346	—	346	248	—	248	(136)	—	(136)
Other changes in fair value	9	10 ^(c)	19	3	5 ^(c)	8	(59)	—	(59)
Loans:									
Changes in instrument-specific credit risk	517	(6) ^(c)	511	322	(4) ^(c)	318	(242)	21 ^(c)	(221)
Other changes in fair value	75	371 ^(c)	446	427	216 ^(c)	643	(1,421)	(794) ^(c)	(2,215)
Other assets	63	—	63	282	(4) ^(d)	278	39	(6) ^(d)	33
Deposits ^(a)	(3,398)	—	(3,398)	(2,582)	—	(2,582)	901	—	901
Federal funds purchased and securities loaned or sold under repurchase agreements	(12)	—	(12)	(121)	—	(121)	181	—	181
Short-term borrowings ^(a)	(922)	—	(922)	(567)	—	(567)	473	—	473
Trading liabilities	(1)	—	(1)	(24)	—	(24)	43	—	43
Beneficial interests issued by consolidated VIEs	—	—	—	—	—	—	(1)	—	(1)
Other liabilities	(11)	—	(11)	(16)	—	(16)	(11)	—	(11)
Long-term debt ^{(a)(b)}	(2,711)	(6) ^{(c)(d)}	(2,717)	(5,875)	(78) ^{(c)(d)}	(5,953)	8,990	98 ^{(c)(d)}	9,088

- (a) Unrealized gains/(losses) due to instrument-specific credit risk (DVA) for liabilities for which the fair value option has been elected are recorded in OCI, while realized gains/(losses) are recorded in principal transactions revenue. Realized gains/(losses) due to instrument-specific credit risk recorded in principal transactions revenue were not material for the years ended December 31, 2024, 2023 and 2022.
- (b) Long-term debt measured at fair value predominantly relates to structured notes. Although the risk associated with the structured notes is actively managed, the gains/(losses) reported in this table do not include the income statement impact of the risk management instruments used to manage such risk.
- (c) Reported in mortgage fees and related income.
- (d) Reported in other income.
- (e) Changes in fair value exclude contractual interest, which is included in interest income and interest expense for all instruments other than certain hybrid financial instruments in CIB. Refer to Note 7 for further information regarding interest income and interest expense.

Determination of instrument-specific credit risk for items for which the fair value option was elected

The following describes how the gains and losses that are attributable to changes in instrument-specific credit risk, were determined.

- Loans and lending-related commitments: For floating-rate instruments, all changes in value are attributed to instrument-specific credit risk. For fixed-rate instruments, an allocation of the changes in value for the period is made between those changes in value that are interest rate-related and changes in value that are credit-related. Allocations are generally based on an analysis of borrower-specific credit spread and recovery information,

where available, or benchmarking to similar entities or industries.

- Long-term debt: Changes in value attributable to instrument-specific credit risk were derived principally from observable changes in the Firm's credit spread as observed in the bond market.
- Securities financing agreements: Generally, for these types of agreements, there is a requirement that collateral be maintained with a market value equal to or in excess of the principal amount loaned; as a result, there would be no adjustment or an immaterial adjustment for instrument-specific credit risk related to these agreements.

Difference between aggregate fair value and aggregate remaining contractual principal balance outstanding

The following table reflects the difference between the aggregate fair value and the aggregate remaining contractual principal balance outstanding as of December 31, 2024 and 2023, for loans, long-term debt and long-term beneficial interests for which the fair value option has been elected.

December 31, (in millions)	2024			2023		
	Contractual principal outstanding	Fair value	Fair value over/(under) contractual principal outstanding	Contractual principal outstanding	Fair value	Fair value over/(under) contractual principal outstanding
Loans						
Nonaccrual loans						
Loans reported as trading assets	\$ 3,429	\$ 464	\$ (2,965)	\$ 2,987	\$ 588	\$ (2,399)
Loans	1,711	1,492	(219)	838	732	(106)
Subtotal	5,140	1,956	(3,184)	3,825	1,320	(2,505)
90 or more days past due and government guaranteed						
Loans ^(a)	50	45	(5)	65	59	(6)
All other performing loans^(b)						
Loans reported as trading assets	12,171	10,852	(1,319)	9,547	7,968	(1,579)
Loans ^(c)	40,342	39,813	(529)	38,948	38,060	(888)
Subtotal	52,513	50,665	(1,848)	48,495	46,028	(2,467)
Total loans	\$ 57,703	\$ 52,666	\$ (5,037)	\$ 52,385	\$ 47,407	\$ (4,978)
Long-term debt						
Principal-protected debt	\$ 57,414 ^(e)	\$ 47,780	\$ (9,634)	\$ 47,768 ^(e)	\$ 38,882	\$ (8,886)
Nonprincipal-protected debt ^(d)	NA	53,000	NA	NA	49,042	NA
Total long-term debt	NA	\$ 100,780	NA	NA	\$ 87,924	NA
Long-term beneficial interests						
Nonprincipal-protected debt ^(d)	NA	\$ 1	NA	NA	\$ 1	NA
Total long-term beneficial interests	NA	\$ 1	NA	NA	\$ 1	NA

(a) These balances are excluded from nonaccrual loans as the loans are insured and/or guaranteed by U.S. government agencies.

(b) There were no performing loans that were ninety days or more past due as of December 31, 2024 and 2023.

(c) Includes loans insured and/or guaranteed by U.S. government agencies less than 90 days past due.

(d) Remaining contractual principal is not applicable to nonprincipal-protected structured notes and long-term beneficial interests. Unlike principal-protected structured notes and long-term beneficial interests, for which the Firm is obligated to return a stated amount of principal at maturity, nonprincipal-protected structured notes and long-term beneficial interests do not obligate the Firm to return a stated amount of principal at maturity, but for structured notes to return an amount based on the performance of an underlying variable or derivative feature embedded in the note. However, investors are exposed to the credit risk of the Firm as issuer for both nonprincipal-protected and principal-protected notes.

(e) Where the Firm issues principal-protected zero-coupon or discount notes, the balance reflects the contractual principal payment at maturity or, if applicable, the contractual principal payment at the Firm's next call date.

At December 31, 2024 and 2023, the contractual amount of lending-related commitments for which the fair value option was elected was \$12.2 billion and \$9.7 billion, respectively, with a corresponding fair value of \$50 million and \$97 million, respectively. Refer to Note 28 for further information regarding off-balance sheet lending-related financial instruments.

Notes to consolidated financial statements

Structured note products by balance sheet classification and risk component

The following table presents the fair value of structured notes, by balance sheet classification and the primary risk type.

(in millions)	December 31, 2024				December 31, 2023			
	Long-term debt	Short-term borrowings	Deposits	Total	Long-term debt	Short-term borrowings	Deposits	Total
Risk exposure								
Interest rate	\$ 46,220	\$ 1,065	\$ 28,871	\$ 76,156	\$ 38,604	\$ 654	\$ 74,526	\$ 113,784
Credit	6,213	1,242	—	7,455	5,444	350	—	5,794
Foreign exchange	2,309	1,058	416	3,783	2,605	941	187	3,733
Equity	44,149	7,881	2,986	55,016	38,685	5,483	2,905	47,073
Commodity	1,331	62	1 ^(a)	1,394	1,862	11	1 ^(a)	1,874
Total structured notes	\$ 100,222	\$ 11,308	\$ 32,274	\$ 143,804	\$ 87,200	\$ 7,439	\$ 77,619	\$ 172,258

(a) Excludes deposits linked to precious metals for which the fair value option has not been elected of \$869 million and \$627 million for the years ended December 31, 2024 and 2023, respectively.

Note 4 – Credit risk concentrations

Concentrations of credit risk arise when a number of clients, counterparties or customers are engaged in similar business activities or activities in the same geographic region, or when they have similar economic features that would cause their ability to meet contractual obligations to be similarly affected by changes in economic conditions.

JPMorganChase regularly monitors various segments of its credit portfolios to assess potential credit risk concentrations and to obtain additional collateral when deemed necessary and permitted under the Firm's agreements. Senior management is significantly involved in the credit approval and review process, and risk levels are adjusted as needed to reflect the Firm's risk appetite.

In the Firm's consumer portfolio, concentrations are managed primarily by product and by U.S. geographic region, with a key focus on trends and concentrations at the portfolio level, where potential credit risk concentrations can be remedied through changes in underwriting policies and portfolio guidelines. Refer to Note 12 for additional information on the geographic composition of the Firm's consumer loan portfolios. In the wholesale portfolio, credit risk concentrations are evaluated primarily by industry and monitored regularly on both an aggregate portfolio level and on an individual client or counterparty basis.

The Firm's wholesale exposure is managed through loan syndications and participations, loan sales, securitizations, credit derivatives, master netting agreements, collateral and other risk-reduction techniques. Refer to Note 12 for additional information on loans.

The Firm does not believe that its exposure to any particular loan product or industry segment results in a significant concentration of credit risk.

Terms of loan products and collateral coverage are included in the Firm's assessment when extending credit and establishing its allowance for credit losses. Refer to Note 13 for additional information on the allowance for credit losses.

Notes to consolidated financial statements

The table below presents both on-balance sheet and off-balance sheet consumer and wholesale credit exposure by the Firm's three credit portfolio segments as of December 31, 2024 and 2023. The wholesale industry of risk category is generally based on the client or counterparty's primary business activity.

December 31, (in millions)	2024				2023			
	Credit exposure ^(h)	On-balance sheet		Off-balance sheet ⁽ⁱ⁾	Credit exposure ^(h)	On-balance sheet		Off-balance sheet ⁽ⁱ⁾
		Loans	Derivatives			Loans	Derivatives	
Consumer, excluding credit card	\$ 437,654	\$ 392,810	\$ —	\$ 44,844	\$ 455,496	\$ 410,093	\$ —	\$ 45,403
Credit card^(a)	1,234,171	232,860	—	1,001,311	1,126,781	211,123	—	915,658
Total consumer^(a)	1,671,825	625,670	—	1,046,155	1,582,277	621,216	—	961,061
Wholesale^(b)								
Real Estate	207,050	169,506	310	37,234	208,261	166,372	420	41,469
Individuals and Individual Entities ^(c)	144,145	130,317	1,259	12,569	145,849	126,339	725	18,785
Asset Managers	135,541	58,720	15,695	61,126	129,574	52,178	9,925	67,471
Consumer & Retail	129,815	46,509	1,608	81,698	127,086	46,274	2,013	78,799
Technology, Media & Telecommunications	84,716	21,449	2,448	60,819	77,296	22,450	2,451	52,395
Industrials	72,530	24,011	2,035	46,484	75,092	26,548	1,335	47,209
Healthcare	64,224	23,243	616	40,365	65,025	23,169	1,577	40,279
Banks & Finance Companies	61,287	40,239	3,890	17,158	57,177	33,941	2,898	20,338
Utilities	35,871	6,172	2,631	27,068	36,061	7,067	3,396	25,598
State & Municipal Govt ^(d)	35,039	19,279	372	15,388	35,986	20,019	442	15,525
Automotive	34,336	17,696	794	15,846	33,977	17,459	428	16,090
Oil & Gas	31,724	7,226	1,153	23,345	34,475	8,480	705	25,290
Insurance	24,267	2,533	9,703	12,031	20,501	2,535	7,138	10,828
Chemicals & Plastics	20,782	6,176	267	14,339	20,773	6,458	441	13,874
Transportation	17,019	5,380	769	10,870	16,060	5,080	555	10,425
Metals & Mining	15,860	4,425	564	10,871	15,508	4,655	274	10,579
Central Govt	13,862	4,715	6,285	2,862	17,704	5,463	10,669	1,572
Securities Firms	9,443	1,878	3,197	4,368	8,689	865	3,285	4,539
Financial Markets Infrastructure	4,446	16	2,410	2,020	4,251	86	2,155	2,010
All other ^(e)	140,873	100,906	4,961	35,006	134,777	97,034	4,032	33,711
Subtotal	1,282,830	690,396	60,967	531,467	1,264,122	672,472	54,864	536,786
Loans held-for-sale and loans at fair value	31,922	31,922	—	—	30,018	30,018	—	—
Receivables from customers ^(f)	51,929	—	—	—	47,625	—	—	—
Total wholesale	1,366,681	722,318	60,967	531,467	1,341,765	702,490	54,864	536,786
Total exposure^(g)	\$3,038,506	\$1,347,988	\$ 60,967	\$1,577,622	\$2,924,042	\$1,323,706	\$ 54,864	\$1,497,847

(a) Also includes commercial card lending-related commitments primarily in CIB.

(b) The industry rankings presented in the table as of December 31, 2023, are based on the industry rankings of the corresponding exposures at December 31, 2024, not actual rankings of such exposures at December 31, 2023.

(c) Individuals and Individual Entities predominantly consists of Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB, and includes exposure to personal investment companies and personal and testamentary trusts.

(d) In addition to the credit risk exposure to states and municipal governments (both U.S. and non-U.S.) at December 31, 2024 and 2023, noted above, the Firm held: \$6.1 billion and \$5.9 billion, respectively, of trading assets; \$17.9 billion and \$21.4 billion, respectively, of AFS securities; and \$9.3 billion and \$9.9 billion, respectively, of HTM securities, issued by U.S. state and municipal governments. Refer to Note 2 and Note 10 for further information.

(e) All other includes: SPEs and Private education and civic organizations, representing approximately 94% and 6%, respectively, at both December 31, 2024 and 2023. Refer to Note 14 for more information on exposures to SPEs.

(f) Receivables from customers reflect held-for-investment margin loans to brokerage clients in CIB, CCB and AWM that are collateralized by assets maintained in the clients' brokerage accounts (including cash on deposit, and primarily liquid and readily marketable debt or equity securities).

(g) Excludes cash placed with banks of \$459.2 billion and \$614.1 billion, at December 31, 2024 and 2023, respectively, which is predominantly placed with various central banks, primarily Federal Reserve Banks.

(h) Credit exposure is net of risk participations and excludes the benefit of credit derivatives used in credit portfolio management activities held against derivative receivables or loans and liquid securities and other cash collateral held against derivative receivables.

(i) Represents lending-related financial instruments.

Note 5 – Derivative instruments

Derivative contracts derive their value from underlying asset prices, indices, reference rates, other inputs or a combination of these factors and may expose counterparties to risks and rewards of an underlying asset or liability without having to initially invest in, own or exchange the asset or liability. JPMorganChase makes markets in derivatives for clients and also uses derivatives to hedge or manage its own risk exposures. Predominantly all of the Firm's derivatives are entered into for market-making or risk management purposes.

Market-making derivatives

The majority of the Firm's derivatives are entered into for market-making purposes. Clients use derivatives to mitigate or modify interest rate, credit, foreign exchange, equity and commodity risks. The Firm actively manages the risks from its exposure to these derivatives by entering into other derivative contracts or by purchasing or selling other financial instruments that partially or fully offset the exposure from client derivatives.

Risk management derivatives

The Firm manages certain market and credit risk exposures using derivative instruments, including derivatives in hedge accounting relationships and other derivatives that are used to manage risks associated with specified assets and liabilities.

The Firm generally uses interest rate derivatives to manage the risk associated with changes in interest rates. Fixed-rate assets and liabilities appreciate or depreciate in market value as interest rates change. Similarly, interest income and expense increase or decrease as a result of variable-rate assets and liabilities resetting to current market rates, and as a result of the repayment and subsequent origination or issuance of fixed-rate assets and liabilities at current market rates. Gains and losses on the derivative instruments related to these assets and liabilities are expected to substantially offset this variability.

Foreign currency derivatives are used to manage the foreign exchange risk associated with certain foreign currency-denominated (i.e., non-U.S. dollar) assets and liabilities and forecasted transactions, as well as the Firm's net investments in certain non-U.S. subsidiaries or branches whose functional currencies are not the U.S. dollar. As a result of fluctuations in foreign currencies, the U.S. dollar-equivalent values of the foreign currency-denominated assets and liabilities or the forecasted revenues or expenses increase or decrease. Gains or losses on the derivative instruments related to these foreign currency-denominated assets or liabilities, or forecasted transactions, are expected to substantially offset this variability.

Commodities derivatives are used to manage the price risk of certain commodities inventories. Gains or losses on these derivative instruments are expected to

substantially offset the depreciation or appreciation of the related inventory.

Credit derivatives are used to manage the counterparty credit risk associated with loans and lending-related commitments. Credit derivatives compensate the purchaser when the entity referenced in the contract experiences a credit event, such as bankruptcy or a failure to pay an obligation when due. Credit derivatives primarily consist of CDS. Refer to the Credit derivatives section on pages 222–224 of this Note for a further discussion of credit derivatives.

Refer to the risk management derivatives gains and losses table on page 221 and the hedge accounting gains and losses tables on pages 218–221 of this Note for more information about risk management derivatives.

Derivative counterparties and settlement types

The Firm enters into OTC derivatives, which are negotiated and settled bilaterally with the derivative counterparty. The Firm also enters into, as principal, certain ETD such as futures and options, and OTC-cleared derivative contracts with CCPs. ETD contracts are generally standardized contracts traded on an exchange and cleared by the CCP, which is the Firm's counterparty from the inception of the transactions. OTC-cleared derivatives are traded on a bilateral basis and then novated to the CCP for clearing.

Derivative clearing services

The Firm provides clearing services for clients in which the Firm acts as a clearing member at certain exchanges and clearing houses. The Firm does not reflect the clients' derivative contracts in its Consolidated Financial Statements. Refer to Note 28 for further information on the Firm's clearing services.

Accounting for derivatives

All free-standing derivatives that the Firm executes for its own account are required to be recorded on the Consolidated balance sheets at fair value.

As permitted under U.S. GAAP, the Firm nets derivative assets and liabilities, and the related cash collateral receivables and payables, when a legally enforceable master netting agreement exists between the Firm and the derivative counterparty. Refer to Note 1 for further discussion of the offsetting of assets and liabilities. The accounting for changes in value of a derivative depends on whether or not the transaction has been designated and qualifies for hedge accounting. Derivatives that are not designated as hedges are reported and measured at fair value through earnings. The tabular disclosures on pages 213–221 of this Note provide additional information on the amount of, and reporting for, derivative assets, liabilities, gains and losses. Refer to Notes 2 and 3 for a further discussion of derivatives embedded in structured notes.

Notes to consolidated financial statements

Derivatives designated as hedges

The Firm applies hedge accounting to certain derivatives executed for risk management purposes – generally interest rate, foreign exchange and commodity derivatives. However, JPMorganChase does not seek to apply hedge accounting to all of the derivatives associated with the Firm's risk management activities. For example, the Firm does not apply hedge accounting to purchased CDS used to manage the credit risk of loans and lending-related commitments, because of the difficulties in qualifying such contracts as hedges. For the same reason, the Firm does not apply hedge accounting to certain interest rate, foreign exchange, and commodity derivatives used for risk management purposes.

To qualify for hedge accounting, a derivative must be highly effective at reducing the risk associated with the exposure being hedged. In addition, for a derivative to be designated as a hedge, the risk management objective and strategy must be documented. Hedge documentation must identify the derivative hedging instrument, the asset or liability or forecasted transaction and type of risk to be hedged, and how the effectiveness of the derivative is assessed prospectively and retrospectively. To assess effectiveness, the Firm uses statistical methods such as regression analysis, nonstatistical methods such as dollar-value comparisons of the change in the fair value of the derivative to the change in the fair value or cash flows of the hedged item, and qualitative comparisons of critical terms and the evaluation of any changes in those terms. The extent to which a derivative has been, and is expected to continue to be, highly effective at offsetting changes in the fair value or cash flows of the hedged item must be assessed and documented at least quarterly. If it is determined that a derivative is not highly effective at hedging the designated exposure, hedge accounting is discontinued.

There are three types of hedge accounting designations: fair value hedges, cash flow hedges and net investment hedges. JPMorganChase uses fair value hedges primarily to hedge fixed-rate long-term debt, AFS securities and certain commodities inventories. For qualifying fair value hedges, the changes in the fair value of the derivative, and in the value of the hedged item for the risk being hedged, are recognized in earnings. Certain amounts excluded from the assessment of effectiveness are recorded in OCI and recognized in earnings over the life of the derivative. If the hedge relationship is terminated, then the adjustment to the hedged item continues to be reported as part of the basis of the hedged item and, for interest-bearing financial instruments, is amortized to earnings as a yield adjustment. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily net interest income and principal transactions revenue.

The Firm employs the portfolio layer method to manage the interest rate risk of portfolios of fixed-rate assets. Throughout the life of the open hedge, basis adjustments are maintained at the portfolio level and are only allocated to individual assets under certain circumstances. These include instances where the portfolio amount falls below the hedged layer amounts, or in cases of voluntary de-designation.

JPMorganChase uses cash flow hedges primarily to hedge the exposure to variability in forecasted cash flows from floating-rate assets and liabilities and foreign currency-denominated revenue and expense. For qualifying cash flow hedges, changes in the fair value of the derivative are recorded in OCI and recognized in earnings as the hedged item affects earnings. Derivative amounts affecting earnings are recognized consistent with the classification of the hedged item – primarily noninterest revenue, net interest income and compensation expense. If the hedge relationship is terminated, then the change in value of the derivative recorded in AOCI is recognized in earnings when the cash flows that were hedged affect earnings. For hedge relationships that are discontinued because a forecasted transaction is expected to not occur according to the original hedge forecast, any related derivative values recorded in AOCI are immediately recognized in earnings.

JPMorganChase uses net investment hedges to protect the value of the Firm's net investments in certain non-U.S. subsidiaries or branches whose functional currencies are not the U.S. dollar. For qualifying net investment hedges, changes in the fair value of the derivatives due to changes in spot foreign exchange rates are recorded in OCI as translation adjustments. Amounts excluded from the assessment of effectiveness are recorded directly in earnings.

The following table outlines the Firm's primary uses of derivatives and the related hedge accounting designation or disclosure category.

Type of Derivative	Use of Derivative	Designation and disclosure	Affected segment or unit	Page reference
Manage specifically identified risk exposures in qualifying hedge accounting relationships:				
• Interest rate	Hedge fixed rate assets and liabilities	Fair value hedge	Corporate	218-219
• Interest rate	Hedge floating-rate assets and liabilities	Cash flow hedge	Corporate	220
• Foreign exchange	Hedge foreign currency-denominated assets and liabilities	Fair value hedge	Corporate	218-219
• Foreign exchange	Hedge foreign currency-denominated forecasted revenue and expense	Cash flow hedge	Corporate	220
• Foreign exchange	Hedge the value of the Firm's investments in non-U.S. dollar functional currency entities	Net investment hedge	Corporate	221
• Commodity	Hedge commodity inventory	Fair value hedge	CIB, AWM	218-219
Manage specifically identified risk exposures not designated in qualifying hedge accounting relationships:				
• Interest rate	Manage the risk associated with mortgage commitments, warehouse loans and MSRs	Specified risk management	CCB	221
• Credit	Manage the credit risk associated with wholesale lending exposures	Specified risk management	CIB, AWM	221
• Interest rate and foreign exchange	Manage the risk associated with certain other specified assets and liabilities	Specified risk management	Corporate, CIB	221
Market-making derivatives and other activities:				
• Various	Market-making and related risk management	Market-making and other	CIB	221
• Various	Other derivatives	Market-making and other	CIB, AWM, Corporate	221

Notes to consolidated financial statements

Notional amount of derivative contracts

The following table summarizes the notional amount of free-standing derivative contracts outstanding as of December 31, 2024 and 2023.

December 31, (in billions)	Notional amounts ^(b)	
	2024	2023
Interest rate contracts		
Swaps	\$ 20,437	\$ 23,251
Futures and forwards	3,067	2,690
Written options	3,067	3,370
Purchased options	3,089	3,362
Total interest rate contracts	29,660	32,673
Credit derivatives^(a)	1,191	1,045
Foreign exchange contracts		
Cross-currency swaps	4,509	4,721
Spot, futures and forwards	7,005	6,957
Written options	1,015	830
Purchased options	984	798
Total foreign exchange contracts	13,513	13,306
Equity contracts		
Swaps	850	639
Futures and forwards	206	157
Written options	914	778
Purchased options	788	698
Total equity contracts	2,758	2,272
Commodity contracts		
Swaps	148	115
Spot, futures and forwards	191	157
Written options	137	130
Purchased options	125	115
Total commodity contracts	601	517
Total derivative notional amounts	\$ 47,723	\$ 49,813

(a) Refer to the Credit derivatives discussion on pages 222–224 for more information on volumes and types of credit derivative contracts.

(b) Represents the sum of gross long and gross short third-party notional derivative contracts.

While the notional amounts disclosed above give an indication of the volume of the Firm's derivatives activity, the notional amounts significantly exceed, in the Firm's view, the possible losses that could arise from such transactions. For most derivative contracts, the notional amount is not exchanged; it is simply a reference amount used to calculate payments.

Impact of derivatives on the Consolidated balance sheets

The following table summarizes information on derivative receivables and payables (before and after netting adjustments) that are reflected on the Firm's Consolidated balance sheets as of December 31, 2024 and 2023, by accounting designation (e.g., whether the derivatives were designated in qualifying hedge accounting relationships or not) and contract type.

Free-standing derivative receivables and payables^(a)

December 31, 2024 (in millions)	Gross derivative receivables				Gross derivative payables			
	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables ^(b)	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables ^(b)
Trading assets and liabilities								
Interest rate	\$ 290,734	\$ —	\$ 290,734	\$ 24,945	\$ 274,226	\$ 2	\$ 274,228	\$ 9,239
Credit	11,087	—	11,087	814	13,796	—	13,796	1,898
Foreign exchange	261,035	1,885	262,920	25,312	253,289	1,278	254,567	15,597
Equity	85,220	—	85,220	5,285	96,139	—	96,139	8,648
Commodity	15,490	136	15,626	4,611	14,415	73	14,488	4,279
Total fair value of trading assets and liabilities	\$ 663,566	\$ 2,021	\$ 665,587	\$ 60,967	\$ 651,865	\$ 1,353	\$ 653,218	\$ 39,661

December 31, 2023 (in millions)	Gross derivative receivables				Gross derivative payables			
	Not designated as hedges	Designated as hedges	Total derivative receivables	Net derivative receivables ^(b)	Not designated as hedges	Designated as hedges	Total derivative payables	Net derivative payables ^(b)
Trading assets and liabilities								
Interest rate	\$ 250,689	\$ 2	\$ 250,691	\$ 26,324	\$ 240,482	\$ —	\$ 240,482	\$ 11,896
Credit	9,654	—	9,654	551	12,038	—	12,038	1,089
Foreign exchange	205,010	765	205,775	18,019	210,623	1,640	212,263	12,620
Equity	57,689	—	57,689	4,928	65,811	—	65,811	9,368
Commodity	15,228	211	15,439	5,042	16,286	92	16,378	5,874
Total fair value of trading assets and liabilities	\$ 538,270	\$ 978	\$ 539,248	\$ 54,864	\$ 545,240	\$ 1,732	\$ 546,972	\$ 40,847

(a) Balances exclude structured notes for which the fair value option has been elected. Refer to Note 3 for further information.

(b) As permitted under U.S. GAAP, the Firm has elected to net derivative receivables and derivative payables and the related cash collateral receivables and payables when a legally enforceable master netting agreement exists.

Notes to consolidated financial statements

Derivatives netting

The following tables present, as of December 31, 2024 and 2023, gross and net derivative receivables and payables by contract and settlement type. Derivative receivables and payables, as well as the related cash collateral from the same counterparty, have been netted on the Consolidated balance sheets where the Firm has obtained an appropriate legal opinion with respect to the master netting agreement. Where such a legal opinion has not been either sought or obtained, amounts are not eligible for netting on the Consolidated balance sheets, and those derivative receivables and payables are shown separately in the tables.

In addition to the cash collateral received and transferred that is presented on a net basis with derivative receivables and payables, the Firm receives and transfers additional collateral (financial instruments and cash). These amounts mitigate counterparty credit risk associated with the Firm's derivative instruments, but are not eligible for net presentation:

- collateral that consists of liquid securities and other cash collateral held at third-party custodians, which are shown separately as "Collateral not nettable on the Consolidated balance sheets" in the tables, up to the fair value exposure amount. For the purpose of this disclosure, the definition of liquid securities is consistent with the definition of high quality liquid assets as defined in the LCR rule;
- the amount of collateral held or transferred that exceeds the fair value exposure at the individual counterparty level, as of the date presented, which is excluded from the tables; and
- collateral held or transferred that relates to derivative receivables or payables where an appropriate legal opinion has not been either sought or obtained with respect to the master netting agreement, which is excluded from the tables.

December 31, (in millions)	2024			2023		
	Gross derivative receivables	Amounts netted on the Consolidated balance sheets	Net derivative receivables	Gross derivative receivables	Amounts netted on the Consolidated balance sheets	Net derivative receivables
U.S. GAAP nettable derivative receivables						
Interest rate contracts:						
OTC	\$ 158,202	\$ (134,791)	\$ 23,411	\$ 176,901	\$ (152,703)	\$ 24,198
OTC-cleared	130,989	(130,810)	179	71,419	(71,275)	144
Exchange-traded ^(a)	190	(188)	2	402	(389)	13
Total interest rate contracts	289,381	(265,789)	23,592	248,722	(224,367)	24,355
Credit contracts:						
OTC	8,680	(8,030)	650	7,637	(7,226)	411
OTC-cleared	2,267	(2,243)	24	1,904	(1,877)	27
Total credit contracts	10,947	(10,273)	674	9,541	(9,103)	438
Foreign exchange contracts:						
OTC	259,608	(236,931)	22,677	203,624	(187,295)	16,329
OTC-cleared	685	(677)	8	469	(459)	10
Exchange-traded ^(a)	34	—	34	6	(2)	4
Total foreign exchange contracts	260,327	(237,608)	22,719	204,099	(187,756)	16,343
Equity contracts:						
OTC	33,269	(30,742)	2,527	25,001	(23,677)	1,324
Exchange-traded ^(a)	51,040	(49,193)	1,847	30,462	(29,084)	1,378
Total equity contracts	84,309	(79,935)	4,374	55,463	(52,761)	2,702
Commodity contracts:						
OTC	8,340	(5,848)	2,492	8,049	(5,084)	2,965
OTC-cleared	126	(84)	42	133	(123)	10
Exchange-traded ^(a)	5,179	(5,083)	96	5,214	(5,190)	24
Total commodity contracts	13,645	(11,015)	2,630	13,396	(10,397)	2,999
Derivative receivables with appropriate legal opinion	658,609	(604,620)	53,989 ^(d)	531,221	(484,384)	46,837 ^(d)
Derivative receivables where an appropriate legal opinion has not been either sought or obtained	6,978		6,978	8,027		8,027
Total derivative receivables recognized on the Consolidated balance sheets	\$ 665,587		\$ 60,967	\$ 539,248		\$ 54,864
Collateral not nettable on the Consolidated balance sheets ^{(b)(c)}			(28,160)			(22,461)
Net amounts			\$ 32,807			\$ 32,403

Notes to consolidated financial statements

December 31, (in millions)	2024			2023		
	Gross derivative payables	Amounts netted on the Consolidated balance sheets	Net derivative payables	Gross derivative payables	Amounts netted on the Consolidated balance sheets	Net derivative payables
U.S. GAAP nettable derivative payables						
Interest rate contracts:						
OTC	\$ 138,215	\$ (130,375)	\$ 7,840	\$ 161,901	\$ (152,467)	\$ 9,434
OTC-cleared	134,555	(134,262)	293	76,007	(75,729)	278
Exchange-traded ^(a)	363	(352)	11	436	(390)	46
Total interest rate contracts	273,133	(264,989)	8,144	238,344	(228,586)	9,758
Credit contracts:						
OTC	11,381	(10,133)	1,248	10,332	(9,313)	1,019
OTC-cleared	1,779	(1,765)	14	1,639	(1,636)	3
Total credit contracts	13,160	(11,898)	1,262	11,971	(10,949)	1,022
Foreign exchange contracts:						
OTC	251,860	(238,292)	13,568	209,386	(199,173)	10,213
OTC-cleared	772	(678)	94	552	(470)	82
Exchange-traded ^(a)	14	—	14	6	—	6
Total foreign exchange contracts	252,646	(238,970)	13,676	209,944	(199,643)	10,301
Equity contracts:						
OTC	44,394	(38,298)	6,096	29,999	(27,360)	2,639
Exchange-traded ^(a)	49,578	(49,193)	385	33,137	(29,083)	4,054
Total equity contracts	93,972	(87,491)	6,481	63,136	(56,443)	6,693
Commodity contracts:						
OTC	6,918	(5,206)	1,712	8,788	(5,192)	3,596
OTC-cleared	84	(84)	—	120	(120)	—
Exchange-traded ^(a)	5,182	(4,919)	263	5,376	(5,192)	184
Total commodity contracts	12,184	(10,209)	1,975	14,284	(10,504)	3,780
Derivative payables with appropriate legal opinion	645,095	(613,557)	31,538	537,679	(506,125)	31,554
Derivative payables where an appropriate legal opinion has not been either sought or obtained	8,123		8,123	9,293		9,293
Total derivative payables recognized on the Consolidated balance sheets	\$ 653,218		\$ 39,661	\$ 546,972		\$ 40,847
Collateral not nettable on the Consolidated balance sheets^{(b)(c)}			(10,163)			(4,547)
Net amounts			\$ 29,498			\$ 36,300

(a) Exchange-traded derivative balances that relate to futures contracts are settled daily.

(b) Includes liquid securities and other cash collateral held at third-party custodians related to derivative instruments where an appropriate legal opinion has been obtained. For some counterparties, the collateral amounts of financial instruments may exceed the derivative receivables and derivative payables balances. Where this is the case, the total amount reported is limited to the net derivative receivables and net derivative payables balances with that counterparty.

(c) Derivative collateral relates only to OTC and OTC-cleared derivative instruments.

(d) Net derivatives receivable included cash collateral netted of \$51.9 billion and \$48.3 billion at December 31, 2024 and 2023, respectively. Net derivatives payable included cash collateral netted of \$60.8 billion and \$70.0 billion at December 31, 2024 and 2023, respectively. Derivative cash collateral relates to OTC and OTC-cleared derivative instruments.

Liquidity risk and credit-related contingent features

In addition to the specific market risks introduced by each derivative contract type, derivatives expose JPMorganChase to credit risk — the risk that derivative counterparties may fail to meet their payment obligations under the derivative contracts and the collateral, if any, held by the Firm proves to be of insufficient value to cover the payment obligation. It is the policy of JPMorganChase to actively pursue, where possible, the use of legally enforceable master netting arrangements and collateral agreements to mitigate derivative counterparty credit risk inherent in derivative receivables.

While derivative receivables expose the Firm to credit risk, derivative payables expose the Firm to liquidity risk, as the derivative contracts typically require the Firm to post cash or securities collateral with counterparties as the fair value of the contracts moves in the counterparties' favor or upon specified downgrades in the Firm's and its subsidiaries' respective credit ratings. Certain derivative contracts also provide for termination of the contract, generally upon a downgrade of either the Firm or the counterparty, at the fair value of the derivative contracts. The following table shows the aggregate fair value of net derivative payables related to OTC and OTC-cleared derivatives that contain contingent collateral or termination features that may be triggered upon a ratings downgrade, and the associated collateral the Firm has posted in the normal course of business, at December 31, 2024 and 2023.

OTC and OTC-cleared derivative payables containing downgrade triggers

(in millions)	December 31, 2024	December 31, 2023
Aggregate fair value of net derivative payables	\$ 15,371	\$ 14,655
Collateral posted	15,204	14,673

The following table shows the impact of a single-notch and two-notch downgrade of the long-term issuer ratings of JPMorgan Chase & Co. and its subsidiaries, predominantly JPMorgan Chase Bank, N.A., at December 31, 2024 and 2023, related to OTC and OTC-cleared derivative contracts with contingent collateral or termination features that may be triggered upon a ratings downgrade. Derivatives contracts generally require additional collateral to be posted or terminations to be triggered when the predefined rating threshold is breached. A downgrade by a single rating agency that does not result in a rating lower than a preexisting corresponding rating provided by another major rating agency will generally not result in additional collateral (except in certain instances in which additional initial margin may be required upon a ratings downgrade), nor in termination payment requirements. The liquidity impact in the table is calculated based upon a downgrade below the lowest current rating of the rating agencies referred to in the derivative contract.

Liquidity impact of downgrade triggers on OTC and OTC-cleared derivatives

(in millions)	December 31, 2024		December 31, 2023	
	Single-notch downgrade	Two-notch downgrade	Single-notch downgrade	Two-notch downgrade
Amount of additional collateral to be posted upon downgrade ^(a)	\$ 119	\$ 1,205	\$ 75	\$ 1,153
Amount required to settle contracts with termination triggers upon downgrade ^(b)	78	458	93	592

(a) Includes the additional collateral to be posted for initial margin.

(b) Amounts represent fair values of derivative payables, and do not reflect collateral posted.

Derivatives executed in contemplation of a sale of the underlying financial asset

In certain instances the Firm enters into transactions in which it transfers financial assets but maintains the economic exposure to the transferred assets by entering into a derivative with the same counterparty in contemplation of the initial transfer. The Firm generally accounts for such transfers as collateralized financing transactions as described in Note 11, but in limited circumstances they may qualify to be accounted for as a sale and a derivative under U.S. GAAP. The amount of such transfers accounted for as a sale where the associated derivative was outstanding was not material at December 31, 2024 and 2023.

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Impact of derivatives on the Consolidated statements of income

The following tables provide information related to gains and losses recorded on derivatives based on their hedge accounting designation or purpose.

Fair value hedge gains and losses

The following tables present derivative instruments, by contract type, used in fair value hedge accounting relationships, as well as pre-tax gains/(losses) recorded on such derivatives and the related hedged items for the years ended December 31, 2024, 2023 and 2022, respectively. The Firm includes gains/(losses) on the hedging derivative in the same line item in the Consolidated statements of income as the related hedged item.

Year ended December 31, 2024 (in millions)	Gains/(losses) recorded in income			Income statement impact of excluded components ^(e)		OCI impact
	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI ^(f)
Contract type						
Interest rate ^{(a)(b)}	\$ 711	\$ (65)	\$ 646	\$ —	\$ 699	\$ —
Foreign exchange ^(c)	(177)	402	225	(532)	225	(115)
Commodity ^(d)	293	(160)	133	—	122	—
Total	\$ 827	\$ 177	\$ 1,004	\$ (532)	\$ 1,046	\$ (115)

Year ended December 31, 2023 (in millions)	Gains/(losses) recorded in income			Income statement impact of excluded components ^(e)		OCI impact
	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI ^(f)
Contract type						
Interest rate ^{(a)(b)}	\$ 1,554	\$ (1,248)	\$ 306	\$ —	\$ 157	\$ —
Foreign exchange ^(c)	722	(483)	239	(601)	239	(134)
Commodity ^(d)	1,227	(706)	521	—	525	—
Total	\$ 3,503	\$ (2,437)	\$ 1,066	\$ (601)	\$ 921	\$ (134)

Year ended December 31, 2022 (in millions)	Gains/(losses) recorded in income			Income statement impact of excluded components ^(e)		OCI impact
	Derivatives	Hedged items	Income statement impact	Amortization approach	Changes in fair value	Derivatives - Gains/(losses) recorded in OCI ^(f)
Contract type						
Interest rate ^{(a)(b)}	\$ (14,352)	\$ 14,047	\$ (305)	\$ —	\$ (262)	\$ —
Foreign exchange ^(c)	(1,317)	1,423	106	(528)	106	130
Commodity ^(d)	106	(70)	36	—	48	—
Total	\$ (15,563)	\$ 15,400	\$ (163)	\$ (528)	\$ (108)	\$ 130

- (a) Primarily consists of hedges of the benchmark (e.g., Secured Overnight Financing Rate (“SOFR”)) interest rate risk of fixed-rate long-term debt and AFS securities. Gains and losses were recorded in net interest income.
- (b) Includes the amortization of income/expense associated with the inception hedge accounting adjustment applied to the hedged item. Excludes the accrual of interest on interest rate swaps and the related hedged items.
- (c) Primarily consists of hedges of the foreign currency risk of long-term debt and AFS securities for changes in spot foreign currency rates. Gains and losses related to the derivatives and the hedged items due to changes in foreign currency rates and the income statement impact of excluded components were recorded primarily in principal transactions revenue and net interest income.
- (d) Consists of overall fair value hedges of physical commodities inventories that are generally carried at the lower of cost or net realizable value (net realizable value approximates fair value). Gains and losses were recorded in principal transactions revenue.
- (e) The assessment of hedge effectiveness excludes certain components of the changes in fair values of the derivatives and hedged items such as forward points on foreign exchange forward contracts, time values and cross-currency basis spreads. Excluded components may impact earnings either through amortization of the initial amount over the life of the derivative or through fair value changes recognized in the current period.
- (f) Represents the change in value of amounts excluded from the assessment of effectiveness under the amortization approach, predominantly cross-currency basis spreads. The amount excluded at inception of the hedge is recognized in earnings over the life of the derivative.

As of December 31, 2024 and 2023, the following amounts were recorded on the Consolidated balance sheets related to certain cumulative fair value hedge basis adjustments that are expected to reverse through the income statement in future periods as an adjustment to yield.

December 31, 2024 (in millions)	Carrying amount of the hedged items ^{(a)(b)}	Cumulative amount of fair value hedging adjustments included in the carrying amount of hedged items: ^(d)		
		Active hedging relationships	Discontinued hedging relationships ^(e)	Total
Assets				
Investment securities - AFS	\$ 203,141 ^(c)	\$ (1,675)	\$ (1,959)	\$ (3,634)
Liabilities				
Long-term debt	\$ 211,288	\$ (3,711)	\$ (9,332)	\$ (13,043)
Beneficial interests issued by consolidated VIEs	\$ 5,312	\$ (30)	\$ (5)	\$ (35)
December 31, 2023 (in millions)	Carrying amount of the hedged items ^{(a)(b)}	Cumulative amount of fair value hedging adjustments included in the carrying amount of hedged items: ^(d)		
		Active hedging relationships	Discontinued hedging relationships ^(e)	Total
Assets				
Investment securities - AFS	\$ 151,752 ^(c)	\$ 549	\$ (2,010)	\$ (1,461)
Liabilities				
Long-term debt	\$ 195,455	\$ (2,042)	\$ (9,727)	\$ (11,769)
Beneficial interests issued by consolidated VIEs	\$ —	\$ —	\$ —	\$ —

- (a) Excludes physical commodities with a carrying value of \$6.2 billion and \$5.6 billion at December 31, 2024 and 2023, respectively, to which the Firm applies fair value hedge accounting. As a result of the application of hedge accounting, these inventories are carried at fair value, thus recognizing unrealized gains and losses in current periods. Since the Firm exits these positions at fair value, there is no incremental impact to net income in future periods.
- (b) Excludes hedged items where only foreign currency risk is the designated hedged risk, as basis adjustments related to foreign currency hedges will not reverse through the income statement in future periods. At December 31, 2024 and 2023, the carrying amount excluded for AFS securities was \$28.7 billion and \$19.3 billion, respectively. At December 31, 2024 and 2023, the carrying amount excluded for long-term debt was \$518 million and zero, respectively.
- (c) Carrying amount represents the amortized cost, net of allowance if applicable. At December 31, 2024 and December 31, 2023, the amortized cost of the portfolio layer method closed portfolios was \$72.8 billion and \$83.9 billion, of which \$41.2 billion and \$68.0 billion was designated as hedged, respectively. The amount designated as hedged is the sum of the notional amounts of all outstanding layers in each portfolio, which includes both spot starting and forward starting layers. At December 31, 2024 and December 31, 2023, the cumulative amount of basis adjustments was \$(1.7) billion and \$(165) million, which is comprised of \$(1.2) billion and \$73 million for active hedging relationships, and \$(566) million and \$(238) million for discontinued hedging relationships, respectively. Refer to Note 10 for additional information.
- (d) Positive (negative) amounts related to assets represent cumulative fair value hedge basis adjustments that will reduce (increase) net interest income in future periods. Positive (negative) amounts related to liabilities represent cumulative fair value hedge basis adjustments that will increase (reduce) net interest income in future periods.
- (e) Represents basis adjustments existing on the balance sheet date associated with hedged items that have been de-designated from qualifying fair value hedging relationships.

Notes to consolidated financial statements

Cash flow hedge gains and losses

The following tables present derivative instruments, by contract type, used in cash flow hedge accounting relationships, and the pre-tax gains/(losses) recorded on such derivatives, for the years ended December 31, 2024, 2023 and 2022, respectively. The Firm includes the gains/(losses) on the hedging derivative in the same line item in the Consolidated statements of income as the change in cash flows on the related hedged item.

Year ended December 31, 2024 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate ^(a)	\$ (2,668)	\$ (3,603)	\$ (935)
Foreign exchange ^(b)	89	(139)	(228)
Total	\$ (2,579)	\$ (3,742)	\$ (1,163)

Year ended December 31, 2023 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate ^(a)	\$ (1,839)	\$ 274	\$ 2,113
Foreign exchange ^(b)	64	209	145
Total	\$ (1,775)	\$ 483	\$ 2,258

Year ended December 31, 2022 (in millions)	Derivatives gains/(losses) recorded in income and other comprehensive income/(loss)		
	Amounts reclassified from AOCI to income	Amounts recorded in OCI	Total change in OCI for period
Contract type			
Interest rate ^(a)	\$ (153)	\$ (7,131)	\$ (6,978)
Foreign exchange ^(b)	(267)	(342)	(75)
Total	\$ (420)	\$ (7,473)	\$ (7,053)

(a) Primarily consists of hedges of SOFR-indexed floating-rate assets. Gains and losses were recorded in net interest income.

(b) Primarily consists of hedges of the foreign currency risk of non-U.S. dollar-denominated revenue and expense. The income statement classification of gains and losses follows the hedged item – primarily noninterest revenue and compensation expense.

The Firm did not experience any forecasted transactions that failed to occur for the years ended 2024, 2023 and 2022.

Over the next 12 months, the Firm expects that approximately \$(1.6) billion (after-tax) of net losses recorded in AOCI at December 31, 2024, related to cash flow hedges will be recognized in income. For cash flow hedges that have been terminated, the maximum length of time over which the derivative results recorded in AOCI will be recognized in earnings is approximately seven years, corresponding to the timing of the originally hedged forecasted cash flows. For open cash flow hedges, the maximum length of time over which forecasted transactions are hedged is approximately seven years. The Firm's longer-dated forecasted transactions relate to core lending and borrowing activities.

Net investment hedge gains and losses

The following table presents hedging instruments, by contract type, that were used in net investment hedge accounting relationships, and the pre-tax gains/(losses) recorded on such instruments for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024		2023		2022	
	Amounts recorded in income ^{(a)(b)}	Amounts recorded in OCI	Amounts recorded in income ^{(a)(b)}	Amounts recorded in OCI	Amounts recorded in income ^{(a)(b)}	Amounts recorded in OCI
Foreign exchange derivatives	\$467	\$4,411	\$384	\$(1,732)	\$(123)	\$3,591

- (a) Certain components of hedging derivatives are permitted to be excluded from the assessment of hedge effectiveness, such as forward points on foreign exchange forward contracts. The Firm elects to record changes in fair value of these amounts directly in other income.
- (b) Excludes amounts reclassified from AOCI to income on the sale or liquidation of hedged entities. During the year ended December 31, 2024, the Firm reclassified a net pre-tax gain of \$89 million to other income/expense. During the year ended December 31, 2023, the Firm reclassified a net pre-tax loss of \$(35) million to other revenue including the impact of the acquisition of CIFM. During the year ended December 31, 2022, the Firm reclassified net pre-tax gains of \$38 million to other income/expense related to the liquidation of certain legal entities. Refer to Note 24 for further information.

Gains and losses on derivatives used for specified risk management purposes

The following table presents pre-tax gains/(losses) recorded on a limited number of derivatives, not designated in hedge accounting relationships, that are used to manage risks associated with certain specified assets and liabilities, including certain risks arising from mortgage commitments, warehouse loans, MSRs, wholesale lending exposures, and foreign currency denominated assets and liabilities.

Year ended December 31, (in millions)	Derivatives gains/(losses) recorded in income		
	2024	2023	2022
Contract type			
Interest rate ^(a)	\$ (425)	\$ (135)	\$ (827)
Credit ^(b)	(604)	(441)	51
Foreign exchange ^(c)	(10)	(2)	(48)
Total	\$ (1,039)	\$ (578)	\$ (824)

- (a) Primarily represents interest rate derivatives used to hedge the interest rate risk inherent in mortgage commitments, warehouse loans and MSRs, as well as written commitments to originate warehouse loans. Gains and losses were recorded predominantly in mortgage fees and related income.
- (b) Relates to credit derivatives used to mitigate credit risk associated with lending exposures in the Firm's wholesale businesses. These derivatives do not include credit derivatives used to mitigate counterparty credit risk arising from derivative receivables, which is included in gains and losses on derivatives related to market-making activities and other derivatives. Gains and losses were recorded in principal transactions revenue.
- (c) Primarily relates to derivatives used to mitigate foreign exchange risk of specified foreign currency-denominated assets and liabilities. Gains and losses were recorded in principal transactions revenue.

Gains and losses on derivatives related to market-making activities and other derivatives

The Firm makes markets in derivatives in order to meet the needs of customers and uses derivatives to manage certain risks associated with net open risk positions from its market-making activities, including the counterparty credit risk arising from derivative receivables. All derivatives not included in the hedge accounting or specified risk management categories above are included in this category. Gains and losses on these derivatives are primarily recorded in principal transactions revenue. Refer to Note 6 for information on principal transactions revenue.

Credit derivatives

Credit derivatives are financial instruments whose value is derived from the credit risk associated with the debt of a third-party issuer (the reference entity) and which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller). Credit derivatives expose the protection purchaser to the creditworthiness of the protection seller, as the protection seller is required to make payments under the contract when the reference entity experiences a credit event, such as a bankruptcy, a failure to pay its obligation or a restructuring. The seller of credit protection receives a premium for providing protection but has the risk that the underlying instrument referenced in the contract will be subject to a credit event.

The Firm is both a purchaser and seller of protection in the credit derivatives market and uses these derivatives for two primary purposes. First, in its capacity as a market-maker, the Firm actively manages a portfolio of credit derivatives by purchasing and selling credit protection, predominantly on corporate debt obligations, to meet the needs of customers. Second, as an end-user, the Firm uses credit derivatives to manage credit risk associated with lending exposures (loans and unfunded commitments) in its wholesale and consumer businesses and derivatives counterparty exposures in its wholesale businesses, and to manage the credit risk arising from certain financial instruments in the Firm's market-making businesses. Following is a summary of various types of credit derivatives.

Credit default swaps

Credit derivatives may reference the credit of either a single reference entity ("single-name"), broad-based index or portfolio. The Firm purchases and sells protection on both single-name and index-reference obligations. Single-name CDS and index CDS contracts are either OTC or OTC-cleared derivative contracts. Single-name CDS are used to manage the default risk of a single reference entity, while index CDS contracts are used to manage the credit risk associated with the broader credit markets or credit market segments. Like the S&P 500 and other market indices, a CDS index consists of a portfolio of CDS across many reference entities. New series of CDS indices are periodically established with a new underlying portfolio of reference entities to reflect changes in the credit markets. If one of the reference entities in the index experiences a credit event, then the reference entity that defaulted is removed from the index. CDS can also be referenced against specific portfolios of reference names or against customized exposure levels: for example, to provide protection against the first \$1 million of realized credit losses in a

\$10 million portfolio of exposure. Such structures are commonly known as tranche CDS.

For both single-name CDS contracts and index CDS contracts, upon the occurrence of a credit event, under the terms of a CDS contract neither party to the CDS contract has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the CDS contract and the fair value of the reference obligation at settlement of the credit derivative contract, also known as the recovery value. The protection purchaser does not need to hold the debt instrument of the underlying reference entity in order to receive amounts due under the CDS contract when a credit event occurs.

Credit-related notes

A credit-related note is a funded derivative with a credit risk component where the issuer of the credit-related note purchases from the note investor credit protection on a reference entity or an index. Under the contract, the investor pays the issuer the par value of the note at the inception of the transaction, and in return, the issuer makes periodic payments to the investor, based on the credit risk of the referenced entity. The issuer also repays the investor the par value of the note at maturity unless the reference entity (or one of the entities that makes up a reference index) experiences a specified credit event. If a credit event occurs, the issuer is not obligated to repay the par value of the note, but rather, the issuer pays the investor the difference between the par value of the note and the fair value of the defaulted reference obligation at the time of settlement. Neither party to the credit-related note has recourse to the defaulting reference entity.

The following tables present a summary of the notional amounts of credit derivatives and credit-related notes the Firm sold and purchased as of December 31, 2024 and 2023. Upon a credit event, the Firm as a seller of protection would typically pay out a percentage of the full notional amount of net protection sold, as the amount actually required to be paid on the contracts takes into account the recovery value of the reference obligation at the time of settlement. The Firm manages the credit risk on contracts to sell protection by purchasing protection with identical or similar underlying reference entities. Other purchased protection referenced in the following tables includes credit derivatives bought on related, but not identical, reference positions (including indices, portfolio coverage and other reference points) as well as protection purchased by CIB through credit-related notes. Other purchased protection also includes credit protection against certain loans in the retained lending portfolio through the issuance of credit derivatives and credit-related notes.

The Firm does not use notional amounts of credit derivatives as the primary measure of risk management for such derivatives, because the notional amount does not take into account the probability of the occurrence of a credit event, the recovery value of the reference obligation, or related cash instruments and economic hedges, each of which reduces, in the Firm's view, the risks associated with such derivatives.

Total credit derivatives and credit-related notes

December 31, 2024 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlyings ^(c)	Net protection (sold)/purchased ^(d)	Other protection purchased ^(e)
Credit derivatives				
Credit default swaps	\$ (450,184)	\$ 474,554	\$ 24,370	\$ 6,858
Other credit derivatives ^(a)	(110,913)	137,927	27,014	10,169
Total credit derivatives	(561,097)	612,481	51,384	17,027
Credit-related notes ^(b)	—	—	—	10,471
Total	\$ (561,097)	\$ 612,481	\$ 51,384	\$ 27,498

December 31, 2023 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlyings ^(c)	Net protection (sold)/purchased ^(d)	Other protection purchased ^(e)
Credit derivatives				
Credit default swaps	\$ (450,172)	\$ 473,823	\$ 23,651	\$ 7,517
Other credit derivatives ^(a)	(38,846)	45,416	6,570	29,206
Total credit derivatives	(489,018)	519,239	30,221	36,723
Credit-related notes ^(b)	—	—	—	9,788
Total	\$ (489,018)	\$ 519,239	\$ 30,221	\$ 46,511

(a) Other credit derivatives predominantly consist of credit swap options and total return swaps.

(b) Predominantly represents Other protection purchased by CIB.

(c) Represents the total notional amount of protection purchased where the underlying reference instrument is identical to the reference instrument on protection sold; the notional amount of protection purchased for each individual identical underlying reference instrument may be greater or lower than the notional amount of protection sold.

(d) Does not take into account the fair value of the reference obligation at the time of settlement, which would generally reduce the amount the seller of protection pays to the buyer of protection in determining settlement value.

(e) Represents protection purchased by the Firm on referenced instruments (single-name, portfolio or index) where the Firm has not sold any protection on the identical reference instrument. Also includes credit protection against certain loans and lending-related commitments in the retained lending portfolio through the issuance of credit derivatives and credit-related notes.

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The following tables summarize the notional amounts by the ratings, maturity profile, and total fair value, of credit derivatives as of December 31, 2024 and 2023, where JPMorganChase is the seller of protection. The maturity profile is based on the remaining contractual maturity of the credit derivative contracts. The ratings profile is based on the rating of the reference entity on which the credit derivative contract is based. The ratings and maturity profile of credit derivatives where JPMorganChase is the purchaser of protection are comparable to the profile reflected below.

Protection sold – credit derivatives ratings^(a)/maturity profile

December 31, 2024 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value of receivables ^(b)	Fair value of payables ^(b)	Net fair value
Risk rating of reference entity							
Investment-grade	\$ (135,950)	\$ (277,052)	\$ (33,379)	\$ (446,381)	\$ 4,593	\$ (904)	\$ 3,689
Noninvestment-grade	(42,149)	(70,525)	(2,042)	(114,716)	1,889	(1,738)	151
Total	\$ (178,099)	\$ (347,577)	\$ (35,421)	\$ (561,097)	\$ 6,482	\$ (2,642)	\$ 3,840
December 31, 2023 (in millions)	<1 year	1–5 years	>5 years	Total notional amount	Fair value of receivables ^(b)	Fair value of payables ^(b)	Net fair value
Risk rating of reference entity							
Investment-grade	\$ (89,981)	\$ (263,834)	\$ (29,470)	\$ (383,285)	\$ 3,659	\$ (1,144)	\$ 2,515
Noninvestment-grade	(31,419)	(69,515)	(4,799)	(105,733)	2,466	(1,583)	883
Total	\$ (121,400)	\$ (333,349)	\$ (34,269)	\$ (489,018)	\$ 6,125	\$ (2,727)	\$ 3,398

(a) The ratings scale is primarily based on external credit ratings defined by S&P and Moody's.

(b) Amounts are shown on a gross basis, before the benefit of legally enforceable master netting agreements including cash collateral netting.

Note 6 – Noninterest revenue and noninterest expense

Noninterest revenue

The Firm records noninterest revenue from certain contracts with customers in investment banking fees, deposit-related fees, asset management fees, commissions and other fees, and components of card income. The related contracts are often terminable on demand and the Firm has no remaining obligation to deliver future services. For arrangements with a fixed term, the Firm may commit to deliver services in the future. Revenue associated with these remaining performance obligations typically depends on the occurrence of future events or underlying asset values, and is not recognized until the outcome of those events or values are known.

Investment banking fees

This revenue category includes debt and equity underwriting and advisory fees. As an underwriter, the Firm helps clients raise capital via public offering and private placement of various types of debt and equity instruments. Underwriting fees are primarily based on the issuance price and quantity of the underlying instruments, and are recognized as revenue typically upon execution of the client's transaction. The Firm also manages and syndicates loan arrangements. Credit arrangement and syndication fees, included within debt underwriting fees, are recorded as revenue after satisfying certain retention, timing and yield criteria.

The Firm also provides advisory services by assisting its clients with mergers and acquisitions, divestitures, restructuring and other complex transactions. Advisory fees are recognized as revenue typically upon execution of the client's transaction.

The following table presents the components of investment banking fees.

Year ended December 31, (in millions)	2024	2023	2022
Underwriting			
Equity	\$ 1,687	\$ 1,149	\$ 975
Debt	3,945	2,610	2,732
Total underwriting	5,632	3,759	3,707
Advisory	3,278	2,760	2,979
Total investment banking fees	\$ 8,910	\$ 6,519	\$ 6,686

Investment banking fees are earned primarily by CIB.

Principal transactions

Principal transactions revenue is driven by many factors, including:

- the bid-offer spread, which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa; and
- realized and unrealized gains and losses on financial instruments and commodities transactions, including those accounted for under the fair value option, primarily used in client-driven market-making activities.
 - Realized gains and losses result from the sale of instruments, closing out or termination of transactions, or interim cash payments.
 - Unrealized gains and losses result from changes in valuation.

In connection with its client-driven market-making activities, the Firm transacts in debt and equity instruments, derivatives and commodities, including physical commodities inventories and financial instruments that reference commodities.

Principal transactions revenue also includes realized and unrealized gains and losses related to:

- derivatives designated in qualifying hedge accounting relationships, primarily fair value hedges of commodity and foreign exchange risk;
- derivatives used for specific risk management purposes, primarily to mitigate credit, foreign exchange and interest rate risks.

Refer to Note 5 for further information on the income statement classification of gains and losses from derivatives activities.

In the financial commodity markets, the Firm transacts in OTC derivatives (e.g., swaps, forwards, options) and ETD that reference a wide range of underlying commodities. In the physical commodity markets, the Firm primarily purchases and sells precious and base metals, natural gas, and may hold other commodities inventories under financing and other arrangements with clients.

The following table presents all realized and unrealized gains and losses recorded in principal transactions revenue. This table excludes interest income and interest expense on trading assets and liabilities, which are an integral part of the overall performance of the Firm's client-driven market-making activities in CIB and fund deployment activities in Treasury and CIO. Refer to Note 7 for further information on interest income and interest expense.

Trading revenue is presented primarily by instrument type. The Firm's client-driven market-making businesses generally utilize a variety of instrument

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types in connection with their market-making and related risk-management activities; accordingly, the trading revenue presented in the table below is not representative of the total revenue of any individual LOB.

Year ended December 31, (in millions)	2024	2023	2022
Trading revenue by instrument type			
Interest rate ^(a)	\$ 3,631	\$ 5,607	\$ 3,010
Credit ^(b)	1,545	1,434	1,412 ^(c)
Foreign exchange	4,874	5,082	5,119
Equity	13,476	10,229	8,068
Commodity	1,194	2,202	2,348
Total trading revenue	24,720	24,554	19,957
Private equity gains/(losses)	67	(94)	(45)
Principal transactions	\$24,787	\$ 24,460	\$ 19,912

(a) Includes the impact of changes in funding valuation adjustments on derivatives.

(b) Includes the impact of changes in credit valuation adjustments on derivatives, net of the associated hedging activities.

(c) Includes net markdowns on held-for-sale positions, primarily unfunded commitments, in the bridge financing portfolio.

Principal transactions revenue is earned primarily by CIB.

Lending- and deposit-related fees

Lending-related fees include fees earned from loan commitments, standby letters of credit, financial guarantees, and other loan-servicing activities. Deposit-related fees include fees earned from performing cash management activities, and providing overdraft and other deposit account services. Deposit-related fees also include the impact of credits earned by clients that reduce such fees. Lending- and deposit-related fees are recognized over the period in which the related service is provided. Refer to Note 28 for further information on lending-related commitments.

The following table presents the components of lending- and deposit-related fees.

Year ended December 31, (in millions)	2024	2023	2022
Lending-related fees	\$ 2,192 ^(a)	\$ 2,365 ^(a)	\$ 1,468
Deposit-related fees	5,414	5,048	5,630
Total lending- and deposit-related fees	\$7,606	\$ 7,413	\$ 7,098

(a) Includes the amortization of the fair value discount on certain acquired lending-related commitments associated with First Republic, predominantly in AWM and CIB. The discount is deferred in other liabilities and recognized on a straight-line basis over the commitment period and was largely recognized in 2023 as the commitments were generally short term. Refer to Note 34 for additional information.

Lending- and deposit-related fees are earned by CCB, CIB and AWM.

Asset management fees

Investment management fees include fees associated with assets the Firm manages on behalf of its clients, including investors in Firm-sponsored funds and owners of separately managed investment accounts. Management fees are typically based on the value of assets under management and are collected and recognized at the end of each period over which the management services are provided and the value of the managed assets is known. The Firm also receives performance-based management fees, which are earned based on exceeding certain benchmarks or other performance targets and are accrued and recognized when the probability of reversal is remote, typically at the end of the related billing period.

All other asset management fees include commissions earned on the sales or distribution of mutual funds to clients. These fees are recorded as revenue at the time the service is rendered or, in the case of certain distribution fees, based on the underlying fund's asset value or investor redemption activity.

The following table presents the components of asset management fees.

Year ended December 31, (in millions)	2024	2023	2022
Asset management fees			
Investment management fees	\$ 17,425	\$ 14,908	\$ 13,765
All other asset management fees	376	312	331
Total asset management fees	\$ 17,801	\$ 15,220	\$ 14,096

Asset management fees earned primarily by AWM and CCB.

Commissions and other fees

This revenue category includes commissions and fees from brokerage and custody services, and other products.

Brokerage commissions represents commissions earned when the Firm acts as a broker, by facilitating its clients' purchases and sales of securities and other financial instruments. Brokerage commissions are collected and recognized as revenue upon occurrence of the client transaction. The Firm reports certain costs paid to third-party clearing houses and exchanges net against commission revenue.

Administration fees predominantly include fees for custody, funds services, securities lending and securities clearance. These fees are recorded as revenue over the period in which the related service is provided.

The following table presents the components of commissions and other fees.

Year ended December 31, (in millions)	2024	2023	2022
Commissions and other fees			
Brokerage commissions and fees	\$ 3,119	\$ 2,820	\$ 2,831
Administration fees	2,526	2,310	2,348
All other commissions and fees ^(a)	1,885	1,706	1,402
Total commissions and other fees	\$ 7,530	\$ 6,836	\$ 6,581

(a) Includes annuity sales commissions, depository receipt-related service fees and travel-related sales commissions, as well as other service fees, which are recognized as revenue when the services are rendered.

Commissions and other fees are earned primarily by CIB, CCB and AWM.

Mortgage fees and related income

This revenue category reflects CCB's Home Lending production and net mortgage servicing revenue.

Production revenue includes fees and income recognized as earned on mortgage loans originated with the intent to sell, and the impact of risk management activities associated with the mortgage pipeline and warehouse loans. Production revenue also includes gains and losses on sales and lower of cost or fair value adjustments on mortgage loans held-for-sale (excluding certain repurchased loans insured by U.S. government agencies), and changes in the fair value of financial instruments measured under the fair value option. Net mortgage servicing revenue includes operating revenue earned from servicing third-party mortgage loans, which is recognized over the period in which the service is provided; changes in the fair value of MSRs; the impact of risk management activities associated with MSRs; and gains and losses on securitization of excess mortgage servicing. Net mortgage servicing revenue also includes gains and losses on sales and lower of cost or fair value adjustments of certain repurchased loans insured by U.S. government agencies.

Refer to Note 15 for further information on risk management activities and MSRs.

Net interest income from mortgage loans is recorded in interest income.

Card income

This revenue category includes interchange and other income from credit and debit card transactions; and fees earned from processing card transactions for merchants, both of which are recognized when purchases are made by a cardholder and presented net of certain transaction-related costs. Card income also includes account origination costs and annual fees, which are deferred and recognized on a straight-line basis over a 12-month period.

Certain credit card products offer the cardholder the ability to earn points based on account activity, which the cardholder can choose to redeem for cash and non-cash rewards. The cost to the Firm related to these proprietary rewards programs varies based on multiple factors including the terms and conditions of the rewards programs, cardholder activity, cardholder reward redemption rates and cardholder reward selections. The Firm maintains a liability for its obligations under its rewards programs and reports the current-period cost as a reduction of card income.

Credit card revenue sharing agreements

The Firm has contractual agreements with numerous co-brand partners that grant the Firm exclusive rights to issue co-branded credit card products and market them to the customers of such partners. These partners endorse the co-brand credit card programs and provide their customer or member lists to the Firm. The partners may also conduct marketing activities and provide rewards redeemable under their own loyalty programs that the Firm will grant to co-brand credit cardholders based on account activity. The terms of these agreements generally range from five to ten years.

The Firm typically makes payments to the co-brand credit card partners based on the cost of partners' marketing activities and loyalty program rewards provided to credit cardholders, new account originations and sales volumes. Payments to partners based on marketing efforts undertaken by the partners are expensed by the Firm as incurred and reported as marketing expense. Payments for partner loyalty program rewards are reported as a reduction of card income when incurred. Payments to partners based on new credit card account originations are accounted for as direct loan origination costs and are deferred and recognized as a reduction of card income on a straight-line basis over a 12-month period. Payments to partners based on sales volumes are reported as a reduction of card income when the related interchange income is earned.

The following table presents the components of card income:

Year ended December 31, (in millions)	2024	2023	2022
Interchange and merchant processing income	\$ 33,847	\$ 31,021	\$ 28,085
Reward costs and partner payments	(26,784)	(24,601)	(22,162)
All other ^(a)	(1,566)	(1,636)	(1,503)
Total card income	\$ 5,497	\$ 4,784	\$ 4,420

(a) Predominantly represents the amortization of account origination costs and annual fees, which are deferred and recognized on a straight-line basis over a 12-month period.

Card income is earned primarily by CCB and CIB.

Notes to consolidated financial statements

Other income

This revenue category includes operating lease income, as well as losses associated with the Firm's tax-oriented investments, predominantly alternative energy equity-method investments in CIB. The losses associated with these tax-oriented investments are more than offset by lower income tax expense from the associated tax credits.

The following table presents certain components of other income:

Year ended December 31, (in millions)	2024	2023	2022
Operating lease income	\$ 2,795	\$ 2,843	\$ 3,654
Losses on tax-oriented investments	(97)	(1,538)	(1,491)
Gain on Visa shares	7,990 ^(b)	—	914 ^(c)
Estimated bargain purchase gain associated with the First Republic acquisition	103	2,775	—
Gain related to the acquisition of CIFM ^(a)	—	339	—

(a) Gain on the original minority interest in CIFM upon the Firm's acquisition of the remaining 51% of the entity.

(b) Relates to the initial gain recognized on May 6, 2024 on the Visa C shares. Refer to Note 2 for additional information.

(c) Relates to the sale of Visa B shares.

Refer to Note 18 for additional information on operating leases.

Proportional Amortization Method: Effective January 1, 2024, as a result of adopting updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, the amortization of certain of the Firm's alternative energy tax-oriented investments that was previously recognized in other income is now recognized in income tax expense, which aligns with the associated tax credits and other tax benefits. Refer to Notes 1, 14 and 25 for additional information.

Noninterest expense

Other expense

Other expense on the Firm's Consolidated statements of income included:

Year ended December 31, (in millions)	2024	2023	2022
Legal expense	\$ 740	\$1,436	\$ 266
FDIC-related expense	1,893 ^(c)	4,203 ^(c)	860
Operating losses	1,417	1,228	1,101
Contribution of Visa shares ^(a)	1,000	—	—
First Republic-related expense ^(b)	777	1,060	—

(a) Represents the contribution of a portion of Visa C shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Note 2 for additional information.

(b) Reflects the expenses classified within other expense, including \$488 million and \$360 million of integration and restructuring costs associated with First Republic for the full years ended December 31, 2024 and 2023, respectively. Additionally, the second quarter of 2023 included payments to the FDIC for the First Republic individuals who were not employees of the Firm until July 2, 2023. Refer to Note 34 for additional information on the First Republic acquisition.

(c) The first quarter of 2024 included an increase of \$725 million to the FDIC special assessment reflecting the FDIC's revised estimate of Deposit Insurance Fund losses. The fourth quarter of 2023 included the \$2.9 billion FDIC special assessment.

Refer to Note 32 for additional information on noninterest revenue and expense by segment.

Note 7 – Interest income and interest expense

Interest income and interest expense are recorded in the Consolidated statements of income and classified based on the nature of the underlying asset or liability.

Interest income and interest expense includes the current-period interest accruals for financial instruments measured at fair value, except for derivatives and certain financial instruments containing embedded derivatives; for those instruments, all changes in fair value including any interest elements, are primarily reported in principal transactions revenue. For financial instruments that are not measured at fair value, the related interest is included within interest income or interest expense, as applicable. Interest income and interest expense also includes the effect of derivatives that qualify for hedge accounting where applicable.

Interest income on loans and securities include the amortization and accretion of purchase premiums and discounts, as well as net deferred fees and costs on loans. These amounts are deferred in loans and investment securities, respectively, and recognized on a level-yield basis.

Refer to Notes 5, 10, 11, 12, and 20 for further information on accounting for interest income and interest expense related to hedge accounting, investment securities, securities financing activities (i.e., securities purchased or sold under resale or repurchase agreements; securities borrowed; and securities loaned), loans and long-term debt, respectively.

The following table presents the components of interest income and interest expense:

Year ended December 31, (in millions)	2024	2023	2022
Interest income			
Loans	\$ 92,353 ^(d)	\$ 83,384 ^(d)	\$ 52,736
Taxable securities	21,947	17,390	10,372
Non-taxable securities ^(a)	1,197	1,336	975
Total investment securities	23,144 ^(d)	18,726 ^(d)	11,347
Trading assets - debt instruments	20,327	15,950	9,053
Federal funds sold and securities purchased under resale agreements	18,299	15,079	4,632
Securities borrowed	9,208	7,983	2,237
Deposits with banks	22,297	21,797	9,039
All other interest-earning assets ^(b)	8,305	7,669	3,763
Total interest income	\$193,933	\$170,588	\$ 92,807
Interest expense			
Interest bearing deposits	\$ 49,559	\$ 40,016	\$ 10,082
Federal funds purchased and securities loaned or sold under repurchase agreements	19,149	13,259	3,721
Short-term borrowings	2,101	1,894	747
Trading liabilities - debt and all other interest-bearing liabilities ^(c)	10,238	9,396	3,246
Long-term debt	18,920	15,803	8,075
Beneficial interest issued by consolidated VIEs	1,383	953	226
Total interest expense	\$101,350	\$ 81,321	\$ 26,097
Net interest income	\$ 92,583	\$ 89,267	\$ 66,710
Provision for credit losses	10,678	9,320	6,389
Net interest income after provision for credit losses	\$ 81,905	\$ 79,947	\$ 60,321

(a) Represents securities that are tax-exempt for U.S. federal income tax purposes.

(b) Includes interest earned on brokerage-related held-for-investment customer receivables, which are classified in accrued interest and accounts receivable, and all other interest-earning assets, which are classified in other assets on the Consolidated balance sheets.

(c) All other interest-bearing liabilities includes interest expense on brokerage-related customer payables.

(d) Includes the accretion of the purchase discount on certain acquired loans and investment securities associated with First Republic. Refer to Note 34 for additional information.

Note 8 – Pension and other postretirement employee benefit plans

The Firm has various defined benefit pension plans and OPEB plans that provide benefits to its employees in the U.S. and certain non-U.S. locations. Substantially all the defined benefit pension plans are closed to new participants. The principal defined benefit pension plan in the U.S., which covered substantially all U.S. employees, was closed to new participants and frozen for existing participants on January 1, 2020, (and January 1, 2019 for new hires on or after December 2, 2017). Interest credits continue to accrue to participants' accounts based on their accumulated balances.

The Firm maintains funded and unfunded postretirement benefit plans that provide medical and life insurance for certain eligible employees and retirees

The following table presents the pretax benefit obligations, plan assets, the net funded status, and the amounts recorded in AOCI on the Consolidated balance sheets for the Firm's significant defined benefit pension and OPEB plans.

As of or for the year ended December 31, (in millions)	2024	2023
Projected benefit obligations	\$ (14,459)	\$ (14,740)
Fair value of plan assets	22,201	22,013
Net funded status	7,742	7,273
Accumulated other comprehensive income/(loss)	(1,649)	(1,517)

The weighted-average discount rate used to value the benefit obligations as of December 31, 2024 and 2023, was 5.49% and 5.16%, respectively.

Gains and losses

Gains or losses resulting from changes in the benefit obligation and the fair value of plan assets are recorded in OCI. Amortization of net gains or losses are recognized as part of the net periodic benefit cost over subsequent periods, if, as of the beginning of the year, the net gain or loss exceeds 10% of the greater of the projected benefit obligation or the fair value of the plan assets. Amortization is generally over the average expected remaining lifetime of plan participants, given the frozen status of most plans. For the year ended

December 31, 2024, the net loss was attributable to lower than expected returns on plan assets, partially offset by projected benefit obligation net gains primarily related to changes in the discount rate. For the year ended December 31, 2023, the net gain was attributable to market-driven increases in the fair value of plan assets, partially offset by changes in the discount rate and interest crediting rate.

The following table presents the net periodic benefit costs reported in the Consolidated statements of income for the Firm's defined benefit pension, defined contribution and OPEB plans, and in other comprehensive income for the defined benefit pension and OPEB plans.

Year ended December 31, (in millions)	2024	2023	2022
Total net periodic defined benefit plan cost/(credit) ^(a)	\$ (462)	\$ (393)	\$ (192) ^(b)
Total defined contribution plans	1,733	1,609	1,408
Total pension and OPEB cost included in noninterest expense	\$ 1,271	\$ 1,216	\$ 1,216
Total recognized in other comprehensive (income)/loss	\$ 131	\$ (421)	\$ 1,459

(a) The service cost component of net periodic defined benefit cost is reported in compensation expense; all other components of net periodic defined benefit costs are reported in other expense in the Consolidated statements of income.

(b) Includes pension settlement losses of \$92 million for the year ended December 31, 2022.

The following table presents the weighted-average actuarial assumptions used to determine the net periodic benefit costs for the defined benefit pension and OPEB plans.

Year ended December 31,	2024	2023	2022
Discount rate	5.16 %	5.14 %	2.54 %
Expected long-term rate of return on plan assets	6.15 %	5.74 %	3.68 %

Plan assumptions

The Firm's expected long-term rate of return is a blended weighted average, by asset allocation of the projected long-term returns for the various asset classes, taking into consideration local market conditions and the specific allocation of plan assets. Returns on asset classes are developed using a forward-looking approach and are not strictly based on historical returns, with consideration given to current market conditions and the portfolio mix of each plan.

The discount rates used in determining the benefit obligations are generally provided by the Firm's actuaries, with the Firm's principal defined benefit pension plan using a rate that was selected by reference to the yields on portfolios of bonds with maturity dates and coupons that closely match the plan's projected annual cash flows.

Investment strategy and asset allocation

The assets of the Firm's defined benefit pension plans are held in various trusts and are invested in well-diversified portfolios of equity and fixed income securities, cash and cash equivalents, and alternative investments. The Firm regularly reviews the asset allocations and asset managers, as well as other factors that could impact the portfolios, which are rebalanced when deemed necessary. As of December 31, 2024, the approved asset allocation ranges by asset class for the Firm's principal defined benefit plan are 41-100% debt securities, 0-40% equity securities, 0-1% real estate, and 0-8% alternatives.

Assets held by the Firm's defined benefit pension and OPEB plans do not include securities issued by JPMorganChase or its affiliates, except through indirect exposures through investments in exchange traded funds, mutual funds and collective investment funds managed by third-parties. The defined benefit pension and OPEB plans hold investments that are sponsored or managed by affiliates of JPMorganChase in the amount of \$1.8 billion as of both December 31, 2024 and 2023.

Fair value measurement of the plans' assets and liabilities

Refer to Note 2 for information on fair value measurements, including descriptions of level 1, 2, and 3 of the fair value hierarchy and the valuation methods employed by the Firm.

Defined benefit pension and OPEB plans assets and liabilities measured at fair value

December 31, (in millions)	2024				2023			
	Level 1 ^(a)	Level 2 ^(b)	Level 3 ^(c)	Total fair value	Level 1 ^(a)	Level 2 ^(b)	Level 3 ^(c)	Total fair value
Assets measured at fair value classified in the fair value hierarchy	\$ 6,910	\$ 9,693	\$ 3,956	\$ 20,559	\$ 6,521	\$ 10,713	\$ 3,124	\$ 20,358
Assets measured at fair value using NAV as a practical expedient				2,101				2,097
Net defined benefit pension plan payables				(459)				(442)
Total fair value of plan assets				\$ 22,201				\$ 22,013

(a) Consists predominantly of equity securities, U.S. federal, state, and local and non-U.S. government debt securities, and cash equivalents.

(b) Consists of corporate debt securities, fund investments, mortgage-backed securities, and U.S. federal, state, and local and non-U.S. government debt securities.

(c) Consists predominantly of corporate-owned life insurance policies.

Notes to consolidated financial statements

Changes in level 3 fair value measurements using significant unobservable inputs

Investments classified in level 3 of the fair value hierarchy increased in 2024 to \$4.0 billion, due to \$536 million of transfers in and \$415 million in unrealized gains, partially offset by \$123 million in settlements. The net increase in 2023 was due to \$400 million in unrealized gains and \$173 million of transfers in, partially offset by \$59 million in settlements.

Estimated future benefit payments

The following table presents benefit payments expected to be paid for the defined benefit pension and OPEB plans for the years indicated.

Year ended December 31, (in millions)		
2025	\$	1,186
2026		1,155
2027		1,134
2028		1,095
2029		1,093
Years 2030–2034		5,229

Note 9 – Employee share-based incentives

Employee share-based awards

In 2024, 2023 and 2022, JPMorganChase granted long-term share-based awards to certain employees under its LTIP, as amended and restated effective May 18, 2021, and subsequently amended effective May 21, 2024. Under the terms of the LTIP, as of December 31, 2024, 81 million shares of common stock were available for issuance through May 2028. The LTIP is the only active plan under which the Firm is currently granting share-based incentive awards. In the following discussion, the LTIP constitutes the Firm's share-based incentive plans.

RSUs are awarded at no cost to the recipient upon their grant. Generally, RSUs are granted annually and vest at a rate of 50% after two years and 50% after three years and are converted into shares of common stock as of the vesting date. In addition, RSUs typically include full-career eligibility provisions, which allow employees to continue to vest upon voluntary termination based on age and/or service-related requirements, subject to post-employment and other restrictions. All RSU awards are subject to forfeiture until vested and contain clawback provisions that may result in cancellation under certain specified circumstances. Predominantly all RSUs entitle the recipient to receive cash payments equivalent to any dividends paid on the underlying common stock during the period the RSUs are outstanding.

Performance share units ("PSUs") are granted annually, and approved by the Firm's Board of Directors, to members of the Firm's Operating Committee under the variable compensation program. PSUs are subject to the Firm's achievement of specified performance criteria over a three-year period. The number of awards that vest can range from zero to 150% of the grant amount. In addition, dividends that accrue during the vesting period are reinvested in dividend equivalent share units. PSUs and the related dividend equivalent share units are converted into shares of common stock after vesting.

Once the PSUs and dividend equivalent share units have vested, the shares of common stock that are delivered, after applicable tax withholding, must be retained for an additional holding period, for a total combined vesting and holding period of approximately five to eight years from the grant date depending on regulations in certain countries.

Under the LTI Plans, stock appreciation rights ("SARs") were granted with an exercise price equal to the fair value of JPMorganChase's common stock on the grant date. SARs expire ten years after the grant date. There were no grants of SARs in 2024, 2023 or 2022.

The Firm separately recognizes compensation expense for each tranche of each award, net of estimated forfeitures, as if it were a separate award with its own vesting date. Generally, for each tranche granted, compensation expense is recognized on a straight-line basis from the grant date until the vesting date of the respective tranche, provided that the employees will not become full-career eligible during the vesting period. For awards with full-career eligibility provisions and awards granted with no future substantive service requirement, the Firm accrues the estimated value of awards expected to be awarded to employees as of the grant date without giving consideration to the impact of post-employment restrictions. For each tranche granted to employees who will become full-career eligible during the vesting period, compensation expense is recognized on a straight-line basis from the grant date until the earlier of the employee's full-career eligibility date or the vesting date of the respective tranche.

The Firm's policy for issuing shares upon settlement of employee share-based incentive awards is to issue either new shares of common stock or treasury shares. During 2024, 2023 and 2022, the Firm settled all of its employee share-based awards by issuing treasury shares.

Refer to Note 23 for further information on the classification of share-based awards for purposes of calculating earnings per share.

Notes to consolidated financial statements

RSUs, PSUs and SARs activity

Generally, compensation expense for RSUs and PSUs is measured based on the number of units granted multiplied by the stock price at the grant date, and for SARs, is measured at the grant date using the Black-Scholes valuation model. Compensation expense for these awards is recognized in net income as described previously. The following table summarizes JPMorganChase's RSUs, PSUs and SARs activity for 2024.

Year ended December 31, 2024 (in thousands, except weighted-average data, and where otherwise stated)	RSUs/PSUs		SARs			
	Number of units	Weighted-average grant date fair value	Number of awards	Weighted-average exercise price	Weighted-average remaining contractual life (in years)	Aggregate intrinsic value
Outstanding, January 1	52,243	\$ 141.31	2,250	\$ 152.19		
Granted	20,020	166.74	—	—		
Exercised or vested	(19,542)	143.02	—	—		
Forfeited	(2,112)	147.41	—	—		
Canceled	NA	NA	—	—		
Outstanding, December 31	50,609	\$ 150.41	2,250	\$ 152.19	6.7	\$ 198,113
Exercisable, December 31	NA	NA	—	—	—	—

The total fair value of RSUs and PSUs that vested during the years ended December 31, 2024, 2023 and 2022, was \$3.5 billion, \$2.5 billion and \$3.2 billion, respectively. There were no SARs exercised in 2024. The total intrinsic value of SARs exercised during the years ended December 31, 2023 and 2022, was \$24 million and \$75 million, respectively.

Compensation expense

The Firm recognized the following noncash compensation expense related to its various employee share-based incentive plans in its Consolidated statements of income.

Year ended December 31, (in millions)	2024	2023	2022
Cost of prior grants of RSUs, PSUs and SARs that are amortized over their applicable vesting periods	\$ 1,622	\$ 1,510	\$ 1,253
Accrual of estimated costs of share-based awards to be granted in future periods, predominantly those to full-career eligible employees	1,882	1,607	1,541
Total noncash compensation expense related to employee share-based incentive plans	\$3,504	\$ 3,117	\$ 2,794

At December 31, 2024, approximately \$963 million (pretax) of compensation expense related to unvested awards had not yet been charged to net income. That cost is expected to be amortized into compensation expense over a weighted-average period of 1.6 years. The Firm does not capitalize any compensation expense related to share-based compensation awards to employees.

Tax benefits

Income tax benefits (including tax benefits from dividends or dividend equivalents) related to share-based incentive arrangements recognized in the Firm's Consolidated statements of income for the years ended December 31, 2024, 2023 and 2022, were \$1.0 billion, \$836 million and \$901 million, respectively.

Note 10 – Investment securities

Investment securities consist of debt securities that are classified as AFS or HTM. Debt securities classified as trading assets are discussed in Note 2. Predominantly all of the Firm's AFS and HTM securities are held by Treasury and CIO in connection with its asset-liability management activities.

AFS securities are carried at fair value on the Consolidated balance sheets. Unrealized gains and losses, after any applicable hedge accounting adjustments or allowance for credit losses, are reported in AOCI. The specific identification method is used to determine realized gains and losses on AFS securities, which are included in investment securities gains/(losses) on the Consolidated statements of income. HTM securities, which the Firm has the intent and ability to hold until maturity, are carried at amortized cost, net of allowance for credit losses, on the Consolidated balance sheets.

For both AFS and HTM securities, purchase discounts or premiums are generally amortized into interest income on a level-yield basis over the contractual life of the security. However, premiums on certain callable debt securities are amortized to the earliest call date.

Notes to consolidated financial statements

The amortized costs and estimated fair values of the investment securities portfolio were as follows for the dates indicated.

December 31, (in millions)	2024				2023			
	Amortized cost ^{(d)(e)}	Gross unrealized gains	Gross unrealized losses	Fair value	Amortized cost ^{(d)(e)}	Gross unrealized gains	Gross unrealized losses	Fair value
Available-for-sale securities								
Mortgage-backed securities:								
U.S. GSEs and government agencies	\$ 95,671	\$ 251	\$ 4,029	\$ 91,893	\$ 88,377	\$ 870	\$ 4,077	\$ 85,170
Residential:								
U.S.	4,242	16	50	4,208	2,086	10	68	2,028
Non-U.S.	600	3	—	603	1,608	4	1	1,611
Commercial	4,115	20	70	4,065	2,930	12	139	2,803
Total mortgage-backed securities	104,628	290	4,149	100,769	95,001	896	4,285	91,612
U.S. Treasury and government agencies	235,495	545	1,261	234,779	58,051	276	522	57,805
Obligations of U.S. states and municipalities	18,337	110	534	17,913	21,243	390	266	21,367
Non-U.S. government debt securities	36,655	94	504	36,245	21,387	254	359	21,282
Corporate debt securities	71	—	1	70	128	—	28	100
Asset-backed securities:								
Collateralized loan obligations	14,887	59	3	14,943	6,769	11	28	6,752
Other	2,125	17	9	2,133	2,804	8	26	2,786
Unallocated portfolio layer fair value basis adjustments ^(a)	(1,153)	—	(1,153)	NA	73	(73)	—	NA
Total available-for-sale securities	411,045	1,115	5,308	406,852	205,456	1,762	5,514	201,704
Held-to-maturity securities^(b)								
Mortgage-backed securities:								
U.S. GSEs and government agencies	97,177	6	13,531	83,652	105,614	39	11,643	94,010
U.S. Residential	8,605	4	904	7,705	9,709	4	970	8,743
Commercial	8,817	24	389	8,452	10,534	13	581	9,966
Total mortgage-backed securities	114,599	34	14,824	99,809	125,857	56	13,194	112,719
U.S. Treasury and government agencies	108,632	—	11,212	97,420	173,666	—	13,074	160,592
Obligations of U.S. states and municipalities	9,310	32	631	8,711	9,945	74	591	9,428
Asset-backed securities:								
Collateralized loan obligations	40,573	84	14	40,643	58,565	47	352	58,260
Other	1,354	2	39	1,317	1,815	1	61	1,755
Total held-to-maturity securities^(c)	274,468	152	26,720	247,900	369,848	178	27,272	342,754
Total investment securities, net of allowance for credit losses	\$ 685,513	\$ 1,267	\$ 32,028	\$ 654,752	\$ 575,304	\$ 1,940	\$ 32,786	\$ 544,458

- (a) Represents the amount of portfolio layer method basis adjustments related to AFS securities hedged in a closed portfolio. Under U.S. GAAP portfolio layer method basis adjustments are not allocated to individual securities, however the amounts impact the unrealized gains or losses in the table for the types of securities being hedged. Refer to Note 1 and Note 5 for additional information.
- (b) The Firm purchased \$4.7 billion, \$4.1 billion and \$33.7 billion of HTM securities for the years ended December 31, 2024, 2023 and 2022, respectively.
- (c) Effective January 1, 2023, the Firm adopted the portfolio layer method hedge accounting guidance which permitted a transfer of HTM securities to AFS upon adoption. The Firm transferred obligations of U.S. states and municipalities with a carrying value of \$7.1 billion resulting in the recognition of \$38 million net pre-tax unrealized losses in AOCI. This transfer was a non-cash transaction. Refer to Note 24 for additional information.
- (d) The amortized cost of investment securities is reported net of allowance for credit losses of \$152 million, \$128 million and \$96 million at December 31, 2024, 2023 and 2022, respectively.
- (e) Excludes \$3.7 billion and \$2.8 billion of accrued interest receivable at December 31, 2024 and 2023, respectively, included in accrued interest and accounts receivable on the Consolidated balance sheets. The Firm generally does not recognize an allowance for credit losses on accrued interest receivable, consistent with its policy to write them off no later than 90 days past due by reversing interest income. The Firm did not reverse through interest income any accrued interest receivable for the years ended December 31, 2024 and 2023.

At December 31, 2024, the investment securities portfolio consisted of debt securities with an average credit rating of AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings). Risk ratings are used to identify the credit quality of securities and differentiate risk within the portfolio. The Firm's internal risk ratings generally align with the qualitative characteristics (e.g., borrower capacity to meet financial commitments and vulnerability to changes in the economic environment) defined by S&P and

Moody's, however the quantitative characteristics (e.g., probability of default ("PD") and loss given default ("LGD")) may differ as they reflect internal historical experiences and assumptions. Risk ratings are assigned at acquisition, reviewed on a regular and ongoing basis by Credit Risk Management and adjusted as necessary over the life of the investment for updated information affecting the issuer's ability to fulfill its obligations.

AFS securities impairment

The following tables present the fair value and gross unrealized losses by aging category for AFS securities at December 31, 2024 and 2023. The tables exclude U.S. Treasury and government agency securities and U.S. GSE and government agency MBS with unrealized losses of \$5.3 billion and \$4.6 billion, at December 31, 2024 and 2023, respectively; changes in the value of these securities are generally driven by changes in interest rates rather than changes in their credit profile given the explicit or implicit guarantees provided by the U.S. government.

Year ended December 31, 2024 (in millions)	Available-for-sale securities with gross unrealized losses											
	Less than 12 months		12 months or more		Total fair value	Total gross unrealized losses						
	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses								
Available-for-sale securities												
Mortgage-backed securities:												
Residential:												
U.S.	\$	1,505	\$	6	\$	925	\$	44	\$	2,430	\$	50
Non-U.S.		—		—		30		—		30		—
Commercial		763		8		1,184		62		1,947		70
Total mortgage-backed securities		2,268		14		2,139		106		4,407		120
Obligations of U.S. states and municipalities		10,037		233		2,412		301		12,449		534
Non-U.S. government debt securities		14,234		234		4,184		270		18,418		504
Corporate debt securities		9		—		30		1		39		1
Asset-backed securities:												
Collateralized loan obligations		2		—		375		3		377		3
Other		214		1		200		8		414		9
Total available-for-sale securities with gross unrealized losses	\$	26,764	\$	482	\$	9,340	\$	689	\$	36,104	\$	1,171

Year ended December 31, 2023 (in millions)	Available-for-sale securities with gross unrealized losses											
	Less than 12 months		12 months or more		Total fair value	Total gross unrealized losses						
	Fair value	Gross unrealized losses	Fair value	Gross unrealized losses								
Available-for-sale securities												
Mortgage-backed securities:												
Residential:												
U.S.	\$	81	\$	—	\$	1,160	\$	68	\$	1,241	\$	68
Non-U.S.		—		—		722		1		722		1
Commercial		228		3		1,775		136		2,003		139
Total mortgage-backed securities		309		3		3,657		205		3,966		208
Obligations of U.S. states and municipalities		2,134		20		2,278		246		4,412		266
Non-U.S. government debt securities		7,145		23		4,987		336		12,132		359
Corporate debt securities		9		—		79		28		88		28
Asset-backed securities:												
Collateralized loan obligations		932		2		3,744		26		4,676		28
Other		208		1		1,288		25		1,496		26
Total available-for-sale securities with gross unrealized losses	\$	10,737	\$	49	\$	16,033	\$	866	\$	26,770	\$	915

Notes to consolidated financial statements

AFS securities are considered impaired if the fair value is less than the amortized cost.

The Firm recognizes impairment losses in earnings if the Firm has the intent to sell the debt security, or if it is more likely than not that the Firm will be required to sell the debt security before recovery of its amortized cost. In these circumstances the impairment loss is recognized in investment securities gains/(losses) in the Consolidated Statements of Income and is equal to the full difference between the amortized cost (net of allowance if applicable) and the fair value of the security.

For impaired debt securities that the Firm has the intent and ability to hold, the securities are evaluated to determine if a credit loss exists. If it is determined that a credit loss exists, that loss is recognized as an allowance for credit losses through the provision for credit losses in the Consolidated Statements of Income, limited by the amount of impairment. Any impairment on debt securities that the Firm has the intent and ability to hold not due to credit losses is recorded in OCI.

Factors considered in evaluating credit losses include adverse conditions specifically related to the industry, geographic area or financial condition of the issuer or underlying collateral of a security; and payment structure of the security.

When assessing securities issued in a securitization for credit losses, the Firm estimates cash flows considering relevant market and economic data, underlying loan-level data, and structural features of the securitization, such as subordination, excess spread, overcollateralization or other forms of credit enhancement, and compares the losses projected for the underlying collateral ("pool losses") against the level of credit enhancement in the securitization structure to determine whether these features are sufficient to absorb the pool losses, or whether a credit loss exists.

For beneficial interests in securitizations that are rated below "AA" at their acquisition, or that can be contractually prepaid or otherwise settled in such a way that the Firm would not recover substantially all of its recorded investment, the Firm evaluates impairment for credit losses when there is an adverse change in expected cash flows.

HTM securities – credit risk

Allowance for credit losses

The allowance for credit losses on HTM securities represents expected credit losses over the remaining expected life of the securities.

The allowance for credit losses on HTM obligations of U.S. states and municipalities and commercial mortgage-backed securities is calculated by applying statistical credit loss factors (estimated PD and LGD)

to the amortized cost. The credit loss factors are derived using a weighted average of five internally developed eight-quarter macroeconomic scenarios, followed by a single year straight-line interpolation to revert to long run historical information for periods beyond the forecast period. Refer to Note 13 for further information on the eight-quarter macroeconomic forecast.

The allowance for credit losses on HTM collateralized loan obligations and U.S. residential mortgage-backed securities is calculated as the difference between the amortized cost and the present value of the cash flows expected to be collected, discounted at the security's effective interest rate. These cash flow estimates are developed based on expectations of underlying collateral performance derived using the eight-quarter macroeconomic forecast and the single year straight-line interpolation, as well as considering the structural features of the security.

The application of different inputs and assumptions into the calculation of the allowance for credit losses is subject to significant management judgment, and emphasizing one input or assumption over another, or considering other inputs or assumptions, could affect the estimate of the allowance for credit losses on HTM securities.

Credit quality indicator

The primary credit quality indicator for HTM securities is the risk rating assigned to each security. At both December 31, 2024 and 2023, all HTM securities were rated investment grade and were current and accruing, with approximately 99% rated at least AA+ (based upon external ratings where available, and where not available, based primarily upon internal risk ratings).

Allowance for credit losses on investment securities

The allowance for credit losses on investment securities was \$152 million, \$128 million and \$96 million as of December 31, 2024, 2023 and 2022, respectively, which included a cumulative-effect adjustment to retained earnings related to the transfer of HTM securities to AFS for the year ended December 31, 2023.

Selected impacts of investment securities on the Consolidated statements of income

Year ended December 31, (in millions)	2024	2023	2022
Realized gains	\$ 593	\$ 622	\$ 198
Realized losses	(1,614)	(3,802)	(2,578)
Investment securities losses	\$(1,021)	\$(3,180)	\$(2,380)
Provision for credit losses	\$ 24	\$ 38	\$ 54

Contractual maturities and yields

The following table presents the amortized cost and estimated fair value at December 31, 2024, of JPMorganChase's investment securities portfolio by contractual maturity.

By remaining maturity December 31, 2024 (in millions)	Due in one year or less	Due after one year through five years	Due after five years through 10 years	Due after 10 years ^(c)	Total
Available-for-sale securities					
Mortgage-backed securities					
Amortized cost	\$ 132	\$ 9,410	\$ 4,059	\$ 91,027	\$ 104,628
Fair value	130	9,345	4,041	87,253	100,769
Average yield ^(a)	3.32 %	4.58 %	5.25 %	4.75 %	4.75 %
U.S. Treasury and government agencies					
Amortized cost	\$ 20,685	\$ 169,162	\$ 38,667	\$ 6,981	\$ 235,495
Fair value	20,730	169,145	38,619	6,285	234,779
Average yield ^(a)	4.50 %	4.57 %	4.96 %	5.53 %	4.65 %
Obligations of U.S. states and municipalities					
Amortized cost	\$ 4	\$ 16	\$ 92	\$ 18,225	\$ 18,337
Fair value	4	16	90	17,803	17,913
Average yield ^(a)	1.59 %	3.95 %	4.46 %	5.32 %	5.32 %
Non-U.S. government debt securities					
Amortized cost	\$ 13,331	\$ 11,769	\$ 7,609	\$ 3,946	\$ 36,655
Fair value	13,327	11,701	7,403	3,814	36,245
Average yield ^(a)	4.24 %	4.32 %	2.82 %	4.15 %	3.96 %
Corporate debt securities					
Amortized cost	\$ 106	\$ 9	\$ 5	\$ —	\$ 120
Fair value	57	9	4	—	70
Average yield ^(a)	14.09 %	4.06 %	4.19 %	— %	12.92 %
Asset-backed securities					
Amortized cost	\$ —	\$ 368	\$ 1,609	\$ 15,035	\$ 17,012
Fair value	—	370	1,619	15,087	17,076
Average yield ^(a)	— %	6.14 %	6.04 %	5.96 %	5.97 %
Total available-for-sale securities					
Amortized cost ^(b)	\$ 34,258	\$ 190,734	\$ 52,041	\$ 135,214	\$ 412,247
Fair value	34,248	190,586	51,776	130,242	406,852
Average yield ^(a)	4.42 %	4.56 %	4.70 %	4.98 %	4.70 %
Held-to-maturity securities					
Mortgage-backed securities					
Amortized cost	\$ 104	\$ 7,994	\$ 6,077	\$ 100,495	\$ 114,670
Fair value	101	7,453	5,352	86,903	99,809
Average yield ^(a)	0.97 %	2.63 %	2.67 %	2.95 %	2.92 %
U.S. Treasury and government agencies					
Amortized cost	\$ 20,083	\$ 40,497	\$ 48,052	\$ —	\$ 108,632
Fair value	19,500	37,715	40,205	—	97,420
Average yield ^(a)	0.43 %	1.23 %	1.25 %	— %	1.09 %
Obligations of U.S. states and municipalities					
Amortized cost	\$ —	\$ 9	\$ 307	\$ 9,026	\$ 9,342
Fair value	—	9	276	8,426	8,711
Average yield ^(a)	— %	4.76 %	3.25 %	3.97 %	3.95 %
Asset-backed securities					
Amortized cost	\$ —	\$ 134	\$ 22,433	\$ 19,360	\$ 41,927
Fair value	—	134	22,466	19,360	41,960
Average yield ^(a)	— %	5.94 %	5.57 %	5.73 %	5.65 %
Total held-to-maturity securities					
Amortized cost ^(d)	\$ 20,187	\$ 48,634	\$ 76,869	\$ 128,881	\$ 274,571
Fair value	19,601	45,311	68,299	114,689	247,900
Average yield ^(a)	0.43 %	1.47 %	2.63 %	3.44 %	2.64 %

- (a) Average yield is computed using the effective yield of each security owned at the end of the period, weighted based on the amortized cost of each security. The effective yield considers the contractual coupon, amortization of premiums and accretion of discounts, and the effect of related hedging derivatives, including closed portfolio hedges. Taxable-equivalent amounts are used where applicable. The effective yield excludes unscheduled principal prepayments; and accordingly, actual maturities of securities may differ from their contractual or expected maturities as certain securities may be prepaid. However, for certain callable debt securities, the average yield is calculated to the earliest call date.
- (b) For purposes of this table, the amortized cost of available-for-sale securities excludes the allowance for credit losses of \$49 million and the portfolio layer fair value hedge basis adjustments of \$(1.2) billion at December 31, 2024. The amortized cost of held-to-maturity securities also excludes the allowance for credit losses of \$103 million at December 31, 2024.
- (c) Substantially all of the Firm's U.S. residential MBS and collateralized mortgage obligations are due in 10 years or more, based on contractual maturity. The estimated weighted-average life, which reflects anticipated future prepayments, is approximately eight years for agency residential MBS, six years for agency residential collateralized mortgage obligations, and five years for nonagency residential collateralized mortgage obligations.

Note 11 – Securities financing activities

JPMorganChase enters into resale, repurchase, securities borrowed and securities loaned agreements (collectively, “securities financing agreements”) primarily to finance the Firm’s inventory positions, acquire securities to cover short sales, accommodate customers’ financing needs, settle other securities obligations and to deploy the Firm’s excess cash.

Securities financing agreements are treated as collateralized financings on the Firm’s Consolidated balance sheets. Where appropriate under applicable accounting guidance, securities financing agreements with the same counterparty are reported on a net basis. Refer to Note 1 for further discussion of the offsetting of assets and liabilities. Fees received and paid in connection with securities financing agreements are recorded over the life of the agreement in interest income and interest expense on the Consolidated statements of income.

The Firm has elected the fair value option for certain securities financing agreements. Refer to Note 3 for further information regarding the fair value option. The securities financing agreements for which the fair value option has been elected are reported within securities purchased under resale agreements, securities loaned or sold under repurchase agreements, and securities borrowed on the Consolidated balance sheets. Generally, for agreements carried at fair value, current-period interest accruals are recorded within interest income and interest expense, with changes in fair value reported in principal transactions revenue. However, for financial instruments containing embedded derivatives that would be separately accounted for in accordance with accounting guidance for hybrid instruments, all changes in fair value, including any interest elements, are reported in principal transactions revenue.

Securities financing agreements not elected under the fair value option are measured at amortized cost. As a result of the Firm’s credit risk mitigation practices described below, the Firm did not hold any allowance for credit losses with respect to resale and securities borrowed arrangements as of December 31, 2024 and 2023.

Credit risk mitigation practices

Securities financing agreements expose the Firm primarily to credit and liquidity risk. To manage these risks, the Firm monitors the value of the underlying securities (predominantly high-quality securities collateral, including government-issued debt and U.S. GSEs and government agencies MBS) that it has received from or provided to its counterparties compared to the value of cash proceeds and exchanged collateral, and either requests additional collateral or returns securities or collateral when appropriate. Margin levels are initially established based upon the counterparty, the type of underlying securities, and the permissible collateral, and are monitored on an ongoing basis.

In resale and securities borrowed agreements, the Firm is exposed to credit risk to the extent that the value of the securities received is less than initial cash principal advanced and any collateral amounts exchanged. In repurchase and securities loaned agreements, credit risk exposure arises to the extent that the value of underlying securities advanced exceeds the value of the initial cash principal received, and any collateral amounts exchanged.

Additionally, the Firm typically enters into master netting agreements and other similar arrangements with its counterparties, which provide for the right to liquidate the underlying securities and any collateral amounts exchanged in the event of a counterparty default. It is also the Firm’s policy to take possession, where possible, of the securities underlying resale and securities borrowed agreements. Refer to Note 29 for further information regarding assets pledged and collateral received in securities financing agreements.

The table below summarizes the gross and net amounts of the Firm's securities financing agreements, as of December 31, 2024 and 2023. When the Firm has obtained an appropriate legal opinion with respect to a master netting agreement with a counterparty and where other relevant netting criteria under U.S. GAAP are met, the Firm nets, on the Consolidated balance sheets, the balances outstanding under its securities financing agreements with the same counterparty. In addition, the Firm exchanges securities and/or cash collateral with its counterparty to reduce the economic exposure with the counterparty, but such collateral is not eligible for net Consolidated balance sheet presentation. Where the Firm has obtained an appropriate legal opinion with respect to the counterparty master netting agreement, such

collateral, along with securities financing balances that do not meet all these relevant netting criteria under U.S. GAAP, is presented in the table below as "Amounts not nettable on the Consolidated balance sheets," and reduces the "Net amounts" presented. Where a legal opinion has not been either sought or obtained, the securities financing balances are presented gross in the "Net amounts" below. In transactions where the Firm is acting as the lender in a securities-for-securities lending agreement and receives securities that can be pledged or sold as collateral, the Firm recognizes the securities received at fair value within other assets and the obligation to return those securities within accounts payable and other liabilities on the Consolidated balance sheets.

	December 31, 2024					
(in millions)	Gross amounts	Amounts netted on the Consolidated balance sheets	Amounts presented on the Consolidated balance sheets	Amounts not nettable on the Consolidated balance sheets ^(b)	Net amounts ^(c)	
Assets						
Securities purchased under resale agreements	\$ 607,154	\$ (312,183)	\$ 294,971	\$ (282,220)	\$	12,751
Securities borrowed	267,917	(48,371)	219,546	(170,702)		48,844
Liabilities						
Securities sold under repurchase agreements	\$ 603,683	\$ (312,183)	\$ 291,500	\$ (249,763)	\$	41,737
Securities loaned and other ^(a)	58,989	(48,371)	10,618	(10,557)		61
	December 31, 2023					
(in millions)	Gross amounts	Amounts netted on the Consolidated balance sheets	Amounts presented on the Consolidated balance sheets	Amounts not nettable on the Consolidated balance sheets ^(b)	Net amounts ^(c)	
Assets						
Securities purchased under resale agreements	\$ 523,308	\$ (247,181)	\$ 276,127	\$ (267,582)	\$	8,545
Securities borrowed	244,046	(43,610)	200,436	(144,543)		55,893
Liabilities						
Securities sold under repurchase agreements	\$ 459,985	\$ (247,181)	\$ 212,804	\$ (182,011)	\$	30,793
Securities loaned and other ^(a)	52,142	(43,610)	8,532	(8,501)		31

(a) Includes securities-for-securities lending agreements of \$5.9 billion and \$5.6 billion at December 31, 2024 and 2023, respectively, accounted for at fair value, where the Firm is acting as lender.

(b) In some cases, collateral exchanged with a counterparty exceeds the net asset or liability balance with that counterparty. In such cases, the amounts reported in this column are limited to the related net asset or liability with that counterparty.

(c) Includes securities financing agreements that provide collateral rights, but where an appropriate legal opinion with respect to the master netting agreement has not been either sought or obtained. At December 31, 2024 and 2023, included \$8.7 billion and \$7.1 billion, respectively, of securities purchased under resale agreements; \$42.9 billion and \$50.7 billion, respectively, of securities borrowed; \$40.9 billion and \$30.0 billion, respectively, of securities sold under repurchase agreements; and securities loaned and other which were not material.

Notes to consolidated financial statements

The tables below present as of December 31, 2024 and 2023 the types of financial assets pledged in securities financing agreements and the remaining contractual maturity of the securities financing agreements.

December 31, (in millions)	Gross liability balance			
	2024		2023	
	Securities sold under repurchase agreements	Securities loaned and other	Securities sold under repurchase agreements	Securities loaned and other
Mortgage-backed securities:				
U.S. GSEs and government agencies	\$ 82,645	\$ —	\$ 71,064	\$ —
Residential - nonagency	2,610	—	2,292	—
Commercial - nonagency	2,344	—	2,669	—
U.S. Treasury, GSEs and government agencies	300,022	759	216,467	1,034
Obligations of U.S. states and municipalities	1,872	—	2,323	—
Non-U.S. government debt	117,614	1,852	97,400	1,455
Corporate debt securities	44,495	4,033	39,247	2,025
Asset-backed securities	4,619	—	2,703	—
Equity securities	47,462	52,345	25,820	47,628
Total	\$ 603,683	\$ 58,989	\$ 459,985	\$ 52,142

December 31, 2024 (in millions)	Remaining contractual maturity of the agreements				
	Overnight and continuous	Up to 30 days	30 – 90 days	Greater than 90 days	Total
Total securities sold under repurchase agreements	\$ 308,392	\$ 171,346	\$ 19,932	\$ 104,013	\$ 603,683
Total securities loaned and other	54,066	1,463	1	3,459	58,989

December 31, 2023 (in millions)	Remaining contractual maturity of the agreements				
	Overnight and continuous	Up to 30 days	30 – 90 days	Greater than 90 days	Total
Total securities sold under repurchase agreements	\$ 259,048	\$ 102,941	\$ 20,960	\$ 77,036	\$ 459,985
Total securities loaned and other	49,610	1,544	—	988	52,142

Transfers not qualifying for sale accounting

At December 31, 2024 and 2023, the Firm held \$805 million and \$505 million, respectively, of financial assets for which the rights have been transferred to third parties; however, the transfers did not qualify as a sale in accordance with U.S. GAAP. These transfers have been recognized as collateralized financing transactions. The transferred assets are recorded in trading assets and loans, and the corresponding liabilities are recorded primarily in short-term borrowings and long-term debt on the Consolidated balance sheets.

Note 12 – Loans

Loan accounting framework

The accounting for a loan depends on management's strategy for the loan. The Firm accounts for loans based on the following categories:

- Originated or purchased loans held-for-investment (i.e., "retained")
- Loans held-for-sale
- Loans at fair value

The following provides a detailed accounting discussion of the Firm's loans by category:

Loans held-for-investment

Originated or purchased loans held-for-investment, including PCD, are recorded at amortized cost, reflecting the principal amount outstanding, net of the following: unamortized deferred loan fees, costs, premiums or discounts; charge-offs; collection of cash; and foreign exchange. Credit card loans also include billed finance charges and fees.

Interest income

Interest income on performing loans held-for-investment is accrued and recognized as interest income at the contractual rate of interest. Purchase price discounts or premiums, as well as net deferred loan fees or costs, are recognized in interest income over the contractual life of the loan as an adjustment of yield.

The Firm classifies accrued interest on loans, including accrued but unbilled interest on credit card loans, in accrued interest and accounts receivables on the Consolidated balance sheets. For credit card loans, accrued interest once billed is then recognized in the loan balances, with the related allowance recorded in the allowance for credit losses. Changes in the allowance for credit losses on accrued interest on credit card loans are recognized in the provision for credit losses and charge-offs are recognized by reversing interest income. For other loans, the Firm generally does not recognize an allowance for credit losses on accrued interest receivables, consistent with its policy to write them off no later than 90 days past due by reversing interest income.

Nonaccrual loans

Nonaccrual loans are those on which the accrual of interest has been suspended. Loans (other than credit card loans and certain consumer loans insured by U.S. government agencies) are placed on nonaccrual status and considered nonperforming when full payment of principal and interest is not expected, regardless of delinquency status, or when principal and interest has been in default for a period of 90 days or more, unless the loan is both well-secured and in the process of collection. A loan is determined to be past due when the minimum payment is not received from

the borrower by the contractually specified due date or for certain loans (e.g., residential real estate loans), when a monthly payment is due and unpaid for 30 days or more. Finally, collateral-dependent loans are typically maintained on nonaccrual status.

On the date a loan is placed on nonaccrual status, all interest accrued but not collected is reversed against interest income. In addition, the amortization of deferred amounts is suspended. Interest income on nonaccrual loans may be recognized as cash interest payments are received (i.e., on a cash basis) if the recorded loan balance is deemed fully collectible; however, if there is doubt regarding the ultimate collectibility of the recorded loan balance, all interest cash receipts are applied to reduce the carrying value of the loan (the cost recovery method). For consumer loans, application of this policy typically results in the Firm recognizing interest income on nonaccrual consumer loans on a cash basis.

A loan may be returned to accrual status when repayment is reasonably assured and there has been demonstrated performance under the terms of the loan or, if applicable, the terms of the restructured loan.

As permitted by regulatory guidance, credit card loans are generally exempt from being placed on nonaccrual status; accordingly, interest and fees related to credit card loans continue to accrue until the loan is charged off or paid in full.

Allowance for loan losses

The allowance for loan losses represents the estimated expected credit losses in the held-for-investment loan portfolio at the balance sheet date and is recognized on the balance sheet as a contra asset, which brings the amortized cost to the net carrying value. Changes in the allowance for loan losses resulting from lending-related activity, macroeconomic variables, changes in credit and other inputs are recorded in the provision for credit losses on the Firm's Consolidated statements of income. Refer to Note 13 for further information on the Firm's accounting policies for the allowance for loan losses.

Charge-offs

Consumer loans are generally charged off or charged down to the lower of the amortized cost or the net realizable value of the underlying collateral (i.e., fair value less estimated costs to sell), with an offset to the allowance for loan losses, upon reaching specified stages of delinquency in accordance with standards established by the FFIEC. Residential real estate loans, unmodified credit card loans and scored business banking loans are generally charged off no later than 180 days past due. Scored auto and closed-end consumer loans, including modified credit card

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accounts placed on a fixed payment plan, are charged off no later than 120 days past due.

Certain consumer loans are charged off or charged down to their net realizable value earlier than the FFIEC charge-off standards in the following circumstances:

- Loans modified to borrowers experiencing financial difficulty that are determined to be collateral-dependent.
- Loans to borrowers who have experienced an event that suggests a loss is either known or highly certain are subject to accelerated charge-off standards (e.g., residential real estate and auto loans are charged off or charged down within 60 days of receiving notification of a bankruptcy filing).
- Auto loans upon repossession of the automobile.

Other than in certain limited circumstances, the Firm typically does not recognize charge-offs on the government-guaranteed portion of loans.

Wholesale loans are charged off when it is highly certain that a loss has been realized. The determination of whether to recognize a charge-off includes many factors, including the prioritization of the Firm's claim in bankruptcy, expectations of the workout/restructuring of the loan and valuation of the borrower's equity or the loan collateral.

When a loan is charged down to the lower of its amortized cost or the estimated net realizable value of the underlying collateral, the determination of the fair value of the collateral depends on the type of collateral (e.g., securities, real estate). In cases where the collateral is in the form of liquid securities, the fair value is based on quoted market prices or broker quotes. For illiquid securities or other financial assets, the fair value of the collateral is generally estimated using a discounted cash flow model.

For residential real estate loans, collateral values are based upon external valuation sources. When it becomes likely that a borrower is either unable or unwilling to pay, the Firm utilizes a broker's price opinion, appraisal and/or an automated valuation model of the home based on an exterior-only valuation ("exterior opinions"), which is then updated at least every 12 months, or more frequently depending on various market factors. As soon as practicable after the Firm receives the property in satisfaction of a debt (e.g., by taking legal title or physical possession), the Firm generally obtains an appraisal based on an inspection that includes the interior of the home ("interior appraisals"). Exterior opinions and interior appraisals are discounted based upon the Firm's experience with actual liquidation values as compared with the estimated values provided by exterior opinions and interior appraisals, considering state-specific factors.

For commercial real estate loans, collateral values are generally based on appraisals from internal and external valuation sources. Collateral values are typically updated every six to twelve months, either by obtaining a new appraisal or by performing an internal analysis, in accordance with the Firm's policies. The Firm also considers both borrower- and market-specific factors, which may result in obtaining appraisal updates or broker price opinions at more frequent intervals.

Loans held-for-sale

Loans held-for-sale are measured at the lower of cost or fair value, with valuation changes recorded in noninterest revenue. For consumer loans, the valuation is performed on a portfolio basis. For wholesale loans, the valuation is performed on an individual loan basis.

Interest income on loans held-for-sale is accrued and recognized based on the contractual rate of interest.

Loan origination fees or costs and purchase price discounts or premiums are deferred in a contra loan account until the related loan is sold. The deferred fees or costs and discounts or premiums are an adjustment to the basis of the loan and therefore are included in the periodic determination of the lower of cost or fair value adjustments and/or the gain or loss recognized at the time of sale.

Because these loans are recognized at the lower of cost or fair value, the Firm's allowance for loan losses and charge-off policies do not apply to these loans. However, loans held-for-sale are subject to the Firm's nonaccrual policies.

Loans at fair value

Loans for which the fair value option has been elected are measured at fair value, with changes in fair value recorded in noninterest revenue.

Interest income on these loans is accrued and recognized based on the contractual rate of interest. Changes in fair value are recognized in noninterest revenue. Loan origination fees are recognized upfront in noninterest revenue. Loan origination costs are recognized in the associated expense category as incurred.

Because these loans are recognized at fair value, the Firm's allowance for loan losses and charge-off policies do not apply to these loans. However, loans at fair value are subject to the Firm's nonaccrual policies.

Refer to Note 3 for further information on the Firm's elections of fair value accounting under the fair value option. Refer to Note 2 and Note 3 for further information on loans carried at fair value and classified as trading assets.

Loan classification changes

Loans in the held-for-investment portfolio that management decides to sell are transferred to the held-for-sale portfolio at the lower of cost or fair value on the date of transfer. Credit-related losses are charged against the allowance for loan losses; non-credit related losses such as those due to changes in interest rates or foreign currency exchange rates are recognized in noninterest revenue.

In the event that management decides to retain a loan in the held-for-sale portfolio, the loan is transferred to the held-for-investment portfolio at amortized cost on the date of transfer. These loans are subsequently assessed for impairment based on the Firm's allowance methodology. Refer to Note 13 for a further discussion of the methodologies used in establishing the Firm's allowance for loan losses.

Loan modifications

The Firm seeks to modify certain loans in conjunction with its loss mitigation activities. Through the modification, JPMorganChase grants one or more concessions to a borrower who is experiencing financial difficulty in order to minimize the Firm's economic loss and avoid foreclosure or repossession of the collateral, and to ultimately maximize payments received by the Firm from the borrower. The concessions granted vary by program and by borrower-specific characteristics, and may include interest rate reductions, term extensions, other-than-insignificant payment delays or principal forgiveness.

Loans, except for credit card loans, reported as FDMs are generally placed on nonaccrual status, although in many cases such loans were already on nonaccrual status prior to modification. These loans may be returned to performing status (the accrual of interest is resumed) if the following criteria are met: (i) the borrower has performed under the modified terms for a minimum of six months and/or six payments, and (ii) the Firm has an expectation that repayment of the modified loan is reasonably assured based on, for example, the borrower's debt capacity and level of future earnings, collateral values, LTV ratios, and other current market considerations. In certain limited and well-defined circumstances in which the loan is current at the modification date, such loans are not placed on nonaccrual status at the time of modification.

The allowance for credit losses associated with FDMs is measured using the Firm's established allowance methodology, which considers the expected default rates for the modified loans. Refer to Note 13 for further discussion.

Foreclosed property

The Firm acquires property from borrowers through loan restructurings, workouts, and foreclosures. Property acquired may include real property (e.g., residential real estate, land, and buildings) and other commercial and personal property (e.g., automobiles, aircraft, railcars, and ships).

The Firm recognizes foreclosed property upon receiving assets in satisfaction of a loan (e.g., by taking legal title or physical possession). For loans collateralized by real property, the Firm generally recognizes the asset received at foreclosure sale or upon the execution of a deed in lieu of foreclosure transaction with the borrower. Foreclosed assets are reported in other assets on the Consolidated balance sheets and initially recognized at fair value less estimated costs to sell. Each quarter the fair value of the acquired property is reviewed and adjusted, if necessary, to the lower of cost or fair value. Subsequent adjustments to fair value are charged/credited to noninterest revenue. Operating expense, such as real estate taxes and maintenance, are charged to other expense.

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Loan portfolio

The Firm's loan portfolio is divided into three portfolio segments, which are the same segments used by the Firm to determine the allowance for loan losses: Consumer, excluding credit card; Credit card; and Wholesale. Within each portfolio segment the Firm monitors and assesses the credit risk in the following classes of loans, based on the risk characteristics of each loan class.

Consumer, excluding credit card	Credit card	Wholesale ^{(c)(d)}
- Residential real estate^(a) - Auto and other^(b)	Credit card loans	- Secured by real estate - Commercial and industrial - Other^(e)

(a) Includes scored mortgage and home equity loans held in CCB and AWM, and scored mortgage loans held in CIB.

(b) Includes scored auto, business banking and consumer unsecured loans as well as overdrafts, primarily in CCB.

(c) Includes loans held in CIB, AWM, Corporate, and risk-rated exposure held in CCB, for which the wholesale methodology is applied when determining the allowance for loan losses.

(d) The wholesale portfolio segment's classes align with loan classifications as defined by the bank regulatory agencies, based on the loan's collateral, purpose, and type of borrower.

(e) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. Refer to Note 14 for more information on SPEs.

The following tables summarize the Firm's loan balances by portfolio segment.

December 31, 2024				
(in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total ^{(a)(b)}
Retained	\$ 376,334	\$ 232,860	\$ 690,396	\$ 1,299,590
Held-for-sale	945	—	6,103	7,048
At fair value	15,531	—	25,819	41,350
Total	\$ 392,810	\$ 232,860	\$ 722,318	\$ 1,347,988

December 31, 2023				
(in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total ^{(a)(b)}
Retained	\$ 397,275	\$ 211,123	\$ 672,472	\$ 1,280,870
Held-for-sale	487	—	3,498	3,985
At fair value	12,331	—	26,520	38,851
Total	\$ 410,093	\$ 211,123	\$ 702,490	\$ 1,323,706

(a) Excludes \$6.6 billion and \$6.8 billion of accrued interest receivable at December 31, 2024 and 2023, respectively. The Firm wrote off accrued interest receivable of \$84 million and \$49 million for the years ended December 31, 2024 and 2023, respectively.

(b) Loans (other than those for which the fair value option has been elected) are presented net of unamortized discounts and premiums and net deferred loan fees or costs. These amounts were not material as of December 31, 2024 and 2023. For the discount associated with First Republic loans, refer to Note 34 on pages 319–321.

The following tables provide information about the carrying value of retained loans purchased, sold and reclassified to held-for-sale during the periods indicated. Loans that were reclassified to held-for-sale and sold in a subsequent period are excluded from the sales line of these tables.

2024				
Year ended December 31, (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	\$ 647 ^{(b)(c)}	\$ —	\$ 1,432	\$ 2,079
Sales	10,440	—	45,147	55,587
Retained loans reclassified to held-for-sale ^(a)	1,656	—	749	2,405

2023				
Year ended December 31, (in millions)	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	\$ 92,205 ^{(b)(c)(d)}	\$ —	\$ 60,300 ^(d)	\$ 152,505
Sales	2,202	—	43,949	46,151
Retained loans reclassified to held-for-sale ^(a)	274	—	1,486	1,760

Year ended December 31, (in millions)	2022			
	Consumer, excluding credit card	Credit card	Wholesale	Total
Purchases	\$ 1,625 ^{(b)(c)}	\$ —	\$ 1,088	\$ 2,713
Sales	2,884	—	41,934	44,818
Retained loans reclassified to held-for-sale ^(a)	229	—	1,055	1,284

(a) Reclassifications of loans to held-for-sale are non-cash transactions.

(b) Includes purchases of residential real estate loans, including the Firm's voluntary repurchases of certain delinquent loans from loan pools as permitted by Government National Mortgage Association ("Ginnie Mae") guidelines for the years ended December 31, 2024, 2023 and 2022. The Firm typically elects to repurchase these delinquent loans as it continues to service them and/or manage the foreclosure process in accordance with applicable requirements of Ginnie Mae, FHA, RHS, and/or VA.

(c) Excludes purchases of retained loans of \$902 million, \$5.1 billion and \$12.4 billion for the years ended December 31, 2024, 2023 and 2022, respectively, which are predominantly sourced through the correspondent origination channel and underwritten in accordance with the Firm's standards.

(d) Includes loans acquired in the First Republic acquisition consisting of \$91.9 billion in Consumer, excluding credit card and \$59.2 billion in Wholesale. Refer to Note 34 for additional information.

Gains and losses on sales of loans

Net gains/(losses) on sales of loans and lending-related commitments (including adjustments to record loans and lending-related commitments held-for-sale at the lower of cost or fair value) recognized in noninterest revenue were \$154 million for the year ended December 31, 2024, of which \$113 million were related to loans. Net gains/(losses) on sales of loans and lending-related commitments were \$56 million for the year ended December 31, 2023, of which \$62 million were related to loans. Net gains/(losses) on sales of loans and lending-related commitments were \$(186) million for the year ended December 31, 2022, of which \$(48) million were related to loans. In addition, the sale of loans may also result in write downs, recoveries or changes in the allowance recognized in the provision for credit losses.

Consumer, excluding credit card loan portfolio

Consumer loans, excluding credit card loans, consist primarily of scored residential mortgages, home equity loans and lines of credit, auto and business banking loans, with a focus on serving the prime consumer credit market. These loans include home equity loans secured by junior liens, prime mortgage loans with an interest-only payment period, and certain payment-option loans that may result in negative amortization.

The following table provides information about retained consumer loans, excluding credit card, by class.

December 31, (in millions)	2024	2023
Residential real estate	\$ 309,513	\$ 326,409
Auto and other	66,821	70,866
Total retained loans	\$ 376,334	\$ 397,275

Delinquency rates are the primary credit quality indicator for consumer loans. Loans that are more than 30 days past due provide an early warning of borrowers who may be experiencing financial difficulties and/or who may be unable or unwilling to repay the loan. As the loan continues to age, it becomes more clear whether the borrower is likely to be unable or unwilling to pay. In the case of residential real estate loans, late-stage delinquencies (greater than 150 days past due) are a strong indicator of loans that will ultimately result in a foreclosure or similar liquidation transaction. In addition to delinquency rates, other credit quality indicators for consumer loans vary based on the class of loan, as follows:

- For residential real estate loans, the current estimated LTV ratio, or the combined LTV ratio in the case of junior lien loans, is an indicator of the potential loss severity in the event of default. Additionally, LTV or combined LTV ratios can provide insight into a borrower's continued willingness to pay, as the delinquency rate of high-LTV loans tends to be greater than that for loans where the borrower has equity in the collateral. The geographic distribution of the loan collateral also provides insight as to the credit quality of the portfolio, as factors such as the regional economy, home price changes and specific events such as natural disasters, will affect credit quality. The borrower's current or "refreshed" FICO score is a secondary credit quality indicator for certain loans, as FICO scores are an indication of the borrower's credit payment history. Thus, a loan to a borrower with a low FICO score (less than 660) is considered to be of higher risk than a loan to a borrower with a higher FICO score. Further, a loan to a borrower with a high LTV ratio and a low FICO score is at greater risk of default than a loan to a borrower that has both a high LTV ratio and a high FICO score.
- For scored auto and business banking loans, geographic distribution is an indicator of the credit performance of the portfolio. Similar to residential real estate loans, geographic distribution provides insights into the portfolio performance based on regional economic activity and events.

Residential real estate

Delinquency is the primary credit quality indicator for retained residential real estate loans. The following tables provide information on delinquency and gross charge-offs.

(in millions, except ratios)	December 31, 2024								
	Term loans by origination year ^(c)						Revolving loans		Total
	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	
Loan delinquency^(a)									
Current	\$ 12,301	\$ 17,280	\$ 61,337	\$ 79,760	\$ 52,289	\$70,270	\$ 6,974	\$ 7,088	\$307,299
30–149 days past due	13	54	139	110	59	747	53	204	1,379
150 or more days past due	—	11	71	68	49	501	8	127	835
Total retained loans	\$ 12,314	\$ 17,345	\$ 61,547	\$ 79,938	\$ 52,397	\$71,518	\$ 7,035	\$ 7,419	\$309,513
% of 30+ days past due to total retained loans ^(b)	0.11 %	0.37 %	0.34 %	0.22 %	0.21 %	1.72 %	0.87 %	4.46 %	0.71 %
Gross charge-offs	\$ —	\$ —	\$ 1	\$ 1	\$ —	\$ 176	\$ 21	\$ 7	\$ 206

(in millions, except ratios)	December 31, 2023								
	Term loans by origination year ^(c)						Revolving loans		Total
	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	
Loan delinquency^(a)									
Current	\$ 23,216	\$ 64,366	\$ 84,496	\$ 55,546	\$ 21,530	\$59,563	\$ 7,479	\$ 8,151	\$324,347
30–149 days past due	33	74	89	70	41	801	49	223	1,380
150 or more days past due	1	10	17	8	21	456	5	164	682
Total retained loans	\$ 23,250	\$ 64,450	\$ 84,602	\$ 55,624	\$ 21,592	\$60,820	\$ 7,533	\$ 8,538	\$326,409
% of 30+ days past due to total retained loans ^(b)	0.15 %	0.13 %	0.13 %	0.14 %	0.29 %	2.04 %	0.72 %	4.53 %	0.63 %
Gross charge-offs	\$ —	\$ —	\$ —	\$ —	\$ 4	\$ 167	\$ 26	\$ 7	\$ 204

(a) Individual delinquency classifications include mortgage loans insured by U.S. government agencies which were not material at December 31, 2024 and 2023.

(b) Excludes mortgage loans that are 30 or more days past due insured by U.S. government agencies which were not material at December 31, 2024 and 2023. These amounts have been excluded based upon the government guarantee.

(c) Purchased loans are included in the year in which they were originated.

Approximately 38% of the total revolving loans are senior lien loans; the remaining balance are junior lien loans. The lien position the Firm holds is considered in the Firm's allowance for credit losses. Revolving loans that have been converted to term loans have higher delinquency rates than those that are still within the revolving period. That is primarily because the fully-amortizing payment that is generally required for those products is higher than the minimum payment options available for revolving loans within the revolving period.

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Nonaccrual loans and other credit quality indicators

The following table provides information on nonaccrual and other credit quality indicators for retained residential real estate loans.

(in millions, except weighted-average data)	December 31, 2024	December 31, 2023
Nonaccrual loans ^{(a)(b)(c)(d)}	\$ 2,984	\$ 3,466
Current estimated LTV ratios^{(e)(f)(g)}		
Greater than 125% and refreshed FICO scores:		
Equal to or greater than 660	\$ 72	\$ 72
Less than 660	3	—
101% to 125% and refreshed FICO scores:		
Equal to or greater than 660	161	223
Less than 660	5	4
80% to 100% and refreshed FICO scores:		
Equal to or greater than 660	4,962	6,491
Less than 660	73	102
Less than 80% and refreshed FICO scores:		
Equal to or greater than 660	294,797	309,251
Less than 660	8,534	9,277
No FICO/LTV available ^(h)	906	989
Total retained loans	\$ 309,513	\$ 326,409
Weighted-average LTV ratio ^{(e)(i)}	47 %	49 %
Weighted-average FICO ^{(f)(i)}	774	770
Geographic region^{(h)(i)}		
California	\$ 120,944	\$ 127,072
New York	46,854	48,815
Florida	21,820	22,778
Texas	14,531	15,506
Massachusetts	13,511	14,213
Colorado	10,465	10,800
Illinois	9,835	10,856
Washington	9,372	9,923
New Jersey	7,554	8,050
Connecticut	6,854	7,163
All other	47,773	51,233
Total retained loans	\$ 309,513	\$ 326,409

- (a) Includes collateral-dependent residential real estate loans that are charged down to the fair value of the underlying collateral less costs to sell. The Firm reports, in accordance with regulatory guidance, residential real estate loans that have been discharged under Chapter 7 bankruptcy and not reaffirmed by the borrower ("Chapter 7 loans") as collateral-dependent nonaccrual loans, regardless of their delinquency status. At December 31, 2024, approximately 8% of Chapter 7 residential real estate loans were 30 days or more past due.
- (b) Mortgage loans insured by U.S. government agencies excluded from nonaccrual loans were not material at December 31, 2024 and 2023.
- (c) Generally, all consumer nonaccrual loans have an allowance. In accordance with regulatory guidance, certain nonaccrual loans that are considered collateral-dependent have been charged down to the lower of amortized cost or the fair value of their underlying collateral less costs to sell. If the value of the underlying collateral improves subsequent to charge down, the related allowance may be negative.
- (d) Interest income on nonaccrual loans recognized on a cash basis was \$160 million and \$180 million for the years ended December 31, 2024 and 2023, respectively.
- (e) Represents the aggregate unpaid principal balance of loans divided by the estimated current property value. Current property values are estimated, at a minimum, quarterly, based on home valuation models using nationally recognized home price index valuation estimates incorporating actual data to the extent available and forecasted data where actual data is not available. Current estimated combined LTV for junior lien home equity loans considers all available lien positions, as well as unused lines, related to the property.
- (f) Refreshed FICO scores represent each borrower's most recent credit score, which is obtained by the Firm on at least a quarterly basis.
- (g) Includes residential real estate loans, primarily held in LLCs in AWM that did not have a refreshed FICO score. These loans have been included in a FICO band based on management's estimation of the borrower's credit quality.
- (h) Included U.S. government-guaranteed loans as of December 31, 2024 and 2023.
- (i) Excludes loans with no FICO and/or LTV data available.
- (j) The geographic regions presented in the table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of residential real estate loans to borrowers experiencing financial difficulty. The Firm's proprietary modification programs as well as government programs, including U.S. GSE programs, that generally provide various modifications to borrowers experiencing financial difficulty including, but not limited to, interest rate reductions, term extensions, other-than-insignificant payment deferral and principal forgiveness that would otherwise have been required under the terms of the original agreement, are considered FDMs.

In addition, the Firm offers trial modifications of residential real estate loans, which generally include a three-month trial payment period during which the borrower makes monthly payments under the proposed modified loan terms. Loans in a trial payment period continue to age and accrue interest in accordance with the original contractual terms. At the completion of a trial period, the loan modification is considered permanent.

Financial effects of FDMs

For the year ended December 31, 2024, retained residential real estate FDMs were \$206 million. The financial effects of the FDMs, which were predominantly in the form of term extensions and interest rate reductions, included extending the weighted-average life of the loans by 15 years, and reducing the weighted-average contractual interest rate from 7.53% to 5.44% for the year ended December 31, 2024.

For the year ended December 31, 2023, retained residential real estate FDMs were \$136 million. The financial effects of the FDMs, which were predominantly in the form of term extensions and interest rate reductions, included extending the weighted-average life of the loans by 20 years, and reducing the weighted-average contractual interest rate from 7.21% to 4.44% for the year ended December 31, 2023.

As of December 31, 2024 and 2023, there were no additional commitments to lend to borrowers experiencing financial difficulty whose loans have been modified as FDMs.

For the years ended December 31, 2024 and 2023, loans subject to a trial modification, where the terms of the loans have not been permanently modified, and Chapter 7 loans were not material.

Payment status of FDMs

The following table provides information on the payment status of retained residential real estate FDMs during the years ended December 31, 2024 and 2023.

Year ended December 31, (in millions)	Amortized cost basis	
	2024	2023
Current	\$ 139	\$ 107
30-149 days past due	47	13
150 or more days past due	20	16
Total	\$ 206	\$ 136

Defaults of FDMs

Retained residential real estate FDMs that defaulted during the year ended December 31, 2024 and that were reported as FDMs in the twelve months prior to the default were \$93 million. Retained residential real estate FDMs that defaulted during the year ended December 31, 2023 and that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance were not material. Refer to Note 1 for further information.

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Nature and extent of TDRs

For periods ending prior to January 1, 2023, modifications of residential real estate loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs. For the year ended December 31, 2022, new TDRs were \$362 million, and there were no additional commitments to lend to borrowers whose residential real estate loans were modified in TDRs.

The Firm's proprietary modification programs as well as government programs, including U.S. GSE programs, generally provide various concessions to financially troubled borrowers including, but not limited to, interest rate reductions, term or payment extensions and delays of principal and/or interest payments that would otherwise have been required under the terms of the original agreement.

The following table provides information about how residential real estate loans were modified in TDRs during the period presented. This table excludes loans with short-term or other insignificant modifications that are not considered concessions.

Year ended December 31,	2022
Number of loans approved for a trial modification	3,902
Number of loans permanently modified	4,182
Concession granted:^(a)	
Interest rate reduction	54 %
Term or payment extension	67
Principal and/or interest deferred	10
Principal forgiveness	1
Other ^(b)	37

(a) Represents concessions granted in permanent modifications as a percentage of the number of loans permanently modified. The sum of the percentages exceeds 100% because predominantly all of the modifications include more than one type of concession. Concessions offered on trial modifications are generally consistent with those granted on permanent modifications.

(b) Includes variable interest rate to fixed interest rate modifications and payment delays that meet the definition of a TDR.

Financial effects of TDRs and defaults

The following table provides information about the financial effects of the various concessions granted in modifications of residential real estate loans and about defaults of certain loans modified in TDRs for the period presented. The following table presents only the financial effects of permanent modifications and does not include temporary concessions offered through trial modifications. This table also excludes loans with short-term or other insignificant modifications that were not considered concessions.

Year ended December 31, (in millions, except weighted - average data)	2022
Weighted-average interest rate of loans with interest rate reductions - before TDR	4.75 %
Weighted-average interest rate of loans with interest rate reductions - after TDR	3.35
Weighted-average remaining contractual term (in years) of loans with term or payment extensions - before TDR	22
Weighted-average remaining contractual term (in years) of loans with term or payment extensions - after TDR	38
Charge-offs recognized upon permanent modification	\$ 1
Principal deferred	16
Principal forgiven	2
Balance of loans that defaulted within one year of permanent modification ^(a)	\$ 147

(a) Represents loans permanently modified in TDRs that experienced a payment default in the periods presented, and for which the payment default occurred within one year of the modification. The dollar amounts presented represent the balance of such loans at the end of the reporting period in which such loans defaulted.

Active and suspended foreclosure

At December 31, 2024 and 2023, the Firm had retained residential real estate loans, excluding those insured by U.S. government agencies, with a carrying value of \$576 million and \$566 million, respectively, that were not included in REO, but were in the process of active or suspended foreclosure.

Auto and other

Delinquency is the primary credit quality indicator for retained auto and other loans. The following tables provide information on delinquency and gross charge-offs.

	December 31, 2024								
	Term loans by origination year						Revolving loans		
						Prior to 2020	Within the revolving period	Converted to term loans	
(in millions, except ratios)	2024	2023	2022	2021	2020	Prior to 2020			Total
Loan delinquency									
Current	\$26,165	\$15,953	\$ 9,201	\$ 7,014	\$ 2,895	\$ 624	\$ 3,714	\$ 148	\$65,714
30–119 days past due	190	283	259	179	53	23	40	34	1,061
120 or more days past due	1	1	—	5	6	—	3	30	46
Total retained loans	\$26,356	\$16,237	\$ 9,460	\$ 7,198	\$ 2,954	\$ 647	\$ 3,757	\$ 212	\$66,821
% of 30+ days past due to total retained loans	0.72 %	1.75 %	2.74	2.50 %	1.76 %	3.55 %	1.14 %	30.19 %	1.64 %
Gross charge-offs	\$ 269	\$ 348	\$ 224	\$ 126	\$ 37	\$ 82	\$ 1	\$ 6	\$ 1,093

	December 31, 2023								
	Term loans by origination year						Revolving loans		
						Prior to 2019	Within the revolving period	Converted to term loans	
(in millions, except ratios)	2023	2022	2021	2020	2019	Prior to 2019			Total
Loan delinquency									
Current	\$30,328	\$14,797	\$12,825	\$ 6,538	\$ 1,777	\$ 511	\$ 2,984	\$ 102	\$69,862
30–119 days past due	276	279	231	78	43	17	19	24	967
120 or more days past due	1	1	7	8	—	—	3	17	37
Total retained loans	\$30,605	\$15,077	\$13,063	\$ 6,624	\$ 1,820	\$ 528	\$ 3,006	\$ 143	\$70,866
% of 30+ days past due to total retained loans	0.91 %	1.86 %	1.75 %	1.15 %	2.36 %	3.22 %	0.73 %	28.67 %	1.39 %
Gross charge-offs	\$ 333	\$ 297	\$ 161	\$ 53	\$ 35	\$ 64	\$ —	\$ 4	\$ 947

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Nonaccrual loans and other credit quality indicators

The following table provides information on nonaccrual and geographic region as a credit quality indicator for retained auto and other consumer loans.

(in millions)	Total Auto and other	
	December 31, 2024	December 31, 2023
Nonaccrual loans^{(a)(b)}	\$ 249	\$ 177
Geographic region^(c)		
California	\$ 10,321	\$ 10,959
Texas	7,772	8,502
Florida	5,428	5,684
New York	4,905	4,938
Illinois	2,890	3,147
New Jersey	2,468	2,609
Pennsylvania	2,012	1,900
Georgia	1,716	1,912
Arizona	1,643	1,779
North Carolina	1,597	1,714
All other	26,069	27,722
Total retained loans	\$ 66,821	\$ 70,866

- (a) Generally, all consumer nonaccrual loans have an allowance. In accordance with regulatory guidance, certain nonaccrual loans that are considered collateral-dependent have been charged down to the lower of amortized cost or the fair value of their underlying collateral less costs to sell. If the value of the underlying collateral improves subsequent to charge down, the related allowance may be negative.
- (b) Interest income on nonaccrual loans recognized on a cash basis was not material for the years ended December 31, 2024 and 2023.
- (c) The geographic regions presented in this table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of auto and other loans to borrowers experiencing financial difficulty.

For the years ended December 31, 2024 and 2023, retained auto and other FDMs were not material.

As of December 31, 2024 and 2023, there were no additional commitments to lend to borrowers modified as FDMs.

For periods ending prior to January 1, 2023, modifications of auto and other loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs. For the year ended December 31, 2022, auto and other TDRs were not material.

Credit card loan portfolio

The credit card portfolio segment includes credit card loans originated and purchased by the Firm.

Delinquency rates are the primary credit quality indicator for credit card loans as they provide an early warning that borrowers may be experiencing difficulties (30 days past due); information on those borrowers that have been delinquent for a longer period of time (90 days past due) is also considered. In addition to delinquency rates, the geographic distribution of the loans provides insight as to the credit quality of the portfolio based on the regional economy.

While the borrower's credit score is another general indicator of credit quality, the Firm does not view credit scores as a primary indicator of credit quality because the borrower's credit score tends to be a

lagging indicator. The distribution of such scores provides a general indicator of credit quality trends within the portfolio; however, the score does not capture all factors that would be predictive of future credit performance. Refreshed FICO score information, which is obtained at least quarterly, for a statistically significant random sample of the credit card portfolio is indicated in other credit quality indicators. FICO is considered to be the industry benchmark for credit scores.

The Firm generally originates new credit card accounts to prime consumer borrowers. However, certain cardholders' FICO scores may decrease over time, depending on the performance of the cardholder and changes in the credit score calculation.

The following tables provide information on delinquency and gross charge-offs.

(in millions, except ratios)	December 31, 2024		
	Within the revolving period	Converted to term loans	Total
Loan delinquency			
Current and less than 30 days past due and still accruing	\$ 226,532	\$ 1,284	\$ 227,816
30–89 days past due and still accruing	2,291	109	2,400
90 or more days past due and still accruing	2,591	53	2,644
Total retained loans	\$ 231,414	\$ 1,446	\$ 232,860
Loan delinquency ratios			
% of 30+ days past due to total retained loans	2.11 %	11.20 %	2.17 %
% of 90+ days past due to total retained loans	1.12	3.67	1.14
Gross charge-offs	\$ 7,951	\$ 247	\$ 8,198
(in millions, except ratios)	December 31, 2023		
	Within the revolving period	Converted to term loans	Total
Loan delinquency			
Current and less than 30 days past due and still accruing	\$ 205,731	\$ 882	\$ 206,613
30–89 days past due and still accruing	2,217	84	2,301
90 or more days past due and still accruing	2,169	40	2,209
Total retained loans	\$ 210,117	\$ 1,006	\$ 211,123
Loan delinquency ratios			
% of 30+ days past due to total retained loans	2.09 %	12.33 %	2.14 %
% of 90+ days past due to total retained loans	1.03	3.98	1.05
Gross charge-offs	\$ 5,325	\$ 166	\$ 5,491

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Other credit quality indicators

The following table provides information on other credit quality indicators for retained credit card loans.

(in millions, except ratios)	December 31, 2024		December 31, 2023	
Geographic region^(a)				
California	\$	36,385	\$	32,652
Texas		24,423		22,086
New York		18,525		16,915
Florida		17,236		15,103
Illinois		12,442		11,364
New Jersey		9,644		8,688
Ohio		6,976		6,424
Colorado		6,962		6,307
Pennsylvania		6,558		6,088
Arizona		5,796		5,209
All other		87,913		80,287
Total retained loans	\$	232,860	\$	211,123
Percentage of portfolio based on carrying value with estimated refreshed FICO scores				
Equal to or greater than 660		85.5 %		85.8 %
Less than 660		14.3		14.0
No FICO available		0.2		0.2

(a) The geographic regions presented in the table are ordered based on the magnitude of the corresponding loan balances at December 31, 2024.

Loan modifications

The Firm grants certain modifications of credit card loans to borrowers experiencing financial difficulty. These modifications may involve placing the customer's credit card account on a fixed payment plan, generally for 60 months, which typically includes reducing the interest rate on the credit card account. If the borrower does not make the contractual payments when due under the modified payment terms, the credit card loan continues to age and will be charged-off in accordance with the Firm's standard charge-off policy. In most cases, the Firm does not reinstate the borrower's line of credit.

Financial effects of FDMs

The following table provides information on retained credit card FDMs.

Year ended December 31, (in millions)	Loan modifications	
	2024	2023
Term extension and interest rate reduction^{(a)(b)}		
Amortized cost basis	\$ 926	\$ 648
% of total modifications to total retained credit card loans	0.40 %	0.31 %
Financial effect of loan modifications	Term extension with a reduction in the weighted average contractual interest rate from 23.64% to 3.20%	
	Term extension with a reduction in the weighted average contractual interest rate from 23.19% to 3.64%	

(a) Term extension includes credit card loans whose terms have been modified under long-term programs by placing the customer's credit card account on a fixed payment plan.

(b) The interest rates represent weighted average at the time of modification.

Payment status of FDMs

The following table provides information on the payment status of retained credit card FDMs during the years ended December 31, 2024 and 2023.

Year ended December 31, (in millions)	Amortized cost basis	
	2024	2023
Current and less than 30 days past due and still accruing	\$ 811	\$ 558
30-89 days past due and still accruing	74	59
90 or more days past due and still accruing	41	31
Total	\$ 926	\$ 648

Defaults of FDMs

Retained credit card FDMs that defaulted during the year ended December 31, 2024 and that were reported as FDMs in the twelve months prior to the default were not material. Retained credit card FDMs that defaulted during the year ended December 31, 2023 and that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance were not material. Refer to Note 1 for further information.

For credit card loans modified as FDMs, payment default is deemed to have occurred when the borrower misses two consecutive contractual payments. Defaulted modified credit card loans remain in the modification program and continue to be charged off in accordance with the Firm's standard charge-off policy.

Financial effects of TDRs and defaults

For periods ending prior to January 1, 2023, modifications of credit card loans where the Firm granted concessions to borrowers who were experiencing financial difficulty were generally accounted for and reported as TDRs. The Firm granted concessions for most of the credit card loans under long-term programs. These concessions involved placing the customer's credit card account on a fixed payment plan, generally for 60 months, and typically included reducing the interest rate on the credit card account. Substantially all modifications under the Firm's long-term programs were considered to be TDRs. Loans with short-term or other insignificant modifications that were not considered concessions were not reported as TDRs.

The following table provides information about the financial effects of the concessions granted on credit card loans modified in TDRs and defaults for the periods presented. For the period disclosed, new enrollments were less than 1% of total retained credit card loans.

Year ended December 31, (in millions, except weighted-average data)	2022
Balance of new TDRs ^(a)	\$ 418
Weighted-average interest rate of loans – before TDR	19.86 %
Weighted-average interest rate of loans – after TDR	4.13
Balance of loans ^(b) that defaulted within one year of modification	\$ 34

(a) Represents the outstanding balance prior to modification.

(b) Represents loans modified in TDRs that experienced a payment default in the periods presented, and for which the payment default occurred within one year of the modification. The amounts presented represent the balance of such loans as of the end of the quarter in which they defaulted.

For credit card loans modified in TDRs, payment default was deemed to have occurred when the borrower missed two consecutive contractual payments. Defaulted modified credit card loans remained in the modification program and continued to be charged off in accordance with the Firm's standard charge-off policy.

Wholesale loan portfolio

Wholesale loans include loans made to a variety of clients, ranging from large corporate and institutional clients to small businesses and high-net-worth individuals.

The primary credit quality indicator for wholesale loans is the internal risk rating assigned to each loan. Risk ratings are used to identify the credit quality of loans and differentiate risk within the portfolio. Risk ratings on loans consider the PD and the LGD. The PD is the likelihood that a loan will default. The LGD is the estimated loss on the loan that would be realized upon the default of the borrower and takes into consideration collateral and structural support for each credit facility.

Management considers several factors to determine an appropriate internal risk rating, including the obligor's debt capacity and financial flexibility, the level of the obligor's earnings, the amount and sources for repayment, the level and nature of contingencies, management strength, and the industry and geography in which the obligor operates. The Firm's internal risk ratings generally align with the qualitative characteristics (e.g., borrower capacity to meet financial commitments and vulnerability to changes in the economic environment) defined by S&P and Moody's, however the quantitative characteristics (e.g., PD and LGD) may differ as they reflect internal historical experiences and assumptions. The Firm generally considers internal ratings with qualitative characteristics equivalent to BBB-/Baa3 or higher as investment grade, and these ratings have a lower PD and/or lower LGD than non-investment grade ratings.

loans, representing management's assessment of the collectibility of principal and interest. Criticized loans have a higher PD than noncriticized loans. The Firm's definition of criticized aligns with the U.S. banking regulatory definition of criticized exposures, which consist of special mention, substandard and doubtful categories. Refer to Note 1 for additional information.

Risk ratings are reviewed on a regular and ongoing basis by Credit Risk Management and are adjusted as necessary for updated information affecting the obligor's ability to fulfill its obligations.

As noted above, the risk rating of a loan considers the industry in which the obligor conducts its operations. As part of the overall credit risk management framework, the Firm focuses on the management and diversification of its industry and client exposures, with particular attention paid to industries with an actual or potential credit concern. Refer to Note 4 for further detail on industry concentrations.

Noninvestment-grade ratings are further classified as noncriticized and criticized, and the criticized portion is further subdivided into performing and nonaccrual

Internal risk rating is the primary credit quality indicator for retained wholesale loans. The following tables provide information on internal risk rating and gross charge-offs for the year ended December 31, 2024.

December 31, (in millions, except ratios)	Secured by real estate		Commercial and industrial		Other ^(a)		Total retained loans	
	2024	2023	2024	2023	2024	2023	2024	2023
Loans by risk ratings								
Investment-grade	\$ 114,280	\$ 120,405	\$ 70,862	\$ 72,624	\$ 286,528	\$ 265,809	\$ 471,670	\$ 458,838
Noninvestment-grade:								
Noncriticized	37,422	34,241	83,191	80,637	72,743	75,178	193,356	190,056
Criticized performing	9,291	7,291	10,977	12,684	1,160	1,257	21,428	21,232
Criticized nonaccrual	1,439	401	1,760	1,221	743	724	3,942	2,346
Total noninvestment-grade	48,152	41,933	95,928	94,542	74,646	77,159	218,726	213,634
Total retained loans	\$ 162,432	\$ 162,338	\$ 166,790	\$ 167,166	\$ 361,174	\$ 342,968	\$ 690,396	\$ 672,472
% of investment-grade to total retained loans	70.36 %	74.17 %	42.49 %	43.44 %	79.33 %	77.50 %	68.32 %	68.23 %
% of total criticized to total retained loans	6.61	4.74	7.64	8.32	0.53	0.58	3.67	3.51
% of criticized nonaccrual to total retained loans	0.89	0.25	1.06	0.73	0.21	0.21	0.57	0.35

(a) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. As of December 31, 2024, predominantly consisted of \$114.8 billion to individuals and individual entities, \$94.0 billion to financial institutions, and \$92.5 billion to SPEs. Refer to Note 14 for more information on SPEs.

Secured by real estate									
December 31, 2024									
(in millions)	Term loans by origination year						Revolving loans		Total
	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	
Loans by risk ratings									
Investment-grade	\$ 10,002	\$ 9,834	\$ 25,284	\$ 22,796	\$ 15,548	\$ 29,488	\$ 1,328	\$ —	\$ 114,280
Noninvestment-grade	4,238	5,366	14,717	8,567	3,462	10,392	1,317	93	48,152
Total retained loans	\$ 14,240	\$ 15,200	\$ 40,001	\$ 31,363	\$ 19,010	\$ 39,880	\$ 2,645	\$ 93	\$ 162,432
Gross charge-offs	\$ 72	\$ 18	\$ 43	\$ 2	\$ 109	\$ 80	\$ —	\$ —	\$ 324

Secured by real estate											
December 31, 2023											
(in millions)	Term loans by origination year						Revolving loans			Total	
	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans			
Loans by risk ratings											
Investment-grade	\$ 10,687	\$ 28,874	\$ 25,784	\$ 16,820	\$ 15,677	\$ 21,108	\$ 1,455	\$ —	\$ 120,405		
Noninvestment-grade	4,477	12,579	7,839	3,840	3,987	7,918	1,291	2	41,933		
Total retained loans	\$ 15,164	\$ 41,453	\$ 33,623	\$ 20,660	\$ 19,664	\$ 29,026	\$ 2,746	\$ 2	\$ 162,338		
Gross charge-offs	\$ 20	\$ 48	\$ 22	\$ —	\$ 23	\$ 78	\$ —	\$ 1	\$ 192		

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Commercial and industrial										
December 31, 2024										
(in millions)	Term loans by origination year						Revolving loans		Total	
	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans		
Loans by risk ratings										
Investment-grade	\$ 11,564	\$ 6,285	\$ 6,588	\$ 3,119	\$ 1,067	\$ 1,139	\$ 41,099	\$ 1	\$ 70,862	
Noninvestment-grade	21,251	11,350	10,942	5,322	783	975	45,181	124	95,928	
Total retained loans	\$ 32,815	\$ 17,635	\$ 17,530	\$ 8,441	\$ 1,850	\$ 2,114	\$ 86,280	\$ 125	\$ 166,790	
Gross charge-offs	\$ 25	\$ 22	\$ 128	\$ 24	\$ 1	\$ 50	\$ 270	\$ 5	\$ 525	

Commercial and industrial										
December 31, 2023										
(in millions)	Term loans by origination year						Revolving loans		Total	
	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans		
Loans by risk ratings										
Investment-grade	\$ 14,875	\$ 10,642	\$ 4,276	\$ 2,291	\$ 1,030	\$ 1,115	\$ 38,394	\$ 1	\$ 72,624	
Noninvestment-grade	18,890	16,444	9,299	1,989	1,144	1,006	45,696	74	94,542	
Total retained loans	\$ 33,765	\$ 27,086	\$ 13,575	\$ 4,280	\$ 2,174	\$ 2,121	\$ 84,090	\$ 75	\$ 167,166	
Gross charge-offs	\$ 25	\$ 8	\$ 110	\$ 55	\$ 2	\$ 12	\$ 259	\$ 8	\$ 479	

Other ^(a)									
December 31, 2024									
(in millions)	Term loans by origination year						Revolving loans		Total
	2024	2023	2022	2021	2020	Prior to 2020	Within the revolving period	Converted to term loans	
Loans by risk ratings									
Investment-grade	\$ 30,484	\$ 17,039	\$ 13,272	\$ 6,288	\$ 8,632	\$ 7,382	\$ 201,949	\$ 1,482	\$ 286,528
Noninvestment-grade	11,784	7,248	5,918	3,296	1,366	1,886	42,954	194	74,646
Total retained loans	\$ 42,268	\$ 24,287	\$ 19,190	\$ 9,584	\$ 9,998	\$ 9,268	\$ 244,903	\$ 1,676	\$ 361,174
Gross charge-offs	\$ —	\$ 38	\$ 3	\$ 36	\$ 40	\$ 50	\$ 6	\$ —	\$ 173

Other ^(a)									
December 31, 2023									
	Term loans by origination year						Revolving loans		Total
	2023	2022	2021	2020	2019	Prior to 2019	Within the revolving period	Converted to term loans	
(in millions)									
Loans by risk ratings									
Investment-grade	\$ 38,338	\$ 18,034	\$ 10,033	\$ 10,099	\$ 3,721	\$ 6,662	\$ 176,728	\$ 2,194	\$ 265,809
Noninvestment-grade	14,054	8,092	6,169	2,172	811	2,001	43,801	59	77,159
Total retained loans	\$ 52,392	\$ 26,126	\$ 16,202	\$ 12,271	\$ 4,532	\$ 8,663	\$ 220,529	\$ 2,253	\$ 342,968
Gross charge-offs	\$ 5	\$ 298	\$ 8	\$ 8	\$ —	\$ 8	\$ 13	\$ —	\$ 340

(a) Includes loans to financial institutions, SPEs, personal investment companies and trusts, individuals and individual entities (predominantly Global Private Bank clients within AWM and J.P. Morgan Wealth Management within CCB), states and political subdivisions, as well as loans to nonprofits. Refer to Note 14 for more information on SPEs.

The following table presents additional information on retained loans secured by real estate, which consists of loans secured wholly or substantially by a lien or liens on real property at origination. Multifamily lending includes financing for acquisition, leasing and construction of apartment buildings. Other commercial lending largely includes financing for acquisition, leasing and construction, largely for office, retail and industrial real estate. Included in secured by real estate loans were \$12.2 billion and \$10.2 billion as of December 31, 2024 and 2023, respectively, of construction and development loans made to finance land development and on-site construction of commercial, industrial, residential, or farm buildings.

December 31, (in millions, except ratios)	Multifamily		Other Commercial		Total retained loans secured by real estate	
	2024	2023	2024	2023	2024	2023
Retained loans secured by real estate	\$ 101,114	\$ 100,725	\$ 61,318	\$ 61,613	\$162,432	\$162,338
Criticized	4,700	3,596	6,030	4,096	10,730	7,692
% of criticized to total retained loans secured by real estate	4.65 %	3.57 %	9.83 %	6.65 %	6.61 %	4.74 %
Criticized nonaccrual	\$ 337	\$ 76	\$ 1,102	\$ 325	\$ 1,439	\$ 401
% of criticized nonaccrual loans to total retained loans secured by real estate	0.33 %	0.08 %	1.80 %	0.53 %	0.89 %	0.25 %

Geographic distribution and delinquency

The following table provides information on the geographic distribution and delinquency for retained wholesale loans.

December 31, (in millions)	Secured by real estate		Commercial and industrial		Other		Total retained loans	
	2024	2023	2024	2023	2024	2023	2024	2023
Loans by geographic distribution^(a)								
Total U.S.	\$ 159,209	\$ 159,499	\$ 127,626	\$ 127,638	\$ 278,077	\$ 262,499	\$ 564,912	\$ 549,636
Total non-U.S.	3,223	2,839	39,164	39,528	83,097	80,469	125,484	122,836
Total retained loans	\$ 162,432	\$ 162,338	\$ 166,790	\$ 167,166	\$ 361,174	\$ 342,968	\$ 690,396	\$ 672,472
Loan delinquency								
Current and less than 30 days past due and still accruing	\$ 159,949	\$ 161,314	\$ 164,104	\$ 164,899	\$ 359,191	\$ 341,128	\$ 683,244	\$ 667,341
30–89 days past due and still accruing	918	473	868	884	1,152	1,090	2,938	2,447
90 or more days past due and still accruing ^(b)	126	150	58	162	88	26	272	338
Criticized nonaccrual	1,439	401	1,760	1,221	743	724	3,942	2,346
Total retained loans	\$ 162,432	\$ 162,338	\$ 166,790	\$ 167,166	\$ 361,174	\$ 342,968	\$ 690,396	\$ 672,472

(a) The U.S. and non-U.S. distribution is determined based predominantly on the domicile of the borrower.

(b) Represents loans that are considered well-collateralized and therefore still accruing interest.

Nonaccrual loans

The following table provides information on retained wholesale nonaccrual loans.

December 31, (in millions)	Secured by real estate		Commercial and industrial		Other		Total retained loans	
	2024	2023	2024	2023	2024	2023	2024	2023
Nonaccrual loans								
With an allowance	\$ 366	\$ 129	\$ 1,362	\$ 776	\$ 555	\$ 492	\$ 2,283	\$ 1,397
Without an allowance ^(a)	1,073	272	398	445	188	232	1,659	949
Total nonaccrual loans^(b)	\$ 1,439	\$ 401	\$ 1,760	\$ 1,221	\$ 743	\$ 724	\$ 3,942	\$ 2,346

(a) When the discounted cash flows or collateral value equals or exceeds the amortized cost of the loan, the loan does not require an allowance. This typically occurs when the loans have been partially charged off and/or there have been interest payments received and applied to the loan balance.

(b) Interest income on nonaccrual loans recognized on a cash basis was \$51 million and \$19 million for the years ended December 31, 2024 and 2023, respectively.

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Loan modifications

The Firm grants certain modifications of wholesale loans to borrowers experiencing financial difficulty.

Financial effects of FDMs

The following tables provide information on retained wholesale loan modifications considered FDMs.

Secured by real estate			
Year ended December 31, 2024			
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Secured by real estate loans	Financial effect of loan modifications
Single modifications			
Term extension	\$ 271	0.17 %	Extended loans by a weighted-average of 21 months
Other-than-insignificant payment deferral	37	0.02	Provided payment deferrals with delayed amounts re-amortized over the remaining tenor
Multiple modifications			
Other-than-insignificant payment deferral and interest rate reduction	46	0.03	Provided payment deferrals with delayed amounts recaptured at maturity and reduced weighted-average contractual interest by 162 bps
Total	\$ 354		

Secured by real estate			
Year ended December 31, 2023			
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Secured by real estate loans	Financial effect of loan modifications
Single modifications			
Term extension	\$ 149	0.09 %	Extended loans by a weighted-average of 14 months
Other-than-insignificant payment deferral	3	—	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining life of the loan
Multiple modifications			
Other-than-insignificant payment deferral and interest rate reduction	5	—	Provided payment deferrals with delayed amounts primarily recaptured at maturity and reduced weighted-average contractual interest by 184 bps
Other ^(a)	3	—	NM
Total	\$ 160		

(a) Includes a loan with multiple modifications.

Commercial and industrial			
Year ended December 31, 2024			
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Commercial and industrial loans	Financial effect of loan modifications
Single modifications			
Term extension	\$ 1,180	0.71 %	Extended loans by a weighted-average of 20 months
Other-than-insignificant payment deferral	464	0.28	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining tenor
Multiple modifications			
Other-than-insignificant payment deferral and term extension	175	0.10	Provided payment deferrals with delayed amounts recaptured at maturity and extended loans by a weighted-average of 18 months
Interest rate reduction and term extension	51	0.03	Reduced weighted-average contractual interest by 434 bps and extended loans by a weighted-average of 36 months
Other ^(a)	30	0.02	NM
Total	\$ 1,900		

(a) Includes loans with single and multiple modifications.

Commercial and industrial			
Year ended December 31, 2023			
(in millions, except ratios)	Amortized cost basis	% of loan modifications to total retained Commercial and industrial loans	Financial effect of loan modifications
Single modifications			
Term extension	\$ 916	0.55 %	Extended loans by a weighted-average of 17 months
Other-than-insignificant payment deferral	402	0.24	Provided payment deferrals with delayed amounts primarily recaptured at the end of the deferral period
Multiple modifications			
Other-than-insignificant payment deferral and term extension	35	0.02	Provided payment deferrals with delayed amounts primarily re-amortized over the remaining life of the loan and extended loans by a weighted-average of 7 months
Interest rate reduction and term extension	1	—	Reduced weighted average contractual interest rate over the life of the loan as a result of converting from variable to fixed rate and extended loans by a weighted-average of 16 months
Other ^(a)	9	—	NM
Total	\$ 1,363		

(a) Include loans with multiple modifications.

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(in millions, except ratios)	Other		
	Year ended December 31, 2024		
	Amortized cost basis	% of loan modifications to total retained Other loans	Financial effect of loan modification
Single modifications			
Term extension	\$ 268	0.07 %	Extended loans by a weighted-average of 28 months
Multiple modifications			
Other-than-insignificant payment deferral and term extension	2	—	Provided payment deferrals with delayed amounts recaptured at maturity and extended loans by a weighted-average of 6 months
Other ^(a)	5	—	NM
Total	\$ 275		

(a) Includes loans with a single modification.

(in millions, except ratios)	Other		
	Year ended December 31, 2023		
	Amortized cost basis	% of loan modifications to total retained Other loans	Financial effect of loan modifications
Single modifications			
Term extension	\$ 355	0.10 %	Extended loans by a weighted-average of 23 months
Multiple modifications			
Other-than-insignificant payment deferral and term extension	245	0.07	Provided payment deferrals with delayed amounts primarily recaptured at the end of the deferral period and extended loans by a weighted-average of 137 months
Other ^(a)	\$ 9	—	NM
Total	\$ 609		

(a) Includes a loan with a single modification.

Payment status of FDMs

The following table provides information on the payment status of retained wholesale FDMs during the year ended December 31, 2024 and 2023.

(in millions)	Amortized cost basis					
	Secured by real estate	Commercial and industrial	Other	Secured by real estate	Commercial and industrial	Other
	Year ended December 31, 2024			Year ended December 31, 2023		
Current and less than 30 days past due and still accruing	\$ 264	\$ 1,215	\$ 240	\$ 118	\$ 947	\$ 400
30-89 days past due and still accruing	3	13	9	2	42	—
Criticized nonaccrual	87	672	26	40	374	209
Total	\$ 354	\$ 1,900	\$ 275	\$ 160	\$ 1,363	\$ 609

Defaults of FDMs

The following table provides information on retained wholesale FDMs that defaulted during the year ended December 31, 2024 that were reported as FDMs in the twelve months prior to the default, and FDMs that defaulted during the year ended December 31, 2023 that were reported as FDMs on or after January 1, 2023, the date that the Firm adopted the changes to the TDR accounting guidance.

(in millions)	Amortized cost basis					
	Secured by real estate	Commercial and industrial	Other	Secured by real estate	Commercial and industrial	Other
	Year ended December 31, 2024			Year ended December 31, 2023		
Term extension	\$ 3	\$ 92	\$ 22	\$ 1	\$ 49	\$ 31
Other-than-insignificant payment deferral	—	118	—	2	—	—
Interest rate reduction and term extension	—	—	—	3	1	—
Total^(a)	\$ 3	\$ 210	\$ 22	\$ 6	\$ 50	\$ 31

(a) Represents FDMs that were 30 days or more past due.

As of December 31, 2024 and 2023, additional unfunded commitments on modified loans to borrowers experiencing financial difficulty were \$1.8 billion at each period in Commercial and industrial, and \$69 million and \$4 million, respectively, in Other loan class. Additional commitments on modified loans to borrowers experiencing financial difficulty whose loans have been modified as FDMs in Secured by real estate were not material at each period.

Nature and extent of TDRs

Prior to January 1, 2023, certain loan modifications were considered TDRs. These loan modifications provided various concessions to borrower who were experiencing financial difficulty. Loans with short-term or other insignificant modifications that were not considered concessions were not TDRs nor were loans for which the Firm elected to suspend TDR accounting guidance under the option provided by the CARES Act.

For the year ended December 31, 2022, new TDRs were \$801 million. New TDRs for the year ended December 31, 2022 reflected extended maturity dates and covenant waivers primarily in the Commercial and Industrial loan class. For the year ended December 31, 2022, the impact of these modifications resulting in new TDRs was not material to the Firm.

As a result of the elimination of the requirement to assess whether a modification is reasonably expected or involves a concession, the population of loans considered FDMs is greater than the population previously considered TDRs.

Note 13 – Allowance for credit losses

The Firm's allowance for credit losses represents management's estimate of expected credit losses over the remaining expected life of the Firm's financial assets measured at amortized cost and certain off-balance sheet lending-related commitments. The allowance for credit losses generally comprises:

- the allowance for loan losses, which covers the Firm's retained loan portfolios (scored and risk-rated),
- the allowance for lending-related commitments, which is presented on the Consolidated balance sheets in accounts payable and other liabilities, and
- the allowance for credit losses on investment securities, which is reflected in investment securities on the Consolidated balance sheets.

The income statement effect of all changes in the allowance for credit losses is recognized in the provision for credit losses.

Determining the appropriateness of the allowance for credit losses is complex and requires significant judgment by management about the effect of matters that are inherently uncertain. At least quarterly, the allowance for credit losses is reviewed by the CRO, the CFO and the Controller of the Firm. Subsequent evaluations of credit exposures, considering the macroeconomic conditions, forecasts and other factors then prevailing, may result in significant changes in the allowance for credit losses in future periods. The Firm's policies used to determine its allowance for loan losses and its allowance for lending-related commitments are described in the following paragraphs. Refer to Note 10 for a description of the policies used to determine the allowance for credit losses on investment securities.

Methodology for allowances for loan losses and lending-related commitments

The allowance for loan losses and allowance for lending-related commitments represents expected credit losses over the remaining expected life of retained loans and lending-related commitments that are not unconditionally cancellable. The Firm does not record an allowance for future draws on unconditionally cancellable lending-related commitments (e.g., credit cards). Expected losses related to accrued interest on credit card loans are considered in the Firm's allowance for loan losses. However, the Firm does not record an allowance on other accrued interest receivables, due to its policy to write these receivables off no later than 90 days past due by reversing interest income.

The expected life of each instrument is determined by considering its contractual term, expected prepayments, cancellation features, and certain extension and call options. The expected life of funded credit card loans is generally estimated by considering expected future payments on the credit card account, and determining how much of those amounts should be allocated to repayments of the funded loan balance (as

of the balance sheet date) versus other account activity. This allocation is made using an approach that incorporates the payment application requirements of the Credit Card Accountability Responsibility and Disclosure Act of 2009, generally paying down the highest interest rate balances first.

The estimate of expected credit losses includes expected recoveries of amounts previously charged off or expected to be charged off, even if such recoveries result in a negative allowance.

Collective and Individual Assessments

When calculating the allowance for loan losses and the allowance for lending-related commitments, the Firm assesses whether exposures share similar risk characteristics. If similar risk characteristics exist, the Firm estimates expected credit losses collectively, considering the risk associated with a particular pool and the probability that the exposures within the pool will deteriorate or default. The assessment of risk characteristics is subject to significant management judgment. Emphasizing one characteristic over another or considering additional characteristics could affect the allowance.

- Relevant risk characteristics for the consumer portfolio include product type, delinquency status, current FICO scores, geographic distribution, and, for collateralized loans, current LTV ratios.
- Relevant risk characteristics for the wholesale portfolio include risk rating, delinquency status, tenor, level and type of collateral, LOB, geography, industry, credit enhancement, product type, facility purpose, and payment terms.

The majority of the Firm's credit exposures share risk characteristics with other similar exposures, and as a result are collectively assessed for impairment ("portfolio-based component"). The portfolio-based component covers consumer loans, performing risk-rated loans and certain lending-related commitments.

If an exposure does not share risk characteristics with other exposures, the Firm generally estimates expected credit losses on an individual basis, considering expected repayment and conditions impacting that individual exposure ("asset-specific component"). The asset-specific component covers collateral-dependent loans and risk-rated loans that have been placed on nonaccrual status.

Portfolio-based component

The portfolio-based component begins with a quantitative calculation that considers the likelihood of the borrower changing delinquency status or moving from one risk rating to another. The quantitative calculation covers expected credit losses over an instrument's expected life and is estimated by applying credit loss factors to the Firm's estimated exposure at default. The credit loss factors incorporate the

probability of borrower default as well as loss severity in the event of default. They are derived using a weighted average of five internally developed macroeconomic scenarios over an eight-quarter forecast period, followed by a single year straight-line interpolation to revert to long run historical information for periods beyond the eight-quarter forecast period. The five macroeconomic scenarios consist of a central, relative adverse, extreme adverse, relative upside and extreme upside scenario, and are updated by the Firm's central forecasting team. The scenarios take into consideration the Firm's macroeconomic outlook, internal perspectives from subject matter experts across the Firm, and market consensus and involve a governed process that incorporates feedback from senior management across LOBs, Corporate Finance and Risk Management.

The quantitative calculation is adjusted to take into consideration model imprecision, emerging risk assessments, trends and other subjective factors that are not yet reflected in the calculation. These adjustments are accomplished in part by analyzing the historical loss experience, including during stressed periods, for each major product or model. Management applies judgment in making this adjustment, including taking into account uncertainties associated with the economic and political conditions, quality of underwriting standards, borrower behavior, credit concentrations or deterioration within an industry, product or portfolio, as well as other relevant internal and external factors affecting the credit quality of the portfolio. In certain instances, the interrelationships between these factors create further uncertainties.

The application of different inputs into the quantitative calculation, and the assumptions used by management to adjust the quantitative calculation, are subject to significant management judgment, and emphasizing one input or assumption over another, or considering other inputs or assumptions, could affect the estimate of the allowance for loan losses and the allowance for lending-related commitments.

Asset-specific component

To determine the asset-specific component of the allowance, collateral-dependent loans (including those loans for which foreclosure is probable) and nonaccrual risk-rated loans in the wholesale portfolio segment are generally evaluated individually.

For collateral-dependent loans, the fair value of collateral less estimated costs to sell, as applicable, is used to determine the charge-off amount for declines in value (to reduce the amortized cost of the loan to the fair value of collateral) or the amount of negative allowance that should be recognized (for recoveries of prior charge-offs associated with improvements in the fair value of the collateral).

For non-collateral dependent loans, the Firm generally measures the asset-specific allowance as the

difference between the amortized cost of the loan and the present value of the cash flows expected to be collected, discounted at the loan's effective interest rate. Subsequent changes in impairment are generally recognized as an adjustment to the allowance for loan losses. The asset-specific component of the allowance for non-collateral dependent loans incorporates the effect of the modification on the loan's expected cash flows including changes in interest rates, principal forgiveness, and other concessions, as well as management's expectation of the borrower's ability to repay under the modified terms.

Estimating the timing and amounts of future cash flows is highly judgmental as these cash flow projections rely upon estimates such as loss severities, asset valuations, the amounts and timing of interest or principal payments (including any expected prepayments) or other factors that are reflective of current and expected market conditions. These estimates are, in turn, dependent on factors such as the duration of current overall economic conditions, industry, portfolio, or borrower-specific factors, the expected outcome of insolvency proceedings as well as, in certain circumstances, other economic factors. All of these estimates and assumptions require significant management judgment and certain assumptions are highly subjective.

Other financial assets

In addition to loans and investment securities, the Firm holds other financial assets that are measured at amortized cost on the Consolidated balance sheets, including credit exposures arising from lending activities subject to collateral maintenance requirements. Management estimates the allowance for other financial assets using various techniques considering historical losses and current economic conditions.

Credit risk arising from lending activities subject to collateral maintenance requirements is generally mitigated by factors such as the short-term nature of the activity, the fair value of collateral held and the Firm's right to call for, and the borrower's obligation to provide additional margin when the fair value of the collateral declines. Because of these mitigating factors, these exposures generally do not require an allowance for credit losses. However, management may also consider other factors such as the borrower's ongoing ability to provide collateral to satisfy margin requirements, or whether collateral is significantly concentrated in an individual issuer or in securities with similar risk characteristics. If in management's judgment, an allowance for credit losses for these exposures is required, the Firm estimates expected credit losses based on the value of the collateral and probability of borrower default.

Notes to consolidated financial statements

Allowance for credit losses and related information

The table below summarizes information about the allowances for credit losses and includes a breakdown of loans and lending-related commitments by impairment methodology. Refer to Note 10 for further information on the allowance for credit losses on investment securities.

(Table continued on next page)

Year ended December 31, (in millions)	2024			
	Consumer, excluding credit card	Credit card	Wholesale	Total
Allowance for loan losses				
Beginning balance at January 1,	\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420
Cumulative effect of a change in accounting principle ^(a)	NA	NA	NA	NA
Gross charge-offs	1,299	8,198	1,022	10,519
Gross recoveries collected	(625)	(1,056)	(200)	(1,881)
Net charge-offs	674	7,142	822	8,638
Provision for loan losses	624	9,292	578	10,494
Other	1	—	68	69
Ending balance at December 31,	\$ 1,807	\$ 14,600	\$ 7,938	\$ 24,345
Allowance for lending-related commitments				
Beginning balance at January 1,	\$ 75	\$ —	\$ 1,899	\$ 1,974
Provision for lending-related commitments	7	—	121	128
Other	—	—	(1)	(1)
Ending balance at December 31,	\$ 82	\$ —	\$ 2,019	\$ 2,101
Total allowance for investment securities	NA	NA	NA	\$ 152
Total allowance for credit losses^(b)	\$ 1,889	\$ 14,600	\$ 9,957	\$ 26,598
Allowance for loan losses by impairment methodology				
Asset-specific ^(c)	\$ (728)	\$ —	\$ 526	\$ (202)
Portfolio-based	2,535	14,600	7,412	24,547
Total allowance for loan losses	\$ 1,807	\$ 14,600	\$ 7,938	\$ 24,345
Loans by impairment methodology				
Asset-specific ^(c)	\$ 2,805	\$ —	\$ 3,912	\$ 6,717
Portfolio-based	373,529	232,860	686,484	1,292,873
Total retained loans	\$ 376,334	\$ 232,860	\$ 690,396	\$ 1,299,590
Collateral-dependent loans				
Net charge-offs	\$ 1	\$ —	\$ 324	\$ 325
Loans measured at fair value of collateral less cost to sell	2,696	—	1,834	4,530
Allowance for lending-related commitments by impairment methodology				
Asset-specific	\$ —	\$ —	\$ 109	\$ 109
Portfolio-based	82	—	1,910	1,992
Total allowance for lending-related commitments^(d)	\$ 82	\$ —	\$ 2,019	\$ 2,101
Lending-related commitments by impairment methodology				
Asset-specific	\$ —	\$ —	\$ 737	\$ 737
Portfolio-based ^(e)	25,608	19	510,254	535,881
Total lending-related commitments	\$ 25,608	\$ 19	\$ 510,991	\$ 536,618

(a) Represents the impact to the allowance for loan losses upon the adoption of the Financial Instruments - Credit Losses: Troubled Debt Restructurings accounting guidance. Refer to Note 1 for further information.

(b) At December 31, 2024 and 2023 and 2022, in addition to the allowance for credit losses in the table above, the Firm also had an allowance for credit losses of \$268 million, \$243 million and \$21 million, respectively, associated with certain accounts receivable in CIB.

(c) Includes collateral-dependent loans, including those for which foreclosure is deemed probable, and nonaccrual risk-rated loans.

(d) The allowance for lending-related commitments is reported in accounts payable and other liabilities on the Consolidated balance sheets.

(e) At December 31, 2024, 2023 and 2022, lending-related commitments excluded \$19.2 billion, \$17.2 billion and \$13.1 billion, respectively, for the consumer, excluding credit card portfolio segment; \$1.0 trillion, \$915.7 billion and \$821.3 billion, respectively, for the credit card portfolio segment; and \$20.5 billion, \$19.7 billion and \$9.8 billion, respectively, for the wholesale portfolio segment, which were not subject to the allowance for lending-related commitments.

(table continued from previous page)

2023				2022			
Consumer, excluding credit card	Credit card	Wholesale	Total	Consumer, excluding credit card	Credit card	Wholesale	Total
\$ 2,040	\$ 11,200	\$ 6,486	\$ 19,726	\$ 1,765	\$ 10,250	\$ 4,371	\$ 16,386
(489)	(100)	2	(587)	NA	NA	NA	NA
1,151	5,491	1,011	7,653	812	3,192	322	4,326
(519)	(793)	(132)	(1,444)	(543)	(789)	(141)	(1,473)
632	4,698	879	6,209	269	2,403	181	2,853
936	6,048	2,484	9,468	543	3,353	2,293	6,189
1	—	21	22	1	—	3	4
\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420	\$ 2,040	\$ 11,200	\$ 6,486	\$ 19,726
\$ 76	\$ —	\$ 2,306	\$ 2,382	\$ 113	\$ —	\$ 2,148	\$ 2,261
(1)	—	(407)	(408)	(37)	—	157	120
—	—	—	—	—	—	1	1
\$ 75	\$ —	\$ 1,899	\$ 1,974	\$ 76	\$ —	\$ 2,306	\$ 2,382
NA	NA	NA	\$ 128	NA	NA	NA	\$ 96
\$ 1,931	\$ 12,450	\$ 10,013	\$ 24,522	\$ 2,116	\$ 11,200	\$ 8,792	\$ 22,204
\$ (876)	\$ —	\$ 392	\$ (484)	\$ (624)	\$ 223	\$ 467	\$ 66
2,732	12,450	7,722	22,904	2,664	10,977	6,019	19,660
\$ 1,856	\$ 12,450	\$ 8,114	\$ 22,420	\$ 2,040	\$ 11,200	\$ 6,486	\$ 19,726
\$ 3,287	\$ —	\$ 2,338	\$ 5,625	\$ 11,978	\$ 796	\$ 2,189	\$ 14,963
393,988	211,123	670,134	1,275,245	288,775	184,379	601,481	1,074,635
\$ 397,275	\$ 211,123	\$ 672,472	\$ 1,280,870	\$ 300,753	\$ 185,175	\$ 603,670	\$ 1,089,598
\$ 6	\$ —	\$ 180	\$ 186	\$ (33)	\$ —	\$ 16	\$ (17)
3,216	—	1,012	4,228	3,585	—	464	4,049
\$ —	\$ —	\$ 89	\$ 89	\$ —	\$ —	\$ 90	\$ 90
75	—	1,810	1,885	76	—	2,216	2,292
\$ 75	\$ —	\$ 1,899	\$ 1,974	\$ 76	\$ —	\$ 2,306	\$ 2,382
\$ —	\$ —	\$ 464	\$ 464	\$ —	\$ —	\$ 455	\$ 455
28,248	—	516,577	544,825	20,423	—	461,688	482,111
\$ 28,248	\$ —	\$ 517,041	\$ 545,289	\$ 20,423	\$ —	\$ 462,143	\$ 482,566

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Discussion of changes in the allowance

The allowance for credit losses as of December 31, 2024 was \$26.9 billion, reflecting a net addition of \$2.1 billion from December 31, 2023.

The net addition to the allowance for credit losses included:

- \$2.1 billion in **consumer**, reflecting:
 - a \$2.2 billion net addition in Card Services, predominantly driven by loan growth, reflecting higher revolving balances, including the seasoning of vintages originated in recent years, partially offset by
 - a \$125 million net reduction in Home Lending in the first quarter of 2024, and
- a net reduction of \$30 million in **wholesale**, reflecting:
 - changes in certain macroeconomic variables, an update to loss assumptions on certain loans in Markets, and a reduction due to charge-offs largely from collateral-dependent loans, predominantly offset by
 - net downgrade activity, primarily in Real Estate, and the impact of incorporating the First Republic portfolio into the Firm's modeled credit loss estimates in the second quarter of 2024.

The Firm's qualitative adjustments continued to include additional weight placed on the adverse scenarios to reflect ongoing uncertainties and downside risks related to the geopolitical and macroeconomic environment.

The Firm's allowance for credit losses is estimated using a weighted average of five internally developed macroeconomic scenarios. The adverse scenarios incorporate more punitive macroeconomic factors than the central case assumptions provided in the table below, resulting in:

- a weighted average U.S. unemployment rate peaking at 5.5% in the fourth quarter of 2025, and
- a weighted average U.S. real GDP level that is 1.9% lower than the central case at the end of the second quarter of 2026.

The following table presents the Firm's central case assumptions for the periods presented:

	Central case assumptions at December 31, 2024		
	2Q25	4Q25	2Q26
U.S. unemployment rate ^(a)	4.5 %	4.3 %	4.3 %
YoY growth in U.S. real GDP ^(b)	2.0 %	1.9 %	1.8 %
	Central case assumptions at December 31, 2023		
	2Q24	4Q24	2Q25
U.S. unemployment rate ^(a)	4.1 %	4.4 %	4.1 %
YoY growth in U.S. real GDP ^(b)	1.8 %	0.7 %	1.0 %

(a) Reflects quarterly average of forecasted U.S. unemployment rate.

(b) The year over year growth in U.S. real GDP in the forecast horizon of the central scenario is calculated as the percentage change in U.S. real GDP levels from the prior year.

Subsequent changes to this forecast and related estimates will be reflected in the provision for credit losses in future periods.

Refer to Note 12 for additional information on the consumer and wholesale credit portfolios.

Refer to Critical Accounting Estimates Used by the Firm on pages 161–164 for further information on the allowance for credit losses and related management judgments.

Note 14 – Variable interest entities

Refer to Note 1 on page 177 for a further description of the Firm’s accounting policies regarding consolidation of and involvement with VIEs.

The following table summarizes the most significant types of Firm-sponsored VIEs by business segment. The Firm considers a “Firm-sponsored” VIE to include any entity where: (1) JPMorganChase is the primary beneficiary of the structure; (2) the VIE is used by JPMorganChase to securitize Firm assets; (3) the VIE issues financial instruments with the JPMorganChase name; or (4) the entity is a JPMorganChase-administered asset-backed commercial paper conduit.

Line of Business	Transaction Type	Activity	2024 Form 10-K page references
CCB	Credit card securitization trusts	Securitization of originated credit card receivables	pages 271–272
	Mortgage securitization trusts	Servicing and securitization of both originated and purchased residential mortgages	pages 272–274
CIB	Mortgage and other securitization trusts	Securitization of both originated and purchased residential and commercial mortgages, and other consumer loans	pages 272–274
	Multi-seller conduits	Assisting clients in accessing the financial markets in a cost-efficient manner and structuring transactions to meet investor needs	page 274
	Municipal bond vehicles	Financing of municipal bond investments	pages 274–275

The Firm’s other business segments and Corporate are also involved with VIEs (both third-party and Firm-sponsored), but to a lesser extent, as follows:

- **Asset & Wealth Management:** AWM sponsors and manages certain funds that are deemed VIEs. As asset manager of the funds, AWM earns a fee based on assets managed; the fee varies with each fund’s investment objective and is competitively priced. For fund entities that qualify as VIEs, AWM’s interests are, in certain cases, considered to be significant variable interests that result in consolidation of the financial results of these entities.
- **Corporate:** Corporate is involved with entities that may meet the definition of VIEs; however these entities are generally subject to specialized investment company accounting, which does not require the consolidation of investments, including VIEs. In addition, Treasury and CIO invest in securities generally issued by third parties which may meet the definition of VIEs (e.g., issuers of asset-backed securities). In general, the Firm does not have the power to direct the significant activities of these entities and therefore does not consolidate these entities. Refer to Note 10 for further information on the Firm’s investment securities portfolio.

In addition, CIB also invests in and provides financing, lending-related services and other services to VIEs sponsored by third parties. Refer to page 276 of this Note for more information on the VIEs sponsored by third parties.

Significant Firm-sponsored VIEs

Credit card securitizations

CCB’s Card Services business may securitize originated credit card loans, primarily through the Chase Issuance Trust (the “Trust”). The Firm’s continuing involvement in credit card securitizations includes servicing the receivables, retaining an undivided seller’s interest in the receivables, retaining certain senior and subordinated securities and maintaining escrow accounts.

The Firm consolidates the assets and liabilities of its sponsored credit card trusts as it is considered to be the primary beneficiary of these securitization trusts based on the Firm’s ability to direct the activities of these VIEs through its servicing responsibilities and other duties, including making decisions as to the receivables that are transferred into those trusts and as to any related modifications and workouts. Additionally, the nature and extent of the Firm’s other

continuing involvement with the trusts, as indicated above, obligates the Firm to absorb losses and gives the Firm the right to receive certain benefits from these VIEs that could potentially be significant.

The underlying securitized credit card receivables and other assets of the securitization trusts are available only for payment of the beneficial interests issued by the securitization trusts; they are not available to pay the Firm’s other obligations or the claims of the Firm’s creditors.

The agreements with the credit card securitization trusts require the Firm to maintain a minimum undivided interest in the credit card trusts (generally 5%). As of December 31, 2024 and 2023, the Firm held undivided interests in Firm-sponsored credit card securitization trusts of \$6.6 billion and \$4.9 billion, respectively. The Firm maintained an average undivided interest in principal receivables owned by

Notes to consolidated financial statements

those trusts of approximately 45% and 65% for the years ended December 31, 2024 and 2023, respectively. The Firm did not retain any senior securities and retained \$1.5 billion of subordinated securities in certain of its credit card securitization trusts at both December 31, 2024 and 2023. The Firm's undivided interests in the credit card trusts and securities retained are eliminated in consolidation.

Firm-sponsored mortgage and other securitization trusts

The Firm securitizes (or has securitized) originated and purchased residential mortgages, commercial mortgages and other consumer loans primarily in its CCB and CIB businesses. Depending on the particular transaction, as well as the respective business involved, the Firm may act as the servicer of the loans and/or retain certain beneficial interests in the securitization trusts.

The following tables present the total unpaid principal amount of assets held in Firm-sponsored private-label securitization entities, including those in which the Firm has continuing involvement, and those that are consolidated by the Firm. Continuing involvement includes servicing the loans, holding senior interests or subordinated interests (including amounts required to be held pursuant to credit risk retention rules), recourse or guarantee arrangements, and derivative contracts. In certain instances, the Firm's only continuing involvement is servicing the loans. The Firm's maximum loss exposure from retained and purchased interests is the carrying value of these interests. Refer to page 279 of this Note for information on the securitization-related loan delinquencies and liquidation losses.

December 31, 2024 (in millions)	Principal amount outstanding			JPMorganChase interest in securitized assets in nonconsolidated VIEs ^{(c)(d)(e)}			
	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	Investment securities	Other financial assets	Total interests held by JPMorgan Chase
Securitization-related^(a)							
Residential mortgage:							
Prime/Alt-A and option ARMs	\$ 71,085	\$ 615	\$ 50,846	\$ 613	\$ 1,850	\$ 614	\$ 3,077
Subprime	8,824	—	1,847	44	19	—	63
Commercial and other ^(b)	186,293	243	125,510	530	5,768	1,074	7,372
Total	\$ 266,202	\$ 858	\$ 178,203	\$ 1,187	\$ 7,637	\$ 1,688	\$ 10,512

December 31, 2023 (in millions)	Principal amount outstanding			JPMorganChase interest in securitized assets in nonconsolidated VIEs ^{(c)(d)(e)}			
	Total assets held by securitization VIEs	Assets held in consolidated securitization VIEs	Assets held in nonconsolidated securitization VIEs with continuing involvement	Trading assets	Investment securities	Other financial assets	Total interests held by JPMorgan Chase
Securitization-related^(a)							
Residential mortgage:							
Prime/Alt-A and option ARMs	\$ 58,570	\$ 675	\$ 39,319	\$ 595	\$ 1,981	\$ 60	\$ 2,636
Subprime	8,881	—	1,312	3	—	—	3
Commercial and other ^(b)	168,042	—	120,262	831	5,638	1,354	7,823
Total	\$ 235,493	\$ 675	\$ 160,893	\$ 1,429	\$ 7,619	\$ 1,414	\$ 10,462

(a) Excludes U.S. GSEs and government agency securitizations and re-securitizations, which are not Firm-sponsored.

(b) Consists of securities backed by commercial real estate loans and non-mortgage-related consumer receivables.

(c) Excludes the following: retained servicing; securities retained from loan sales and securitization activity related to U.S. GSEs and government agencies; interest rate and foreign exchange derivatives primarily used to manage interest rate and foreign exchange risks of securitization entities; senior securities of \$256 million and \$52 million at December 31, 2024 and 2023, respectively, and subordinated securities which were not material at December 31, 2024 and 2023, which the Firm purchased in connection with CIB's secondary market-making activities.

(d) Includes interests held in re-securitization transactions.

(e) At both December 31, 2024 and 2023, 77% of the Firm's retained securitization interests, which are predominantly carried at fair value and include amounts required to be held pursuant to credit risk retention rules, were risk-rated "A" or better, on an S&P-equivalent basis. The retained interests in prime residential mortgages consisted of \$2.9 billion and \$2.5 billion of investment-grade retained interests at December 31, 2024 and 2023, respectively, and \$216 million and \$88 million of noninvestment-grade retained interests at December 31, 2024 and 2023, respectively. The retained interests in commercial and other securitization trusts consisted of \$6.0 billion and \$6.1 billion of investment-grade retained interests, and \$1.4 billion and \$1.7 billion of noninvestment-grade retained interests at December 31, 2024 and 2023, respectively.

Residential mortgage

The Firm securitizes residential mortgage loans originated by CCB, as well as residential mortgage loans purchased from third parties by either CCB or CIB. CCB generally retains servicing for all residential mortgage loans it originated or purchased, and for certain mortgage loans purchased by CIB. For securitizations of loans serviced by CCB, the Firm has the power to direct the significant activities of the VIE because it is responsible for decisions related to loan modifications and workouts. CCB may also retain an interest upon securitization.

In addition, CIB engages in underwriting and trading activities involving securities issued by Firm-sponsored securitization trusts. As a result, CIB at times retains senior and/or subordinated interests (including residual interests and amounts required to be held pursuant to credit risk retention rules) in residential mortgage securitizations at the time of securitization, and/or reacquires positions in the secondary market in the normal course of business. In certain instances, as a result of the positions retained or reacquired by CIB or held by Treasury and CIO or CCB, when considered together with the servicing arrangements entered into by CCB, the Firm is deemed to be the primary beneficiary of certain securitization trusts.

The Firm does not consolidate residential mortgage securitizations (Firm-sponsored or third-party-sponsored) when it is not the servicer (and therefore does not have the power to direct the most significant activities of the trust) or does not hold a beneficial interest in the trust that could potentially be significant to the trust.

Commercial mortgages and other consumer securitizations

CIB originates and securitizes commercial mortgage loans, and engages in underwriting and trading activities involving the securities issued by securitization trusts. CIB may retain unsold senior and/or subordinated interests (including amounts required to be held pursuant to credit risk retention rules) in commercial mortgage securitizations at the time of securitization but, generally, the Firm does not service commercial loan securitizations. Treasury and CIO may choose to invest in these securitizations as well. For commercial mortgage securitizations the power to direct the significant activities of the VIE generally is held by the servicer or investors in a specified class of securities ("controlling class"). The Firm generally does not retain an interest in the controlling class in its sponsored commercial mortgage securitization transactions.

Re-securitizations

The Firm engages in certain re-securitization transactions in which debt securities are transferred to a VIE in exchange for new beneficial interests. These transfers occur in connection with both U.S. GSEs and government agency sponsored VIEs, which are backed by residential mortgages. The Firm's consolidation analysis is largely dependent on the Firm's role and interest in the re-securitization trusts.

The following table presents the principal amount of securities transferred to re-securitization VIEs.

Year ended December 31, (in millions)	2024	2023	2022
Transfers of securities to VIEs			
U.S. GSEs and government agencies	\$ 44,456	\$ 18,864	\$ 16,128

Most re-securitizations with which the Firm is involved are client-driven transactions in which a specific client or group of clients is seeking a specific return or risk profile. For these transactions, the Firm has concluded that the decision-making power of the entity is shared between the Firm and its clients, considering the joint effort and decisions in establishing the re-securitization trust and its assets, as well as the significant economic interest the client holds in the re-securitization trust; therefore the Firm does not consolidate the re-securitization VIE.

The Firm did not transfer any private label securities to re-securitization VIEs during 2024, 2023 and 2022, and retained interests in any such Firm-sponsored VIEs as of December 31, 2024 and 2023 were not material.

Additionally, the Firm may invest in beneficial interests of third-party-sponsored re-securitizations and generally purchases these interests in the secondary market. In these circumstances, the Firm does not have the unilateral ability to direct the most significant activities of the re-securitization trust, either because it was not involved in the initial design of the trust, or the Firm was involved with an independent third-party sponsor and demonstrated shared power over the creation of the trust; therefore, the Firm does not consolidate the re-securitization VIE.

Notes to consolidated financial statements

The following table presents information on the Firm's interests in nonconsolidated re-securitization VIEs.

December 31, (in millions)	Nonconsolidated re-securitization VIEs	
	2024	2023
U.S. GSEs and government agencies		
Interest in VIEs	\$ 3,219	\$ 3,371

As of December 31, 2024 and 2023, the Firm did not consolidate any U.S. GSE and government agency re-securitization VIEs or any Firm-sponsored private-label re-securitization VIEs.

Multi-seller conduits

Multi-seller conduit entities are separate bankruptcy remote entities that provide secured financing, collateralized by pools of receivables and other financial assets, to customers of the Firm. The conduits fund their financing facilities through the issuance of highly rated commercial paper. The primary source of repayment of the commercial paper is the cash flows from the pools of assets. In most instances, the assets are structured with deal-specific credit enhancements provided to the conduits by the customers (i.e., sellers) or other third parties. Deal-specific credit enhancements are generally structured to cover a multiple of historical losses expected on the pool of assets, and are typically in the form of overcollateralization provided by the seller. The deal-specific credit enhancements mitigate the Firm's potential losses on its agreements with the conduits.

To ensure timely repayment of the commercial paper, and to provide the conduits with funding to provide financing to customers in the event that the conduits do not obtain funding in the commercial paper market, each asset pool financed by the conduits has a minimum 100% deal-specific liquidity facility associated with it provided by JPMorgan Chase Bank, N.A. JPMorgan Chase Bank, N.A. also provides the multi-seller conduit vehicles with uncommitted program-wide liquidity facilities and program-wide credit enhancement in the form of standby letters of credit. The amount of program-wide credit enhancement required is based upon commercial paper issuance and approximates 10% of the outstanding balance of commercial paper.

The Firm consolidates its Firm-administered multi-seller conduits, as the Firm has both the power to direct the significant activities of the conduits and a potentially significant economic interest in the conduits. As administrative agent and in its role in structuring transactions, the Firm makes decisions regarding asset types and credit quality, and manages the commercial paper funding needs of the conduits. The Firm's interests that could potentially be significant to the VIEs include the fees received as administrative agent and liquidity and program-wide

credit enhancement provider, as well as the potential exposure created by the liquidity and credit enhancement facilities provided to the conduits.

In the normal course of business, JPMorganChase makes markets in and invests in commercial paper issued by the Firm-administered multi-seller conduits. The Firm held \$2.9 billion and \$9.8 billion of the commercial paper issued by the Firm-administered multi-seller conduits at December 31, 2024 and 2023, respectively, which have been eliminated in consolidation. The Firm's investments reflect the Firm's funding needs and capacity and were not driven by market illiquidity. Other than the amounts required to be held pursuant to credit risk retention rules, the Firm is not obligated under any agreement to purchase the commercial paper issued by the Firm-administered multi-seller conduits.

Deal-specific liquidity facilities, program-wide liquidity and credit enhancement provided by the Firm have been eliminated in consolidation. The Firm or the Firm-administered multi-seller conduits provide lending-related commitments to certain clients of the Firm-administered multi-seller conduits. The unfunded commitments were \$10.3 billion and \$10.8 billion at December 31, 2024 and 2023, respectively, and are reported as off-balance sheet lending-related commitments in other unfunded commitments to extend credit. Refer to Note 28 for more information on off-balance sheet lending-related commitments.

Municipal bond vehicles

Municipal bond vehicles or tender option bond ("TOB") trusts allow institutions to finance their municipal bond investments at short-term rates. In a typical TOB transaction, the trust purchases highly rated municipal bond(s) of a single issuer and funds the purchase by issuing two types of securities: (1) puttable floating-rate certificates ("floaters") and (2) inverse floating-rate residual interests ("residuals"). The floaters are typically purchased by money market funds or other short-term investors and may be tendered, with requisite notice, to the TOB trust. The residuals are retained by the investor seeking to finance its municipal bond investment. TOB transactions where the residual is held by a third-party investor are typically known as customer TOB trusts, and non-customer TOB trusts are transactions where the Residual is retained by the Firm. Customer TOB trusts are sponsored by a third party. The Firm serves as sponsor for all non-customer TOB transactions. The Firm may provide various services to a TOB trust, including remarketing agent, liquidity or tender option provider, and/or sponsor.

J.P. Morgan Securities LLC may serve as a remarketing agent on the floaters for TOB trusts. The remarketing agent is responsible for establishing the periodic variable rate on the floaters, conducting the initial

placement and remarketing tendered floaters. The remarketing agent may, but is not obligated to, make markets in floaters. Floaters held by the Firm were not material during 2024 and 2023.

JPMorgan Chase Bank, N.A. or J.P. Morgan Securities LLC often serves as the sole liquidity or tender option provider for the TOB trusts. The liquidity provider's obligation to perform is conditional and is limited by certain events ("Termination Events"), which include bankruptcy or failure to pay by the municipal bond issuer or credit enhancement provider, an event of taxability on the municipal bonds or the immediate downgrade of the municipal bond to below investment grade. In addition, the liquidity provider's exposure is typically further limited by the high credit quality of the underlying municipal bonds, the excess collateralization in the vehicle, or, in certain

transactions, the reimbursement agreements with the Residual holders.

Holders of the floaters may "put," or tender, their floaters to the TOB trust. If the remarketing agent cannot successfully remarket the floaters to another investor, the liquidity provider either provides a loan to the TOB trust for the TOB trust's purchase of the floaters, or it directly purchases the tendered floaters.

TOB trusts are considered to be variable interest entities. The Firm consolidates non-customer TOB trusts because as the Residual holder, the Firm has the right to make decisions that significantly impact the economic performance of the municipal bond vehicle, and it has the right to receive benefits and bear losses that could potentially be significant to the municipal bond vehicle.

Consolidated VIE assets and liabilities

The following table presents information on assets and liabilities related to VIEs consolidated by the Firm as of December 31, 2024 and 2023.

December 31, 2024 (in millions)	Assets				Liabilities		
	Trading assets	Loans	Other ^(c)	Total assets ^(d)	Beneficial interests in VIE assets ^(e)	Other ^(f)	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	\$ —	\$ 13,531	\$ 168	\$ 13,699	\$ 5,312	\$ 10	\$ 5,322
Firm-administered multi-seller conduits	1	20,383	133	20,517	18,228	26	18,254
Municipal bond vehicles	3,388	—	22	3,410	3,617	15	3,632
Mortgage securitization entities ^(a)	—	630	8	638	115	48	163
Other	496	1,966 ^(b)	350	2,812	51	355	406
Total	\$ 3,885	\$ 36,510	\$ 681	\$ 41,076	\$ 27,323	\$ 454	\$ 27,777
December 31, 2023 (in millions)	Assets				Liabilities		
	Trading assets	Loans	Other ^(c)	Total assets ^(d)	Beneficial interests in VIE assets ^(e)	Other ^(f)	Total liabilities
VIE program type							
Firm-sponsored credit card trusts	\$ —	\$ 9,460	\$ 117	\$ 9,577	\$ 2,998	\$ 6	\$ 3,004
Firm-administered multi-seller conduits	1	27,372	194	27,567	17,781	30	17,811
Municipal bond vehicles	2,056	—	22	2,078	2,116	11	2,127
Mortgage securitization entities ^(a)	—	693	8	701	125	57	182
Other	113	86	250	449	—	159	159
Total	\$ 2,170	\$ 37,611	\$ 591	\$ 40,372	\$ 23,020	\$ 263	\$ 23,283

(a) Includes residential mortgage securitizations.

(b) Primarily includes consumer loans in CIB.

(c) Includes assets classified as cash and other assets on the Consolidated balance sheets.

(d) The assets of the consolidated VIEs included in the program types above are used to settle the liabilities of those entities. The assets and liabilities include third-party assets and liabilities of consolidated VIEs and exclude intercompany balances that eliminate in consolidation.

(e) The interest-bearing beneficial interest liabilities issued by consolidated VIEs are classified on the Consolidated balance sheets as "Beneficial interests issued by consolidated VIEs". The holders of these beneficial interests generally do not have recourse to the general credit of JPMorganChase. Included in beneficial interests in VIE assets are long-term beneficial interests of \$5.5 billion and \$3.1 billion at December 31, 2024 and 2023, respectively.

(f) Includes liabilities classified as accounts payable and other liabilities on the Consolidated balance sheets.

Notes to consolidated financial statements

VIEs sponsored by third parties

The Firm enters into transactions with VIEs structured by other parties. These include, for example, acting as a derivative counterparty, liquidity provider, investor, underwriter, placement agent, remarketing agent, trustee or custodian. These transactions are conducted at arm's-length, and individual credit decisions are based on the analysis of the specific VIE, taking into consideration the quality of the underlying assets. Where the Firm does not have the power to direct the activities of the VIE that most significantly impact the VIE's economic performance, or a variable interest that could potentially be significant, the Firm generally does not consolidate the VIE, but it records and reports these positions on its Consolidated balance sheets in the same manner it would record and report positions in respect of any other third-party transaction.

Tax credit vehicles

The Firm holds investments in unconsolidated tax credit vehicles, which are limited partnerships and similar entities that own and operate affordable housing, alternative energy, and other projects. These entities are primarily considered VIEs. A third party is typically the general partner or managing member and has control over the significant activities of the tax credit vehicles, and accordingly the Firm does not consolidate tax credit vehicles. The Firm generally invests in these partnerships as a limited partner and earns a return primarily through the receipt of tax credits allocated to the projects. The maximum loss exposure, represented by equity investments and funding commitments, was \$35.2 billion and \$35.1 billion, of which \$15.0 billion and \$14.7 billion was unfunded at December 31, 2024 and 2023, respectively. The Firm assesses each project and to reduce the risk of loss, may withhold varying amounts of its capital investment until the project qualifies for tax credits. Refer to Note 28 for more information on off-balance sheet lending-related commitments.

Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance which expanded the types of tax-oriented investments, beyond affordable housing tax credit investments, that the Firm can elect on a program by program basis, to be accounted for using the proportional amortization method. Refer to Notes 1, 6 and 25 for further information.

The proportional amortization method requires the cost of eligible investments, within an elected program, be amortized in proportion to the tax benefits received with the resulting amortization reported directly in income tax expense, which aligns with the associated tax credits and other tax benefits. Investments must meet certain criteria to be eligible,

including that substantially all of the return is from income tax credits and other income tax benefits.

In addition, under this method deferred taxes are generally not recorded as the investment is now amortized in proportion to the income tax credits and other income tax benefits received. Delayed equity contributions that are unconditional and legally binding or conditional and probable of occurring are recorded in other liabilities with a corresponding increase in the carrying value of the investment. The guidance also requires a reevaluation of eligible investments when significant modifications or events occur that result in a change in the nature of the investment or a change in the Firm's relationship with the underlying project. During the period, there were no significant modifications or events that resulted in a change in the nature of an eligible investment or a change in the Firm's relationship with the underlying project.

The following table provides information on tax-oriented investments for which the Firm elected to apply the proportional amortization method.

Year ended December 31, (in millions)	Alternative energy and affordable housing programs ^(d)		
	2024	2023	2022
Programs for which the Firm elected proportional amortization:			
Carrying value ^(a)	\$ 31,978	\$ 14,644	\$ 12,052
Tax credits and other tax benefits ^(b)	6,379	2,044	1,786
Investments that qualify to be accounted for using proportional amortization:			
Amortization losses recognized as a component of income tax expense	(5,018)	(1,561)	(1,353)
Non-income-tax-related gains/ (losses) and other returns received that are recognized outside of income tax expense ^(c)	142	(1)	(1)

- (a) Recorded in Other assets on the Consolidated balance sheets. Excludes programs to which the Firm does not apply the proportional amortization method, such as historic tax credit and new market tax credit programs.
- (b) Reflected in Income tax expense on the Consolidated statements of income and Operating activities on the Consolidated statements of cash flows.
- (c) Recorded in Other income on the Consolidated statements of income and Operating activities on the Consolidated statements of cash flows.
- (d) As of December 31, 2023 and 2022 represents eligible affordable housing investments.

Customer municipal bond vehicles (TOB trusts)

The Firm may provide various services to customer TOB trusts, including remarketing agent, liquidity or tender option provider. In certain customer TOB transactions, the Firm, as liquidity provider, has entered into a reimbursement agreement with the Residual holder. In those transactions, upon the termination of the vehicle, the Firm has recourse to the third-party Residual holders for any shortfall. The Firm does not have any intent to protect Residual holders from potential losses on any of the underlying municipal bonds. The Firm does not consolidate customer TOB trusts, since the Firm does not have the power to make decisions that significantly impact the economic performance of the municipal bond vehicle.

The Firm's maximum exposure as a liquidity provider to customer TOB trusts at December 31, 2024 and 2023, was \$5.8 billion and \$5.1 billion, respectively. The fair value of assets held by such VIEs at December 31, 2024 and 2023 was \$8.1 billion and \$7.3 billion, respectively.

Loan securitizations

The Firm has securitized and sold a variety of loans, including residential mortgages, credit card receivables, commercial mortgages and other consumer loans. The purposes of these securitization transactions were to satisfy investor demand and to generate liquidity for the Firm.

For loan securitizations in which the Firm is not required to consolidate the trust, the Firm records the transfer of the loan receivable to the trust as a sale when all of the following accounting criteria for a sale are met: (1) the transferred financial assets are legally isolated from the Firm's creditors; (2) the transferee or beneficial interest holder can pledge or exchange the transferred financial assets; and (3) the Firm does not maintain effective control over the transferred financial assets (e.g., the Firm cannot repurchase the transferred assets before their maturity and it does not have the ability to unilaterally cause the holder to return the transferred assets).

For loan securitizations accounted for as a sale, the Firm recognizes a gain or loss based on the difference between the value of proceeds received (including cash, beneficial interests, or servicing assets received) and the carrying value of the assets sold. Gains and losses on securitizations are reported in noninterest revenue.

Notes to consolidated financial statements

Securitization activity

The following table provides information related to the Firm's securitization activities for the years ended December 31, 2024, 2023 and 2022, related to assets held in Firm-sponsored securitization entities that were not consolidated by the Firm, and where sale accounting was achieved at the time of the securitization.

Year ended December 31, (in millions)	2024		2023		2022	
	Residential mortgage ^(d)	Commercial and other ^(e)	Residential mortgage ^(d)	Commercial and other ^(e)	Residential mortgage ^(d)	Commercial and other ^(e)
Principal securitized	\$ 19,988	\$ 17,683	\$ 7,678	\$ 3,901	\$ 10,218	\$ 9,036
All cash flows during the period:^(a)						
Proceeds received from loan sales as financial instruments ^{(b)(c)}	\$ 19,870	\$ 17,346	\$ 7,251	\$ 3,896	\$ 9,783	\$ 8,921
Servicing fees collected	35	35	24	5	62	2
Cash flows received on interests	405	1,303	325	425	489	285

(a) Excludes re-securitization transactions.

(b) Primarily includes Level 2 assets.

(c) The carrying value of the loans accounted for at fair value approximated the proceeds received upon loan sale.

(d) Represents prime mortgages. Excludes loan securitization activity related to U.S. GSEs and government agencies.

(e) Includes commercial mortgages and auto loans.

Key assumptions used to value retained interests originated during the year are shown in the table below.

Year ended December 31,	2024	2023	2022
Residential mortgage retained interest:			
Weighted-average life (in years)	4.3	9.6	10.8
Weighted-average discount rate	7.1 %	4.8 %	4.0 %
Commercial and other retained interest:			
Weighted-average life (in years)	4.5	3.0	5.9
Weighted-average discount rate	6.2 %	4.6 %	2.9 %

Loans and excess MSR sold to U.S. government-sponsored enterprises and loans in securitization transactions pursuant to Ginnie Mae guidelines

In addition to the amounts reported in the securitization activity tables above, the Firm, in the normal course of business, sells originated and purchased mortgage loans and certain originated excess MSR on a nonrecourse basis, predominantly to U.S. GSEs. These loans and excess MSR are sold primarily for the purpose of securitization by the U.S. GSEs, who provide certain guarantee provisions (e.g., credit enhancement of the loans). The Firm also sells loans into securitization transactions pursuant to Ginnie Mae guidelines; these loans are typically insured or guaranteed by another U.S. government agency. The Firm does not consolidate the securitization vehicles underlying these transactions as it is not the primary beneficiary. For a limited number of loan sales, the Firm is obligated to share a portion of the credit risk associated with the sold loans with the purchaser. Refer to Note 28 for additional information about the Firm's loan sales- and securitization-related indemnifications and Note 15 for additional information about the impact of the Firm's sale of certain excess MSR.

The following table summarizes the activities related to loans sold to the U.S. GSEs, and loans in securitization transactions pursuant to Ginnie Mae guidelines.

Year ended December 31, (in millions)	2024	2023	2022
Carrying value of loans sold	\$ 25,765	\$ 19,906	\$ 48,891
Proceeds received from loan sales as cash	\$ 2,380	\$ 300	\$ 22
Proceeds from loan sales as securities ^{(a)(b)}	23,178	19,389	48,096
Total proceeds received from loan sales^(c)	\$ 25,558	\$ 19,689	\$ 48,118
Gains/(losses) on loan sales ^{(d)(e)}	\$ —	\$ —	\$ (25)

(a) Includes securities from U.S. GSEs and Ginnie Mae that are generally sold shortly after receipt or retained as part of the Firm's investment securities portfolio.

(b) Included in level 2 assets.

(c) Excludes the value of MSR retained upon the sale of loans.

(d) Gains/(losses) on loan sales include the value of MSR.

(e) The carrying value of the loans accounted for at fair value approximated the proceeds received upon loan sale.

Options to repurchase delinquent loans

In addition to the Firm's obligation to repurchase certain loans due to material breaches of representations and warranties as discussed in Note 28, the Firm also has the option to repurchase delinquent loans that it services for Ginnie Mae loan pools, as well as for other U.S. government agencies under certain arrangements. The Firm typically elects to repurchase delinquent loans from Ginnie Mae loan pools as it continues to service them and/or manage the foreclosure process in accordance with the applicable requirements, and such loans continue to be insured or guaranteed. When the Firm's repurchase option becomes exercisable, such loans must be reported on the Consolidated balance sheets as a loan with a corresponding liability. Refer to Note 12 for additional information.

The following table presents loans the Firm repurchased or had an option to repurchase, real estate owned, and foreclosed government-guaranteed residential mortgage loans recognized on the Firm's Consolidated balance sheets as of December 31, 2024 and 2023. Substantially all of these loans and real estate are insured or guaranteed by U.S. government agencies.

December 31, (in millions)	2024	2023
Loans repurchased or option to repurchase ^(a)	\$ 577	\$ 597
Real estate owned	6	8
Foreclosed government-guaranteed residential mortgage loans ^(b)	10	22

(a) Primarily all of these amounts relate to loans that have been repurchased from Ginnie Mae loan pools.

(b) Relates to voluntary repurchases of loans, which are included in accrued interest and accounts receivable.

Loan delinquencies and liquidation losses

The table below includes information about components of and delinquencies related to nonconsolidated securitized financial assets held in Firm-sponsored private-label securitization entities, in which the Firm has continuing involvement as of December 31, 2024 and 2023. For loans sold or securitized where servicing is the Firm's only form of continuing involvement, the Firm generally experiences a loss only if the Firm was required to repurchase a delinquent loan or foreclosed asset due to a breach in representations and warranties associated with its loan sale or servicing contracts.

As of or for the year ended December 31, (in millions)	Securitized assets		90 days past due		Net liquidation losses / (recoveries)	
	2024	2023	2024	2023	2024	2023
Securitized loans						
Residential mortgage:						
Prime/ Alt-A & option ARMs	\$ 50,846	\$ 39,319	\$ 501	\$ 440	\$ 10	\$ 14
Subprime	1,847	1,312	113	131	2	5
Commercial and other	125,510	120,262	1,715	2,874	77	60
Total loans securitized	\$ 178,203	\$ 160,893	\$ 2,329	\$ 3,445	\$ 89	\$ 79

Note 15 – Goodwill, mortgage servicing rights, and other intangible assets**Goodwill**

Goodwill is recorded upon completion of a business combination as the difference between the purchase price and the fair value of the net assets acquired, and can be adjusted up to one year from the acquisition date as additional information pertaining to facts and circumstances that existed as of the acquisition date is obtained about the fair value of assets acquired and liabilities assumed. Subsequent to initial recognition, goodwill is not amortized but is tested for impairment during the fourth quarter of each fiscal year, or more often if events or circumstances, such as adverse changes in the business climate, indicate that there may be an impairment.

The goodwill associated with each business combination is allocated to the related reporting units, which are generally determined based on how the Firm's businesses are managed and how they are reviewed. The following table presents goodwill attributed to the reportable business segments and Corporate.

December 31, (in millions)	2024	2023	2022
Consumer & Community Banking	\$ 32,116	\$ 32,116	\$ 32,121
Commercial & Investment Bank	11,236	11,251	10,993
Asset & Wealth Management	8,521	8,582	7,902
Corporate	692	685	646
Total goodwill	\$ 52,565	\$ 52,634	\$ 51,662

The following table presents changes in the carrying amount of goodwill.

(in millions)	2024	2023	2022
Balance at beginning of period	\$ 52,634	\$ 51,662	\$ 50,315
Changes during the period from:			
Business combinations ^(a)	29	917	1,426
Other ^(b)	(98)	55	(79)
Balance at December 31,	\$ 52,565	\$ 52,634	\$ 51,662

(a) For 2024, includes estimated goodwill associated with the acquisition of LayerOne Financial in CIB. For 2023, predominantly represents estimated goodwill associated with the acquisition of the remaining 51% interest in CIFM in AWM and the acquisition of Aumni Inc., predominantly in CIB. For 2022, represents estimated goodwill associated with the acquisitions of Global Shares PLC in AWM, Frosch Travel Group, LLC and Figg, Inc. in CCB, and Renovite Technologies, Inc. and Volkswagen Payments S.A. in CIB.

(b) Primarily foreign currency adjustments.

Goodwill impairment testing

The Firm's goodwill was not impaired at December 31, 2024, 2023 and 2022.

The goodwill impairment test is generally performed by comparing the current fair value of each reporting unit with its carrying value. If the fair value is in excess of the carrying value, then the reporting unit's goodwill is considered not to be impaired. If the fair value is less than the carrying value, then an impairment is recognized for the amount by which the reporting unit's carrying value exceeds its fair value, up to the amount of goodwill allocated to that reporting unit.

The Firm uses the reporting units' allocated capital plus goodwill and other intangible assets as a proxy for the carrying values of equity for the reporting units in the goodwill impairment testing. Reporting unit equity is determined on a similar basis as the allocation of capital to the LOBs which takes into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. LOB's allocated capital levels are incorporated into the Firm's annual budget process, which is reviewed by the Firm's Board of Directors and Operating Committee.

The primary method the Firm uses to estimate the fair value of its reporting units is the income approach. This approach projects cash flows for the forecast period and uses the perpetuity growth method to calculate terminal values. These cash flows and terminal values, which are based on the reporting units' annual budgets and forecasts are then discounted using an appropriate discount rate. The discount rate used for each reporting unit represents an estimate of the cost of equity for that reporting unit and is determined considering the Firm's overall estimated cost of equity (estimated using the Capital Asset Pricing Model), as adjusted for the risk characteristics specific to each reporting unit (for example, for higher levels of risk or uncertainty associated with the business or management's forecasts and assumptions). To assess the reasonableness of the discount rates used for each reporting unit, management compares the discount rate to the estimated cost of equity for publicly traded institutions with similar businesses and risk characteristics. In addition, the weighted average cost of equity (aggregating the various reporting units) is compared with the Firm's overall estimated cost of equity for reasonableness. The valuations derived from the discounted cash flow analysis are then compared with market-based trading and transaction multiples for relevant competitors. Trading and transaction comparables are used as general indicators to assess the overall reasonableness of the estimated fair values, although precise conclusions generally cannot be drawn due to the differences that

naturally exist between the Firm's businesses and competitor institutions.

The Firm also takes into consideration a comparison between the aggregate fair values of the Firm's reporting units and JPMorganChase's market capitalization. In evaluating this comparison, the Firm considers several factors, including (i) a control premium that would exist in a market transaction, (ii) factors related to the level of execution risk that would exist at the Firmwide level that do not exist at the reporting unit level and (iii) short-term market volatility and other factors that do not directly affect the value of individual reporting units.

Unanticipated declines in business performance, increases in credit losses, increases in capital requirements, as well as deterioration in economic or market conditions, adverse regulatory or legislative changes or increases in the estimated market cost of equity, could cause the estimated fair values of the Firm's reporting units to decline in the future, which could result in a material impairment loss to earnings in a future period related to some portion of the associated goodwill.

Mortgage servicing rights

MSRs represent the fair value of expected future cash flows for performing servicing activities for others. The fair value considers estimated future servicing fees and ancillary revenue, offset by estimated costs to service the loans, and generally declines over time as net servicing cash flows are received, effectively amortizing the MSR asset against contractual servicing and ancillary fee income. MSRs are either purchased from third parties or recognized upon sale or securitization of mortgage loans if servicing is retained.

As permitted by U.S. GAAP, the Firm has elected to account for its MSRs at fair value. The Firm treats its MSRs as a single class of servicing assets based on the availability of market inputs used to measure the fair value of its MSR asset and its treatment of MSRs as one aggregate pool for risk management purposes. The Firm estimates the fair value of MSRs using an option-adjusted spread ("OAS") model, which projects MSR cash flows over multiple interest rate scenarios in conjunction with the Firm's prepayment model, and then discounts these cash flows at risk-adjusted rates. The model considers portfolio characteristics, contractually specified servicing fees, prepayment assumptions, delinquency rates, costs to service, late charges and other ancillary revenue, and other economic factors. The Firm compares fair value estimates and assumptions to observable market data where available, and also considers recent market activity and actual portfolio experience.

Notes to consolidated financial statements

The fair value of MSRs is sensitive to changes in interest rates, including their effect on prepayment speeds. MSRs typically decrease in value when interest rates decline because declining interest rates tend to increase prepayments and therefore reduce the expected life of the net servicing cash flows that comprise the MSR asset. Conversely, securities (e.g., mortgage-backed securities), and certain derivatives (e.g., those for which the Firm receives fixed-rate

interest payments) increase in value when interest rates decline. JPMorganChase uses combinations of derivatives and securities to manage the risk of changes in the fair value of MSRs. The intent is to offset any interest-rate related changes in the fair value of MSRs with changes in the fair value of the related risk management instruments.

The following table summarizes MSR activity for the years ended December 31, 2024, 2023 and 2022.

As of or for the year ended December 31, (in millions, except where otherwise noted)	2024	2023	2022
Fair value at beginning of period	\$ 8,522	\$ 7,973	\$ 5,494
MSR activity:			
Originations of MSRs	325	253	798
Purchase of MSRs ^(a)	601	1,028	1,400
Disposition of MSRs ^(b)	(21)	(188)	(822)
Net additions/(dispositions)	905	1,093	1,376
Changes due to collection/realization of expected cash flows	(1,068)	(1,011)	(936)
Changes in valuation due to inputs and assumptions:			
Changes due to market interest rates and other ^(c)	670	424	2,022
Changes in valuation due to other inputs and assumptions:			
Projected cash flows (e.g., cost to service)	102	(22)	14
Discount rates	14	14	—
Prepayment model changes and other ^(d)	(24)	51	3
Total changes in valuation due to other inputs and assumptions	92	43	17
Total changes in valuation due to inputs and assumptions	762	467	2,039
Fair value at December 31,	\$ 9,121	\$ 8,522	\$ 7,973
Change in unrealized gains/(losses) included in income related to MSRs held at December 31,	\$ 762	\$ 467	\$ 2,039
Contractual service fees, late fees and other ancillary fees included in income	1,606	1,590	1,535
Third-party mortgage loans serviced at December 31, (in billions)	652	632	584
Servicer advances, net of an allowance for uncollectible amounts, at December 31 ^(e)	577	659	758

(a) Includes purchase price adjustments associated with MSRs purchased, primarily as a result of loans that prepaid within 90 days of settlement, allowing the Firm to recover the purchase price.

(b) Includes excess MSRs transferred to agency-sponsored trusts in exchange for stripped mortgage-backed securities ("SMBS"). In each transaction, a portion of the SMBS was acquired by third parties at the transaction date; the Firm acquired the remaining balance of those SMBS as trading securities.

(c) Represents both the impact of changes in estimated future prepayments due to changes in market interest rates, and the difference between actual and expected prepayments.

(d) Represents changes in prepayments other than those attributable to changes in market interest rates.

(e) Represents amounts the Firm pays as the servicer (e.g., scheduled principal and interest, taxes and insurance), which will generally be reimbursed within a short period of time after the advance from future cash flows from the trust or the underlying loans. The Firm's credit risk associated with these servicer advances is minimal because reimbursement of the advances is typically senior to all cash payments to investors. In addition, the Firm maintains the right to stop payment to investors if the collateral is insufficient to cover the advance. However, certain of these servicer advances may not be recoverable if they were not made in accordance with applicable rules and agreements.

The following table presents the components of mortgage fees and related income (including the impact of MSR risk management activities) for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024	2023	2022
CCB mortgage fees and related income			
Production revenue	\$ 627	\$ 421	\$ 497
Net mortgage servicing revenue:			
Operating revenue:			
Loan servicing revenue	1,659	1,634	1,582
Changes in MSR asset fair value due to collection/realization of expected cash flows	(1,067)	(1,011)	(936)
Total operating revenue	592	623	646
Risk management:			
Changes in MSR asset fair value due to market interest rates and other ^(a)	670	424	2,022
Other changes in MSR asset fair value due to other inputs and assumptions in model ^(b)	92	43	17
Change in derivative fair value and other	(603)	(336)	(1,946)
Total risk management	159	131	93
Total net mortgage servicing revenue	751	754	739
Total CCB mortgage fees and related income	1,378	1,175	1,236
All other	23	1	14
Mortgage fees and related income	\$ 1,401	\$ 1,176	\$ 1,250

- (a) Represents both the impact of changes in estimated future prepayments due to changes in market interest rates, and the difference between actual and expected prepayments.
- (b) Represents the aggregate impact of changes in model inputs and assumptions such as projected cash flows (e.g., cost to service), discount rates and changes in prepayments other than those attributable to changes in market interest rates (e.g., changes in prepayments due to changes in home prices).

Changes in fair value based on variations in assumptions generally cannot be easily extrapolated, because the relationship of the change in the assumptions to the change in fair value are often highly interrelated and may not be linear. In the following table, the effect that a change in a particular assumption may have on the fair value is calculated without changing any other assumption. In reality, changes in one factor may result in changes in another, which would either magnify or counteract the impact of the initial change.

The table below outlines the key economic assumptions used to determine the fair value of the Firm's MSRs at December 31, 2024 and 2023, and outlines the sensitivities of those fair values to immediate adverse changes in those assumptions, as defined below.

December 31, (in millions, except rates)	2024	2023
Weighted-average prepayment speed assumption (constant prepayment rate)	6.19 %	6.29 %
Impact on fair value of 10% adverse change	\$ (209)	\$ (206)
Impact on fair value of 20% adverse change	(406)	(401)
Weighted-average option adjusted spread ^(a)	5.97 %	6.10 %
Impact on fair value of 100 basis points adverse change	\$ (391)	\$ (369)
Impact on fair value of 200 basis points adverse change	(751)	(709)

(a) Includes the impact of operational risk and regulatory capital.

Notes to consolidated financial statements

Other intangible assets

The Firm's finite-lived and indefinite-lived other intangible assets are initially recorded at their fair value primarily upon completion of a business combination. Subsequently, the Firm's finite-lived intangible assets, including core deposit intangibles, customer relationship intangibles, and certain other intangible assets, are amortized over their useful lives, estimated based on the expected future economic benefits to the Firm of the intangible asset. The Firm's intangible assets with indefinite lives, such as asset management contracts, are not subject to amortization and are assessed periodically for impairment.

As of December 31, 2024 and 2023, the gross carrying values of other intangible assets were \$3.8 billion and \$4.2 billion, respectively, and the accumulated amortization was \$879 million and \$994 million, respectively.

As of December 31, 2024 and 2023, the net carrying values consist of finite-lived intangible assets of \$1.7 billion and \$2.0 billion, respectively, as well as indefinite-lived intangible assets, which are not subject to amortization, of \$1.2 billion at both periods.

As of December 31, 2024, other intangible assets reflected core deposit and certain wealth management customer relationship intangibles related to the First Republic acquisition, and asset management contracts related to the Firm's acquisition of the remaining 51% interest in CIFM. Refer to Note 34 for additional information on the First Republic acquisition.

For the years ended December 31, 2024 and 2023, amortization expense was \$339 million and \$315 million, respectively.

The following table presents estimated future amortization expense.

December 31, (in millions)	Finite-lived intangible assets
2025	\$ 288
2026	285
2027	284
2028	267
2029	261

Impairment testing

The Firm's finite-lived and indefinite-lived other intangible assets are assessed for impairment annually or more often if events or changes in circumstances indicate that the asset might be impaired. Once the Firm determines that an impairment exists for an intangible asset, the impairment is recognized in other expense.

Note 16 – Premises and equipment

Premises and equipment includes land carried at cost, as well as buildings, leasehold improvements, internal-use software and furniture and equipment carried at cost less accumulated depreciation and amortization. The Firm's operating lease right-of-use assets are also included in Premises and equipment. Refer to Note 18 for a further discussion of the Firm's right-of-use assets.

The following table presents certain components of Premises and equipment.

December 31, (in millions)	2024	2023
Land, buildings and leasehold improvements	\$ 16,874	\$ 14,862
Right-of-use assets ^(a)	7,930	7,917
Other premises and equipment ^(b)	7,419	7,378
Total premises and equipment	\$ 32,223	\$ 30,157

(a) Excluded \$564 million and \$514 million of right-of-use assets that were recorded in Other assets at December 31, 2024 and 2023, respectively.

(b) Other premises and equipment is comprised of internal-use software and furniture and equipment.

JPMorganChase computes depreciation using the straight-line method over the estimated useful life for buildings and furniture and equipment. The Firm depreciates leasehold improvements over the lesser of the remainder of the lease term or the estimated useful life. The Firm also capitalizes certain costs associated with the acquisition or development of internal-use software. Once the software is ready for its intended use, these costs are amortized on a straight-line basis over the software's expected useful life. The estimated useful lives range from 10 to 50 years for buildings and leasehold improvements, and 3 to 10 years for internal-use software and furniture and equipment.

Impairment is assessed when events or changes in circumstances indicate that the carrying value of an asset may not be fully recoverable.

Note 17 – Deposits

As of December 31, 2024 and 2023, noninterest-bearing and interest-bearing deposits were as follows:

December 31, (in millions)	2024	2023
U.S. offices		
Noninterest-bearing (included \$28,904 and \$75,393 at fair value) ^(a)	\$ 592,500	\$ 643,748
Interest-bearing (included \$1,101 and \$573 at fair value) ^(a)	1,345,914	1,303,100
Total deposits in U.S. offices	1,938,414	1,946,848
Non-U.S. offices		
Noninterest-bearing (included \$2,255 and \$1,737 at fair value) ^(a)	26,806	23,097
Interest-bearing (included \$1,508 and \$681 at fair value) ^(a)	440,812	430,743
Total deposits in non-U.S. offices	467,618	453,840
Total deposits	\$2,406,032	\$ 2,400,688

(a) Includes structured notes classified as deposits for which the fair value option has been elected. Refer to Note 3 for further discussion.

As of December 31, 2024 and 2023, time deposits in denominations that met or exceeded the insured limit were as follows:

December 31, (in millions)	2024	2023
U.S. offices	\$ 149,239	\$ 132,654
Non-U.S. offices ^(a)	92,639	90,187
Total	\$ 241,878	\$ 222,841

(a) Represents all time deposits in non-U.S. offices as these deposits typically exceed the insured limit.

As of December 31, 2024, the remaining maturities of interest-bearing time deposits were as follows:

December 31, (in millions)	U.S.	Non-U.S.	Total
2025	\$222,676	\$ 89,427	\$ 312,103
2026	749	87	836
2027	482	2	484
2028	149	18	167
2029	314	721	1,035
After 5 years	162	129	291
Total	\$224,532	\$ 90,384	\$ 314,916

Note 18 - Leases

Firm as lessee

At December 31, 2024, JPMorganChase and its subsidiaries were obligated under a number of noncancellable leases, predominantly operating leases for premises and equipment used primarily for business purposes. These leases generally have terms of 20 years or less, determined based on the contractual maturity of the lease, and include periods covered by options to extend or terminate the lease when the Firm is reasonably certain that it will exercise those options. All leases with lease terms greater than twelve months are reported as a lease liability with a corresponding right-of-use ("ROU") asset. None of these lease agreements impose restrictions on the Firm's ability to pay dividends, engage in debt or equity financing transactions or enter into further lease agreements. Certain of these leases contain escalation clauses that will increase rental payments based on maintenance, utility and tax increases, which are non-lease components. The Firm elected not to separate lease and non-lease components of a contract for its real estate leases. As such, real estate lease payments represent payments on both lease and non-lease components.

Operating lease liabilities and ROU assets are recognized at the lease commencement date based on the present value of the future minimum lease payments over the lease term. The future lease payments are discounted at a rate that estimates the Firm's collateralized borrowing rate for financing instruments of a similar term and are included in accounts payable and other liabilities. The operating lease ROU assets, predominantly included in premises and equipment, also include any lease prepayments made, plus initial direct costs incurred, less any lease incentives received. Rental expense associated with operating leases is recognized on a straight-line basis over the lease term, and generally included in occupancy expense in the Consolidated statements of income.

The carrying values of the Firm's operating leases were as follows:

December 31, (in millions, except where otherwise noted)	2024	2023
Right-of-use assets	\$ 8,494	\$ 8,431
Lease liabilities	8,900	8,833
Weighted average remaining lease term (in years)	8.3	8.4
Weighted average discount rate	4.24 %	4.01 %

Supplemental cash flow information

Cash paid for amounts included in the measurement of lease liabilities - operating cash flows	\$ 1,734	\$ 1,662
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Supplemental non-cash information

Right-of-use assets obtained in exchange for operating lease obligations	\$ 1,565	\$ 2,094
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Year ended December 31, (in millions)	2024	2023
Rental expense		
Gross rental expense	\$ 2,231	\$ 2,079
Sublease rental income	(41)	(72)
Net rental expense	\$ 2,190	\$ 2,007

The following table presents future payments under operating leases as of December 31, 2024.

Year ended December 31, (in millions)	
2025	1,709
2026	1,553
2027	1,412
2028	1,248
2029	1,048
After 2029	3,721
Total future minimum lease payments	10,691
Less: Imputed interest	(1,791)
Total	\$ 8,900

In addition to the table above, as of December 31, 2024, the Firm had additional future operating lease commitments of \$887 million that were signed but had not yet commenced. These operating leases will commence between 2025 and 2027 with lease terms up to 21 years.

Firm as lessor

The Firm provides auto and equipment lease financing to its customers through lease arrangements with lease terms that may contain renewal, termination and/or purchase options. The Firm's lease financings are predominantly auto operating leases. These assets subject to operating leases are recognized in other assets on the Firm's Consolidated balance sheets and are depreciated on a straight-line basis over the lease term to reduce the asset to its estimated residual value. Depreciation expense is included in technology, communications and equipment expense in the Consolidated statements of income. The Firm's lease income is generally recognized on a straight-line basis over the lease term and is included in other income in the Consolidated statements of income.

On a periodic basis, the Firm assesses leased assets for impairment, and if the carrying amount of the leased asset exceeds the undiscounted cash flows from the lease payments and the estimated residual value upon disposition of the leased asset, an impairment is recognized.

The risk of loss on auto and equipment leased assets relating to the residual value of the leased assets is monitored through projections of the asset residual values at lease origination and periodic review of residual values, and is mitigated through arrangements with certain manufacturers or lessees.

The following table presents the carrying value of assets subject to leases reported on the Consolidated balance sheets.

December 31, (in millions)	2024	2023
Carrying value of assets subject to operating leases, net of accumulated depreciation	\$ 12,988	\$ 10,663
Accumulated depreciation	2,509	3,288

The following table presents the Firm's operating lease income and the related depreciation expense on the Consolidated statements of income.

Year ended December 31, (in millions)	2024	2023	2022
Operating lease income	\$ 2,795	\$ 2,843	\$ 3,654
Depreciation expense	1,685	1,778	2,475

The following table presents future receipts under operating leases as of December 31, 2024.

Year ended December 31, (in millions)	
2025	\$ 2,381
2026	1,707
2027	704
2028	47
2029	5
After 2029	4
Total future minimum lease receipts	\$ 4,848

Note 19 – Accounts payable and other liabilities

Accounts payable and other liabilities consist of brokerage payables, which include payables to customers and payables related to security purchases that did not settle, as well as other accrued expenses, such as compensation accruals, credit card rewards liability, merchant servicing payables, operating lease liabilities, accrued interest payables, income tax payables and litigation reserves.

The following table presents the components of accounts payable and other liabilities.

December 31, (in millions)	2024	2023
Brokerage payables	\$ 153,153	\$ 161,960
Other payables and liabilities ^(a)	127,519	128,347
Total accounts payable and other liabilities	\$ 280,672	\$ 290,307

(a) Includes credit card rewards liability of \$14.4 billion and \$13.2 billion at December 31, 2024 and 2023, respectively.

The credit card rewards liability represents the estimated cost of rewards points earned and expected to be redeemed by cardholders. The liability is accrued as the cardholder earns the benefit and is reduced when the cardholder redeems points. The redemption rate and cost per point assumptions are key assumptions to estimate the liability and the current period impact is recognized in Card Income.

Refer to Notes 7, 18, 25 and 30 for additional information on accrued interest, operating lease liabilities, income taxes and litigation reserves, respectively.

Note 20 – Long-term debt

JPMorganChase issues long-term debt denominated in various currencies, predominantly U.S. dollars, with both fixed and variable interest rates. Included in senior and subordinated debt below are various equity-linked or other indexed instruments, which the Firm has elected to measure at fair value. Changes in fair value are recorded in principal transactions revenue in the Consolidated statements of income, except for unrealized gains/(losses) due to DVA which are recorded in OCI. The following table is a summary of long-term debt carrying values (including unamortized premiums and discounts, issuance costs, valuation adjustments and fair value adjustments, where applicable) by remaining contractual maturity as of December 31, 2024.

By remaining maturity at December 31, (in millions, except rates)		2024				2023
		Under 1 year	1-5 years	After 5 years	Total	Total
Parent company						
Senior debt:	Fixed rate	\$ 7,112	\$ 90,132	\$ 117,667	\$ 214,911	\$ 200,984
	Variable rate	255	6,838	1,562	8,655	8,105
	Interest rates ^(f)	3.04 %	3.40 %	4.02 %	3.71 %	3.32 %
Subordinated debt:	Fixed rate	\$ 302	\$ 5,582	\$ 8,573	\$ 14,457	\$ 17,725
	Variable rate	—	—	—	—	—
	Interest rates ^(f)	7.75 %	4.72 %	4.69 %	4.76 %	4.62 %
	Subtotal	\$ 7,669	\$ 102,552	\$ 127,802	\$ 238,023	\$ 226,814
Subsidiaries						
Federal Home Loan Banks advances:	Fixed rate	\$ 7,582	\$ 1,651	\$ 24	\$ 9,257	\$ 23,246
	Variable rate	4,000	16,000	—	20,000	18,000
	Interest rates ^(f)	4.42 %	4.84 %	5.91 %	4.67 %	4.89 %
Purchase Money Note ^(a) :	Fixed rate	\$ —	\$ 49,208	\$ —	\$ 49,208	\$ 48,989
	Interest rates ^(f)	— %	3.40 %	— %	3.40 %	3.40 %
Senior debt:	Fixed rate	\$ 2,361	\$ 16,695	\$ 7,489	\$ 26,545	\$ 20,745
	Variable rate	19,350	30,981	6,451	56,782	52,048
	Interest rates ^(f)	5.39 %	5.18 %	1.44 %	3.81 %	3.91 %
Subordinated debt:	Fixed rate	\$ —	\$ —	\$ —	\$ —	\$ 255
	Variable rate	—	—	—	—	—
	Interest rates ^(f)	— %	— %	— %	— %	8.25 %
	Subtotal	\$ 33,293	\$ 114,535	\$ 13,964	\$ 161,792	\$ 163,283
Junior subordinated debt:	Fixed rate	\$ —	\$ —	\$ 488	\$ 488	\$ 518
	Variable rate	—	421	694	1,115	1,210
	Interest rates ^(f)	— %	5.35 %	7.01 %	6.58 %	7.14 %
	Subtotal	\$ —	\$ 421	\$ 1,182	\$ 1,603	\$ 1,728
Total long-term debt^{(b)(c)(d)}		\$ 40,962	\$ 217,508	\$ 142,948	\$ 401,418	\$ 391,825
Long-term beneficial interests:						
	Fixed rate	\$ 999	\$ 4,313	\$ —	\$ 5,312	\$ 2,998
	Variable rate	—	27	139	166	125
	Interest rates ^(f)	3.97 %	4.82 %	2.92 %	4.62 %	4.69 %
Total long-term beneficial interests^(e)		\$ 999	\$ 4,340	\$ 139	\$ 5,478	\$ 3,123

(a) Reflects the Purchase Money Note associated with First Republic. Refer to Note 34 for additional information.

(b) Included long-term debt of \$80.9 billion and \$93.0 billion secured by assets totaling \$185.5 billion and \$218.5 billion at December 31, 2024 and 2023, respectively. The amount of long-term debt secured by assets does not include amounts related to hybrid instruments.

(c) Included \$100.8 billion and \$87.9 billion of long-term debt accounted for at fair value at December 31, 2024 and 2023, respectively.

(d) Included \$13.5 billion and \$12.5 billion of outstanding zero-coupon notes at December 31, 2024 and 2023, respectively. The aggregate principal amount of these notes at their respective maturities is \$50.2 billion and \$47.9 billion, respectively. The aggregate principal amount reflects the contractual principal payment at maturity, which may exceed the contractual principal payment at the Firm's next call date, if applicable.

(e) Included on the Consolidated balance sheets in beneficial interests issued by consolidated VIEs. Also included amounts accounted for at fair value which were not material as of December 31, 2024 and 2023. Excluded short-term commercial paper and other short-term beneficial interests of \$21.8 billion and \$19.9 billion at December 31, 2024 and 2023, respectively.

(f) The interest rates shown are the weighted average of contractual rates in effect at December 31, 2024 and 2023, respectively, including non-U.S. dollar fixed- and variable-rate issuances, which excludes the effects of the associated derivative instruments used in hedge accounting relationships, if applicable. The interest rates shown exclude structured notes accounted for at fair value.

(g) As of December 31, 2024, long-term debt in the aggregate of \$297.1 billion was redeemable at the option of JPMorganChase, in whole or in part, prior to maturity, based on the terms specified in the respective instruments.

(h) The aggregate carrying values of debt that matures in each of the five years subsequent to 2024 is \$41.0 billion in 2025, \$64.5 billion in 2026, \$32.7 billion in 2027, \$93.4 billion in 2028 and \$26.9 billion in 2029.

Notes to consolidated financial statements

The weighted-average contractual interest rates for total long-term debt excluding structured notes accounted for at fair value were 3.82% and 3.65% as of December 31, 2024 and 2023, respectively. In order to modify exposure to interest rate and currency exchange rate movements, JPMorganChase utilizes derivative instruments, primarily interest rate and cross-currency interest rate swaps, in conjunction with some of its debt issuances. The use of these instruments modifies the Firm's interest expense on the associated debt. The modified weighted-average interest rates for total long-term debt, including the effects of related derivative instruments, were 5.15% and 5.20% as of December 31, 2024 and 2023, respectively.

JPMorgan Chase & Co. has guaranteed certain long-term debt of its subsidiaries, including structured notes. These guarantees rank pari passu with the Firm's other unsecured and unsubordinated indebtedness. The amount of such guaranteed long-term debt and structured notes was \$41.2 billion and \$41.1 billion at December 31, 2024 and 2023, respectively.

The Firm's unsecured debt does not contain requirements that would call for an acceleration of payments, maturities or changes in the structure of the existing debt, provide any limitations on future borrowings or require additional collateral, based on unfavorable changes in the Firm's credit ratings, financial ratios, earnings or stock price.

Note 21 – Preferred stock

At December 31, 2024 and 2023, JPMorganChase was authorized to issue 200 million shares of preferred stock, in one or more series, with a par value of \$1 per share. In the event of a liquidation or dissolution of the Firm, JPMorganChase's preferred stock then outstanding takes precedence over the Firm's common stock with respect to the payment of dividends and the distribution of assets.

The following is a summary of JPMorganChase's non-cumulative preferred stock outstanding as of December 31, 2024 and 2023, and the quarterly dividend declarations for the years ended December 31, 2024, 2023 and 2022.

	Shares ^(a)		Carrying value (in millions)		Issue date	Contractual rate in effect at December 31, 2024	Earliest redemption date ^(b)	Floating annualized rate ^(c)	Dividend declared per share ^(d)		
	December 31,		December 31,						Year ended December 31,		
	2024	2023	2024	2023					2024	2023	2022
Fixed-rate:											
Series DD	169,625	169,625	1,696	1,696	9/21/2018	5.750	12/1/2023	NA	575.00	575.00	575.00
Series EE	185,000	185,000	1,850	1,850	1/24/2019	6.000	3/1/2024	NA	600.00	600.00	600.00
Series GG	90,000	90,000	900	900	11/7/2019	4.750	12/1/2024	NA	475.00	475.00	475.00
Series JJ	150,000	150,000	1,500	1,500	3/17/2021	4.550	6/1/2026	NA	455.00	455.00	455.00
Series LL	185,000	185,000	1,850	1,850	5/20/2021	4.625	6/1/2026	NA	462.52	462.52	462.52
Series MM	200,000	200,000	2,000	2,000	7/29/2021	4.200	9/1/2026	NA	420.00	420.00	420.00
Fixed-to-floating rate:											
Series I	—	—	\$ —	\$ —	4/23/2008	— %	4/30/2018	— %	\$ —	\$ —	\$ 375.03
Series Q	—	150,000	—	1,500	4/23/2013	—	5/1/2023	SOFR + 3.25	220.45	801.41 ^(g)	515.00
Series R	—	150,000	—	1,500	7/29/2013	—	8/1/2023	SOFR + 3.30	221.70	756.73 ^(h)	600.00
Series S	—	200,000	—	2,000	1/22/2014	—	2/1/2024	SOFR + 3.78	233.70 ^(e)	675.00	675.00
Series U	—	100,000	—	1,000	3/10/2014	—	4/30/2024	SOFR + 3.33	153.13	612.50	612.50
Series V	—	—	—	—	6/9/2014	—	7/1/2019	—	—	—	340.91
Series X	—	160,000	—	1,600	9/23/2014	—	10/1/2024	SOFR + 3.33	457.50	610.00	610.00
Series CC	125,750	125,750	1,258	1,258	10/20/2017	SOFR + 2.58	11/1/2022	SOFR + 2.58	812.73	804.08	526.27 ⁽ⁱ⁾
Series FF	—	225,000	—	2,250	7/31/2019	—	8/1/2024	SOFR + 3.38	250.00	500.00	500.00
Series HH	300,000	300,000	3,000	3,000	1/23/2020	4.600	2/1/2025	SOFR + 3.125	460.00	460.00	460.00
Series II	150,000	150,000	1,500	1,500	2/24/2020	4.000	4/1/2025	SOFR + 2.745	400.00	400.00	400.00
Series KK	200,000	200,000	2,000	2,000	5/12/2021	3.650	6/1/2026	CMT + 2.85	365.00	365.00	365.00
Series NN	250,000	NA	2,496	NA	3/12/2024	6.875	6/1/2029	CMT + 2.737	494.63 ^(f)	NA	NA
Total preferred stock	2,005,375	2,740,375	\$20,050	\$ 27,404							

(a) Represented by depositary shares.

(b) Each series of fixed-to-floating rate preferred stock converts to a floating rate at the earliest redemption date.

(c) Effective June 30, 2023, CME Term SOFR became the replacement reference rate for fixed-to-floating rate preferred stock issued by the Firm that formerly referenced U.S. dollar LIBOR. References in the table to "SOFR" mean a floating annualized rate equal to three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spreads noted. References to "CMT" mean a floating annualized rate equal to the five-year Constant Maturity Treasury ("CMT") rate plus the spreads noted.

(d) Dividends on preferred stock are discretionary and non-cumulative. When declared, dividends are declared quarterly. Dividends are payable quarterly on fixed-rate preferred stock. Dividends are payable semiannually on fixed-to-floating rate preferred stock while at a fixed rate, and payable quarterly after converting to a floating rate.

(e) The dividend rate for Series S preferred stock became floating and payable quarterly starting on February 1, 2024; prior to which the dividend rate was fixed at 6.75% or \$337.50 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on February 1, 2024 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.78%.

(f) The initial dividend declared was prorated based on the number of days outstanding for the period. Dividends were declared quarterly thereafter at the contractual rate.

(g) The dividend rate for Series Q preferred stock became floating and payable quarterly starting on May 1, 2023; prior to which the dividend rate was fixed at 5.15% or \$257.50 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.25%.

(h) The dividend rate for Series R preferred stock became floating and payable quarterly starting on August 1, 2023; prior to which the dividend rate was fixed at 6.00% or \$300.00 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 3.30%.

(i) The dividend rate for Series CC preferred stock became floating and payable quarterly starting on November 1, 2022; prior to which the dividend rate was fixed at 4.625% or \$231.25 per share payable semiannually. The dividend rate for each quarterly dividend period commencing on August 1, 2023 was three-month term SOFR (plus a spread adjustment of 0.26% per annum) plus the spread of 2.58%.

Notes to consolidated financial statements

Each series of preferred stock has a liquidation value and redemption price per share of \$10,000, plus accrued but unpaid dividends. The aggregate liquidation value was \$20.2 billion at December 31, 2024.

Issuances

On February 4, 2025, the Firm issued \$3.0 billion of fixed-rate reset non-cumulative preferred stock, Series OO.

On March 12, 2024, the Firm issued \$2.5 billion of fixed-rate reset non-cumulative preferred stock, Series NN.

Redemptions

On February 1, 2025, the Firm redeemed all \$3.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series HH.

On October 1, 2024, the Firm redeemed all \$1.6 billion of its fixed-to-floating rate non-cumulative preferred stock, Series X.

On August 1, 2024, the Firm redeemed all \$2.3 billion of its fixed-to-floating rate non-cumulative preferred stock, Series FF.

On May 1, 2024, the Firm redeemed all \$5.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series Q, Series R and Series S.

On April 30, 2024, the Firm redeemed all \$1.0 billion of its fixed-to-floating rate non-cumulative preferred stock, Series U.

Redemption rights

Each series of the Firm's preferred stock may be redeemed on any dividend payment date on or after the earliest redemption date for that series. All outstanding preferred stock series may also be redeemed following a "capital treatment event," as described in the terms of each series. Any redemption of the Firm's preferred stock is subject to non-objection from the Board of Governors of the Federal Reserve System (the "Federal Reserve").

Note 22 – Common stock

At December 31, 2024 and 2023, JPMorganChase was authorized to issue 9.0 billion shares of common stock with a par value of \$1 per share.

Common shares issued which were reissued from treasury by the Firm during the years ended December 31, 2024, 2023 and 2022 were as follows.

Year ended December 31, (in millions)	2024	2023	2022
Total issued – balance at January 1	4,104.9	4,104.9	4,104.9
Treasury – balance at January 1	(1,228.3)	(1,170.7)	(1,160.8)
Repurchase	(91.7)	(69.5)	(23.1)
Reissuance:			
Employee benefits and compensation plans	11.9	10.9	12.0
Employee stock purchase plans	0.8	1.0	1.2
Total reissuance	12.7	11.9	13.2
Total treasury – balance at December 31	(1,307.3)	(1,228.3)	(1,170.7)
Outstanding at December 31	2,797.6	2,876.6	2,934.2

On June 28, 2024, the Firm announced that its Board of Directors had authorized a new \$30 billion common share repurchase program, effective July 1, 2024. Through June 30, 2024, the Firm was authorized to purchase up to \$30 billion of common shares under its previously-approved common share repurchase program that was announced on April 13, 2022.

The following table sets forth the Firm's repurchases of common stock for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions)	2024	2023	2022 ^(b)
Total number of shares of common stock repurchased	91.7	69.5	23.1
Aggregate purchase price of common stock repurchases ^(a)	\$18,841	\$9,898	\$ 3,122

(a) Excludes excise tax and commissions. As part of the Inflation Reduction Act of 2022, a 1% excise tax is imposed on net share repurchases commencing January 1, 2023.

(b) In the second half of 2022, the Firm temporarily suspended share repurchases, which it resumed in the first quarter of 2023 under its common share repurchase program.

The Board of Directors' authorization to repurchase common shares is utilized at management's discretion. The \$30 billion common share repurchase program approved by the Board of Directors does not establish specific price targets or timetables. Management determines the amount and timing of common share repurchases based on various factors, including market conditions; legal and regulatory considerations affecting the amount and timing of repurchase activity; the Firm's capital position (taking into account goodwill and intangibles); organic capital generation; current and proposed future capital requirements; and other investment opportunities. The amount of common shares that the Firm repurchases in any period may be substantially more or less than the amounts estimated or actually repurchased in prior periods, reflecting the dynamic nature of the decision-making process. The Firm's common share repurchases may be suspended by management at any time; and may be executed through open market purchases or privately negotiated transactions, or utilizing Rule 10b5-1 plans, which are written trading plans that the Firm may enter into from time to time under Rule 10b5-1 of the Securities Exchange Act of 1934 and which allow the Firm to repurchase its common shares during periods when it may otherwise not be repurchasing common shares — for example, during internal trading blackout periods.

As of December 31, 2024, approximately 58.8 million shares of common stock were reserved for issuance under various employee incentive, compensation, option and stock purchase plans, and directors' compensation plans.

Note 23 – Earnings per share

Basic earnings per share (“EPS”) is calculated using the two-class method. Under the two-class method, all earnings (distributed and undistributed) are allocated to common stock and participating securities.

JPMorganChase grants RSUs under its share-based compensation programs, predominantly all of which entitle recipients to receive nonforfeitable dividends during the vesting period on a basis equivalent to dividends paid to holders of the Firm’s common stock. These unvested RSUs meet the definition of participating securities based on their respective rights to receive nonforfeitable dividends, and they are treated as a separate class of securities in computing basic EPS. Participating securities are not included as incremental shares in computing diluted EPS; refer to Note 9 for additional information.

Diluted EPS incorporates the potential impact of contingently issuable shares, including awards which require future service as a condition of delivery of the underlying common stock. Diluted EPS is calculated under both the two-class and treasury stock methods, and the more dilutive amount is reported. For each of the periods presented in the table below, diluted EPS calculated under the two-class method was more dilutive.

The following table presents the calculation of net income applicable to common stockholders and basic and diluted EPS for the years ended December 31, 2024, 2023 and 2022.

Year ended December 31, (in millions, except per share amounts)	2024	2023	2022
Basic earnings per share			
Net income	\$ 58,471	\$ 49,552	\$ 37,676
Less: Preferred stock dividends	1,259	1,501	1,595
Net income applicable to common equity	57,212	48,051	36,081
Less: Dividends and undistributed earnings allocated to participating securities	344	291	189
Net income applicable to common stockholders	\$ 56,868	\$ 47,760	\$ 35,892
Total weighted-average basic shares outstanding	2,873.9	2,938.6	2,965.8
Net income per share	\$ 19.79	\$ 16.25	\$ 12.10
Diluted earnings per share			
Net income applicable to common stockholders	\$ 56,868	\$ 47,760	\$ 35,892
Total weighted-average basic shares outstanding	2,873.9	2,938.6	2,965.8
Add: Dilutive impact of unvested PSUs, nondividend-earning RSUs and SARs	5.1	4.5	4.2
Total weighted-average diluted shares outstanding	2,879.0	2,943.1	2,970.0
Net income per share	\$ 19.75	\$ 16.23	\$ 12.09

Note 24 – Accumulated other comprehensive income/(loss)

AOCI includes the after-tax change in unrealized gains and losses on investment securities, foreign currency translation adjustments (including the impact of related derivatives), fair value changes of excluded components on fair value hedges, cash flow hedging activities, net gain/(loss) related to the Firm's defined benefit pension and OPEB plans, and fair value option-elected liabilities arising from changes in the Firm's own credit risk (DVA).

Year ended December 31, (in millions)	Unrealized gains/(losses) on investment securities	Translation adjustments, net of hedges	Fair value hedges	Cash flow hedges	Defined benefit pension and OPEB plans	DVA on fair value option elected liabilities	Accumulated other comprehensive income/(loss)
Balance at December 31, 2021	\$ 2,640	\$ (934)	\$ (131)	\$ (296)	\$ (210)	\$ (1,153)	\$ (84)
Net change	(11,764)	(611)	98	(5,360)	(1,241)	1,621	(17,257)
Balance at December 31, 2022	\$ (9,124) ^(a)	\$ (1,545)	\$ (33)	\$ (5,656)	\$ (1,451)	\$ 468	\$ (17,341)
Net change	5,381	329	(101)	1,724	373	(808)	6,898
Balance at December 31, 2023	\$ (3,743) ^(a)	\$ (1,216)	\$ (134)	\$ (3,932)	\$ (1,078)	\$ (340)	\$ (10,443)
Net change	(87)	(858)	(87)	(882)	(63)	(36)	(2,013)
Balance at December 31, 2024	\$ (3,830) ^(a)	\$ (2,074)	\$ (221)	\$ (4,814)	\$ (1,141)	\$ (376)	\$ (12,456)

(a) Included after-tax net unamortized unrealized gains/(losses) of \$(651) million, \$(895) million, and \$(1.3) billion for the years ended 2024, 2023 and 2022, respectively, related to AFS securities that have been transferred to HTM. As of December 31, 2023 included after-tax net unamortized unrealized gains/(losses) of \$(29) million related to HTM securities that have been transferred to AFS as permitted by the new hedge accounting guidance adopted on January 1, 2023. Refer to Note 10 for further information.

The following table presents the pre-tax and after-tax changes in the components of OCI.

Year ended December 31, (in millions)	2024			2023			2022		
	Pre-tax	Tax effect	After-tax	Pre-tax	Tax effect	After-tax	Pre-tax	Tax effect	After-tax
Unrealized gains/(losses) on investment securities:									
Net unrealized gains/(losses) arising during the period	\$ (1,135)	\$ 274	\$ (861)	\$ 3,891	\$ (922)	\$ 2,969	\$ (17,862)	\$ 4,290	\$ (13,572)
Reclassification adjustment for realized (gains)/losses included in net income ^(a)	1,021	(247)	774	3,180	(768)	2,412	2,380	(572)	1,808
Net change	(114)	27	(87)	7,071	(1,690)	5,381	(15,482)	3,718	(11,764)
Translation adjustments ^(b):									
Translation	(4,385)	250	(4,135)	1,714	(95)	1,619	(3,574)	265	(3,309)
Hedges	4,322	(1,045)	3,277	(1,697)	407	(1,290)	3,553	(855)	2,698
Net change	(63)	(795)	(858)	17	312	329	(21)	(590)	(611)
Fair value hedges, net change ^(c)	(115)	28	(87)	(134)	33	(101)	130	(32)	98
Cash flow hedges:									
Net unrealized gains/(losses) arising during the period	(3,742)	904	(2,838)	483	(114)	369	(7,473)	1,794	(5,679)
Reclassification adjustment for realized (gains)/losses included in net income ^(d)	2,579	(623)	1,956	1,775	(420)	1,355	420	(101)	319
Net change	(1,163)	281	(882)	2,258	(534)	1,724	(7,053)	1,693	(5,360)
Defined benefit pension and OPEB plans, net change ^(e)	(131)	68	(63)	421	(48)	373	(1,459)	218	(1,241)
DVA on fair value option elected liabilities, net change	(45)	9	(36)	(1,066)	258	(808)	2,141	(520)	1,621
Total other comprehensive income/(loss)	\$ (1,631)	\$ (382)	\$ (2,013)	\$ 8,567	\$ (1,669)	\$ 6,898	\$ (21,744)	\$ 4,487	\$ (17,257)

(a) The pre-tax amount is reported in Investment securities gains/(losses) in the Consolidated statements of income.

(b) Reclassifications of pre-tax realized gains/(losses) on translation adjustments and related hedges are reported in other income/expense in the Consolidated statements of income. During the year ended December 31, 2024, the Firm reclassified a net pre-tax gain of \$7 million to other income/expense, of which \$89 million gain related to net investment hedges and \$(82) million loss related to cumulative translation adjustments. During the year ended December 31, 2023, the Firm reclassified a net pre-tax loss of \$(3) million. During the year ended December 31, 2022, the Firm reclassified a net pre-tax loss of \$(8) million.

(c) Represents changes in fair value of cross-currency swaps attributable to changes in cross-currency basis spreads, which are excluded from the assessment of hedge effectiveness and recorded in other comprehensive income. The initial cost of cross-currency basis spreads is recognized in earnings as part of the accrual of interest on the cross-currency swaps.

(d) The pre-tax amounts are primarily recorded in noninterest revenue, net interest income and compensation expense in the Consolidated statements of income.

(e) During the year ended December 31, 2022, a remeasurement of the Firm's U.S. principal defined benefit plan in the third quarter, was required as a result of a pension settlement. The remeasurement resulted in a net decrease of \$1.4 billion in pre-tax AOCI.

Note 25 – Income taxes

JPMorganChase and its eligible subsidiaries file a consolidated U.S. federal income tax return. JPMorganChase uses the asset and liability method to provide for income taxes on all transactions recorded in the Consolidated Financial Statements. This method requires that income taxes reflect the expected future tax consequences of temporary differences between the carrying amounts of assets or liabilities for book and tax purposes. Accordingly, a deferred tax asset or liability for each temporary difference is determined based on the tax rates that the Firm expects to be in effect when the underlying items of income and expense are realized. JPMorganChase's expense for income taxes includes the current and deferred portions of that expense. A valuation allowance is established to reduce deferred tax assets to the amount the Firm expects to realize.

Due to the inherent complexities arising from the nature of the Firm's businesses, and from conducting business and being taxed in a substantial number of jurisdictions, significant judgments and estimates are required to be made. Agreement of tax liabilities between JPMorganChase and the many tax jurisdictions in which the Firm files tax returns may not be finalized for several years. Thus, the Firm's final tax-related assets and liabilities may ultimately be different from those currently reported.

Effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance, under the modified retrospective method. Refer to Notes 1, 6 and 14 for additional information.

Effective tax rate and expense

The following table presents a reconciliation of the applicable statutory U.S. federal income tax rate to the effective tax rate.

Effective tax rate

Year ended December 31,	2024	2023	2022
Statutory U.S. federal tax rate	21.0 %	21.0 %	21.0 %
Increase/(decrease) in tax rate resulting from:			
U.S. state and local income taxes, net of U.S. federal income tax benefit	3.1	2.8	3.5
Tax-exempt income	(0.7)	(0.9)	(0.9)
Non-U.S. earnings	1.4	1.5	0.4
Business tax credits	(2.4)	(4.4)	(5.4)
Other, net	(0.3)	(0.4)	(0.2)
Effective tax rate	22.1 %	19.6 % ^(a)	18.4 %

(a) Income tax expense associated with the First Republic acquisition was reflected in the estimated bargain purchase gain, which resulted in a reduction in the Firm's effective tax rate.

The following table reflects the components of income tax expense/(benefit) included in the Consolidated statements of income.

Income tax expense/(benefit)

Year ended December 31, (in millions)	2024	2023	2022
Current income tax expense/(benefit)			
U.S. federal	\$ 7,091	\$ 8,973	\$ 5,606
Non-U.S.	4,753	4,355	2,992
U.S. state and local	2,762	3,266	2,630
Total current income tax expense/(benefit)	14,606	16,594	11,228
Deferred income tax expense/(benefit)			
U.S. federal	1,771	(3,475)	(2,004)
Non-U.S.	72	35	(154)
U.S. state and local	161	(1,094)	(580)
Total deferred income tax expense/(benefit)	2,004	(4,534)	(2,738)
Total income tax expense	\$ 16,610	\$ 12,060	\$ 8,490

Total income tax expense includes \$314 million, \$68 million and \$331 million of tax benefits in 2024, 2023, and 2022, respectively, resulting from the resolution of tax audits.

Tax effect of items recorded in stockholders' equity

The preceding table does not reflect the tax effect of certain items that are recorded each period directly in stockholders' equity, which are predominantly reflected in OCI as disclosed in Note 24. For the year ended December 31, 2024, stockholders' equity reflected the tax effect associated with the Firm's adoption of the Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method guidance. For the year ended December 31, 2023, stockholders' equity reflected the tax effect associated with the Firm's adoption of the TDR accounting guidance. Both of the respective adoptions were recognized in retained earnings. Refer to Note 1, 6 and 14 for further information.

Results from U.S. and non-U.S. earnings

The following table presents the U.S. and non-U.S. components of income before income tax expense.

Year ended December 31, (in millions)	2024	2023	2022
U.S.	\$ 59,472	\$ 46,868	\$ 34,626
Non-U.S. ^(a)	15,609	14,744	11,540
Income before income tax expense	\$ 75,081	\$ 61,612	\$ 46,166

(a) For purposes of this table, non-U.S. income is defined as income generated from operations located outside the U.S.

The Firm will recognize any U.S. income tax expense it may incur on global intangible low tax income as income tax expense in the period in which the tax is incurred.

Deferred taxes

Deferred income tax expense/(benefit) reflects the differences between assets and liabilities measured for financial reporting purposes versus income tax return purposes. Deferred tax assets are recognized if, in management's judgment, their realizability is determined to be more likely than not. If a deferred tax asset is determined to be unrealizable, a valuation allowance is established. The significant components of deferred tax assets and liabilities are reflected in the following table, the net deferred tax assets are reflected in other assets on the Firm's Consolidated balance sheets.

December 31, (in millions)	2024	2023
Deferred tax assets		
Allowance for loan losses	\$ 6,117	\$ 5,809
Employee benefits	1,165	1,247
Accrued expenses and other	8,881	9,887 ^(a)
Depreciation and amortization	386	—
Non-U.S. operations	948	860
Tax attribute carryforwards	352	290
Gross deferred tax assets	17,849	18,093
Valuation allowance	(249)	(183)
Deferred tax assets, net of valuation allowance	\$ 17,600	\$ 17,910
Deferred tax liabilities		
Depreciation and amortization	\$ —	\$ 779
Mortgage servicing rights, net of hedges	1,912	1,794
Leasing transactions	2,249	2,254
Other, net	1,264	2,935
Gross deferred tax liabilities	5,425	7,762
Net deferred tax assets	\$ 12,175	\$ 10,148

(a) Includes the estimated net deferred tax asset associated with the First Republic acquisition.

JPMorganChase has recorded deferred tax assets of \$352 million at December 31, 2024 in connection with tax attribute carryforwards. State and local capital loss carryforwards were \$914 million, U.S. federal NOL carryforwards were \$496 million, non-U.S. NOL carryforwards were \$958 million, and other U.S. federal tax attributes were \$111 million. If not utilized, a portion of the U.S. federal NOL carryforwards and other U.S. federal tax attributes will expire between 2026 and 2036 whereas others have an unlimited carryforward period. Similarly, certain non-U.S. NOL carryforwards will expire between 2026 and 2041 whereas others have an unlimited carryforward period. The state and local capital loss carryforwards will expire in 2026.

The valuation allowance at December 31, 2024, was predominantly driven by certain non-U.S. deferred tax assets, including NOL carryforwards.

Notes to consolidated financial statements

Unrecognized tax benefits

At December 31, 2024, 2023 and 2022, JPMorganChase's unrecognized tax benefits, excluding related interest expense and penalties, were \$6.2 billion, \$5.4 billion and \$5.0 billion, respectively, of which \$4.4 billion, \$3.9 billion and \$3.8 billion, respectively, if recognized, would reduce the annual effective tax rate. Included in the amount of unrecognized tax benefits are certain items that would not affect the effective tax rate if they were recognized in the Consolidated statements of income. These unrecognized items include the tax effect of certain temporary differences, the portion of gross state and local unrecognized tax benefits that would be offset by the benefit from associated U.S. federal income tax deductions, and the portion of gross non-U.S. unrecognized tax benefits that would have offsets in other jurisdictions. JPMorganChase evaluates the need for changes in unrecognized tax benefits based on its anticipated tax return filing positions as part of its U.S. federal and state and local tax returns. In addition, the Firm is presently under audit by a number of taxing authorities, most notably by the Internal Revenue Service, as summarized in the Tax examination status table below. The evaluation of unrecognized tax benefits as well as the potential for audit settlements make it reasonably possible that over the next 12 months the gross balance of unrecognized tax benefits may increase or decrease by as much as approximately \$2.4 billion. The change in the unrecognized tax benefit would result in a payment or income statement recognition.

The following table presents a reconciliation of the beginning and ending amount of unrecognized tax benefits.

(in millions)	2024	2023	2022
Balance at January 1,	\$ 5,401	\$ 5,043	\$ 4,636
Increases based on tax positions related to the current period	1,721	1,440	1,234
Increases based on tax positions related to prior periods	92	37	123
Decreases based on tax positions related to prior periods	(907)	(1,110)	(824)
Decreases related to cash settlements with taxing authorities	(148)	(9)	(126)
Balance at December 31,	\$ 6,159	\$ 5,401	\$ 5,043

After-tax interest expense/(benefit) and penalties related to income tax liabilities recognized in income tax expense were \$288 million, \$229 million and \$141 million in 2024, 2023 and 2022, respectively.

At December 31, 2024 and 2023, in addition to the liability for unrecognized tax benefits, the Firm had accrued \$1.7 billion and \$1.6 billion, respectively, for income tax-related interest and penalties.

Tax examination status

JPMorganChase is continually under examination by the Internal Revenue Service, by taxing authorities throughout the world, and by many state and local jurisdictions throughout the U.S. The following table summarizes the status of tax years that remain subject to income tax examination of JPMorganChase and its consolidated subsidiaries by significant jurisdictions as of December 31, 2024.

	Periods under examination	Status
JPMorganChase – U.S.	2011 – 2013	Field examination of amended returns; certain matters at Appellate level
JPMorganChase – U.S.	2014 – 2020	Field examination of original and amended returns; certain matters at Appellate level
JPMorganChase – New York City	2015 – 2018	Field Examination
JPMorganChase – U.K.	2017 – 2022	Field examination of certain select entities

Note 26 – Restricted cash, other restricted assets and intercompany funds transfers

Restricted cash and other restricted assets

Certain of the Firm's cash and other assets are restricted as to withdrawal or usage. These restrictions are imposed by various regulatory authorities based on the particular activities of the Firm's subsidiaries.

The business of JPMorgan Chase Bank, N.A. is subject to examination and regulation by the OCC. The Bank is a member of the U.S. Federal Reserve System, and its deposits in the U.S. are insured by the FDIC, subject to applicable limits.

The Firm is required to maintain cash reserves at certain non-US central banks.

The Firm is also subject to rules and regulations established by other U.S. and non U.S. regulators. As part of its compliance with the respective regulatory requirements, the Firm's broker-dealer activities are subject to certain restrictions on cash and other assets.

The following table presents the components of the Firm's restricted cash:

December 31, (in billions)	2024	2023
Segregated for the benefit of securities and cleared derivative customers	18.7	10.3
Cash reserves at non-U.S. central banks and held for other general purposes	8.8	9.3
Total restricted cash^(a)	\$ 27.5	\$ 19.6

(a) Comprises \$26.1 billion and \$18.2 billion in deposits with banks, and \$1.4 billion and \$1.4 billion in cash and due from banks on the Consolidated balance sheets as of December 31, 2024 and 2023, respectively.

Also, as of December 31, 2024 and 2023, the Firm had the following other restricted assets:

- Cash and securities pledged with clearing organizations for the benefit of customers of \$40.7 billion and \$40.5 billion, respectively.
- Securities with a fair value of \$26.8 billion and \$20.5 billion, respectively, were also restricted in relation to customer activity.

Intercompany funds transfers

Restrictions imposed by U.S. federal law prohibit JPMorgan Chase Bank, N.A., and its subsidiaries, from lending to JPMorgan Chase & Co. ("Parent Company") and certain of its affiliates unless the loans are secured in specified amounts. Such secured loans provided by any banking subsidiary to the Parent Company or to any particular affiliate, together with certain other transactions with such affiliate (collectively referred to as "covered transactions"), must be made on terms and conditions that are consistent with safe and sound banking practices. In addition, unless collateralized with cash or US Government debt obligations, covered transactions are generally limited to 10% of the banking subsidiary's total capital, as determined by the risk-based capital guidelines; the aggregate amount of covered transactions between any banking subsidiary and all of its affiliates is limited to 20% of the banking subsidiary's total capital.

The Parent Company's two principal subsidiaries are JPMorgan Chase Bank, N.A. and JPMorgan Chase Holdings LLC, an intermediate holding company (the "IHC"). The IHC generally holds the stock of JPMorgan Chase's subsidiaries other than JPMorgan Chase Bank, N.A. and its subsidiaries. The IHC also owns other assets and provides intercompany loans to the Parent Company. The Parent Company is obligated to contribute to the IHC substantially all the net proceeds received from securities issuances (including issuances of senior and subordinated debt securities and of preferred and common stock).

The principal sources of income and funding for the Parent Company are dividends from JPMorgan Chase Bank, N.A. and dividends and extensions of credit from the IHC. In addition to dividend restrictions set forth in statutes and regulations, the Federal Reserve, the OCC and the FDIC have authority under the Financial Institutions Supervisory Act to prohibit or to limit the payment of dividends by the banking organizations they supervise, including the Parent Company and its subsidiaries that are banks or bank holding companies, if, in the banking regulator's opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the banking organization. The IHC is prohibited from paying dividends or extending credit to the Parent Company if certain capital or liquidity "thresholds" are breached or if limits are otherwise imposed by the Parent Company's management or Board of Directors.

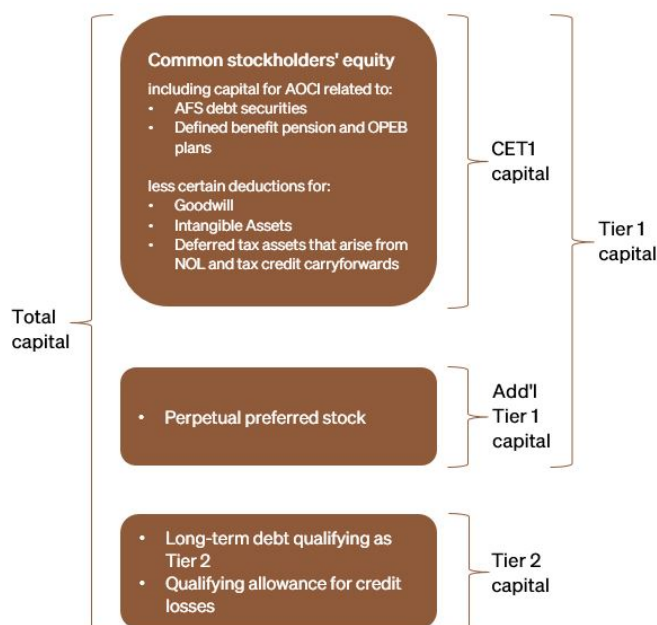
At January 1, 2025, the Parent Company's banking subsidiaries could pay, in the aggregate, approximately \$15.5 billion in dividends to their respective bank holding companies without the prior approval of their relevant banking regulators. The capacity to pay dividends in 2025 will be supplemented by the banking subsidiaries' earnings during the year.

Note 27 – Regulatory capital

The Federal Reserve establishes capital requirements, including well-capitalized standards, for the Firm as a consolidated financial holding company. The OCC establishes similar minimum capital requirements and standards for the Firm's principal IDI subsidiary, JPMorgan Chase Bank, N.A.

The capital rules under Basel III establish minimum capital ratios and overall capital adequacy standards for large and internationally active U.S. bank holding companies and banks, including the Firm and JPMorgan Chase Bank, N.A. Under the rules currently in effect, two comprehensive approaches are prescribed for calculating RWA: a standardized approach ("Basel III Standardized"), and an advanced approach ("Basel III Advanced"). For each of these risk-based capital ratios, the capital adequacy of the Firm and JPMorgan Chase Bank, N.A. is evaluated against the lower of the Standardized or Advanced approaches compared to their respective regulatory capital ratio requirements.

The three components of regulatory capital under the Basel III rules and their primary drivers are as illustrated below:



Under the risk-based capital and leverage-based guidelines of the Federal Reserve, JPMorgan Chase & Co. is required to maintain minimum ratios for CET1 capital, Tier 1 capital, Total capital, Tier 1 leverage and the SLR. Failure to meet these minimum requirements could cause the Federal Reserve to take action. JPMorgan Chase Bank, N.A. is also subject to these capital requirements established by its primary regulators.

The following table presents the risk-based regulatory capital ratio requirements and well-capitalized ratios to which the Firm and JPMorgan Chase Bank, N.A. were subject as of December 31, 2024 and 2023.

	Standardized capital ratio requirements		Advanced capital ratio requirements		Well-capitalized ratios	
	BHC ^{(a)(b)}	IDI ^(c)	BHC ^{(a)(b)}	IDI ^(c)	BHC ^(d)	IDI ^(e)
Risk-based capital ratios						
CET1 capital	12.3 %	7.0 %	11.5 %	7.0 %	NA	6.5 %
Tier 1 capital	13.8	8.5	13.0	8.5	6.0 %	8.0
Total capital	15.8	10.5	15.0	10.5	10.0	10.0

Note: The table above is as defined by the regulations issued by the Federal Reserve, OCC and FDIC and to which the Firm and JPMorgan Chase Bank, N.A. are subject.

- (a) Represents the regulatory capital ratio requirements applicable to the Firm. The CET1, Tier 1 and Total capital ratio requirements each include a respective minimum requirement plus a GSIB surcharge of 4.5% as calculated under Method 2; plus a 3.3% SCB for Basel III Standardized ratios and a fixed 2.5% capital conservation buffer for Basel III Advanced ratios. The countercyclical buffer is currently set to 0% by the federal banking agencies.
- (b) For the period ended December 31, 2023, the CET1, Tier 1, and Total capital ratio requirements under Basel III Standardized applicable to the Firm were 11.4%, 12.9%, and 14.9%, respectively; the Basel III Advanced CET1, Tier 1, and Total capital ratio requirements applicable to the Firm were 11.0%, 12.5%, and 14.5%, respectively.
- (c) Represents requirements for JPMorgan Chase Bank, N.A. The CET1, Tier 1 and Total capital ratio requirements include a fixed capital conservation buffer requirement of 2.5% that is applicable to JPMorgan Chase Bank, N.A. JPMorgan Chase Bank, N.A. is not subject to the GSIB surcharge.
- (d) Represents requirements for bank holding companies pursuant to regulations issued by the Federal Reserve.
- (e) Represents requirements for JPMorgan Chase Bank, N.A. pursuant to regulations issued under the FDIC Improvement Act.

The following table presents the leverage-based regulatory capital ratio requirements and well-capitalized ratios to which the Firm and JPMorgan Chase Bank, N.A. were subject as of December 31, 2024 and 2023.

	Capital ratio requirements ^(a)		Well-capitalized ratios	
	BHC	IDI	BHC ^(b)	IDI
Leverage-based capital ratios				
Tier 1 leverage	4.0 %	4.0 %	NA	5.0 %
SLR	5.0	6.0	NA	6.0

Note: The table above is as defined by the regulations issued by the Federal Reserve, OCC and FDIC and to which the Firm and JPMorgan Chase Bank, N.A. are subject.

- (a) Represents minimum SLR requirement of 3.0%, as well as supplementary leverage buffer requirements of 2.0% and 3.0% for BHC and JPMorgan Chase Bank, N.A., respectively.
- (b) The Federal Reserve's regulations do not establish well-capitalized thresholds for these measures for BHCs.

CECL Regulatory Capital Transition

Beginning January 1, 2022, the \$2.9 billion CECL capital benefit, provided by the Federal Reserve in response to the COVID-19 pandemic, is being phased out at 25% per year over a three-year period. As of December 31, 2024 and 2023, the Firm's CET1 capital reflected the remaining benefit of \$720 million and \$1.4 billion, respectively, associated with the CECL capital transition provisions.

Similarly, as of January 1, 2024, the Firm has phased out 75% of the other CECL capital transition provisions which impacted Tier 2 capital, adjusted average assets, total leverage exposure and RWA, as applicable.

The following tables present risk-based capital metrics under both the Basel III Standardized and Basel III Advanced approaches and leverage-based capital metrics for JPMorgan Chase & Co. and JPMorgan Chase Bank, N.A. As of December 31, 2024 and 2023, JPMorgan Chase & Co. and JPMorgan Chase Bank, N.A. were well-capitalized and met all capital requirements to which each was subject.

December 31, 2024 (in millions, except ratios)	Basel III Standardized		Basel III Advanced	
	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.
Risk-based capital metrics: ^(a)				
CET1 capital	\$ 275,513	\$ 275,732	\$ 275,513	\$ 275,732
Tier 1 capital	294,881	275,737	294,881	275,737
Total capital	325,589	296,041	311,898	282,328
Risk-weighted assets	1,757,460	1,718,777	1,740,429	1,594,072
CET1 capital ratio	15.7 %	16.0 %	15.8 %	17.3 %
Tier 1 capital ratio	16.8	16.0	16.9	17.3
Total capital ratio	18.5	17.2	17.9	17.7

December 31, 2023 (in millions, except ratios)	Basel III Standardized		Basel III Advanced	
	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.
Risk-based capital metrics: ^(a)				
CET1 capital	\$ 250,585	\$ 262,030	\$ 250,585	\$ 262,030
Tier 1 capital	277,306	262,032	277,306	262,032
Total capital	308,497	281,308	295,417	268,392
Risk-weighted assets	1,671,995	1,621,789	1,669,156	1,526,952
CET1 capital ratio	15.0 %	16.2 %	15.0 %	17.2 %
Tier 1 capital ratio	16.6	16.2	16.6	17.2
Total capital ratio	18.5	17.3	17.7	17.6

(a) The capital metrics reflect the CECL capital transition provisions.

(b) Includes the impacts of certain assets associated with First Republic to which the Standardized approach has been applied as permitted by the transition provisions in the U.S. capital rules.

Three months ended (in millions, except ratios)	December 31, 2024		December 31, 2023	
	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.	JPMorgan Chase & Co.	JPMorgan Chase Bank, N.A.
Leverage-based capital metrics: ^(a)				
Adjusted average assets ^(b)	\$ 4,070,499	\$ 3,491,283	\$ 3,831,200	\$ 3,337,842
Tier 1 leverage ratio	7.2 %	7.9 %	7.2 %	7.9 %
Total leverage exposure	\$ 4,837,568	\$ 4,246,516	\$ 4,540,465	\$ 4,038,739
SLR	6.1 %	6.5 %	6.1 %	6.5 %

(a) The capital metrics reflect the CECL capital transition provisions.

(b) Adjusted average assets, for purposes of calculating the leverage ratios, includes quarterly average assets adjusted for on-balance sheet assets that are subject to deduction from Tier 1 capital, predominantly goodwill, inclusive of estimated equity method goodwill, and other intangible assets.

Note 28 – Off-balance sheet lending-related financial instruments, guarantees, and other commitments

JPMorganChase provides lending-related financial instruments (e.g., commitments and guarantees) to address the financing needs of its customers and clients. The contractual amount of these financial instruments represents the maximum possible credit risk to the Firm should the customer or client draw upon the commitment or the Firm be required to fulfill its obligation under the guarantee, and should the customer or client subsequently fail to perform according to the terms of the contract. Most of these commitments and guarantees have historically been refinanced, extended, cancelled, or expired without being drawn or a default occurring. As a result, the total contractual amount of these instruments is not, in the Firm's view, representative of its expected future credit exposure or funding requirements.

To provide for expected credit losses in wholesale and certain consumer lending-related commitments, an allowance for credit losses on lending-related commitments is maintained. Refer to Note 13 for further information regarding the allowance for credit losses on lending-related commitments.

The following table summarizes the contractual amounts and carrying values of off-balance sheet lending-related financial instruments, guarantees and other commitments at December 31, 2024 and 2023. The amounts in the table below for credit card, home equity and certain scored business banking lending-related commitments represent the total available credit for these products. The Firm has not experienced, and does not anticipate, that all available lines of credit for these commitments will be utilized at the same time. The Firm can reduce or cancel these commitments, in accordance with the contract, or to the extent otherwise permitted by law, including when there has been a demonstrable decline in the creditworthiness of the borrower or significant decrease in the value of underlying property.

Off-balance sheet lending-related financial instruments, guarantees and other commitments

By remaining maturity as of December 31, (in millions)	Contractual amount						Carrying value ^{(h)(i)}	
	2024					2023	2024	2023
	Expires in 1 year or less	Expires after 1 year through 3 years	Expires after 3 years through 5 years	Expires after 5 years	Total	Total		
Lending-related								
Consumer, excluding credit card:								
Residential Real Estate ^(a)	\$ 10,838	\$ 7,240	\$ 4,601	\$ 7,670	\$ 30,349	\$ 30,125	\$ 534 ⁽ⁱ⁾	\$ 678 ⁽ⁱ⁾
Auto and other	10,833	14	5	3,643	14,495	15,278	37 ⁽ⁱ⁾	148 ⁽ⁱ⁾
Total consumer, excluding credit card	21,671	7,254	4,606	11,313	44,844	45,403	571	826
Credit card ^(b)	1,001,311	—	—	—	1,001,311	915,658	—	—
Total consumer^(c)	1,022,982	7,254	4,606	11,313	1,046,155	961,061	571	826
Wholesale:								
Other unfunded commitments to extend credit ^(d)	101,500	199,878	172,066	24,993	498,437	503,526	2,608 ⁽ⁱ⁾	2,797 ⁽ⁱ⁾
Standby letters of credit and other financial guarantees ^(d)	15,825	8,506	3,780	565	28,676	28,872	473	479
Other letters of credit ^(d)	3,958	278	21	97	4,354	4,388	37	37
Total wholesale^(c)	121,283	208,662	175,867	25,655	531,467	536,786	3,118	3,313
Total lending-related	\$1,144,265	\$ 215,916	\$ 180,473	\$ 36,968	\$1,577,622	\$1,497,847	\$ 3,689	\$ 4,139
Other guarantees and commitments								
Securities lending indemnification agreements and guarantees ^(e)	\$ 310,046	\$ —	\$ —	\$ —	\$ 310,046	\$ 283,664	\$ —	\$ —
Derivatives qualifying as guarantees	902	343	9,890	38,493	49,628	54,562	113	89
Unsettled resale and securities borrowed agreements	115,939	—	—	—	115,939	95,106	2	—
Unsettled repurchase and securities loaned agreements	66,986	—	—	—	66,986	60,724	(2)	—
Loan sale and securitization-related indemnifications:								
Mortgage repurchase liability	NA	NA	NA	NA	NA	NA	45	76
Loans sold with recourse	NA	NA	NA	NA	1,189	803	23	24
Exchange & clearing house guarantees and commitments ^(h)	401,486	—	—	—	401,486	265,887	—	—
Other guarantees and commitments ^(g)	10,652	425	435	884	12,396	15,074	28	38

(a) Includes certain commitments to purchase loans from correspondents.

(b) Also includes commercial card lending-related commitments primarily in CIB.

(c) Predominantly all consumer and wholesale lending-related commitments are in the U.S.

(d) As of December 31, 2024 and 2023, reflected the contractual amount net of risk participations totaling \$85 million and \$88 million, respectively, for other unfunded commitments to extend credit; \$9.5 billion and \$8.2 billion, respectively, for standby letters of credit and other financial guarantees; and \$556 million and \$589 million, respectively, for other letters of credit. In regulatory filings with the Federal Reserve these commitments are shown gross of risk participations.

(e) As of December 31, 2024 and 2023, collateral held by the Firm in support of securities lending indemnification agreements was \$328.7 billion and \$300.3 billion, respectively. Securities lending collateral primarily consists of cash, G7 government securities, and securities issued by U.S. GSEs and government agencies.

(f) As of December 31, 2024 and 2023, includes guarantees to the Fixed Income Clearing Corporation under the sponsored member repo program and commitments and guarantees associated with the Firm's membership in certain clearing houses.

(g) As of December 31, 2024 and 2023, primarily includes unfunded commitments to purchase secondary market loans, other equity investment commitments, and unfunded commitments related to certain tax-oriented equity investments, and reflects the impact of adopting updates to the Accounting for Investments in Tax Credit Structures guidance effective January 1, 2024.

(h) For lending-related products, the carrying value includes the allowance for lending-related commitments and the guarantee liability; for derivative-related products, and lending-related commitments for which the fair value option was elected, the carrying value represents the fair value.

(i) For lending-related commitments, the carrying value also includes fees and any purchase discounts or premiums that are deferred and recognized in accounts payable and other liabilities on the Consolidated balance sheets. Deferred amounts for revolving commitments and commitments not expected to fund, are amortized to lending- and deposit-related fees on a straight line basis over the commitment period. For all other commitments the deferred amounts remain deferred until the commitment funds or is sold.

(j) As of December 31, 2024 and 2023, includes fair value adjustments associated with First Republic for residential real estate lending-related commitments totaling \$459 million and \$630 million, respectively; for auto and other lending-related commitments totaling \$37 million and \$148 million, respectively; and for other unfunded commitments to extend credit totaling \$699 million and \$1.1 billion, respectively. Refer to Note 34 for additional information.

Notes to consolidated financial statements

Other unfunded commitments to extend credit

Other unfunded commitments to extend credit generally consist of commitments for working capital and general corporate purposes, extensions of credit to support commercial paper facilities and bond financings in the event that those obligations cannot be remarketed to new investors, as well as committed liquidity facilities to clearing organizations. The Firm also issues commitments under multipurpose facilities which could be drawn upon in several forms, including the issuance of a standby letter of credit.

Guarantees

U.S. GAAP requires that a guarantor recognize, at the inception of a guarantee, a liability in an amount equal to the fair value of the obligation undertaken in issuing the guarantee. U.S. GAAP defines a guarantee as a contract that contingently requires the guarantor to pay the guaranteed party based upon: (a) changes in an underlying asset, liability or equity security of the guaranteed party; or (b) a third party's failure to perform under a specified agreement. The Firm considers the following off-balance sheet arrangements to be guarantees under U.S. GAAP: standby letters of credit and other financial guarantees, securities lending indemnifications, certain indemnification agreements included within third-party contractual arrangements, certain derivative contracts and the guarantees under the sponsored member repo program.

As required by U.S. GAAP, the Firm initially records guarantees at the inception date fair value of the non-contingent obligation assumed (e.g., the amount of consideration received or the net present value of the premium receivable). For these obligations, the Firm records this fair value amount in other liabilities with an offsetting entry recorded in cash (for premiums received), or other assets (for premiums receivable). Any premium receivable recorded in other assets is

reduced as cash is received under the contract, and the fair value of the liability recorded at inception is amortized into income as lending and deposit-related fees over the life of the guarantee contract. The lending-related contingent obligation is recognized based on expected credit losses in addition to, and separate from, any non-contingent obligation.

Non-lending-related contingent obligations are recognized when the liability becomes probable and reasonably estimable. These obligations are not recognized if the estimated amount is less than the carrying amount of any non-contingent liability recognized at inception (adjusted for any amortization). Examples of non-lending-related contingent obligations include indemnifications provided in sales agreements, where a portion of the sale proceeds is allocated to the guarantee, which adjusts the gain or loss that would otherwise result from the transaction. For these indemnifications, the initial liability is amortized to income as the Firm's risk is reduced (i.e., over time or when the indemnification expires).

The contractual amount and carrying value of guarantees and indemnifications are included in the table on page 303.

For additional information on the guarantees, see below.

Standby letters of credit and other financial guarantees

Standby letters of credit and other financial guarantees are conditional lending commitments issued by the Firm to guarantee the performance of a client or customer to a third party under certain arrangements, such as commercial paper facilities, bond financings, acquisition financings, trade financings and similar transactions.

The following table summarizes the contractual amount and carrying value of standby letters of credit and other financial guarantees and other letters of credit arrangements as of December 31, 2024 and 2023.

Standby letters of credit, other financial guarantees and other letters of credit

December 31, (in millions)	2024		2023	
	Standby letters of credit and other financial guarantees	Other letters of credit	Standby letters of credit and other financial guarantees	Other letters of credit
Investment-grade ^(a)	\$ 20,443	\$ 3,380	\$ 19,694	\$ 3,552
Noninvestment-grade ^(a)	8,233	974	9,178	836
Total contractual amount	\$ 28,676	\$ 4,354	\$ 28,872	\$ 4,388
Allowance for lending-related commitments	\$ 94	\$ 37	\$ 110	\$ 37
Guarantee liability	379	—	369	—
Total carrying value	\$ 473	\$ 37	\$ 479	\$ 37
Commitments with collateral	\$ 16,805	\$ 357	\$ 16,861	\$ 539

(a) The ratings scale is based on the Firm's internal risk ratings. Refer to Note 12 for further information on internal risk ratings.

Securities lending indemnifications

Through the Firm's securities lending program, counterparties' securities, via custodial and non-custodial arrangements, may be lent to third parties. As part of this program, the Firm provides an indemnification in the lending agreements which protects the lender against the failure of the borrower to return the lent securities. To minimize its liability under these indemnification agreements, the Firm obtains cash or other highly liquid collateral with a market value exceeding 100% of the value of the securities on loan from the borrower. Collateral is marked to market daily to help assure that collateralization is adequate. Additional collateral is called from the borrower if a shortfall exists, or collateral may be released to the borrower in the event of overcollateralization. If a borrower defaults, the Firm would use the collateral held to purchase replacement securities in the market or to credit the lending client or counterparty with the cash equivalent thereof.

The cash collateral held by the Firm may be invested on behalf of the client in indemnified resale agreements, whereby the Firm indemnifies the client against the loss of principal invested. To minimize its liability under these agreements, the Firm obtains collateral with a market value exceeding 100% of the principal invested.

Derivatives qualifying as guarantees

The Firm transacts in certain derivative contracts that have the characteristics of a guarantee under U.S. GAAP. These contracts include written put options that require the Firm to purchase assets upon exercise by the option holder at a specified price by a specified date in the future. The Firm may enter into written put option contracts in order to meet client needs, or for other trading purposes. The terms of written put options are typically five years or less.

Derivatives deemed to be guarantees also includes stable value contracts, commonly referred to as "stable value products", that require the Firm to make a payment of the difference between the market value and the book value of a counterparty's reference portfolio of assets in the event that market value is less than book value and certain other conditions have been met. Stable value products are transacted in order to allow investors to realize investment returns with less volatility than an unprotected portfolio. These contracts are typically longer-term or may have no stated maturity, but allow the Firm to elect to terminate the contract under certain conditions.

The notional value of derivative guarantees generally represents the Firm's maximum exposure. However, exposure to certain stable value products is contractually limited to a substantially lower percentage of the notional amount.

The fair value of derivative guarantees reflects the probability, in the Firm's view, of whether the Firm will be required to perform under the contract. The Firm reduces exposures to these contracts by entering into offsetting transactions, or by entering into contracts that hedge the market risk related to the derivative guarantees.

The following table summarizes the derivatives qualifying as guarantees as of December 31, 2024 and 2023.

(in millions)	December 31, 2024	December 31, 2023
Notional amounts		
Derivative guarantees	\$ 49,628	\$ 54,562
Stable value contracts with contractually limited exposure	32,939	32,488
Maximum exposure of stable value contracts with contractually limited exposure	1,740	1,652
Fair value		
Derivative payables	113	89

In addition to derivative contracts that meet the characteristics of a guarantee, the Firm is both a purchaser and seller of credit protection in the credit derivatives market. Refer to Note 5 for a further discussion of credit derivatives.

Unsettled securities financing agreements

In the normal course of business, the Firm enters into resale and securities borrowed agreements. At settlement, these commitments result in the Firm advancing cash to and receiving securities collateral from the counterparty. The Firm also enters into repurchase and securities loaned agreements. At settlement, these commitments result in the Firm receiving cash from and providing securities collateral to the counterparty. Such agreements settle at a future date. These agreements generally do not meet the definition of a derivative, and therefore, are not recorded on the Consolidated balance sheets until settlement date. These agreements predominantly have regular-way settlement terms. Refer to Note 11 for a further discussion of securities financing agreements.

Loan sales- and securitization-related indemnifications

Mortgage repurchase liability

In connection with the Firm's mortgage loan sale and securitization activities with U.S. GSEs the Firm has made representations and warranties that the loans sold meet certain requirements, and that may require the Firm to repurchase mortgage loans and/or indemnify the loan purchaser if such representations and warranties are breached by the Firm.

Private label securitizations

The liability related to repurchase demands associated with private label securitizations is separately evaluated by the Firm in establishing its litigation reserves.

Refer to Note 30 for additional information regarding litigation.

Loans sold with recourse

The Firm provides servicing for mortgages and certain commercial lending products on both a recourse and nonrecourse basis. In nonrecourse servicing, the principal credit risk to the Firm is the cost of temporary servicing advances of funds (i.e., normal servicing advances). In recourse servicing, the servicer agrees to share credit risk with the owner of the mortgage loans, such as Fannie Mae or Freddie Mac or a private investor, insurer or guarantor. Losses on recourse servicing predominantly occur when foreclosure sales proceeds of the property underlying a defaulted loan are less than the sum of the outstanding principal balance, plus accrued interest on the loan and the cost of holding and disposing of the underlying property. The Firm's securitizations are predominantly nonrecourse, thereby effectively transferring the risk of future credit losses to the purchaser of the mortgage-backed securities issued by the trust. The unpaid principal balance of loans sold with recourse as well as the carrying value of the related liability that the Firm has recorded in accounts payable and other liabilities on the Consolidated balance sheets, which is representative of the Firm's view of the likelihood it will have to perform under its recourse obligations, are disclosed in the table on page 303.

Other off-balance sheet arrangements

Indemnification agreements – general

In connection with issuing securities to investors outside the U.S., the Firm may agree to pay additional amounts to the holders of the securities in the event that, due to a change in tax law, certain types of withholding taxes are imposed on payments on the securities. The terms of the securities may also give the Firm the right to redeem the securities if such additional amounts are payable. The Firm may also enter into indemnification clauses such as in connection with the licensing of software to clients ("software licensees") or when it sells a business or assets to a third party ("third-party purchasers"), pursuant to which it indemnifies software licensees for claims of liability or damages that may occur subsequent to the licensing of the software, or third-party purchasers for losses they may incur due to actions taken by the Firm prior to the sale of the business or assets. It is difficult to estimate the Firm's maximum exposure under these indemnification arrangements, since this would require an assessment of future changes in tax law and future claims that may

be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to be remote.

Merchant charge-backs

Under the rules of payment networks, in its role as a merchant acquirer, the Firm's Merchant Services business in CIB Payments, retains a contingent liability for disputed processed credit and debit card transactions that result in a charge-back to the merchant. If a dispute is resolved in the cardholder's favor, the Firm will (through the cardholder's issuing bank) credit or refund the amount to the cardholder and will charge back the transaction to the merchant. If the Firm is unable to collect the amount from the merchant, the Firm will bear the loss for the amount credited or refunded to the cardholder. The Firm mitigates this risk by withholding future settlements, retaining cash reserve accounts or obtaining other collateral. In addition, the Firm recognizes a valuation allowance that covers the payment or performance risk related to charge-backs.

Clearing Services – Client Credit Risk

The Firm provides clearing services for clients by entering into securities purchases and sales and derivative contracts with CCPs, including ETDs such as futures and options, as well as OTC-cleared derivative contracts. As a clearing member, the Firm stands behind the performance of its clients, collects cash and securities collateral (margin) as well as any settlement amounts due from or to clients, and remits them to the relevant CCP or client in whole or part. There are two types of margin: variation margin is posted on a daily basis based on the value of clients' derivative contracts and initial margin is posted at inception of a derivative contract, generally on the basis of the potential changes in the variation margin requirement for the contract.

As a clearing member, the Firm is exposed to the risk of nonperformance by its clients, but is not liable to clients for the performance of the CCPs. Where possible, the Firm seeks to mitigate its risk to the client through the collection of appropriate amounts of margin at inception and throughout the life of the transactions. The Firm can also cease providing clearing services if clients do not adhere to their obligations under the clearing agreement. In the event of nonperformance by a client, the Firm would close out the client's positions and access available margin. The CCP would utilize any margin it holds to make itself whole, with any remaining shortfalls required to be paid by the Firm as a clearing member.

The Firm reflects its exposure to nonperformance risk of the client through the recognition of margin receivables from clients and margin payables to CCPs; the clients' underlying securities or derivative contracts are not reflected in the Firm's Consolidated Financial Statements.

It is difficult to estimate the Firm's maximum possible exposure through its role as a clearing member, as this would require an assessment of transactions that clients may execute in the future. However, based upon historical experience, and the credit risk mitigants available to the Firm, management believes it is unlikely that the Firm will have to make any material payments under these arrangements and the risk of loss is expected to be remote.

Refer to Note 5 for information on the derivatives that the Firm executes for its own account and records in its Consolidated Financial Statements.

Exchange & Clearing House Memberships

The Firm is a member of several securities and derivative exchanges and clearing houses, both in the U.S. and other countries, and it provides clearing services to its clients. Membership in some of these organizations requires the Firm to pay a pro rata share of the losses incurred by the organization as a result of the default of another member. Such obligations vary with different organizations. These obligations may be limited to the amount (or a multiple of the amount) of the Firm's contribution to the guarantee fund maintained by a clearing house or exchange as part of the resources available to cover any losses in the event of a member default. Alternatively, these obligations may also include a pro rata share of the residual losses after applying the guarantee fund. Additionally, certain clearing houses require the Firm as a member to pay a pro rata share of losses that may result from the clearing house's investment of guarantee fund contributions and initial margin, unrelated to and independent of the default of another member. Generally a payment would only be required should such losses exceed the resources of the clearing house or exchange that are contractually required to absorb the losses in the first instance. In certain cases, it is difficult to estimate the Firm's maximum possible exposure under these membership agreements, since this would require an assessment of future claims that may be made against the Firm that have not yet occurred. However, based on historical experience, management expects the risk of loss to the Firm to be remote. Where the Firm's maximum possible exposure can be estimated, the amount is disclosed in the table on page 303, in the Exchange & clearing house guarantees and commitments line.

Sponsored member repo program

The Firm acts as a sponsoring member to clear eligible overnight and term resale and repurchase agreements through the Government Securities Division of the Fixed Income Clearing Corporation ("FICC") on behalf of clients that become sponsored members under the FICC's rules. The Firm also guarantees to the FICC the prompt and full payment and performance of its sponsored member clients' respective obligations under the FICC's rules. The Firm minimizes its liability under these guarantees by obtaining a security interest in the cash or high-quality securities collateral that the clients place with the clearing house; therefore, the Firm expects the risk of loss to be remote. The Firm's maximum possible exposure, without taking into consideration the associated collateral, is included in the Exchange & clearing house guarantees and commitments line on page 303. Refer to Note 11 for additional information on credit risk mitigation practices on resale agreements and the types of collateral pledged under repurchase agreements.

Guarantees of subsidiaries

In the normal course of business, the Parent Company may provide counterparties with guarantees of certain of the trading and other obligations of its subsidiaries on a contract-by-contract basis, as negotiated with the Firm's counterparties. The obligations of the subsidiaries are included on the Firm's Consolidated balance sheets or are reflected as off-balance sheet commitments; therefore, the Parent Company has not recognized a separate liability for these guarantees. The Firm believes that the occurrence of any event that would trigger payments by the Parent Company under these guarantees is remote.

The Parent Company has guaranteed certain long-term debt and structured notes of its subsidiaries, including JPMorgan Chase Financial Company LLC ("JPMFC"), a 100%-owned finance subsidiary. All securities issued by JPMFC are fully and unconditionally guaranteed by the Parent Company and no other subsidiary of the Parent Company guarantees these securities. These guarantees, which rank *pari passu* with the Firm's unsecured and unsubordinated indebtedness, are not included in the table on page 303 of this Note. Refer to Note 20 for additional information.

Note 29 – Pledged assets and collateral

Pledged assets

The Firm pledges financial assets that it owns to maintain potential borrowing capacity at discount windows with Federal Reserve banks, various other central banks and FHLBs. Additionally, the Firm pledges assets for other purposes, including to collateralize repurchase and other securities financing agreements, to cover short sales and to collateralize derivative contracts and deposits. Certain of these pledged assets may be sold or repledged or otherwise used by the secured parties and are parenthetically identified on the Consolidated balance sheets as assets pledged.

The following table presents the carrying value of the Firm's pledged assets.

December 31, (in billions)	2024	2023
Assets that may be sold or repledged or otherwise used by secured parties	\$ 152.5	\$ 145.0
Assets that may not be sold or repledged or otherwise used by secured parties	297.9	244.2
Assets pledged at Federal Reserve banks and FHLBs	724.0	675.6
Total pledged assets	\$ 1,174.4	\$ 1,064.8

Total pledged assets do not include assets of consolidated VIEs; these assets are used to settle the liabilities of those entities. Refer to Note 14 for additional information on assets and liabilities of consolidated VIEs. Refer to Note 11 for additional information on the Firm's securities financing activities. Refer to Note 20 for additional information on the Firm's long-term debt. The significant components of the Firm's pledged assets were as follows.

December 31, (in billions)	2024	2023
Investment securities	\$ 89.6	\$ 108.6
Loans	740.9	681.7
Trading assets and other	343.9	274.5
Total pledged assets	\$ 1,174.4	\$ 1,064.8

Collateral

The Firm accepts financial assets as collateral that it is permitted to sell or repledge, deliver or otherwise use. This collateral is generally obtained under resale and other securities financing agreements, prime brokerage-related held-for-investment customer receivables and derivative contracts. Collateral is generally used under repurchase and other securities financing agreements, to cover short sales, and to collateralize derivative contracts and deposits.

The following table presents the fair value of collateral accepted.

December 31, (in billions)	2024	2023
Collateral permitted to be sold or repledged, delivered, or otherwise used	\$ 1,544.0	\$ 1,303.9
Collateral sold, repledged, delivered or otherwise used	1,210.7	982.8

Note 30 – Litigation

Contingencies

As of December 31, 2024, the Firm and its subsidiaries and affiliates are defendants or respondents in numerous evolving legal proceedings, including private proceedings, public proceedings, government investigations, regulatory enforcement matters, and the matters described below. The litigations range from individual actions involving a single plaintiff to class action lawsuits with potentially millions of class members. Investigations and regulatory enforcement matters involve both formal and informal proceedings, by both governmental agencies and self-regulatory organizations. These legal proceedings are at varying stages of adjudication, arbitration or investigation, and involve each of the Firm's lines of business and several geographies and a wide variety of claims (including common law tort and contract claims and statutory antitrust, securities and consumer protection claims), some of which present novel legal theories.

The Firm believes the estimate of the aggregate range of reasonably possible losses, in excess of reserves established, for its legal proceedings is from \$0 to approximately \$1.4 billion at December 31, 2024. This estimated aggregate range of reasonably possible losses was based upon information available as of that date for those proceedings in which the Firm believes that an estimate of reasonably possible loss can be made. For certain matters, the Firm does not believe that such an estimate can be made, as of that date. The Firm's estimate of the aggregate range of reasonably possible losses involves significant judgment, given:

- the number, variety and varying stages of the proceedings, including the fact that many are in preliminary stages,
- the existence in many such proceedings of multiple defendants, including the Firm, whose share of liability (if any) has yet to be determined,
- the numerous yet-unresolved issues in many of the proceedings, including issues regarding class certification and the scope of many of the claims, and
- the uncertainty of the various potential outcomes of such proceedings, including where the Firm has made assumptions concerning future rulings by the court or other adjudicator, or about the behavior or incentives of adverse parties or regulatory authorities, and those assumptions prove to be incorrect.

In addition, the outcome of a particular proceeding may be a result which the Firm did not take into account in its estimate because the Firm had deemed the likelihood of that outcome to be remote. Accordingly, the Firm's estimate of the aggregate range of reasonably possible losses will change from time to time, and actual losses may vary significantly.

Set forth below are descriptions of the Firm's material legal proceedings.

1MDB Litigation. J.P. Morgan (Suisse) SA was named as a defendant in a civil litigation filed in May 2021 in Malaysia by 1Malaysia Development Berhad ("1MDB"), a Malaysian state-owned and controlled investment fund. The claim alleges "dishonest assistance" against J.P. Morgan (Suisse) SA in relation to payments of \$300 million and \$500 million, from 2009 and 2010, respectively, received from 1MDB and paid into an account at J.P. Morgan (Suisse) SA held by 1MDB PetroSaudi Limited, a joint venture company between 1MDB and PetroSaudi Holdings (Cayman) Limited. In March 2024, the Court upheld the Firm's challenge to the validity of service and the Malaysian Court's jurisdiction to hear the claim. That decision has been appealed by 1MDB. In August 2023, the Court denied an application by 1MDB to discontinue its claim with permission to re-file a new claim in the future. That decision was appealed by both 1MDB and the Firm, and an appeals court is scheduled to hear both appeals in November 2025. In its appeal, the Firm seeks to prevent any claim from continuing.

In addition, in November 2023, the Federal Office of the Attorney General (OAG) in Switzerland notified J.P. Morgan (Suisse) SA that it is conducting an investigation into possible criminal liability in connection with transactions arising from J.P. Morgan (Suisse) SA's relationship with the 1MDB PetroSaudi joint venture and its related persons for the period September 2009 through August 2015. The OAG investigation is ongoing.

Amrapali. India's Enforcement Directorate ("ED") is investigating J.P. Morgan India Private Limited in connection with investments made in 2010 and 2012 by two offshore funds formerly managed by JPMorganChase entities into residential housing projects developed by the Amrapali Group ("Amrapali") relating to delays in delivering or failure to deliver residential units. In August 2021, the ED issued an order fining J.P. Morgan India Private Limited approximately \$31.5 million, and the Firm is appealing that order. Relatedly, in July 2019, the Supreme Court of India issued an order making preliminary findings that Amrapali and other parties, including unspecified JPMorganChase entities and the offshore funds that had invested in the projects, violated certain criminal currency control and money laundering provisions, and ordered the ED to conduct a further inquiry. The Firm is responding to and cooperating with the inquiry.

Foreign Exchange Investigations and Litigation. The Firm previously reported settlements with certain government authorities relating to its foreign exchange ("FX") sales and trading activities and controls related to those activities. Among those resolutions, in May 2015, the Firm pleaded guilty to a single violation of federal antitrust law. The Department of Labor ("DOL") granted the Firm exemptions that permit the Firm and its

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affiliates to continue to rely on the Qualified Professional Asset Manager exemption under the Employee Retirement Income Security Act (“ERISA”) through the ten-year disqualification period following the antitrust plea. The only remaining FX-related governmental inquiry is a South Africa Competition Commission matter which is currently pending before the South Africa Competition Tribunal.

With respect to civil litigation matters, some FX-related individual and putative class actions filed outside the U.S., including in the U.K., Israel, the Netherlands, Brazil and Australia, remain. In July 2023, the U.K. Court of Appeal overturned the Competition Appeal Tribunal's earlier denial of a request for class certification on an opt-out basis. The defendants have appealed this decision to the U.K. Supreme Court. In Israel, a settlement in principle has been reached on the putative class action, which remains subject to court approval.

Interchange Litigation. Groups of merchants and retail associations filed a series of class action complaints alleging that Visa and Mastercard, as well as certain banks, conspired to set the price of credit and debit card interchange fees and enacted related rules in violation of antitrust laws.

In September 2018, the parties settled the class action seeking monetary relief, with the defendants collectively contributing approximately \$6.2 billion. The settlement has been approved by the United States District Court for the Eastern District of New York and affirmed on appeal. Based on the percentage of merchants that opted out of the settlement, \$700 million has been returned to the defendants from the settlement escrow. A separate class action seeking injunctive relief continues, and in September 2021, the District Court granted plaintiffs' motion for class certification in part, and denied the motion in part. In June 2024, the District Court denied preliminary approval of a settlement of the injunctive class action in which Visa and Mastercard agreed to certain changes to their respective network rules and system-wide reductions in interchange rates for U.S.-based merchants. The parties are considering next steps.

Of the merchants who opted out of the damages class settlement, certain merchants filed individual actions raising similar allegations against Visa and Mastercard, as well as against the Firm and other banks. While some of those actions remain pending, the defendants have reached settlements with the merchants who opted out representing over 70% of the combined Mastercard-branded and Visa-branded payment card sales volume.

LIBOR and Other Benchmark Rate Investigations and Litigation. JPMorganChase has responded to inquiries from various governmental agencies and entities around the world relating primarily to the British Bankers Association's (“BBA”) London Interbank Offered Rate (“LIBOR”) for various currencies and the European Banking Federation's Euro Interbank Offered

Rate (“EURIBOR”). The Swiss Competition Commission's investigation relating to EURIBOR, to which the Firm and one other bank remain subject, continues. The Firm appealed a December 2016 decision by the European Commission against the Firm and other banks finding an infringement of European antitrust rules relating to EURIBOR. In December 2023, the European General Court annulled the fine imposed by the European Commission, but exercised its discretion to re-impose a fine in an identical amount. In March 2024, the Firm filed an appeal of this decision with the Court of Justice of the European Union.

In addition, the Firm has been named as a defendant along with other banks in various individual and putative class actions related to benchmark rates, including U.S. dollar LIBOR. In actions related to U.S. dollar LIBOR during the period that it was administered by the BBA, the United States District Court for the Southern District of New York granted class certification of antitrust claims related to bonds and interest rate swaps sold directly by the defendants, including the Firm. The Firm has obtained dismissal of certain actions and resolved certain other actions, and as to all remaining actions has moved for summary judgment. In addition, a lawsuit filed by a group of individual plaintiffs asserting antitrust claims, alleging that the Firm and other defendants were engaged in an unlawful agreement to set U.S. dollar LIBOR and conspired to monopolize the market for LIBOR-based consumer loans and credit cards was dismissed in October 2023 and affirmed on appeal by the United States Court of Appeals for the Ninth Circuit in December 2024. The Firm has resolved all non-U.S. dollar LIBOR actions.

Russian Litigation. The Firm is obligated to comply with international sanctions laws, which mandate the blocking of certain assets. These laws apply when assets associated with individuals, companies, products or services are within the scope of the sanctions. The Firm has faced actual and threatened litigation in Russia seeking payments that the Firm cannot make under, and is contractually excused from paying as a result of, relevant sanctions laws. In claims involving the Firm and claims filed against other financial institutions, Russian courts have disregarded the parties' contractual agreements concerning forum selection and did not recognize foreign sanctions laws as a basis for not making payment. Russian courts have entered judgment against the Firm in a number of claims, including one for \$439 million, and a judgment has been executed against assets held onshore by the Firm in Russia. The total amount of the judgments exceeds the total amount of available assets that the Firm holds in Russia. The Firm continues to appeal the Russian courts' decisions, and judgments may not be executed while on appeal. Russian courts have also ordered interim freezes of Firm assets in Russia (including, among other things, funds in bank accounts, securities, shares in authorized capital, and certain trademarks, of the named defendants) pending a determination of

certain underlying claims against the Firm. The Firm has challenged claims being pursued in the Russian courts and related freeze orders in other jurisdictions provided for by the parties' contractual forum selections. If further claims are enforced despite the actions taken by the Firm to challenge the claims and orders and to seek the proper application of law, the Firm's assets in Russia could be seized in full, and certain client assets could also be seized, or the Firm could be prevented from complying with its obligations.

SEC Inquiries. In October 2024, the Firm entered into settlements with the SEC to resolve inquiries related to, among other things, conflict disclosures concerning the selection of portfolio managers and the timing of the Firm's liquidation of shares distributed in-kind to certain investment vehicles that invest in third-party managed private funds and certain other matters. The resolutions required the Firm to pay a combined \$151 million in civil penalties and voluntary payments to customers. The Firm continues to cooperate in connection with SEC inquiries concerning the aggregation of accounts for fee billing and various other matters.

Shareholder Litigation. Several shareholder putative class actions, as well as shareholder derivative actions purporting to act on behalf of the Firm, have been filed against the Firm, its Board of Directors and certain of its current and former officers.

Certain of these shareholder suits relate to historical trading practices by former employees in the precious metals and U.S. treasuries markets and related conduct which were the subject of the Firm's resolutions with the DOJ, CFTC and SEC in September 2020, and fiduciary activities that were separately the subject of a resolution between JPMorgan Chase Bank, N.A. and the OCC in November 2020. One of these shareholder derivative suits was filed in the Supreme Court of the State of New York in May 2022, asserting breach of fiduciary duty and unjust enrichment claims relating to the historical trading practices and related conduct and fiduciary activities which were the subject of the resolutions described above. In December 2022, the court granted defendants' motion to dismiss this action in full, and in July 2023, the plaintiff filed an appeal, which remains pending.

A second shareholder derivative action relating to the historical trading practices and related conduct was filed in the United States District Court for the Eastern District of New York in December 2022. Defendants have moved to dismiss the complaint.

Zelle Network Litigation. In December 2024, the Consumer Financial Protection Bureau ("CFPB") filed a complaint against Early Warning Services, LLC ("EWS"), Bank of America, N.A., Wells Fargo Bank, N.A. and JPMorgan Chase Bank, N.A. in the United States District Court for the District of Arizona. The CFPB alleges that EWS and the defendant banks have failed to take sufficient efforts to prevent fraud on the Zelle network. The defendants will file a response to the complaint.

* * *

In addition to the various legal proceedings discussed above, JPMorganChase and its subsidiaries are named as defendants or are otherwise involved in a substantial number of other legal proceedings. The Firm believes it has meritorious defenses to the claims asserted against it in its currently outstanding legal proceedings and it intends to defend itself vigorously. Additional legal proceedings may be initiated from time to time in the future.

The Firm has established reserves for several hundred of its currently outstanding legal proceedings. In accordance with the provisions of U.S. GAAP for contingencies, the Firm accrues for a litigation-related liability when it is probable that such a liability has been incurred and the amount of the loss can be reasonably estimated. The Firm evaluates its outstanding legal proceedings each quarter to assess its litigation reserves, and makes adjustments in such reserves, upward or downward, as appropriate, based on management's best judgment after consultation with counsel. The Firm's legal expense was \$740 million, \$1.4 billion and \$266 million for the years ended December 31, 2024, 2023 and 2022, respectively. There is no assurance that the Firm's litigation reserves will not need to be adjusted in the future.

In view of the inherent difficulty of predicting the outcome of legal proceedings, particularly where the claimants seek very large or indeterminate damages, or where the matters present novel legal theories, involve a large number of parties or are in early stages of discovery, the Firm cannot state with confidence what will be the eventual outcomes of the currently pending matters, the timing of their ultimate resolution or the eventual losses, fines, penalties or consequences related to those matters. JPMorganChase believes, based upon its current knowledge and after consultation with counsel, consideration of the material legal proceedings described above and after taking into account its current litigation reserves and its estimated aggregate range of possible losses, that the other legal proceedings currently pending against it should not have a material adverse effect on the Firm's consolidated financial condition. The Firm notes, however, that in light of the uncertainties involved in such proceedings, there is no assurance that the ultimate resolution of these matters will not significantly exceed the reserves it has currently accrued or that a matter will not have material reputational consequences. As a result, the outcome of a particular matter may be material to JPMorganChase's operating results for a particular period, depending on, among other factors, the size of the loss or liability imposed and the level of JPMorganChase's income for that period.

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Note 31 – International operations

The following table presents income statement and balance sheet-related information for JPMorganChase by major international geographic area. The Firm defines international activities for purposes of this footnote presentation as business transactions that involve clients residing outside of the U.S., and the information presented below is based predominantly on the domicile of the client, the location from which the client relationship is managed, booking location or the location of the trading desk. However, many of the Firm's U.S. operations serve international businesses.

As the Firm's operations are highly integrated, estimates and subjective assumptions have been made to apportion revenue and expense between U.S. and international operations. These estimates and assumptions are consistent with the allocations used for the Firm's segment reporting as set forth in Note 32.

The Firm's long-lived assets for the periods presented are not considered by management to be significant in relation to total assets. The majority of the Firm's long-lived assets are located in the U.S.

As of or for the year ended December 31, (in millions)	Revenue ^(b)	Expense ^(c)	Income before income tax expense	Net income	Total assets
2024					
Europe/Middle East/Africa	\$ 22,353	\$ 12,843	\$ 9,510	\$ 6,713	\$ 552,407 ^(d)
Asia-Pacific	11,995	6,922	5,073	3,615	296,430
Latin America/Caribbean	3,885	1,895	1,990	1,512	73,631
Total international	38,233	21,660	16,573	11,840	922,468
North America ^(a)	139,323	80,815	58,508	46,631	3,080,346
Total	\$ 177,556	\$ 102,475	\$ 75,081	\$ 58,471	\$ 4,002,814
2023					
Europe/Middle East/Africa	\$ 20,974	\$ 11,947	\$ 9,027	\$ 6,402	\$ 529,335 ^(d)
Asia-Pacific	10,605	6,550	4,055	2,709	251,588
Latin America/Caribbean	3,294	1,971	1,323	994	83,003
Total international	34,873	20,468	14,405	10,105	863,926
North America ^(a)	123,231	76,024	47,207	39,447	3,011,467
Total	\$ 158,104	\$ 96,492	\$ 61,612	\$ 49,552	\$ 3,875,393
2022					
Europe/Middle East/Africa	\$ 18,765	\$ 11,754	\$ 7,011	\$ 5,158	\$ 558,430 ^(d)
Asia-Pacific	10,025	6,763	3,262	2,119	281,479
Latin America/Caribbean	3,178	1,697	1,481	1,156	78,673
Total international	31,968	20,214	11,754	8,433	918,582
North America ^(a)	96,727	62,315	34,412	29,243	2,747,161
Total	\$ 128,695	\$ 82,529	\$ 46,166	\$ 37,676	\$ 3,665,743

(a) Substantially reflects the U.S.

(b) Revenue is composed of net interest income and noninterest revenue.

(c) Expense is composed of noninterest expense and the provision for credit losses.

(d) Total assets for the U.K. were approximately \$369 billion, \$352 billion and \$357 billion at December 31, 2024, 2023 and 2022, respectively.

Note 32 – Business segments & Corporate

Business segment reorganization: Effective in the second quarter of 2024, the Firm reorganized its reportable business segments by combining the former Corporate & Investment Bank and Commercial Banking business segments to form one reportable segment, the Commercial & Investment Bank. As a result of the reorganization, the Firm has three reportable business segments – Consumer & Community Banking, Commercial & Investment Bank, and Asset & Wealth Management – with the remaining activities in Corporate.

Adoption of accounting standard — Segment Reporting — Improvements to Reportable Segment Disclosures:

This guidance was adopted retrospectively for the Firm's annual Consolidated Financial Statements for the year ended December 31, 2024. The adoption of this guidance requires additional reportable segment disclosures, primarily relating to significant segment expenses and the chief operating decision maker ("CODM"). Adoption of this guidance did not result in changes to the identification of the Firm's reportable business segments, or of its CODM.

The Firm is managed on an LOB basis. The business segments are determined based on the products and services provided, or the type of customer served, and they reflect the manner in which financial information is evaluated by the Firm's Operating Committee, whose members act collectively as the Firm's CODM. Segment results are presented on a managed basis. Refer to Segment & Corporate results in this footnote for a further discussion of JPMorganChase's reportable business segments and Corporate.

The following is a description of each of the Firm's reportable business segments, and the products and services that they provide to their respective client bases, as well as a description of Corporate activities.

Consumer & Community Banking

Consumer & Community Banking offers products and services to consumers and small businesses through bank branches, ATMs, digital (including mobile and online) and telephone banking. CCB is organized into Banking & Wealth Management (including Consumer Banking, Business Banking and J.P. Morgan Wealth Management), Home Lending (including Home Lending Production, Home Lending Servicing and Real Estate Portfolios) and Card Services & Auto. Banking & Wealth Management offers deposit, investment and lending products, cash management, payments and services. Home Lending includes mortgage origination and servicing activities, as well as portfolios consisting of residential mortgages and home equity loans. Card Services issues credit cards and offers travel services. Auto originates and services auto loans and leases.

Commercial & Investment Bank

The Commercial & Investment Bank is comprised of the Banking & Payments and Markets & Securities Services businesses. These businesses offer investment banking, lending, payments, market-making, financing, custody and securities products and services to a global base of corporate and institutional clients. Banking & Payments offers products and services in all major capital markets, including advising on corporate strategy and structure, capital-raising in equity and debt markets, and loan origination and syndication. Banking & Payments also provides services that enable clients to manage payments globally across liquidity and account solutions, commerce solutions, clearing, trade, and working capital. Markets & Securities Services includes Markets, which is a global market-maker across products, including cash and derivative instruments, and also offers sophisticated risk management solutions, lending, prime brokerage, clearing and research. Markets & Securities Services also includes Securities Services, a leading global custodian that provides custody, fund services, liquidity and trading services, and data solutions products.

Asset & Wealth Management

Asset & Wealth Management, with client assets of \$5.9 trillion, is a global leader in investment and wealth management.

Asset Management

Offers multi-asset investment management solutions across equities, fixed income, alternatives and money market funds to institutional and retail investors providing for a broad range of clients' investment needs.

Global Private Bank

Provides retirement products and services, brokerage, custody, estate planning, lending, deposits and investment management to high net worth clients.

The majority of AWM's client assets are in actively managed portfolios.

Corporate

Corporate consists of Treasury and Chief Investment Office ("CIO") and Other Corporate. Treasury and CIO is predominantly responsible for measuring, monitoring, reporting and managing the Firm's liquidity, funding, capital, structural interest rate and foreign exchange risks.

Other Corporate includes staff functions and expense that is centrally managed as well as certain Firm initiatives and activities not solely aligned to a specific LOB. The major Other Corporate functions include Real Estate, Technology, Legal, Corporate Finance, Human Resources, Internal Audit, Risk Management, Compliance, Control Management, Corporate Responsibility and various Other Corporate groups.

Description of business segment reporting methodology

Results of the reportable business segments are intended to present each segment as if it were a stand-alone business. The management reporting process that derives business segment results includes the allocation of certain income and expense items. The Firm periodically assesses the assumptions, methodologies and reporting classifications used for segment reporting, and therefore further refinements may be implemented in future periods. The Firm also assesses the level of capital required for each LOB on at least an annual basis. The Firm's LOBs also provide various business metrics which are utilized by the Firm and its investors and analysts in assessing performance.

Revenue sharing

When business segments or businesses within each segment join efforts to sell products and services to the Firm's clients and customers, the participating businesses may agree to share revenue from those transactions. Revenue is generally recognized in the segment responsible for the related product or service, with allocations to the other segments or businesses involved in the transaction. The segment and business results reflect these revenue-sharing agreements.

Expense allocation

Where business segments use services provided by Corporate support units, or another business segment, the costs of those services are allocated to the respective business segments. The expense is generally allocated based on the actual cost and use of services provided. In contrast, certain costs and investments related to Corporate that are not currently utilized by any LOB are not allocated to the business segments and are retained in Corporate. Expense retained in Corporate generally includes costs that would not be incurred if the segments were stand-alone businesses, and other items not solely aligned with a particular reportable business segment.

Funds transfer pricing

Funds transfer pricing ("FTP") is the process by which the Firm allocates interest income and expense to the LOBs and Other Corporate and transfers the primary interest rate risk and liquidity risk to Treasury and CIO.

The funds transfer pricing process considers the interest rate and liquidity risk characteristics of assets and liabilities and off-balance sheet products.

Periodically, the methodology and assumptions utilized in the FTP process are adjusted to reflect economic conditions and other factors, which may impact the allocation of net interest income to the segments. Effective in the fourth quarter of 2024, the Firm updated its FTP with respect to consumer deposits, which resulted in an increase in the funding benefit reflected within CCB's net interest income that is fully offset in Corporate, with no effect on the Firm's net interest income.

Foreign exchange risk

Foreign exchange risk is transferred from the LOBs and Other Corporate to Treasury and CIO for certain revenues and expenses. Treasury and CIO manages these risks centrally and reports the impact of foreign exchange rate movements related to the transferred risk in its results.

Debt expense and preferred stock dividend allocation

As part of the FTP process, almost all of the cost of the credit spread component of outstanding unsecured long-term debt and preferred stock dividends is allocated to the reportable business segments, while the balance of the cost is retained in Corporate. The methodology to allocate the cost of unsecured long-term debt and preferred stock dividends to the business segments is aligned with the relevant regulatory capital requirements and funding needs of the LOBs, as applicable. The allocated cost of unsecured long-term debt is included in a business segment's net interest income, and net income is reduced by preferred stock dividends, to arrive at a business segment's net income applicable to common equity.

Capital allocation

Each LOB and Corporate is allocated capital by taking into consideration a variety of factors including capital levels of similarly rated peers and applicable regulatory capital requirements. ROE is measured and internal targets for expected returns are established as key measures of an LOB's performance.

The Firm's current equity allocation methodology incorporates Basel III Standardized RWA and the GSIB surcharge, both under rules currently in effect, as well as a simulation of capital depletion in a severe stress environment. At least annually, the assumptions, judgments and methodologies used to allocate capital are reassessed and, as a result, the capital allocated to the LOBs and Corporate may change.

Segment & Corporate results

The following table provides a summary of results for the Firm's reportable business segments and Corporate activities as of or for the years ended December 31, 2024, 2023 and 2022, on a managed basis. The Firm's definition of managed basis starts with the reported U.S. GAAP results and includes certain reclassifications to present total net revenue for the Firm as a whole (and for each of the reportable business segments and Corporate) on an FTE basis. Accordingly, revenue from investments that receive tax credits and tax-exempt securities is presented in the managed results on a basis comparable to taxable investments and securities. This allows management to assess the comparability of revenue from year-to-year arising from both taxable and tax-exempt

sources. The corresponding income tax impact related to tax-exempt items is recorded within income tax expense/(benefit). These adjustments have no impact on net income as reported by the Firm as a whole or by the each of the LOBs and Corporate.

The Operating Committee reviews segment results including net interest income, noninterest revenue, noninterest expense, provision for credit losses and net income on a managed basis. The Operating Committee uses these measures to evaluate segment performance and to make key operating decisions, including resource and capital allocations.

Segment & Corporate results and reconciliation^(a)

(Table continued on next page)

As of or for the year ended December 31, (in millions, except ratios)	Consumer & Community Banking			Commercial & Investment Bank			Asset & Wealth Management		
	2024	2023	2022	2024	2023	2022	2024	2023	2022
Noninterest revenue	\$ 16,649	\$ 15,118	\$ 14,886 ^(e)	\$ 48,253	\$ 43,809	\$ 39,538 ^(b)	\$ 15,023	\$ 13,560	\$ 12,507
Net interest income	54,858	55,030	39,928	21,861	20,544	20,097	6,555	6,267	5,241
Total net revenue	71,507	70,148	54,814	70,114	64,353	59,635	21,578	19,827	17,748
Provision for credit losses	9,974	6,899	3,813	762	2,091	2,426	(68)	159	128
Compensation expense ^(b)	17,045	15,171	13,092	18,191	17,105	16,214	7,984	7,115	6,336
Noncompensation expense ^{(c)(d)}	20,991	19,648	18,116	17,162	16,867	15,855	6,430	5,665	5,493
Total noninterest expense	38,036	34,819	31,208^(e)	35,353	33,972	32,069^(b)	14,414	12,780	11,829
Income/(loss) before income tax expense/(benefit)	23,497	28,430	19,793	33,999	28,290	25,140	7,232	6,888	5,791
Income tax expense/(benefit)	5,894	7,198	4,877 ^(e)	9,153	8,018	6,002 ^(b)	1,811	1,661	1,426
Net income/(loss)	\$ 17,603	\$ 21,232	\$ 14,916	\$ 24,846	\$ 20,272	\$ 19,138	\$ 5,421	\$ 5,227	\$ 4,365
Average equity	\$ 54,500	\$ 54,349	\$ 50,000	\$ 132,000	\$ 137,507	\$ 128,000	\$ 15,500	\$ 16,671	\$ 17,000
Total assets	650,268	642,951	514,085	1,773,194	1,638,493	1,591,402	255,385	245,512	232,037
Return on equity	32 %	38 %	29 %	18 %	14 %	14 %	34 %	31 %	25 %
Overhead ratio	53	50	57	50	53	54	67	64	67

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(Table continued from previous page)

As of or for the year ended December 31, (in millions, except ratios)	Corporate			Reconciling Items ^(a)			Total		
	2024	2023	2022	2024	2023	2022	2024	2023	2022
Noninterest revenue	\$ 7,608 ^(f)	\$ 132	\$ (1,798)	\$ (2,560)	\$ (3,782)	\$ (3,148)	\$ 84,973 ^(f)	\$ 68,837	\$ 61,985
Net interest income	9,786	7,906	1,878	(477)	(480)	(434)	92,583	89,267	66,710
Total net revenue	17,394	8,038	80	(3,037)	(4,262)	(3,582)	177,556	158,104	128,695
Provision for credit losses	10	171	22	—	—	—	10,678	9,320	6,389
Total noninterest expense ^(d)	3,994 ^(g)	5,601	1,034	—	—	—	91,797 ^(g)	87,172	76,140
Income/(loss) before income tax expense/(benefit)	13,390	2,266	(976)	(3,037)	(4,262)	(3,582)	75,081	61,612	46,166
Income tax expense/(benefit)	2,789	(555)	(233)	(3,037)	(4,262)	(3,582)	16,610	12,060	8,490
Net income/(loss)	\$ 10,601	\$ 2,821	\$ (743)	\$ —	\$ —	\$ —	\$ 58,471	\$ 49,552	\$ 37,676
Average equity	\$ 110,370	\$ 73,529	\$ 58,068	\$ —	\$ —	\$ —	\$ 312,370	\$ 282,056	\$ 253,068
Total assets	1,323,967	1,348,437	1,328,219	NA	NA	NA	4,002,814	3,875,393	3,665,743
Return on equity	NM	NM	NM	NM	NM	NM	18 %	17 %	14 %
Overhead ratio	NM	NM	NM	NM	NM	NM	52	55	59

- (a) Segment results on a managed basis reflect revenue on a FTE basis with the corresponding income tax impact recorded within income tax expense/(benefit). These adjustments are eliminated in reconciling items to arrive at the Firm's reported U.S. GAAP results. In addition, effective January 1, 2024, the Firm adopted updates to the Accounting for Investments in Tax Credit Structures guidance, under the modified retrospective method. Refer to Notes 1, 6, 14 and 25 for additional information.
- (b) Excludes expense related to services provided by Corporate support units, which is allocated from Corporate to each respective reportable business segment, as applicable, through noncompensation expense.
- (c) Reflects occupancy; technology, communications and equipment; professional and outside services; marketing; and other expense. Refer to Note 6 for additional information on other expense.
- (d) Certain services are provided by Corporate and used by each of the reportable business segments. The costs of these services, including compensation-related costs, are allocated from Corporate to the respective reportable business segments, with the allocations recorded in noncompensation expense.
- (e) In the first quarter of 2023, the allocations of revenue and expense to CCB associated with a Merchant Services revenue sharing agreement were discontinued and are now retained in Payments in CIB. Prior-period amounts have been revised to conform with the current presentation.
- (f) Included a \$7.9 billion net gain related to Visa shares recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.
- (g) Included a \$1.0 billion contribution of Visa shares to the JPMorgan Chase Foundation recorded in the second quarter of 2024. Refer to Notes 2 and 6 for additional information.

Note 33 – Parent Company

The following tables present Parent Company-only financial statements.

Statements of income and comprehensive income

Year ended December 31, (in millions)	2024	2023	2022
Income			
Dividends from subsidiaries and affiliates:			
Bank and bank holding company	\$ 37,000	\$ 61,000	\$ 40,500
Non-bank	—	—	—
Interest income from subsidiaries	1,228	1,166	498
Other income/(expense) from subsidiaries:			
Bank and bank holding company	555	1,801	(3,497)
Non-bank	172	250	335
Other income/(expense)	1,252	(654)	5,271
Total income	40,207	63,563	43,107
Expense			
Interest expense/(income) to subsidiaries and affiliates ^(a)	7,433	2,258	22,731
Other interest expense/(income) ^(a)	8,068	11,714	(14,658)
Noninterest expense	3,280	3,431	2,817
Total expense	18,781	17,403	10,890
Income before income tax benefit and undistributed net income of subsidiaries	21,426	46,160	32,217
Income tax benefit	1,264	1,525	1,260
Equity in undistributed net income of subsidiaries	35,781	1,867	4,199
Net income	\$ 58,471	\$ 49,552	\$ 37,676
Other comprehensive income/(loss), net	(2,013)	6,898	(17,257)
Comprehensive income	\$ 56,458	\$ 56,450	\$ 20,419

Balance sheets

December 31, (in millions)	2024	2023
Assets		
Cash and due from banks	\$ 38	\$ 42
Deposits with banking subsidiaries	9,762	9,804
Trading assets	43,214	3,198
Advances to, and receivables from, subsidiaries:		
Bank and bank holding company	142	152
Non-bank	79	21
Investments (at equity) in subsidiaries and affiliates:		
Bank and bank holding company	603,044	568,472
Non-bank	1,238	1,045
Other assets	12,097	8,962
Total assets	\$669,614	\$ 591,696
Liabilities and stockholders' equity		
Borrowings from, and payables to, subsidiaries and affiliates	\$ 72,881	\$ 22,777
Short-term borrowings	—	999
Other liabilities	12,349	11,500
Long-term debt ^{(b)(c)}	239,626	228,542
Total liabilities^(c)	324,856	263,818
Total stockholders' equity	344,758	327,878
Total liabilities and stockholders' equity	\$669,614	\$ 591,696

Notes to consolidated financial statements

Statements of cash flows

Year ended December 31, (in millions)	2024	2023	2022
Operating activities			
Net income	\$ 58,471	\$ 49,552	\$ 37,676
Less: Net income of subsidiaries and affiliates	72,781	62,868	44,699
Parent company net loss	(14,310)	(13,316)	(7,023)
Cash dividends from subsidiaries and affiliates	37,000	61,000	40,500
Other operating adjustments	(44,671)	9,412	(23,747)
Net cash provided by/(used in) operating activities	(21,981)	57,096	9,730
Investing activities			
Net change in:			
Advances to and investments in subsidiaries and affiliates, net	—	(25,000)	—
All other investing activities, net	21	25	31
Net cash provided by/(used in) investing activities	21	(24,975)	31
Financing activities			
Net change in:			
Borrowings from subsidiaries and affiliates	49,902	(2,249)	(4,491)
Short-term borrowings	(999)	—	—
Proceeds from long-term borrowings	44,997	19,398	41,389
Payments of long-term borrowings	(29,753)	(25,105)	(18,294)
Proceeds from issuance of preferred stock	2,500	—	—
Redemption of preferred stock	(9,850)	—	(7,434)
Treasury stock repurchased	(18,830)	(9,824)	(3,162)
Dividends paid	(14,783)	(13,463)	(13,562)
All other financing activities, net	(1,270)	(879)	(1,205)
Net cash provided by/(used in) financing activities	21,914	(32,122)	(6,759)
Net increase/(decrease) in cash and due from banks and deposits with banking subsidiaries	(46)	(1)	3,002
Cash and due from banks and deposits with banking subsidiaries at the beginning of the year	9,846	9,847	6,845
Cash and due from banks and deposits with banking subsidiaries at the end of the year	\$ 9,800	\$ 9,846	\$ 9,847
Cash interest paid	\$ 14,851	\$ 13,742	\$ 7,462
Cash income taxes paid, net ^(d)	6,252	10,291	6,941

(a) Includes interest expense for intercompany derivative hedges on the Firm's LTD and related fair value adjustments, which is offset by related amounts in Other interest expense/(income).

- (b) At December 31, 2024, long-term debt that contractually matures in 2025 through 2029 totaled \$7.7 billion, \$29.3 billion, \$20.2 billion, \$35.0 billion, and \$18.5 billion, respectively.
- (c) Refer to Notes 20 and 28 for information regarding the Parent Company's guarantees of its subsidiaries' obligations.
- (d) Represents payments, net of refunds, made by the Parent Company to various taxing authorities and includes taxes paid on behalf of certain of its subsidiaries that are subsequently reimbursed. The reimbursements were \$5.0 billion, \$13.2 billion, and \$11.3 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

Note 34 – Business combinations

On May 1, 2023, JPMorganChase acquired certain assets and assumed certain liabilities of First Republic Bank (the "First Republic acquisition") from the Federal Deposit Insurance Corporation ("FDIC"), as receiver. The acquisition resulted in a bargain purchase gain, which represents the excess of the estimated fair value of the net assets acquired above the purchase price.

The Firm has determined that this acquisition constitutes a business combination under U.S. GAAP. Accordingly, the initial recognition of the assets acquired and liabilities assumed were generally measured at their estimated fair values as of May 1, 2023. The determination of those fair values required management to make certain market-based assumptions about expected future cash flows, discount rates and other valuation inputs at the time of the acquisition. The Firm believes that the fair value estimates of the assets acquired and liabilities assumed provide a reasonable basis for determining the estimated bargain purchase gain.

The First Republic acquisition resulted in a preliminary estimated bargain purchase gain of \$2.7 billion. The final bargain purchase gain of \$2.9 billion reflects adjustments of \$103 million and \$63 million for the years ended December 31, 2024 and 2023, respectively, made during the one-year measurement period, as permitted by U.S. GAAP, to finalize management's fair value estimates for the assets acquired and liabilities assumed. As of December 31, 2024, certain matters related to the final settlement remained outstanding between the Firm and the FDIC.

On January 17, 2025, the Firm reached an agreement with the FDIC with respect to certain outstanding items. As a result of the agreement, the Firm made a payment of \$609 million to the FDIC on January 31, 2025 and reduced its additional payable to the FDIC, which will result in a gain of approximately \$600 million to be recorded in other income in the first quarter of 2025.

In connection with the First Republic acquisition, the Firm and the FDIC entered into two shared-loss agreements with respect to certain loans and lending-related commitments (the "shared-loss assets"): the Commercial Shared-Loss Agreement ("CSLA") and the Single-Family Shared-Loss Agreement ("SFSLA"). The CSLA covers 80% of credit losses, on a pari passu basis, over 5 years with a subsequent 3-year recovery period for certain acquired commercial loans and other real estate exposure. The SFSLA covers 80% of credit losses, on a pari passu basis, for 7 years for certain acquired loans secured by mortgages on real property or shares in cooperative property constituting a primary residence. The indemnification assets, which represent the fair value of the CSLA and SFSLA on the acquisition date, are reflected in the total assets acquired.

As part of the consideration paid, JPMorganChase issued a five-year, \$50 billion secured note to the FDIC (the "Purchase Money Note"). The Purchase Money Note bears interest at a fixed rate of 3.4% and is secured by certain of the acquired loans. The Purchase Money Note is prepayable upon notice to the holder.

The Firm had placed a \$5 billion deposit with First Republic Bank on March 16, 2023, as part of \$30 billion of deposits provided by a consortium of large U.S. banks. The Firm's \$5 billion deposit was effectively settled as part of the acquisition and the associated allowance for credit losses was released upon closing. The Firm subsequently repaid the remaining \$25 billion of deposits to the consortium of banks, including accrued interest through the payment date on May 9, 2023.

Notes to consolidated financial statements

The computation of the purchase price, the fair values of the assets acquired and liabilities assumed as part of the First Republic acquisition and the related bargain purchase gain are presented below, which reflects adjustments made during the measurement period to the acquisition-date fair value of the net assets acquired. The measurement period ended on April 30, 2024.

(in millions)	Fair value purchase price allocation as of May 1, 2023	
Purchase price consideration		
Amounts paid/due to the FDIC, net of cash acquired ^(a)	\$	13,555
Purchase Money Note (at fair value) ^(b)		48,848
Settlement of First Republic deposit and other related party transactions ^(c)		5,447
Contingent consideration - Shared-loss agreements		15
Purchase price consideration	\$	67,865
Assets		
Securities	\$	30,285
Loans		153,242
Core deposit and customer relationship intangibles		1,455
Indemnification assets - Shared-loss agreements		675
Accounts receivable and other assets ^(d)		6,740
Total assets acquired	\$	192,397
Liabilities		
Deposits	\$	87,572
FHLB advances		27,919
Lending-related commitments		2,614
Accounts payable and other liabilities ^(d)		2,792
Deferred tax liabilities		757
Total liabilities assumed	\$	121,654
Fair value of net assets acquired	\$	70,743
Gain on acquisition, after income taxes	\$	2,878

(a) Net of cash acquired of \$680 million, and including disputed amounts with the FDIC as of April 30, 2024.

(b) As part of the consideration paid, JPMorganChase issued a five-year, \$50 billion secured note to the FDIC (the "Purchase Money Note").

(c) Includes \$447 million of securities financing transactions with First Republic Bank that were effectively settled on the acquisition date.

(d) Other assets include \$1.2 billion in tax-oriented investments and \$683 million of lease right-of-use assets. Other liabilities include the related tax-oriented investment liabilities of \$669 million and lease liabilities of \$748 million.

The following describes the accounting policies and fair value methodologies generally used by the Firm for the following assets acquired and liabilities assumed: core deposit and customer relationship intangibles, shared-loss agreements and the related indemnification assets, Purchase Money Note, and FHLB advances.

For further discussion of the Firm's accounting policies and valuation methodologies, refer to Notes 2 and 3 for fair value measurement, Note 10 for investment securities, Note 12 for loans, Note 17 for deposits, and Note 28 for lending-related commitments.

Core deposit and customer relationship intangibles

Core deposit and certain wealth management customer relationship intangibles were acquired as part of the First Republic acquisition. The core deposit intangible of \$1.3 billion was valued by discounting estimated after-tax cost savings over the remaining useful life of the deposits using the favorable source of funds method. The after-tax cost savings were estimated based on the difference between the cost of maintaining the core deposit base relative to the cost of next best alternative funding sources available to market participants. The customer relationship intangibles of \$180 million were valued by discounting estimated after-tax earnings over their remaining useful lives using the multi-period excess earnings

method. Both intangible asset valuations utilized assumptions that the Firm believes a market participant would use to estimate fair values, such as growth and attrition rates, projected fee income as well as related costs to service the relationships, and discount rates. The core deposit and customer relationship intangibles are amortized over a projected period of future cash flows of approximately 7 years. Refer to Note 15 for further discussion on other intangible assets.

Indemnification assets - Shared-loss agreements

The indemnification assets represent forecasted recoveries from the FDIC associated with the shared-loss assets over the respective shared-loss recovery periods. The indemnification assets were recorded at fair value in other assets on the Consolidated balance sheets on the acquisition date. The fair values of the indemnification assets were estimated based on the timing of the forecasted losses underlying the related allowance for credit losses. The subsequent quarterly remeasurement of the indemnification assets is based on changes in the amount and timing of forecasted losses in the allowance for credit losses associated with the shared-loss assets and is recorded in other income. Under certain circumstances, the Firm may be required to make a payment to the FDIC upon termination of the shared-loss agreements based on the level of actual losses and recoveries on the shared-loss assets. The estimated potential future payment is reflected as contingent consideration as part of the purchase price consideration.

Loans

The following table presents the unpaid principal balance ("UPB") and fair values of the loans acquired as of May 1, 2023, and reflects adjustments made during the measurement period to the acquisition-date fair value of the loans acquired.

(in millions)	May 1, 2023	
	UPB	Fair value
Residential real estate	\$ 106,240	\$ 92,053
Auto and other	3,093	2,030
Total consumer	109,333	94,083
Secured by real estate	37,117	33,602
Commercial & industrial	4,332	3,932
Other	23,499	21,625
Total wholesale	64,948	59,159
Total loans	\$ 174,281	\$ 153,242

Unaudited pro forma condensed combined financial information

The following table presents certain unaudited pro forma financial information for the year ended December 31, 2023 and 2022 as if the First Republic acquisition had occurred on January 1, 2022, including recognition of the estimated bargain purchase gain of \$2.8 billion and the provision for credit losses of \$1.2 billion. Additional adjustments include the interest on the Purchase Money Note and the impact of amortizing and accreting certain estimated fair value adjustments related to intangible assets, loans and lending-related commitments.

The Firm expects to achieve operating cost savings and other business synergies resulting from the acquisition that are not reflected in the pro forma amounts. The pro forma information is not necessarily indicative of the historical results of operations had the acquisition occurred on January 1, 2022, nor is it indicative of the results of operations in future periods.

(in millions)	Year ended December 31,	
	2023	2022
Noninterest revenue	\$ 65,816	\$ 66,510
Net interest income	90,856	71,005
Net income	48,665	41,089

Purchase Money Note and FHLB advances

The Purchase Money Note is recorded in long-term debt on the Consolidated balance sheets. The fair value of the Purchase Money Note was estimated based on a discounted cash flow methodology and incorporated estimated market discount rates.

The FHLB advances assumed in the acquisition are recorded in short-term borrowings and in long-term debt. The fair values of the FHLB advances were based on a discounted cash flow methodology and considered the observed FHLB advance issuance rates.

Supplementary Information: Distribution of assets, liabilities and stockholders' equity; interest rates and interest differentials

Consolidated average balance sheets, interest and rates

Provided below is a summary of JPMorganChase's consolidated average balances, interest and rates on a taxable-equivalent basis for the years 2022 through 2024. Income computed on a taxable-equivalent basis is the income reported in the Consolidated statements

of income, adjusted to present interest income and rates earned on assets exempt from income taxes (i.e., federal taxes) on a basis comparable with other taxable investments. The incremental tax rate used for calculating the taxable-equivalent adjustment was approximately 24% in 2024, 2023 and 2022.

(Table continued on next page)

(Unaudited)		2024	
Year ended December 31, (Taxable-equivalent interest and rates; in millions, except rates)	Average balance ^(f)	Interest ^(f)	Rate
Assets			
Deposits with banks	\$ 490,205	\$ 22,297	4.55 %
Federal funds sold and securities purchased under resale agreements	359,197	18,299	5.09
Securities borrowed	209,744	9,208	4.39
Trading assets – debt instruments	456,029	20,373	4.47
Taxable securities	583,329	21,947	3.76
Non-taxable securities ^(a)	27,912	1,393	4.99
Total investment securities	611,241	23,340	3.82 ⁽ⁱ⁾
Loans	1,322,425	92,588 ^(h)	7.00
All other interest-earning assets ^{(b)(c)}	88,726	8,305	9.36
Total interest-earning assets	3,537,567	194,410	5.50
Allowance for loan losses	(22,877)		
Cash and due from banks	22,591		
Trading assets – equity and other instruments	208,534		
Trading assets – derivative receivables	57,005		
Goodwill, MSRs and other intangible assets	64,393		
All other noninterest-earning assets	218,709		
Total assets	\$ 4,085,922		
Liabilities			
Interest-bearing deposits	\$ 1,748,050	\$ 49,559	2.84 %
Federal funds purchased and securities loaned or sold under repurchase agreements	363,820	19,149	5.26
Short-term borrowings	39,593	2,101	5.31
Trading liabilities – debt and all other interest-bearing liabilities ^{(d)(e)}	314,054	10,238	3.26
Beneficial interests issued by consolidated VIEs	26,515	1,383	5.22
Long-term debt	344,346	18,920	5.49
Total interest-bearing liabilities	2,836,378	101,350	3.57
Noninterest-bearing deposits	638,592		
Trading liabilities – equity and other instruments ^(e)	32,025		
Trading liabilities – derivative payables	39,497		
All other liabilities, including the allowance for lending-related commitments	203,006		
Total liabilities	3,749,498		
Stockholders' equity			
Preferred stock	24,054		
Common stockholders' equity	312,370		
Total stockholders' equity	336,424 ^(g)		
Total liabilities and stockholders' equity	\$ 4,085,922		
Interest rate spread			1.93 %
Net interest income and net yield on interest-earning assets		\$ 93,060	2.63

(a) Represents securities that are tax-exempt for U.S. federal income tax purposes.

(b) Includes brokerage-related held-for-investment customer receivables, which are classified in accrued interest and accounts receivable, and all other interest-earning assets, which are classified in other assets on the Consolidated Balance Sheets.

(c) The rates reflect the impact of interest earned on cash collateral where the cash collateral has been netted against certain derivative payables.

(d) All other interest-bearing liabilities include brokerage-related customer payables.

(e) The combined balance of trading liabilities – debt and equity instruments was \$185.4 billion, \$153.3 billion and \$138.1 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

(f) Includes the effect of derivatives that qualify for hedge accounting where applicable. Taxable-equivalent amounts are used, also where applicable. Refer to Note 5 for additional information on hedge accounting.

(Table continued from previous page)

2023			2022		
Average balance ^(f)	Interest ^(f)	Rate	Average balance ^(f)	Interest ^(f)	Rate
\$ 499,396	\$ 21,797	4.36 %	\$ 670,773	\$ 9,039	1.35 %
317,159	15,079	4.75	307,150	4,632	1.51
193,228	7,983	4.13	205,516	2,237	1.09
376,928	16,001	4.25	283,108	9,097	3.21
573,914	17,390	3.03	626,122	10,372	1.66
30,886	1,560	5.05	27,863	1,224	4.39
604,800	18,950	3.13 ⁽ⁱ⁾	653,985	11,596	1.77 ⁽ⁱ⁾
1,248,076	83,589 ^(h)	6.70	1,100,318	52,877 ^(h)	4.81
86,121	7,669	8.90	128,229	3,763	2.93
3,325,708	171,068	5.14	3,349,079	93,241	2.78
(20,762)			(17,399)		
24,853			27,601		
160,087			140,778		
64,227			78,606		
63,212			59,467		
204,899			215,408		
\$ 3,822,224			\$ 3,853,540		
\$ 1,698,529	\$ 40,016	2.36 %	\$ 1,748,666	\$ 10,082	0.58 %
256,086	13,259	5.18	242,762	3,721	1.53
37,468	1,894	5.05	46,063	747	1.62
286,605	9,396	3.28	268,019	3,246	1.21
18,648	953	5.11	11,208	226	2.02
296,433	15,803	5.33	250,080	8,075	3.23
2,593,769	81,321	3.14	2,566,798	26,097	1.02
660,538			719,249		
30,501			39,155		
46,355			57,388		
181,601			185,989		
3,512,764			3,568,579		
27,404			31,893		
282,056			253,068		
309,460 ^(g)			284,961 ^(g)		
\$ 3,822,224			\$ 3,853,540		
		2.00 %			1.76 %
\$ 89,747		2.70	\$ 67,144		2.00

(g) The ratio of average stockholders' equity to average assets was 8.2%, 8.1% and 7.4% for the years ended December 31, 2024, 2023 and 2022, respectively. The return on average stockholders' equity, based on net income, was 17.4%, 16.0% and 13.2% for the years ended December 31, 2024, 2023 and 2022, respectively.

(h) Included fees and commissions on loans of \$3.6 billion, \$2.2 billion and \$1.8 billion for the years ended December 31, 2024, 2023 and 2022, respectively.

(i) The annualized rate for securities based on amortized cost was 3.79%, 3.09% and 1.75% for the years ended December 31, 2024, 2023 and 2022, respectively, and does not give effect to changes in fair value that are reflected in AOCI.

Within the Consolidated average balance sheets, interest and rates summary, the principal amounts of nonaccrual loans have been included in the average loan balances used to determine the average interest rate earned on loans. Refer to Note 12 for additional information on nonaccrual loans, including interest accrued.

Interest rates and interest differential analysis of net interest income – U.S. and non-U.S.

Presented below is a summary of interest and rates segregated between U.S. and non-U.S. operations for the years 2022 through 2024. The segregation of U.S. and non-U.S. components is based on the location of the office recording the transaction.

(Table continued on next page)

(Unaudited) Year ended December 31, (Taxable-equivalent interest and rates; in millions, except rates)	2024		
	Average balance	Interest	Rate
Interest-earning assets			
Deposits with banks:			
U.S.	\$ 284,913	\$ 15,157	5.32 %
Non-U.S.	205,292	7,140	3.48
Federal funds sold and securities purchased under resale agreements:			
U.S.	193,210	10,686	5.53
Non-U.S.	165,987	7,613	4.59
Securities borrowed:			
U.S.	150,251	7,330	4.88
Non-U.S.	59,493	1,878	3.16
Trading assets – debt instruments:			
U.S.	309,568	13,579	4.39
Non-U.S.	146,461	6,794	4.64
Investment securities:			
U.S.	567,784	21,458	3.78
Non-U.S.	43,457	1,882	4.33
Loans:			
U.S.	1,211,978	85,621	7.06
Non-U.S.	110,447	6,967	6.31
All other interest-earning assets, predominantly U.S. ^(a)	88,726	8,305	9.36
Total interest-earning assets	3,537,567	194,410	5.50
Interest-bearing liabilities			
Interest-bearing deposits:			
U.S.	1,307,000	33,173	2.54
Non-U.S.	441,050	16,386	3.72
Federal funds purchased and securities loaned or sold under repurchase agreements:			
U.S.	294,476	15,949	5.42
Non-U.S.	69,344	3,200	4.61
Trading liabilities – debt, short-term and all other interest-bearing liabilities:			
U.S.	222,710	8,289	3.72
Non-U.S.	130,937	4,050	3.09
Beneficial interests issued by consolidated VIEs, predominantly U.S.	26,515	1,383	5.22
Long-term debt:			
U.S.	338,166	18,760	5.55
Non-U.S.	6,180	160	2.59
Total interest-bearing liabilities	2,836,378	101,350	3.57
Noninterest-bearing liabilities ^(b)	701,189		
Total investable funds	\$ 3,537,567	\$ 101,350	2.86 %
Net interest income and net yield:		\$ 93,060	2.63 %
U.S.		80,913	2.92
Non-U.S.		12,147	1.58
Percentage of total assets and liabilities attributable to non-U.S. operations:			
Assets			24.3
Liabilities			20.5

(a) The rates reflect the impact of interest earned on cash collateral where that cash collateral has been netted against certain derivative payables.

(b) Represents the amount of noninterest-bearing liabilities funding interest-earning assets.

Refer to the “Net interest income” discussion in Consolidated Results of Operations on pages 59–62 for further information.

(Table continued from previous page)

2023			2022		
Average balance	Interest	Rate	Average balance	Interest	Rate
\$ 296,784	\$ 15,348	5.17 %	\$ 456,366	\$ 7,418	1.63 %
202,612	6,449	3.18	214,407	1,621	0.76
155,304	8,330	5.36	130,213	2,191	1.68
161,855	6,749	4.17	176,937	2,441	1.38
133,805	6,239	4.66	142,736	1,811	1.27
59,423	1,744	2.93	62,780	426	0.68
248,541	10,721	4.31	170,975	5,414	3.17
128,387	5,280	4.11	112,133	3,683	3.28
568,505	17,469	3.07	623,285	10,994	1.76
36,295	1,481	4.08	30,700	602	1.96
1,137,162	76,884	6.76	985,187	48,953	4.97
110,914	6,705	6.05	115,131	3,924	3.41
86,121	7,669	8.90	128,229	3,763	2.93
3,325,708	171,068	5.14	3,349,079	93,241	2.78
1,290,110	26,253	2.03	1,358,322	7,026	0.52
408,419	13,763	3.37	390,344	3,056	0.78
197,049	10,639	5.40	173,016	3,083	1.78
59,037	2,620	4.44	69,746	638	0.91
205,388	7,774	3.79	194,570	2,384	1.23
118,685	3,516	2.96	119,512	1,609	1.35
18,648	953	5.11	11,208	226	2.02
293,218	15,749	5.37	246,670	8,026	3.25
3,215	54	1.68	3,410	49	1.44
2,593,769	81,321	3.14	2,566,798	26,097	1.02
731,939			782,281		
\$ 3,325,708	\$ 81,321	2.45 %	\$ 3,349,079	\$ 26,097	0.78 %
	\$ 89,747	2.70 %		\$ 67,144	2.00 %
	77,923	3.01		58,950	2.27
	11,824	1.61		8,194	1.09
		24.7			24.9
		20.2			20.6

Changes in net interest income, volume and rate analysis

The table below presents an attribution of net interest income between volume and rate. The attribution between volume and rate is calculated using annual average balances for each category of assets and liabilities shown in the table and the corresponding annual rates (refer to pages 322-325 for more information on average balances and rates). In this analysis, when the change cannot be isolated to either volume or rate, it has been allocated to volume. The annual rates include the impact of changes in market rates, as well as the impact of any change in composition of the various products within each category of asset or liability. This analysis is calculated separately for each category without consideration of the relationship between categories (for example, the net spread between the rates earned on assets and the rates paid on liabilities that fund those assets). As a result, changes in the granularity or groupings considered in this analysis would produce a different attribution result, and due to the complexities involved, precise allocation of changes in interest rates between volume and rates is inherently complex and judgmental.

(Unaudited)	2024 versus 2023			2023 versus 2022		
	Increase/(decrease) due to change in:		Net change	Increase/(decrease) due to change in:		Net change
	Volume	Rate		Volume	Rate	
Year ended December 31, (On a taxable-equivalent basis; in millions)						
Interest-earning assets						
Deposits with banks:						
U.S.	\$ (636)	\$ 445	\$ (191)	\$ (8,225)	\$ 16,155	\$ 7,930
Non-U.S.	83	608	691	(361)	5,189	4,828
Federal funds sold and securities purchased under resale agreements:						
U.S.	2,092	264	2,356	1,347	4,792	6,139
Non-U.S.	184	680	864	(629)	4,937	4,308
Securities borrowed:						
U.S.	797	294	1,091	(411)	4,839	4,428
Non-U.S.	(3)	137	134	(95)	1,413	1,318
Trading assets – debt instruments:						
U.S.	2,659	199	2,858	3,358	1,949	5,307
Non-U.S.	834	680	1,514	666	931	1,597
Investment securities:						
U.S.	(47)	4,036	3,989	(1,690)	8,165	6,475
Non-U.S.	310	91	401	228	651	879
Loans:						
U.S.	5,326	3,411	8,737	10,296	17,635	27,931
Non-U.S.	(26)	288	262	(258)	3,039	2,781
All other interest-earning assets, predominantly U.S.	240	396	636	(3,749)	7,655	3,906
Change in interest income	11,813	11,529	23,342	477	77,350	77,827
Interest-bearing liabilities						
Interest-bearing deposits:						
U.S.	340	6,580	6,920	(1,284)	20,511	19,227
Non-U.S.	1,194	1,429	2,623	597	10,110	10,707
Federal funds purchased and securities loaned or sold under repurchase agreements:						
U.S.	5,271	39	5,310	1,293	6,263	7,556
Non-U.S.	480	100	580	(480)	2,462	1,982
Trading liabilities – debt, short-term and all other interest-bearing liabilities:						
U.S.	659	(144)	515	409	4,981	5,390
Non-U.S.	380	154	534	(17)	1,924	1,907
Beneficial interests issued by consolidated VIEs, predominantly U.S.						
	409	21	430	381	346	727
Long-term debt:						
U.S.	2,483	528	3,011	2,494	5,229	7,723
Non-U.S.	77	29	106	(3)	8	5
Change in interest expense	11,293	8,736	20,029	3,390	51,834	55,224
Change in net interest income	\$ 520	\$ 2,793	\$ 3,313	\$ (2,913)	\$ 25,516	\$ 22,603

Glossary of Terms and Acronyms

2023 Form 10-K: Annual report on Form 10-K for the year ended December 31, 2023, filed with the U.S. Securities and Exchange Commission.

ABS: Asset-backed securities

Active foreclosures: Loans referred to foreclosure where formal foreclosure proceedings are ongoing. Includes both judicial and non-judicial states.

AFS: Available-for-sale

ALCO: Asset Liability Committee

Alternative assets “Alternatives”: The following types of assets constitute alternative investments - hedge funds, currency, real estate, private equity and other investment funds designed to focus on nontraditional strategies

Amortized cost: Amount at which a financing receivable or investment is originated or acquired, adjusted for accretion or amortization of premium, discount, and net deferred fees or costs, collection of cash, charge-offs, foreign exchange, and fair value hedge accounting adjustments. For AFS securities, amortized cost is also reduced by any impairment losses recognized in earnings. Amortized cost is not reduced by the allowance for credit losses, except where explicitly presented net.

AOCI: Accumulated other comprehensive income/ (loss)

ARM: Adjustable rate mortgage(s)

AUC: “Assets under custody”: Represents assets held directly or indirectly on behalf of clients under safekeeping, custody and servicing arrangements.

AUM: “Assets under management”: Represent assets managed by AWM on behalf of its Private Banking, Institutional and Retail clients. Includes “Committed capital not Called.”

Auto loan and lease origination volume: Dollar amount of auto loans and leases originated.

AWM: Asset & Wealth Management

Beneficial interests issued by consolidated VIEs: Represents the interest of third-party holders of debt, equity securities, or other obligations, issued by VIEs that JPMorganChase consolidates.

Benefit obligation: Refers to the projected benefit obligation for pension plans and the accumulated postretirement benefit obligation for OPEB plans.

BHC: Bank holding company

BWM: Banking & Wealth Management

Bridge Financing Portfolio: A portfolio of held-for-sale unfunded loan commitments and funded loans. The unfunded commitments include both short-term bridge loan commitments that will ultimately be replaced by longer term financing as well as term loan commitments. The funded loans include term loans and funded revolver facilities.

CB: Commercial Banking

CCAR: Comprehensive Capital Analysis and Review

CCB: Consumer & Community Banking

CCB Consumer customer: A unique individual that has financial ownership or decision-making power with respect to accounts; excludes customers under the age of 18. Where a customer uses the same identifier as both a Consumer and a Small business, the customer is included in both metrics.

CCB Small business customer: A unique business or legal entity that has financial ownership or decision-making power with respect to accounts. Where a customer uses the same identifier as both a Consumer and a Small business, the customer is included in both metrics.

CCO: Chief Compliance Officer

CCP: “Central counterparty” is a clearing house that interposes itself between counterparties to contracts traded in one or more financial markets, becoming the buyer to every seller and the seller to every buyer and thereby ensuring the future performance of open contracts. A CCP becomes a counterparty to trades with market participants through novation, an open offer system, or another legally binding arrangement.

CDS: Credit default swaps

CECL: Current Expected Credit Losses

CEO: Chief Executive Officer

CET1 Capital: Common equity Tier 1 capital

CFO: Chief Financial Officer

CFP: Contingency funding plan

CFTC: Commodity Futures Trading Commission

CIB: Commercial & Investment Bank

CIO: Chief Investment Office

Client assets: Represent assets under management as well as custody, brokerage, administration and deposit accounts.

Client deposits and other third-party liabilities:

Deposits, as well as deposits that are swept to on-balance sheet liabilities (e.g., commercial paper, federal funds purchased and securities loaned or sold under repurchase agreements) as part of client cash management programs.

Client investment assets: Represent assets under management as well as custody, brokerage and annuity accounts, and deposits held in investment accounts.

CLO: Collateralized loan obligations

CLTV: Combined loan-to-value

CMT: Constant Maturity Treasury

Collateral-dependent: A loan is considered to be collateral-dependent when repayment of the loan is expected to be provided substantially through the operation or sale of the collateral when the borrower is

Glossary of Terms and Acronyms

experiencing financial difficulty, including when foreclosure is deemed probable based on borrower delinquency.

Commercial Card: provides a wide range of payment services to corporate and public sector clients worldwide through the commercial card products. Services include procurement, corporate travel and entertainment, expense management services, and business-to-business payment solutions.

Credit derivatives: Financial instruments whose value is derived from the credit risk associated with the debt of a third-party issuer (the reference entity) which allow one party (the protection purchaser) to transfer that risk to another party (the protection seller). Upon the occurrence of a credit event by the reference entity, which may include, among other events, the bankruptcy or failure to pay its obligations, or certain restructurings of the debt of the reference entity, neither party has recourse to the reference entity. The protection purchaser has recourse to the protection seller for the difference between the face value of the CDS contract and the fair value at the time of settling the credit derivative contract. The determination as to whether a credit event has occurred is generally made by the relevant International Swaps and Derivatives Association ("ISDA") Determinations Committee.

Criticized: Criticized loans, lending-related commitments and derivative receivables that are classified as special mention, substandard and doubtful categories for regulatory purposes and are generally consistent with a rating of CCC+/Caa1 and below, as defined by S&P and Moody's.

CRO: Chief Risk Officer

CRR: Capital Requirements Regulation

CTC: CIO, Treasury and Corporate

Custom lending: Loans to AWM's Global Private Bank clients, including loans to private investment funds and loans that are collateralized by nontraditional asset types, such as art work, aircraft, etc.

CVA: Credit valuation adjustment

Debit and credit card sales volume: Dollar amount of card member purchases, net of returns.

Deposit margin: Represents net interest income expressed as a percentage of average deposits.

Distributed denial-of-service attack: The use of a large number of remote computer systems to electronically send a high volume of traffic to a target website to create a service outage at the target. This is a form of cyberattack.

Dodd-Frank Act: Wall Street Reform and Consumer Protection Act

DVA: Debit valuation adjustment

EC: European Commission

Eligible HQLA: Eligible high-quality liquid assets, for purposes of calculating the LCR, is the amount of unencumbered HQLA that satisfy certain operational considerations as defined in the LCR rule.

Eligible LTD: Long-term debt satisfying certain eligibility criteria

Embedded derivatives: are implicit or explicit terms or features of a financial instrument that affect some or all of the cash flows or the value of the instrument in a manner similar to a derivative. An instrument containing such terms or features is referred to as a "hybrid." The component of the hybrid that is the non-derivative instrument is referred to as the "host." For example, callable debt is a hybrid instrument that contains a plain vanilla debt instrument (i.e., the host) and an embedded option that allows the issuer to redeem the debt issue at a specified date for a specified amount (i.e., the embedded derivative). However, a floating rate instrument is not a hybrid composed of a fixed-rate instrument and an interest rate swap.

EPS: Earnings per share

ERISA: Employee Retirement Income Security Act of 1974

ETD: "Exchange-traded derivatives": Derivative contracts that are executed on an exchange and settled via a central clearing house.

EU: European Union

Expense categories:

- Volume- and/or revenue-related expenses generally correlate with changes in the related business/transaction volume or revenue. Examples include commissions and incentive compensation within the LOBs, depreciation expense related to operating lease assets, and brokerage expense related to trading transaction volume.
- Investments in the business include expenses associated with supporting medium- to longer-term strategic plans of the Firm. Examples include front office growth, market expansion, initiatives in technology (including related compensation), marketing, and acquisitions.
- Structural expenses are those associated with the day-to-day cost of running the Firm and are expenses not included in the above two categories. Examples include employee salaries and benefits, certain other incentive compensation, and costs related to real estate.

Fannie Mae: Federal National Mortgage Association

FASB: Financial Accounting Standards Board

FCA: Financial Conduct Authority

FCC: Firmwide Control Committee

FDIC: Federal Deposit Insurance Corporation

FDM: "Financial difficulty modification" applies to loan modifications effective January 1, 2023, and is deemed

Glossary of Terms and Acronyms

to occur when the Firm modifies specific terms of the original loan agreement. The following types of modifications are considered FDMs: principal forgiveness, interest rate reduction, other-than-insignificant payment delay, term extension or a combination of these modifications.

Federal Reserve: The Board of the Governors of the Federal Reserve System

FFIEC: Federal Financial Institutions Examination Council

FHA: Federal Housing Administration

FHLB: Federal Home Loan Bank

FICC: The Fixed Income Clearing Corporation

FICO score: A measure of consumer credit risk provided by credit bureaus, typically produced from statistical models by Fair Isaac Corporation utilizing data collected by the credit bureaus.

FINRA: Financial Industry Regulatory Authority

Firm: JPMorgan Chase & Co.

Forward points: Represents the interest rate differential between two currencies, which is either added to or subtracted from the current exchange rate (i.e., “spot rate”) to determine the forward exchange rate.

FRC: Firmwide Risk Committee

Freddie Mac: Federal Home Loan Mortgage Corporation

Free standing derivatives: a derivative contract entered into either separate and apart from any of the Firm’s other financial instruments or equity transactions. Or, in conjunction with some other transaction and is legally detachable and separately exercisable.

FSB: Financial Stability Board

FTE: Fully taxable equivalent

FVA: Funding valuation adjustment

FX: Foreign exchange

G7: Group of Seven nations: Countries in the G7 are Canada, France, Germany, Italy, Japan, the U.K. and the U.S.

G7 government securities: Securities issued by the government of one of the G7 nations.

Ginnie Mae: Government National Mortgage Association

GSIB: Global systemically important banks

HELOC: Home equity line of credit

Home equity – senior lien: Represents loans and commitments where JPMorganChase holds the first security interest on the property.

Home equity – junior lien: Represents loans and commitments where JPMorganChase holds a security

interest that is subordinate in rank to other liens.

HQLA: “High-quality liquid assets” consist of cash and certain high-quality liquid securities as defined in the LCR rule.

HTM: Held-to-maturity

IBOR: Interbank Offered Rate

ICAAP: Internal capital adequacy assessment process

IDI: Insured depository institutions

IHC: JPMorgan Chase Holdings LLC, an intermediate holding company

Indirect tax expense: Refers to taxes that are imposed on goods and services rather than on income. Examples of indirect taxes include value-added tax (“VAT”) and sales tax, among others.

Investment-grade: An indication of credit quality based on JPMorganChase’s internal risk assessment. The Firm considers ratings of BBB-/Baa3 or higher as investment-grade.

IPO: Initial public offering

ISDA: International Swaps and Derivatives Association

JPMorganChase: JPMorgan Chase & Co.

JPMorgan Chase Bank, N.A.: JPMorgan Chase Bank, National Association

JPMorgan Chase Foundation or the Firm’s

Foundation: A not-for-profit organization that makes contributions for charitable and educational purposes.

J.P. Morgan Securities: J.P. Morgan Securities LLC

JPMSE: J.P. Morgan SE

LCR: Liquidity coverage ratio

LDA: Loss Distribution Approach

LGD: Loss given default

LIBOR: London Interbank Offered Rate

LLC: Limited Liability Company

LOB: Line of business

LOB CROs: Line of Business and CTC Chief Risk Officers

LTIP: Long-term incentive plan

LTV: “Loan-to-value”: For residential real estate loans, the relationship, expressed as a percentage, between the principal amount of a loan and the appraised value of the collateral (i.e., residential real estate) securing the loan.

Origination date LTV ratio

The LTV ratio at the origination date of the loan. Origination date LTV ratios are calculated based on the actual appraised values of collateral (i.e., loan-level data) at the origination date.

Current estimated LTV ratio

An estimate of the LTV as of a certain date. The current estimated LTV ratios are calculated using estimated collateral values derived from a nationally recognized

Glossary of Terms and Acronyms

home price index measured at the metropolitan statistical area ("MSA") level. These MSA-level home price indices consist of actual data to the extent available and forecasted data where actual data is not available. As a result, the estimated collateral values used to calculate these ratios do not represent actual appraised loan-level collateral values; as such, the resulting LTV ratios are necessarily imprecise and should therefore be viewed as estimates.

Combined LTV ratio

The LTV ratio considering all available lien positions, as well as unused lines, related to the property. Combined LTV ratios are used for junior lien home equity products.

Macro businesses: the macro businesses include Rates, Currencies and Emerging Markets, Fixed Income Financing and Commodities in CIB's Fixed Income Markets.

Managed basis: A non-GAAP presentation of Firmwide financial results that includes reclassifications to present revenue on a fully taxable-equivalent basis. Management also uses this financial measure at the segment level, because it believes this provides information to enable investors to understand the underlying operational performance and trends of the particular business segment and facilitates a comparison of the business segment with the performance of competitors.

Markets: consists of CIB's Fixed Income Markets and Equity Markets businesses.

Master netting agreement: A single agreement with a counterparty that permits multiple transactions governed by that agreement to be terminated or accelerated and settled through a single payment in a single currency in the event of a default (e.g., bankruptcy, failure to make a required payment or securities transfer or deliver collateral or margin when due).

MBS: Mortgage-backed securities

MD&A: Management's discussion and analysis

Measurement alternative: Measures equity securities without readily determinable fair values at cost less impairment (if any), plus or minus observable price changes from an identical or similar investment of the same issuer.

Merchant Services: offers merchants payment processing capabilities, fraud and risk management, data and analytics, and other payments services. Through Merchant Services, merchants of all sizes can accept payments via credit and debit cards and payments in multiple currencies.

MEV: Macroeconomic variable

Moody's: Moody's Investor Services

Mortgage origination channels:

Retail – Borrowers who buy or refinance a home through direct contact with a mortgage banker employed by the Firm using a branch office, the Internet or by phone. Borrowers are frequently referred to a mortgage banker by a banker in a Chase branch, real estate brokers, home builders or other third parties.

Correspondent – Banks, thrifts, other mortgage banks and other financial institutions that sell closed loans to the Firm.

Mortgage product types:

Alt-A

Alt-A loans are generally higher in credit quality than subprime loans but have characteristics that would disqualify the borrower from a traditional prime loan. Alt-A lending characteristics may include one or more of the following: (i) limited documentation; (ii) a high CLTV ratio; (iii) loans secured by non-owner occupied properties; or (iv) a debt-to-income ratio above normal limits. A substantial proportion of the Firm's Alt-A loans are those where a borrower does not provide complete documentation of his or her assets or the amount or source of his or her income.

Option ARMs

The option ARM real estate loan product is an adjustable-rate mortgage loan that provides the borrower with the option each month to make a fully amortizing, interest-only or minimum payment. The minimum payment on an option ARM loan is based on the interest rate charged during the introductory period. This introductory rate is usually significantly below the fully indexed rate. The fully indexed rate is calculated using an index rate plus a margin. Once the introductory period ends, the contractual interest rate charged on the loan increases to the fully indexed rate and adjusts monthly to reflect movements in the index. The minimum payment is typically insufficient to cover interest accrued in the prior month, and any unpaid interest is deferred and added to the principal balance of the loan. Option ARM loans are subject to payment recast, which converts the loan to a variable-rate fully amortizing loan upon meeting specified loan balance and anniversary date triggers.

Prime

Prime mortgage loans are made to borrowers with good credit records who meet specific underwriting requirements, including prescriptive requirements related to income and overall debt levels. New prime mortgage borrowers provide full documentation and generally have reliable payment histories.

Subprime

Subprime loans are loans that, prior to mid-2008, were offered to certain customers with one or more high risk characteristics, including but not limited to: (i) unreliable or poor payment histories; (ii) a high LTV ratio of greater than 80% (without borrower-paid mortgage insurance); (iii) a high debt-to-income ratio; (iv) an occupancy type for the loan is other than the borrower's

Glossary of Terms and Acronyms

primary residence; or (v) a history of delinquencies or late payments on the loan.

MREL: Minimum requirements for own funds and eligible liabilities

MSR: Mortgage servicing rights

Multi-asset: Any fund or account that allocates assets under management to more than one asset class.

NA: Data is not applicable or available for the period presented.

NAV: Net Asset Value

Net Capital Rule: Rule 15c3-1 under the Securities Exchange Act of 1934.

Net charge-off/(recovery) rate: Represents net charge-offs/(recoveries) (annualized) divided by average retained loans for the reporting period.

Net interchange income includes the following components:

- **Interchange income:** Fees earned by credit and debit card issuers on sales transactions.
- **Rewards costs:** The cost to the Firm for points earned by cardholders enrolled in credit card rewards programs generally tied to sales transactions.
- **Partner payments:** Payments to co-brand credit card partners based on the cost of loyalty program rewards earned by cardholders on credit card transactions.

Net mortgage servicing revenue: Includes operating revenue earned from servicing third-party mortgage loans, which is recognized over the period in which the service is provided; changes in the fair value of MSRs; the impact of risk management activities associated with MSRs; and gains and losses on securitization of excess mortgage servicing. Net mortgage servicing revenue also includes gains and losses on sales and lower of cost or fair value adjustments of certain repurchased loans insured by U.S. government agencies.

Net revenue rate: Represents Card Services net revenue (annualized) expressed as a percentage of average loans for the period.

Net yield on interest-earning assets: The average rate for interest-earning assets less the average rate paid for all sources of funds.

NFA: National Futures Association

NM: Not meaningful

NOL: Net operating loss

Nonaccrual loans: Loans for which interest income is not recognized on an accrual basis. Loans (other than credit card loans and certain consumer loans insured by U.S. government agencies) are placed on nonaccrual status when full payment of principal and interest is not expected, regardless of delinquency status, or when principal and interest have been in default for a period

of 90 days or more unless the loan is both well-secured and in the process of collection. Collateral-dependent loans are typically maintained on nonaccrual status.

Nonperforming assets: Nonperforming assets include nonaccrual loans, nonperforming derivatives and certain assets acquired in loan satisfactions, predominantly real estate owned and other commercial and personal property.

NSFR: Net Stable Funding Ratio

OAS: Option-adjusted spread

OCC: Office of the Comptroller of the Currency

OCI: Other comprehensive income/(loss)

OPEB: Other postretirement employee benefit

Operating losses: Primarily refer to fraud losses associated with customer deposit accounts, credit and debit cards; exclude legal expense.

Over-the-counter (“OTC”) derivatives: Derivative contracts that are negotiated, executed and settled bilaterally between two derivative counterparties, where one or both counterparties is a derivatives dealer.

Over-the-counter cleared (“OTC-cleared”) derivatives: Derivative contracts that are negotiated and executed bilaterally, but subsequently settled via a central clearing house, such that each derivative counterparty is only exposed to the default of that clearing house.

Overhead ratio: Noninterest expense as a percentage of total net revenue.

Parent Company: JPMorgan Chase & Co.

Participating securities: Represents unvested share-based compensation awards containing nonforfeitable rights to dividends or dividend equivalents (collectively, “dividends”), which are included in the earnings per share calculation using the two-class method. JPMorganChase grants RSUs to certain employees under its share-based compensation programs, which entitle the recipients to receive nonforfeitable dividends during the vesting period on a basis equivalent to the dividends paid to holders of common stock. These unvested awards meet the definition of participating securities. Under the two-class method, all earnings (distributed and undistributed) are allocated to each class of common stock and participating securities, based on their respective rights to receive dividends.

PCAOB: Public Company Accounting Oversight Board

PCD: “Purchased credit deteriorated” assets represent acquired financial assets that as of the date of acquisition have experienced a more-than-insignificant deterioration in credit quality since origination, as determined by the Firm.

PD: Probability of default

Pillar 1: The Basel framework consists of a three “Pillar” approach. Pillar 1 establishes minimum capital

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requirements, defines eligible capital instruments, and prescribes rules for calculating RWA.

Pillar 3: The Basel framework consists of a three “Pillar” approach. Pillar 3 encourages market discipline through disclosure requirements which allow market participants to assess the risk and capital profiles of banks.

PRA: Prudential Regulation Authority

Preferred stock dividends: reflects dividends declared and deemed dividends upon redemption of preferred stock

Pre-provision profit/(loss): Represents total net revenue less noninterest expense. The Firm believes that this financial measure is useful in assessing the ability of a lending institution to generate income in excess of its provision for credit losses.

Pre-tax margin: Represents income before income tax expense divided by total net revenue, which is, in management’s view, a comprehensive measure of pretax performance derived by measuring earnings after all costs are taken into consideration. It is one basis upon which management evaluates the performance of AWM against the performance of their respective competitors.

Principal transactions revenue: Principal transactions revenue is driven by many factors, including:

- the bid-offer spread, which is the difference between the price at which a market participant is willing and able to sell an instrument to the Firm and the price at which another market participant is willing and able to buy it from the Firm, and vice versa; and
- realized and unrealized gains and losses on financial instruments and commodities transactions, including those accounted for under the fair value option, primarily used in client-driven market-making activities.
 - Realized gains and losses result from the sale of instruments, closing out or termination of transactions, or interim cash payments.
 - Unrealized gains and losses result from changes in valuation.

In connection with its client-driven market-making activities, the Firm transacts in debt and equity instruments, derivatives and commodities, including physical commodities inventories and financial instruments that reference commodities.

Principal transactions revenue also includes realized and unrealized gains and losses related to:

- derivatives designated in qualifying hedge accounting relationships, primarily fair value hedges of commodity and foreign exchange risk;
- derivatives used for specific risk management purposes, primarily to mitigate credit, foreign exchange and interest rate risks.

Production revenue: Includes fees and income recognized as earned on mortgage loans originated with the intent to sell, and the impact of risk management activities associated with the mortgage pipeline and warehouse loans. Production revenue also includes gains and losses on sales and lower of cost or fair value adjustments on mortgage loans held-for-sale (excluding certain repurchased loans insured by U.S. government agencies), and changes in the fair value of financial instruments measured under the fair value option.

PSU(s): Performance share units

Regulatory VaR: Daily aggregated VaR calculated in accordance with regulatory rules.

REO: Real estate owned

Reported basis: Financial statements prepared under U.S. GAAP, which excludes the impact of taxable-equivalent adjustments.

Retained loans: Loans that are held-for-investment (i.e., excludes loans held-for-sale and loans at fair value).

Revenue wallet: Proportion of fee revenue based on estimates of investment banking fees generated across the industry (i.e., the revenue wallet) from investment banking transactions in M&A, equity and debt underwriting, and loan syndications. Source: Dealogic, a third-party provider of investment banking competitive analysis and volume-based league tables for the above noted industry products.

RHS: Rural Housing Service of the U.S. Department of Agriculture

ROA: Return on assets

ROE: Return on equity

ROTCE: Return on tangible common equity

ROU assets: Right-of-use assets

RSU(s): Restricted stock units

RWA “Risk-weighted assets”: Basel III establishes two comprehensive approaches for calculating RWA (a Standardized approach and an Advanced approach) which include capital requirements for credit risk, market risk, and in the case of Basel III Advanced, also operational risk. Key differences in the calculation of credit risk RWA between the Standardized and Advanced approaches are that for Basel III Advanced, credit risk RWA is based on risk-sensitive approaches which largely rely on the use of internal credit models and parameters, whereas for Basel III Standardized, credit risk RWA is generally based on supervisory risk-weightings which vary primarily by counterparty type and asset class. Market risk RWA is calculated on a generally consistent basis between Basel III Standardized and Basel III Advanced.

S&P: Standard and Poor’s

SAR as it pertains to Hong Kong: Special Administrative Region

Glossary of Terms and Acronyms

SAR(s) as it pertains to employee stock awards: Stock appreciation rights

SCB: Stress capital buffer

Scored portfolios: Consumer loan portfolios that predominantly include residential real estate loans, credit card loans, auto loans to individuals and certain small business loans.

SEC: U.S. Securities and Exchange Commission

Securities financing agreements: Include resale, repurchase, securities borrowed and securities loaned agreements

Securitized Products Group: Comprised of Securitized Products and tax-oriented investments.

Seed capital: Initial JPMorgan capital invested in products, such as mutual funds, with the intention of ensuring the fund is of sufficient size to represent a viable offering to clients, enabling pricing of its shares, and allowing the manager to develop a track record. After these goals are achieved, the intent is to remove the Firm's capital from the investment.

Shelf securities: Securities registered with the SEC under a shelf registration statement that have not been issued, offered or sold. These securities are not included in league tables until they have actually been issued.

Single-name: Single reference-entities

SLR: Supplementary leverage ratio

SMBS: Stripped mortgage-backed securities

SOFR: Secured Overnight Financing Rate

SPEs: Special purpose entities

Structural interest rate risk: Represents interest rate risk of the non-trading assets and liabilities of the Firm.

Structured notes: Structured notes are financial instruments whose cash flows are linked to the movement in one or more indexes, interest rates, foreign exchange rates, commodities prices, prepayment rates, underlying reference pool of loans or other market variables. The notes typically contain embedded (but not separable or detachable) derivatives. Contractual cash flows for principal, interest, or both can vary in amount and timing throughout the life of the note based on non-traditional indexes or non-traditional uses of traditional interest rates or indexes.

Suspended foreclosures: Loans referred to foreclosure where formal foreclosure proceedings have started but are currently on hold, which could be due to bankruptcy or loss mitigation. Includes both judicial and non-judicial states.

Taxable-equivalent basis: In presenting results on a managed basis, the total net revenue for each of the reportable business segments and Corporate, and the Firm as a whole, is presented on a tax-equivalent basis. Accordingly, revenue from investments that receive tax

credits and tax-exempt securities is presented in managed basis results on a level comparable to taxable investments and securities; the corresponding income tax impact related to tax-exempt items is recorded within income tax expense.

TBVPS: Tangible book value per share

TCE: Tangible common equity

TDR: "Troubled debt restructuring" applies to loan modifications granted prior to January 1, 2023 and is deemed to occur when the Firm modifies the original terms of a loan agreement by granting a concession to a borrower that is experiencing financial difficulty. Loans with short-term and other insignificant modifications that are not considered concessions are not TDRs.

TLAC: Total Loss Absorbing Capacity

U.K.: United Kingdom

Unaudited: Financial statements and/or information that have not been subject to auditing procedures by an independent registered public accounting firm.

U.S.: United States of America

U.S. GAAP: Accounting principles generally accepted in the U.S.

U.S. government agencies: U.S. government agencies include, but are not limited to, agencies such as Ginnie Mae and FHA, and do not include Fannie Mae and Freddie Mac which are U.S. government-sponsored enterprises ("U.S. GSEs"). In general, obligations of U.S. government agencies are fully and explicitly guaranteed as to the timely payment of principal and interest by the full faith and credit of the U.S. government in the event of a default.

U.S. GSE(s): "U.S. government-sponsored enterprises" are quasi-governmental, privately-held entities established or chartered by the U.S. government to serve public purposes as specified by the U.S. Congress to improve the flow of credit to specific sectors of the economy and provide certain essential services to the public. U.S. GSEs include Fannie Mae and Freddie Mac, but do not include Ginnie Mae or FHA. U.S. GSE obligations are not explicitly guaranteed as to the timely payment of principal and interest by the full faith and credit of the U.S. government.

U.S. Treasury: U.S. Department of the Treasury

VA: U.S. Department of Veterans Affairs

VaR: "Value-at-risk" is a measure of the dollar amount of potential loss from adverse market moves in an ordinary market environment.

VCG: Valuation Control Group

VGf: Valuation Governance Forum

VIEs: Variable interest entities

Warehouse loans: Consist of prime mortgages originated with the intent to sell that are accounted for at fair value and classified as loans.

Signatures

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on behalf of the undersigned, thereunto duly authorized.

JPMorgan Chase & Co.
(Registrant)

By: /s/ JAMES DIMON
(James Dimon
Chairman and Chief Executive Officer)

February 14, 2025

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacity and on the date indicated. JPMorgan Chase & Co. does not exercise the power of attorney to sign on behalf of any Director.

	Capacity	Date
<u>/s/ JAMES DIMON</u> (James Dimon)	Director, Chairman and Chief Executive Officer (Principal Executive Officer)	
<u>/s/ LINDA B. BAMMANN</u> (Linda B. Bammann)	Director	
<u>/s/ STEPHEN B. BURKE</u> (Stephen B. Burke)	Director	
<u>/s/ TODD A. COMBS</u> (Todd A. Combs)	Director	
<u>/s/ ALICIA BOLER DAVIS</u> (Alicia Boler Davis)	Director	
<u>/s/ ALEX GORSKY</u> (Alex Gorsky)	Director	February 14, 2025
<u>/s/ MELLODY HOBSON</u> (Melody Hobson)	Director	
<u>/s/ PHEBE N. NOVAKOVIC</u> (Phebe N. Novakovic)	Director	
<u>/s/ VIRGINIA M. ROMETTY</u> (Virginia M. Rometty)	Director	
<u>/s/ BRAD D. SMITH</u> (Brad D. Smith)	Director	
<u>/s/ MARK A. WEINBERGER</u> (Mark A. Weinberger)	Director	
<u>/s/ JEREMY BARNUM</u> (Jeremy Barnum)	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	
<u>/s/ ELENA KORABLINA</u> (Elena Korablina)	Managing Director and Firmwide Controller (Principal Accounting Officer)	

Exhibit 4.6

DESCRIPTION OF SECURITIES OF JPMORGAN CHASE & CO. REGISTERED PURSUANT TO SECTION 12 OF THE SECURITIES EXCHANGE ACT OF 1934

As of the filing date of the Annual Report on Form 10-K to which this Exhibit is attached (the “Form 10-K”), the following outstanding securities issued by JPMorgan Chase & Co. are registered pursuant to Section 12 of the Securities Exchange Act of 1934: (i) common stock; (ii) six series of preferred stock represented by depositary shares; (iii) JPMorgan Chase & Co.’s guarantee of the Alerian MLP Index ETNs due January 28, 2044 issued by JPMorgan Chase Financial Company LLC; and (iv) JPMorgan Chase & Co.’s guarantee of the Callable Fixed Rate Notes due June 10, 2032 issued by JPMorgan Chase Financial Company LLC. All references herein to “JPMorganChase,” “we” or “us” are to JPMorgan Chase & Co.

DESCRIPTION OF COMMON STOCK

The following summary is not complete. You should refer to the applicable provisions of our Restated Certificate of Incorporation and our By-laws, each of which are incorporated by reference as Exhibits to the Form 10-K, and to the Delaware General Corporation Law (“DGCL”), for a complete statement of the terms and rights of our common stock, par value \$1.00 per share, which we refer to herein as common stock. We encourage you to read our Restated Certificate of Incorporation, which we refer to herein as our certificate of incorporation, By-laws and the relevant provisions of the DGCL for additional information.

Authorized Shares

We are authorized to issue up to 9,000,000,000 shares of common stock.

Dividends

Holders of common stock are entitled to receive dividends if, as and when declared by our board of directors out of funds legally available for payment, subject to the rights of holders of our preferred stock.

Voting Rights

Each holder of common stock is entitled to one vote per share. Subject to the rights, if any, of the holders of any series of preferred stock under its applicable certificate of designations and applicable law, all voting rights are vested in the holders of shares of our common stock. Holders of shares of our common stock have noncumulative voting rights, which means that the holders of more than 50% of the shares voting for the election of directors can elect 100% of the directors and the holders of the remaining shares will not be able to elect any directors.

Rights Upon Liquidation

In the event of our voluntary or involuntary liquidation, dissolution or winding-up, the holders of our common stock will be entitled to share equally in any of our assets available for distribution after we have paid in full all of our debts and after the holders of all series of our outstanding preferred stock have received their liquidation preferences in full.

Miscellaneous

The issued and outstanding shares of common stock are fully paid and nonassessable. Holders of shares of our common stock are not entitled to preemptive rights or to the benefit of any sinking funds. Our common stock is not convertible into shares of any other class of our capital stock. Computershare Inc. is the transfer agent, registrar and dividend disbursement agent for our common stock.

Listing

Our common stock is listed on the New York Stock Exchange (“NYSE”) under the trading symbol “JPM”.

DESCRIPTION OF LISTED PREFERRED STOCK

The following summary is not complete. You should refer to our certificate of incorporation and to the Certificate of Designations, Powers, Preferences and Rights relating to each series of Listed Preferred Stock (as defined below), which we refer to herein as a certificate of designations, for the complete terms of that series of preferred stock. Copies of our certificate of incorporation and the certificate of designations for each series of Listed Preferred Stock are incorporated by reference as Exhibits to the Form 10-K. We encourage you to read our certificate of incorporation and the relevant certificates of designations for additional information.

Authorized Shares

Under our certificate of incorporation, our board of directors is authorized, without further stockholder action, to issue up to 200,000,000 shares of preferred stock, \$1 par value per share, which we refer herein to as preferred stock, in one or more series, and to determine the voting powers and the designations, preferences and relative, participating, optional or other special rights, and qualifications, limitations or restrictions of each series. We may amend our certificate of incorporation to increase or decrease the number of authorized shares of preferred stock in a manner permitted by our certificate of incorporation and the DGCL.

Outstanding Preferred Stock

As of the filing date of the Form 10-K, we have 11 series of preferred stock issued and outstanding. The shares of each series of our preferred stock are represented by depositary shares, with each depositary share representing a fractional interest in a share of preferred stock of the relevant series. Of the 11 series of our issued and outstanding preferred stock, depositary shares representing the following six series of preferred stock are registered pursuant to Section 12 of the Securities Exchange Act of 1934, with each depositary share representing a 1/400th interest in a share of preferred stock of the relevant series:

- a. 5.75% Non-Cumulative Preferred Stock, Series DD;
- b. 6.00% Non-Cumulative Preferred Stock, Series EE;
- c. 4.75 Non-Cumulative Preferred Stock, Series GG;
- d. 4.55% Non-Cumulative Preferred Stock, Series JJ;
- e. 4.625% Non-Cumulative Preferred Stock, Series LL; and
- f. 4.20% Non-Cumulative Preferred Stock, Series MM.

We refer to the above six series of preferred stock herein collectively as the “Listed Preferred Stock”.

The Listed Preferred Stock is fully paid and nonassessable.

The terms of the depositary shares are summarized below under “Description of Depositary Shares”.

Ranking

The Listed Preferred Stock ranks, as to payment of dividends and distribution of assets upon our liquidation, dissolution or winding-up, on a parity with any series of preferred stock ranking on a parity with the Listed Preferred Stock and senior to our common stock and to any series of preferred stock ranking junior to the Listed Preferred Stock. The Listed Preferred Stock is subordinate to our existing and future indebtedness.

Dividend Rights

Holders of the Listed Preferred Stock are entitled to receive, when, as and if declared by our board of directors or any duly authorized committee of our board, cash dividends at the rates and on the dates described below under “Specific Terms of Listed Preferred Stock”. We will pay each dividend to the holders of record as they appear on our stock register on record dates determined by our board of directors or a duly authorized committee of our board. Dividends on the Listed Preferred Stock are noncumulative. If a dividend is not declared on any series of Listed Preferred Stock, because the dividends are noncumulative, then the right of holders of that series to receive that dividend will be lost, and we will have no obligation to pay the dividend for that dividend period, whether or not dividends are declared for any future dividend period.

We may not declare or pay or set aside for payment full dividends on any series of preferred stock ranking, as to dividends, equally with or junior to a series of Listed Preferred Stock unless we have previously declared and paid or set aside for payment, or we contemporaneously declare and pay or set aside for payment, full dividends on that series of Listed Preferred Stock for the most recently completed dividend period. When dividends are not paid in full on a particular series of Listed Preferred Stock and any other series of preferred stock ranking on a parity as to dividends with that series, all dividends declared and paid upon the shares of that series of Listed Preferred Stock and any other series of preferred stock ranking on a parity as to dividends with the that series will be declared and paid pro rata. For purposes of calculating the pro rata allocation of partial dividend payments, we will allocate dividend payments based on the ratio between the then-current dividends due on shares of that Listed Preferred Stock and (i) in the case of any series of non-cumulative preferred stock ranking on a parity as to dividends with that Listed Preferred Stock, the aggregate of the current and unpaid dividends due on such series of preferred stock and (ii) in the case of any series of cumulative preferred stock ranking on a parity as to dividends with that Listed Preferred Stock, the aggregate of the current and accumulated and unpaid dividends due on such series of preferred stock.

In addition, unless full dividends on all outstanding shares of the Listed Preferred Stock have been declared and paid or a sum sufficient for the payment thereof set aside for such payment in respect of the applicable most recently completed dividend period, with respect to a particular series of Listed Preferred Stock:

- a. no dividend (other than a dividend in common stock or in any other capital stock ranking junior to that Listed Preferred Stock as to dividends and upon liquidation, dissolution or winding-up) will be declared or paid or a sum sufficient for the payment thereof set aside for such payment or other distribution declared or made upon our common stock or upon any other capital stock ranking junior to that Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up, and
- b. no common stock or other capital stock ranking junior or equally with that Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up will be redeemed, purchased or otherwise acquired for any consideration (or any moneys be paid to or made available for a sinking fund for the redemption of any shares of any such capital stock) by us, except:
 - i. by conversion into or exchange for capital stock ranking junior to that Listed Preferred Stock;
 - ii. as a result of reclassification into capital stock ranking junior to that Listed Preferred Stock;
 - iii. through the use of the proceeds of a substantially contemporaneous sale of shares of capital stock ranking junior to that Listed Preferred Stock or, in the case of capital stock ranking on a parity with that Listed Preferred Stock, through the use of the proceeds of a substantially contemporaneous sale of other shares of capital stock ranking on a parity with that Listed Preferred Stock;
 - iv. in the case of capital stock ranking on a parity with that Listed Preferred Stock, pursuant to pro rata offers to purchase all or a pro rata portion of the shares of that Listed Preferred Stock and such capital stock ranking on a parity with that Listed Preferred Stock;
 - v. in connection with the satisfaction of our obligations pursuant to any contract entered into in the ordinary course prior to the beginning of the most recently completed dividend period; or
 - vi. any purchase, redemption or other acquisition of capital stock ranking junior to that Listed Preferred Stock pursuant to any of our or our subsidiaries' employee, consultant or director incentive or benefit plans or arrangements (including any employment, severance or consulting arrangements) adopted before or after the issuance of that Listed Preferred Stock).

However, the foregoing will not restrict our ability or the ability of any of our affiliates to engage in underwriting, stabilization, market-making or similar transactions in our capital stock in the ordinary course of business. Subject to the conditions described above, and not otherwise, dividends (payable in cash, capital stock, or otherwise), as may be determined by our board of directors or a duly authorized committee of our board, may be declared and paid on our common stock and any other capital stock ranking junior to or on a parity with the Listed Preferred Stock from time to time out of any assets legally available for such payment, and the holders of the Listed Preferred Stock will not be entitled to participate in those dividends.

As used herein, "junior to a series of Listed Preferred Stock" and like terms refer to our common stock and any other class or series of our capital stock over which the Listed Preferred Stock has preference or priority, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require; "parity preferred stock" and "on a parity with a series of Listed Preferred Stock" and like terms refer to any class or series of our capital stock that ranks on a parity with the shares of a particular series of Listed Preferred Stock, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require; and "senior to a series of Listed Preferred Stock" and like terms refer to any class or series of our capital stock that ranks senior to a particular series of Listed Preferred Stock, either as to dividends or upon liquidation, dissolution or winding-up, or both, as the context may require.

We will compute the amount of dividends payable by annualizing the applicable dividend rate and dividing by the number of dividend periods in a year, except that the amount of dividends payable for any period greater or less than a full dividend period, other than the initial dividend period, will be computed on the basis of a 360-day year consisting of twelve 30-day months and, for any period less than a full month, the actual number of days elapsed in the period. Dollar amounts resulting from that calculation will be rounded to the nearest cent, with one-half cent being rounded upward.

Rights Upon Liquidation

In the event of our voluntary or involuntary liquidation, dissolution or winding-up, holders of each series of Listed Preferred Stock will be entitled to receive and to be paid out of our assets legally available for distribution to our stockholders the amount of \$10,000 per share, plus any declared and unpaid dividends, without accumulation of undeclared dividends, before we make any distribution of assets to the holders of our common stock or any other class or series of shares ranking junior to the Listed Preferred Stock of such series. After the payment to such holders of the full preferential amounts to which they are entitled, such holders will have no right or claim to any of our remaining assets.

If, upon our voluntary or involuntary liquidation, dissolution or winding-up, we fail to pay in full the amounts payable with respect to a particular series of Listed Preferred Stock, and any stock having the same rank as that series, the holders of that series and of that other stock will share ratably in any such distribution of our assets in proportion to the full respective distributions to which they are entitled. For any series of Listed Preferred Stock, neither the sale of all or substantially all of our property or business, nor our merger or consolidation into or with any other entity will be considered a liquidation, dissolution or winding-up, voluntary or involuntary, of us.

Because we are a holding company, our rights and the rights of our creditors and our stockholders, including the holders of the Listed Preferred Stock, to participate in the assets of any of our subsidiaries upon that subsidiary's liquidation, dissolution, winding-up or recapitalization may be subject to the prior claims of that subsidiary's creditors, except to the extent that we are a creditor with recognized claims against the subsidiary.

Holders of the Listed Preferred Stock are subordinate to all of our indebtedness and to other non-equity claims on us and our assets, including in the event that we enter into a receivership, insolvency, liquidation or similar proceeding. In addition, holders of the Listed Preferred Stock may be fully subordinated to interests held by the U.S. government in the event that we enter into a receivership, insolvency, liquidation or similar proceeding.

Redemption

We may redeem each series of Listed Preferred Stock on the dates and at the redemption prices set forth below under "Specific Terms of Listed Preferred Stock". In addition, we may redeem each series of Listed Preferred Stock in whole, but not in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depository share), plus any declared and unpaid dividends, following the occurrence of a capital treatment event. For these purposes, "capital treatment event" means the good faith determination by JPMorganChase that, as a result of any:

- a. amendment to, or change or any announced prospective change in, the laws or regulations of the United States or any political subdivision of or in the United States that is enacted or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock;
- b. proposed change in those laws or regulations that is announced or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock; or
- c. official administrative decision or judicial decision or administrative action or other official pronouncement interpreting or applying those laws or regulations that is announced or becomes effective after the initial issuance of any shares of such series of Listed Preferred Stock,

there is more than an insubstantial risk that JPMorganChase will not be entitled to treat an amount equal to the full liquidation amount of all shares of such series of Listed Preferred Stock then outstanding as "additional Tier 1 capital" (or its equivalent) for purposes of the capital adequacy guidelines or regulations of the appropriate federal banking agency, as then in effect and applicable, for as long as any share of such series of Listed Preferred Stock is outstanding.

Redemption of any Listed Preferred Stock may be subject to our receipt of any required approvals from the Federal Reserve Board or any other regulatory authority.

If we elect to redeem shares of a series of Listed Preferred Stock, we will provide notice by first class mail, postage prepaid, addressed to the holders of record of such shares to be redeemed. Such mailing will be at least 30 days and not more than 60 days before the date fixed for redemption. Any notice so mailed will be conclusively presumed to have been duly given, whether or not the holder receives such notice, but failure to duly give such notice by mail, or any defect in such notice or in the mailing thereof, to any holder of shares of the series designated for redemption will not affect the validity of the proceedings for the redemption of any other shares of that series. Each notice of redemption will state:

- a. the redemption date;
- b. the number of shares of the series of Listed Preferred Stock to be redeemed and, if fewer than all the shares held by such holder are to be redeemed, the number of such shares to be redeemed from such holder;
- c. the redemption price;
- d. the place or places where the certificates representing such shares are to be surrendered for payment of the redemption price; and
- e. that dividends on the shares to be redeemed will cease to accrue on the redemption date.

Notwithstanding the foregoing, if the series of Listed Preferred Stock is held in book-entry form through The Depository Trust Company, or "DTC", we may give such notice in any manner permitted or required by DTC. For each series of Listed Preferred Stock, neither the holders of a series nor the holders of the related depository shares have the right to require redemption of such series of Listed Preferred Stock.

In the case of any redemption of only part of the shares a series of Listed Preferred Stock at the time outstanding, the shares of the series to be redeemed will be selected either pro rata from the holders of record of that series in proportion to the number of shares held by such holders or by lot. From and after the redemption date, dividends will cease to

accumulate on the shares of Listed Preferred Stock called for redemption up to the redemption date and all rights of the holders of those shares, except the right to receive the redemption price, will cease.

In the event that we fail to pay full dividends, including accumulated but unpaid dividends, if any, on any series of Listed Preferred Stock, we may not redeem that series in part and we may not purchase or acquire any shares of that series, except by a purchase or exchange offer made on the same terms to all holders of that series.

Preemptive and Conversion Rights

The Listed Preferred Stock is not subject to any preemptive rights and is not convertible into property or shares of any other class or series of our capital stock.

Depository, Transfer Agent, and Registrar

Computershare Inc. is the depository, transfer agent and registrar for each series of the Listed Preferred Stock and the related depository shares.

Voting Rights

Except as indicated below or except as expressly required by applicable law, the holders of the Listed Preferred Stock are not entitled to vote. Each share of Listed Preferred Stock a series is entitled to one vote on matters on which holders of that series are entitled to vote. The voting power of each series of Listed Preferred Stock depends on the number of shares in that series, and not on the aggregate liquidation preference or initial offering price of the shares of that series.

If, at any time or times, the equivalent of an aggregate of six quarterly dividends, whether or not consecutive, for any series of Listed Preferred Stock has not been paid, the number of directors constituting our board of directors will be automatically increased by two and the holders of each outstanding series of Listed Preferred Stock with such voting rights, together with holders of such other shares of any other class or series of parity preferred stock outstanding at the time upon which like voting rights have been conferred and are exercisable, which we refer to as “voting parity stock,” voting together as a class, will be entitled to elect those additional two directors, which we refer to as “preferred directors,” at that annual meeting and at each subsequent annual meeting of stockholders until full dividends have been paid for at least four quarterly consecutive dividend periods. At that time such right will terminate, except as expressly provided in the applicable certificate of designations or by law, subject to revesting. Upon any termination of the right of the holders of the Listed Preferred Stock and voting parity stock as a class to vote for directors as provided above, the preferred directors will cease to be qualified as directors, the term of office of all preferred directors then in office will terminate immediately and the authorized number of directors will be reduced by the number of preferred directors elected. Any preferred director may be removed and replaced at any time, with cause as provided by law or without cause by the affirmative vote of the holders of shares of the Listed Preferred Stock, voting together as a class with the holders of shares of voting parity stock, to the extent the voting rights of such holders described above are then exercisable. Any vacancy created by removal with or without cause may be filled only as described in the preceding sentence. If the office of any preferred director becomes vacant for any reason other than removal, the remaining preferred director may choose a successor who will hold office for the unexpired term in respect of which such vacancy occurred.

So long as any shares of a particular series of Listed Preferred Stock remains outstanding, we will not, without the affirmative vote of the holders of at least 66 2/3% in voting power of that series and any voting parity stock, voting together as a class, authorize, create or issue any capital stock ranking senior to that series as to dividends or upon liquidation, dissolution or winding-up, or reclassify any authorized capital stock into any such shares of such capital stock or issue any obligation or security convertible into or evidencing the right to purchase any such shares of capital stock. So long as any shares of a particular series of Listed Preferred Stock remain outstanding, we will not, without the affirmative vote of the holders of at least 66 2/3% in voting power of that series, amend, alter or repeal any provision of the applicable certificate of designations or our certificate of incorporation, including by merger, consolidation or otherwise, so as to adversely affect the powers, preferences or special rights of that series.

Notwithstanding the foregoing, none of the following will be deemed to adversely affect the powers, preferences or special rights of any series of Listed Preferred Stock:

- a. any increase in the amount of authorized common stock or authorized preferred stock, or any increase or decrease in the number of shares of any series of preferred stock, or the authorization, creation and issuance of other classes or series of capital stock, in each case ranking on a parity with or junior to that series of Listed Preferred Stock as to dividends or upon liquidation, dissolution or winding-up;
- b. a merger or consolidation of JPMorganChase with or into another entity in which the shares of that series remain outstanding; and
- c. a merger or consolidation of JPMorganChase with or into another entity in which the shares of the that series are converted into or exchanged for preference securities of the surviving entity or any entity, directly or indirectly,

controlling such surviving entity and such new preference securities have powers, preferences and special rights that are not materially less favorable than that series;

provided that if the amendment would adversely affect such series but not any other series of preferred stock outstanding, then the amendment will only need to be approved by holders of at least two-thirds of the shares of the series of Listed Preferred Stock adversely affected.

In exercising the voting rights described above or when otherwise granted voting rights by operation of law or by us, each share of Listed Preferred Stock with respect to a series will be entitled to one vote (equivalent to 1/400th of a vote per relevant depositary share).

If we redeem or call for redemption all outstanding shares of a series of Listed Preferred Stock and irrevocably deposit in trust sufficient funds to effect such redemption, at or prior to the time when the act with respect to which such vote would otherwise be required or upon which the holders of such series will be entitled to vote will be effected, the voting provisions described above will not apply.

Our board of directors may also from time to time, without notice to or consent of holders of a series of Listed Preferred Stock, issue additional shares of such series. Delaware law provides that the holders of preferred stock will have the right to vote separately as a class on any amendment to our certificate of incorporation (including any certificate of designations) that would increase or decrease the aggregate number of authorized shares of such class, increase or decrease the par value of such class or adversely affect the powers, preferences and special rights of the shares of preferred stock. Notwithstanding the foregoing, as permitted by law, our certificate of incorporation provides that any increase or decrease in our authorized capital stock may be adopted by the affirmative vote of holders of capital stock representing not less than a majority of the voting power represented by the outstanding shares of our capital stock entitled to vote. If any proposed amendment would alter or change the powers, preferences or special rights of one or more series of preferred stock so as to affect them adversely, but would not so affect the entire class of preferred stock, only the shares of the series so affected will be considered a separate class for purposes of this vote on the amendment. This right is in addition to any voting rights that may be provided for in our certificate of incorporation (including any certificate of designations).

Under regulations adopted by the Federal Reserve Board, if the holders of any series of our preferred stock become entitled to vote for the election of directors because dividends on that series are in arrears, that series may then be deemed a “class of voting securities.” In such a case, a holder of 25% or more of the series, or a holder of 5% or more if that holder would also be considered to exercise a “controlling influence” over JPMorganChase, may then be subject to regulation as a bank holding company in accordance with the Bank Holding Company Act. In addition, (1) any other bank holding company may be required to obtain the prior approval of the Federal Reserve Board to acquire or retain 5% or more of that series, and (2) any person other than a bank holding company may be required to provide notice to the Federal Reserve Board prior to acquiring or retaining 10% or more of that series.

Description of Depositary Shares

The following summary of the terms of the depositary shares representing each series of the Listed Preferred Stock is not complete. You should refer to each of the deposit agreements among us, the depositary, and the holders from time to time of the depositary receipts evidencing the depositary shares relating to each series of the Listed Preferred Stock for the complete terms of those depositary shares. Each of those deposit agreements has been filed as an exhibit to a Current Report on Form 8-K filed in connection with the issuance of the depositary shares representing each series of the Listed Preferred Stock.

General. Each depositary share represents a 1/400th interest in a share of the relevant series of Listed Preferred Stock, and is evidenced by depositary receipts. In connection with the issuance of each series of Listed Preferred Stock, we deposited shares of that series of Listed Preferred Stock with Computershare Inc., as depositary under the deposit agreement relating to that series of Listed Preferred Stock. Subject to the terms of each deposit agreement, the depositary shares are entitled to all the powers, preferences and special rights of the relevant series of Listed Preferred Stock, as applicable, in proportion to the applicable fraction of a share of Listed Preferred Stock those depositary shares represent.

Dividends and Other Distributions. Each dividend payable on a depositary share will be in an amount equal to 1/400th of the dividend declared and payable on the related share of the Listed Preferred Stock.

The depositary will distribute all dividends and other cash distributions received on the relevant series of Listed Preferred Stock to the holders of record of the related depositary receipts in proportion to the number of depositary shares held by each holder. In the event of a distribution other than in cash, the depositary will distribute property received by it to the holders of record of the depositary receipts as nearly as practicable in proportion to the number of depositary shares held by each holder, unless the depositary determines that this distribution is not feasible, in which case the depositary may,

with our approval, adopt a method of distribution that it deems practicable, including the sale of the property and distribution of the net proceeds of that sale to the holders of the depositary receipts.

Record dates for the payment of dividends and other matters relating to the depositary shares will be the same as the corresponding record dates for the related shares of Listed Preferred Stock.

The amount paid as dividends or otherwise distributable by the depositary with respect to the depositary shares or the underlying Listed Preferred Stock will be reduced by any amounts required to be withheld by us or the depositary on account of taxes or other governmental charges.

Redemption of Depositary Shares. If we redeem a series of Listed Preferred Stock, in whole or from time to time in part, the corresponding depositary shares also will be redeemed with the proceeds received by the depositary from the redemption of the Listed Preferred Stock held by the depositary. The redemption price per depositary share will be $1/400^{\text{th}}$ of the redemption price per share payable with respect to the Listed Preferred Stock, plus any declared and unpaid dividends, without accumulation of undeclared dividends.

If we redeem shares of a series of Listed Preferred Stock held by the depositary, the depositary will redeem, as of the same redemption date, the number of depositary shares representing those shares of the Listed Preferred Stock so redeemed. If we redeem less than all of the outstanding depositary shares, the depositary will select pro rata or by lot those depositary shares to be redeemed. The depositary will mail notice of redemption to record holders of the depositary receipts not less than 30 and not more than 60 days prior to the date fixed for redemption of the Listed Preferred Stock and the related depositary shares.

The redemption of depositary shares that are held in book-entry form through DTC will be effected in accordance with the applicable procedures of DTC.

Voting the Listed Preferred Stock. Because each depositary share represents a $1/400^{\text{th}}$ interest in a share of Listed Preferred Stock, holders of depositary receipts will be entitled to $1/400^{\text{th}}$ of a vote per depositary share under those limited circumstances in which holders of the Listed Preferred Stock are entitled to a vote.

When the depositary receives notice of any meeting at which the holders of a series of Listed Preferred Stock are entitled to vote, the depositary will mail the information contained in the notice to the record holders of the depositary shares relating to that Listed Preferred Stock. Each record holder of the depositary shares on the record date, which will be the same date as the record date for the applicable Listed Preferred Stock, may instruct the depositary to vote the amount of the Listed Preferred Stock represented by the holder's depositary shares. To the extent practicable, the depositary will vote the amount of the Listed Preferred Stock represented by depositary shares in accordance with the instructions it receives. We will agree to take all actions that the depositary determines are necessary to enable the depositary to vote as instructed. If the depositary does not receive specific instructions from the holders of any depositary shares representing the Listed Preferred Stock, it will abstain from voting with respect to such shares.

Withdrawal of Listed Preferred Stock. Underlying shares of Listed Preferred Stock may be withdrawn from the depositary arrangement upon surrender of depositary receipts at the depositary's office and upon payment of the taxes, charges and fees provided for in the deposit agreement. Subject to the terms of the relevant deposit agreement, the holder of depositary receipts will receive the appropriate number of shares of Listed Preferred Stock represented by such depositary shares. Only whole shares of Listed Preferred Stock may be withdrawn; if a holder holds an amount other than a whole multiple of 400 depositary shares, the depositary will deliver along with the withdrawn shares of Listed Preferred Stock a new depositary receipt evidencing the excess number of depositary shares. Holders of withdrawn shares of Listed Preferred Stock will not be entitled to redeposit such shares or to receive depositary shares.

Form and Notices. Each series of Listed Preferred Stock was issued in registered form to the depositary, and the depositary shares representing that Listed Preferred Stock were issued in book-entry only form through DTC. The depositary will forward to the holders of depositary shares all reports, notices, and communications from us that are delivered to the depositary and that we are required to furnish to the holders of the Listed Preferred Stock.

Amendment and Termination of the Deposit Agreement. We and the depositary may amend any form of depositary receipt evidencing depositary shares and any provision of any deposit agreement at any time regarding any depositary shares. However, any amendment that materially and adversely alters the rights of the holders of depositary shares representing a particular series of Listed Preferred Stock or would be materially and adversely inconsistent with the rights granted to holders of that underlying Listed Preferred Stock pursuant to our certificate of incorporation will not be effective unless the amendment has been approved by the holders of at least a majority of the related depositary shares then outstanding. The deposit agreement relating to the depositary shares representing a particular series of Listed Preferred Stock may be terminated by us or by the depositary only if:

- a. all such outstanding depositary shares have been redeemed; or

- b. there has been a final distribution of the relevant underlying Listed Preferred Stock in connection with our liquidation, dissolution or winding up and the preferred stock has been distributed to the holders of depositary receipts.

Charges of Depositary. We will pay all transfer and other taxes and governmental charges arising solely from the existence of the depositary arrangements regarding any depositary shares. We also pay charges of the depositary in connection with the initial deposit of each series of Listed Preferred Stock and any redemption of the Listed Preferred Stock. Holders of depositary receipts will pay transfer and other taxes and governmental charges and other charges with respect to their depositary receipts as expressly provided in the deposit agreement.

Resignation and Removal of Depositary. With respect to the depositary shares representing each series of Listed Preferred Stock, the depositary may resign at any time by delivering a notice to us of its election to do so. We may remove the depositary at any time. Any such resignation or removal will take effect upon the appointment of a successor depositary and its acceptance of its appointment. We must appoint a successor depositary within 60 days after delivery of the notice of resignation or removal.

Miscellaneous. The depositary will forward to holders of applicable depositary receipts all reports and communications from us that we deliver to the depositary and that we are required to furnish to the holders of the relevant Listed Preferred Stock.

Neither we nor the depositary will be liable if either of us is prevented or delayed by law or any circumstance beyond our control in performing our respective obligations under any deposit agreement. Our obligations and those of the depositary will be limited to performing in good faith our respective duties under any deposit agreement. Neither we nor the depositary will be obligated to prosecute or defend any legal proceeding relating to any depositary shares or Listed Preferred Stock unless satisfactory indemnity is furnished. We and the depositary may rely upon written advice of counsel or accountants, or upon information provided by persons presenting preferred stock for deposit, holders of depositary receipts or other persons we believe to be competent, and on documents we believe to be genuine.

Specific Terms of Listed Preferred Stock

5.75% Non-Cumulative Preferred Stock, Series DD

On September 21, 2018, we issued an aggregate of 169,625 shares of 5.75% Non-Cumulative Preferred Stock, Series DD, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series DD Preferred Stock”). Shares of the Series DD Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series DD Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 5.75% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on December 1, 2018. Dividends on the Series DD Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series DD Preferred Stock may be redeemed on any dividend payment date on or after December 1, 2023, in whole or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series DD Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series DD Preferred Stock are listed on the NYSE under the trading symbol “JPM PR D”.

6.00% Non-Cumulative Preferred Stock, Series EE

On January 24, 2019, we issued an aggregate of 185,000 shares of 6.00% Non-Cumulative Preferred Stock, Series EE, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series EE Preferred Stock”). Shares of the Series EE Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series EE Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 6.00% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on June 1, 2019. Dividends on the Series EE Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series EE Preferred Stock may be redeemed on any dividend payment date on or after March 1, 2024, in whole or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series EE Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series EE Preferred Stock are listed on the NYSE under the trading symbol “JPM PR C”.

4.75% Non-Cumulative Preferred Stock, Series GG

On November 7, 2019, we issued an aggregate of 90,000 shares of 4.75% Non-Cumulative Preferred Stock, Series GG, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series GG Preferred Stock”). Shares of the Series GG Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series GG Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.75% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on March 1, 2020. Dividends on the Series GG Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series GG Preferred Stock may be redeemed on any dividend payment date on or after December 1, 2024, in whole or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series GG Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series GG Preferred Stock are listed on the NYSE under the trading symbol “JPM PR J”.

4.55% Non-Cumulative Preferred Stock, Series JJ

On March 17, 2021, we issued an aggregate of 150,000 shares of 4.55% Non-Cumulative Preferred Stock, Series JJ, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series JJ Preferred Stock”). Shares of the Series JJ Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series JJ Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.55% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on June 1, 2021. Dividends on the Series JJ Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series JJ Preferred Stock may be redeemed on any dividend payment date on or after June 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series JJ Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series JJ Preferred Stock are listed on the NYSE under the trading symbol “JPM PR K”.

4.625% Non-Cumulative Preferred Stock, Series LL

On May 20, 2021, we issued an aggregate of 185,000 shares of 4.625% Non-Cumulative Preferred Stock, Series LL, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series LL Preferred Stock”). Shares of the Series LL Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series LL Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.625% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on September 1, 2021. Dividends on the Series LL Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series LL Preferred Stock may be redeemed on any dividend payment date on or after June 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series LL Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series LL Preferred Stock are listed on the NYSE under the trading symbol “JPM PR L”.

4.20% Non-Cumulative Preferred Stock, Series MM

On July 29, 2021, we issued an aggregate of 200,000 shares of 4.20% Non-Cumulative Preferred Stock, Series MM, \$1 par value, with a liquidation preference of \$10,000 per share (the “Series MM Preferred Stock”). Shares of the Series MM

Preferred Stock are represented by depositary shares, each representing a 1/400th interest in a share of preferred stock of the series.

Dividends. Dividends on the Series MM Preferred Stock are payable when, as, and if declared by our board of directors or a duly authorized committee of our board, at a rate of 4.20% per annum, payable quarterly in arrears, on March 1, June 1, September 1 and December 1 of each year, beginning on December 1, 2021. Dividends on the Series MM Preferred Stock are neither mandatory nor cumulative.

Redemption. The Series MM Preferred Stock may be redeemed on any dividend payment date on or after September 1, 2026, in whole at any time or from time to time in part, at a redemption price equal to \$10,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends. We may also redeem the Series MM Preferred Stock following the occurrence of a “capital treatment event,” as described above.

Listing. The depositary shares representing the Series MM Preferred Stock are listed on the NYSE under the trading symbol “JPM PR M”.

DESCRIPTION OF JPMORGAN CHASE FINANCIAL COMPANY LLC’S THE ALERIAN MLP INDEX ETNS DUE JANUARY 28, 2044, FULLY AND UNCONDITIONALLY GUARANTEED BY JPMORGAN CHASE & CO.

The following description of Alerian MLP Index ETNs due January 28, 2044 (the “Alerian ETNs”) is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to the indenture dated February, 2016 (as may be amended or supplemented from time to time, the “2016 Indenture”), among JPMorgan Chase Financial Company LLC, as issuer (“JPMorgan Financial” or the “Issuer”), JPMorganChase, as guarantor (the “Guarantor”), and Deutsche Bank Trust Company Americas, as trustee (the “Trustee”), which is incorporated by reference as an Exhibit to the Form 10-K. We encourage you to read the 2016 Indenture for additional information.

General

The Alerian ETNs were originally issued on January 30, 2024 and are part of a series of JPMorgan Financial’s debt securities entitled “Global Medium-Term Notes, Series A” (the “Series A Notes”) that JPMorgan Financial may issue under the 2016 Indenture from time to time. For more information about the Series A Notes, please see the section titled “— Description of JPMorgan Chase Financial Company LLC’s Callable Fixed Rate Notes Due June 10, 2032, Fully and Unconditionally Guaranteed by JPMorgan Chase & Co. — General Terms of the Series A Notes” below.

The Alerian ETNs are unsecured and unsubordinated obligations of JPMorgan Financial, the payment of which is fully and unconditionally guaranteed by JPMorganChase, the Guarantor. The Alerian ETNs will rank *pari passu* with all of JPMorgan Financial’s other unsecured and unsubordinated obligations. JPMorgan Chase & Co.’s guarantee of the Alerian ETNs will rank *pari passu* with all of JPMorgan Chase & Co.’s other unsecured and unsubordinated obligations. The Alerian ETNs do not guarantee any return of principal at, or prior to, maturity or upon early redemption or repurchase. **Any payment on the Alerian ETNs is subject to the credit risk of JPMorgan Financial, as issuer of the Alerian ETNs, and JPMorgan Chase & Co., as guarantor of the Alerian ETNs.**

The Alerian ETNs are not bank deposits and are not insured by the Federal Deposit Insurance Corporation or by any other governmental agency, nor are they obligations of, or guaranteed by, a bank.

Unless otherwise specified, references herein to “holders” mean those in whose names the Alerian ETNs are registered on the books that JPMorgan Financial or the Trustee, or any successor trustee, as applicable, maintains for this purpose, and not those who own beneficial interests in the Alerian ETNs (registered in street name or otherwise).

Key Terms of the Alerian ETNs

General

Principal Amount: \$26 per Alerian ETN (equal to the Initial VWAP Level *divided* by ten, rounded to the nearest cent).

Index: The return on the Alerian ETNs is linked to the performance of the Alerian MLP Index[®] (Bloomberg ticker: AMZ), which we refer to as the Index, as measured by its VWAP level, and to cash distributions on its components. See “— Terms Relating to Closing Intrinsic Note Value — VWAP Level” below.

The Index measures the composite performance of energy-oriented Master Limited Partnerships, or MLPs, that earn the majority of their cash flows from qualified activities involving energy commodities (the “Index Components”) using a capped, float-adjusted, capitalization-weighted methodology. MLPs are limited partnerships primarily engaged in the exploration, marketing, mining, processing, production, refining, storage, or transportation of any mineral or natural resource. The Index is calculated and maintained by VettaFi LLC (“VettaFi”).

Coupon Payments: For each Alerian ETN a holder holds on a Coupon Record Date, the holder will receive on the immediately following Coupon Payment Date an amount in cash equal to the Coupon Amount, if any, as of the immediately preceding Coupon Valuation Date.

Coupon Amount: The Coupon Amount as of any Coupon Valuation Date will equal:

- a. the Reference Distribution Amount with respect to that Coupon Valuation Date, *minus*
- b. the Accrued Investor Fee with respect to that Coupon Valuation Date,

provided that the Coupon Amount will not be less than \$0.

If the Reference Distribution Amount on a Coupon Valuation Date is less than the Accrued Investor Fee on that Coupon Valuation Date, an amount equal to the excess of the Accrued Investor Fee over the Reference Distribution Amount (the “Investor Fee Shortfall”) will be included in the Accrued Investor Fee with respect to the next Coupon Valuation Date.

The Accrued Investor Fee will reduce each Coupon Amount. In addition, no Coupon Payment will be payable with respect to a Coupon Valuation Date if the Reference Distribution Amount is less than the Accrued Investor Fee, even if that Reference Distribution Amount is positive. The holders are not guaranteed any Coupon Payments.

The Coupon Amount is subject to adjustment in the event of a split or reverse split of the Alerian ETNs as described under “— Split or Reverse Split of the Alerian ETNs” below.

Payment at Maturity: For each Alerian ETN, unless earlier repurchased or redeemed, holders will receive at maturity a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Final Valuation Date. If that amount is less than or equal to zero, the payment at maturity will be \$0.

Issuer Redemption: On any Business Day on or after July 26, 2024, the Issuer may, in its sole discretion, redeem the Alerian ETNs, in whole or in part. If the Issuer exercises its right to redeem the Alerian ETNs prior to maturity, for each Alerian ETN that is redeemed, holders will receive on the Redemption Settlement Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Redemption Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be \$0.

Weekly Repurchase: On a weekly basis, holders may request that we repurchase a minimum of 50,000 Alerian ETNs (subject to adjustment in the event of a split or reverse split of the Alerian ETNs) if holders comply with the required procedures, subject to a repurchase fee of 0.125%. For each Alerian ETN that is repurchased, holders will receive on the relevant Repurchase Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Repurchase Valuation Date *minus* the Repurchase Fee Amount with respect to the Repurchase Valuation Date.

If the amount calculated above is less than or equal to zero, the payment upon early repurchase will be \$0.

Terms Relating to Closing Intrinsic Note Value

Closing Intrinsic Note Value: As of any date of determination, an amount per Alerian ETN equal to:

- a. the Principal Amount, multiplied by the Index Ratio as of that date, *plus*
- b. the Coupon Amount as of that date, calculated as if that date were a Coupon Valuation Date (the “Stub Coupon Amount”), *minus*
- c. any Investor Fee Shortfall determined in calculating that Stub Coupon Amount.

In addition, if the Coupon Ex-Date with respect to the Coupon Amount as of the immediately preceding Coupon Valuation Date has not yet occurred, the Closing Intrinsic Note Value will also include that Coupon Amount (an “Unpaid Coupon Amount”).

For purposes of determining the Closing Intrinsic Note Value over any Measurement Period, the Index Ratio is determined based on the arithmetic average of the VWAP Levels over that Measurement Period, and the date of determination for purposes of determining the Coupon Amount, any Investor Fee Shortfall and any Unpaid Coupon Amount is the final day of that Measurement Period.

The Closing Intrinsic Note Value is not the closing price or any other trading price of the Alerian ETNs in the secondary market and is not intended as a price or quotation, or as an offer or solicitation for the purchase or sale of the Alerian ETNs or as a recommendation to transact in the Alerian ETNs at the stated price. The trading price of the Alerian ETNs at any time may vary significantly from the Closing Intrinsic Note Value due to, among other things, imbalances of supply and demand (including as a result of any decision of ours to issue, stop issuing or resume issuing additional Alerian ETNs), lack of liquidity, transaction costs, credit considerations and bid-offer spreads.

If the Reference Distribution Amount used to calculate the Stub Coupon Amount on any Index Business Day is less than the Accrued Investor Fee used to calculate that Stub Coupon Amount, the resulting Investor Fee Shortfall will be deducted in calculating the Closing Intrinsic Note Value on that Index Business Day. Accordingly, the payment at maturity or upon early repurchase or redemption will be reduced by the amount of any Investor Fee Shortfall reflected in the relevant Closing Intrinsic Note Value.

Index Ratio: As of any date of determination, the Index Ratio is equal to:

$$\frac{\text{VWAP Level}}{\text{Initial VWAP Level}}$$

provided that, solely for purposes of determining the Closing Intrinsic Note Value in connection with any payment at maturity or upon early repurchase or redemption, the Index Ratio determined over the relevant Measurement Period is equal to:

$$\frac{\text{Final VWAP Level}}{\text{Initial VWAP Level}}$$

Initial VWAP Level: 260.0267 which is the arithmetic average of the VWAP Levels determined over a period of five Index Business Days ending on the Inception Date.

Final VWAP Level: With respect to a Measurement Period, the arithmetic average of the VWAP Levels on the five Index Business Days in that Measurement Period, as calculated by the Note Calculation Agent.

Measurement Period: With respect to the Final Valuation Date or any Repurchase Valuation Date or Redemption Valuation Date, the five Index Business Days starting from and including the Final Valuation Date or that Repurchase Valuation Date or Redemption Valuation Date, as applicable (or, if that day is not an Index Business Day, the five Index Business Days immediately following that day).

The Measurement Period is subject to postponement in the event of a market disruption event and as described under “— Postponement of an Averaging Date.”

VWAP Level: On any Index Business Day, as calculated by the VWAP Calculation Agent, the sum of the products for each Index Component of:

- a. the VWAP of that Index Component as of that day; and
- b. the Index Units of that Index Component as of that day, *divided* by the Index Divisor as of that day.

The calculation of the VWAP Level may be modified in circumstances described under “— Postponement of an Averaging Date” and “— Discontinuation of an Index; Alternation of Method of Calculation” below. The official closing level of the Index may vary significantly from the VWAP Level.

VWAP: With respect to each Index Component, as of any date of determination, the volume-weighted average price of one share of that Index Component as determined by the VWAP Calculation Agent based on the Primary Exchange for that Index Component.

Index Units: With respect to each Index Component, as of any date of determination, the number of units of that Index Component included in the Index for purposes of the calculation of the official level of the Index by the Index Calculation Agent.

Index Divisor: As of any date of determination, the divisor used in the calculation of the official level of the Index by the Index Calculation Agent.

Inception Date: January 26, 2024

Initial Issue Date: January 30, 2024

Final Valuation Date: January 20, 2044

Maturity Date: January 28, 2044. The Maturity Date is subject to postponement in the event of a market disruption event and as described under “— Postponement of a Payment Date” below.

Terms Relating to Coupon Payments

Accrued Investor Fee: The Accrued Investor Fee accrues at a rate of 0.85% per annum each day. In addition, the Accrued Investor Fee carries forward any shortfall if the Reference Distribution Amount determined in connection with any Coupon Payment is less than the Accrued Investor Fee at that time.

With respect to each Coupon Valuation Date, the Accrued Investor Fee is an amount equal to:

- a. the Periodic Investor Fee with respect to that Coupon Valuation Date, *plus*
- b. the Investor Fee Shortfall, if any, as of the immediately preceding Coupon Valuation Date, if any.

Periodic Investor Fee: With respect to each Coupon Valuation Date, an amount equal to the product of

- a. the investor fee of 0.85% per annum;
- b. the Principal Amount multiplied by the Index Ratio as of the immediately preceding Index Business Day that is not a Disrupted Day for any Index Component; and
- c. the day count fraction, calculated using a 30/360 day count convention as described under “— Additional Terms — Day Count Fraction” below.

Reference Distribution Amount: With respect to each Coupon Valuation Date, an amount equal to the sum of the gross cash distributions that a Reference Holder would have been entitled to receive in respect of each Index Component held by that Reference Holder on the “record date” with respect to that Index Component, for those cash distributions whose “ex-dividend date” occurs during the Coupon Accrual Period for that Coupon Valuation Date.

Notwithstanding the foregoing, with respect to cash distributions for an Index Component that are scheduled to be paid prior to the applicable Coupon Ex-Date, if the issuer of that Index Component fails to pay the distribution to holders of that Index Component by the scheduled payment date for that distribution, that distribution will be assumed to be zero for the purposes of calculating the applicable Reference Distribution Amount.

Reference Holder: As of any date of determination, a hypothetical holder of a number of shares of each Index Component equal to:

- a. the Index Units of that Index Component as of that date, *divided by*
- b. the Index Divisor as of that date *multiplied by* 10,

provided that solely for purposes of determining the Reference Distribution Amount included in any Stub Coupon Amount payable at maturity or upon early repurchase or redemption, the Reference Holder will be deemed to hold four-fifths, three-fifths, two-fifths and one-fifth of the shares of each Index Component it would otherwise hold on the second, third, fourth and fifth Index Business Days, respectively, in the relevant Measurement Period.

The Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount and the Reference Holder are each subject to adjustment in the event of a split or reverse split of the Alerian ETNs as described under “— Split or Reverse Split of the Alerian ETNs” below.

Coupon Accrual Period: With respect to each Coupon Valuation Date, the period from but excluding the immediately preceding Coupon Valuation Date (or, in the case of the first Coupon Valuation Date, from but excluding November 15, 2023) to and including that Coupon Valuation Date.

Coupon Valuation Date: The first Index Business Day occurring on or after the 15th of February, May, August and November of each calendar year during the term of the Alerian ETNs, beginning on February 15, 2024.

Coupon Ex-Date: With respect to a Coupon Amount, the first Exchange Business Day on which the Alerian ETNs trade without the right to receive that Coupon Amount. Under current NYSE Arca practice, the Coupon Ex-Date will generally be the first Exchange Business Day immediately preceding the applicable Coupon Record Date; however, beginning May 28, 2024, under NYSE Arca practice, the Coupon Ex-Date is expected to generally be the applicable Coupon Record Date. For purposes of this paragraph, “Exchange Business Day” means any day on which the primary exchange or market for trading of the Alerian ETNs is scheduled to be open for trading.

Coupon Record Date: The 9th Index Business Day following each Coupon Valuation Date.

Coupon Payment Date: The 15th Index Business Day following each Coupon Valuation Date, subject to postponement in the event of a market disruption event as described under “— Postponement of a Payment Date” below.

Terms Relating to Issuer Redemption

Early Redemption: On any Business Day on or after July 26, 2024, the Issuer may, in its sole discretion, redeem the Alerian ETNs, in whole or in part. If the Issuer exercises its right to redeem the Alerian ETNs, the Issuer will deliver an irrevocable redemption notice (the “Redemption Notice”) to DTC (the holder of the master note evidencing the Alerian ETNs) at least

five Business Days prior to the Redemption Valuation Date specified in the Redemption Notice. If fewer than all the Alerian ETNs are to be redeemed, the Issuer will specify in the Redemption Notice the principal amount of the Alerian ETNs to be redeemed, and the Trustee will select the Alerian ETNs to be redeemed pro rata, by lot or in such manner as it deems appropriate and fair.

Payment upon Early Redemption: If the Issuer exercises its right to redeem any Alerian ETNs prior to maturity, for each Alerian ETNs selected for redemption by the Trustee, holders will receive on the Redemption Settlement Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Redemption Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be \$0.

Redemption Valuation Date: The date specified as the Redemption Valuation Date in the Redemption Notice.

Redemption Settlement Date: Unless otherwise specified in the Redemption Notice, the day that follows the final day in the Measurement Period with respect to the Redemption Valuation Date by a number of Business Days corresponding to the standard settlement cycle, which is currently two Business Days and which is expected to be one Business Day beginning May 28, 2024. In no event will the Redemption Notice specify a Redemption Settlement Date that follows the final day in the Measurement Period by more than five Business Days.

Terms Relating to Weekly Repurchase Right

Early Repurchase: On a weekly basis, holders may request that we repurchase a minimum of 50,000 Alerian ETNs (subject to adjustment in the event of a split or reverse split of the Alerian ETNs) if holders comply with the procedures described under “— Repurchase Procedures” below and unless holders have delivered a Redemption Notice to DTC to redeem all of the outstanding notes. The Issuer may from time to time, in its sole discretion, reduce the minimum number of the Alerian ETNs required for an early repurchase on a consistent basis for all holders of the Alerian ETNs, but the Issuer is under no obligation to do so.

Payment upon Early Repurchase: Subject to holders’ compliance with the required procedures, for each Alerian ETN that is repurchased, holders will receive on the relevant Repurchase Date a cash payment equal to the Closing Intrinsic Note Value determined over the Measurement Period with respect to the Repurchase Valuation Date *minus* the Repurchase Fee Amount with respect to the Repurchase Valuation Date. If that amount is less than or equal to zero, the payment upon early redemption will be \$0.

Repurchase Fee Amount: With respect to any Repurchase Valuation Date, an amount per Alerian ETN in cash equal to 0.125% of the Closing Intrinsic Note Value with respect to that Repurchase Valuation Date (but excluding any Unpaid Coupon Amount included in that Closing Intrinsic Note Value).

Repurchase Valuation Date: The last Index Business Day of each week, generally Friday.

Repurchase Date: Unless otherwise specified in the Issuer’s acknowledgement, the day that follows the final day in the Measurement Period with respect to the Repurchase Valuation Date by a number of Business Days corresponding to the standard settlement cycle, which is currently two Business Days and which is expected to be one Business Day beginning May 28, 2024. In no event will the Issuer’s acknowledgement specify a Repurchase Date that follows the final day in the Measurement Period by more than five Business Days.

Repurchase Notice: A repurchase notice in the form specified by the Issuer.

Repurchase Procedures: In order to request that the Issuer repurchase a holder’s Alerian ETN, holders must instruct their broker or other person through which they hold Alerian ETNs to take the following steps:

- a. send a completed Repurchase Notice to the Issuer via email at ETN_Repurchase@jpmorgan.com by no later than 4:00 p.m., New York City time, on the Business Day immediately preceding the applicable Repurchase Valuation Date;
- b. instruct the holder’s DTC custodian to book a delivery versus payment trade with respect to that holder’s Alerian ETNs on the final day in the Measurement Period with respect to the relevant Repurchase Valuation Date at a price equal to the amount payable upon early repurchase of the Alerian ETNs; and
- c. cause the holder’s DTC custodian to deliver the trade as booked for settlement via DTC at or prior to 10:00 a.m., New York City time, on the relevant Repurchase Date.

Different brokerage firms may have different deadlines for accepting instructions from their customers. Accordingly, holders should consult the brokerage firm through which they own their interest in the Alerian ETNs in respect of those deadlines.

Once delivered, a Repurchase Notice may not be revoked. If the Issuer does not receive a holder’s Repurchase Notice by the deadline, such holder’s Repurchase Notice will not be effective. The Issuer or its affiliate must acknowledge receipt of the Repurchase Notice on the same Business Day for it to be effective, such acknowledgment will be deemed to evidence

its acceptance of a holder's repurchase request. The Note Calculation Agent will, in its sole discretion, resolve any questions that may arise as to the validity of a Repurchase Notice and the timing of receipt of a Repurchase Notice.

Additional Terms

Business Day: Any day other than a day on which the banking institutions in the City of New York are authorized or required by law, regulation or executive order to close or a day on which transactions in dollars are not conducted.

Index Business Day: Any day on which the Primary Exchange and the Related Exchange with respect to each Index Component are scheduled to be open for trading.

Primary Exchange: With respect to each Index Component, the primary exchange or market of trading of that Index Component.

Related Exchange: With respect to each Index Component, each exchange or quotation system where trading has a material effect (as determined by the Note Calculation Agent) on the overall market for futures or options contracts relating to that Index Component.

Disrupted Day: With respect to an Index Component, a day on which the Primary Exchange or any Related Exchange with respect to that Index Component fails to open for trading during its regular trading session or on which a market disruption event (as described under “— Market Disruption Events”) with respect to that Index Component has occurred or is continuing, and, in each case, the occurrence of which is determined by the Note Calculation Agent to have a material effect on the VWAP Level.

Index Sponsor and Index Calculation Agent: VettaFi

Note Calculation Agent, VWAP Calculation Agent and Published ETN Value Calculation Agent: J.P. Morgan Securities LLC (“JPMS”), one of our and JPMorgan Financial’s affiliates, will act as the Note Calculation Agent. Solactive AG will act as the VWAP Calculation Agent and the Published ETN Value Calculation Agent. We and JPMorgan Financial may appoint a different Note Calculation Agent, VWAP Calculation Agent or Published ETN Value Calculation Agent from time to time without holders’ consent and without notifying holders.

The Note Calculation Agent will make all necessary calculations and determinations in connection with the Alerian ETNs, including calculations and determinations relating to any payments on the Alerian ETNs and the assumptions used to determine the pricing of the Alerian ETNs, other than determinations to be made by the VWAP Calculation Agent. The VWAP Calculation Agent will determine the VWAP Level and the VWAP of each Index Component on each Index Business Day, except as described below under “— Postponement of an Averaging Date.” The Published ETN Value Calculation Agent is responsible for calculating the Daily Closing Intrinsic Note Value, the Intraday Intrinsic Note Value and the Interim Coupon or Shortfall Amount for purposes of publication. All determinations made by the Note Calculation Agent, the VWAP Calculation Agent or the Published ETN Value Calculation Agent will be at the sole discretion of the Note Calculation Agent, the VWAP Calculation Agent or the Published ETN Value Calculation Agent, as applicable, and will, in the absence of manifest error, be conclusive for all purposes and binding on holders, us and JPMorgan Financial.

The Note Calculation Agent will provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, of any amount payable on the Alerian ETNs at or prior to 11:00 a.m., New York City time, on the date on which payment is to be made.

All values with respect to calculations in connection with the Alerian ETNs will be rounded to the nearest one hundred-thousandth, with five one-millionths rounded upward (e.g., 0.876545 would be rounded to 0.87655). Notwithstanding the foregoing, all dollar amounts related to determination of any payment on the Alerian ETNs per note will be rounded to the nearest ten-thousandth, with five one hundred-thousandths rounded upward (e.g., 0.76545 would be rounded up to 0.7655), and all dollar amounts payable, if any, on the aggregate principal amount of Alerian ETNs per holder will be rounded to the nearest cent, with one-half cent rounded upward.

Trustee: Deutsche Bank Trust Company Americas

Day Count Fraction: With respect to each Coupon Valuation Date, the day count fraction is calculated as follows:

$$\frac{[360 \times (Y2 - Y1) + [30 \times (M2 - M1)]] + (D2 - D1)}{360}$$

where:

- a. “Y1” is the year, expressed as a number, in which the first day of the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
- b. “Y2” is the year, expressed as a number, in which the day immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date falls;

- c. “M1” is the calendar month, expressed as a number, in which the first day of the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
- d. “M2” is the calendar month, expressed as a number, in which the day immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date falls;
- e. “D1” is the first calendar day, expressed as a number, of the Coupon Accrual Period with respect to that Coupon Valuation Date, unless that number would be 31, in which case D1 will be 30; and
- f. “D2” is the calendar day, expressed as a number, immediately following the last day included in the Coupon Accrual Period with respect to that Coupon Valuation Date, unless that number would be 31 and D1 is greater than 29, in which case D2 will be 30.

Split or Reverse Split of the Alerian ETNs

The Issuer may initiate a split or reverse split of the Alerian ETNs at any time. The Issuer will determine the ratio of such split or reverse split, as applicable, using relevant market indicia in its sole discretion. If the Issuer decides to initiate a split or reverse split, as applicable, the Issuer will issue a notice to holders of the Alerian ETNs and a press release announcing the split or reverse split and the ratio and specifying the effective date of the split or reverse split, which will be at least three Business Days after the date on which the split or reverse split, as applicable, is announced, which we refer to as the “announcement date.” The record date for a share split or reverse split will be the Business Day immediately preceding the effective date.

If the Alerian ETNs undergo a split or reverse split, the Issuer will adjust the terms of the Alerian ETNs as may be necessary or desirable to effectuate that split or reverse split, as applicable, including, without limitation, the Principal Amount, any Coupon Amount, the Closing Intrinsic Note Value, the Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount, the number of shares held by a Reference Holder and the minimum number of Alerian ETNs holders may request that the Issuer repurchase. For example, if the Alerian ETNs undergo a 4-for-1 split, each Alerian ETN holder who holds one Alerian ETN via DTC prior to the split will, after the split, hold four Alerian ETNs, and the Principal Amount, any Coupon Amount, the Closing Intrinsic Note Value, the Accrued Investor Fee, the Periodic Investor Fee, the Reference Distribution Amount and the number of shares held by a Reference Holder will be adjusted as may be necessary or desirable to equal 1/4 of their respective values that would have prevailed in the absence of the 4-for-1 split. In addition, the minimum number of Alerian ETNs holders may request that the Issuer repurchase will be increased to equal 4 times its prior value. These adjustments may be applied retroactively for purposes of adjusting any Coupon Amount that has been determined but not yet paid.

In the case of a reverse split, holders who hold a number of Alerian ETNs that is not evenly divisible by the relevant ratio will receive the same treatment as all other holders for the maximum number of Alerian ETNs they hold that is evenly divisible by the relevant ratio, and the Issuer will have the right to compensate holders for their remaining or “partial” Alerian ETNs in a manner determined by the Note Calculation Agent in its sole discretion. The Issuer’s current intention is to provide holders with a cash payment for their partial Alerian ETNs in an amount equal to the appropriate percentage of the Closing Intrinsic Note Value (which may be calculated using an Index Ratio that reflects the average VWAP Level over a measurement period of five Index Business Days) on a specified Index Business Day no later than 20 Index Business Days following the announcement date, with payment to be made on a specified Business Day no later than five Business Days following the date on which the amount of the payment is determined.

Postponement of a Payment Date

In this section, we refer to each Coupon Payment Date and the Maturity Date as “Payment Dates.” If any scheduled Payment Date is not a Business Day, then that Payment Date will be the next succeeding Business Day following the scheduled Payment Date. If, due to a market disruption event or otherwise, any Averaging Date referenced in the determination of a payment on the Alerian ETNs that will or may be payable on any Payment Date is postponed so that it falls less than three Business Days prior to that scheduled Payment Date, that Payment Date will be the third Business Day following the latest such Averaging Date, as postponed. If any Payment Date is adjusted as the result of a non-Business Day, a market disruption event or otherwise, any amount payable on that Payment Date will be made on that Payment Date as postponed, with the same force and effect as if that Payment Date had not been postponed, but no interest will accrue or be payable as a result of the delayed payment.

Postponement of an Averaging Date

The Final VWAP Level used to determine the Closing Intrinsic Note Value in connection with any payment at maturity or upon early repurchase or redemption will reflect the arithmetic average of the VWAP Levels on the five Index Business Days in the relevant Measurement Period, and the VWAP Level on each Index Business Day is calculated by reference to the VWAP and the Index Units of each Index Component, as well as the Index Divisor. In this section, we refer to each of the five Index Business Days in a Measurement Period as an “Averaging Date.” As set forth below, if any Index Component is disrupted on any Averaging Date, determinations with respect to that Index Component for purposes of calculating the

VWAP Level applicable to that Averaging Date will be postponed. In addition, determinations with respect to that disrupted Index Component for purposes of calculating the VWAP Level applicable to each following Averaging Date will also be postponed so as to avoid using the same VWAP of an Index Component on a given Index Business Day to calculate the VWAP Levels for more than one Averaging Date.

If any Averaging Date is an Unavailable Day for any Index Component, the VWAP Level on that Averaging Date will be deemed to be the Adjusted VWAP Level with respect to that Averaging Date, and that Averaging Date will be postponed to the earliest date on which that Adjusted VWAP Level can be calculated as set forth below. For the avoidance of doubt, no Averaging Date will be postponed to a date that is after the applicable Final Disrupted Averaging Date.

The “Adjusted VWAP Level” with respect to an Averaging Date that is an Unavailable Day for any Index Component will be determined by the VWAP Calculation Agent and will be calculated in accordance with the formula for calculating the VWAP Level, using:

- a. with respect to each Index Component for which the originally scheduled Averaging Date is not an Unavailable Day, the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case as of the originally scheduled Averaging Date; and
- b. with respect to each Index Component for which the originally scheduled Averaging Date is an Unavailable Day, the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case, as of the first Index Business Day immediately following the originally scheduled Averaging Date that is not an Unavailable Day with respect to that Index Component, provided that, if each day from and including the originally scheduled Averaging Date to and including the applicable Final Disrupted Averaging Date is an Unavailable Day with respect to that Index Component, the VWAP of that Index Component will be determined by the VWAP Calculation Agent in good faith based on its assessment of the market value of that Index Component on that Final Disrupted Averaging Date and the Index Units of that Index Component and the Index Divisor will each be determined by the VWAP Calculation Agent based on its good faith estimate.

An Index Business Day will be an “Unavailable Day” with respect to an Index Component if it is a Disrupted Day for that Index Component or a day that is a Prior Valuation Date with respect to that Index Component and the relevant Averaging Date.

A “Disrupted Day” means, with respect to an Index Component, a day on which the Primary Exchange or any Related Exchange with respect to that Index Component fails to open for trading during its regular trading session or on which a Market Disruption Event with respect to that Index Component has occurred or is continuing, and, in each case, the occurrence of which is determined by the Note Calculation Agent to have a material effect on the VWAP Level.

An Index Business Day will be a “Prior Valuation Date” with respect to an Index Component and an Averaging Date if the VWAP and the Index Units of that Index Component, and the Index Divisor, in each case as of that Index Business Day, were used in calculating the Adjusted VWAP Level with respect to an earlier occurring originally scheduled Averaging Date within the same Measurement Period as that Averaging Date.

With respect to an Averaging Date, the “Final Disrupted Averaging Date” means the third Index Business Day after that Averaging Date, as originally scheduled.

Market Disruption Events

With respect to an Index Component, a “Market Disruption Event” means, the occurrence or existence of a condition specified below:

- a. any suspension, absence or limitation of trading in that Index Component on the Primary Exchange with respect to that Index Component, whether by reason of movements in price exceeding limits permitted by the Primary Exchange or otherwise;
- b. any suspension, absence or limitation of trading in futures or options contracts related to that Index Component, if available, on any Related Exchange with respect to that Index Component, whether by reason of movements in price exceeding limits permitted by that Related Exchange or otherwise;
- c. any event that disrupts or impairs the ability of market participants in general to effect transactions in, or obtain market values for, that Index Component or futures or options contracts relating to that Index Component; or
- d. the closure on any Index Business Day of the Primary Exchange or any Related Exchange with respect to that Index Component prior to its Scheduled Closing Time unless the earlier closing time is announced by the Primary Exchange or that Related Exchange, as applicable, at least one hour prior to the earlier of (a) the actual closing time for the regular trading session on the Primary Exchange or that Related Exchange, as applicable, on that Index Business Day and (b) the submission deadline for orders to be entered into the system for execution at the close of trading on that Index Business Day for the Primary Exchange or that Related Exchange, as applicable;

in each case determined by the Note Calculation Agent in its sole discretion; and

- a. a determination by the Note Calculation Agent in its sole discretion that the applicable event described above materially interfered with the Issuer's ability or the ability of any of its affiliates to adjust or unwind all or a material portion of any hedge with respect to the Alerian ETNs.

For purposes of determining whether a Market Disruption Event with respect to an Index Component has occurred:

- a. a limitation on the hours or number of days of trading will not constitute a Market Disruption Event if it results from an announced change in the regular business hours of the Primary Exchange or any Related Exchange; and
- b. limitations pursuant to the rules of the Primary Exchange or any Relevant Exchange similar to NYSE Rule 80B (or any applicable rule or regulation enacted or promulgated by any other self-regulatory organization or any government agency of scope similar to NYSE Rule 80B as determined by the Note Calculation Agent) on trading during significant market fluctuations will constitute a suspension, absence or material limitation of trading.

"Scheduled Closing Time" means, with respect to the Primary Exchange or the Related Exchange, on any Index Business Day, the scheduled weekday closing time of the Primary Exchange or such Related Exchange on such Index Business Day, without regard to after hours or any other trading outside of the regular trading session hours.

Discontinuation of the Index; Alteration of Method of Calculation

If the Index Sponsor discontinues publication of, or otherwise fails to publish, the Index, or if the Index Sponsor does not make the Index Components, their Index Units or the Index Divisor available to the VWAP Calculation Agent, and the Index Sponsor or another entity publishes a successor or substitute index that the Note Calculation Agent determines, in its sole discretion, to be comparable to the discontinued Index and for which the index components, their Index Units and the index divisor, or equivalent information, is available to the VWAP Calculation Agent (such index being referred to in this section as a "successor index"), then the VWAP Level on any subsequent relevant date will be determined by reference to the VWAPs of the index components of that successor index, their Index Units and the index divisor of that successor index, or equivalent information, published with respect to that day and a fixed adjustment factor determined by the Note Calculation Agent upon the selection of that successor index to account for the difference in the levels of the Index and that successor index at that time for purposes of maintaining comparability between prior VWAP Levels and succeeding VWAP Levels.

Upon any selection by the Note Calculation Agent of a successor index, the Note Calculation Agent will cause written notice thereof to be promptly furnished to the Trustee, us, the Issuer and DTC, as holder of the Alerian ETNs.

Payment upon an Event of Default

In case an event of default with respect to the Alerian ETNs shall have occurred and be continuing, the amount declared due and payable per Alerian ETN upon any acceleration of the Alerian ETNs will be determined by the Note Calculation Agent and will be an amount in cash equal to the amount payable at maturity per Alerian ETN calculated as if the date of acceleration were (a) the final day of the Measurement Period with respect to the Final Valuation Date and (b) the Final Disrupted Averaging Date for the final day of the Measurement Period with respect to the Final Valuation Date (if the date of acceleration is a Disrupted Day). Under these circumstances, the Measurement Period with respect to the Final Valuation Date will consist of the date of acceleration and the four immediately preceding Index Business Days.

The amount determined as described above will constitute the final payment on the Alerian ETNs, and no additional amounts will accrue with respect to the Alerian ETNs following the date of acceleration.

If the maturity of the Alerian ETNs is accelerated because of an event of default as described above, the Issuer will, or will cause the Note Calculation Agent to, provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, and to DTC, as holder of the Alerian ETNs, of the cash amount due with respect to the Alerian ETNs as promptly as possible and in no event later than two Business Days after the date of acceleration.

Listing

The Alerian ETNs have been listed on the NYSE Arca under the ticker symbol "AMJB." No assurance can be given as to the continued listing for the term of the Alerian ETNs, or the liquidity or trading market for the Alerian ETNs. The Issuer is not required to maintain a listing on NYSE Arca or any other exchange.

Book-Entry Only Issuance — The Depository Trust Company

DTC will act as securities depository for the Alerian ETNs. The Issuer will issue Alerian ETNs only in fully registered form as book-entry notes. The Alerian ETNs will be represented by one or more permanent global notes deposited with, or on behalf of, DTC and registered in the name of DTC or its nominee. These certificates name DTC or its nominee as the owner of the Alerian ETNs. DTC maintains a computerized system that will reflect the interests held by its participants in the

global notes. An investor's beneficial interest will be reflected in the records of DTC's direct or indirect participants through an account maintained by the investor with its broker/dealer, bank, trust company or other representative.

The Alerian ETNs will initially be represented by a type of global note referred to as a master note. A master note represents multiple securities that may be issued at different times and that may have different terms. Unless otherwise specified, in connection with each issuance of the Alerian ETNs, the Trustee and/or paying agent will, in accordance with instructions from the Issuer, make appropriate entries or notations in its records relating to the master note representing the Alerian ETNs to indicate that the master note evidences the Alerian ETNs of that issuance.

References to "holders" mean those who own Alerian ETNs registered in their own names, on the books that the Issuer or the paying agent maintain for this purpose, and not those who own beneficial interests in Alerian ETNs registered in street name or in Alerian ETNs issued in book-entry form through DTC.

Denominations

Book-entry interests in Alerian ETNs will be issued in minimum denominations equal to the Principal Amount and in integral multiples thereof.

Registrar, Transfer Agent and Paying Agent

Payment of amounts due at maturity on the Alerian ETNs will be payable and the transfer of the Alerian ETNs will be registrable at the principal corporate trust office of The Bank of New York Mellon in the City of New York.

The Bank of New York Mellon or one of its affiliates will act as registrar and transfer agent for the Alerian ETNs. The Bank of New York Mellon will also act as paying agent for the Alerian ETNs and may designate additional paying agents.

Registration of transfers of the Alerian ETNs will be effected without charge by or on behalf of The Bank of New York Mellon but upon payment (with the giving of such indemnity as The Bank of New York Mellon may require) in respect of any tax or other governmental charges that may be imposed in relation to it.

Reopening Issuances

In Issuer's sole discretion, and without providing holders notice or obtaining their consent, the Issuer may decide to issue and sell additional Alerian ETNs from time to time. These further issuances, if any, will be consolidated to form a single class with the originally issued Alerian ETNs, will have the same CUSIP number and will trade interchangeably with the originally issued Alerian ETNs immediately upon settlement.

However, the Issuer is under no obligation to issue or sell additional Alerian ETNs at any time, and, if the Issuer does sell additional Alerian ETNs, the Issuer may limit or restrict those sales, and the Issuer may stop and subsequently resume selling additional Alerian ETNs at any time. If the Issuer limits, restricts or stops sales of such additional Alerian ETNs, or if the Issuer subsequently resumes sales of such additional Alerian ETNs, the liquidity and trading price of the Alerian ETNs in the secondary market could be materially and adversely affected. Unless the Issuer indicates otherwise, if the Issuer suspends selling additional Alerian ETNs, the Issuer reserves the right to resume selling additional Alerian ETNs at any time, which might result in the reduction or elimination of any premium in the trading price. If holders pay a premium for the Alerian ETNs above the Closing Intrinsic Note Value and the Intraday Intrinsic Note Value, holders could incur significant losses if holders sell their Alerian ETNs at a time when the premium is no longer present in the market.

A suspension of additional issuances of the Alerian ETNs could result in a significant reduction in the number of outstanding Alerian ETNs if noteholders subsequently exercise their right to have the Alerian ETNs repurchased by the Issuer. Accordingly, the number of outstanding Alerian ETNs, and their liquidity, could vary substantially over the term of the Alerian ETNs.

The Alerian ETNs do not limit the Issuer's ability to incur other indebtedness or to issue other securities. Also, the Issuer is not subject to financial or similar restrictions by the terms of the Alerian ETNs.

The Issuer has no obligation to take holders' interests into account when deciding whether to issue additional Alerian ETNs. In addition, the Issuer is under no obligation to reopen any series of notes or to issue any additional Alerian ETNs.

General Terms of the Series A Notes

For a summary of the material provisions of the 2016 Indenture and the Series A Notes and guarantees issued thereunder, including provisions on events of defaults, modification of the 2016 Indenture and governing law, please see "— Description of JPMorgan Chase Financial Company LLC's Callable Fixed Rate Notes Due June 10, 2032, Fully and Unconditionally Guaranteed by JPMorgan Chase & Co. — General Terms of the Series A Notes" below.

DESCRIPTION OF JPMORGAN CHASE FINANCIAL COMPANY LLC'S CALLABLE FIXED RATE NOTES DUE JUNE 10, 2032, FULLY AND UNCONDITIONALLY GUARANTEED BY JPMORGAN CHASE & CO.

The following description of the Callable Fixed Rate Notes due June 10, 2032 (the "Callable Notes") is a summary and does not purport to be complete. It is subject to and qualified in its entirety by reference to the indenture dated February 19, 2016 (as may be amended or supplemented from time to time, the "2016 Indenture"), among JPMorgan Chase Financial Company LLC, as issuer ("JPMorgan Financial" or the "Issuer"), JPMorganChase, as guarantor (the "Guarantor"), and Deutsche Bank Trust Company Americas, as trustee (the "Trustee"), which is incorporated by reference as an Exhibit to the Form 10-K. We encourage you to read the 2016 Indenture for additional information.

General

As of December 31, 2024, \$5,000,000 aggregate principal amount of the Callable Notes were outstanding.

The Callable Notes are unsecured and unsubordinated obligations of JPMorgan Financial, the payment of which is fully and unconditionally guaranteed by JPMorgan Chase & Co. The Callable Notes will rank *pari passu* with all of the Issuer's other unsecured and unsubordinated obligations. JPMorgan Chase & Co.'s guarantee of the Callable Notes will rank *pari passu* with all of JPMorgan Chase & Co.'s other unsecured and unsubordinated obligations. **Any payment on the Callable Notes issued by JPMorgan Financial is subject to the credit risk of JPMorgan Finance Callable Notes, and the credit risk of JPMorgan Chase & Co., as guarantor of the Callable Notes.**

The Callable Notes are part of a series of the Issuer's debt securities entitled "Global Medium-Term Notes, Series A" (the "Series A Notes") that the Issuer may issue under the 2016 Indenture from time to time. For more information about the Series A Notes, please see the section titled "— General Terms of the Series A Notes" below.

The Callable Notes are not bank deposits and are not insured by the Federal Deposit Insurance Corporation or by any other governmental agency, nor are they obligations of, or guaranteed by, a bank.

Unless otherwise specified, references herein to "holders" mean those in whose names the Callable Notes are registered on the books that the Issuer or the Trustee, or any successor trustee, as applicable, maintain for this purpose, and not those who own beneficial interests in the Callable Notes (registered in street name or otherwise).

Key Terms of the Callable Notes

We issued the Callable Notes on June 10, 2021, in minimum denominations of \$1,000 and in integral multiples of \$1,000 thereafter. The Maturity Date of the Callable Notes is June 10, 2032. Interest on the Callable Notes is payable semiannually on June 10th and December 10th of each year, beginning on December 10, 2021 to and including the Maturity Date (each, an "Interest Payment Date"), subject to any earlier redemption, at the Interest Rate of 2.60% per annum.

On June 10, 2031 (the "Redemption Date"), the Issuer may redeem the holders' Callable Notes, in whole but not in part, at a price equal to the principal amount being redeemed *plus* any accrued and unpaid interest. Any accrued and unpaid interest on the Callable Notes redeemed will be paid to the person who is the holder of record on such Callable Notes at the close of business one (1) business day prior to the Redemption Date. To redeem the Callable Notes, the Issuer will deliver notice to DTC, as holder of the Callable Notes, at least 5 business days and not more than 15 business days prior to the Redemption Date.

Calculation Agent

JPMS (the "Calculation Agent") will make all necessary calculations and determinations in connection with the Callable Notes, including calculations and determinations relating to any payments on the Callable Notes.

Payment upon an Event of Default

In case an event of default with respect to the Callable Notes shall have occurred and be continuing, the amount declared due and payable per \$1,000 principal amount note upon any acceleration of the Callable Notes will be determined by the Calculation Agent and will be an amount in cash equal to \$1,000 per \$1,000 principal amount note *plus* accrued and unpaid interest, calculated as if the date of acceleration were the Maturity Date. In such case, interest will be calculated on the basis of a 360-day year and the actual number of days in such adjusted Interest Period and will be based on the Interest Rate on the applicable date immediately preceding such adjusted Interest Period.

If the maturity of the Callable Notes is accelerated because of an event of default as described above, the Issuer will, or will cause the Calculation Agent to, provide written notice to the Trustee at its New York office, on which notice the Trustee may conclusively rely, and to DTC of the cash amount due with respect to the Callable Notes as promptly as possible and in no event later than two business days after the date of acceleration.

Listing

The Callable Notes are listed and admitted to trading on the NYSE under the trading symbol “JPM/32.” No assurance can be given as to the continued listing for the term of the Callable Notes, or the liquidity or trading market for the Callable Notes.

Book-Entry Only Issuance — The Depository Trust Company

DTC will act as securities depository for the Callable Notes. The Callable Notes have been issued only as fully registered securities registered in the name of Cede & Co. (DTC’s nominee). One or more fully registered global note certificates, representing the total aggregate principal amount of the Callable Notes, have been issued and have been deposited with DTC. We will not issue definitive notes in exchange for the global notes except in limited circumstances.

Registrar, Transfer Agent and Paying Agent

The Bank of New York Mellon or one of its affiliates will act as registrar and transfer agent for the Callable Notes. The Bank of New York Mellon will also act as paying agent for the Callable Notes and may designate additional paying agents.

Reopening Issuances

The Issuer may, in its sole discretion, “reopen” the Callable Notes based upon market conditions at that time. These further issuances, if any, will be consolidated with, have the same CUSIP number as and trade interchangeably with the respective originally issued Callable Notes immediately upon settlement and, consequently, will increase the aggregate principal amount of such outstanding Callable Notes. The price of any additional offering will be determined at the time of pricing of that offering.

General Terms of the Series A Notes

In this “General Terms of the Series A Notes” section, all references to the “debt securities” refer to Series A Notes issued by JPMorgan Chase Financial Company LLC.

The following description of the terms of the debt securities contains certain general terms that may apply to the debt securities, including the Callable Notes and the Alerian ETNs.

We have summarized below the material provisions of the 2016 Indenture and the debt securities and guarantees issued under the 2016 Indenture.

These descriptions are only summaries, and each investor should refer to the 2016 Indenture, which describes completely the terms and definitions summarized below and contains additional information regarding the debt securities issued under it. Where appropriate, we use parentheses to refer you to the particular sections of the 2016 Indenture. Any reference to particular sections or defined terms of the 2016 Indenture in any statement under this heading qualifies the entire statement and incorporates by reference the applicable section or definition into that statement.

The debt securities will be the Issuer’s direct, unsecured general obligations, the payment on which is fully and unconditionally guaranteed by the Guarantor, and will have the same rank in liquidation as all of the Issuer’s other unsecured and unsubordinated debt.

The Guarantor is a holding company that holds the stock of JPMorgan Chase & Co. Bank, National Association (the “Bank”), and JPMorgan Chase & Co. Holdings LLC (the “IHC”), its “intermediate holding company.” The Guarantor conducts substantially all of its operations through subsidiaries, including the Bank and the IHC. As a result, claims of the holders of the debt securities against the Guarantor under the guarantee will generally have a junior position to claims of creditors of the Guarantor’s subsidiaries. Claims of the Guarantor’s subsidiaries’ creditors other than the Guarantor include substantial amounts of deposit liabilities, long-term debt and other liabilities. In addition, the Guarantor is obligated to contribute to the IHC substantially all the net proceeds that the Guarantor receives from the issuance of securities (including the Alerian ETNs), and the ability of the Bank and the IHC to make payments to the Guarantor is limited. As a result of these arrangements, the Guarantor’s ability to make various payments is dependent on its receiving dividends from the Bank and dividends and extensions of credit from the IHC. These limitations could affect the Guarantor’s ability to pay interest on its debt securities, redeem or repurchase outstanding securities and fulfill its other payment obligations, including payment obligations under the guarantees of our debt securities.

Events of Default and Waivers

An “Event of Default” with respect to a series of debt securities issued under the 2016 Indenture is defined in the 2016 Indenture as:

- a. default in the payment of interest on any debt securities of that series and continuance of such default for 30 days;

- b. default in the payment of principal or other amounts payable on any debt securities of that series when due, at maturity, upon redemption, by declaration, or otherwise;
- c. default in the performance, or breach, of any other covenants or warranties applicable to the Issuer contained in the 2016 Indenture applicable to that series, and continuation of such default or breach for 90 days after written notice has been given by the Trustee to the Issuer and the Guarantor or given by holders of at least 25% in aggregate principal amount of the outstanding securities of all series affected thereby to the Issuer, the Guarantor and the Trustee;
- d. certain events of the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation, whether voluntary or involuntary;
- e. the guarantee ceases to be in full force and effect, other than in accordance with the 2016 Indenture, or the Guarantor denies or disaffirms its obligations under the guarantee, *provided* that no Event of Default with respect to the guarantee will occur as a result of, or because it is related directly or indirectly to, the insolvency of the Guarantor or the commencement of proceedings under Title 11 of the United States Code, or the appointment of a receiver for the Guarantor under Title II of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, or the Federal Deposit Insurance Corporation having separately repudiated the Guarantee in receivership, or the commencement of or certain other events of the Guarantor's bankruptcy, insolvency, resolution, receivership, winding up or liquidation; or
- f. any other event of default provided in the applicable supplemental indentures to the 2016 Indenture or form of security. (Section 5.01)

If an Event of Default occurs and is continuing because of a default in the payment of principal, interest or other amounts payable on the debt securities, a failure in the performance, or breach, of any covenant or agreement applicable to the Issuer, the guarantee ceasing to be in full force and effect, or any other event of default provided in the applicable supplemental indentures to the 2016 Indenture or form of security, either the Trustee or the holders of not less than 25% in aggregate principal amount of the debt securities of such series then outstanding, treated as one class, by written notice to the Issuer and the Guarantor, may declare the principal of all outstanding debt securities of such series and any interest accrued thereon, to be due and payable immediately. If a default due to specified events of the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation, occurs and is continuing, the principal of all outstanding debt securities and any interest accrued thereon will automatically, and without any declaration or other action on the part of the Trustee or any holder, become immediately due and payable. Subject to certain conditions such declarations may be annulled and past defaults may be waived by the holders of a majority in aggregate principal amount of the outstanding debt securities of the series affected. (Sections 5.01 and 5.10)

Events of bankruptcy, insolvency, resolution, receivership, winding up or liquidation relating to the Guarantor will not constitute an Event of Default with respect to any series of debt securities. In addition, failure by the Guarantor to perform any of its covenants or warranties (other than a payment default) will not constitute an Event of Default with respect to any series of debt securities. Therefore, events of bankruptcy, resolution, receivership, insolvency, winding up or liquidation relating to the Guarantor (in the absence of any such event occurring with respect to the Issuer) will not permit any of the debt securities to be declared due and payable and the Trustee is not authorized to exercise any remedy against the Issuer or the Guarantor upon the occurrence or continuation of these events with respect to the Guarantor. Instead, even if an event of bankruptcy, insolvency, resolution, receivership, winding up or liquidation relating to the Guarantor has occurred, the Trustee and the holders of debt securities of a series will not be able to declare the relevant debt securities to be immediately due and payable unless there is an Event of Default with respect to that series as described above, such as the Issuer's bankruptcy, insolvency, receivership, winding up or liquidation or a payment default by the Issuer or the Guarantor on the relevant debt securities. **The value holders receive on any series of debt securities may be significantly less than what holders would have otherwise received had the Issuer's debt securities been declared due and payable immediately or the Trustee been authorized to exercise any remedy against the Issuer or the Guarantor upon the occurrence or continuation of these events with respect to the Guarantor.**

An Event of Default with respect to one series of debt securities does not necessarily constitute an Event of Default with respect to any other series of debt securities. The 2016 Indenture requires the Trustee to provide notice of default with respect to the debt securities within 90 days, unless the default is cured, but provides that the Trustee may withhold notice to the holders of the debt securities of any default if the board of directors, the executive committee, or a trust committee of directors or Trustees and/or responsible officers of the Trustee determines in good faith that it is in the interest of the holders of the debt securities of the applicable series to do so. The Trustee may not withhold notice of a default in the payment of principal of, interest on or any other amounts due under, such debt securities. (Section 5.11)

The 2016 Indenture provides that the holders of a majority in aggregate principal amount of outstanding debt securities of each series affected, with all such series voting as a single class, may direct the time, method, and place of conducting any proceeding for any remedy available to the Trustee, or exercising any trust or power conferred on the Trustee. The Trustee may decline to act if the direction is contrary to law and in certain other circumstances set forth in the 2016

Indenture. (Section 5.09) The Trustee is not obligated to exercise any of its rights or powers under the 2016 Indenture at the request or direction of the holders of debt securities unless the holders offer the Trustee security or indemnity satisfactory to it against the costs, expenses and liabilities incurred therein or thereby. (Section 6.02(d))

No holder of any debt security of any affected series has the right to institute any action for remedy unless such holder has previously given to the Trustee written notice of default, the Trustee has failed to take action for 60 days after the holders of not less than 25% in aggregate principal amount of the debt securities of each affected series make written request upon the Trustee to institute such action and have offered reasonable indemnity in connection with the same and the holders of a majority in aggregate principal amount of the debt securities of each affected series (voting as a single class) have not given direction to the Trustee that is inconsistent with the written request referred to above. (Section 5.06)

However, the right of any holder of a debt security or coupon to receive payment of the principal of and interest on that debt security or coupon on or after its due date, or to institute suit for the enforcement of any such payment, may not be impaired or affected without the consent of that holder. (Section 5.07)

The 2016 Indenture requires the Issuer and the Guarantor to file annually with the Trustee a written statement as to whether or not the Issuer or the Guarantor, as the case may be, has knowledge of a default. (Section 3.05)

The Issuer, the Guarantor and certain of their affiliates have a wide range of banking relationships with Deutsche Bank Trust Company Americas and The Bank of New York Mellon. If an actual or potential event of default occurs with respect to any debt securities, the Trustee may be considered to have a conflicting interest for purposes of the Trust Indenture Act of 1939, as amended. In that case, the Trustee may be required to resign under the 2016 Indenture, and the Issuer would be required to appoint a successor trustee. For this purpose, a "potential" event of default means an event that would be an event of default if the requirements for giving the Issuer default notice or for the default having to exist for a specific period of time were disregarded.

Modification of the 2016 Indenture

The 2016 Indenture contains provisions permitting the Issuer, the Guarantor and the Trustee to modify the 2016 Indenture or the rights of the holders of debt securities with the consent of the holders of not less than a majority in aggregate principal amount of each outstanding series of debt securities affected by the modification. Each holder of an affected debt security must consent to a modification that would:

- a. extend the final maturity date of the principal of, or of any interest on, or other amounts payable under any debt security;
- b. reduce the principal amount of, rate of interest on, or any other amounts due under any debt security;
- c. change the currency or currency unit of payment of any debt security or certain provisions of the 2016 Indenture applicable to debt securities in foreign currencies;
- d. change the method in which amounts of payments of principal, interest or other amounts are determined on any debt security;
- e. reduce any amount payable upon redemption of any debt security;
- f. impair the right of a holder to institute suit for the payment of a debt security or, if the debt securities provide, any right of repurchase at the option of the holder of a debt security;
- g. reduce the percentage of debt securities of any series, the consent of the holders of which is required for any modification; or
- h. make any change in the guarantee that would adversely affect the holders of the debt securities of such series or release the Guarantor from the guarantee other than pursuant to the terms of the 2016 Indenture. (Section 8.02)

The 2016 Indenture also permits the Issuer, the Guarantor and the Trustee to amend the 2016 Indenture in certain circumstances without the consent of the holders of debt securities to evidence the Issuer's or the Guarantor's merger or the replacement of the Trustee, to cure any ambiguity or to correct or supplement any defective or inconsistent provision, to make any change to the 2016 Indenture or the Issuer's debt securities that the Issuer deems necessary or desirable and that does not materially and adversely affect the interests of holders of the debt securities and for certain other purposes. (Section 8.01)

Consolidations, Mergers, Sales and Transfers of Assets

Neither the Issuer nor the Guarantor may merge or consolidate with any other entity or sell, convey or transfer all or substantially all of their respective assets to any other entity, unless:

- a. with respect to the Issuer:
 - i. either the Issuer is the continuing company in the case of a merger or consolidation or the successor entity in the case of a merger or consolidation (including an affiliate of the Guarantor) or the entity to whom those assets are sold, conveyed or transferred in the case of a sale, conveyance or transfer is a

United States corporation or limited liability company that expressly assumes the due and punctual payment of the principal of, any interest on, or any other amounts due under the debt securities and the due and punctual performance and observance of all the covenants and conditions of the 2016 Indenture binding upon the Issuer, and

- ii. no Event of Default and no event which, with notice or lapse of time or both, would become an Event of Default has occurred or would be continuing, immediately after the merger or consolidation, or the sale, conveyance or transfer, and

a. with respect to the Guarantor:

- i. either the Guarantor is the continuing corporation in the case of a merger or consolidation or the successor corporation in the case of a merger or consolidation or the entity to whom those assets are sold, conveyed or transferred in the case of a sale, conveyance or transfer is a United States corporation that expressly assumes the full and unconditional guarantee of the full and punctual payment of the principal of, any interest on, or any other amounts due under the debt securities and the due and punctual performance and observance of all the covenants and conditions of the 2016 Indenture binding upon the Guarantor, and
- ii. no Event of Default and no event which, with notice or lapse of time or both, would become an Event of Default has occurred or would be continuing, immediately after the merger or consolidation, or the sale, conveyance or transfer. (Sections 9.01 and 9.02)

Any transfer of material assets of the Guarantor to any other entity that occurs as a result of, or because it is related directly or indirectly to, any proceedings relative to the Guarantor under Title 11 of the United States Code or under a receivership under Title II of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 or under any other applicable federal or state bankruptcy, insolvency, resolution or other similar law will be deemed to be a sale, conveyance or transfer of all or substantially all of the Guarantor's assets.

There are no covenants or other provisions in the 2016 Indenture that would afford holders of debt securities additional protection in the event of a recapitalization transaction involving the Issuer or the Guarantor, a change of control of the Issuer or the Guarantor or a highly leveraged transaction involving the Issuer or the Guarantor. The merger covenant described above would apply only if the recapitalization transaction, change of control or highly leveraged transaction were structured to include a merger or consolidation of the Issuer or the Guarantor or a sale or conveyance of all or substantially all of the Issuer's or the Guarantor's assets. However, the Issuer may provide specific protections, such as a put right or increased interest, for particular debt securities, which the Issuer would describe in the applicable prospectus supplement.

JPMorganChase Guarantee

The Guarantor will fully and unconditionally guarantee the full and punctual payment of the principal of, interest on, and all other amounts payable under the debt securities when the same becomes due and payable, whether at maturity, upon redemption, repurchase at the option of the holders of the applicable debt securities or upon acceleration. If for any reason the Issuer does not make any required payment in respect of the Issuer's debt securities when due, the Guarantor will on demand pay the unpaid amount at the same place and in the same manner that applies to payments made by the Issuer under the 2016 Indenture. The guarantee is of payment and not of collection. (Section 14.01)

The Guarantor's obligations under the guarantee are unconditional and absolute. However, (1) the Guarantor will not be liable for any amount of payment that the Issuer is excused from making or any amount in excess of the amount actually due and owing by the Issuer, and (2) any defense or counterclaims available to the Issuer (except those resulting solely from, or on account of, the Issuer's insolvency or the Issuer's status as debtor or subject of a bankruptcy or insolvency proceeding) will also be available to the Guarantor to the same extent as these defense or counterclaims are available to the Issuer, whether or not asserted by the Issuer. (Section 14.02)

Concerning the Trustee, Paying Agent, Registrar and Transfer Agent

We, the Guarantor and certain of their affiliates have a wide range of banking relationships with Deutsche Bank Trust Company Americas, The Bank of New York Mellon and The Bank of New York Mellon, London Branch. The Bank of New York Mellon and, for notes settled through Euroclear Bank SA/NV or Clearstream Banking, S.A., Luxembourg, The Bank of New York Mellon, London Branch, will be the paying agents, authenticating agents, registrars and transfer agents for debt securities issued under the 2016 Indenture.

Deutsche Bank Trust Company Americas is initially serving as the Trustee for the debt securities issued under our 2016 Indenture, to which JPMorganChase acts as a guarantor, the warrants issued under our warrant indenture, to which JPMorganChase acts as a guarantor, and the debt securities issued under JPMorganChase's indenture.

Governing Law and Judgments

The debt securities and the 2016 Indenture, including the guarantee, will be governed by, and construed in accordance with, the laws of the State of New York. (Section 11.08)

Exhibit 10.19

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN TERMS AND CONDITIONS OF JANUARY 21, 2025 RESTRICTED STOCK UNIT AWARD OPERATING COMMITTEE

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 (“Grant Date”) awarding Restricted Stock Units (“RSUs”) pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan (“Plan”). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in “Definitions” below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as “JPMorgan Chase” and together with its subsidiaries as the “Firm”.

Form and Purpose of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

The purpose of this award is to motivate your future performance for services to be provided during the vesting period and to align your interests with those of the Firm and its shareholders.

Dividend Equivalents

If dividends are paid on Common Stock while RSUs under this award are outstanding, you will be paid an amount equal to the dividend paid on one share of Common Stock, multiplied by the number of RSUs outstanding under this award as of the dividend record date.

Vesting Period

The period from the Grant Date to the last vesting date is the “vesting period” (see subsections captioned “--Amendment” pursuant to which the Firm may extend the vesting period and “--No Ownership Rights/Other Limitations” pursuant to which the Firm may place restrictions on delivered shares of Common Stock following a vesting date).

Protection-Based Vesting

This award is intended and expected to vest on the vesting date(s), provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability”. However, vesting and the number of RSUs in which you vest are subject to these terms and conditions (including, but not limited to, the section captioned “Remedies” and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you during the vesting period (“At Risk RSUs”) may be cancelled if the Chief Executive Officer of JPMorgan Chase (“CEO”) determines in his or her sole discretion that cancellation of all or portion of the At Risk RSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm’s performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are net income, total net revenue, return on equity, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.

- For any calendar year ending during the vesting period, JPMorgan Chase's annual pre-tax pre-provision income at the Firm level is negative.
- Awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

In the event that your employment terminates due to "Job Elimination", "Full Career Eligibility", "Government Office" or "Disability" thereby entitling you to continued vesting in your award (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply to your At Risk RSUs pursuant to the subsection captioned "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity".

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Compensation and Management Development Committee of the Board of Directors of JPMorgan Chase ("Committee"). In the case of an award to the CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

<https://about.jpmorganchase.com/about/governance/corporate-governance-principles>

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Termination of Employment

Except as explicitly set forth below under the subsections captioned "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability" or under the section captioned "Death", any RSUs outstanding under this award will be cancelled effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Your Obligations" and "Remedies"), you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

Job Elimination

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- the Firm terminated your employment because your job was eliminated, and
- after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
- you satisfied the Release/Certification Requirements set forth below.

Full Career Eligibility

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
 - the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
 - you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
 - continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
-

- you satisfied the Release/Certification Requirements set forth below, and
- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
 - work in a government, education or Not-for-Profit Organization (as defined below),
 - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below).

- **Government Office**

In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**

In the event that

- your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third-party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
- you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for "Full Career Eligibility", and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm, that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, complied with all other terms of the Award Agreement. (See section captioned "Your Obligations".)

Death

If you die while you are eligible to vest in RSUs under this award, the RSUs will immediately vest and will be distributed in shares of Common Stock (after applicable tax withholding) to your designated beneficiary on file with the Firm's Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation. Any shares will be distributed no

later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm's receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned "--Confidentiality & Non-Solicitation", "--False Statements", "--Cooperation", "--Compliance with Award Agreement" and "--Notice Period."

- **Confidentiality & Non-Solicitation**

During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**

If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "--Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Bonus Recoupment", and "Termination of Employment", up to 100% of your outstanding RSUs under this award may be cancelled if the Firm in its sole discretion determines that:

- Any of the following detrimental and risk-related conduct has occurred:
 - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
 - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
 - this award was based on a material misrepresentation by you, or
 - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
 - your employment was terminated for Cause (see section captioned "Definitions" below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
 - you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
 - you have failed to return the required forms specified under the section captioned "Release/Certification" by the specified deadline, or
 - you have violated any of the provisions as set forth above in the section captioned "Your Obligations".
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To the extent provided under the subsection captioned “--Amendment” below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. JPMorgan Chase is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also subsection captioned “--No Ownership Rights/Other Limitations”.

- **Recovery**

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:

- Payment may be required with respect to any shares distributed within the three year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
 - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
 - you violated any of the provisions as set forth above in the section captioned “Your Obligations”, or
 - you violated the employment restrictions set forth in the subsection “--Full Career Eligibility” or “--Government Office” following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above “Detrimental Conduct, Risk Related and Other Cancellation/Recapture” subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm’s employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase’s rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm’s intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under “Bonus Recoupment”.

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in “Your Obligations” herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in “Your Obligations”; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent

authorities with respect to the vested shares, dividend equivalents and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award, shares that vest under this award, and/or dividend equivalents, and (iii) retaining shares that vest under this award or dividend equivalents until you pay any taxes associated with the award, vested shares and/or the dividend equivalents directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in “No Ownership Rights/Other Limitations” described below.

No Ownership Rights/Other Limitations: RSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the RSUs have vested. Shares will be issued in accordance with JPMorgan Chase’s procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding RSUs has been suspended, then you agree that such RSUs may be cancelled in whole or part. (See Sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “Termination of Employment” and “Remedies”, as well as the subsection captioned “--Amendment” permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the “Right to Set Off” subsection above). (See Sections captioned “Bonus Recoupment” and “Remedies”.)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is “at-will” and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares and cash hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered to you during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following (i) each applicable vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award and (ii) each date that JPMorgan

Chase specifies for payment of dividends declared on its Common Stock, shall be the payment date(s) for purposes of distributing dividend equivalent payments.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any RSUs outstanding under this award for such corporate events.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase RSUs, the Firm may accelerate the distribution of all or part of your outstanding award effective on or before the required divestiture date; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled "--Job Elimination" or "--Full Career Eligibility", then acceleration shall apply, to the extent required, to the full outstanding award.

Notwithstanding accelerated distribution pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of each applicable vesting date, including, but not limited to, repayment obligations set forth in the section captioned "Remedies" and the employment restrictions in the sections captioned "Protection-Based Vesting" and "Government Office Requirements" and the subsection "--Full Career Eligibility".

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on

your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of RSUs, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Outstanding Awards: The Administrative provisions set forth above shall apply to any award of RSUs outstanding as of the date hereof, and such awards are hereby amended.

Definitions

"Cause" means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm's business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

"Financial Services Company" means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
 - insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the foregoing,
 - financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other
-

similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,

- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
 - financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
 - advising on, or investing in, private equity or real estate funds or ventures, or
 - any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,
- however, subject to disclosure and confirmation of the following, it shall not be considered a “Financial Services Company” for the purpose of this Award Agreement if such family office is solely funded by you and/or your family’s own personal funds without external investors directly or indirectly.

“Firmwide Financial Threshold” means a cumulative return on tangible common equity for calendar years 2025, 2026 and 2027 of not less than 15%. Cumulative return on tangible common equity means (i) the sum of the Firm’s reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

“Government Office” means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

“Line of Business” means a business unit of the Firm (or one or more business units designated below under the definition “Line of Business Financial Threshold” of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

“Line of Business Financial Threshold” means the financial threshold set forth below for the following Lines of Business based on the Firm’s management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income ¹
Card and Auto	Annual negative pre-tax pre-loan loss reserve income ²
Commercial & Investment Bank	Annual negative pre-tax pre-provision income ¹ for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income ²
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income ¹ at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income ²
¹ Pre-tax pre-provision income means Revenue less Expenses	
² Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

“Not-for-Profit Organization” means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals.

“Recognized Service” means the period of service as an employee set forth in the Firm’s applicable service-related policies.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:

“Government Office Requirements” does not apply to you if you satisfy the subsection captioned “--Full Career Eligibility” as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:

Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days’ advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned “Definitions”), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm’s sole discretion.
- Confirmation, in a form satisfactory to the firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See Section captioned “Definitions”).

Portion of Your Awards Subject to Continued Vesting:

Subject to the conditions below, the percentage of your outstanding awards that will continue to vest in accordance with this award’s original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of each award subject to continued vesting above is referred to as the “CV Award” and the portion not subject to continued vesting will be cancelled on the date your employment terminates.

Conditions for Continued Vesting of Awards:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to receive in full your CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be entitled to retain any portion of the CV Award with a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

Satisfaction of Conditions:

If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”) will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above “Conditions for Continued Vesting of Awards”, any outstanding RSUs under each CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as RSUs on the date you failed to satisfy the “Condition for Continued Vesting of Awards” but for their accelerated distribution (as described in the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF JANUARY 21, 2025
RESTRICTED STOCK UNIT AWARD
OPERATING COMMITTEE

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 (“Grant Date”) awarding Restricted Stock Units (“RSUs”) pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan (“Plan”). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in “Definitions” below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as “JPMorgan Chase” and together with its subsidiaries as the “Firm”.

Form and Purpose of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

The purpose of this award is to motivate your future performance for services to be provided during the vesting period and to align your interests with those of the Firm and its shareholders.

Dividend Equivalents

This award is not eligible for dividend equivalent payments.

Vesting Period

The period from the Grant Date to the last vesting date is the “vesting period” (see subsections captioned “--Amendment” pursuant to which the Firm may extend the vesting period and “--No Ownership Rights/Other Limitations” pursuant to which the Firm may place restrictions on delivered shares of Common Stock following a vesting date).

Protection-Based Vesting

This award is intended and expected to vest on the vesting date(s), provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability”. However, vesting and the number of RSUs in which you vest are subject to these terms and conditions (including, but not limited to, the section captioned “Remedies” and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you during the vesting period (“At Risk RSUs”) may be cancelled if the Chief Executive Officer of JPMorgan Chase (“CEO”) determines in his or her sole discretion that cancellation of all or portion of the At Risk RSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm’s performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are net income, total net revenue, return on equity, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase’s annual pre-tax pre-provision income at the Firm level is negative.
- Awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

In the event that your employment terminates due to “Job Elimination”, “Full Career Eligibility”, “Government Office” or “Disability” thereby entitling you to continued vesting in your award (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply to your At Risk RSUs pursuant to the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Compensation and Management Development Committee of the Board of Directors of JPMorgan Chase (“Committee”). In the case of an award to the CEO, all such determinations shall be made by the Committee and ratified by the Board.

Holding Requirement

As of each vesting date, you shall be entitled to a distribution equal to the Fair Market Value of the number of RSUs vesting on such date, less the number being withheld to satisfy tax withholding obligations. You agree that the distribution made to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such distribution for a twelve-month period commencing with the vesting date. Such restrictions shall lapse in event of your death.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled “Other Matters”:

<https://about.jpmorganchase.com/about/governance/corporate-governance-principles>

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

EMEA Malus and Clawback Policy – Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect in My Rewards through the following link: <https://myrewards.jpmorganchase.com>

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm’s financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions.

Termination of Employment

Except as explicitly set forth below under the subsections captioned “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability” or under the section captioned “Death”, any RSUs outstanding under this award will be cancelled effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “EMEA Malus and Clawback Policy – Identified Staff”, “Your Obligations” and “Remedies”), you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- the Firm terminated your employment because your job was eliminated, and
- after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and

- you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
- your Recognized Service (as defined below) on your date of termination equaled or exceeded 15 years, or your combined Recognized Service with the Firm and external professional experience (as attested by you to the Firm) equaled or exceeded 30 years, and
- you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
- continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
- you satisfied the Release/Certification Requirements set forth below, and
- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
 - work in a government, education or Not-for-Profit Organization (as defined below),
 - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the service requirement.

Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below).

- **Government Office**

In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

- **Disability**

In the event that

- your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
- you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written

notification), advise that you are seeking to be treated as an individual eligible for “Full Career Eligibility”, and receive written consent to such continued vesting,

- with respect to “Full Career Eligibility” and “Government Office”, it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm, that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to “Disability”, you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, complied with all other terms of the Award Agreement. (See section captioned “Your Obligations”).

Death

If you die while you are eligible to vest in RSUs under this award, the RSUs will immediately vest and will be distributed in shares of Common Stock (after applicable tax withholding) to your designated beneficiary on file with the Firm’s Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation. Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm’s receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned “—Confidentiality & Non-Solicitation”, “—False Statements”, “--Cooperation”, “--Compliance with Award Agreement” and “--Notice Period.”

• **Confidentiality & Non-Solicitation**

During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm’s then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual’s employment terminated because his or her job was eliminated, or the individual’s employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm’s confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm’s policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm’s business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. “Confidential Information” includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm’s private and internal information through telephonic, intranet or internet means.

For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this “Confidentiality and Non-Solicitation” provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment

or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use “Confidential Information” in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm’s management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else’s employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm’s management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**

If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment (“Notice Period”), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned “--Full Career Eligibility” in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

In addition to the cancellation provisions described under the sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “EMEA Malus and Clawback Policy - Identified Staff”, and “Termination of Employment”, up to 100% of your outstanding RSUs under this award may be cancelled if the Firm in its sole discretion determines that:

- Any of the following detrimental and risk-related conduct has occurred:

- you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
- this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
- this award was based on a material misrepresentation by you, or
- you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
- your employment was terminated for Cause (see section captioned “Definitions” below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
- you have failed to return the required forms specified under the section captioned “Release/Certification” by the specified deadline, or
- you have violated any of the provisions as set forth above in the section captioned “Your Obligations”.

To the extent provided under the subsection captioned “--Amendment” below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. JPMorgan Chase is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also subsection captioned “--No Ownership Rights/Other Limitations”.

• **Recovery**

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:

- Payment may be required with respect to any shares distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
 - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
 - you violated any of the provisions as set forth above in the section captioned “Your Obligations”, or
 - you violated the employment restrictions set forth in the subsection “--Full Career Eligibility” or “--Government Office” following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above “Detrimental Conduct, Risk Related and Other Cancellation/Recapture” subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm’s employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase’s rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm’s intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under “Bonus Recoupment” and “EMEA Malus and Clawback Policy - Identified Staff”.

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in “Your Obligations” herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in “Your Obligations”; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and/or vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in “No Ownership Rights/Other Limitations” described below.

No Ownership Rights/Other Limitations: RSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the RSUs have vested. Shares will be issued in accordance with JPMorgan Chase’s procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding RSUs has been suspended, then you agree that such RSUs may be cancelled in whole or part. (See Sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “EMEA Malus and Clawback Policy - Identified Staff”, “Termination of Employment” and “Remedies”, as well as the subsection captioned “--Amendment” permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the “Right to Set Off” subsection above). (See Sections captioned “Bonus Recoupment” and “Remedies”.)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is “at-will” and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered to you during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following each applicable vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any RSUs outstanding under this award for such corporate events.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under "No Ownership Rights/Other Limitations", in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase RSUs, the Firm may accelerate the distribution of all or part of your outstanding award effective on or before the required divestiture date; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the "Remedies" section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned "Government Office Requirements", acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is

outstanding as a result of the subsections entitled “--Job Elimination” or “--Full Career Eligibility”, then acceleration shall apply, to the extent required, to the full outstanding award.

Notwithstanding accelerated distribution pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of each applicable vesting date, including, but not limited to, repayment obligations set forth in the section captioned “Remedies” and the employment restrictions in the sections captioned “Protection-Based Vesting” and “Government Office Requirements” and the subsection “--Full Career Eligibility”.

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm’s legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm’s By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of RSUs, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Outstanding Awards: The Administrative provisions set forth above shall apply to any award of RSUs outstanding as of the date hereof, and such awards are hereby amended.

Definitions

“Cause” means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm’s business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

“Financial Services Company” means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
 - insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
 - financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks, and hedge funds,
 - issuing, trading, or selling instruments representing interests in pools of assets or in derivatives instruments,
 - financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
 - advising on, or investing in, private equity or real estate funds or ventures, or
 - any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services.
- however, subject to disclosure and confirmation of the following, it shall not be considered a “Financial Services Company” for the purpose of this Award Agreement if such family office is solely funded by you and/or your family’s own personal funds without external investors directly or indirectly.

“Firmwide Financial Threshold” means a cumulative return on tangible common equity for calendar years 2025, 2026 and 2027 of not less than 15%. Cumulative return on tangible common equity means (i) the sum of the Firm’s reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

“Government Office” means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

“Line of Business” means a business unit of the Firm (or one or more business units designated below under the definition “Line of Business Financial Threshold” of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

“Line of Business Financial Threshold” means the financial threshold set forth below for the following Lines of Business based on the Firm’s management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income ¹
Card and Auto	Annual negative pre-tax pre-loan loss reserve income ²
Commercial & Investment Bank	Annual negative pre-tax pre-provision income ¹ for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income ²
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income ¹ at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income ²
¹ Pre-tax pre-provision income means Revenue less Expenses	
² Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

“Not-for-Profit Organization” means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for religious, charitable,

scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals.

“Recognized Service” means the period of service as an employee set forth in the Firm’s applicable service-related policies.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:

“Government Office Requirements” does not apply to you if you satisfy the subsection captioned “--Full Career Eligibility” as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:

Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days’ advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned “Definitions”), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm’s sole discretion.
- Confirmation, in a form satisfactory to the firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See Section captioned “Definitions”).

Portion of Your Awards Subject to Continued Vesting:

Subject to the conditions below, the percentage of your outstanding awards that will continue to vest in accordance with this award’s original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of each award subject to continued vesting above is referred to as the “CV Award” and the portion not subject to continued vesting will be cancelled on the date your employment terminates.

Conditions for Continued Vesting of Awards:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to receive in full your CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be entitled to retain any portion of the CV Award with a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

Satisfaction of Conditions:

If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”) will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above “Conditions for Continued Vesting of Awards”, any outstanding RSUs under each CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as RSUs on the date you failed to satisfy the “Condition for Continued Vesting of Awards” but for their accelerated distribution (as described in the subsection captioned “Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF _____, 20__
PERFORMANCE SHARE UNIT AWARD
OPERATING COMMITTEE
(Protection-Based Vesting Provisions)

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of _____ (“Grant Date”) awarding performance share units (“PSUs”) pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan (“Plan”). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement and Plan. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in “Definitions” below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as “JPMorgan Chase”, and together with its subsidiaries as the “Firm”.

Form and Purpose of Award

Each PSU represents a non-transferable right to receive one share of Common Stock as of the vesting date as set forth in your Award Agreement.

The purpose of this award is to further emphasize sustained long-term performance and to align your interests with those of the Firm and its shareholders.

Number to Vest on the Vesting Date

Subject to any cancellation in whole or part of your award pursuant to these terms and conditions:

Performance calculation: On the vesting date set forth in your Award Agreement, you will vest in a number of PSUs derived by multiplying the Target Award Number by the Award Payout Percentage determined using the Performance Table. See sections captioned “Calculation of Performance Ranking” and “Definitions”.

You will also vest in additional shares of Common Stock as calculated under the section captioned, “Reinvested Dividend Equivalent Share Units”. Delivery of vested shares to your account will be made not later than the date specified in the last sentence of the subsection captioned “Section 409A Compliance”.

Reinvested Dividend Equivalent Share Units

If dividends are paid on Common Stock during the Vesting Period while the award is outstanding, you will receive on the vesting date additional units representing shares of Common Stock as calculated in this section. The number, if any, will be based on the dividends that would have been paid during the Vesting Period as of each dividend payment date on the actual number of shares of Common Stock distributable to you resulting from the vesting of the PSUs, if any, and treated as reinvested in additional shares of Common Stock on each dividend payment based on the Fair Market Value of one share of Common Stock on each dividend payment date (“Reinvested Dividend Equivalent Share Units”).

Holding Requirement

As of the vesting date set forth in your Award Agreement, you shall be entitled to be issued a number of shares of the Common Stock of JPMorgan Chase equal to the number of PSUs, if any, plus any additional Reinvested Dividend Equivalent Share Units, vesting on such date, less the number withheld to satisfy tax withholding obligations. The net number of shares issued to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such shares for a two-year period commencing as of the vesting date and ending as of the second anniversary of the vesting date. Such restrictions shall only lapse, prior to the expiration of the two-year holding period, in the event of your death or for an accelerated distribution for ethics or conflict reasons. See section captioned, “Death” and subsection captioned, “Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity”.

Calculation of Performance Ranking

For purposes of the Performance Ranking, the ranking of the Firm and of each Performance Company for the Performance Period shall be determined and calculated by the Calculation Agent, using the definitions of “Average Tangible Common Equity” (if otherwise applicable), “Calculated PSUs”, “Firm Reported ROTCE”, “Performance Table” and “ROTCE” as set forth in the “Definitions” section of these terms and conditions. See section captioned “Definitions”. Except for Firm’s externally published Reported ROTCE, calculations will be expressed as a decimal to the second place (i.e. xx.yy%), rounded to the nearest hundredth. See section captioned, “Definitions--Performance Table” in the event of a tie. All performance-based calculations as set forth herein are binding and conclusive on you and your successors.

Capital Ratio Performance Threshold

Unvested PSUs are subject to reduction if the Firm’s Common Equity Tier 1 (CET1) capital ratio at any year end falls below a predetermined threshold of _____%.

If the Firm’s CET1 capital ratio at any year end during the Performance Period is below this predetermined threshold, up to one-third of the Target Award Number of PSUs will be subject to downward adjustment by the Compensation & Management Development Committee of the Board of Directors of JPMorgan Chase (“Committee”) for each such year.

Vesting Period

The period from the Grant Date to the vesting date is the “Vesting Period”. (See “Administrative Provision--Amendment” pursuant to which the Firm may extend the vesting period and “No Ownership Rights/Other Limitations” pursuant to which the Firm may place restrictions on delivered shares of Common Stock following the vesting date and section captioned, “Holding Period” above.)

Protection-Based Vesting

This award is intended and expected to vest on the vesting date, provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability”. However, vesting and the number of PSUs that will vest are subject to these terms and conditions (including, but not limited to, sections captioned “Number to Vest on the Vesting Date”, “Capital Ratio Performance Threshold”, “Remedies”, and the following protection-based vesting provision).

Up to a total of fifty percent of your award (including any associated Reinvested Dividend Equivalent Share Units) that would otherwise be distributable to you on the vesting date (“At Risk PSUs”) may be cancelled if the Chief Executive Officer of JPMorgan Chase (“CEO”) determines in his or her sole discretion that cancellation of all or portion of the At Risk PSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm’s performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are: net income, total net revenue, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase’s annual pre-tax pre-provision income at the Firm level is negative.
- RSU awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

For avoidance of doubt, cancellation of the At Risk PSUs, in whole or part, for one or more of the above factors may occur prior to the end of the Performance Period and the maximum number of At Risk PSUs subject to cancellation prior to the end of the Performance Period will be up to fifty percent of the Target Award Number.

In the event that your employment terminates due to “Job Elimination”, “Full Career Eligibility”, “Government Office” or “Disability” thereby entitling you to continued vesting in your award, (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Committee. In the case of an award to any current or former CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded

to you for performance year 20XX and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled "Other Matters":

<https://about.jpmorganchase.com/about/governance/corporate-governance-principles>

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Recovery of Erroneously Awarded Incentive-Based Compensation

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation (or successor policy) as in effect from time to time as it applies to this award. You can access this policy as currently in effect by clicking the following link to the Firmwide Policy & Standard Portal, or alternatively typing go/dogma then searching for the Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation.

[Firmwide Policy & Standard Portal > Recovery of Erroneously Awarded Incentive-Based Compensation Policy](#)

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation (or successor policy).

Termination of Employment

Except as explicitly set forth below under the subsections captioned "--Job Elimination", "--Full Career Eligibility", "--Government Office" or "--Disability" or under the section captioned "Death", this award (for avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units) will be cancelled in full effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned "Protection-Based Vesting", "Number to Vest on the Vesting Date", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation", "Your Obligations" and "Remedies") you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- the Firm terminated your employment because your job was eliminated, and
- after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
- you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
- the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
- you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
- continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
- you satisfied the Release/Certification Requirements set forth below, and

- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
 - work in a government, education or Not-for-Profit Organization (as defined below),
 - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm's sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

Additional advance notice requirements may apply for employees subject to notice period policies (see "Notice Period" below.)

• **Government Office**

In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned "Government Office Requirements". See also definition of Government Office in the section captioned "Definitions".

• **Disability**

In the event that

- your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase-sponsored local country plan (in either case, "LTD Plan"), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm's written consent to such treatment, and
- you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to "Full Career Eligibility", prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for Full Career Eligibility, and receive written consent to such continued vesting,
- with respect to "Full Career Eligibility" and "Government Office", it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to the vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to "Disability", you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, otherwise complied with all other terms of the Award Agreement. (See section captioned "Your Obligations" below.)

Death

If you die while you are eligible to vest in this award, your designated beneficiary on file with the Firm's Stock Administration Department, or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally, then to your estate unless otherwise required by local legislation, may be entitled to receive a

distribution of a number of shares of Common Stock associated with your award. The Award Payout Percentage in the case of death is based on the “Number to Vest on the Vesting Date” calculation described above using the average performance of all completed calendar years, multiplied by one-third of the Target Award Number of PSUs for each completed calendar year in the Performance Period, and using the Award Payout Percentage equal to 100 percent for any remaining calendar years in the Performance Period.

In addition, your beneficiary or your estate shall receive additional shares of Common Stock, i.e., Reinvested Dividend Equivalent Share Units, as set forth in the section captioned, “Reinvested Dividend Equivalent Share Units” but based on dividend equivalents up to the date of your death.

Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm’s receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned “--Confidentiality & Non-Solicitation”, “--False Statements”, “--Cooperation”, “--Compliance with Award Agreement”, and “--Notice Period”.

- **Confidentiality & Non-Solicitation**

During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm’s then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual’s employment terminated because his or her job was eliminated, or the individual’s employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm’s confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm’s policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm’s business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. “Confidential Information” includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm’s private and internal information through telephonic, intranet or internet means.

For Employees with a Work location in California, you agree that the restrictions in the first paragraph of the “Confidentiality and Non-Solicitation” provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use “Confidential Information” in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm’s management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**

If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment ("Notice Period"), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned "-- Full Career Eligibility" in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

In addition to the cancellation provisions described under the sections captioned "Protection-Based Vesting", "Bonus Recoupment", "Recovery of Erroneously Awarded Incentive-Based Compensation" and "Termination of Employment", up to 100% of your outstanding PSUs under this award (for the avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units as well as the Calculated PSUs) may be cancelled if the Firm in its sole discretion determines that:

- Any of the following detrimental and risk-related conduct has occurred:
 - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
 - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
 - this award was based on a material misrepresentation by you, or

- you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
- your employment was terminated for Cause (see section captioned “Definitions” below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
- you have failed to return the required forms specified under the section captioned “Release/Certification” by the specified deadline, or
- you have violated any of the provisions as set forth above in the section captioned “Your Obligations”.

To the extent provided under the subsection captioned “--Amendment” below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. The Firm is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also “Administrative Provisions—No Ownership Rights/Other Limitations”.

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date or acceleration date) of the gross number of shares previously distributed, including vested shares subject to the Holding Requirements, under this award as follows:

- Payment may be required with respect to any shares of Common Stock distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
 - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment;
 - you violated any of the provisions as set forth above in the section captioned “Your Obligations,” or
 - you violated the employment restrictions set forth in the subsection “Full Career Eligibility” or “Government Office” following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above “Detrimental Conduct, Risk Related and Other Cancellation/Recapture” subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm’s employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase’s rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm’s intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under “Bonus Recoupment” or “Recovery of Erroneously Awarded Incentive-Based Compensation”.

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in “Your Obligations” herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in “Your Obligations”; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt or obligation owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in “No Ownership Rights/Other Limitations” described below.

No Ownership Rights/Other Limitations: PSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the number of PSUs have been determined, if any, and have vested. Shares will be issued in accordance with JPMorgan Chase’s procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding PSUs has been suspended, then you agree that such PSUs may be cancelled in whole or part. (See Sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation”, “Termination of Employment” and “Remedies”, as well as the subsection captioned “—Amendment” permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the “Right to Set Off” subsection above). (See sections captioned “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation” and “Remedies”.)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is “at-will” and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered during the first

calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following the vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any PSUs outstanding under this award for such corporate events.

Other Equitable Adjustments: The Committee may make adjustments (up or down) to the award as it deems to be equitable, to maintain the intended economics of the award in light of changed circumstances, which may include unusual or non-recurring events affecting the Firm (or the Performance Companies) or its financial statements in each case resulting from changes in accounting methods, practices or policies, changes in capital structure by reason of legal or regulatory requirements and such other changed circumstances, as the Committee may deem appropriate.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under “No Ownership Rights/Other Limitations”, in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase PSUs, the Firm may accelerate the distribution of all or part of your outstanding award, including Reinvested Dividend Equivalent Share Units, effective on or before the required divestiture date and waive the Holding Requirement; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the “Remedies” section above. The time period for recovery shall be determined by the originally scheduled vesting date or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned “Government Office Requirements”, acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled “Job Elimination” or “Full Career Eligibility”, then acceleration shall apply, to the extent required, to the full outstanding award. Subject to the two foregoing sections, the number of shares of

Common Stock to be received on acceleration shall be determined using the methodology set forth under the section captioned “Death”.

To the extent you have vested shares under this award subject to the Holding Requirement and become subject to divestiture requirement as forth herein, the Firm may waive the holding period to the extent required.

Notwithstanding an accelerated distribution or waiver of the Holding Requirement pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the original vesting period and shares had been distributed as scheduled as of the vesting date, including, but not limited to, repayment obligations set forth in the section captioned “Remedies” and the employment restrictions in the sections captioned “Protection-Based Vesting” and “Government Office Requirements” and the subsection “Full Career Eligibility”.

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm’s legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm’s By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of restricted stock units or of performance-based share units, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Definitions

“Average Tangible Common Equity” means annual average common stockholders’ equity less annual average goodwill and annual average identifiable intangible assets. Annual averages of the components of Average Tangible Common Equity will be calculated using quarterly balances as reported in publicly available financial disclosures. In the event that quarterly balances are not available, annual year end balances will be used. This calculation is used solely for purposes of the Performance Ranking.

“Award Payout Percentage” means the applicable percentage specified in the Performance Table.

“Calculated PSUs” means the number of PSUs determined by multiplying the Target Award Number (after giving effect to any cancellation thereof, in whole or in part) by the Award Payout Percentage corresponding to the Firm’s Performance Ranking based on the three-year average performance for the Performance Period (both percentage and ranking, as set forth in the Performance Table); provided that if the average of the Firm’s Reported ROTCE, expressed as a decimal rounded to the nearest hundredth, for the Performance Period either equals or exceeds _____% or is less than _____%, _____ percent or _____, respectively as the case may be, shall be substituted for the Performance Period’s Award Payout Percentage in calculating the number of PSUs to distribute. For avoidance of doubt, any cancellation of this award (in whole or in part) during the Performance Period will reduce the Target Award Number.

“Calculation Agent” means a third party entity not owned or controlled by the Firm, such as an accounting or consulting firm, retained from time to time by the Director of Human Resources or his/her delegate.

“Cause” means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm’s business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

“Financial Services Company” means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the foregoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,
- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,
- however, subject to disclosure and confirmation of the following, it shall not be considered a “Financial Services Company” for the purpose of this Award Agreement if such family office is solely funded by you and/or your family’s own personal funds without external investors directly or indirectly.

“Firm Reported ROTCE” means the Firm’s percentage return on tangible common equity for each year in the Performance Period.

“Firmwide Financial Threshold” means a cumulative return on tangible common equity for calendar years 20XX, 20XX and 20XX of not less than _____. Cumulative return on tangible common equity means (i) the sum of the Firm’s reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

“Government Office” means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

“Line of Business” means a business unit of the Firm (or one or more business units designated below under the definition “Line of Business Financial Threshold” of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

“Line of Business Financial Threshold” means the financial threshold set forth below: for the following Lines of Business based on the Firm’s management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income ¹
Card and Auto	Annual negative pre-tax pre-loan loss reserve income ²
Commercial & Investment Bank	Annual negative pre-tax pre-provision income ¹ for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income ²
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income ¹ at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income ²
¹ Pre-tax pre-provision income means Revenue less Expenses ² Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

“Not-for-Profit Organization” means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for purposes identical to those applicable to Section 501(c)(3) organization.

“Performance Companies” mean the following institutions which have business activities that overlap with a significant portion of the Firm’s revenue mix: _____, and _____.

If, during the Performance Period, one or more Performance Companies shall merge, engage in a spin-off or otherwise experience a material change in its revenue mix or business activities or its existence or its primary businesses shall terminate or cease due to receivership, bankruptcy, sale, or otherwise, then the Committee may eliminate such institution from the list of Performance Companies or make such other equitable adjustments, such as adding an acquirer or a new company to the list of Performance Companies, as it deems appropriate, with any such changes having effect for purposes of all calculations hereunder on a prospective basis from the date the applicable change is made.

“Performance Period” means calendar years 20XX, 20XX and 20XX.

“Performance Ranking” means the ranking of the average ROTCE of the Firm as compared to the ranking of the average ROTCE of the Performance Companies as specified in the Performance Table for the Performance Period.

“Performance Table” means the table used in the calculation of PSUs for the Performance Period as set forth below:

Firm Reported ROTCE (average performance)	Award Payout Percentage	Firm Performance Ranking (average performance)	Award Payout Percentage
>= _____%	_____%	#1	_____%
		#2	_____%
		#3	_____%
		#4	_____%
_____ % to _____ %	Pay by relative ROTCE scale	#5	_____%
		#6	_____%
		#7	_____%
		#8	_____%
		#9	_____%
< _____%	_____%	#10 +	_____%

If, after the calculation of the Performance Ranking, there is a tie, the tie shall be disregarded for purposes of determining the Award Payout Percentage. For example, in the case of a tie for the fourth ranking between the Firm and a Performance Company, the Firm shall be treated as having satisfied that ranking. In the case of that same tie among Performance Companies, the fourth and fifth rankings will be deemed to have been satisfied.

“Recognized Service” means the period of service as an employee set forth in the Firm’s applicable service-related policies.

“ROTCE” means for the Firm and each of the Performance Companies a percentage derived by, for each year in the Performance Period, dividing (i) annual earnings from continuing operations less dividends on preferred stock as set forth in published financial disclosures by (ii) the Average Tangible Common Equity for the year. If, prior to the end of the vesting period, the Firm or any Performance Company restates its published financial statements for any year in the Performance Period, ROTCE for that year shall be recalculated for the Firm or Performance Company with the Performance Ranking adjusted, if necessary. This calculation is used solely for purposes of the Performance Ranking.

“Target Award Number” means the number of PSUs designated as such in the Award Agreement.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:

“Government Office Requirements” does not apply to you if you satisfy the subsection captioned “--Full Career Eligibility” as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:

Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days’ advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned “Definitions”), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm’s sole discretion.
- Confirmation, in a form satisfactory to the Firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See section captioned “Definitions”).

Portion of Your Award Subject to Continued Vesting:

Subject to the conditions below, the percentage of this award that will continue to vest in accordance with this award’s original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of this award subject to continued vesting above is referred to as the "CV Award" and the portion not subject to continued vesting will be cancelled as of the date your employment terminates.

Conditions for Continued Vesting of Award:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to be eligible to receive the CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be eligible to receive the CV Award if it has a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

For avoidance of doubt, the performance criteria and protection-based vesting set forth in these terms and conditions continue to apply to a CV Award.

Satisfaction of Conditions:

If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned "--Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity") will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above "Conditions for Continued Vesting of Awards", any outstanding PSUs under the CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as PSUs on the date you failed to satisfy the "Conditions for Continued Vesting of Award" but for their accelerated distribution (as described in subsection captioned, "Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity"). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF _____, 20__
PERFORMANCE SHARE UNIT AWARD
OPERATING COMMITTEE
(Protection-Based Vesting Provisions)

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of _____ (“Grant Date”) awarding performance share units (“PSUs”) pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan (“Plan”). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement and Plan. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in “Definitions” below or elsewhere in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as “JPMorgan Chase”, and together with its subsidiaries as the “Firm”.

Form and Purpose of Award

Each PSU represents a non-transferable right to receive one share of Common Stock following each vesting date as set forth in your Award Agreement.

The purpose of this award is to further emphasize sustained long-term performance and to align your interests with those of the Firm and its shareholders.

Number of Performance Share Units at End of Performance Period

Subject to any cancellation in whole or part of your award pursuant to these terms and conditions:

Performance calculation: The number of PSUs at the end of the Performance Period will be derived by multiplying the Target Award Number by the Award Payout Percentage determined using the Performance Table.

The number of PSUs determined above will be subject to the Qualitative Performance Factor (as detailed below), which if the Committee determines that such an adjustment is appropriate, will be applied following the end of each year during the Performance Period, to adjust downward one-third of the Target Award Number of PSUs for each calendar year in the Performance Period. Additionally, the Committee, in its discretion, may make a qualitative performance assessment based on the entire three-year Performance Period and apply the Qualitative Performance Factor to the entire number of PSUs determined above.

See sections captioned “Calculation of Performance Ranking” and “Definitions”.

Delivery of vested shares of common stock to your account will be made not later than the date specified in the last sentence of the subsection captioned “Section 409A Compliance”.

Reinvested Dividend Equivalent Share Units

This award is not eligible for reinvested dividend equivalent share units.

Holding Requirement

The net number of shares of Common Stock (after tax and all other lawful withholdings) in which you have vested, if any, as of the vesting date set forth in your Award Agreement will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging or otherwise encumbering such shares for (i) a twelve month period measured from each vesting date; and (ii) a two-year period for such shares vesting on _____, with the holding periods running concurrently. Such restrictions shall only lapse, prior to the expiration of the two-year holding period, in the event of your death or for an accelerated distribution for ethics or conflict reasons. See section captioned, “Death” and subsection captioned, “Accelerated Distribution for Ethics or Conflict Reasons Resulting from Employment by a Government Entity”.

Calculation of Performance Ranking

For purposes of the Performance Ranking, the ranking of the Firm and of each Performance Company for the Performance Period shall be determined and calculated by the Calculation Agent, using the definitions of “Average Tangible Common Equity” (if otherwise applicable), “Calculated PSUs”, “Firm Reported ROTCE”, “Performance Table” and “ROTCE” as set forth in the “Definitions” section of these terms and conditions. See section captioned “Definitions”. Except for Firm’s externally published Reported ROTCE, calculations will be expressed as a decimal to the second place (i.e. xx.yy%), rounded to the nearest hundredth. See section captioned, “Definitions--Performance Table” in the event of a tie. All performance-based calculations as set forth herein are binding and conclusive on you and your successors.

Capital Ratio Performance Threshold

Unvested PSUs are subject to reduction if the Firm’s Common Equity Tier 1 (CET1) capital ratio at any year end falls below a predetermined threshold of ____%.

If the Firm’s CET1 capital ratio at any year end during the Performance Period is below this predetermined threshold, up to one-third of the Target Award Number of PSUs will be subject to downward adjustment by the Compensation & Management Development Committee of the Board of Directors of JPMorgan Chase (“Committee”) for each such year.

Qualitative Performance

Determination of Qualitative Performance Factor. Annually during the Performance Period, the Committee will formally assess your qualitative performance based on four broad categories: (1) Client/Customer Focus; (2) Risk, Controls & Conduct; (3) Teamwork & Leadership; and (4) Business Results. If the Committee determines that your performance “Meets” expectations, no downward adjustment to one-third of the Target Award Number of PSUs for that year shall take place (and the Qualitative Performance Factor shall be 100%). If the Committee determines that your performance did “Not Meet” expectations, the Committee shall determine whether a downward adjustment is appropriate, and if so, to what extent. A downward adjustment could result in a Qualitative Performance Factor of between 0% and 99%, depending on the circumstances. During the Performance Period, a 0% Performance Factor for each year in the Performance Period would reduce your Target Award Number of PSUs to zero, resulting in the cancellation of award with no shares vesting.

Additionally, the Committee may, in its sole discretion, make such assessment of your qualitative performance based on your performance during the entire three-year Performance Period and apply the Qualitative Performance Factor to the entire number of PSUs determined under section captioned “Number of Performance Share Units at the end of the Performance Period”. In the case of a Qualitative Performance Factor of 0%, the award would be cancelled.

The assessment will have regard to feedback solicited from the Chair of the UK Remuneration Committee to incorporate qualitative performance against local regulatory responsibilities as a “Senior Manager” of the relevant UK-regulated entities.

The Qualitative Performance Factor shall only be applied, if applicable, in respect of a period of your employment with the Firm, or as soon as administratively practical.

Protection-Based Vesting

This award is intended and expected to vest on each vesting date set forth in your Award Agreement, provided that you are continuously employed by the Firm through such vesting date, or you meet the requirements for continued vesting described under the subsections “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability”. However, vesting and the number of PSUs that will vest are subject to these terms and conditions (including, but not limited to, sections captioned “Number of Performance Share Units at End of Performance Period”, “Capital Ratio Performance Threshold”, “Remedies”, and the following protection-based vesting provision).

Up to a total of fifty percent of your award that would otherwise be distributable to you as of any vesting date (“At Risk PSUs”) may be cancelled if the Chief Executive Officer of JPMorgan Chase (“CEO”) determines in his or her sole discretion that cancellation of all or portion of the At Risk PSUs is appropriate in light of any one or a combination of the following factors:

- Your performance in relation to the priorities for your position, or the Firm’s performance in relation to the priorities for which you share responsibility as a member of the Operating Committee, have been unsatisfactory for a sustained period of time. Among the factors the CEO may consider in assessing performance are: net income, total net revenue, earnings per share and capital ratios of the Firm, both on an absolute basis and, as appropriate, relative to peer firms.
- For any calendar year ending during the vesting period, JPMorgan Chase’s annual pre-tax pre-provision income at the Firm level is negative.

- RSU awards granted to participants in a Line of Business for which you exercise, or during the vesting period exercised, direct or indirect responsibility, were in whole or in part cancelled because the Line of Business did not meet its annual Line of Business Financial Threshold.
- The Firm does not meet the Firmwide Financial Threshold.

For avoidance of doubt, cancellation of the At Risk PSUs, in whole or part, for one or more of the above factors may occur prior to the end of the Performance Period and the maximum number of At Risk PSUs subject to cancellation prior to the end of the Performance Period will be up to fifty percent of the Target Award Number.

In the event that your employment terminates due to “Job Elimination”, “Full Career Eligibility”, “Government Office” or “Disability” thereby entitling you to continued vesting in your award, (or potentially acceleration due to satisfaction of the Government Office Requirements), the cancellation circumstances described above will continue to apply.

Any determination above with respect to protection-based vesting provisions is subject to ratification by the Committee. In the case of an award to any current or former CEO, all such determinations shall be made by the Committee and ratified by the Board.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 20_____ and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled “Other Matters”:

<https://about.jpmorganchase.com/about/governance/corporate-governance-principles>

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

Recovery of Erroneously Awarded Incentive-Based Compensation

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Recovery of Erroneously Awarded Incentive-Based Compensation Policy (or successor policy) as in effect from time to time as it applies to this award. You can access this policy as currently in effect by clicking the following link to the Firmwide Policy & Standard Portal, or alternatively typing go/dogma then searching for the Recovery of Erroneously Awarded Incentive-Based Compensation Policy.

[Firmwide Policy & Standard Portal > Recovery of Erroneously Awarded Incentive-Based Compensation Policy](#)

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Recovery of Erroneously Awarded Incentive-Based Compensation Policy (or successor policy).

EMEA Malus and Clawback Policy - Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year _____ and to this award.

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm’s financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions. You can access this policy as currently in effect in My Rewards through the following link: <https://myrewards.jpmorganchase.com>

Termination of Employment

Except as explicitly set forth below under the subsections captioned “--Job Elimination”, “--Full Career Eligibility”, “--Government Office” or “--Disability” or under the section captioned “Death”, this award will be cancelled in full effective on the date your employment with the Firm terminates for any reason.

Subject to these terms and conditions (including, but not limited to, sections captioned “Protection-Based Vesting”, “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation”, “EMEA Malus and Clawback Policy – Identified Staff”, “Your Obligations” and “Remedies”) you will be eligible to continue to vest (as you otherwise would vest if you were still employed by JPMorgan Chase) with respect to your award in accordance with its terms and conditions following the termination of your employment if one of the following circumstances applies to you:

- **Job Elimination**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- the Firm terminated your employment because your job was eliminated, and
- after you are notified that your job will be eliminated, you provided such services as requested by the Firm in a cooperative and professional manner, and
- you satisfied the Release/Certification Requirements set forth below.

- **Full Career Eligibility**

In the event that the Director of Human Resources or nominee in his or her sole discretion determines that

- (i) you voluntarily terminated your employment with the Firm, and (ii) had completed at least five years of Recognized Service (as defined below) on your date of termination with at least three years of continuous service with the Firm immediately preceding your termination date or if you are over 60 years of age with at least three years of continuous service with the Firm immediately preceding your termination date, and
- the sum of your age and Recognized Service (as defined below) on your date of termination equaled or exceeded 60 and
- you provided at least 180 days advance written notice to the Firm of your intention to voluntarily terminate your employment under this provision, during which notice period you provided such services as requested by the Firm in a cooperative and professional manner and you did not perform any services for any other employer, and
- continued vesting shall be appropriate, which determination shall be made prior to your termination and will be based on your performance and conduct (before and after providing notice), and
- you satisfied the Release/Certification Requirements set forth below, and
- for 36 months from the date of grant of this award you do not either perform services in any capacity (such as an employee, contractor, consultant, advisor, or self-employed individual, whether paid or unpaid) for a Financial Services Company (as defined below) or work in your profession (whether or not for a Financial Services Company); provided that you may do one or more of the following:
 - work in a government, education or Not-for-Profit Organization (as defined below),
 - subject to disclosure and confirmation of the following, serve on the board of directors of a public or non-public company that is not a Financial Services Company.

After receipt of such advance written notice, the Firm may choose to have you continue to provide services during such 180-day period as a condition to continued vesting or shorten the length of the 180-day period at the Firm’s sole discretion, but to a date no earlier than the date you would otherwise meet the age and service requirements.

Additional advance notice requirements may apply for employees subject to notice period policies (see “Notice Period” below.)

- **Government Office**

In the event that you voluntarily terminate your employment with the Firm to accept a Government Office or become a candidate for an elective Government Office, as described at the end of these terms and conditions under the section captioned “Government Office Requirements”. See also definition of Government Office in the section captioned “Definitions”.

- **Disability**

In the event that

- your employment with the Firm terminates because (i) you are unable to return to work while you are receiving benefits under the JPMorgan Chase Long Term Disability Plan, or for non-U.S. employees, under the equivalent JPMorgan Chase-sponsored local country plan (in either case, “LTD Plan”), or (ii) if you are not covered by a LTD Plan, you are unable to return to work due to a long-term disability that would qualify for benefits under the applicable LTD Plan, as determined by the Firm or a third party designated by the Firm; provided that you (x) request in writing continued vesting due to such disability within 30 days of the date your employment terminates, and (y) provide any requested supporting documentation and (z) receive the Firm’s written consent to such treatment, and
- you satisfied the Release/Certification Requirements set forth below.

Release/Certification

To qualify for continued vesting after termination of your employment under any of the foregoing circumstances:

- you must timely execute and deliver a release of claims in favor of the Firm, having such form and terms as the Firm shall specify,
- with respect to “Full Career Eligibility”, prior to the termination of your employment, you must confirm with management that you meet the eligibility criteria (including providing at least 180 days advance written notification), advise that you are seeking to be treated as an individual eligible for Full Career Eligibility, and receive written consent to such continued vesting,
- with respect to “Full Career Eligibility” and “Government Office”, it is your responsibility to (i) notify the Firm within 15 days after the date you are no longer in compliance with the employment restrictions (as described herein) or (ii) take the appropriate steps to certify to the Firm prior to each vesting date while the employment restrictions are outstanding, on the authorized form of, and by the due date set by, the Firm that you have complied with the employment restrictions applicable to you (as described herein) from your date of termination of employment through the applicable vesting date,
- with respect to “Disability”, you must satisfy the notice and documentation described above and receive written consent to such continued vesting, and
- in all cases, otherwise complied with all other terms of the Award Agreement. (See section captioned “Your Obligations” below.)

Death

If you die while you are eligible to vest in this award, your designated beneficiary on file with the Firm’s Stock Administration Department (or your estate or if no beneficiary has been designated or survives you or if beneficiary designation is not recognized locally) may be entitled to receive a distribution of a number of shares of Common Stock associated with your award.

Should you die after the end of the Performance Period, your beneficiary will receive shares of Common Stock equal to any outstanding PSUs.

Should you die during the Performance Period, your Beneficiary will receive shares of Common Stock based on the average performance of all completed calendar years, multiplied by one-third of the Target Award Number of PSUs for each completed calendar year in the Performance Period, and using the Award Payout Percentage equal to 100 percent for any remaining calendar years in the Performance Period.

Any shares will be distributed no later than the end of the calendar year immediately following the calendar year which contains your date of death; however, our administrative practice is to register such shares in the name of your beneficiary or estate within 60 days of the Firm’s receipt of any required documentation.

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned “--Confidentiality & Non-Solicitation”, “--False Statements”, “--Cooperation”, “--Compliance with Award Agreement”, and “--Notice Period”.

Confidentiality & Non-Solicitation

During your employment by the Firm and for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm’s then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written

notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. "Confidential Information" includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means. .

For Employees with a Work location in California, you agree that the restrictions in the first paragraph of the "Confidentiality and Non-Solicitation" provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use "Confidential Information" in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys, or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of the law or from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate, and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third party who might be affected thereby, including your prospective employer.

- **Notice Period**

If you are subject to a notice period or become subject to a notice period after the Grant Date, whether by contract or by policy, that requires you to provide advance written notice of your intention to terminate your employment (“Notice Period”), then as consideration for this award and continued employment, you will provide the Firm with the necessary advance written notice that applies to you, as specified by such contract or policy.

After receipt of your notice, the Firm may choose to have you continue to provide services during the applicable Notice Period or may place you on a paid leave for all or part of the applicable Notice Period. During the Notice Period, you shall continue to devote your full time and loyalty to the Firm by providing services in a cooperative and professional manner and not perform any services for any other employer and shall receive your base salary and certain benefits until your employment terminates. You and the Firm may mutually agree to waive or modify the length of the Notice Period.

Regardless of whether a Notice Period applies to you, you must comply with the 180-day advance notice period described under the subsection captioned “-- Full Career Eligibility” in the event you wish to terminate employment under that same subsection.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

In addition to the cancellation provisions described under the sections captioned “Protection-Based Vesting”, “Qualitative Performance Factor”, “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation”, “EMEA Malus and Clawback Policy – Identified Staff” and “Termination of Employment”, up to 100% of your outstanding PSUs under this award (for the avoidance of doubt, including any associated Reinvested Dividend Equivalent Share Units as well as the Calculated PSUs) may be cancelled if the Firm in its sole discretion determines that:

- Any of the following detrimental and risk-related conduct has occurred:
 - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
 - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
 - this award was based on a material misrepresentation by you, or
 - you improperly or with gross negligence failed to identify, raise or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
 - your employment was terminated for Cause (see section captioned “Definitions” below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.
- you have failed to comply with any of the advance notice/cooperation requirements or employment restrictions applicable to your termination of employment, or
- you have failed to return the required forms specified under the section captioned “Release/Certification” by the specified deadline, or
- you have violated any of the provisions as set forth above in the section captioned “Your Obligations”.

To the extent provided under the subsection captioned “--Amendment” below, JPMorgan Chase reserves the right to suspend vesting of this award and/or distribution of shares under this award, including, without limitation, during any period that JPMorgan Chase is evaluating whether this award is subject to cancellation and/or recovery and/or whether the conditions for distributions of shares under this award are satisfied. The Firm is not responsible for any price fluctuations during any period of suspension and, if applicable, suspended units will be reinstated consistent with Plan administration procedures. See also “Administrative Provisions—No Ownership Rights/Other Limitations”.

In addition, you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date or acceleration date) of the gross number of shares of Common Stock previously distributed, including vested shares subject to the Holding Requirements, under this award as follows:

- Payment may be required with respect to any shares of Common Stock distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
 - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment;
 - you violated any of the provisions as set forth above in the section captioned “Your Obligations”; or
 - you violated the employment restrictions set forth in the subsection “Full Career Eligibility” or “Government Office” following the termination of your employment.
- In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above “Detrimental Conduct, Risk Related and Other Cancellation/Recapture” subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm’s employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase’s rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm’s intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation” or “EMEA Malus and Clawback Policy – Identified Staff”.

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in “Your Obligations” herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in “Your Obligations”; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account

agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt or obligation owed by you under this award without your consent. This restriction on the Firm does not apply to accounts described and authorized in “No Ownership Rights/Other Limitations” described below.

No Ownership Rights/Other Limitations: PSUs do not convey the rights of ownership of Common Stock and do not carry voting rights. No shares of Common Stock will be issued to you until after the number of PSUs have been determined, if any, and have vested. Shares will be issued in accordance with JPMorgan Chase’s procedures for issuing stock. By accepting this award, you authorize the Firm, in its sole discretion, to establish on your behalf a brokerage account in your name with the Firm or book-entry account with our stock plan administrator and/or transfer agent and deliver to that account any vested shares derived from the award. You also acknowledge that should there be a determination that the cancellation provisions of this award apply during the period when the vesting of any outstanding PSUs has been suspended, then you agree that such PSUs may be cancelled in whole or part. (See Sections captioned “Protection-Based Vesting”, “Qualitative Performance Factor”, “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation”, “EMEA Malus and Clawback Policy – Identified Staff”, “Termination of Employment” and “Remedies”, as well as the subsection captioned “—Amendment” permitting suspension of vesting.)

With respect to any applicable vesting date, JPMorgan Chase may impose for any reason, as of such vesting date for such period as it may specify in its sole discretion, such restrictions on the Common Stock to be issued to you as it may deem appropriate, including, but not limited to, restricting the sale, transfer, pledging, assignment, hedging or encumbrance of such shares of Common Stock. Such restrictions described in the last sentence shall not impact your right to vote or receive dividends with respect to the Common Stock. By accepting this award, you acknowledge that during such specified period should there be a determination that the recovery provisions of this award apply, then you agree that you may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares subject to such restrictions (notwithstanding the limitation set forth in the “Right to Set Off” subsection above). (See sections captioned “Bonus Recoupment”, “Recovery of Erroneously Awarded Incentive-Based Compensation” and “Remedies”.)

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is “at-will” and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Notwithstanding anything herein to the contrary, if you (i) are subject to taxation under the Code, (ii) are a specified employee as defined in the JPMorgan Chase 2005 Deferred Compensation Plan and (iii) have incurred a separation from service (as defined in that Plan with the exception of death) and if any units/shares under this award represent deferred compensation as defined in Section 409A and such shares are distributable (under the terms of this award) within six months following, and as a result of your separation from service, then those shares will be delivered during the first calendar month after the expiration of six full months from date of your separation from service. Further, if your award is not subject to a substantial risk of forfeiture as defined by regulations issued under Section 409A of the Code, then the remainder of each calendar year immediately following each vesting date set forth in your Award Agreement shall be a payment date for purposes of distributing the vested portion of the award.

Change in Outstanding Shares: In the event of any change in the outstanding shares of Common Stock by reason of any stock dividend or split, recapitalization, issuance of a new class of common stock, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to stockholders of Common Stock other than regular cash dividends, the Committee will make an equitable substitution or proportionate adjustment, in the number or kind of shares of Common Stock or other securities issued or reserved for issuance pursuant to the Plan and to any PSUs outstanding under this award for such corporate events.

Other Equitable Adjustments: Except for the “Qualitative Performance Factor”, the Committee may make adjustments (up or down) to the award as it deems to be equitable, to maintain the intended economics of the award in light of changed

circumstances, which may include unusual or non-recurring events affecting the Firm (or the Performance Companies) or its financial statements in each case resulting from changes in accounting methods, practices or policies, changes in capital structure by reason of legal or regulatory requirements and such other changed circumstances, as the Committee may deem appropriate.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under “No Ownership Rights/Other Limitations”, in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity: Upon receipt of satisfactory evidence that applicable United States federal, state, local, foreign or supranational ethics or conflict of interest laws or regulations require you to divest your interest in JPMorgan Chase PSUs, the Firm may accelerate the distribution of all or part of your outstanding award, effective on or before the required divestiture date and waive the Holding Requirement; provided that no accelerated distribution shall occur if the Firm determines that such acceleration will violate Section 409A of the Code. Accelerated distribution under this paragraph does not impact the dates as set forth in the “Remedies” section above. The time period for recovery shall be determined by the originally scheduled vesting date set forth in your Award Agreement or distribution date prior to any acceleration event.

If you have voluntarily terminated your employment and have satisfied the requirements of the section captioned “Government Office Requirements”, acceleration shall apply (to extent required) to the percentage of your outstanding award that would continue to vest under that section. In the case of a termination of employment where the award is outstanding as a result of the subsections entitled “Job Elimination” or “Full Career Eligibility”, then acceleration shall apply, to the extent required, to the full outstanding award. Subject to the two foregoing sections, the number of shares of Common Stock to be received on acceleration shall be determined using the methodology set forth under the section captioned “Death”.

To the extent you have vested shares under this award subject to the Holding Requirement and become subject to divestiture requirement as forth herein, the Firm may waive the holding period to the extent required.

Notwithstanding an accelerated distribution or waiver of the Holding Requirement pursuant to the foregoing, you will remain subject to the applicable terms of your Award Agreement as if your award had remained outstanding for the duration of the vesting period and shares had been distributed as scheduled as of each vesting date, including, but not limited to, repayment obligations set forth in the section captioned “Remedies” and the employment restrictions in the sections captioned “Protection-Based Vesting” and “Government Office Requirements” and the subsection “Full Career Eligibility”.

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm's legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm's By-Laws in the event of such a suit by the Firm.

Non-transferability: Neither this award or any other outstanding awards of restricted stock units or of performance-based share units, nor your interests or rights in any such awards, shall be assigned, pledged, transferred, hedged, hypothecated or subject to any lien. An award may be transferred following your death by will, the laws of descent or by a beneficiary designation on file with the Firm.

Definitions

"Average Tangible Common Equity" means annual average common stockholders' equity less annual average goodwill and annual average identifiable intangible assets. Annual averages of the components of Average Tangible Common Equity will be calculated using quarterly balances as reported in publicly available financial disclosures. In the event that quarterly balances are not available, annual year end balances will be used. This calculation is used solely for purposes of the Performance Ranking.

"Award Payout Percentage" means the applicable percentage specified in the Performance Table.

"Calculated PSUs" means the number of PSUs determined by multiplying the Target Award Number (after giving effect to any cancellation thereof, in whole or in part) by the Award Payout Percentage corresponding to the Firm's Performance Ranking based on the three-year average performance for the Performance Period (both percentage and ranking, as set forth in the Performance Table); provided that if the average of the Firm's Reported ROTCE, expressed as a decimal rounded to the nearest hundredth, for the Performance Period either equals or exceeds _____% or is less than _____%, _____ percent or _____, respectively as the case may be, shall be substituted for the Performance Period's Award Payout Percentage in calculating the number of PSUs to distribute. For avoidance of doubt, any cancellation of this award (in whole or in part) during the Performance Period will reduce the Target Award Number.

"Calculation Agent" means a third-party entity not owned or controlled by the Firm, such as an accounting or consulting firm, retained from time to time by the Director of Human Resources or his/her delegate.

“Cause” means a determination by the Firm that your employment terminated as a result of your (i) violation of any law, rule or regulation (including rules of self-regulatory bodies) related to the Firm’s business, (ii) indictment or conviction of a felony, (iii) commission of a fraudulent act, (iv) violation of the JPMorgan Chase Code of Conduct or other Firm policies or misconduct related to your duties to the Firm (other than immaterial and inadvertent violations or misconduct), (v) grossly inadequate performance of the duties associated with your position or job function or failure to follow reasonable directives of your manager, or (vi) any act or failure to act that is injurious to the interests of the Firm or its relationship with a customer, client or an employee.

“Financial Services Company” means a business enterprise that engages in any of the following services (itself or through an affiliate or subsidiary), regardless of whether such services are the principle strategy or revenue-generating activity:

- commercial or retail banking, including, but not limited to, commercial, institutional and personal trust, custody and/or lending and processing services, internet banking, originating and servicing mortgages, issuing and servicing credit cards, payment servicing or processing or merchant services,
- insurance, including but not limited to, guaranteeing against loss, harm, damage, illness, disability, or death, providing and issuing annuities, acting as principal, agent or broker for purpose of the forgoing,
- financial, investment or economic advisory services, including but not limited to, investment banking services (such as advising on mergers or dispositions, underwriting, dealing in, or making a market in securities or other similar activities), brokerage services, investment management services, asset management services, foreign exchange services, interbank networks and hedge funds,
- issuing, trading or selling instruments representing interests in pools of assets or in derivatives instruments,
- financial technology companies, such as those selling blockchain services, or offering or selling financial products/services,
- advising on, or investing in, private equity or real estate funds or ventures, or
- any similar activities that the Director of Human Resources or nominee determines in his or her sole discretion constitute financial services,
- however, subject to disclosure and confirmation of the following, it shall not be considered a “Financial Services Company” for the purpose of this Award Agreement if such family office is solely funded by you and/or your family’s own personal funds without external investors directly or indirectly.

“Firm Reported ROTCE” means the Firm’s percentage return on tangible common equity for each year in the Performance Period.

“Firmwide Financial Threshold” means a cumulative return on tangible common equity for calendar years 20XX, 20XX and 20XX of not less than _____. Cumulative return on tangible common equity means (i) the sum of the Firm’s reported net income for all three calendar years, divided by (ii) reported year-end tangible equity averaged over the three years.

“Government Office” means (i) a full-time position in an elected or appointed office in local, state, or federal government (including equivalent positions outside the U.S. or in a supranational organization), not reasonably anticipated to be a full-career position, or (ii) conducting a bona fide full-time campaign for such an elective public office after formally filing for candidacy, where it is customary and reasonably necessary to campaign full-time for the office.

“Line of Business” means a business unit of the Firm (or one or more business units designated below under the definition “Line of Business Financial Threshold” of the Commercial & Investment Bank). All Corporate Functions (including the functions of the Chief Investment Office) are considered a single Line of Business.

“Line of Business Financial Threshold” means the financial threshold set forth below: for the following Lines of Business based on the Firm’s management reporting system:

Asset & Wealth Management	Annual negative pre-tax pre-provision income ¹
Card and Auto	Annual negative pre-tax pre-loan loss reserve income ²
Commercial & Investment Bank	Annual negative pre-tax pre-provision income ¹ for CIB overall or annual negative allocated product revenues (excluding XVA) for: • Fixed Income • Equities • Securities Services • Global Investment Banking • Payments
Consumer Banking, J.P. Morgan Wealth Management and Business Banking	Annual negative pre-tax pre-loan loss reserve income ²
Corporate Functions (including Chief Investment Office)	Annual negative pre-tax pre-provision income ¹ at the Firm level
Home Lending	Annual negative pre-tax pre-loan loss reserve income ²
¹ Pre-tax pre-provision income means Revenue less Expenses ² Pre-tax pre-loan loss reserve income means Revenue less (Expenses plus Net Charge-offs)	

“Not-for-Profit Organization” means an entity exempt from tax under state law and under Section 501(c)(3) of the Code. Section 501(c)(3) only includes entities organized and operated exclusively for religious, charitable, scientific, testing for public safety, literary or educational purposes, or to foster national or international amateur sports competition or for the prevention of cruelty to children or animals. Not-for-Profit Organization shall also mean entities outside the United States exempt from local and national tax laws because they are organized and operated exclusively for purposes identical to those applicable to Section 501(c)(3) organization.

“Performance Companies” mean the following institutions which have business activities that overlap with a significant portion of the Firm’s revenue mix: _____, and _____.

If, during the Performance Period, one or more Performance Companies shall merge, engage in a spin-off or otherwise experience a material change in its revenue mix or business activities or its existence or its primary businesses shall terminate or cease due to receivership, bankruptcy, sale, or otherwise, then the Committee may eliminate such institution from the list of Performance Companies or make such other equitable adjustments, such as adding an acquirer or a new company to the list of Performance Companies, as it deems appropriate, with any such changes having effect for purposes of all calculations hereunder on a prospective basis from the date the applicable change is made.

“Performance Period” means calendar years 20XX, 20XX and 20XX.

“Performance Ranking” means the ranking of the average ROTCE of the Firm as compared to the ranking of the average ROTCE of the Performance Companies as specified in the Performance Table for the Performance Period.

“Performance Table” means the table used in the calculation of PSUs for the Performance Period as set forth below:

Firm Reported ROTCE (average performance)	Award Payout Percentage		Firm Performance Ranking (average performance)	Award Payout Percentage
>= _____%	_____%		#1	_____%
			#2	_____%
			#3	_____%
_____% to _____%	Pay by relative ROTCE scale	---	#4	_____%
			#5	_____%
			#6	_____%
			#7	_____%
< _____%	_____%		#8	_____%
			#9	_____%
			#10 +	_____%

If, after the calculation of the Performance Ranking, there is a tie, the tie shall be disregarded for purposes of determining the Award Payout Percentage. For example, in the case of a tie for the fourth ranking between the Firm and a Performance Company, the Firm shall be treated as having satisfied that ranking. In the case of that same tie among Performance Companies, the fourth and fifth rankings will be deemed to have been satisfied.

“Recognized Service” means the period of service as an employee set forth in the Firm’s applicable service-related policies.

“ROTCE” means for the Firm and each of the Performance Companies a percentage derived by, for each year in the Performance Period, dividing (i) annual earnings from continuing operations less dividends on preferred stock as set forth in published financial disclosures by (ii) the Average Tangible Common Equity for the year. If, prior to the end of the vesting period, the Firm or any Performance Company restates its published financial statements for any year in the Performance Period, ROTCE for that year shall be recalculated for the Firm or Performance Company with the Performance Ranking adjusted, if necessary. This calculation is used solely for purposes of the Performance Ranking.

“Target Award Number” means the number of PSUs designated as such in the Award Agreement.

Government Office Requirements

You may be eligible to continue vesting in all or part of your award if you voluntarily resign to accept a Government Office (as defined above) or to become a candidate for an elective Government Office.

Full Career Eligibility:

“Government Office Requirements” does not apply to you if you satisfy the subsection captioned “--Full Career Eligibility” as of the date that you voluntarily terminate your employment with the Firm.

Eligibility:

Eligibility for continued vesting is conditioned on your providing the Firm:

- At least 60 days’ advance written notice of your intention to resign to accept or pursue a Government Office (see section captioned “Definitions”), during which period you must perform in a cooperative and professional manner services requested by the Firm and not provide services for any other employer. The Firm may elect to shorten this notice period at the Firm’s sole discretion.
- Confirmation, in a form satisfactory to the Firm, that vesting in this award pursuant to this provision would not violate any applicable law, regulation or rule.
- Documentation in a form satisfactory to the Firm that your resignation is for the purpose of accepting a Government Office or becoming a candidate for a Government Office. (See section captioned “Definitions”).

Portion of Your Award Subject to Continued Vesting:

Subject to the conditions below, the percentage of this award that will continue to vest in accordance with this award’s original schedule will be based on your years of continuous service completed with the Firm immediately preceding your termination date, as follows:

- 50% if you have at least 3 but less than 4 years of continuous service,
- 75% if you have at least 4 but less than 5 years of continuous service, or
- 100% if you have 5 or more years of continuous service.

The portion of this award subject to continued vesting above is referred to as the “CV Award” and the portion not subject to continued vesting will be cancelled as of the date your employment terminates.

Conditions for Continued Vesting of Award:

- You must remain in a non-elective Government Office for two or more years after your employment with the Firm terminates to be eligible to receive the CV Award; provided that if your non-elective Government Office is for a period less than two years, you will be eligible to receive the CV Award if it has a vesting date during your period of Government Service; or
- In the case of resignation from the Firm to campaign for an elective Government Office, your name must be on the primary or final public ballot for the election. (If you are not elected, see below for employment restrictions.)

For avoidance of doubt, the performance criteria and protection-based vesting set forth in these terms and conditions continue to apply to a CV Award.

Satisfaction of Conditions:

If your service in a Government Office ends two years or more after your employment with the Firm terminates, or in the case of resignation from the Firm to campaign for a Government Office, your name is on the primary or final public ballot for the election and you are not elected, any CV Awards then outstanding and any such awards that would have then been outstanding but for an accelerated distribution of shares (as described in the subsection captioned “--Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”) will be subject for the remainder of the applicable vesting period to the same terms and conditions of this Award Agreement, including employment restrictions during the vesting period, as if you had resigned from the Firm having met the requirements for Full Career Eligibility.

Failure to Satisfy Conditions:

If you do not satisfy the above “Conditions for Continued Vesting of Awards”, any outstanding PSUs under the CV Award will be cancelled. You also will be required to repay the Fair Market Value of the number of shares (before tax and other withholdings) of Common Stock distributed to you that would have been outstanding as PSUs on the date you failed to satisfy the “Conditions for Continued Vesting of Award” but for their accelerated distribution (as described in subsection

captioned, “Accelerated Distribution for Ethics or Conflict Reasons Resulting From Employment by a Government Entity”). Fair Market Value for this purpose will be determined as the date that the shares were distributed.

JPMORGAN CHASE & CO. LONG-TERM INCENTIVE PLAN
TERMS AND CONDITIONS OF JANUARY 21, 2025
RESTRICTED STOCK UNIT AWARD
Identified Staff – Retained Award

Award Agreement

These terms and conditions are made part of the Award Agreement dated as of January 21, 2025 (“Grant Date”) awarding Restricted Stock Units (“RSUs”) pursuant to the terms of the JPMorgan Chase & Co. Long-Term Incentive Plan (“Plan”). To the extent the terms of the Award Agreement (all references to which will include these terms and conditions) conflict with the Plan, the Plan will govern. The Award Agreement, the Plan and Prospectus supersede any other agreement, whether written or oral, that may have been entered into by the Firm and you relating to this award.

This award was granted on the Grant Date subject to the Award Agreement. **Unless you decline by the deadline and in the manner specified in the Award Agreement, you will have agreed to be bound by these terms and conditions, effective as of the Grant Date.** If you decline the award, it will be cancelled as of the Grant Date.

Capitalized terms that are not defined in the Award Agreement will have the same meaning as set forth in the Plan.

JPMorgan Chase & Co. will be referred to throughout the Award Agreement as “JPMorgan Chase” and together with its subsidiaries as the “Firm”.

Form of Award

Each RSU represents a non-transferable right to receive one share of Common Stock as of the applicable vesting date as set forth in your Award Agreement.

Holding Requirement

As of the vesting date, you shall be entitled to a distribution equal to the Fair Market Value of the number of RSUs vesting on such date, less the number being withheld to satisfy tax withholding obligations. You agree that the distribution made to you will be held in an account in your name with restrictions preventing you from transferring, assigning, hedging, selling, pledging, or otherwise encumbering such distribution for a twelve-month period commencing with the vesting date. Such restrictions shall lapse in event of your death.

Bonus Recoupment

In consideration of the grant of this award, you agree that you are subject to the JPMorgan Chase Bonus Recoupment Policy (or successor policy) as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award. You can access this policy as currently in effect by clicking the following link to the JPMorgan Chase & Co. Corporate Governance Principles web page and scrolling to the Bonus Recoupment Policy located under the section titled “Other Matters”:

<https://about.jpmorganchase.com/about/governance/corporate-governance-principles>

For the avoidance of doubt, nothing in these terms and conditions in any way limits the rights of the Firm under the JPMorgan Chase Bonus Recoupment Policy (or successor policy).

EMEA Malus and Clawback Policy – Identified Staff

In consideration of grant of this award, and without prejudice to any other provision of this Award Agreement, you agree that you are subject to the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff (and any applicable supplement(s) to that policy) or successor policy as in effect from time to time as it applies both to the cash incentive compensation awarded to you for performance year 2024 and to this award.

The provisions of the JPMorgan Chase EMEA Malus and Clawback Policy - Identified Staff set out the terms and conditions applying to the grant of this award which ensure that the Firm is able to meet its regulatory obligations to operate malus (reduce) and/or clawback (recover) to awards in certain circumstances. These include, but are not limited to, where (i) there is a material downturn in the Firm’s financial performance or (ii) where the Firm is required to hold more capital. The circumstances in which the events at (i) and (ii) would occur are analogous to some of the circumstances considered under the existing Firmwide terms and conditions, in particular the Bonus Recoupment Policy and the Protection Based Vesting provisions. You can access this policy as currently in effect in My Rewards through the following link: <https://myrewards.jpmorganchase.com>

Your Obligations

In consideration of the grant of this award, you agree to comply with and be bound by the obligations set forth below next to the subsections captioned “--Confidentiality & Non-Solicitation”, “--False Statements”, “--Cooperation”, “--Compliance with Award Agreement” and “--Notice Period.”

- **Confidentiality & Non-Solicitation**

During your employment by the Firm and for the one year period following the termination of your employment, you will not directly or indirectly, whether on your own behalf or on behalf of any other party, without the prior written consent of the Director of Human Resources: (A) solicit, induce or encourage any of the Firm's then current employees to leave the Firm or to apply for employment elsewhere, unless such current employee has received official, written notice that his or her employment will be terminated due to job elimination, (B) hire any employee or former employee who was employed by the Firm at the date your employment terminated, unless the individual's employment terminated because his or her job was eliminated, or the individual's employment with the Firm has been terminated for more than six months, (C) to the fullest extent enforceable under applicable law, solicit or induce or attempt to induce to leave the Firm, or divert or attempt to divert from doing business with the Firm, any then current customers, suppliers or other persons or entities that were serviced by you or whose names became known to you by virtue of your employment with the Firm, or otherwise interfere with the relationship between the Firm and such customers, suppliers or other persons or entities. This does not apply to publicly known institutional customers that you service after your employment with the Firm without the use of the Firm's confidential or proprietary information.

These restrictions do not apply to authorized actions you take in the normal course of your employment with the Firm, such as employment decisions with respect to employees you supervise or business referrals in accordance with the Firm's policies.

You will not, either during your employment with the Firm or thereafter, directly or indirectly use or disclose to anyone any Confidential Information (as defined herein) related to the Firm's business or its customers, except as explicitly permitted by the JPMorgan Chase Code of Conduct and applicable policies or law or legal process. “Confidential Information” includes but is not limited to: (i) information received by the Firm from third parties under confidential conditions; (ii) intellectual property and trade secrets, technical, product, business, financial, or development information from the Firm, the use or disclosure of which reasonably might be construed to be contrary to the interest of the Firm; or (iii) other proprietary information or data, including, but not limited to, customer lists. In addition, following your termination of employment, you will not, without prior written authorization, access the Firm's private and internal information through telephonic, intranet or internet means.

For Employees with a Work Location in California, you agree that the restrictions in the first paragraph of this “Confidentiality and Non-Solicitation” provision shall not apply to you. However, you agree that, as a condition of receipt of this award, you will not, for the longer of the (i) one year period following the termination of your employment or, (ii) if your award is not cancelled as of your termination date, the three year period from Grant Date, use “Confidential Information” in order to solicit or otherwise interfere in the relationship between the Firm and any current Firm customers, suppliers, or other persons or entities whose names become known to you by virtue of your employment with the Firm and in connection with your access to Confidential Information.

Nothing in this award precludes you from reporting to the Firm's management or directors, the government, a regulator, a self-regulatory agency, your attorneys or a court, conduct you believe to be in violation of the law or concerns of any known or suspected Code of Conduct violation. It is also not intended to prevent you from responding truthfully to questions or requests from the government, a regulator or in a court of law.

If you are required by law or requested to provide information to any private party, including the news media, related to your or anyone else's employment with the Firm, you will, in advance of providing any response (to the extent lawfully permitted), and within five days of receiving any such legal demand or request, provide written notice to the Firm. Additionally, you agree to cooperate with the Firm in connection with the request for such information to the extent lawfully permitted.

- **False Statements**

You will not, either during your employment with the Firm or thereafter, make any untrue statements, such that they are made with knowledge of their falsity or with reckless disregard for their truth or falsity, about the Firm, its employees, officers, directors or shareholders as a group in verbal, written, electronic or any other form. This shall not preclude you from reporting to the Firm's management or directors regarding conduct you believe to be in violation of

the law or preclude you from providing information to or cooperating with any government, regulator or law enforcement agency.

- **Cooperation**

You will cooperate with any Firm investigation, inquiry, or litigation, and provide full and accurate information to the Firm and its counsel with respect to any matter that relates to issues or events about which you may have knowledge or information, subject to reimbursement for actual, appropriate and reasonable out-of-pocket expenses incurred by you. This Agreement does not restrict you from communicating with any federal, state, or local government, regulatory, or law enforcement agency or otherwise participating in any investigation or proceeding that may be conducted by any such agency, including providing documents or other information without notice to the Firm.

- **Compliance with Award Agreement**

You will provide the Firm with any information reasonably requested to determine compliance with the Award Agreement, and you authorize the Firm to disclose the terms of the Award Agreement to any third-party who might be affected thereby, including your prospective employer.

Remedies

- **Detrimental Conduct, Risk Related and Other Cancellation/Recapture**

Notwithstanding any terms of this Award Agreement to the contrary, JPMorgan Chase reserves the right to recover from you up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed (including shares withheld for tax purposes) under this award if the Firm in its sole discretion determines that:

- Any of the following detrimental and risk-related conduct has occurred:
 - you engaged in conduct detrimental to the Firm insofar as it causes material financial or reputational harm to the Firm or its business activities, or
 - this award was based on materially inaccurate performance metrics, whether or not you were responsible for the inaccuracy, or
 - this award was based on a material misrepresentation by you, or
 - you improperly or with gross negligence failed to identify, raise, or assess, in a timely manner and as reasonably expected, risks and/or concerns with respect to risks material to the Firm or its business activities, or
 - your employment was terminated for Cause (see section captioned “Definitions” below) or, in the case of a determination after the termination of your employment, that your employment could have been terminated for Cause.

- **Recovery**

You may be required to pay the Firm up to an amount equal to the Fair Market Value (determined as of the applicable vesting date) of the gross number of shares of Common Stock previously distributed under this award as follows:

- Payment may be required with respect to any shares distributed within the three-year period prior to a notice-of-recovery under this section, if the Firm in its sole discretion determines that:
 - you committed a fraudulent act, or engaged in knowing and willful misconduct related to your employment, or
 - you violated any of the provisions as set forth above in the section captioned “Your Obligations”, or
- In addition, payment may be required with respect to any shares distributed within the one-year period prior to notice-of-recovery under this section, if the Firm in its sole discretion determines appropriate as a result of the detrimental and risk-related conduct listed in the above “Detrimental Conduct, Risk Related and Other Cancellation/Recapture” subsection.

Notice-of-recovery under this subsection is a written (including electronic) notice from the Firm to you either requiring payment under this subsection or stating that JPMorgan Chase is evaluating requiring payment under this subsection. Without limiting the foregoing, notice-of-recovery will be deemed provided if the Firm makes a good faith attempt to provide written (including electronic) notice at your last known address maintained in the Firm’s employment records. For the avoidance of doubt, a notice-of-recovery that the Firm is evaluating requiring payment under this subsection shall preserve JPMorgan Chase’s rights to require payment as set forth above in all respects and the Firm shall be under no obligation to complete its evaluation other than as the Firm may determine in its sole discretion.

For purposes of this subsection, shares distributed under this award include shares withheld for tax purposes. However, it is the Firm’s intention that you only be required to pay the amounts under this subsection with respect to shares that are or may be retained by you following a determination of tax liability and that you will not be required to

pay amounts with respect to shares representing irrevocable tax withholdings or tax payments previously made (whether by you or the Firm) that you will not be able to recover, recapture or reclaim (including as a tax credit, refund or other benefit). Accordingly, JPMorgan Chase will not require you to pay any amount that the Firm or its nominee in his or her sole discretion determines is represented by such withholdings or tax payments.

Payment may be made in shares of Common Stock or in cash. You agree that any repayment will be a lawful recovery under the terms and conditions of your Award Agreement and is not to be construed in any manner as a penalty.

Nothing in the section in any way limits your obligations under “Bonus Recoupment” and “EMEA Malus and Clawback Policy - Identified Staff”.

- **Right to an Injunction**

You acknowledge that a violation or attempted violation of any of the provisions set forth in “Your Obligations” herein will cause immediate and irreparable damage to the Firm, and therefore agree that the Firm shall be entitled as a matter of right to an injunction, from any court of competent jurisdiction, restraining any violation or further violation of any of the provisions set forth in “Your Obligations”; such right to an injunction, however, shall be cumulative and in addition to whatever other remedies the Firm may have under law or equity.

Administrative Provisions

Withholding Taxes: As a result of legal and/or tax obligations the Firm, in its sole discretion, may (i) retain from each distribution the number of shares of Common Stock required to satisfy applicable tax obligations or (ii) implement any other desirable or necessary procedures, so that appropriate withholding and other taxes are paid to the competent authorities with respect to the vested shares and the award. This may include but is not limited to (i) a market sale of a number of such shares on your behalf substantially equal to the withholding or other taxes, (ii) to the extent required by law, withhold from cash compensation, an amount equal to any withholding obligation with respect to the award and shares that vest under this award, and (iii) retaining shares that vest under this award until you pay any taxes associated with the award and/or vested shares directly to the competent authorities.

Right to Set Off: Although the Firm expects to settle this award in share(s) of Common Stock as of the applicable vesting date, as set forth in your Award Agreement, the Firm may, to the maximum extent permitted by applicable law (including Section 409A of the Code to the extent it is applicable to you), retain for itself funds or the Common Stock resulting from any vesting of this award to satisfy any obligation or debt that you owe to the Firm. Notwithstanding any account agreement with the Firm to the contrary, the Firm will not recoup or recover any amount owed from any funds or unrestricted securities held in your name and maintained at the Firm pursuant to such account agreement to satisfy any obligation or debt owed by you under this award without your consent.

Binding Agreement: The Award Agreement will be binding upon any successor in interest to JPMorgan Chase, by merger or otherwise.

Not a Contract of Employment: Nothing contained in the Award Agreement constitutes a contract of employment or continued employment. Employment is “at-will” and may be terminated by either you or JPMorgan Chase for any reason at any time. This award does not confer any right or entitlement to, nor does the award impose any obligation on the Firm to provide, the same or any similar award in the future and its value is not compensation for purposes of determining severance.

Section 409A Compliance: To the extent that Section 409A of the Code is applicable to this award, distributions of shares hereunder are intended to comply with Section 409A of the Code, and the Award Agreement, including these terms and conditions, shall be interpreted in a manner consistent with such intent.

Interpretation/Administration: The Committee or its nominee has sole and complete authority to interpret and administer this Award Agreement, including, without limitation, the power to (i) interpret the Plan and the terms of this Award Agreement; (ii) determine the reason for termination of employment; (iii) determine application of the post-employment obligations and cancellation and recovery provisions; (iv) decide all claims arising with respect to this award; and (v) delegate such authority as it deems appropriate. Any determination contemplated hereunder by the Committee, the Firm, the Director of Human Resources or their respective nominees shall be binding on all parties.

Notwithstanding anything herein to the contrary, the determinations of the Director of Human Resources, the Firm, the Committee and their respective nominees under the Plan and the Award Agreements are not required to be uniform. By way of clarification, the Committee, the Firm, the Director of Human Resources and their respective nominees shall be entitled to make non-uniform and selective determinations and modifications under Award Agreements and the Plan.

Amendment: The Committee or its nominee reserves the right to amend this Award Agreement in any manner, at any time and for any reason; provided, however, that no such amendment shall materially adversely affect your rights under this Award Agreement without your consent except to the extent that the Committee or its nominee considers advisable to (x) comply with applicable laws or changes in or interpretation of applicable laws, regulatory requirements and accounting rules or standards and/or (y) make a change in a scheduled vesting date or impose the restrictions described above under “No Ownership Rights/Other Limitations”, in either case, to the extent permitted by Section 409A of the Code if it is applicable to you. This Award Agreement may not be amended except in writing signed by the Director of Human Resources of JPMorgan Chase.

Severability: If any portion of the Award Agreement is determined by the Firm to be unenforceable in any jurisdiction, any court or arbitrator of competent jurisdiction or the Director of Human Resources may reform the relevant provisions (e.g., as to length of service, time, geographical area, or scope) to the extent the Firm (or court/arbitrator) considers necessary to make the provision enforceable under applicable law.

Use of Personal Data: By accepting this award, you acknowledge that the Firm may process your personal data for the purposes of providing you this award (to include registration of shares and units or establishing a brokerage account on your behalf) and disclosing to third parties, such as service providers or tax and regulatory authorities (e.g., for compensation reporting and payroll tax withholding purposes). Additionally, you agree that the Firm may transfer your personal data to jurisdictions that do not afford protections equivalent to the protections in the country in which we collected your data. Where applicable law provides a right to terminate the foregoing authorization, you may do so at any time, except with respect to tax and regulatory reporting and the Firm’s legal and regulatory obligations. In the event you terminate this authorization, your award will be cancelled.

Governing Law: This award shall be governed by and construed in accordance with the laws of the State of New York, without regard to conflicts of law principles.

Choice of Forum: By accepting this award under the Plan, you agree (and have agreed) that to the extent not otherwise subject to arbitration under an arbitration agreement between you and the Firm, any dispute arising directly or indirectly in connection with this award or the Plan shall be submitted to arbitration in accordance with the rules of the American Arbitration Association if so elected by the Firm in its sole discretion. In the event such a dispute is not subject to arbitration for any reason, you agree to accept the exclusive jurisdiction and venue of the United States District Court for the Southern District of New York with respect to any judicial proceeding in connection with this award or the Plan. You waive, to the fullest extent permitted by law, any objection to personal jurisdiction or to the laying of venue of such dispute and further agree not to commence any action arising out of or relating to this award or the Plan in any other forum.

Waiver of Jury Trial/Class Claims: By accepting this award, you agree, with respect to any claim brought in connection with your employment with the Firm in any forum (i) to waive the right to a jury trial and (ii) that any judicial proceeding or arbitration claim will be brought on an individual basis, and you hereby waive any right to submit, initiate, or participate in a representative capacity or as a plaintiff, claimant or member in a class action, collective action, or other representative or joint action.

Litigation: By accepting any award under the Plan, you agree (and have agreed) that in any action or proceeding by the Firm (other than a derivative suit in the right of the Firm) to enforce the terms and conditions of this Award Agreement or any other Award Agreement where the Firm is the prevailing party, the Firm shall be entitled to recover from you its reasonable attorney fees and expenses incurred in such action or proceeding. In addition, you agree that you are not entitled to, and agree not to seek, advancement of attorney fees and indemnification under the Firm’s By-Laws in the event of such a suit by the Firm.

Exhibit 19

Insider Trading Policy – Firmwide

Effective Date: January 31, 2025

1. Summary or Rationale

This Insider Trading Policy – Firmwide (“Policy”) provides guidelines to promote compliance by Firm Personnel and their Connected Persons with the provisions of the U.S. federal securities laws that prohibit persons who are in possession of Firm Material Nonpublic Information from: (i) trading in Firm Securities; or (ii) impermissibly providing Firm Material Nonpublic Information to other persons who may trade on the basis of that information (i.e., “tipping”).

This Policy applies to Directors, Employees and Non-Employees (each, a “Firm Person” and, collectively, “Firm Personnel”), as well as their Connected Persons, and extends to the securities accounts of such persons and any other securities account that such persons could be expected to influence or control.

This Policy applies to all Firm Personnel and their Connected Persons in all jurisdictions in accordance with and subject to Firm policies, even if the activities prohibited in this Policy are not illegal in the jurisdiction where any particular Firm Person or Connected Person is located.

In addition, neither JPMC nor any Firm subsidiary may trade in Firm Securities while in possession of Firm Material Nonpublic Information, consistent with applicable law.

2. Defined Terms

CEO	The Chief Executive Officer of JPMC
CFO	The Chief Financial Officer of JPMC
Closed Window Period	Any period outside of an Open Window Period
Connected Persons	The following individuals and entities who, based on their relationship with a Firm Person, are subject to provisions of this Policy: • The spouse, domestic partner or minor children (even if financially independent) of a Firm Person • Anyone to whom a Firm Person provides significant financial support or for whom a Firm Person, or anyone listed above, has or shares the power, directly or indirectly, to make investment decisions • Any corporations, partnerships, trusts or other entities that are controlled by a Firm Person and/or anyone listed above

Corporate Officers	Members of the JPMC Operating Committee and the Firmwide Controller
ESPP	JPMorgan Chase Employee Stock Purchase Plan
Exchange Act	The Securities Exchange Act of 1934, as amended
Firm	JPMC and its direct and indirect subsidiaries
Firm MNPI	Material Nonpublic Information relating to the Firm or Firm Securities
Firm Personnel	The following persons are Firm Personnel (each a Firm Person) within the meaning of this Policy: • Directors: Any member of the Board of Directors of JPMC • Employees: Any person directly employed by the Firm • Non-Employees: Contingent workers and other types of non-employee personnel (including any director of a Firm subsidiary) who are not directly employed by the Firm but have access to JPMC facilities and/or internal JPMC systems
Firm Securities	Firm Securities means: • Securities and other financial instruments (such as common stock, preferred stock and bonds) issued by JPMC or any Firm subsidiary • Options, put or call options, forward contracts, warrants, swaps (including credit default swaps) and other derivative securities tied to any of the above Baskets, indices and exchange traded funds that (i) consist of 30 or fewer components and (ii) are comprised of a considerable portion (greater than 3%) of the above are also deemed to be Firm Securities for purposes of this Policy
GC	The General Counsel of JPMC
GRAT	Grantor Retained Annuity Trust
JPMC	JPMorgan Chase & Co.
Material Nonpublic Information or MNPI	Information about an issuer of financial instruments (including private companies that issue publicly traded financial instruments) that is not known by the public, but if it were, is reasonably likely to affect the market price of the financial instruments to which the information relates or is likely to influence a reasonable investor to buy or sell those financial instruments, or would be likely to be used as part their investment decision regarding those financial instruments, as applicable

Open Window Period	Generally, each period beginning the day after public release of JPMC's quarterly earnings and ending on the 15th day of the following month, provided that an Open Window Period may be closed or suspended at any time if circumstances require such action with respect to some or all Window List Personnel
Policy	This Insider Trading Policy – Firmwide
SAR	Stock Appreciation Right
Trading or Transactions	For purposes of this Policy, the terms “trading” or “transactions” (and variations thereof) include any self-directed (i) purchases or sales of, or any other transactions in, Firm Securities; (ii) making or modifying any instructions or investment options relating to any transaction in Firm Securities; and (iii) gifts of Firm Securities
Window List Personnel	Firm Personnel whose position, the nature of their work or their access to information at the Firm indicates that they have or may have access to Firm MNPI Firm Personnel are notified when they become Window List Personnel.
10b5-1 Plan	A plan that meets the requirements set forth in Rule 10b5-1(c)(1) under the Exchange Act

3. Prohibition on Insider Trading

Insider trading is illegal and strictly prohibited by the Firm. Firm Personnel and their Connected Persons must never trade or engage in any transaction in Firm Securities while in possession of Firm MNPI. In addition, neither JPMC nor any Firm subsidiary may trade in Firm Securities when in possession of Firm MNPI, consistent with applicable law. When JPMC possesses or may be deemed to possess Firm MNPI, it will generally observe a “blackout period” during which it will not trade in Firm Securities. However, the Firm may repurchase its common stock under a 10b5-1 Plan during the blackout period preceding each quarterly earnings announcement.

4. Prohibition on Unauthorized Disclosure

Firm Personnel and their Connected Persons must not impermissibly provide Firm MNPI to any person who may trade on the basis of that information (i.e., “tipping”). In addition, Firm MNPI must not be shared with any person unless the disclosure complies with the Firm’s policies concerning the safeguarding of confidential information, including the Code of Conduct.

5. Policies Applicable to Window List Personnel

5.1 Window List Personnel

Firm Personnel who have or may have access to Firm MNPI are designated as Window List Personnel. Firm Personnel are notified via email when they become Window List Personnel.

Window List Personnel and their Connected Persons are prohibited from trading in Firm Securities during any Closed Window Period. During any Open Window Period, Window List Personnel and their Connected Persons are permitted to trade in Firm Securities after obtaining preapproval from the Compliance Department. Any such approval is valid for the length indicated on the approval notification which is generally the end of the next business day, unless otherwise stipulated. An Open Window Period may be closed or suspended at any time if circumstances require such action with respect to some or all persons designated as Window List Personnel. If preapproval for a transaction is requested during a Closed Window Period, the Window List Person making the request will receive a denial notification from Compliance and will not be authorized to execute the trade. In extremely rare circumstances, an exception may be granted in consultation with the Window List Person's manager, Compliance and Legal.

Non-Window List Personnel are not subject to the preapproval requirements described in this Section 5.1 and in Section 6.2 or the minimum holding period requirements described in Section 7.3, but regional or line of business-specific restrictions may apply.

For additional requirements for Corporate Officers and Directors, see Section 6.

5.2 Window Covered Transactions

Window List Personnel and their Connected Persons may only engage in the following types of transactions in Firm Securities, subject to compliance with preapproval requirements and the requirement that the transaction be conducted during an Open Window Period:

- Direct purchases, sales and gifts of JPMC common stock or other Firm Securities, including self-directed sales to meet a margin call;
- Liquidating holdings of Firm Securities under the ESPP or the Firm's deferred compensation plan;
- Placing, cancelling or amending limit orders (i.e., an order to buy or sell a stock at a specific price or better) with respect to Firm Securities;
- Entering into, amending or terminating 10b5-1 Plans with respect to Firm Securities; and

- Use of previously-owned JPMC shares or share withholding for option exercises, unless as part of a 10b5-1 Plan entered into during an Open Window Period.

Additional restrictions apply to JPMC Employee Stock Options and JPMC Stock Appreciation Rights.

The following transactions are not considered to be restricted transactions for Window List Personnel and their Connected Persons and are not required to be preapproved:

- Dividend reinvestments and automatic purchases of JPMC common stock in accordance with previously made employee benefit elections (e.g., the ESPP) entered into (or amended) and preapproved during an Open Window Period;
- Transactions under a 10b5-1 Plan entered into (or amended) and preapproved during an Open Window Period; and
- Limit orders and derivatives that are executed during a Closed Window Period that were placed during an Open Window Period.

6. Policies Applicable to Corporate Officers and Directors

6.1 Transactions that are Required to be Pursuant to a 10b5-1 Plan

When a Corporate Officer or Director wishes to engage in an open market sale of Firm Securities, the transaction must be made pursuant to a 10b5-1 Plan.

When a Corporate Officer or Director wishes to (i) exercise and hold a SAR or option, (ii) make a charitable gift of Firm Securities, (iii) create a GRAT relating to Firm Securities or (iv) transfer Firm Securities to or from a GRAT or another trust, the transaction requires pre-approval under this Policy but is not required to be pursuant to a 10b5-1 Plan.

A Corporate Officer or Director may enter into a 10b5-1 Plan subject to compliance with the Firm's other share retention policies. To enter into a 10b5-1 Plan, a Corporate Officer or Director must comply with the preapproval requirements described in Section 6.2 of this Policy.

All 10b5-1 Plans must comply with all of the requirements set forth in Rule 10b5-1(c) and may be entered into, amended or terminated only during an Open Window Period.

6.2 Pre-approval for Firm Securities Transactions and Entry into 10b5-1 Plans

During an Open Window Period, Corporate Officers and Directors are permitted to trade in Firm Securities after obtaining the preapprovals described below.

- Corporate Officers and Directors must receive preapproval from the CEO and the GC for any transactions in Firm Securities that require

preapproval under Section 5 and for entry into, amendment of or termination of a 10b5-1 Plan under Section 6.1.

- In the case of transactions or 10b5-1 Plans entered into by the GC, preapproval is required from the CEO,
- in the case of transactions or 10b5-1 Plans entered into by the Firmwide Controller, approval is required from the GC and the CFO, and
- in the case of transactions or 10b5-1 Plans entered into by the CEO, approval is required from the GC.

The preapprovals described above for Corporate Officers are submitted to the Compliance Department for its preapproval, which is valid for the length indicated on the approval notification which is generally the end of the next business day, unless otherwise stipulated.

7. Other Prohibitions

Other restrictions that are applicable to transactions in Firm Securities are described below.

7.1 Prohibited Trading Activity

Trading with respect to Firm Securities is prohibited for Window List Personnel and all other Firm Personnel who are subject to personal account dealing restrictions under Firm policies in the following:

- Forwards and futures (all asset classes)
- Swaps (all asset classes)
- Uncovered option strategies except Long-Term Equity Anticipation Securities (LEAPS)
- Short sales

Except as set forth in Section 7.2, direct hedging strategies (e.g., covered calls) with respect to Firm Securities are permitted but must be proportionate to the risk (i.e., fully covered with underlying shares).

All Employees are prohibited from hedging or pledging unvested shares of JPMC common stock received as compensation.

7.2 Anti-Hedging/Anti-Pledging

Shares of JPMC common stock owned outright or through deferred compensation by a member of the JPMC Operating Committee or Director may not be held in margin accounts or otherwise pledged as collateral, nor may the economic risk of such shares be hedged.

7.3 Minimum Holding Period

Window List Personnel and their Connected Persons are required to hold Firm Securities for a minimum holding period (specific to each region and/or line of business) prior to transacting in those Firm Securities.

Vested shares of JPMC common stock or shares of JPMC common stock received in connection with purchases under the ESPP are deemed to have satisfied the minimum holding period requirement.

8. Post-termination Transactions

If a Firm Person is aware of Firm MNPI when their employment, directorship or other service relationship with the Firm terminates, that Firm Person and their Connected Persons may not trade in Firm Securities or disclose any Firm Material Nonpublic Information until such information has become public or is no longer material.

9. Personal Responsibility for Compliance with this Policy and Potential Disciplinary Action

This Policy is designed to reduce the risk of violations of the insider trading provisions of the U.S. federal securities laws, but compliance with this Policy is not an assurance that an insider trading violation will not be found to have occurred. The ultimate responsibility for adhering to this Policy and avoiding improper trading rests exclusively with Firm Personnel and their Connected Persons. Any action on the part of Firm Personnel and their Connected Persons pursuant to this Policy (or otherwise) does not insulate an individual from liability under applicable securities laws and this Policy does not constitute legal advice in any way. The existence of a personal financial emergency does not excuse anyone subject to this Policy from compliance with this Policy and the laws prohibiting insider trading.

In addition to potential criminal and civil liability, failure to comply with the requirements of this Policy may result in the receipt of Policy reminders or violations which may lead to disciplinary action, up to and including termination of employment (subject to local laws and regulations). The Firm also reserves the right to instruct an individual to cancel any trade at his or her expense.

For further information or questions regarding this Policy, Firm Personnel may reach out to the Compliance Department or Office of the Secretary. Firm Personnel should also refer to the Code of Conduct and other related Firm policies.

Exhibit 21

JPMorgan Chase & Co.

List of subsidiaries

While there are a number of subsidiaries that are required to be reported for various purposes to bank regulators, the following is a list of JPMorgan Chase & Co.'s significant legal entity subsidiaries as of December 31, 2024, as defined by SEC rules. The list includes the parent company of significant subsidiaries even if the parent company did not meet the definition of a significant subsidiary. Excluded from the list are subsidiaries that, if considered in the aggregate, would not constitute a significant subsidiary under SEC rules as of December 31, 2024.

Also included in the list are certain subsidiaries that have been designated as material legal entities for resolution planning purposes under the Dodd-Frank Act that did not meet the definition of a significant subsidiary under SEC rules.

December 31, 2024 Name	Organized Under The Laws Of
JPMorgan Chase Bank, National Association	United States
Paymentech, LLC	United States
J.P. Morgan International Finance Limited	United States
JPMorgan Securities Japan Co., Ltd.	Japan
J.P. Morgan Capital Holdings Limited	United Kingdom
J.P. Morgan Securities plc	United Kingdom
J.P. Morgan SE	Germany
JPMorgan Chase Holdings LLC	United States
J.P. Morgan Services India Private Limited	India
JPMorgan Asset Management Holdings Inc.	United States
JPMorgan Distribution Services, Inc.	United States
JPMorgan Asset Management International Limited	United Kingdom
JPMorgan Asset Management (UK) Limited	United Kingdom
JPMorgan Asset Management Holdings (Luxembourg) S.à r.l.	Luxembourg
JPMorgan Asset Management (Europe) S.à r.l.	Luxembourg
J.P. Morgan Investment Management Inc.	United States
J.P. Morgan Broker-Dealer Holdings Inc.	United States
J.P. Morgan Securities LLC	United States

Exhibit 22.2
JPMorgan Chase & Co.

JPMorgan Chase & Co. guarantee of subsidiary issuances

Securities	Guarantor
JPMorgan Chase Financial Company LLC issued, from time to time, its Global Medium-Term Notes, Series A, under the Indenture dated February 19, 2016 ("Series A Notes"), that are each fully and unconditionally guaranteed by JPMorgan Chase & Co. In addition, JPMorgan Chase Financial Company LLC may issue, from time to time, debt securities (including its Series A Notes) and warrants that are each fully and unconditionally guaranteed by JPMorgan Chase & Co. under the Registration Statement on Form S-3 (Registration Statement Nos. 333-270004 and 333-270004-01), which was declared effective on April 13, 2023.	JPMorgan Chase & Co.

Exhibit 23

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We hereby consent to the incorporation by reference in the Registration Statements on:

Form S-3

(No. 333-270004)

(No. 333-270004-01)

(No. 333-263304)

Form S-8

(No. 333-272306)

(No. 333-272303)

(No. 333-272302)

(No. 333-272299)

(No. 333-219702)

(No. 333-219701)

(No. 333-219699)

(No. 333-185584)

(No. 333-185582)

(No. 333-185581)

(No. 333-175681)

(No. 333-158325)

(No. 333-142109)

(No. 333-125827)

(No. 333-112967)

of JPMorgan Chase & Co. or JPMorgan Chase Financial Company LLC of our report dated February 14, 2025 relating to the financial statements and the effectiveness of internal control over financial reporting, which appears in this Form 10-K.

/s/ PricewaterhouseCoopers LLP

New York, New York

February 14, 2025

Exhibit 31.1
JPMorgan Chase & Co.

CERTIFICATION

I, James Dimon, certify that:

1. I have reviewed this Annual Report on Form 10-K of JPMorgan Chase & Co.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 14, 2025

/s/ James Dimon

James Dimon
Chairman and Chief Executive Officer

Exhibit 31.2
JPMorgan Chase & Co.

CERTIFICATION

I, Jeremy Barnum, certify that:

1. I have reviewed this Annual Report on Form 10-K of JPMorgan Chase & Co.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):
 - (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 14, 2025

/s/ Jeremy Barnum

Jeremy Barnum
Executive Vice President and Chief Financial Officer

Exhibit 32
JPMorgan Chase & Co.

**CERTIFICATION PURSUANT TO 18 U.S.C. SECTION 1350, AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002**

In connection with the Annual Report of JPMorgan Chase & Co. on Form 10-K for the period ended December 31, 2024 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), each of the undersigned officers of JPMorgan Chase & Co., certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that:

1. The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934; and
2. The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of JPMorgan Chase & Co.

Date: February 14, 2025

By: /s/ James Dimon

James Dimon
Chairman and Chief Executive Officer

Date: February 14, 2025

By: /s/ Jeremy Barnum

Jeremy Barnum
Executive Vice President and Chief Financial Officer

This certification accompanies this Annual Report and shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, or otherwise subject to the liability of that Section.

A signed original of this written statement required by Section 906 has been provided to, and will be retained by, JPMorgan Chase & Co. and furnished to the Securities and Exchange Commission or its staff upon request.

Exhibit 97

Policy for the Recovery of Erroneously Awarded Incentive-Based Compensation

Effective Date: October 2, 2023

1. Summary or Rationale

In accordance with the applicable rules of The New York Stock Exchange (the “NYSE”) Listed Company Manual, Section 10D of the Securities Exchange Act of 1934, as amended (“Exchange Act”) and Rule 10D-1 thereunder (“Rule 10D-1”), JPMorganChase (the “Firm”) has adopted this policy (this “Policy”) to provide for the recovery of Erroneously Awarded Incentive-Based Compensation from Executive Officers in the event of a Restatement. All capitalized terms used and not otherwise defined herein shall have the meanings set forth in Section 5, below.

2. Scope

	Subject to	Role to Play
Lines of Business	All	
Corporate Functions	All	Human Resources
Locations	All	
Legal Entities	All	

2.1. Included in Scope

This Policy applies to Incentive-Based Compensation received by a person:

- after beginning service as an Executive Officer,
- who served as an Executive Officer at any time during the performance period for that Incentive-Based Compensation,
- while the Firm has a class of securities listed on a national securities exchange, and
- during the Recovery Period.

Notwithstanding the look-back requirement, the Firm is only required to apply this Policy to Incentive-Based Compensation received on or after October 2, 2023. This Policy will continue to apply after an employee has terminated.

For purposes of this Policy, Incentive-Based Compensation shall be deemed “received” in the Firm’s fiscal period during which the Financial Reporting Measure specified in the Incentive-Based Compensation award is attained, even if the payment or grant of the Incentive-Based Compensation occurs after the end of that period.

3. Changes from Previous Version

This is the first version of this Policy.

4. Policy Statement

The Firm shall recover reasonably promptly the Erroneously Awarded Incentive-Based Compensation in the event that the Firm is required to prepare a Restatement.

For clarity, the Firm's obligation to recover Erroneously Awarded Incentive-Based Compensation under this Policy is not dependent on if or when a Restatement is filed.

The Firm shall recover Erroneously Awarded Incentive-Based Compensation in compliance with this Policy except to the extent provided under Section 4.1.2 below.

4.1 Recovery of Erroneously Awarded Incentive-Based Compensation

4.1.1 In the event of a Restatement, the Firm will reasonably promptly recover the Erroneously Awarded Incentive-Based Compensation received in accordance with the NYSE rules and Rule 10D-1 as follows:

- For purposes of determining the relevant Recovery Period, the date that the Firm is required to prepare the Restatement ("Restatement Date") is the earlier to occur of:
 - the date the Board of Directors of the Firm (the "Board"), a committee of the Board, or the officer or officers of the Firm authorized to take such action if Board action is not required, concludes, or reasonably should have concluded, that the Firm is required to prepare a Restatement, or
 - the date a court, regulator, or other legally authorized body directs the Firm to prepare a Restatement.
- The Compensation & Management Development Committee of the Board (the "Committee") composed of independent directors responsible for executive compensation, or in the absence of such committee, a majority of the independent directors serving on the Board, shall determine the Erroneously Awarded Incentive-Based Compensation received by each Executive Officer during the relevant Recovery Period.
- The amount of Incentive-Based Compensation subject to recovery under this Policy is the amount of Incentive-Based Compensation received that exceeds the amount of Incentive-Based Compensation that otherwise would have been received had it been determined based on the restated amounts, computed without regard to any taxes paid (the "Erroneously Awarded Incentive-Based Compensation").
- For Incentive-Based Compensation based on (or derived from) the Firm's stock price or total shareholder return, where the Erroneously Awarded Incentive-Based Compensation is not subject to mathematical recalculation directly from the information in the applicable Restatement:
 - The amount to be recovered shall be based on a reasonable estimate of the effect of the Restatement on the Firm's stock price or total shareholder return upon which the Incentive-Based Compensation was received; and
 - The Firm shall maintain documentation of the determination of such reasonable estimate and provide such documentation to the NYSE.
- The Committee shall have discretion to determine the appropriate means of recovering Erroneously Awarded Incentive-Based Compensation based on the

particular facts and circumstances. Notwithstanding the foregoing, except as set forth in 4.1.2 below, in no event may the Firm settle for an amount that is less than the Erroneously Awarded Incentive-Based Compensation that is subject to recovery under this Policy.

- To the extent that the Executive Officer has already reimbursed the Firm for any Erroneously Awarded Incentive-Based Compensation received under any duplicative recovery obligations established by the Firm or applicable law, it shall be appropriate for any such reimbursed amount to be credited to the Erroneously Awarded Incentive-Based Compensation that is subject to recovery under this Policy.

1. Notwithstanding anything herein to the contrary, the Firm shall not be required to take the actions contemplated by Section 4.1.1 above if it is determined by the Committee that recovery would be impracticable and any of the following conditions are met:

- The direct expenses paid to a third party to assist in enforcing this Policy would exceed the amount to be recovered. Before making this determination, the Firm shall make a reasonable attempt to recover the Erroneously Awarded Incentive-Based Compensation, document such attempt(s) and provide such documentation to the NYSE;
- Recovery would violate home country law where that law was adopted prior to November 28, 2022, provided that, before determining that it would be impracticable to recover any Erroneously Awarded Incentive-Based Compensation based on violation of home country law, the Firm shall obtain an opinion of home country counsel, acceptable to the NYSE, that recovery would result in such a violation and a copy of the opinion shall be provided to the NYSE; or
- Recovery would likely cause an otherwise tax-qualified retirement plan, under which benefits are broadly available to employees of the Firm, to fail to meet the requirements of Section 401(a)(13) or Section 411(a) of the Internal Revenue Code of 1986, as amended, and regulations thereunder.

4.2 Disclosure Requirements

The Firm shall file all disclosures with respect to this Policy and recoveries under this Policy in accordance with the requirements of the U.S. Federal securities laws, including the disclosure required by the applicable Securities and Exchange Commission (“SEC”) filings.

4.3 Prohibition of Indemnification

Notwithstanding the terms of any indemnification arrangement or insurance policy with any individual covered by this Policy, the Firm shall not indemnify any Executive Officer or former Executive Officer against the loss of Erroneously Awarded Incentive-Based Compensation, including any payment or reimbursement for the cost of insurance obtained by any such covered individual to fund amounts recoverable under this Policy.

4.4 Administration; Amendment; Termination

All determinations under this Policy shall be made by the Committee, including determinations regarding how any recovery under this Policy is effected. Any

determinations of the Committee shall be final, binding and conclusive and need not be uniform with respect to each individual covered by this Policy.

The Committee may amend this Policy from time to time and may terminate this Policy at any time, in its discretion.

4.5 Other Recovery Rights

Any right of recovery under this Policy is in addition to, and not in lieu of, any other remedies or rights of recovery that may be available to the Firm under applicable law, regulation or rule or pursuant to the terms of any policy of the Firm or any provision in any employment agreement, equity award agreement, compensation plan, or any other agreement or arrangement.

5. Defined Terms

Executive Officer	An Executive Officer is the Firm's president, principal financial officer, principal accounting officer (or if there is no such accounting officer, the controller), any vice-president of the Firm in charge of a principal business unit, division, or function (such as sales, administration, or finance), any other officer who performs a policy-making function, or any other person who performs similar policy-making functions for the Firm. Policy-making function is not intended to include policy-making functions that are not significant. Identification of an Executive Officer for purposes of this Policy will include at a minimum executive officers identified pursuant to 17 CFR 229.401(b).
Financial Reporting Measure	Any measure that is determined and presented in accordance with the accounting principles used in preparing the Firm's financial statements, and any measures that are derived wholly or in part from such measure. Stock price and total shareholder return are also Financial Reporting Measures. A Financial Reporting Measure need not be presented within the Firm's financial statements or included in a filing with the SEC.
Incentive-Based Compensation	Any compensation that is granted, earned or vested based wholly or in part upon the attainment of a Financial Reporting Measure.
Recovery Period	With respect to any accounting restatement, the three completed fiscal years immediately preceding the Restatement Date, and if the Firm changes its fiscal year, any transition period of less than nine months within or immediately following those three completed fiscal years.
Restatement	An accounting restatement due to the material noncompliance of the Firm with any financial reporting requirement under the securities laws, including any required accounting restatement to correct an error in previously issued financial statements that is material to the previously issued financial statements (a "Big R" restatement), or that would result in a material misstatement if the error were corrected in the current period or left uncorrected in the current period (a "little r" restatement).